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Diener et al.

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(45) **Date of Patent: Sep. 9, 2025**

(54) **NATRIURETIC PEPTIDE RECEPTOR 1
ANTIBODIES AND METHODS OF USE**

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patent is extended or adjusted under 35
U.S.C. 154(b) by 66 days.

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US 2023/0365674 A1 Nov. 16, 2023

Related U.S. Application Data

(62) Division of application No. 17/348,949, filed on Jun.
16, 2021, now abandoned, which is a division of
application No. 16/897,935, filed on Jun. 10, 2020,
now Pat. No. 11,066,469.

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12, 2019.

(51) **Int. Cl.**

C07K 16/28 (2006.01)

C12N 15/63 (2006.01)

A61K 39/00 (2006.01)

A61K 39/395 (2006.01)

A61K 45/06 (2006.01)

A61P 9/00 (2006.01)

A61P 13/12 (2006.01)

C07K 14/58 (2006.01)

C07K 14/72 (2006.01)

(52) **U.S. Cl.**

CPC **C07K 16/2869** (2013.01); **C12N 15/63**
(2013.01); **A61K 39/3955** (2013.01); **A61K**
2039/505 (2013.01); **A61K 45/06** (2013.01);
A61P 9/00 (2018.01); **A61P 13/12** (2018.01);
C07K 14/58 (2013.01); **C07K 14/72** (2013.01);
C07K 2317/50 (2013.01); **C07K 2317/54**
(2013.01); **C07K 2317/55** (2013.01); **C07K**
2317/56 (2013.01); **C07K 2317/565** (2013.01);
C07K 2317/622 (2013.01); **C07K 2317/75**
(2013.01)

(58) **Field of Classification Search**

CPC . **C07K 16/2869**; **A61K 39/3955**; **C12N 15/63**
See application file for complete search history.

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(74) *Attorney, Agent, or Firm* — Sansun Yeh

(57) **ABSTRACT**

This disclosure relates to anti-Natriuretic Peptide Receptor 1
(NPR1) antibodies including agonist antibodies which are
able to activate the NPR1 receptor, pharmaceutical compo-
sitions comprising the same, and methods of treatment
comprising the same.

12 Claims, 40 Drawing Sheets

Specification includes a Sequence Listing.

(56)

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FIG. 1

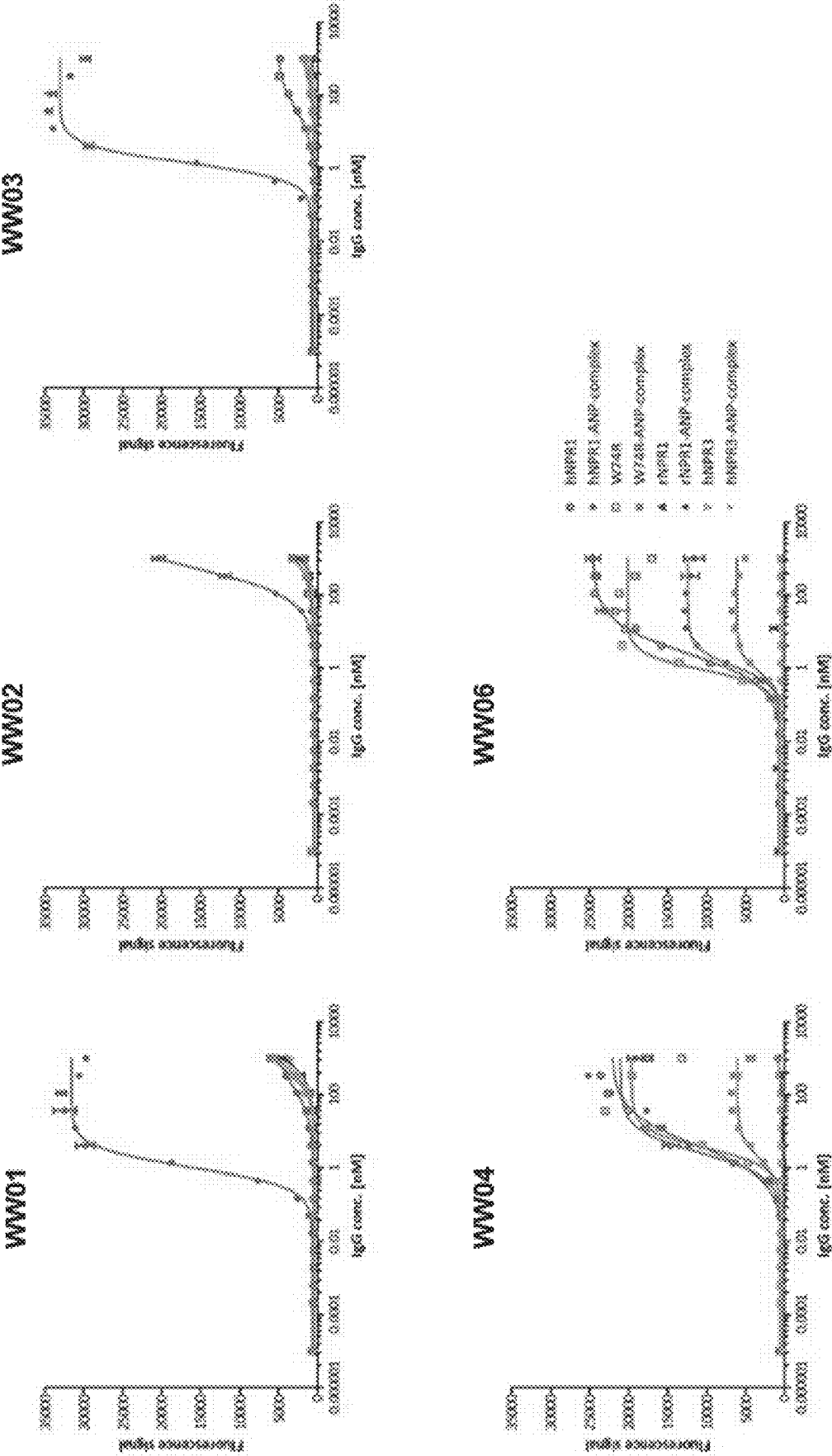


FIG. 2

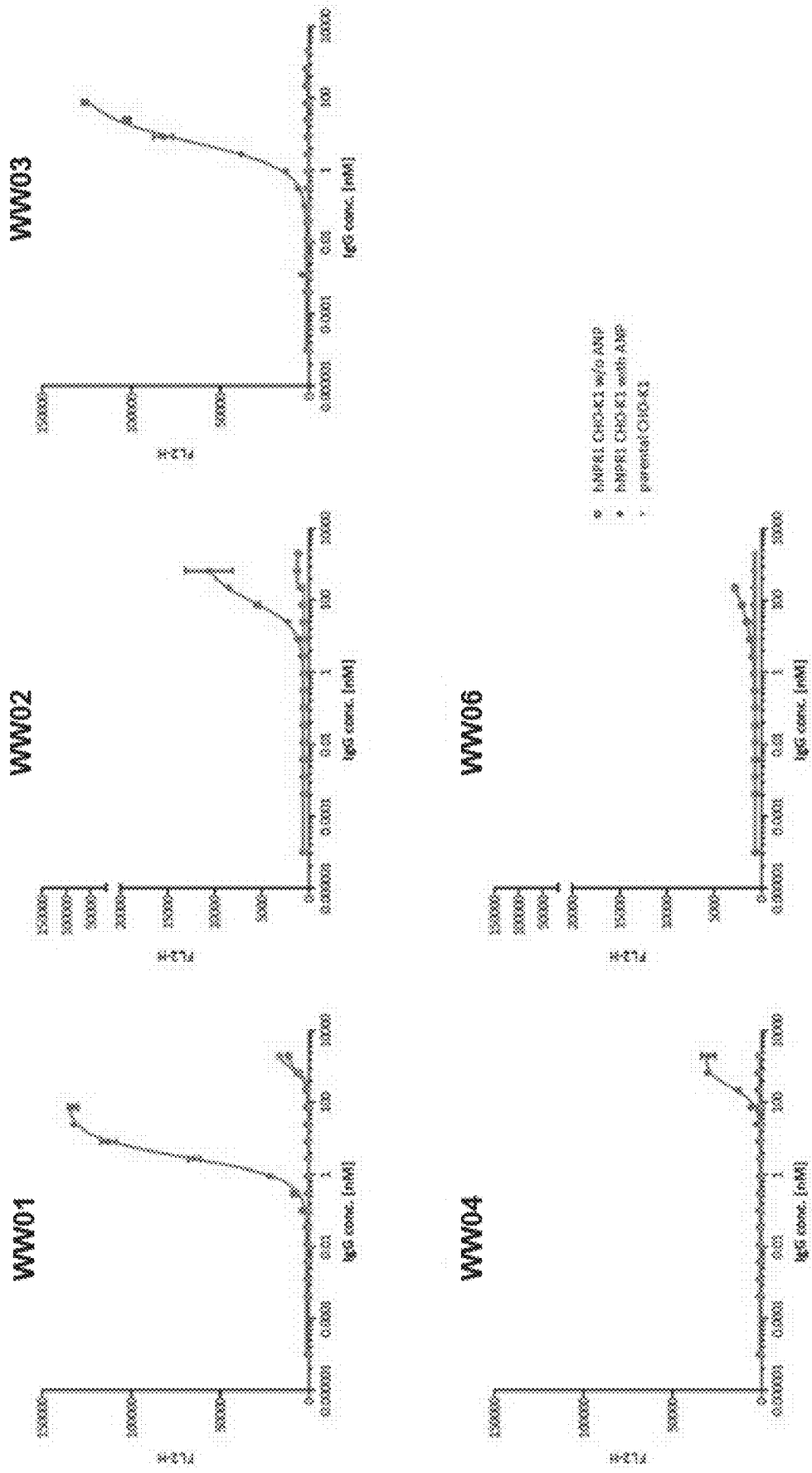


FIG. 3

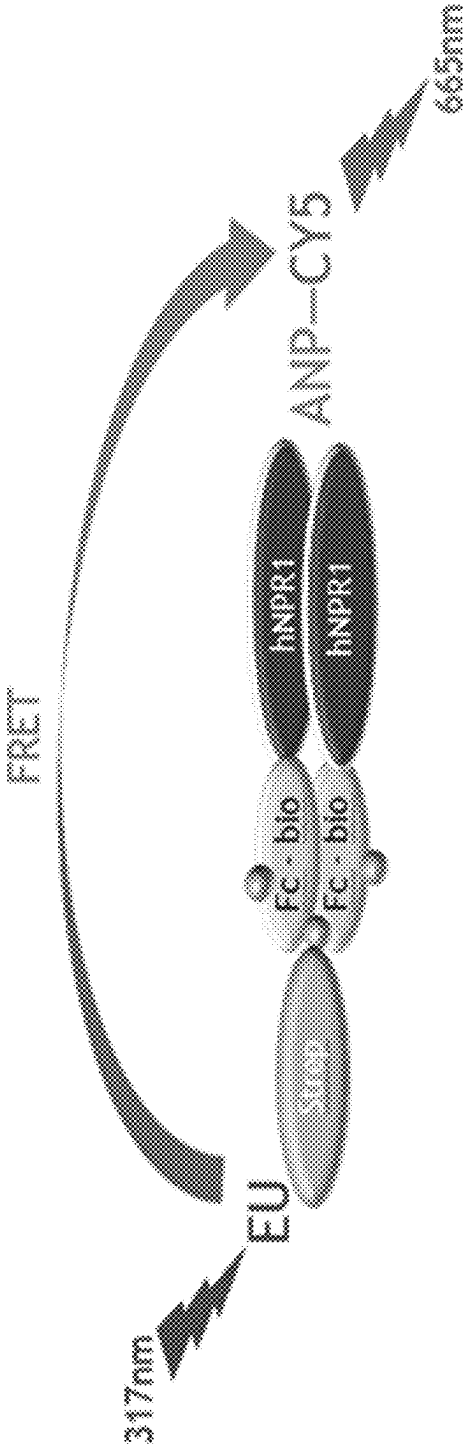


FIG. 4

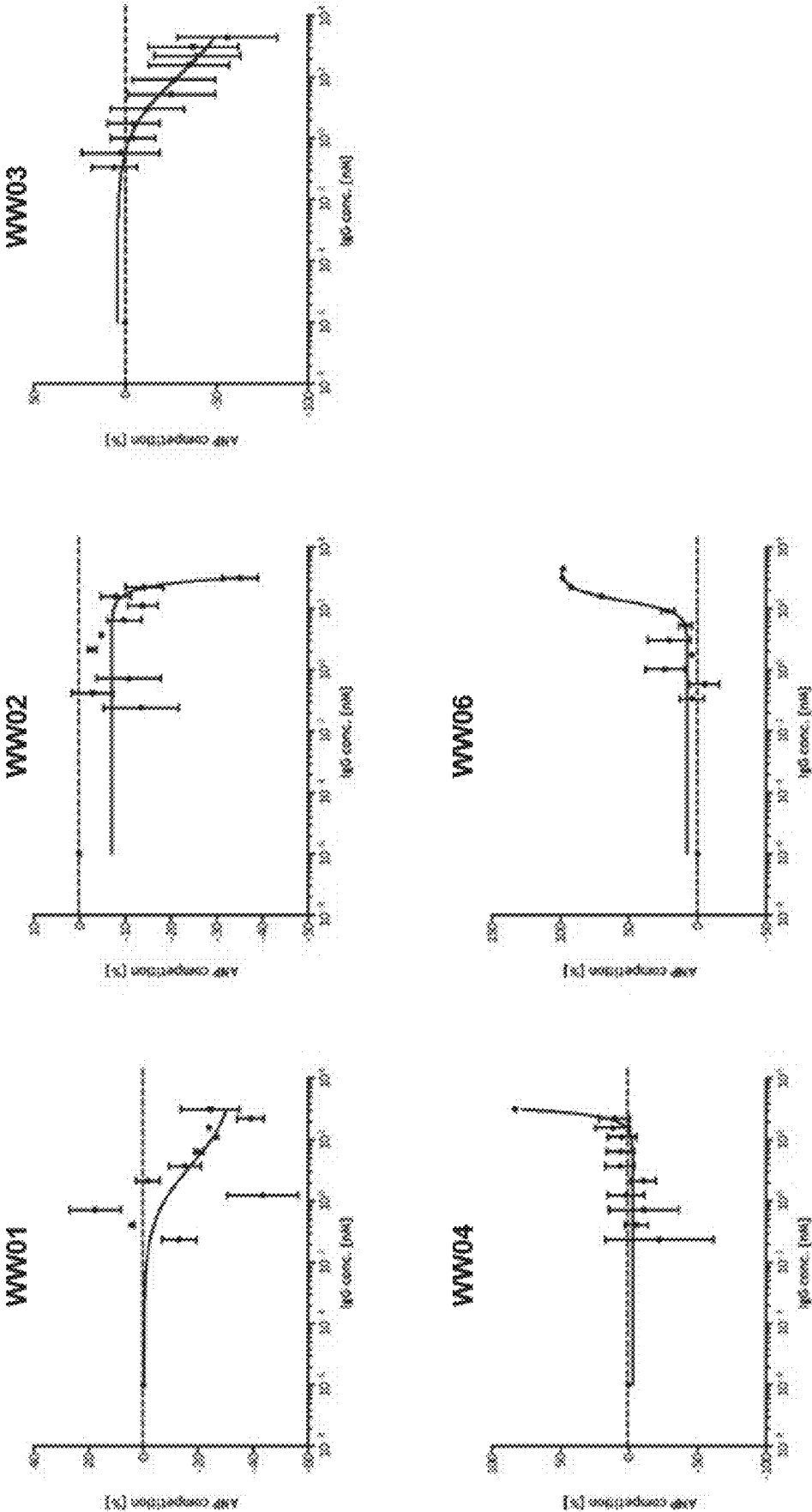


FIG. 5

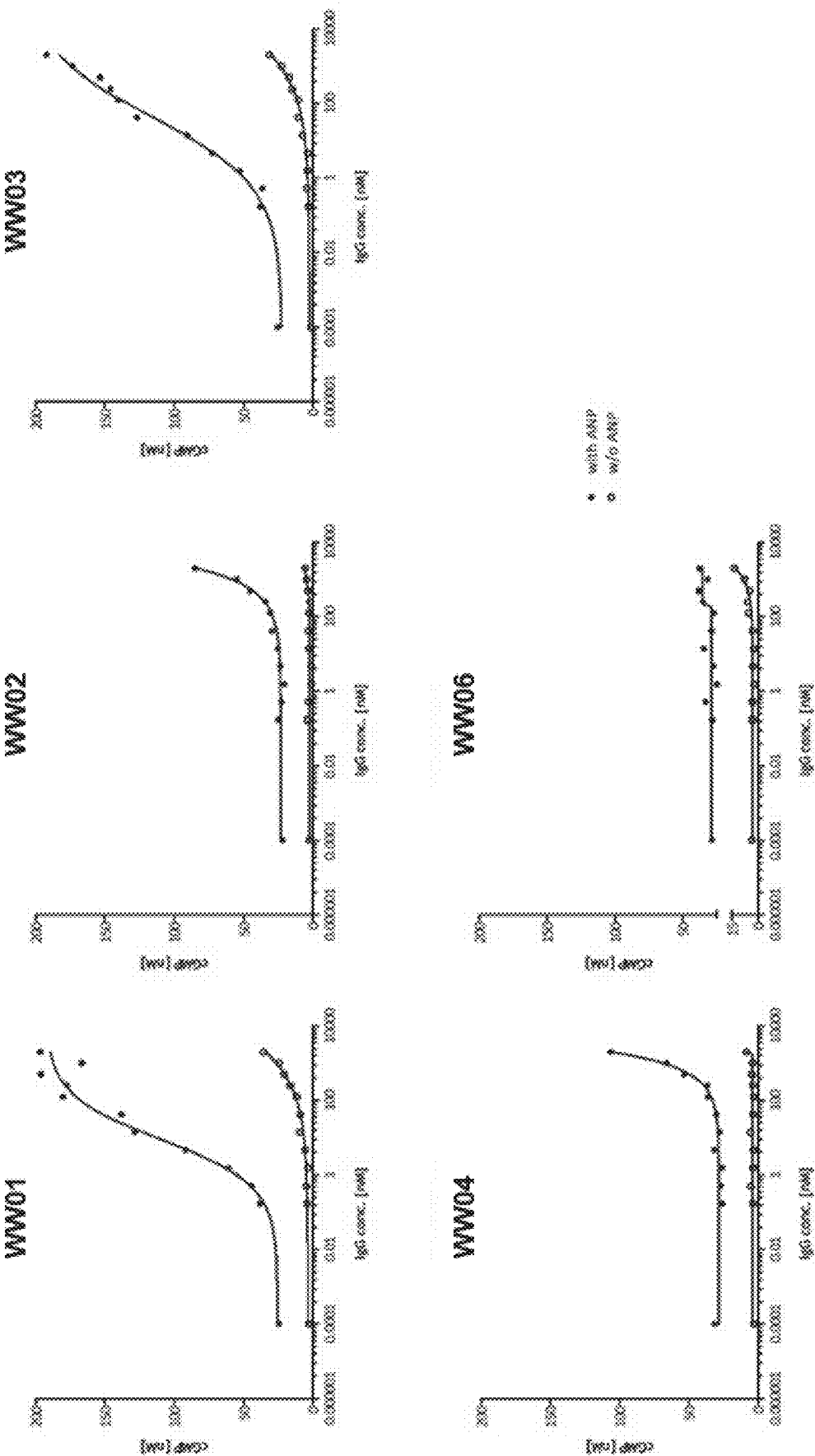


FIG. 6

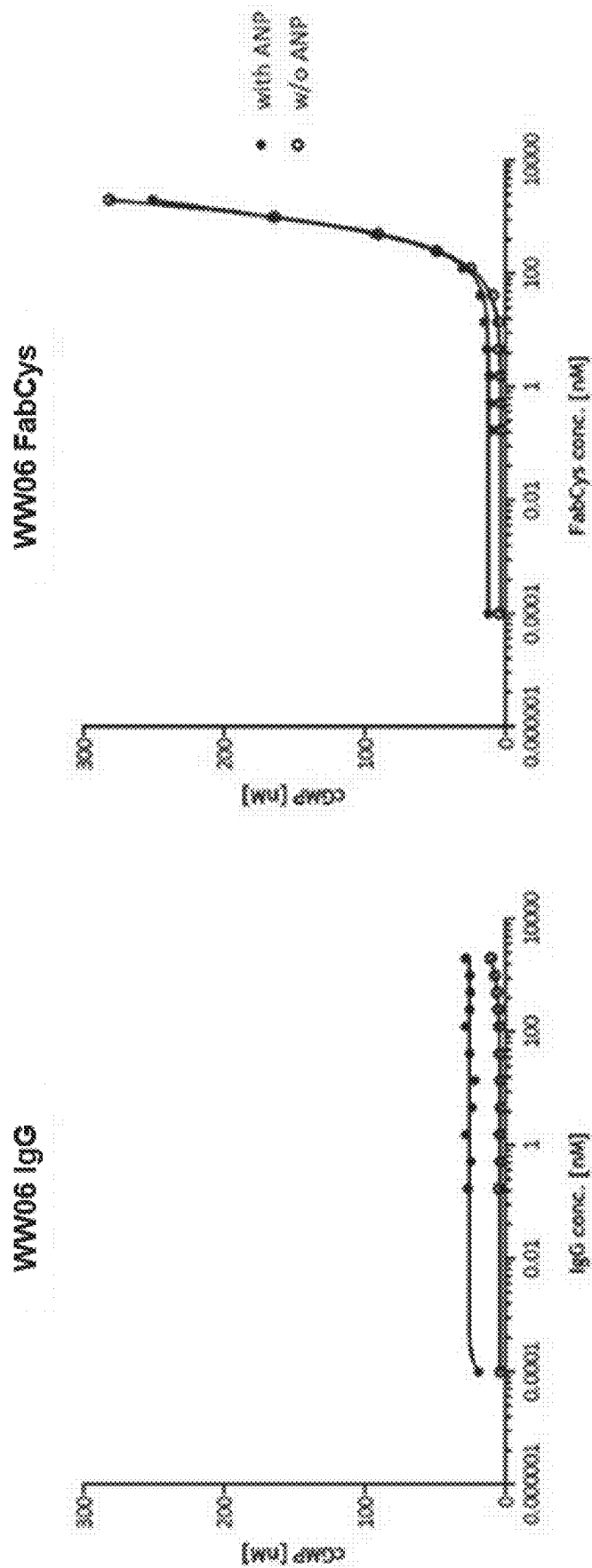


FIG. 7

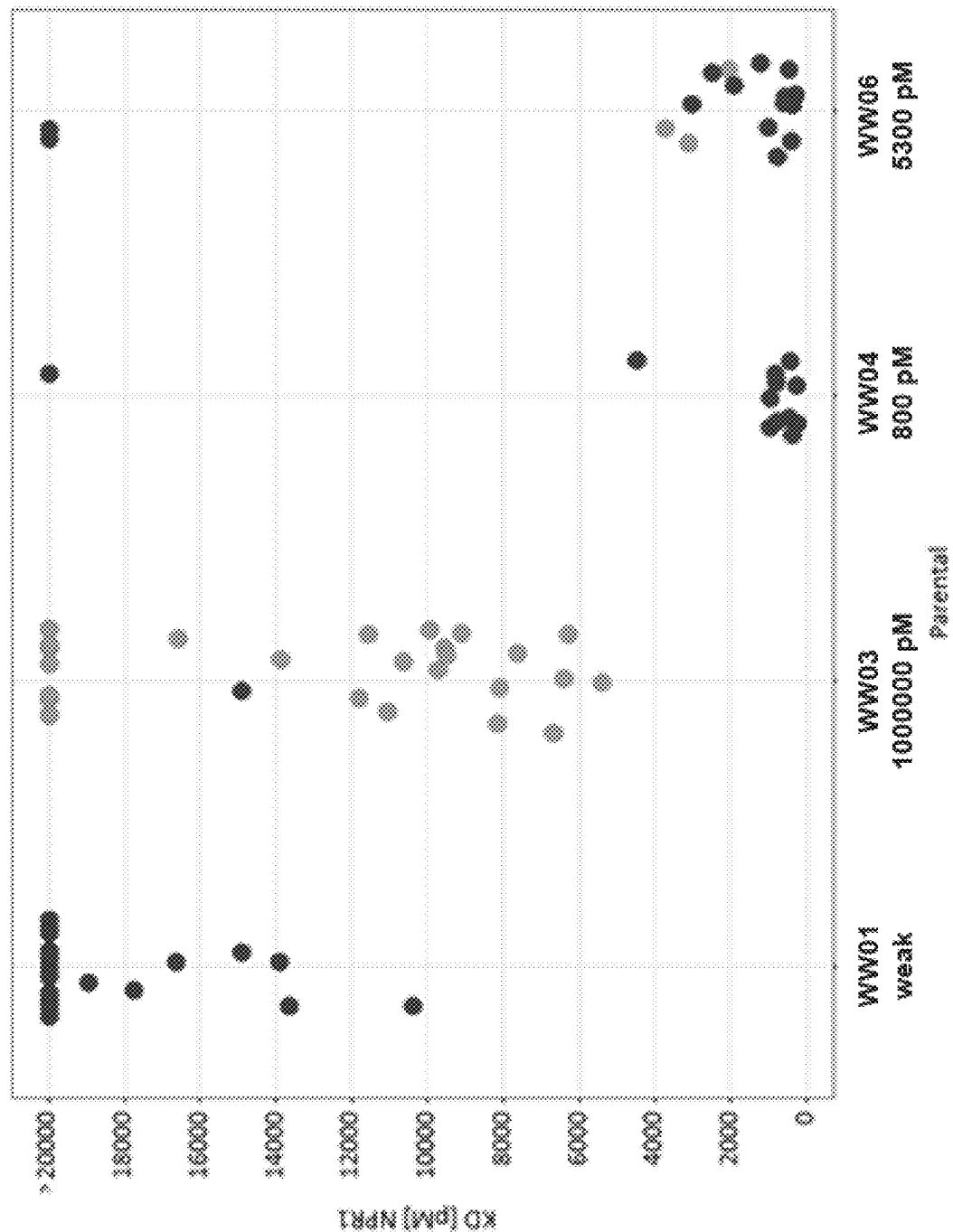


FIG. 8

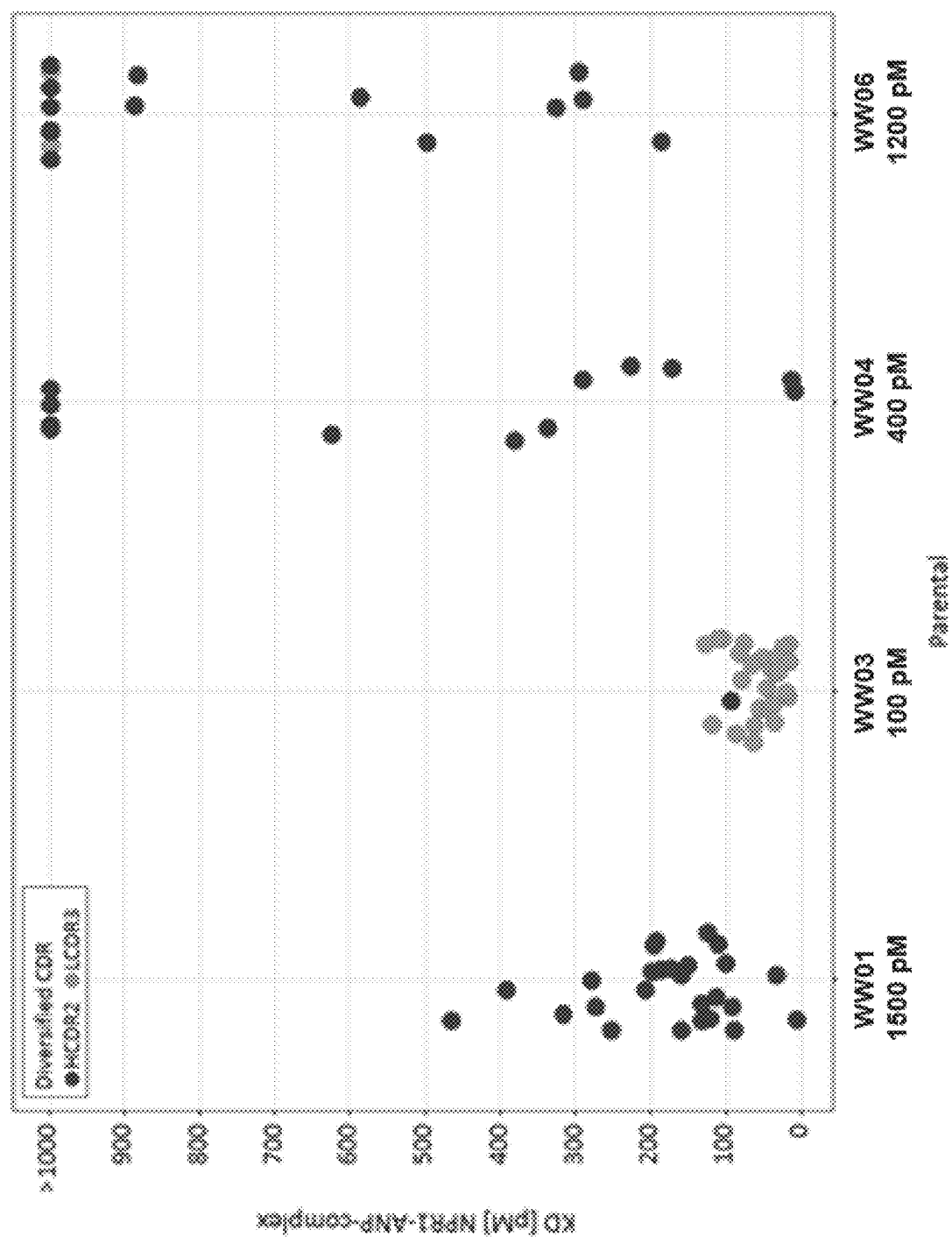
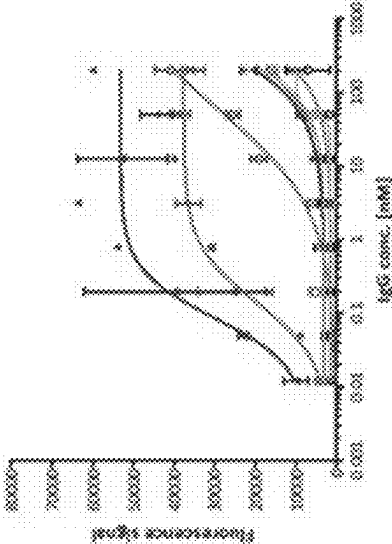
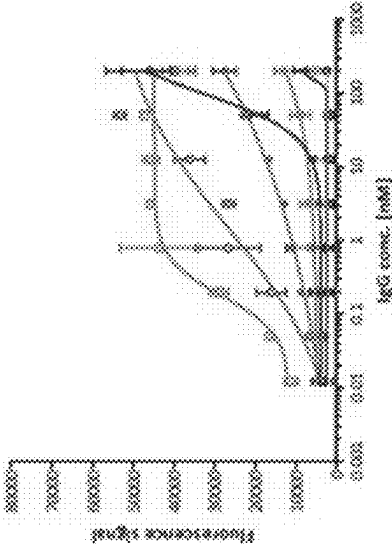


FIG. 9

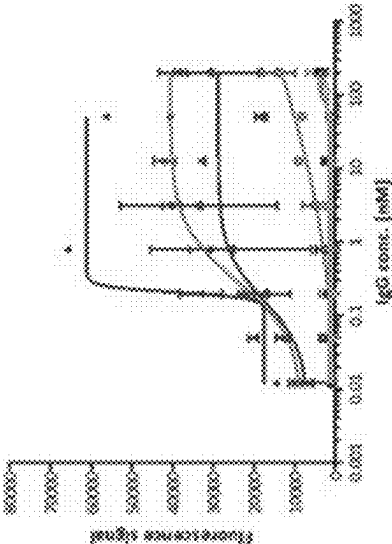
XX01



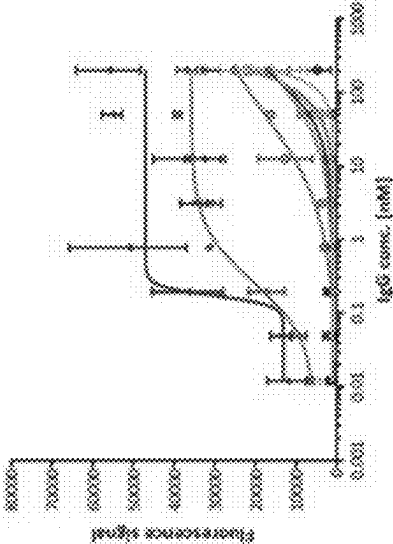
XX02



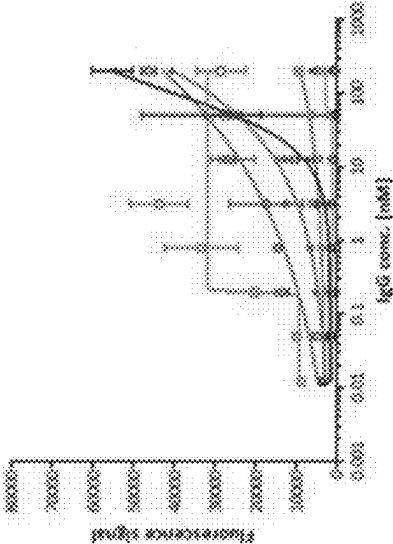
XX03



XX04



XX05



XX06

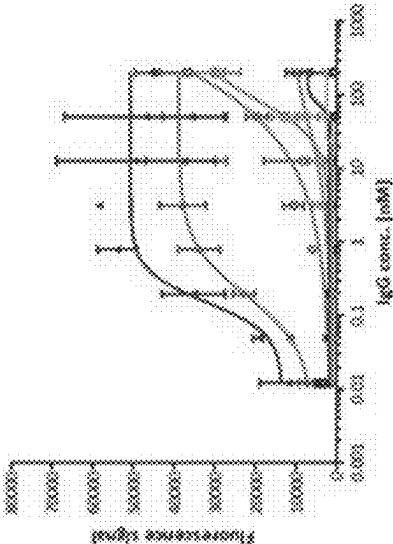
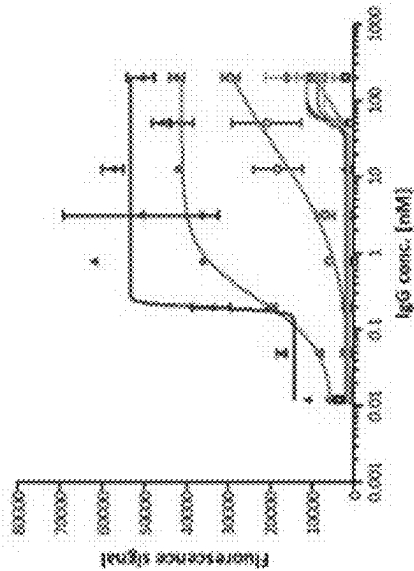
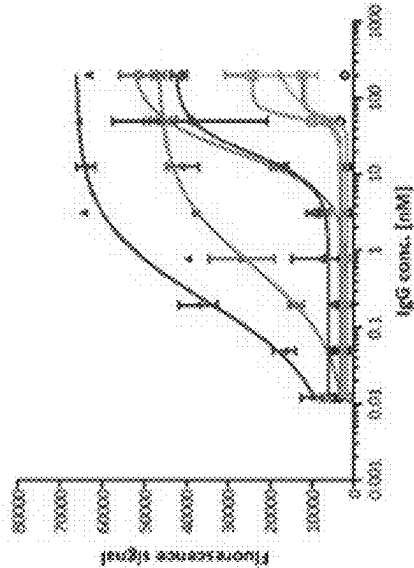


FIG. 9 (cont.)

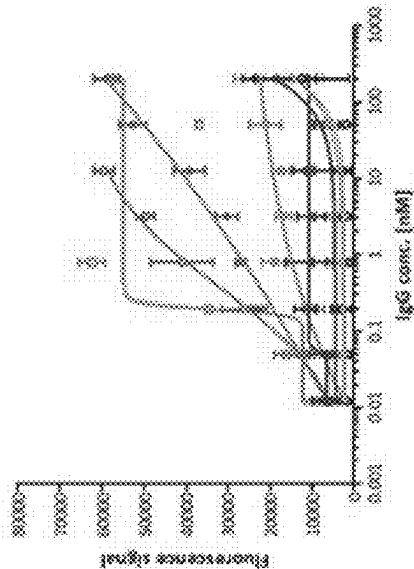
XX07



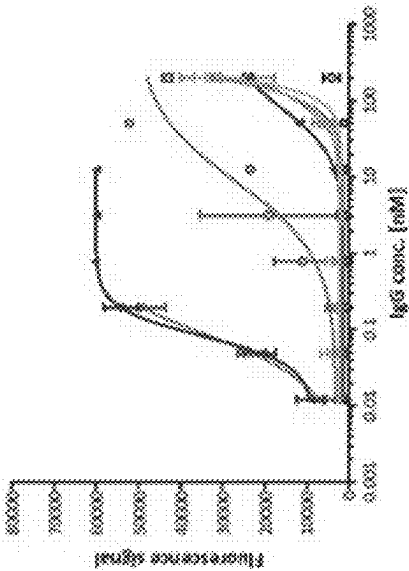
XX08



XX10



XX12



- BNPRI
- BNPRI + ANP
- BNPRI-W74R
- × BNPRI-W74R + ANP
- ▲ rBNPRI
- △ rBNPRI + ANP
- ▽ BNPRI3
- ▽ BNPRI3 + ANP
- ◇ BSA

FIG. 10

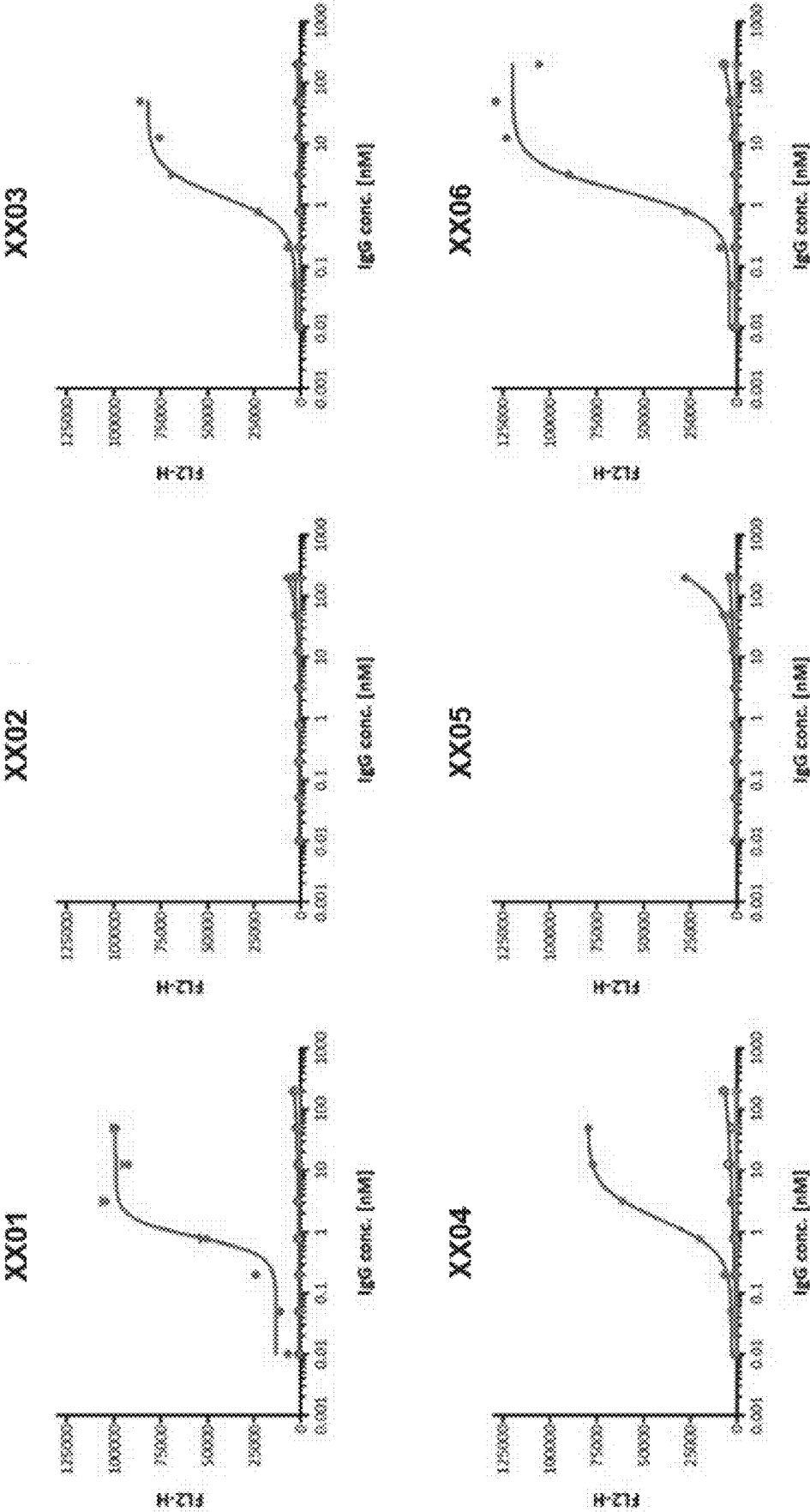


FIG. 10 (cont.)

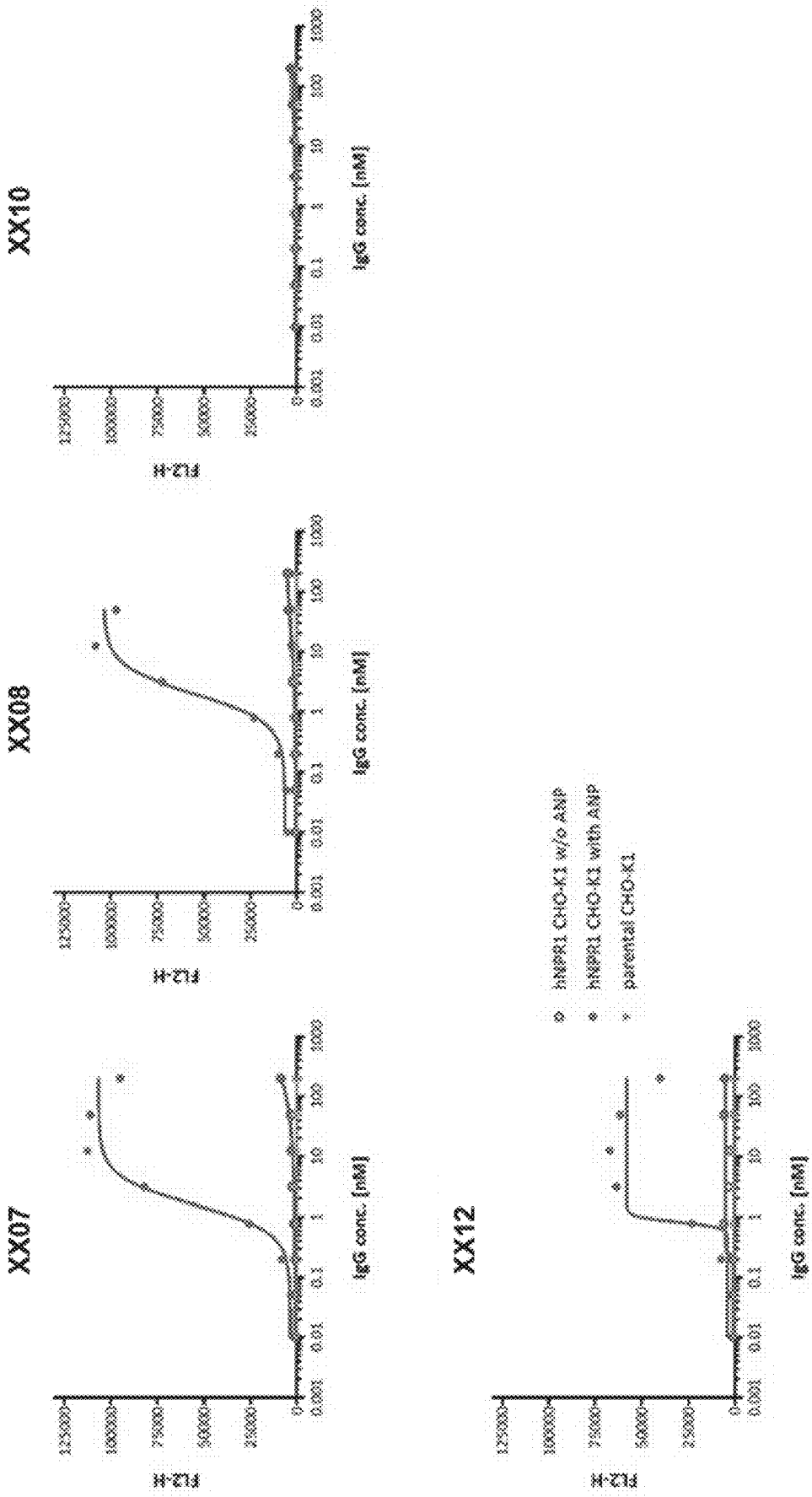


FIG. 11

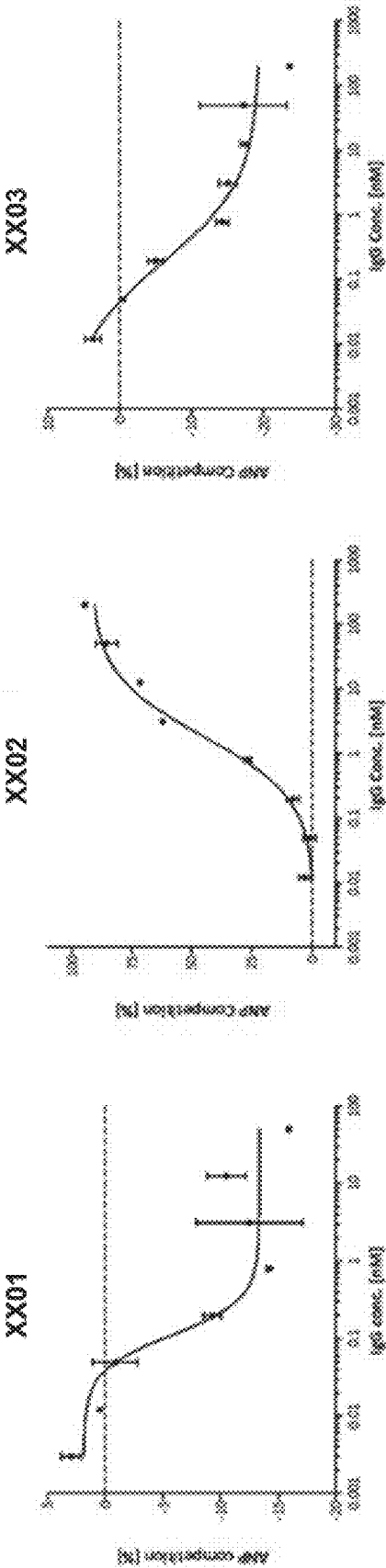


FIG. 11 (cont.)

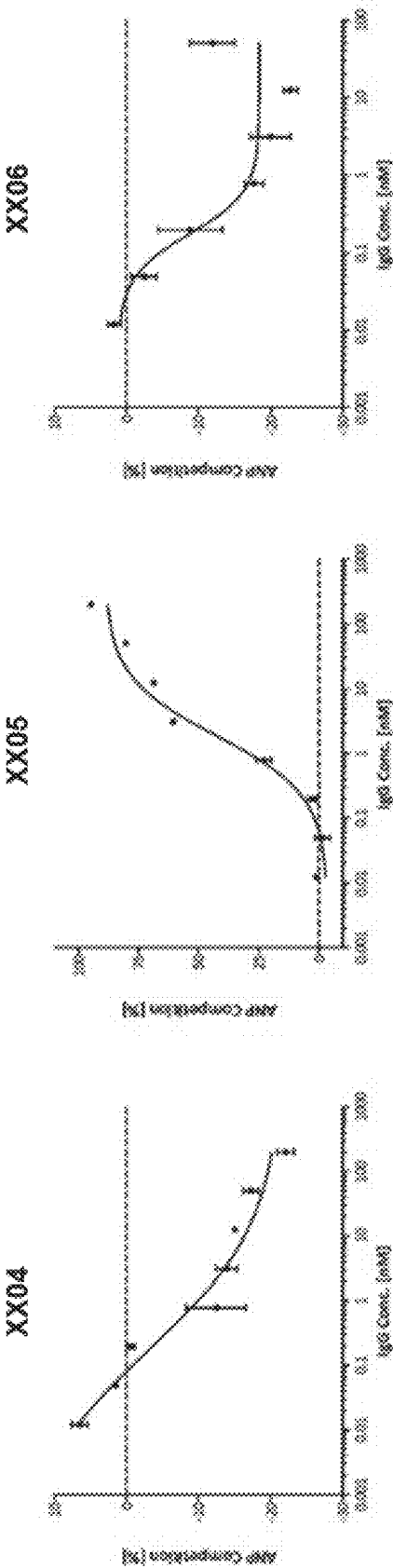


FIG. 11 (cont.)

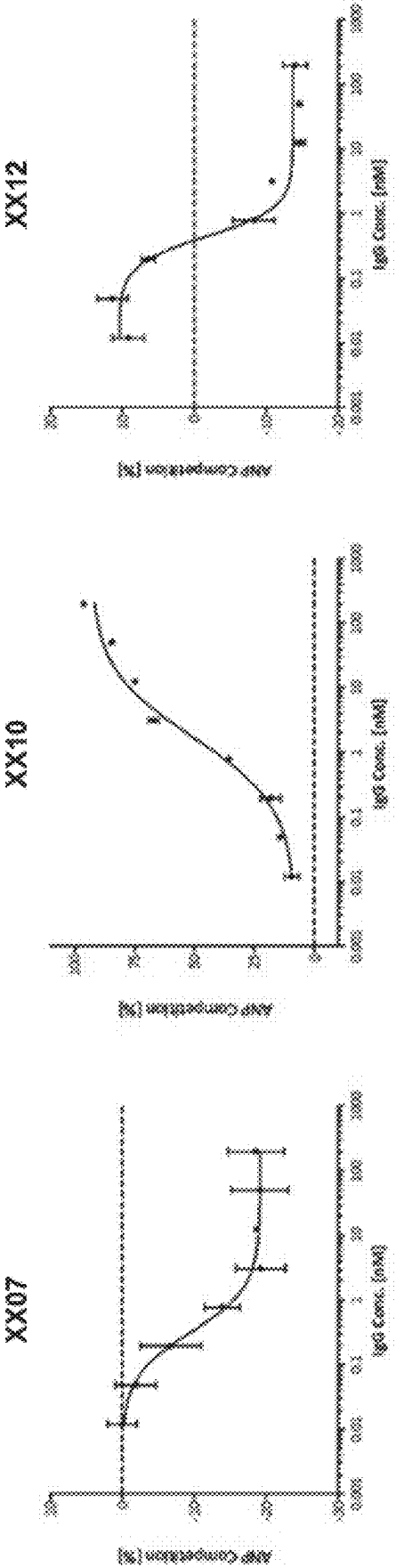


FIG. 12

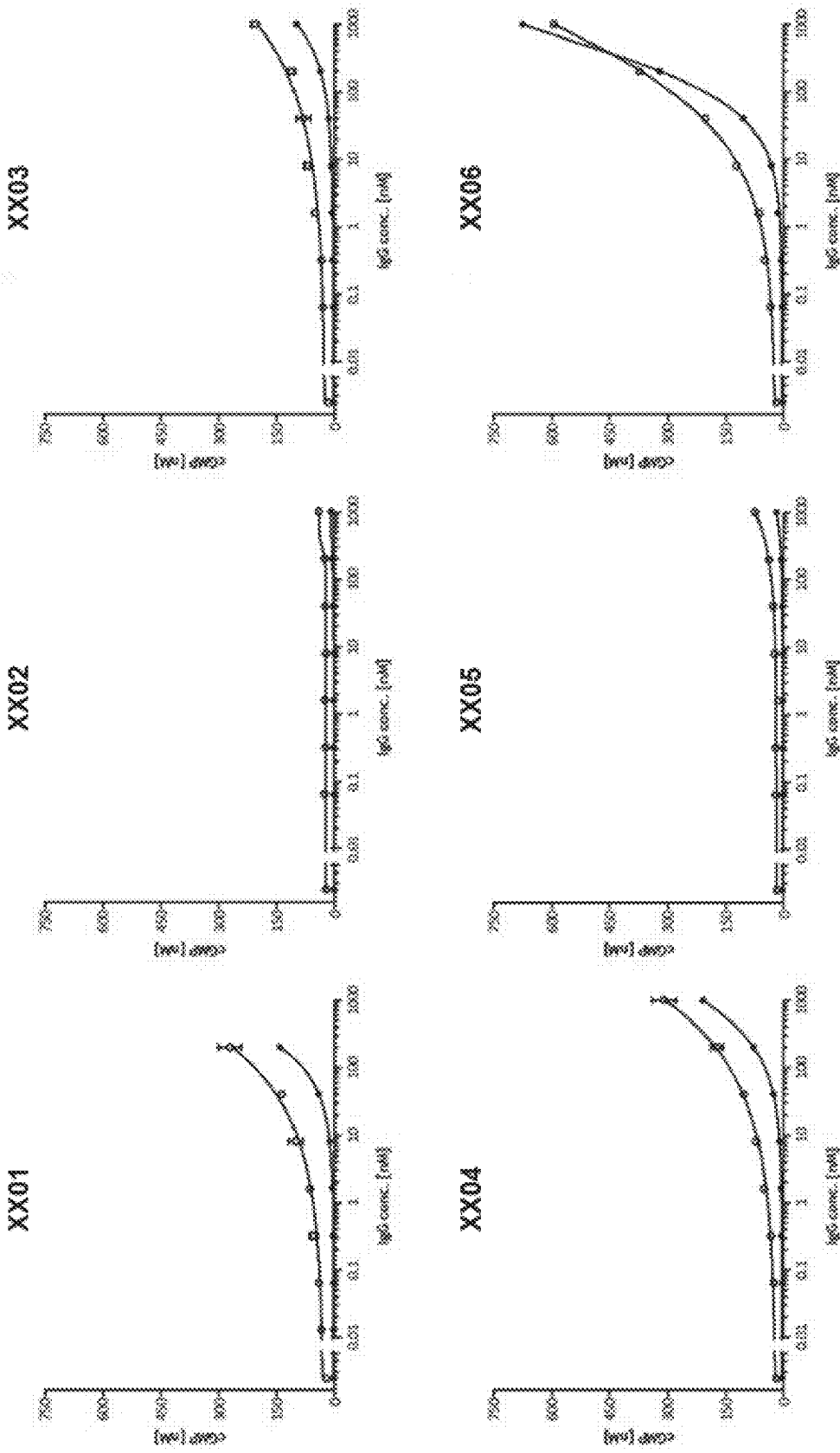


FIG. 12 (cont.)

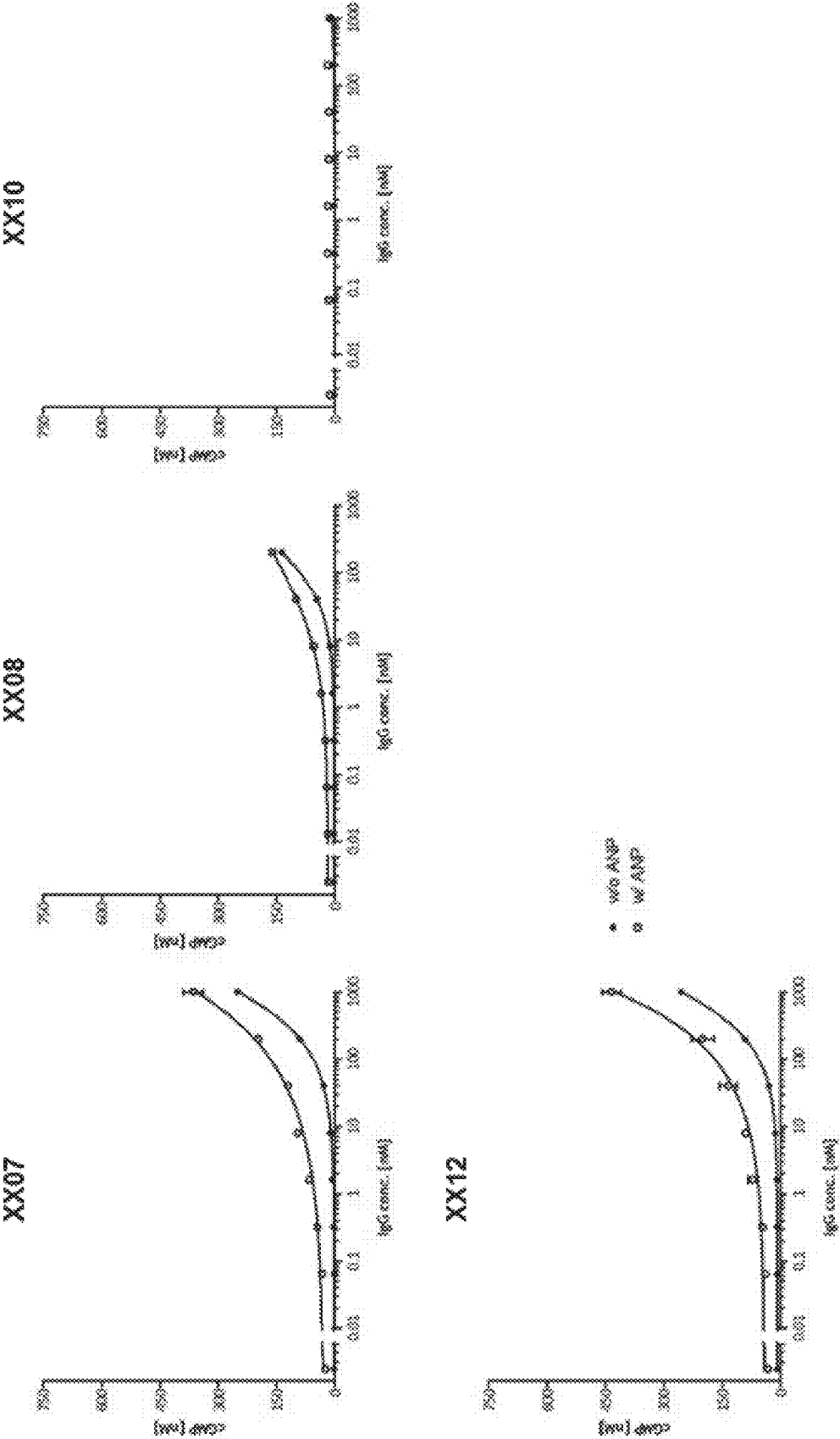


FIG. 13

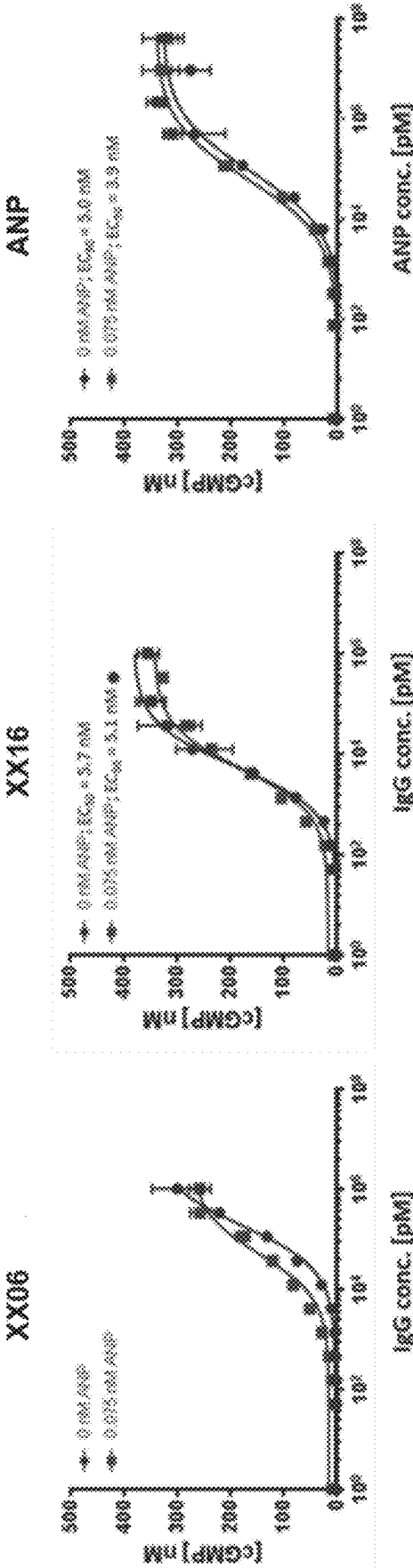


FIG. 14

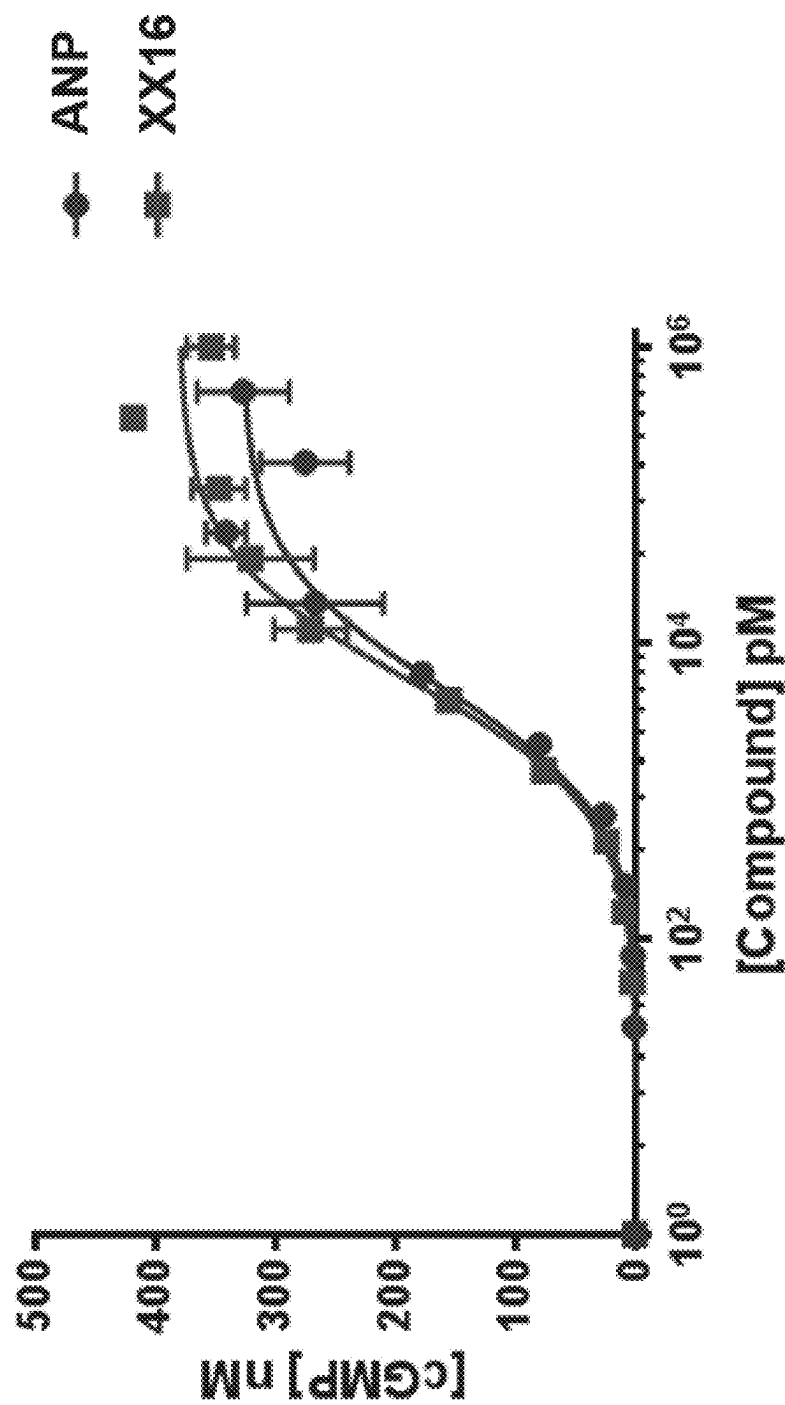


FIG. 15

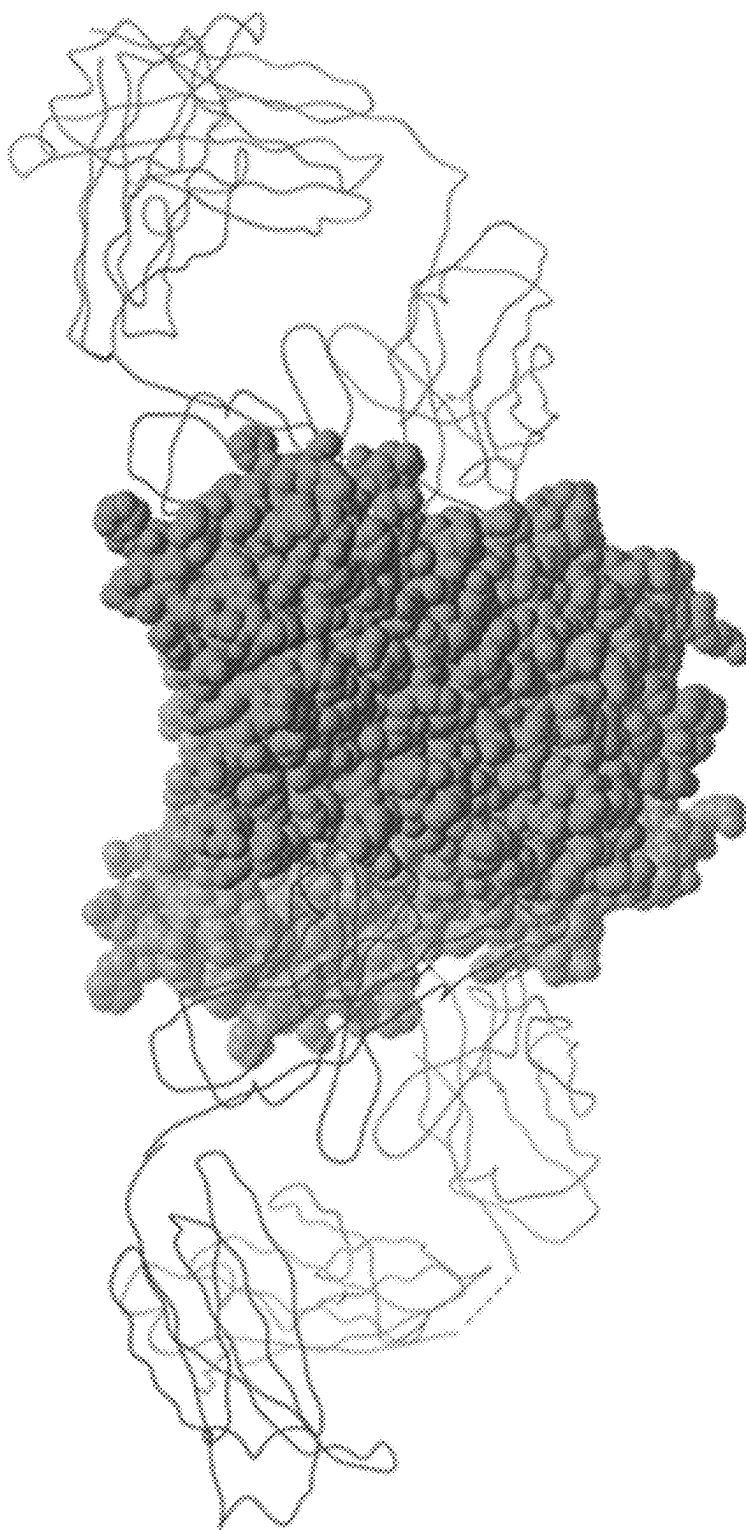


FIG. 16

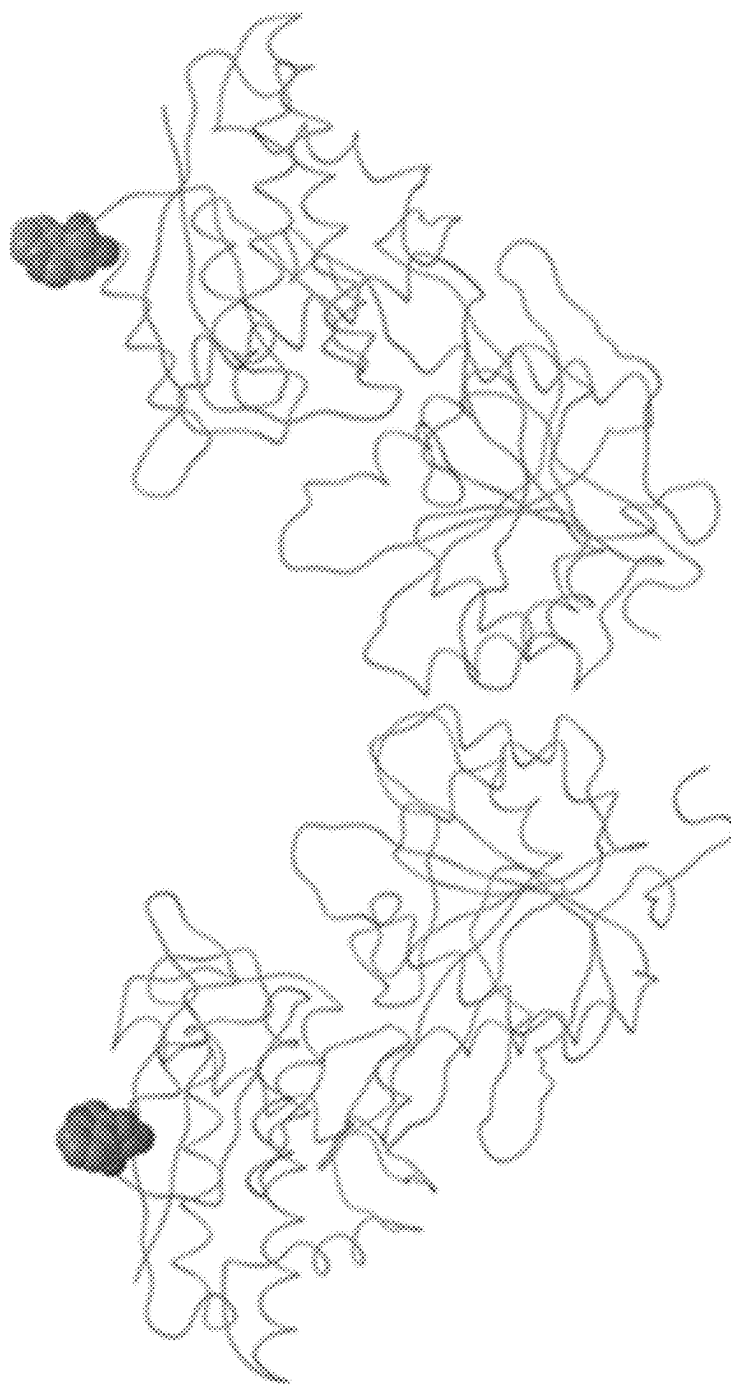


FIG. 17

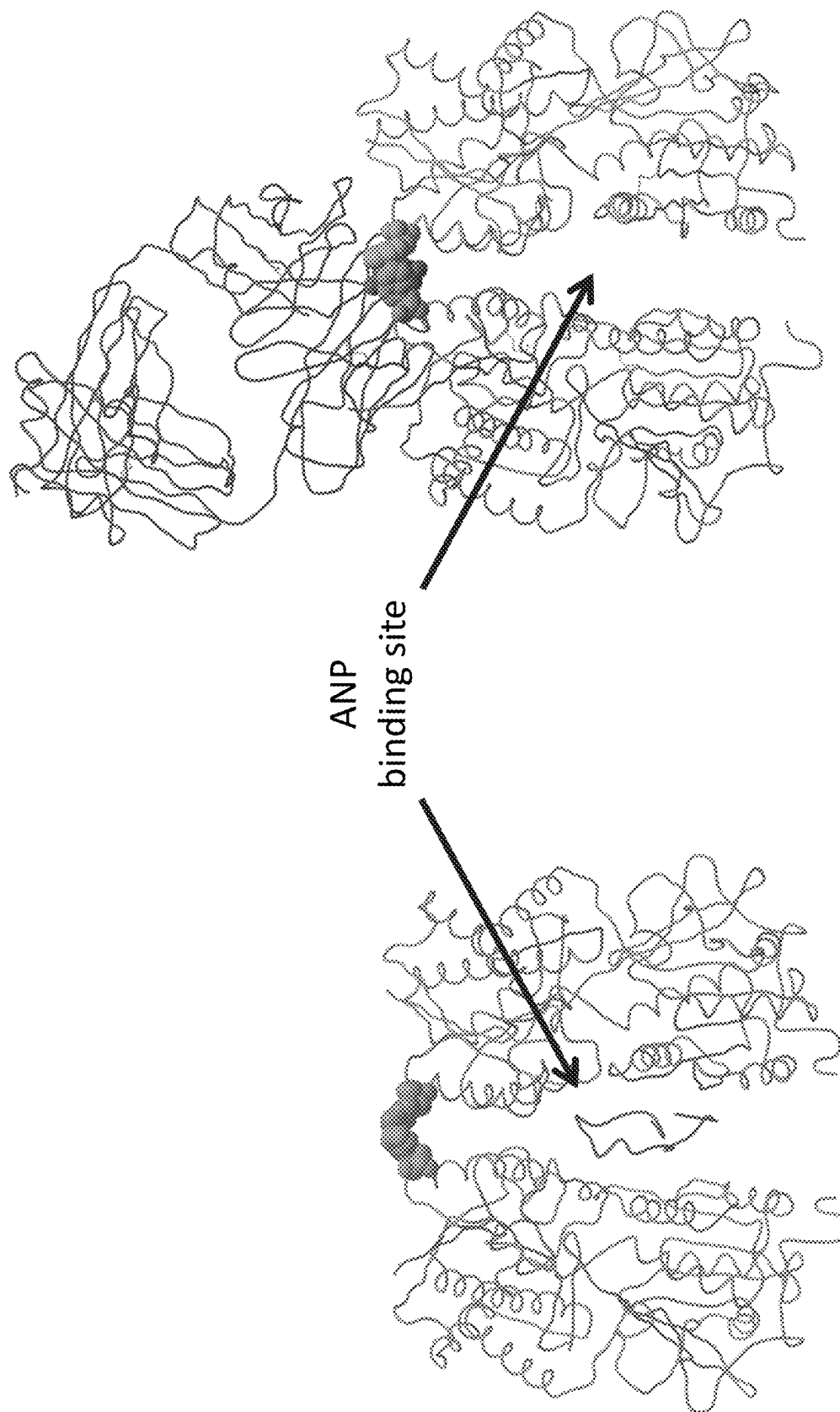


FIG. 18

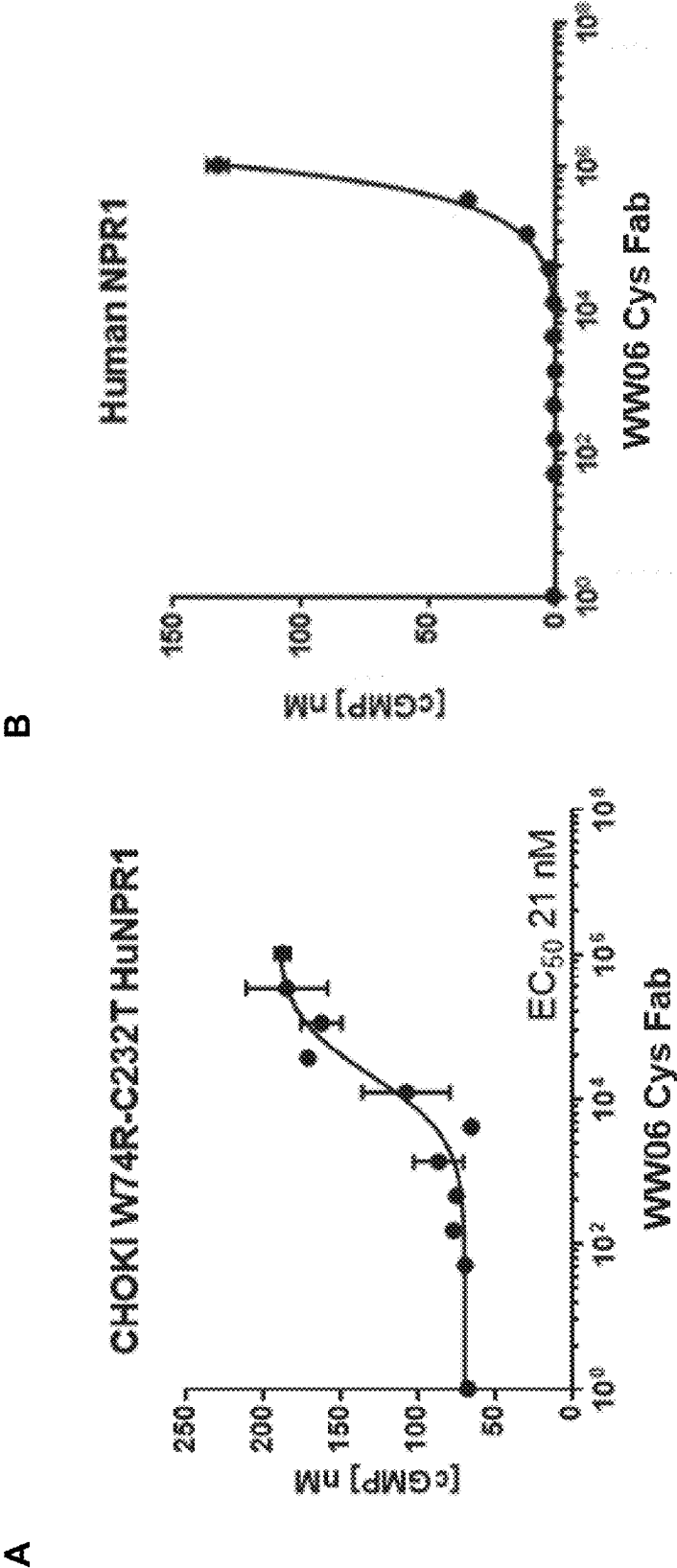


FIG. 19

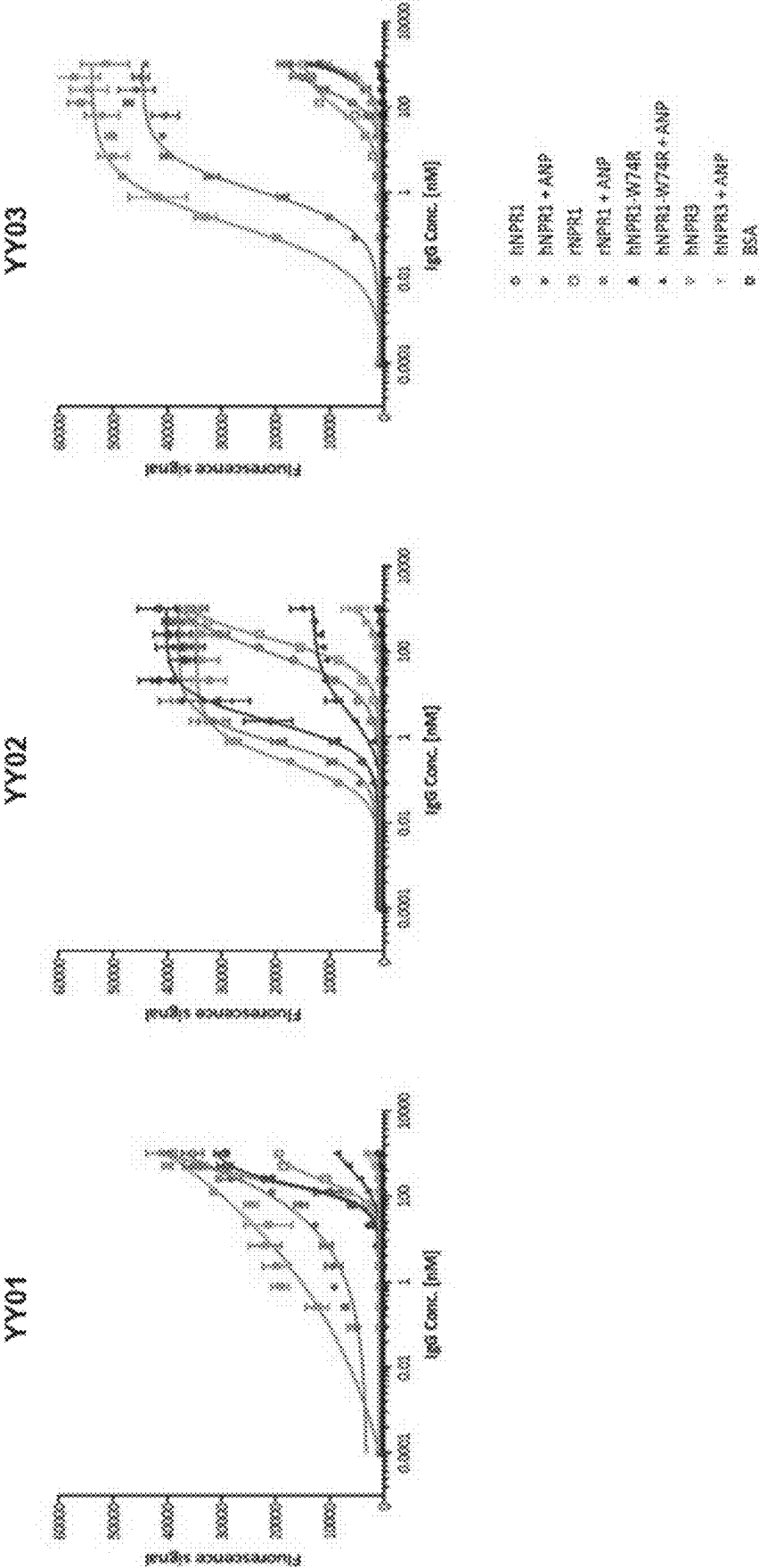


FIG. 19 (cont.)

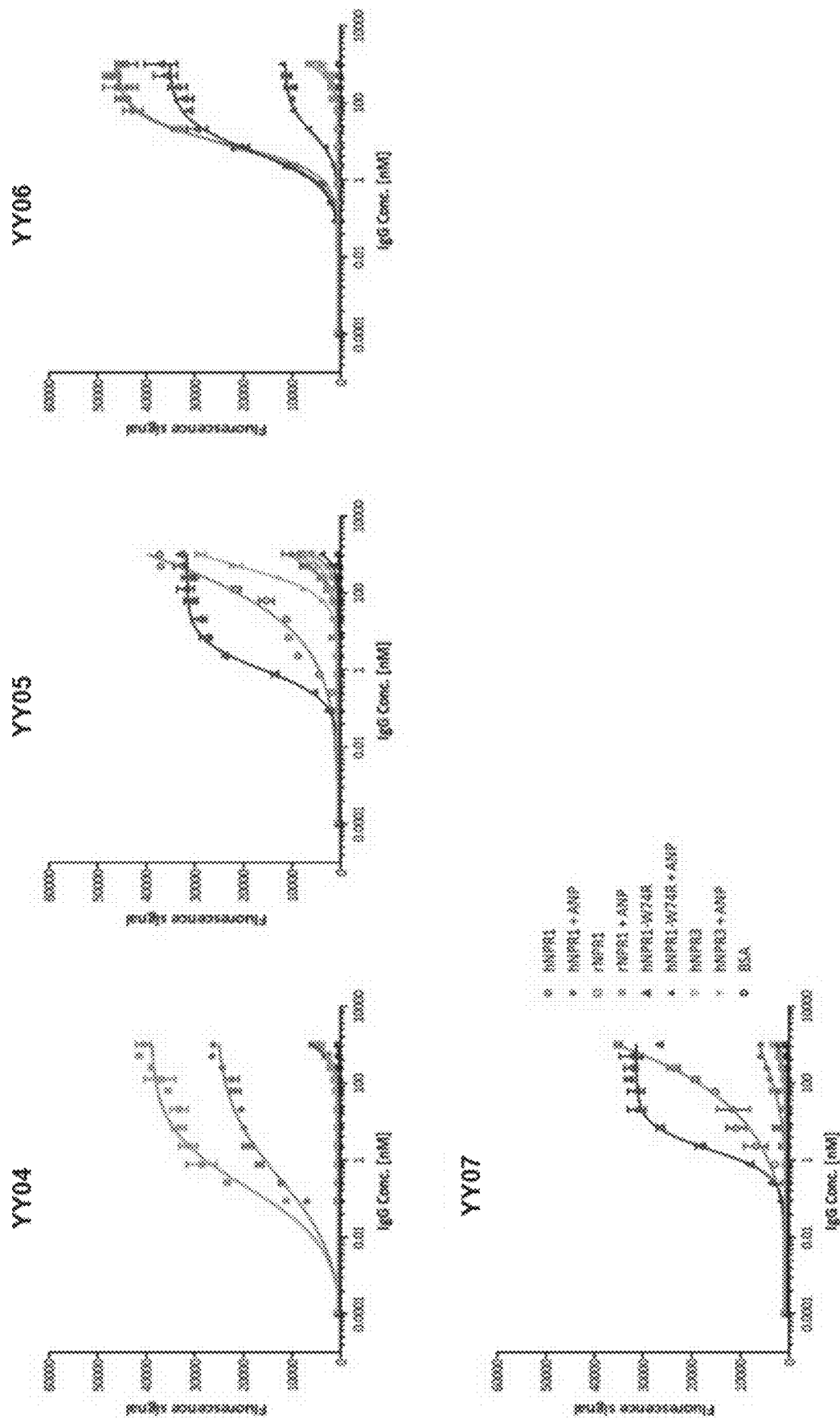


FIG. 20 (cont.)

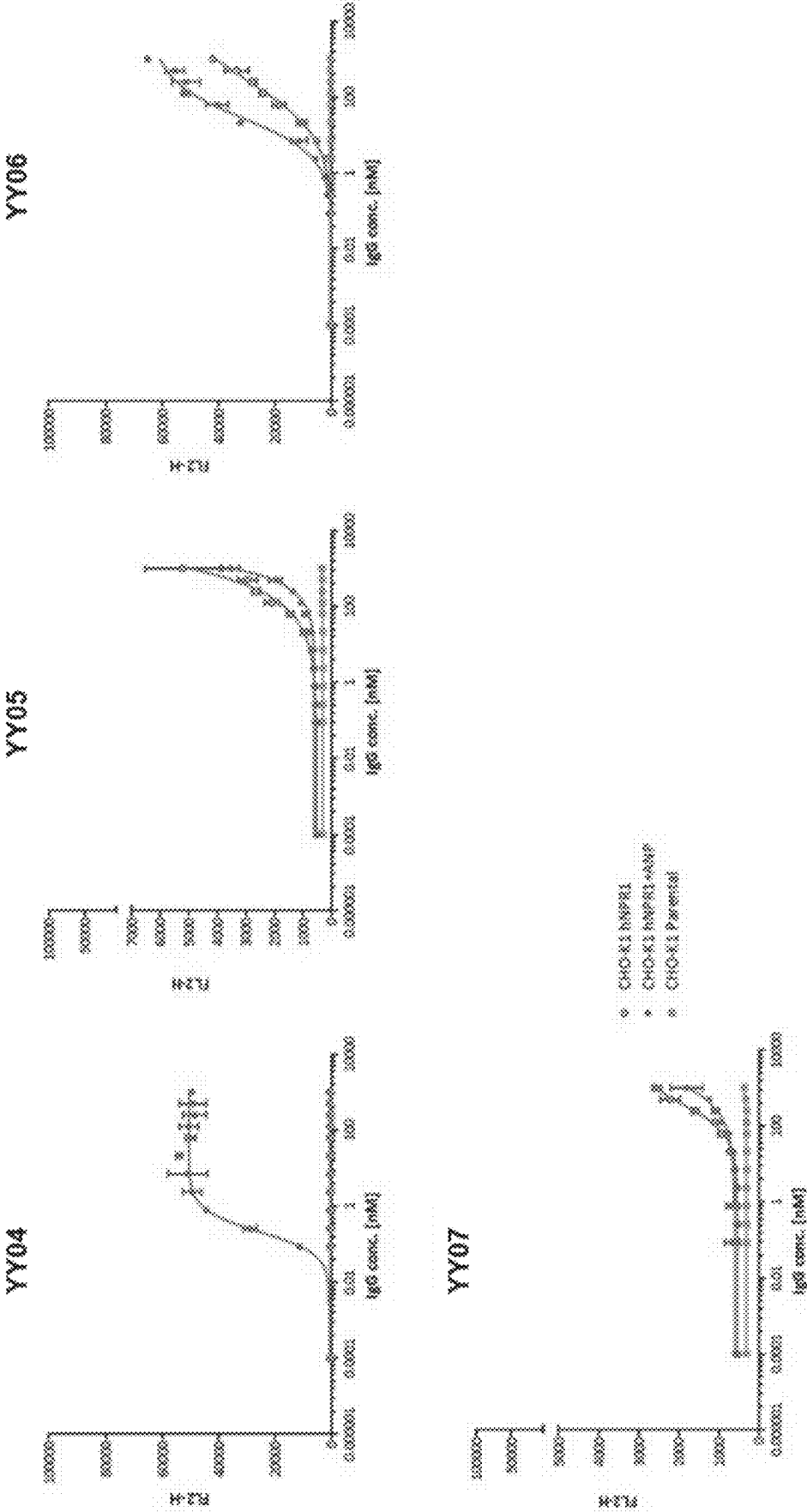


FIG. 21

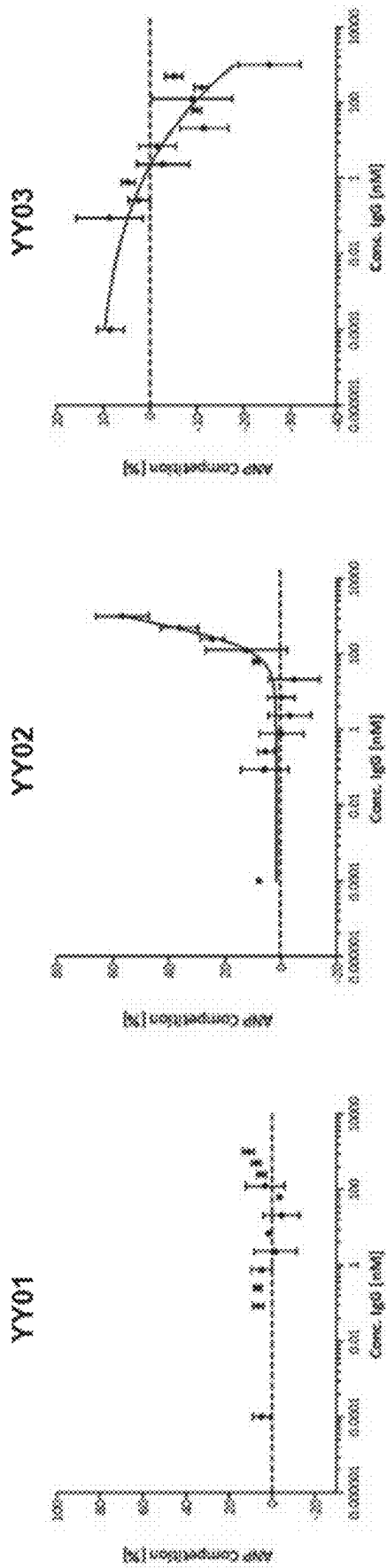


FIG. 21 (cont.)

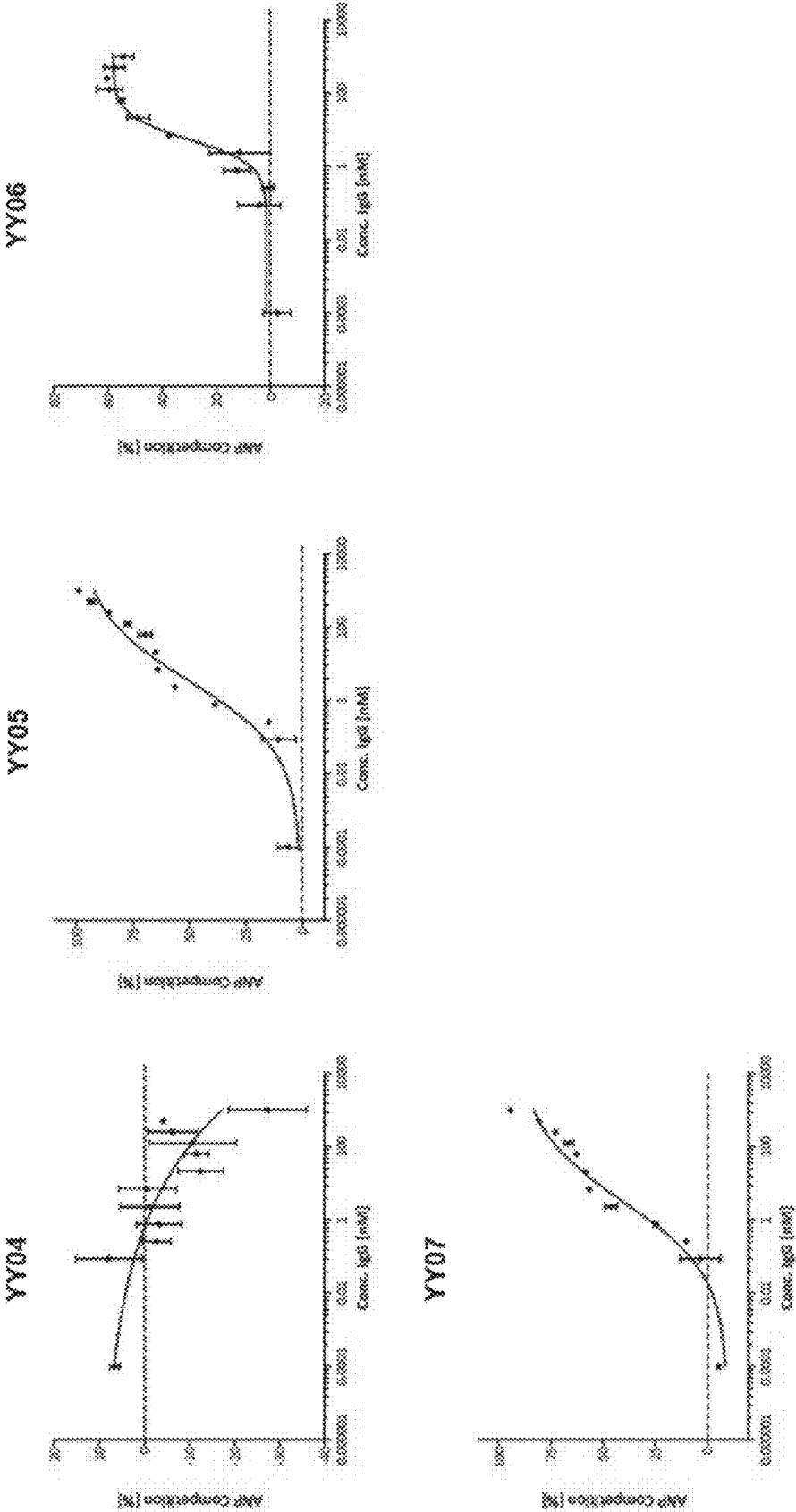


FIG. 22

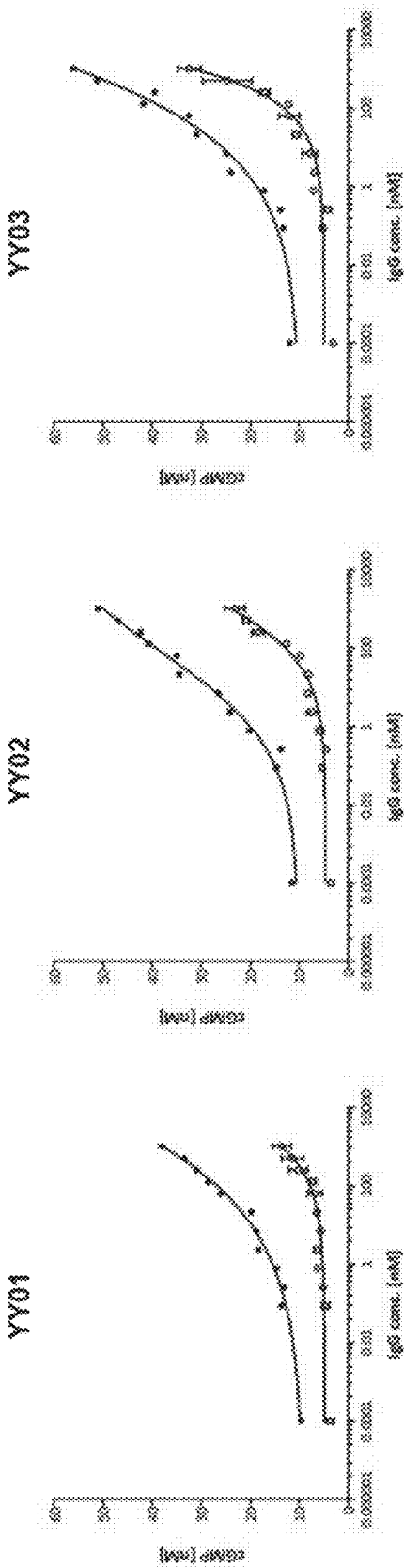


FIG. 22 (cont.)

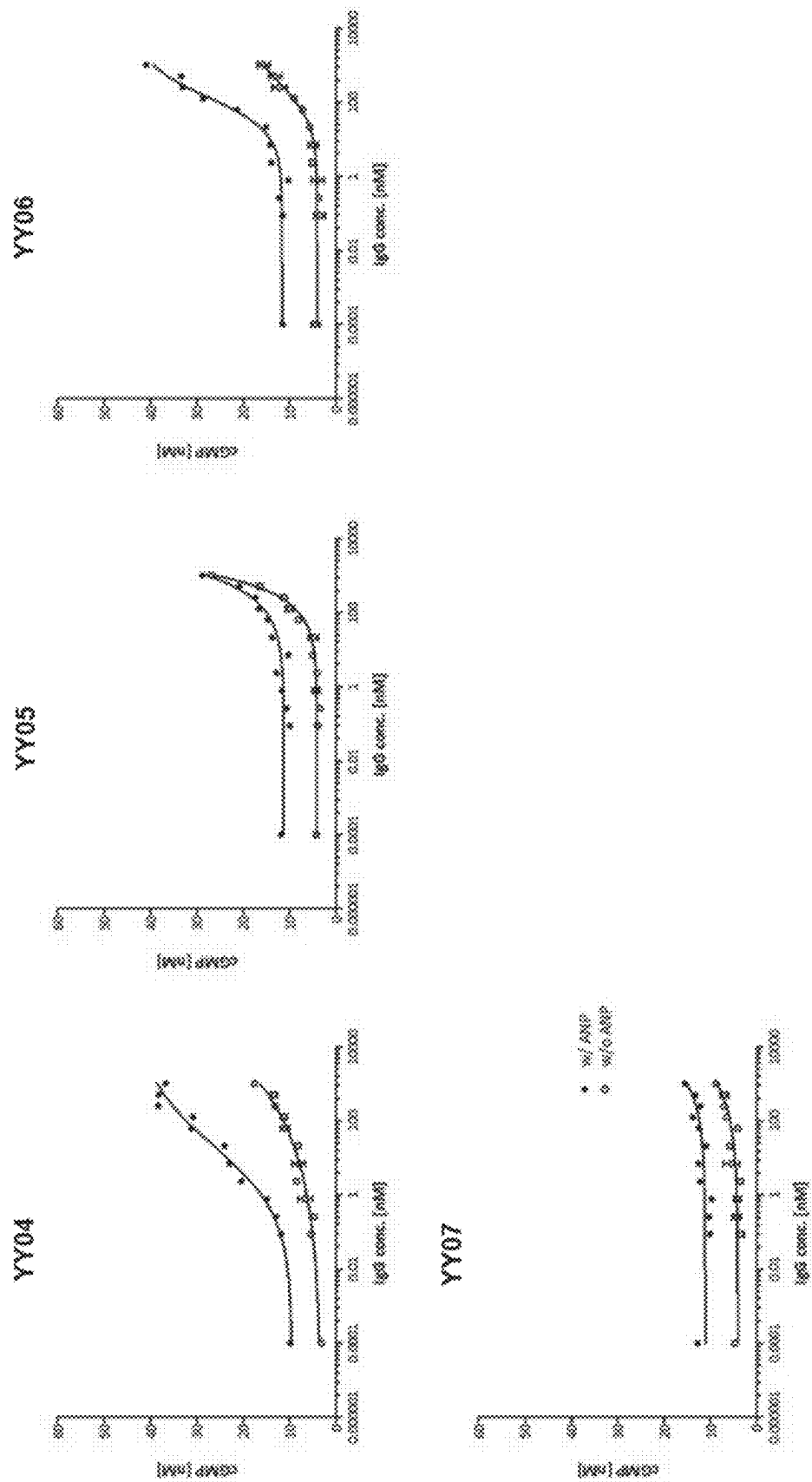


FIG. 23

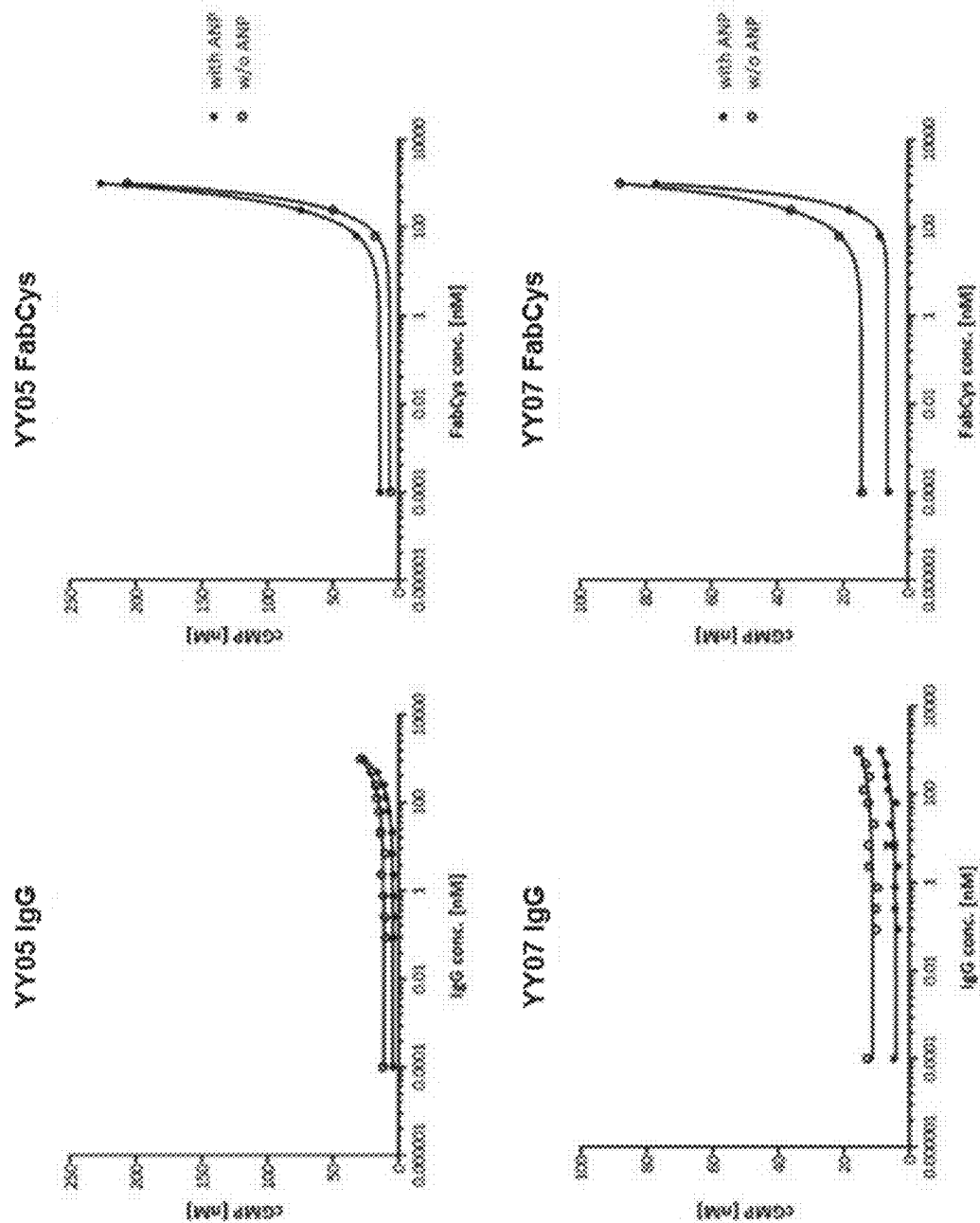


FIG. 24

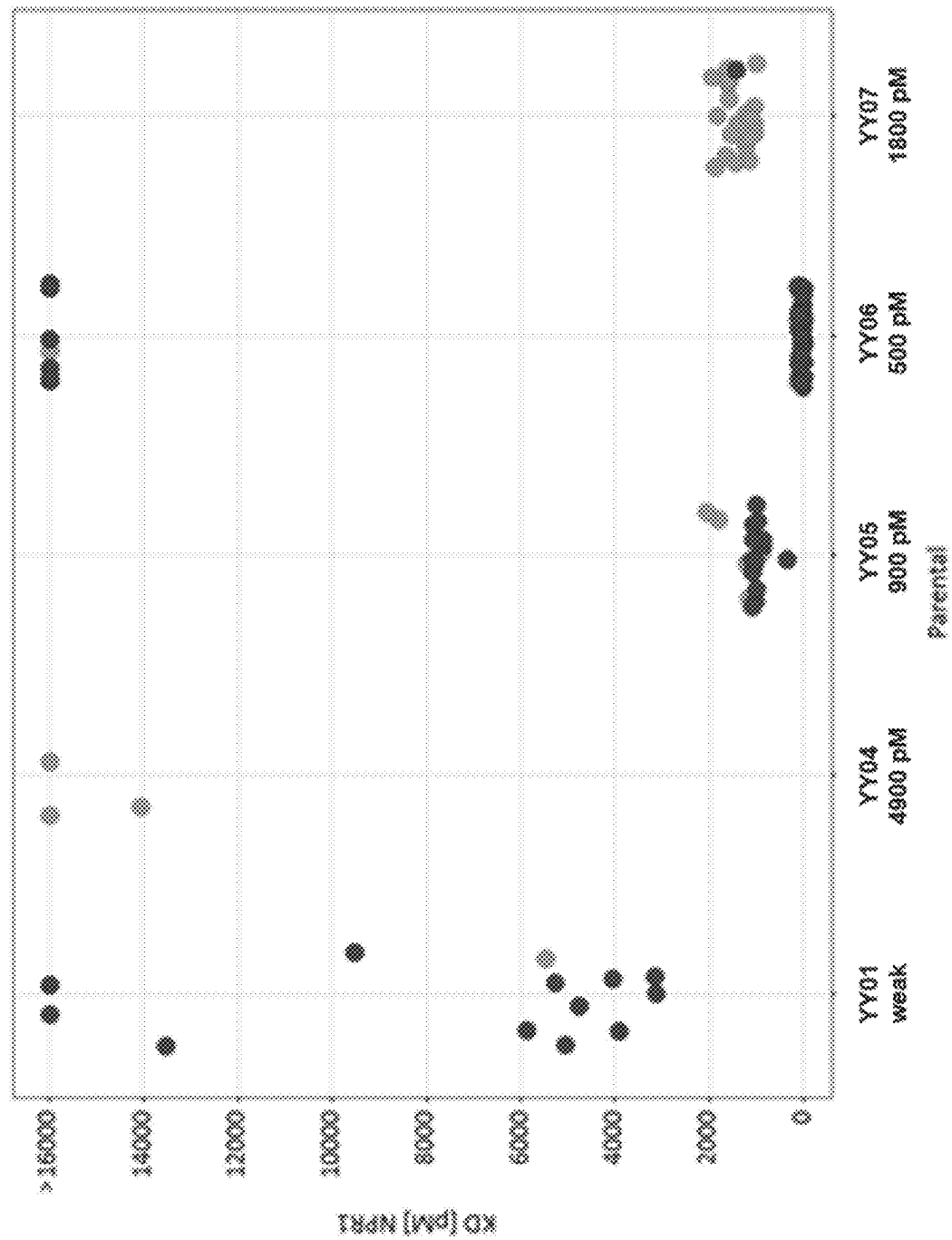


FIG. 25

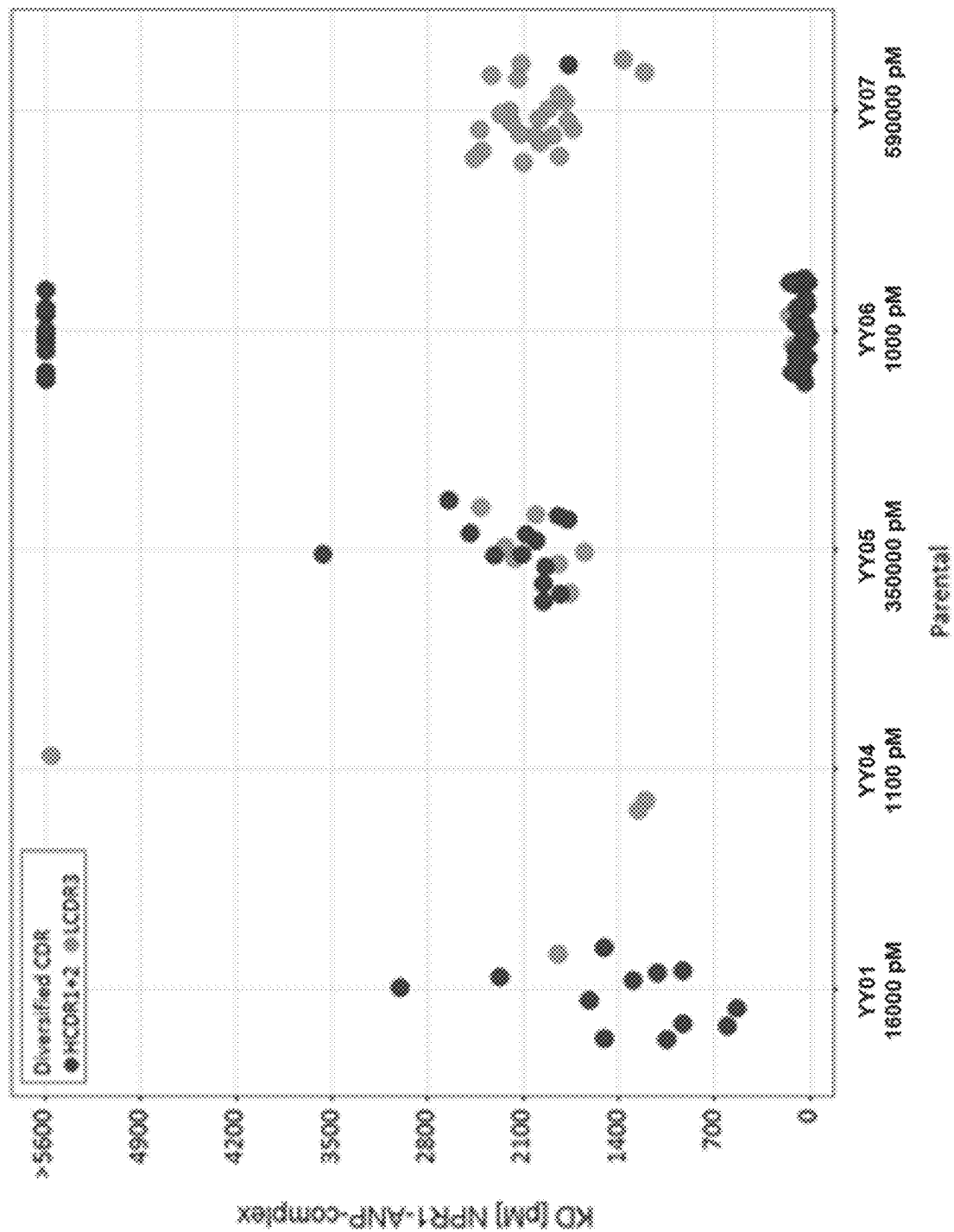


FIG. 26

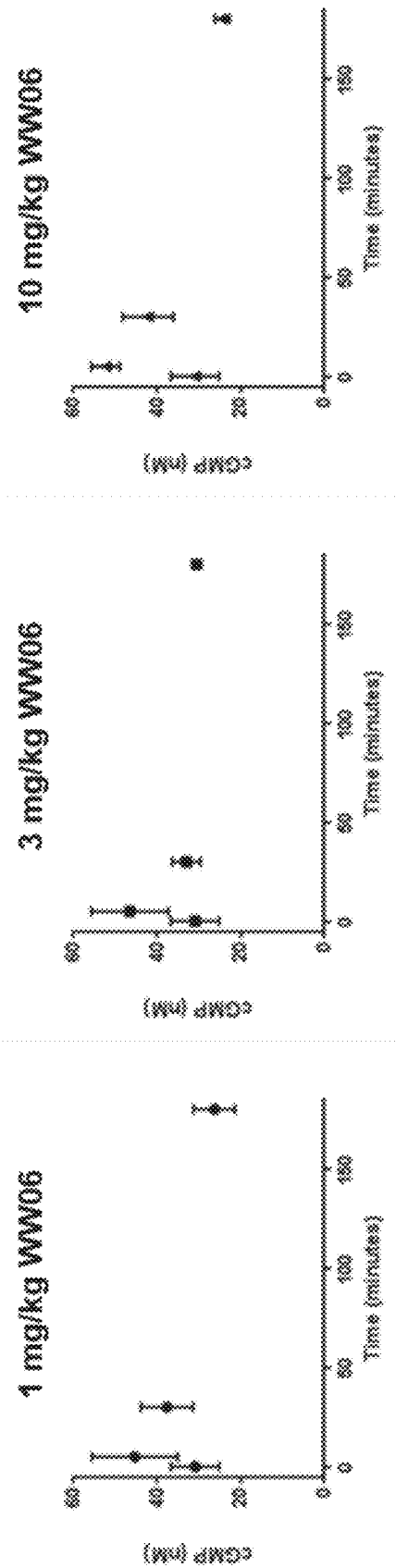


FIG. 27

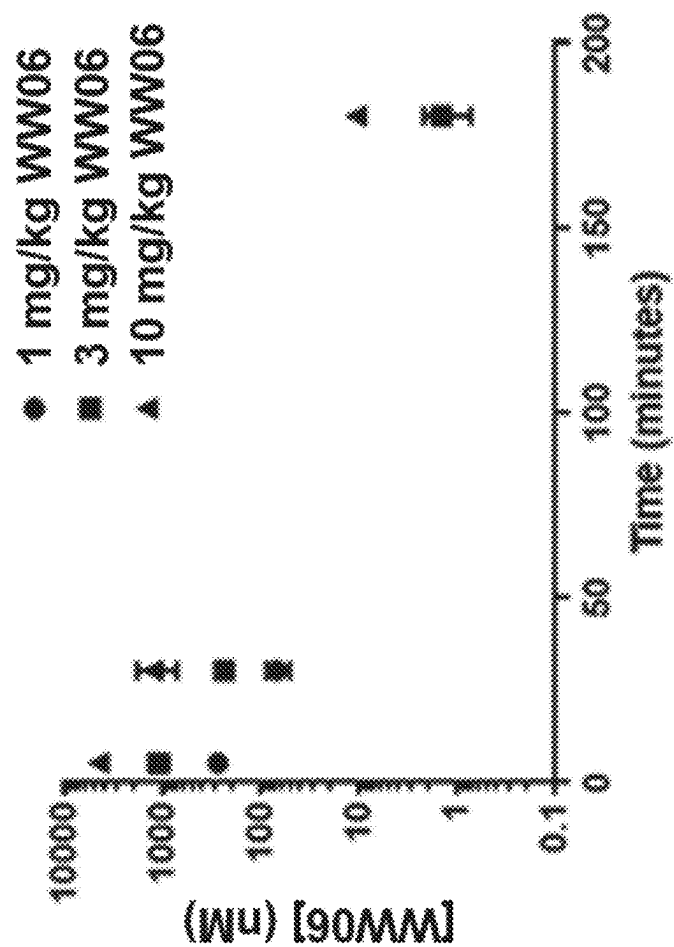


FIG. 28

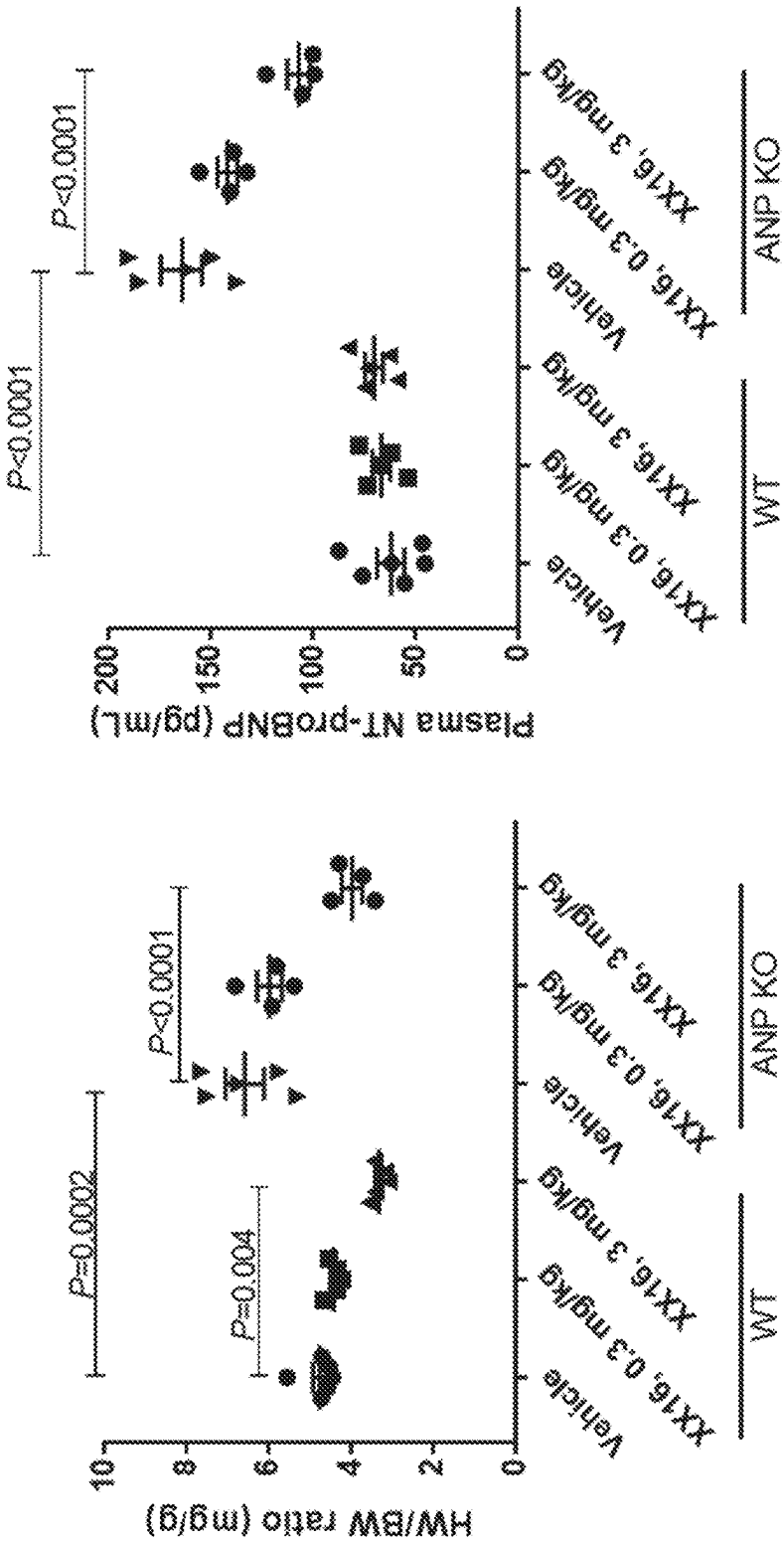


FIG. 29

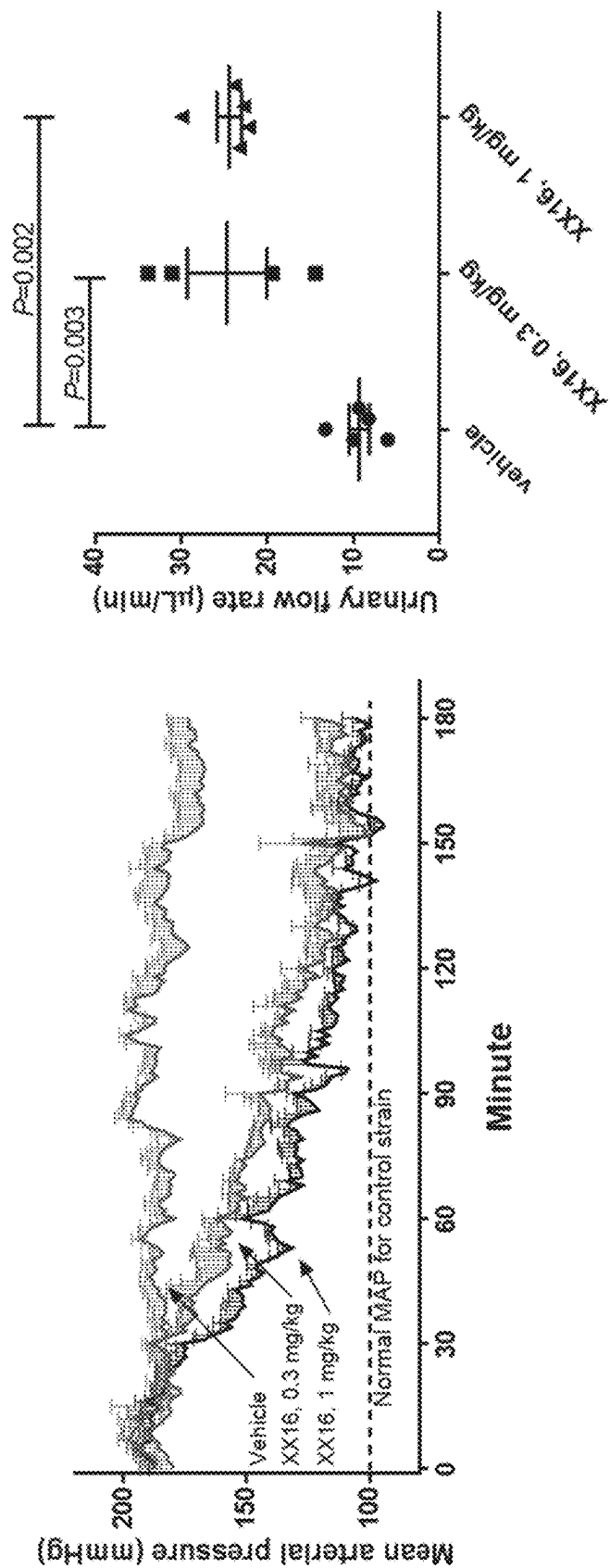


FIG. 30

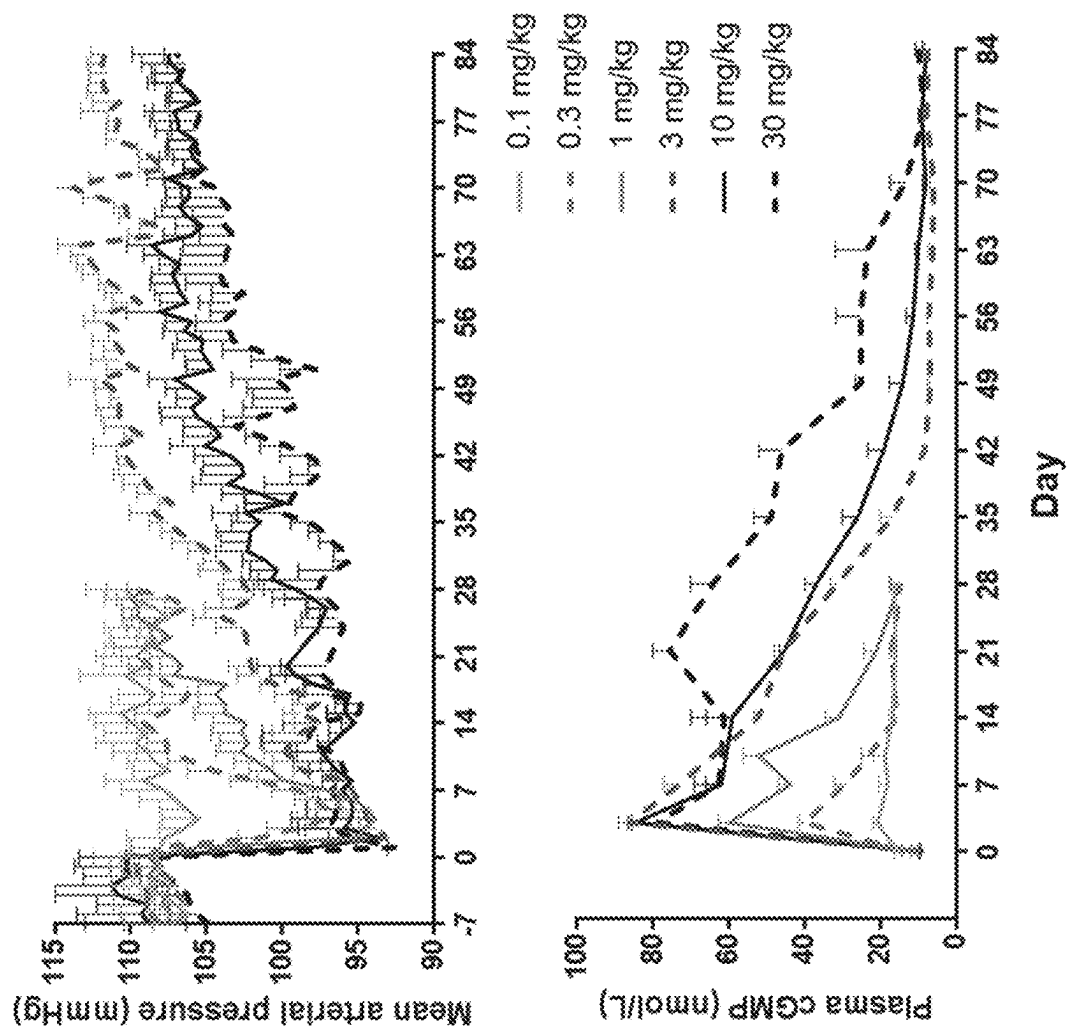
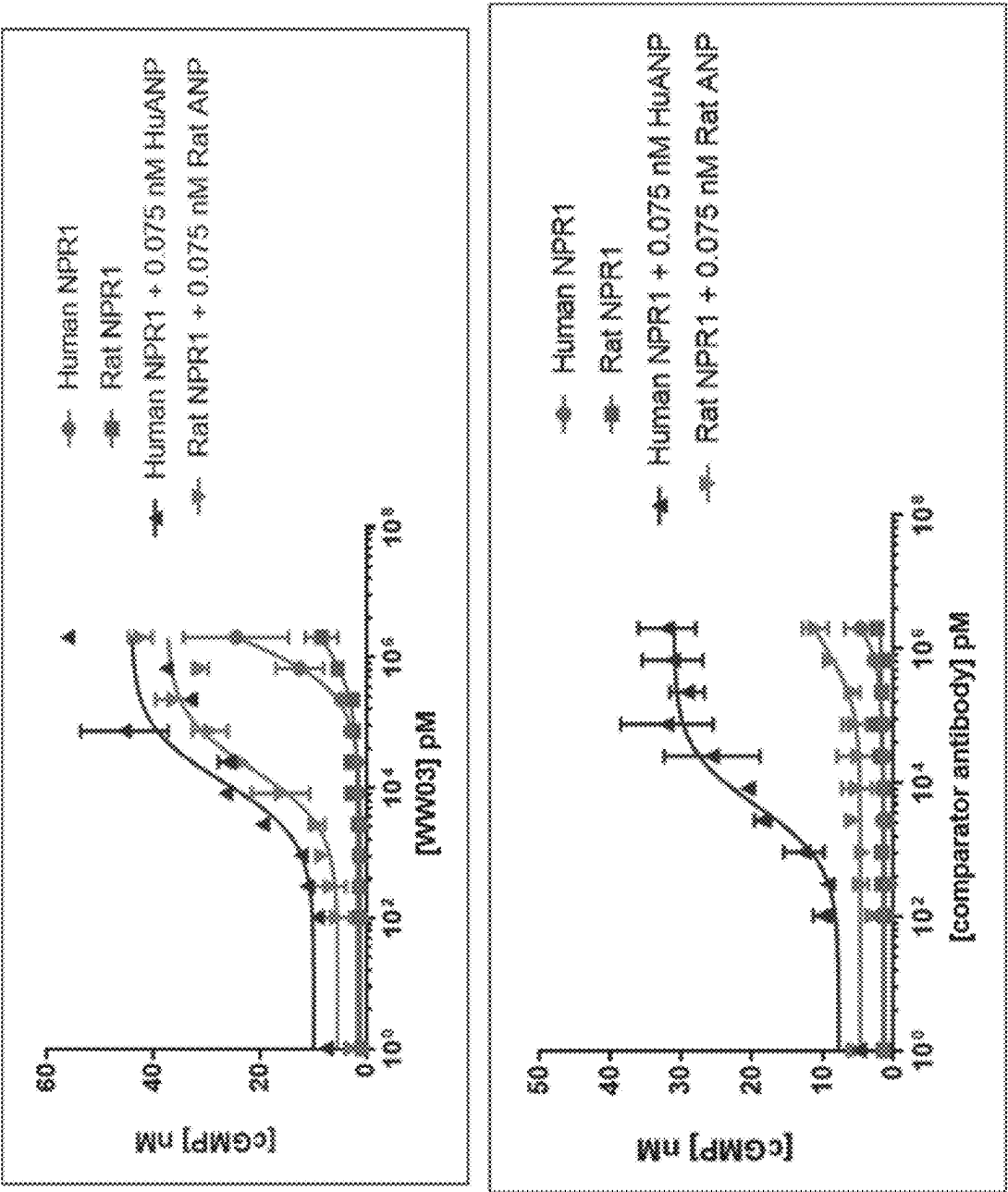


FIG. 31



1

NATRIURETIC PEPTIDE RECEPTOR 1 ANTIBODIES AND METHODS OF USE

RELATED APPLICATIONS

This application is a divisional of U.S. patent application Ser. No. 17/348,949, filed Jun. 16, 2021, which is a direct filing of U.S. patent application Ser. No. 16/897,935, filed Jun. 10, 2020, which claims the benefit of and priority to U.S. Provisional Application No. 62/860,508 filed Jun. 12, 2019, the entire contents of each of which are incorporated herein by reference in its entirety.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically herewith and is hereby incorporated by reference in its entirety. This sequence listing is named PAT058321-US-DIV02_SL.

BACKGROUND

Heart failure is a major public health problem concerning more than 20 million patients around the world (Orso et al., 2014, Expert Opin Pharmacother. 15 (13): 1849-1861) and is associated with high morbidity (Ibebuogu et al., 2011, Circulation. Heart failure 4 (2): 114-120). Natriuretic Peptide Receptor 1 (NPR1; also known as NPRA) is a receptor guanylate cyclase, which is activated by Atrial Natriuretic Peptide (ANP) resulting in lowering of blood pressure and blood volume (Chen & Burnett, 2006, European Heart Journal Supplements 8 (Suppl E): E18-E25; Ibebuogu et al. 2011, supra; Mani et al. 2015, Bioscience Reports 35 (5): e00260). ANP binding induces dimerization and twisting of the receptor that induces activation of the guanylate cyclase domain and conversion of GTP into cGMP (Misono et al., 2011, The FEBS journal 278 (11): 1818-1829). ANP is cleared by NPR3, a natriuretic peptide receptor that lacks the guanylate cyclase domain, and degraded by Neutral Endopeptidase (NEP) (Chen & Burnett 2006, supra; Schmitt et al., 2003, Clin Sci (Lond). 105 (2): 141-160). Certain antibodies against NPR1 have been described, for example, in WO2010/065293 (including antibody 5591-IgG). However, these antibodies appeared to have no functional activity in the absence of ANP in vitro and no functional activity in vivo.

It has been shown that an increase in ANP may be beneficial for patients with heart failure with reduced ejection fraction (outbound pumping of blood by heart). See McMurray et al., N. Engl. J. Med.; Vol. 371, No. 11, pp 993-1004 (2014); and Nogue et al., Eur J Heart Fail. 2019 May; 21 (5): 598-605. However, there is a need for further longer acting agents that have an alternative mode of action to supplement or replace existing therapies.

SUMMARY OF THE DISCLOSURE

Herein is demonstrated that it is possible to activate NPR1 by the use of agonistic anti-NPR1 antibodies or antigen binding fragments thereof. Furthermore, the present disclosure demonstrates that there are two types of such antibodies. While one type binds to NPR1 and competes with ANP binding (yet still activates NPR1; hereinafter "ANP competitive" anti-NPR1 antibodies), the second type is able to bind and activate NPR1 while not competing with ANP (hereinafter "ANP non-competitive" anti-NPR1 antibodies). Such antibodies (e.g., ANP non-competitive anti-NPR1 anti-

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bodies) may be used to bolster the body's natural system and/or existing treatment rationales. Furthermore, certain NPR1 agonist antibodies that are able to activate NPR1 in the absence of ANP have been found to be functionally equivalent to ANP.

The antibodies of the instant application show in vivo activity in both mouse and rat. Furthermore, the unique epitope binding of the antibodies described herein has been demonstrated using crystal structure data.

Thus the disclosure provides anti-NPR1 antibodies (e.g., human monoclonal antibodies) or antigen-binding fragments thereof that (i) bind to natriuretic peptide receptor 1 (NPR1); and (ii) are capable of activating NPR1 in the absence of ANP. Such antibodies are agonistic anti-NPR1 antibodies. In some embodiments of the invention, the disclosure also provides anti-NPR1 antibodies or antigen binding fragments thereof that (i) bind to natriuretic peptide receptor 1 (NPR1); and (ii) activate NPR1 in the absence of ANP. In some embodiments of the invention, the disclosure also provides antibodies or antigen binding fragments thereof that (i) bind to natriuretic peptide receptor 1 (NPR1); and (ii) activate NPR1 in both the presence and absence of ANP. Also provided are nucleic acids encoding said antibodies, vectors comprising said nucleic acids, host cells comprising said nucleic acids and/or vectors, and methods of manufacture of said antibodies using said nucleic acids, vectors and/or host cells. Also provided are pharmaceutical compositions and combinations comprising said antibodies, nucleic acids, vectors or host cells, as well as methods of treatment using said antibodies, nucleic acids, vectors, host cells or pharmaceutical compositions. The use of said antibodies, nucleic acids, vectors, host cells or pharmaceutical compositions or combinations in treating disease is also disclosed herein.

Thus, in one aspect of the invention, herein is provided an isolated antibody or antigen binding fragment that (i) binds to natriuretic peptide receptor 1 (NPR1); and (ii) is capable of activating NPR1 in the absence of atrial natriuretic peptide (ANP). In some embodiments of the invention, the isolated antibody or antigen binding fragment does not bind to and/or does not activate natriuretic peptide receptor 2 (NPR2) and/or natriuretic peptide receptor 3 (NPR3). In some embodiments of the invention, the isolated antibody or antigen binding fragment binds to (a) human NPR1; and (b) mouse NPR1 and/or rat NPR1.

In some embodiments of the invention, the antibody or antigen binding fragment binds to (a) human NPR1; and (b) cyno NPR1. In some embodiments of the invention, the antibody or antigen binding fragment is ANP non-competitive. In some embodiments of the invention, the antibody or antigen binding fragment is ANP competitive. In some embodiments of the invention, the antibody or antigen binding fragment is capable of stabilizing the ANP-NPR1 complex.

In some embodiments of the invention, the antibody or antigen binding fragment binds to an epitope within amino acids 99-133 of SEQ ID NO: 1, e.g., within a region of human NPR1 encompassed by amino acids 99-133 of SEQ ID NO: 1. In some embodiments of the invention, the antibody or antigen binding fragment binds to an epitope comprising at least two amino acid residues within amino acids 99-133 of SEQ ID NO: 1. In some embodiments of the invention, the antibody or antigen binding fragment binds to an epitope comprising at least 3, 4, 5, 6, 7, or 8 amino acid residues within amino acids 99-133 of SEQ ID NO: 1.

In some embodiments of the invention, the antibody or antigen binding fragment binds to an epitope within amino

In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), wherein: (a) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28; HCDR2 comprises or consists of an amino acid sequence as set forth in X₁IX₂SX₃GX₄YX₅X₆YADSVKG (SEQ ID NO: 429), wherein X₁ is A or V, X₂ is S or E, X₃ is D or K, X₄ is S or N, X₅ is I or T, and X₆ is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41; LCDR2 comprises

or consists of an amino acid sequence as set forth in SEQ ID NO: 42; and LCDR3 comprises or consists of an amino acid sequence as set forth in Y₁QY₂Y₃Y₄Y₅PRT (SEQ ID NO: 430); wherein Y₁ is M or Q, Y₂ is S, E, T, or I, Y₃ is Y or W, Y₄ is E, V, R, A, T, or M, and Y₅ is K, V, R, or A; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31; HCDR2 comprises or consists of an amino acid sequence as set forth in X₁IX₂SX₃GX₄YX₅X₆YADSVKG (SEQ ID NO: 429), wherein X₁ is A or V, X₂ is S or E, X₃ is D or K, X₄ is S or N, X₅ is I or T, and X₆ is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42; and LCDR3 comprises or consists of an amino acid sequence as set forth in Y₁QY₂Y₃Y₄Y₅PRT (SEQ ID NO: 430); wherein Y₁ is M or Q, Y₂ is S, E, T, or I, Y₃ is Y or W, Y₄ is E, V, R, A, T, or M, and Y₅ is K, V, R, or A; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 32; HCDR2 comprises or consists of an amino acid sequence as set forth in X₁SX₂GX₃Y (SEQ ID NO: 431), wherein X₁ is S or E, X₂ is D or K, or X₃ is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 44; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45; and LCDR3 comprises or consists of an amino acid sequence as set forth in Y₁Y₂Y₃Y₄PR (SEQ ID NO: 432); wherein Y₁ is S, E, T, or I, Y₂ is Y or W, Y₃ is E, V, R, A, T, or M, and Y₄ is K, V, R, or A; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 34; HCDR2 comprises or consists of an amino acid sequence as set forth in IX₁SX₂GX₃YX₄ (SEQ ID NO: 433), wherein X₁ is S or E, X₂ is D or K, X₃ is S or N, and X₄ is I or T, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 36; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 47; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45; and LCDR3 comprises or consists of an amino acid sequence as set forth in Y₁QY₂Y₃Y₄Y₅PRT (SEQ ID NO: 430); wherein Y₁ is M or Q, Y₂ is S, E, T, or I, Y₃ is Y or W, Y₄ is E, V, R, A, T, or M, and Y₅ is K, V, R, or A; (b) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28; HCDR2 comprises or consists of an amino acid sequence as set forth in X₁IX₂SX₃GX₄YX₅X₆YADSVKG (SEQ ID NO: 429), wherein X₁ is A or V, X₂ is S or E, X₃ is D or K, X₄ is S or N, X₅ is I or T, and X₆ is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42; and LCDR3 comprises or consists of an amino acid sequence as set forth in QY₁WY₂Y₃PRT (SEQ ID NO: 434); wherein Y₁ is S, E, T, or I, Y₂ is V, R, A, T, or M, and Y₃ is K, V, R, or A; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31; HCDR2 comprises or consists of an amino acid sequence as set forth in X₁IX₂SX₃GX₄YX₅X₆YADSVKG (SEQ ID NO: 429), wherein X₁ is A or V, X₂ is S or E, X₃ is D or K, X₄ is S or N, X₅ is I or T, and X₆ is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42; and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 43.

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comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in QQY₁WY₂Y₃PRT (SEQ ID NO: 434); wherein Y₁ is S, E, T, or I, Y₂ is V, R, A, T, or M, and Y₃ is K, V, R, or A; (d) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX₁THYIH (SEQ ID NO: 436), wherein X₁ is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in SIY₁Y₂Y₃GY₄Y₅TY₆YADSVKG (SEQ ID NO: 437), wherein Y₁ is S or G, Y₂ is S or G, Y₃ is S or Q, Y₄ is S, Q, or G, Y₅ is S, N, or M, and Y₆ is Y or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 17, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 18, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 7, HCDR2 comprises or consists of an amino acid sequence as set forth in SIY₁Y₂Y₃GY₄Y₅TY₆YADSVKG (SEQ ID NO: 437), wherein Y₁ is S or G, Y₂ is S or G, Y₃ is S or Q, Y₄ is S, Q, or G, Y₅ is S, N, or M, and Y₆ is Y or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 17, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 18, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX₁TH (SEQ ID NO: 438), wherein X₁ is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in Y₁Y₂Y₃GY₄Y₅ (SEQ ID NO: 439), wherein Y₁ is S or G, Y₂ is S or G, Y₃ is S or Q, Y₄ is S, Q, or G, and Y₅ is S, N, or M, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 20, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 22; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX₁THY (SEQ ID NO: 440), wherein X₁ is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in IY₁Y₂Y₃GY₄Y₅T (SEQ ID NO: 441), wherein Y₁ is S or G, Y₂ is S or G, Y₃ is S or Q, Y₄ is S, Q, or G, and Y₅ is S, N, or M, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 12, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 23, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (e) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX₁THYIH (SEQ ID NO: 436), wherein X₁ is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in SISY₁SGY₂Y₃TTYADSVKG (SEQ ID NO: 442), wherein Y₁ is S or G, Y₂ is S or Q, and Y₃ is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 17, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 18, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 7, HCDR2 comprises or consists of an amino acid sequence as set forth in

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(SEQ ID NO: 452), wherein X_1 is S or R, X_2 is W or Y, and X_3 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in $Y_1IY_2QY_3Y_4Y_5EY_6Y_7YVESVKG$ (SEQ ID NO: 446), wherein Y_1 is S or N, Y_2 is K or H, Y_3 is S, Q, or H, Y_4 is G or A, Y_5 is S, H, or L, Y_6 is Tor K, and Y_7 is Y, K, or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in $X_1YX_2IX_3$ (SEQ ID NO: 454), wherein X_1 is S or R, X_2 is W or Y, and X_3 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in $Y_1IY_2QY_3Y_4Y_5EY_6Y_7YVESVKG$ (SEQ ID NO: 446), wherein Y_1 is S or N, Y_2 is K or H, Y_3 is S, Q, or H, Y_4 is G or A, Y_5 is S, H, or L, Y_6 is Tor K, and Y_7 is Y, K, or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX₁Y (SEQ ID NO: 455), wherein X_1 is S or R, HCDR2 comprises or consists of an amino acid sequence as set forth in $Y_1QY_2Y_3Y_4E$ (SEQ ID NO: 449), wherein Y_1 is K or H, Y_2 is S, Q, or H, Y_3 is G or A, and Y_4 is S, H, or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 240, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 242; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX₁YX₂ (SEQ ID NO: 457), wherein X_1 is S or R, and X_2 is W or Y, HCDR2 comprises or consists of an amino acid sequence as set forth in $IY_1QY_2Y_3Y_4EY_5$ (SEQ ID NO: 451), wherein Y_1 is K or H, Y_2 is S, Q, or H, Y_3 is G or A, Y_4 is S, H, or L, and Y_5 is T or K, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 232, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; or (i) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX₁YX₂IX₃ (SEQ ID NO: 452), wherein X_1 is S or R, X_2 is W or Y, and X_3 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in $SIHQY_1Y_2Y_3EY_4Y_5YVESVKG$ (SEQ ID NO: 453), wherein Y_1 is Q or H, Y_2 is G or A, Y_3 is H or L, Y_4 is T or K, and Y_5 is K or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in $X_1YX_2IX_3$ (SEQ ID NO: 454), wherein X_1 is S or R, X_2 is W or Y, and X_3 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in $SIHQY_1Y_2Y_3EY_4Y_5YVESVKG$ (SEQ ID NO: 453), wherein Y_1 is Q or H, Y_2 is G or A, Y_3 is H or L, Y_4 is T or

K, and Y_5 is K or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX₁Y (SEQ ID NO: 455), wherein X_1 is S or R, HCDR2 comprises or consists of an amino acid sequence as set forth in $HQY_1Y_2Y_3E$ (SEQ ID NO: 456), wherein Y_1 is Q or H, Y_2 is G or A, and Y_3 is H or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 240, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 242; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX₁YX₂ (SEQ ID NO: 457), wherein X_1 is S or R, and X_2 is W or Y, HCDR2 comprises or consists of an amino acid sequence as set forth in $IHQY_1Y_2Y_3EY_4$ (SEQ ID NO: 458), wherein Y_1 is Q or H, Y_2 is G or A, Y_3 is H or L, and Y_4 is T or K, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 232, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239.

In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), wherein: (a) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 310, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 311, HCDR3 comprises or consists of an amino acid sequence as set forth in $GX_1X_2X_3GX_4LGFDH$ (SEQ ID NO: 459), wherein X_1 is A or S, X_2 is V or L, X_3 is A or P, and X_4 is Q or L, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 320, LCDR2 comprises or consists of an amino acid sequence as set forth in $GNSNRPY_1$ (SEQ ID NO: 460), wherein Y_1 is S or N, and LCDR3 comprises or consists of an amino acid sequence as set forth in $QSYZ_1Z_2Z_3Z_4Z_5Z_6Z_7V$ (SEQ ID NO: 461), wherein Z_1 is Y, D, or G, Z_2 is T, S, or A, Z_3 is S, P, or F, Z_4 is S, T, or P, Z_5 is H, S, or R, Z_6 is G, S, or F, and Z_7 is P, S, or V; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 229, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 311, HCDR3 comprises or consists of an amino acid sequence as set forth in $GX_1X_2X_3GX_4LGFDH$ (SEQ ID NO: 459), wherein X_1 is A or S, X_2 is V or L, X_3 is A or P, and X_4 is Q or L, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 320, LCDR2 comprises or consists of an amino acid sequence as set forth in $GNSNRPY_1$ (SEQ ID NO: 460), wherein Y_1 is S or N, and LCDR3 comprises or consists of an amino acid sequence as set forth in $QSYZ_1Z_2Z_3Z_4Z_5Z_6Z_7V$ (SEQ ID NO: 461), wherein Z_1 is Y, D, or G, Z_2 is T, S, or A, Z_3 is S, P, or F, Z_4 is S, T, or P, Z_5 is H, S, or R, Z_6 is G, S, or F, and Z_7 is P, S, or V; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 80, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 313,

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HCDR3 comprises or consists of an amino acid sequence as set forth in $GX_1X_2X_3GX_4LGFDH$ (SEQ ID NO: 459), wherein X_1 is A or S, X_2 is V or L, X_3 is A or P, and X_4 is Q or L, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 323, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 324, and LCDR3 comprises or consists of an amino acid sequence as set forth in $YZ_1Z_2Z_3Z_4Z_5Z_6Z_7$ (SEQ ID NO: 462), wherein Z_1 is Y, D, or G, Z_2 is T, S, or A, Z_3 is S, P, or F, Z_4 is S, T, or P, Z_5 is H, S, or R, Z_6 is G, S, or F, and Z_7 is P, S, or V; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 82, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 314, HCDR3 comprises or consists of an amino acid sequence as set forth in $ARGX_1X_2X_3GX_4LGFDH$ (SEQ ID NO: 463), wherein X_1 is A or S, X_2 is V or L, X_3 is A or P, and X_4 is Q or L, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 326, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 324, and LCDR3 comprises or consists of an amino acid sequence as set forth in $QSYZ_1Z_2Z_3Z_4Z_5Z_6Z_7V$ (SEQ ID NO: 461), wherein Z_1 is Y, D, or G, Z_2 is T, S, or A, Z_3 is S, P, or F, Z_4 is S, T, or P, Z_5 is H, S, or R, Z_6 is G, S, or F, and Z_7 is P, S, or V; (b) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in $GFTFX_1X_2YAX_3X_4$ (SEQ ID NO: 464), wherein X_1 is S or G, X_2 is S or T, X_3 is I or M, and X_4 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in $Y_1ISY_2Y_3GY_4Y_5Y_6Y_7YAY_8SVKG$ (SEQ ID NO: 465), wherein Y_1 is A or S, Y_2 is A, S, or G, Y_3 is S or H, Y_4 is G or Y, Y_5 is S or Y, Y_6 is T or A, Y_7 is Y, R, or N, and Y_8 is E or G, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in $X_1YAX_2X_3$ (SEQ ID NO: 466), wherein X_1 is S or T, X_2 is I or M, and X_3 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in $Y_1ISY_2Y_3GY_4Y_5Y_6Y_7YAY_8SVKG$ (SEQ ID NO: 465), wherein Y_1 is A or S, Y_2 is A, S, or G, Y_3 is S or H, Y_4 is G or Y, Y_5 is S or Y, Y_6 is T or A, Y_7 is Y, R, or N, and Y_8 is E or G, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in $GFTFX_1X_2Y$ (SEQ ID NO: 467), wherein X_1 is S or G, and X_2 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in $SY_1Y_2GY_3Y_4$ (SEQ ID NO: 468), wherein Y_1 is A, S, or G, Y_2 is S or H, Y_3 is G or Y, and Y_4 is S or Y, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 340, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 342; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in $GFTFX_1X_2YA$ (SEQ ID NO: 469), wherein X_1 is S or G, and X_2 is S or T, HCDR2 comprises or consists of an amino acid sequence as

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set forth in $ISY_1Y_2GY_3Y_4T$ (SEQ ID NO: 470), wherein Y_1 is S or G, Y_2 is S or H, Y_3 is G or Y, and Y_4 is S or Y, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 332, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 343, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; or (c) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in $GFTFX_1X_2YAX_3X_4$ (SEQ ID NO: 464), wherein X_1 is S or G, X_2 is S or T, X_3 is I or M, and X_4 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in $SISY_1Y_2GY_3Y_4YAY_5SVKG$ (SEQ ID NO: 471), wherein Y_1 is A or S, Y_2 is S or H, Y_3 is T or A, Y_4 is R or N, and Y_5 is E or G, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in $X_1YAX_2X_3$ (SEQ ID NO: 466), wherein X_1 is S or T, X_2 is I or M, and X_3 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in $SISY_1Y_2GY_3Y_4YAY_5SVKG$ (SEQ ID NO: 471), wherein Y_1 is A or S, Y_2 is S or H, Y_3 is T or A, Y_4 is R or N, and Y_5 is E or G, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in $GFTFX_1X_2Y$ (SEQ ID NO: 467), wherein X_1 is S or G, and X_2 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in $SY_1Y_2GY_3Y_4$ (SEQ ID NO: 472), wherein Y_1 is A or S, and Y_2 is S or H, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 340, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 342; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in $GFTFX_1X_2YA$ (SEQ ID NO: 469), wherein X_1 is S or G, and X_2 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in ISY_1Y_2G (SEQ ID NO: 473), wherein Y_1 is A, S, or G, and Y_2 is S or H, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 332, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 343, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339.

In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), wherein: (a) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 29, 119, and 190, HCDR3 comprises or

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consists of an amino acid sequence as set forth in SEQ ID NO: 22; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 10, 114, and 167, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 11, 102, and 153, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 12, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 23, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; or (c) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 226, 367, and 378, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 227, 368, and 379, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 229, 369, and 380, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 227, 368, and 379, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 32, 370, and 381, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 230, 371, and 382, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 240, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 242; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 34, 372, and 383, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 231, 373, and 384, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 232, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239.

In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), wherein: (a) (i) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 310, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 311, HCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOS: 312 and 348, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 320, LCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOS: 321 and 354, and LCDR3 comprises or

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(HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 46 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 43 (LCDR3); (w) (I) SEQ ID NO: 367 (HCDR1), SEQ ID NO: 368 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (II) SEQ ID NO: 369 (HCDR1), SEQ ID NO: 368 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (III) SEQ ID NO: 370 (HCDR1), SEQ ID NO: 371 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 242 (LCDR3); or (IV) SEQ ID NO: 372 (HCDR1), SEQ ID NO: 373 (HCDR2), SEQ ID NO: 232 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 239 (LCDR3); (x) (I) SEQ ID NO: 378 (HCDR1), SEQ ID NO: 379 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (II) SEQ ID NO: 380 (HCDR1), SEQ ID NO: 379 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (III) SEQ ID NO: 381 (HCDR1), SEQ ID NO: 382 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 242 (LCDR3); or (IV) SEQ ID NO: 383 (HCDR1), SEQ ID NO: 384 (HCDR2), SEQ ID NO: 232 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 239 (LCDR3); (y) (I) SEQ ID NO: 226 (HCDR1), SEQ ID NO: 227 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 227 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 30 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 242 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 231 (HCDR2), SEQ ID NO: 232 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 239 (LCDR3); (z) (I) SEQ ID NO: 270 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 282 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 283 (LCDR3); (II) SEQ ID NO: 273 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 282 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 283 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 274 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 284 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 285 (LCDR3); or (IV) SEQ ID NO: 275 (HCDR1), SEQ ID NO: 276 (HCDR2), SEQ ID NO: 277 (HCDR3), SEQ ID NO: 286 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 283 (LCDR3); or (aa) (I) SEQ ID NO: 291 (HCDR1), SEQ ID NO: 292 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 304 (LCDR3); (II) SEQ ID NO: 294 (HCDR1), SEQ ID NO: 292 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 304 (LCDR3); (III) SEQ ID NO: 295 (HCDR1), SEQ ID NO: 296 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 305 (LCDR3); or (IV)

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SEQ ID NO: 297 (HCDR1), SEQ ID NO: 298 (HCDR2), SEQ ID NO: 299 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 304 (LCDR3).

In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (a) (I) SEQ ID NO: 52 (HCDR1), SEQ ID NO: 53 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 65 (LCDR1), SEQ ID NO: 66 (LCDR2), and SEQ ID NO: 67 (LCDR3); (II) SEQ ID NO: 55 (HCDR1), SEQ ID NO: 53 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 65 (LCDR1), SEQ ID NO: 66 (LCDR2), and SEQ ID NO: 67 (LCDR3); (III) SEQ ID NO: 56 (HCDR1), SEQ ID NO: 57 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 68 (LCDR1), SEQ ID NO: 69 (LCDR2), and SEQ ID NO: 70 (LCDR3); or (IV) SEQ ID NO: 58 (HCDR1), SEQ ID NO: 59 (HCDR2), SEQ ID NO: 60 (HCDR3), SEQ ID NO: 71 (LCDR1), SEQ ID NO: 69 (LCDR2), and SEQ ID NO: 67 (LCDR3); (b) (I) SEQ ID NO: 76 (HCDR1), SEQ ID NO: 77 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 89 (LCDR1), SEQ ID NO: 90 (LCDR2), and SEQ ID NO: 91 (LCDR3); (II) SEQ ID NO: 79 (HCDR1), SEQ ID NO: 77 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 89 (LCDR1), SEQ ID NO: 90 (LCDR2), and SEQ ID NO: 91 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 81 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 92 (LCDR1), SEQ ID NO: 93 (LCDR2), and SEQ ID NO: 94 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 83 (HCDR2), SEQ ID NO: 84 (HCDR3), SEQ ID NO: 95 (LCDR1), SEQ ID NO: 93 (LCDR2), and SEQ ID NO: 91 (LCDR3); (c) (I) SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 361 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 361 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 362 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 349 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 361 (LCDR3); (d) (I) SEQ ID NO: 270 (HCDR1), SEQ ID NO: 389 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 273 (HCDR1), SEQ ID NO: 389 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 390 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 275 (HCDR1), SEQ ID NO: 391 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3); (e) (I) SEQ ID NO: 407 (HCDR1), SEQ ID NO: 408 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 409 (HCDR1), SEQ ID NO: 408 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 410 (HCDR1), SEQ ID NO: 411 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 412 (HCDR1), SEQ ID NO: 413 (HCDR2),

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SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3); (f) (I) SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 321 (LCDR2), and SEQ ID NO: 322 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 321 (LCDR2), and SEQ ID NO: 322 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 325 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 315 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 322 (LCDR3); (g) (I) SEQ ID NO: 270 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 273 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 274 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 275 (HCDR1), SEQ ID NO: 276 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3); or (h) (I) SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 355 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 355 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 356 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 349 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 355 (LCDR3).

In some embodiments of the invention, the antibody or antigen binding fragment comprises: (a) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 136; (b) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 136; (c) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 128; (d) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 128; (e) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 147; (f) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 147; (g) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 174; (h) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 174; (i) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an

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In some embodiments of the invention, the antibody or antigen binding fragment comprises: (a) a heavy chain comprising an amino acid sequence of SEQ ID NO: 203, and a light chain comprising an amino acid sequence of SEQ ID NO: 138; (b) a heavy chain comprising an amino acid sequence of SEQ ID NO: 208, and a light chain comprising an amino acid sequence of SEQ ID NO: 138; (c) a heavy chain comprising an amino acid sequence of SEQ ID NO: 203, and a light chain comprising an amino acid sequence of SEQ ID NO: 130; (d) a heavy chain comprising an amino acid sequence of SEQ ID NO: 208, and a light chain comprising an amino acid sequence of SEQ ID NO: 130; (e) a heavy chain comprising an amino acid sequence of SEQ ID NO: 203, and a light chain comprising an amino acid sequence of SEQ ID NO: 149; (f) a heavy chain comprising an amino acid sequence of SEQ ID NO: 208, and a light chain comprising an amino acid sequence of SEQ ID NO: 149; (g) a heavy chain comprising an amino acid sequence of SEQ ID NO: 203, and a light chain comprising an amino acid sequence of SEQ ID NO: 176; (h) a heavy chain comprising an amino acid sequence of SEQ ID NO: 208, and a light chain comprising an amino acid sequence of SEQ ID NO: 176; (i) a heavy chain comprising an amino acid sequence of SEQ ID NO: 203, and a light chain comprising an amino acid sequence of SEQ ID NO: 182; (j) a heavy chain comprising an amino acid sequence of SEQ ID NO: 208, and a light chain comprising an amino acid sequence of SEQ ID NO: 182; (k) a heavy chain comprising an amino acid sequence of SEQ ID NO: 203, and a light chain comprising an amino acid sequence of SEQ ID NO: 188; (l) a heavy chain comprising an amino acid sequence of SEQ ID NO: 208, and a light chain comprising an amino acid sequence of SEQ ID NO: 188; (m) a heavy chain comprising an amino acid sequence of SEQ ID NO: 105, and a light chain comprising an amino acid sequence of SEQ ID NO:

26; (n) a heavy chain comprising an amino acid sequence of SEQ ID NO: 108, and a light chain comprising an amino acid sequence of SEQ ID NO: 26; (o) a heavy chain comprising an amino acid sequence of SEQ ID NO: 117, and a light chain comprising an amino acid sequence of SEQ ID NO: 26; (p) a heavy chain comprising an amino acid sequence of SEQ ID NO: 124, and a light chain comprising an amino acid sequence of SEQ ID NO: 50; (q) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 130; (r) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 138; (s) a heavy chain comprising an amino acid sequence of SEQ ID NO: 141, and a light chain comprising an amino acid sequence of SEQ ID NO: 138; (t) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 149; (u) a heavy chain comprising an amino acid sequence of SEQ ID NO: 156, and a light chain comprising an amino acid sequence of SEQ ID NO: 26; (v) a heavy chain comprising an amino acid sequence of SEQ ID NO: 159, and a light chain comprising an amino acid sequence of SEQ ID NO: 26; (w) a heavy chain comprising an amino acid sequence of SEQ ID NO: 163, and a light chain comprising an amino acid sequence of SEQ ID NO: 26; (x) a heavy chain comprising an amino acid sequence of SEQ ID NO: 170, and a light chain comprising an amino acid sequence of SEQ ID NO: 26; (y) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 176; (z) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 182; (aa) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 188; (bb) a heavy chain comprising an amino acid sequence of SEQ ID NO: 195, and a light chain comprising an amino acid sequence of SEQ ID NO: 176; (dd) a heavy chain comprising an amino acid sequence of SEQ ID NO: 15, and a light chain comprising an amino acid sequence of SEQ ID NO: 26; (ee) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 50; (ff) a heavy chain comprising an amino acid sequence of SEQ ID NO: 376, and a light chain comprising an amino acid sequence of SEQ ID NO: 246; (gg) a heavy chain comprising an amino acid sequence of SEQ ID NO: 387, and a light chain comprising an amino acid sequence of SEQ ID NO: 246; (hh) a heavy chain comprising an amino acid sequence of SEQ ID NO: 235, and a light chain comprising an amino acid sequence of SEQ ID NO: 246; (ii) a heavy chain comprising an amino acid sequence of SEQ ID NO: 280, and a light chain comprising an amino acid sequence of SEQ ID NO: 289; or (jj) a heavy chain comprising an amino acid sequence of SEQ ID NO: 302, and a light chain comprising an amino acid sequence of SEQ ID NO: 308.

In some embodiments of the invention, the antibody or antigen binding fragment comprises: (a) a heavy chain comprising an amino acid sequence of SEQ ID NO: 63, and a light chain comprising an amino acid sequence of SEQ ID NO: 74; (b) a heavy chain comprising an amino acid sequence of SEQ ID NO: 87, and a light chain comprising an amino acid sequence of SEQ ID NO: 98; (c) a heavy

chain comprising an amino acid sequence of SEQ ID NO: 352, and a light chain comprising an amino acid sequence of SEQ ID NO: 365; (d) a heavy chain comprising an amino acid sequence of SEQ ID NO: 394, and a light chain comprising an amino acid sequence of SEQ ID NO: 346; (e) a heavy chain comprising an amino acid sequence of SEQ ID NO: 416, and a light chain comprising an amino acid sequence of SEQ ID NO: 346; (f) a heavy chain comprising an amino acid sequence of SEQ ID NO: 318, and a light chain comprising an amino acid sequence of SEQ ID NO: 329; (g) a heavy chain comprising an amino acid sequence of SEQ ID NO: 335, and a light chain comprising an amino acid sequence of SEQ ID NO: 346; or (h) a heavy chain comprising an amino acid sequence of SEQ ID NO: 352, and a light chain comprising an amino acid sequence of SEQ ID NO: 359.

In some embodiments of the invention, the antibody or antigen binding fragment is an antigen binding fragment selected from the group consisting of a Fab, Fab', F(ab')₂, Fv, single domain antibody (dAb), and a single chain variable fragment (scFv). In some embodiments of the invention, the antibody or antigen binding fragment is an antigen binding fragment selected from the group consisting of a Fab, Fab', Fv, single domain antibody (dAb), and a single chain variable fragment (scFv).

In some embodiments of the invention, the antibody or antigen binding fragment is monoclonal. In some embodiments of the invention, the antibody or antigen binding fragment is fully human. In some embodiments of the invention, the antibody or antigen binding fragment is an IgG antibody. In some embodiments of the invention, the antibody or antigen binding fragment is an IgG1 antibody. In some embodiments of the invention, the antibody or antigen binding fragment is an IgG1 antibody having a kappa light chain. In some embodiments of the invention, the antibody or antigen binding fragment is a fully human antibody of the IgG1 isotype and has a kappa light chain.

In some embodiments of the invention, the antibody or antigen binding fragment additionally has mutations in the Fc region according to the EU index of Kabat, wherein the mutations comprise at least D265A and P329A.

In some embodiments of the invention, the antibody or antigen binding fragment additionally has mutations in the Fc region according to the EU index of Kabat, wherein the mutations comprise at least L234A and L235A.

In some embodiments of the invention, the antibody or antigen binding fragment is therapeutic.

In some embodiments of the invention, the antibody or antigen binding fragment binds to the same epitope on human NPR1 as any of the antibodies or antigen binding fragments or groups defined herein (e.g., XX16). In some embodiments of the invention, the antibody or antigen binding fragment competes for binding to human NPR1 with any of the antibodies or antigen binding fragments or groups defined herein (e.g., XX16).

In one aspect of the invention, provided herein is an isolated nucleic acid or nucleic acids encoding the amino acid sequence of any of the antibodies or antigen binding fragments or groups defined herein. In one aspect of the invention, provided herein is a vector comprising the isolated nucleic acid(s). In one aspect of the invention, provided herein is a host cell comprising the isolated nucleic acid(s) or the vector.

In one aspect of the invention, provided herein is a method of producing any of the antibodies or antigen binding fragments described herein, comprising culturing the host cell described herein under conditions suitable to

produce the antibody or antigen binding fragment. In some embodiments of the invention, the method additionally comprises purification of the antibody or antigen binding fragment.

In one aspect of the invention, provided herein is pharmaceutical composition comprising a purified antibody or antigen binding fragment produced by the method described herein and a pharmaceutically acceptable carrier.

In one aspect of the invention, provided herein is pharmaceutical composition comprising any of the antibodies or antigen binding fragments described herein and a pharmaceutically acceptable carrier.

In one aspect of the invention, provided herein is pharmaceutical composition comprising: a) means for binding natriuretic peptide receptor 1 (NPR1) and activating NPR1 in the absence of ANP; and b) a pharmaceutically acceptable excipient. In some embodiments of the invention, said means for binding and activating is ANP non-competitive. In some embodiments of the invention, said means for binding and activating is ANP competitive. In some embodiments of the invention, said means for binding and activating is additionally capable of stabilizing the ANP-NPR1 complex. In some embodiments of the invention, the composition further comprises an additional therapeutic agent.

In some embodiments of the invention, the additional therapeutic agent is selected from an ACE (angiotensin-converting-enzyme) inhibitor, an angiotensin receptor blocker (ARB), a neprilysin inhibitor, a beta blocker, a diuretic, a calcium channel blocker, a cardiac glycoside, a sodium-glucose co-transporter 2 inhibitor (SGLT2i), and combinations thereof. In some embodiments of the invention, the additional therapeutic agent is selected from enalapril, benazepril, captopril, fosinopril, lisinopril, moexipril, perindopril, quinapril, ramipril, trandolapril, valsartan, azilsartan, candesartan, eprosartan, irbesartan, losartan, olmesartan, telmisartan, sacubitril, bisoprolol, carvedilol, propranolol, metoprolol, metoprolol tartrate, metoprolol succinate, thiazide diuretics, loop diuretics, potassium-sparing diuretics, amlodipine, cleveldipine, diltiazem, felodipine, isradipine, nicardipine, nifedipine, nisoldipine, verapamil, a digitalis glycoside, canagliflozin, dapagliflozin, empagliflozin, ertugliflozin, and combinations thereof. In some embodiments of the invention, the additional therapeutic agent is selected from chlorothiazide, chlorthalidone, hydrochlorothiazide, indapamide, metolazone, bumetanide, ethacrynic acid, furosemide, torsemide, amiloride, eplerenone, spironolactone, triamterene, digoxin, and combinations thereof. In some embodiments of the invention, the additional therapeutic agent is an angiotensin receptor-neprilysin inhibitor (ARNi).

In some embodiments of the invention, the additional therapeutic agent is selected from a corticosteroid, a leukotriene modifier, a bronchodilator, and combinations thereof. In some embodiments of the invention, the additional therapeutic agent is selected from fluticasone, budesonide, mometasone, beclomethasone, ciclesonide, fluticasone furoate, prednisone, methylprednisolone, montelukast, zafirlukast, zileuton, a long-acting beta agonist, a short-acting beta agonist, theophylline and ipratropium, and combinations thereof. In some embodiments of the invention, the additional therapeutic agent is selected from salmeterol, formoterol, albuterol, and levalbuterol, and combinations thereof.

In some embodiments of the invention, the additional therapeutic agent is selected from a beta-adrenoceptor antagonist, a carbonic anhydrase inhibitor, an alpha 2-adrenoceptor agonist, a parasympathomimetic, a prostaglandin

analog, a rho kinase inhibitor, and combinations thereof, and combinations thereof. In some embodiments of the invention, the additional therapeutic agent is selected from timolol, levobunolol, metipranolol, carteolol, betaxolol, acetazolamide, dorzolamide, brinzolamide, methazolamide, brimonidine, apraclonidine, a cholinomimetic, latanoprost, latanoprostene bunod, travoprost, bimatoprost, tafluprost, netarsudil and ripasudil, and combinations thereof.

In one aspect of the invention, provided herein is a method of treating a disorder or a disease associated with natriuretic peptide receptor activity in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of any of the antibodies or antigen binding fragments described herein or a pharmaceutical composition or combination as described herein.

In one aspect of the invention, provided herein is a method of treating a cardiovascular disorder in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of any one of the antibodies or antigen binding fragments thereof described herein or any one of the pharmaceutical compositions or combinations described herein.

In some embodiments of the invention, the cardiovascular disorder is selected from: hypertension, peripheral vascular disease, heart failure, coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy, diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, and myocardial infarction (MI).

In one aspect of the invention, provided herein is a method of treating heart failure, hypertrophic cardiomyopathy (HCM), hypertension, preeclampsia, asthma, glaucoma, and/or cytokine release syndrome in a subject in need thereof, comprising administering to the subject a therapeutically effective amount of any one of the antibodies or antigen binding fragments thereof described herein or any one of the pharmaceutical compositions or combinations described herein.

In some embodiments of the invention, the subject has heart failure, wherein the heart failure is selected from a heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure. In some embodiments of the invention, the subject has hypertrophic cardiomyopathy, wherein the hypertrophic cardiomyopathy is ventricular hypertrophy. In some embodiments of the invention, the subject has hypertension, wherein the hypertension is selected from resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, and pulmonary arterial hypertension. In some embodiments of the invention, the subject has hypertension, wherein the hypertension is selected from resistant hypertension or hypertensive heart disease.

In one aspect of the invention, provided herein is a method of treating a kidney disorder in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of any one of the antibodies or antigen binding fragments thereof described herein or any one of the pharmaceutical compositions or combinations described herein. In some embodiments of the invention, the kidney disorder is selected from: diabetic renal insufficiency, non-diabetic renal insufficiency, renal

failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD).

In one aspect of the invention, provided herein is a use of any one of the antibodies or antigen binding fragments thereof described herein or any one of the pharmaceutical compositions or combinations described herein, for the manufacture of a medicament for the treatment of a disorder or disease associated with natriuretic peptide receptor activity in a subject in need of such treatment.

In one aspect of the invention, provided herein is a use of any one of the antibodies or antigen binding fragments thereof described herein or any one of the pharmaceutical compositions or combinations described herein, for the manufacture of a medicament for the treatment of a cardiovascular disorder in a subject in need of such treatment.

In some embodiments of the invention, the cardiovascular disorder is selected from: hypertension, peripheral vascular disease, heart failure, coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy, diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, and myocardial infarction (MI).

In one aspect of the invention, provided herein is a use of any one of the antibodies or antigen binding fragments thereof described herein or any one of the pharmaceutical compositions or combinations described herein, for the manufacture of a medicament for the treatment of heart failure, hypertrophic cardiomyopathy (HCM), hypertension, preeclampsia, asthma, glaucoma, and/or cytokine release syndrome in a subject in need of such treatment.

In some embodiments of the invention, the subject has heart failure, and the heart failure is selected from a heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure. In some embodiments of the invention, the subject has hypertrophic cardiomyopathy, wherein the hypertrophic cardiomyopathy is ventricular hypertrophy. In some embodiments of the invention, the subject has hypertension, wherein the hypertension is selected from resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, and pulmonary arterial hypertension. In some embodiments of the invention, the subject has hypertension, wherein the hypertension is selected from resistant hypertension or hypertensive heart disease.

In one aspect of the invention, provided herein is a use of any one of the antibodies or antigen binding fragments thereof described herein or any one of the pharmaceutical compositions or combinations described herein, for the manufacture of a medicament for the treatment of a kidney disorder in a subject in need of such treatment. In some embodiments of the invention, the kidney disorder is selected from: diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary

renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD).

In one aspect of the invention, provided herein is an antibody or antigen binding fragment thereof described herein or any one of the pharmaceutical compositions or combinations described herein, for use in the treatment of a disorder or disease associated with natriuretic peptide receptor activity in a subject in need of such treatment.

In one aspect of the invention, provided herein is an antibody or antigen binding fragment thereof described herein or any one of the pharmaceutical compositions or combinations described herein, for use in the treatment of a cardiovascular disorder in a subject in need of such treatment.

In some embodiments of the invention, the cardiovascular disorder is selected from: hypertension, peripheral vascular disease, heart failure, coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy, diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, and myocardial infarction (MI).

In one aspect of the invention, provided herein is an antibody or antigen binding fragment thereof described herein or any one of the pharmaceutical compositions or combinations described herein, for use in the treatment of heart failure, hypertrophic cardiomyopathy (HCM), hypertension, preeclampsia, asthma, glaucoma, and/or cytokine release syndrome in a subject in need of such treatment.

In some embodiments of the invention, the subject has heart failure, and the heart failure is selected from a heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure. In some embodiments of the invention, the subject has hypertrophic cardiomyopathy, wherein the hypertrophic cardiomyopathy is ventricular hypertrophy. In some embodiments of the invention, the subject has hypertension, wherein the hypertension is selected from resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, and pulmonary arterial hypertension. In some embodiments of the invention, the subject has hypertension, wherein the hypertension is selected from resistant hypertension or hypertensive heart disease.

In one aspect of the invention, provided herein is an antibody or antigen binding fragment thereof described herein or any one of the pharmaceutical compositions or combinations described herein, for use in the treatment of a kidney disorder in a subject in need of such treatment. In some embodiments of the invention, the kidney disorder is selected from: diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD).

In one aspect of the invention, provided herein is a method of treating a disorder or a disease associated with

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natriuretic peptide receptor activity in a subject in need thereof, comprising administering a pharmaceutical composition comprising: means for binding natriuretic peptide receptor 1 (NPR1) and activating NPR1 in the absence of ANP; and a pharmaceutically acceptable excipient.

In one aspect of the invention, provided herein is a method of treating a cardiovascular disorder in a subject in need thereof, comprising administering a pharmaceutical composition comprising: means for binding natriuretic peptide receptor 1 (NPR1) and activating NPR1 in the absence of ANP; and a pharmaceutically acceptable excipient. In some embodiments of the invention, the cardiovascular disorder is selected from: hypertension, peripheral vascular disease, heart failure, coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy, diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, and myocardial infarction (MI).

In one aspect of the invention, provided herein is a method of treating heart failure, hypertrophic cardiomyopathy (HCM), hypertension, preeclampsia, asthma, glaucoma, and/or cytokine release syndrome in a subject in need thereof, comprising administering a pharmaceutical composition comprising: means for binding natriuretic peptide receptor 1 (NPR1) and activating NPR1 in the absence of ANP; and a pharmaceutically acceptable excipient. In some embodiments of the invention, the subject has heart failure, wherein the heart failure is selected from a heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure. In some embodiments of the invention, the subject has hypertrophic cardiomyopathy, wherein the hypertrophic cardiomyopathy is ventricular hypertrophy. In some embodiments of the invention, the subject has hypertension, wherein the hypertension is selected from resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, and pulmonary arterial hypertension. In some embodiments of the invention, the subject has hypertension, wherein the hypertension is selected from resistant hypertension or hypertensive heart disease.

In one aspect of the invention, provided herein is a method of treating a kidney disorder in a subject in need thereof, comprising administering a pharmaceutical composition comprising: means for binding natriuretic peptide receptor 1 (NPR1) and activating NPR1 in the absence of ANP; and a pharmaceutically acceptable excipient. In some embodiments of the invention, the kidney disorder is selected from: diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD). In some embodiments of the invention, said means for binding and activating is ANP non-competitive. In some embodiments of the invention, said means for binding and activating is ANP competitive. In some embodiments of the invention, said means for binding and activating is additionally capable of stabilizing the ANP-NPR1 complex.

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In one aspect of the invention, herein is provided an isolated anti-NPR1 antibody or antigen binding fragment, wherein the antibody or antigen binding fragment thereof binds to a conformational epitope of human NPR1, and wherein the conformational epitope comprises at least one amino acid residue within each of (i) amino acids 99-103 of SEQ ID NO: 1, (ii) 105-111 of SEQ ID NO: 1, (iii) 131-134 of SEQ ID NO: 1, and additionally binds to amino acid 375 and/or 378 of SEQ ID NO: 1.

In one aspect of the invention, herein is provided an isolated anti-NPR1 antibody or antigen binding fragment, wherein the antibody or antigen binding fragment thereof binds to a conformational epitope within NPR1, and wherein the conformational epitope comprises at least one amino acid residue within each of (i) amino acids 188-198 of SEQ ID NO: 1, (ii) 201-208 of SEQ ID NO: 1, (iii) 215-238 of SEQ ID NO: 1, and (iv) 294-297 of SEQ ID NO: 1, optionally wherein the antibody or antigen binding fragment thereof binds to at least amino acids 188, 192, 194, 197, 201, 208, and 219 of SEQ ID NO: 1.

In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), wherein: (a) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28; HCDR2 comprises or consists of an amino acid sequence as set forth in $X_1IX_2SX_3GX_4YX_5X_6YADSVK$ (SEQ ID NO: 429), wherein X_1 is A or V, X_2 is S or E, X_3 is D or K, X_4 is S or N, X_5 is I or T, and X_6 is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42; and LCDR3 comprises or consists of an amino acid sequence as set forth in $Y_1QY_2Y_3Y_4Y_5PRT$ (SEQ ID NO: 430); wherein Y_1 is M or Q, Y_2 is S, E, T, or I, Y_3 is Y or W, Y_4 is E, V, R, A, T, or M, and Y_5 is K, V, R, or A; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31; HCDR2 comprises or consists of an amino acid sequence as set forth in $X_1IX_2SX_3GX_4YX_5X_6YADSVK$ (SEQ ID NO: 429), wherein X_1 is A or V, X_2 is S or E, X_3 is D or K, X_4 is S or N, X_5 is I or T, and X_6 is Y or F; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42; and LCDR3 comprises or consists of an amino acid sequence as set forth in $Y_1QY_2Y_3Y_4Y_5PRT$ (SEQ ID NO: 430); wherein Y_1 is M or Q, Y_2 is S, E, T, or I, Y_3 is Y or W, Y_4 is E, V, R, A, T, or M, and Y_5 is K, V, R, or A; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 32; HCDR2 comprises or consists of an amino acid sequence as set forth in $X_1SX_2GX_3Y$ (SEQ ID NO: 431), wherein X_1 is S or E, X_2 is D or K, or X_3 is S or N; HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30; LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 44; LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45; and LCDR3 comprises or consists of an amino acid sequence as set forth in $Y_1Y_2Y_3Y_4PR$ (SEQ ID NO: 432); wherein Y_1 is S, E, T, or I, Y_2 is Y or W, Y_3 is E, V, R, A, T, or M, and Y_4 is K, V, R, or A; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 34; HCDR2

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is S or G, Y₂ is S or G, Y₃ is S or Q, Y₄ is S, Q, or G, and Y₅ is S, N, or M, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 20, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 22; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX₁THY (SEQ ID NO: 440), wherein X₁ is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in IY₁Y₂Y₃GY₄Y₅ST (SEQ ID NO: 441), wherein Y₁ is S or G, Y₂ is S or G, Y₃ is S or Q, Y₄ is S, Q, or G, and Y₅ is S, N, or M, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 12, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 23, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; or (e) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX₁THYIH (SEQ ID NO: 436), wherein X₁ is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in SISY₁SGY₂Y₃TYADSVKG (SEQ ID NO: 442), wherein Y₁ is S or G, Y₂ is S or Q, and Y₃ is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 17, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 18, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 7, HCDR2 comprises or consists of an amino acid sequence as set forth in SISY₁SGY₂Y₃TYADSVKG (SEQ ID NO: 442), wherein Y₁ is S or G, Y₂ is S or Q, and Y₃ is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 17, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 18, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX₁TH (SEQ ID NO: 438), wherein X₁ is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in SY₁SGY₂Y₃ (SEQ ID NO: 443), wherein Y₁ is S or G, Y₂ is S or Q, and Y₃ is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 20, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 22; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX₁THY (SEQ ID NO: 440), wherein X₁ is N, S, or Q, HCDR2 comprises or consists of an amino acid sequence as set forth in ISY₁SGY₂Y₃T (SEQ ID NO: 444), wherein Y₁ is S or G, Y₂ is S or Q, and Y₃ is S or N, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 12, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 23, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19.

In some aspects and embodiments of the invention, herein is provided an isolated anti-NPR1 antibody or antigen binding fragment, wherein the antibody or antigen binding

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fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), and wherein: (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 29, 119, and 190, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 43, 126, 134, 145, 172, 178, and 184; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 29, 119, and 190, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 43, 126, 134, 145, 172, 178, and 184; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 32, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 33, 120, and 191, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 44, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 46, 127, 135, 146, 173, 179, and 185; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 34, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 35, 121, and 192, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 36, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 47, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 43, 126, 134, 145, 172, 178, and 184.

In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (a) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3); (b) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41

[illegible]

(HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 184 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 185 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 184 (LCDR3); (n) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 191 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 192 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3); (o) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 191 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 173 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 192 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 172 (LCDR3); (p) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 46 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 43 (LCDR3).

In some embodiments of the invention, the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 76 (HCDR1), SEQ ID NO: 77 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 89 (LCDR1), SEQ ID NO: 90 (LCDR2), and SEQ ID NO: 91 (LCDR3); (II) SEQ ID NO: 79 (HCDR1), SEQ ID NO: 77 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 89 (LCDR1), SEQ ID NO: 90 (LCDR2), and SEQ ID NO: 91 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 81 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 92 (LCDR1), SEQ ID NO: 93 (LCDR2), and SEQ ID NO: 94 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 83 (HCDR2), SEQ ID NO: 84 (HCDR3), SEQ ID NO: 95 (LCDR1), SEQ ID NO: 93 (LCDR2), and SEQ ID NO: 91 (LCDR3).

In some embodiments of the invention, the antibody or antigen binding fragment comprises: (a) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an

amino acid sequence of SEQ ID NO: 136; (b) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 136; or (c) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 37, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 48.

In some embodiments of the invention, the antibody or antigen binding fragment comprises: a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 85, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 96.

In some embodiments of the invention, the antibody or antigen binding fragment comprises: (a) a heavy chain comprising an amino acid sequence of SEQ ID NO: 203, and a light chain comprising an amino acid sequence of SEQ ID NO: 138; (b) a heavy chain comprising an amino acid sequence of SEQ ID NO: 208, and a light chain comprising an amino acid sequence of SEQ ID NO: 138; or (c) a heavy chain comprising an amino acid sequence of SEQ ID NO: 39, and a light chain comprising an amino acid sequence of SEQ ID NO: 50.

In some embodiments of the invention, the antibody or antigen binding fragment comprises: a heavy chain comprising an amino acid sequence of SEQ ID NO: 87, and a light chain comprising an amino acid sequence of SEQ ID NO: 98.

In some embodiments of the invention, the antigen binding fragment is selected from the group consisting of a Fab, Fab', F(ab')₂, Fv, single domain antibody (dAb), and a single chain variable fragment (scFv). In some embodiments of the invention, the antibody or antigen binding fragment is therapeutic.

In some aspects of the invention, provided herein is an isolated nucleic acid or nucleic acids encoding the amino acid sequence of any of the antibodies or antigen binding fragments described herein. In some aspects of the invention, provided herein is a vector comprising any of the isolated nucleic acid(s) described herein. In some aspects of the invention, provided herein is a host cell comprising any of the isolated nucleic acid(s) described herein. In some aspects of the invention, provided herein is a method of producing an isolated anti-NPR1 antibody or antigen binding fragment, comprising culturing any of the host cells described herein under conditions suitable for producing the antibody or antigen binding fragment.

In some aspects of the invention, provided herein is a pharmaceutical composition comprising any of the antibodies or antigen binding fragments described herein and a pharmaceutically acceptable carrier. In some embodiments of the invention, the composition further comprises an additional therapeutic agent. In some embodiments of the invention, the additional therapeutic agent is selected from an ACE (angiotensin-converting-enzyme) inhibitor, an angiotensin receptor blocker (ARB), a neprilysin inhibitor, a beta blocker, a diuretic, a calcium channel blocker, a cardiac glycoside, a sodium-glucose co-transporter 2 inhibitor (SGLT2i), an angiotensin receptor-neprilysin inhibitor (ARNi), a corticosteroid, a leukotriene modifier, a bronchodilator, a beta-adrenoceptor antagonist, a carbonic anhydrase inhibitor, an alpha 2-adrenoceptor agonist, a parasympathomimetic, a prostaglandin analog, a rho kinase inhibitor, and combinations thereof. In some embodiments of the invention, the additional therapeutic agent is selected from enalapril, benazepril, captopril, fosinopril, lisinopril, moexipril, perindopril, quinapril, ramipril, trandolapril, valsartan, azilsartan, candesartan, eprosartan, irbesartan, losartan, olme-

sartan, telmisartan, sacubitril, bisoprolol, carvedilol, propranolol, metoprolol, metoprolol tartrate, metoprolol succinate, thiazide diuretics, loop diuretics, potassium-sparing diuretics, amlodipine, clevidipine, diltiazem, felodipine, isradipine, nicardipine, nifedipine, nisoldipine, verapamil, a digitalis glycoside, canagliflozin, dapagliflozin, empagliflozin, ertugliflozin, chlorothiazide, chlorthalidone, hydrochlorothiazide, indapamide, metolazone, bumetanide, ethacrynic acid, furosemide, torsemide, amiloride, eplerenone, spironolactone, triamterene, digoxin, fluticasone, budesonide, mometasone, beclomethasone, ciclesonide, fluticasone furoate, prednisone, methylprednisolone, montelukast, zafirlukast, zileuton, a long-acting beta agonist, a short-acting beta agonist, theophylline, ipratropium, salmeterol, formoterol, albuterol, levalbuterol, timolol, levobunolol, metipranolol, carteolol, betaxolol, acetazolamide, dorzolamide, brinzolamide, methazolamide, brimonidine, apraclonidine, a cholinomimetic, latanoprost, latanoprostene bunod, travoprost, bimatoprost, tafluprost, netarsudil, ripasudil, and combinations thereof.

In some aspects of the invention, provided herein is a method of treating a disorder or a disease associated with natriuretic peptide receptor activity in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of any of the antibodies or antigen binding fragments described herein.

In some aspects of the invention, provided herein is a method of treating a cardiovascular disorder in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of any of the antibodies or antigen binding fragments described herein. In some embodiments of the invention, the cardiovascular disorder is selected from: hypertension, peripheral vascular disease, heart failure, coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy, diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, and myocardial infarction (MI).

In some aspects of the invention, provided herein is a method of treating heart failure, hypertrophic cardiomyopathy (HCM), hypertension, preeclampsia, asthma, glaucoma, and/or cytokine release syndrome in a subject in need thereof, comprising administering to the subject a therapeutically effective amount of any of the antibodies or antigen binding fragments described herein. In some embodiments of the invention, the subject has heart failure, and the heart failure is selected from a heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure. In some embodiments of the invention, the subject has hypertrophic cardiomyopathy, and wherein the hypertrophic cardiomyopathy is ventricular hypertrophy. In some embodiments of the invention, the subject has hypertension, and the hypertension is selected from resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, and pulmonary arterial hypertension. In some embodiments of the invention, the subject has hypertension, and the hypertension is selected from resistant hypertension or hypertensive heart disease.

In some aspects of the invention, provided herein is a method of treating a kidney disorder in a subject in need thereof, the method comprising administering to the subject

a therapeutically effective amount of any of the antibodies or antigen binding fragments described herein. In some embodiments of the invention, the kidney disorder is selected from: diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD).

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a set of graphs displaying the results of antibody candidates WW01, WW02, WW03, WW04, and WW06 binding to the following antigens (ELISA analysis): human NPR1, constitutively active human NPR1 mutant (W74R), rat NPR1, and human NPR3 (counter target) both in the absence of and presence of a 250 fold molar excess of ANP.

FIG. 2 is a set of graphs displaying the results of flow cytometry analysis of antibody candidates WW01, WW02, WW03, WW04, and WW06 for binding to human NPR1 expressing CHO-K1 cells in the absence and presence of a saturating concentration of ANP and on parental CHO-K1 cells.

FIG. 3 is a graphical representation of a Fluorescence Resonance Energy Transfer (FRET)-based assay in which the NPR1-specific antibodies competed with ANP for binding to NPR1. In this FRET based assay, Eu-labeled Streptavidin (for measurement of IgGs) or Eu-labeled anti-hFc antibody (for measurement of FabCys) was used as an energy donor, while Cy5-labeled ANP was used as an acceptor.

FIG. 4 is a set of graphs displaying the results of ANP competition analyses of candidates WW01, WW02, WW03, WW04, and WW06 using the FRET-based assay depicted in FIG. 3.

FIG. 5 is a set of graphs depicting the results of functional activity analyses of candidates WW01, WW02, WW03, WW04, and WW06 in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells. Results represent the cellular production of cGMP [nM] in the absence or presence of 0.075 nM ANP.

FIG. 6 is a set of graphs depicting the results of functional activity analyses of candidate WW06 in IgG or FabCys format in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells. Results represent the cellular production of cGMP [nM] in the absence or presence of 0.075 nM ANP.

FIG. 7 demonstrates the results of SET screening (hNPR1 affinity) of 82 HCDR2 or LCDR3 unique improved HuCAL® derivatives based on the parental antibody.

FIG. 8 demonstrates the results of SET screening (hNPR1-ANP complex affinity) of 82 HCDR2 or LCDR3 unique improved HuCAL® derivatives based on the parental antibody.

FIG. 9 is a set of graphs displaying the results of antibody candidates XX01-XX08, XX10, and XX12 binding to the following antigens (ELISA analysis): human NPR1, constitutively active human NPR1 mutant (W74R), rat NPR1, and human NPR3 (counter target) in the absence of or in the presence of a 250 fold molar excess of ANP.

FIG. 10 is a set of graphs displaying the results of flow cytometry analysis of antibody candidates XX01-XX08, XX10, and XX12 for binding to human NPR1 expressing

CHO-K1 cells in the absence or presence of a saturating concentration of ANP and on parental CHO-K1 cells.

FIG. 11 is a set of graphs displaying the results of ANP competition analyses of candidates XX01-XX07, XX10, and XX12 using the FRET-based assay depicted in FIG. 3.

FIG. 12 is a set of graphs depicting the results of functional activity analyses of candidates XX01-XX08, XX10, and XX12 in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells. Results represent the cellular production of cGMP [nM] in the absence or presence of 0.075 nM ANP.

FIG. 13 is a set of graphs depicting the results of functional activity analyses of candidates XX06 and XX16 alongside natural ligand ANP in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells. Results represent the cellular production of cGMP [nM] in the absence or presence of 0.075 nM ANP.

FIG. 14 is a graph depicting the results of functional activity analyses of candidate XX16 alongside natural ligand ANP in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells. Results represent the cellular production of cGMP [nM] as a function of either ANP or XX16.

FIG. 15 is a graphical representation of the crystal structure of Fab06 (the WW06 antibody in Fab format) in complex with hNPR1 extracellular domain.

FIG. 16 is a graphical representation of the conformation of the hNPR1 extracellular domain as it would be in complex with Fab06. The Fab06 were removed from this representation to more clearly reveal the conformation of hNPR1 induced by Fab binding. W74R is shown in space filling.

FIG. 17 is a graphical representation of the crystal structure of hNPR1 extracellular domain in complex with ANP (left) and the crystal structure of Fab16 (the XX16 antibody in Fab format) in complex with hNPR1 extracellular domain (right) with W74 shown in space filling.

FIG. 18 is a set of graphs depicting the results of functional activity analyses of candidate WW06 in a cellular cGMP production assay using when tested on CHO-K1 cells expressing hNPR1 W74R/C232T (constitutively active mutant; panel A) compared to WT hNPR1 (panel B).

FIG. 19 is a set of graphs displaying the results of antibody candidates YY01-YY07 binding to the following antigens (ELISA analysis): human NPR1, constitutively active human NPR1 mutant (W74R), rat NPR1, and human NPR3 (counter target) in the absence of or presence of a 250 fold molar excess of ANP.

FIG. 20 is a set of graphs displaying the results of flow cytometry analysis of antibody candidates YY01-YY07 for binding to human NPR1 expressing CHO-K1 cells in the absence or presence of a saturating concentration of ANP and on parental CHO K1 cells.

FIG. 21 is a set of graphs displaying the results of ANP competition analyses of candidates YY01-YY07 using the FRET-based assay depicted in FIG. 3.

FIG. 22 is a set of graphs depicting the results of functional activity analyses of candidates YY01-YY07 in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells. Results represent the cellular production of cGMP [nM] in the absence or presence of 0.075 nM ANP.

FIG. 23 is a set of graphs depicting the results of functional activity analyses of candidates YY05 and YY07 in IgG or FabCys format in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells. Results rep-

resent the cellular production of cGMP [nM] in the absence or presence of 0.075 nM ANP.

FIG. 24 demonstrates the results of SET screening (hNPR1 affinity) of 112 HCDR1/2 or LCDR3 unique improved Ylanthia® derivatives based on the parental antibody.

FIG. 25 demonstrates the results of SET screening (hNPR1-ANP complex affinity) of 112 HCDR1/2 or LCDR3 unique improved Ylanthia® derivatives based on the parental antibody.

FIG. 26 is a set of graphs representing the plasma cGMP (nM) concentration over time in hNPR1 Tg mice which were intravenously administered 1 mg/kg, 3 mg/kg, or 10 mg/kg of the WW06 Fab.

FIG. 27 is a graph representing the concentration of antibody over time in hNPR1 Tg mice which were intravenously administered 1 mg/kg, 3 mg/kg, or 10 mg/kg of the WW06 Fab.

FIG. 28 is a set of graphs demonstrating the results of subcutaneous administration of vehicle, 0.3 mg/kg XX16, or 3 mg/kg XX16 once every two weeks in wild type or ANP knockout (ANP KO) mice on heart weight/body weight ratio (left) and plasma NT-proBNP levels (pg/mL; right). Measurements were taken two weeks after the second administration.

FIG. 29 is a set of graphs demonstrating the results of a single intravenous administration of vehicle, 0.3 mg/kg XX16, or 1 mg/kg XX16 on blood pressure (mean arterial pressure; left) and urinary flow rate (right) in hypertensive rats (spontaneous hypertensive rat stroke prone, SHRsp). Measurements were taken three hours after the intravenous administration.

FIG. 30 is a set of graphs demonstrating the results of a single subcutaneous administration of 0.1, 0.3, 1, 3, 10, or 30 mg/kg XX16 in telemetry implanted normal rats on mean arterial pressure (MAP; top) and plasma cGMP (bottom) over time.

FIG. 31 is a set of graphs depicting the results of functional activity analyses of candidate WW03 and a comparator antibody (antibody 5591-IgG from PCT Application No. WO2010/065293A1) in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells or rat NPR1 expressing rat cells. Results represent the cellular production of cGMP [nM] in the absence or presence of 0.075 nM ANP (human or rat).

GENERAL DEFINITIONS

In order that the present disclosure may be more readily understood, certain terms are first defined. Additional definitions are set forth throughout the detailed description as required.

Throughout the description and claims of this specification, the words “comprise” and “contain” and variations of the words, for example “comprising” and “comprises”, mean “including but not limited to”, and do not exclude other components, integers, or steps. Moreover the singular encompasses the plural unless the context otherwise requires: in particular, where the indefinite article is used, the specification is to be understood as contemplating plurality as well as singularity, unless the context requires otherwise.

As used herein, “NPR1” and “NPR1 protein” refers to Natriuretic Peptide Receptor 1. This protein is also known as Atrial natriuretic peptide receptor type A (ANP-A, ANPR-A or NPR-A) and Guanylate cyclase A (GC-A). In some embodiments, the NPR1 referred to is human NPR1. In some embodiments the human NPR1 is has UniProt acces-

sion number P16066 or GenBank Accession number EAW53284.1 (SEQ ID NO: 1). In some embodiments, the NPR1 referred to is mouse (*Mus musculus*) NPR1. In some embodiments the mouse NPR1 has NCBI Reference Sequence number NP_032753.5 (SEQ ID NO: 2). In some 5 embodiments, the NPR1 referred to is rat (*Rattus norvegicus*) NPR1. In some embodiments the rat NPR1 has NCBI

Reference Sequence number NP_036745.1 (SEQ ID NO: 3). Exemplary NPR1 proteins are shown in Table 1. Where a constitutively active or W74R mutant is discussed herein, this mutant refers to Trp at amino acid 74 of the mature human NPR1 protein, which may also be referred to as the Trp at amino acid 106 of the hNPR1 protein shown in SEQ ID NO: 1.

TABLE 1

NPR1 Protein Sequences		
SEQ ID NO: Animal	Sequence (amino acid)	
1 Human	MPGPRRPAQSRLRLLLLLLLPPLLLLLRGSHAGNLTAVVPLPLANTSYPWS WARVGPVELALAQVKARPDLLPGWTVRVLGSSSENALGVCSDTAAPLA AVDLKWEHNPVFLGPGCVYAAAPVGRFTHAWRVPLLTAGAPALGFGV KDEYALTTRAGPSYAKLGFVAALHRRLLGWERQALMLYAYRPGDEEHC FFLVEGLFMRVDRRLNITVDHLEFAEDDLSHYTRLLRTPMRKGRVIYICSS PDAFRTLMMLALEAGLCGEDYVFFHLDFGQSLQGGQGPAPRRPWERG QDVSAQAFQAAKIIITYKDPDNPEYLEFLKQLKHLAYEQNFMTMEDGLVN TTPASFHDGLLLYVQAVTETLAHGGTVTDGENITQRMWNRSFQGVTYGLK IDSSGDRDTPSLWMDMPETGAFRVVLNNGTSQELMAVSGRKLWPLG YPPDIPKCGFDNEDPACNQDHLSTLEVLALVGSLSLLGIL- IVSFFIYRKMQ LEKELASELWVRWEDVEPSSLERHLRSAGSRLTSGRGSNYGSLLTTEG QFQVFAKTAYYKGNLVAVKRVNRKRIELTRKVLPELKHMRDVQNEHLTR FVGACTDPPNICILTEYCPRGSQDILENESITLDWMFRYSLTNDI- VKGMLF LHNGAICSHGNLKSSNCVVDGRFVLKITDYGLESFRDLDEPQGHVTYAKK LWTAPELLRMASPPVRGSQAGDVYSFGIILQEIALRSGVFHVEGLDL- SPKEI IERVTRGEQPPFRPSLALQSHLEELGLLMQRCWAEDPQERPPFQIIRLTLR KFNRENSNILDNLLSRMEQYANNLEELVEERTQAYLEEKRAEALLYQI LPHSVAEQLKRGETVQAEAFDSVTIYFSDIVGFTALSAESTPMQVVTLLND LYTCFDAVIDNFDVYKVETIGDAYMVVSGLPVRNGRLHACEVARMALAL LDAVRSFRIRHRPQEQLRLRIGIHTGPVCAGVVGLKMPRYCLFGDVTNTA SRMESNGEALKIHLSSSETKAVLEEFDFEELRGDVEKMGKGVRTYWL LGERGSSTRG	
2 Mouse	MPGSRRVRPRRLRALLLLPPLLLLLRSGHASDLTVAVVPLPLTNTSYPWSWAR VGPVELALGRVKARPDLLPGWTVRMLGSSSNAAGVCSDTAAPLA AVDLKWEHNPVFLGPGCVYAAAPVGRFTHAWRVPLLTAGAPALGIGVDE YALTRTGP SHVKLGDFVTALHRRLLGWEHQALVLYADRLGDDRCPFFIV EGLYMRVRERLNI TVNHQEFVEGDDHYTKLLRTVQRKGRVIYICSSPDA FRNLMLLALDAGLTGEDYVFFHLDFGQSLQGAQGPVPRKPWERDDGQ DRRARQAFQAAKIIITYKEPDNPEYLEFLKQLKLLADKKFNFTMEDGLKNI IPASFHDGLLLYVQAVTETLAQGGTVTDGENITQRMWNRSFQGVTYGLKI DRNGDRDTPSLWMDMPETGAFRVVLNNGTSQELMAVSEHRLYWLPLG YPPDIPKCGFDNEDPACNQDHFSTLEVLA- LVGSLSLVSVFLIVSFFIYRKMQ LEKELVSELWVRWEDLQPSSELERHLRSAGSRLTSGRGSNYGSLLTTEG QFQVFAKTAYYKGNLVAVKRVNRKRIELTRKVLPELKHMRDVQNEHLTR FVGACTDPPNICILTEYCPRGSQDILENESITLDWMFRYSLTNDI- VKGMLF LHNGAIGSHGNLKSSNCVVDGRFVLKITDYGLESFRDPEPQGHVTLFAKK LWTAPELLRMASPPARGSQAGDVYSFGIILQEIALRSGVFYVEGLDL- SPKEI IERVTRGEQPPFRPSMDLQSHLEELGQLMQRCWAEDPQERPPFQIIRLALR KFNKENSNILDNLLSRMEQYANNLEELVEERTQAYLEEKRAEALLYQI LPHSVAEQLKRGETVQAEAFDSVTIYFSDIVGFTALSAESTPMQVVTLLND LYTCFDAVIDNFDVYKVETIGDAYMVVSGLPVRNGQLHAREVARMALAL LDAVRSFRIRHRPQEQLRLRIGIHTGPVCAGVVGLKMPRYCLFGDVTNTA SRMESNGEALRIHLSSSETKAVLEEFDFEELRGDVEKMGKGVRTYWL LGERGCSTRG	
3 Rat	MPGSRRVRPRRLRALLLLPPLLLLLRGGHASDLTVAVVPLPLTNTSYPWSWAR VGPVELALARVKARPDLLPGWTVRMLGSSSNAAGVCSDTAAPLA AVDLKWEHNPVFLGPGCVYAAAPVGRFTHAWRVPLLTAGAPALGIGVDE YALTRTGP SHVKLGDFVTALHRRLLGWEHQALVLYADRLGDDRCPFFIV EGLYMRVRERLNI TVNHQEFVEGDDHYTKLLRAVRKGRVIYICSSPDA FRNLMLLALNAGLTGEDYVFFHLDFGQSLKSAQGLVPQKPWERGDGQD RSARQAFQAAKIIITYKEPDNPEYLEFLKQLKLLADKKFNFTVEDGLKNIIP ASFHDGLLLYVQAVTETLAQGGTVTDGENITQRMWNRSFQGVTYGLKID RINGDRDTPSLWMDMPETGAFRVVLNNGTSQELMAVSEHKLYWLPLGY PPPDVPKCGFDNEDPACNQDHFSTLEVLALVGSLSLIS- FLIVSFFIYRKML EKELVSELWVRWEDLQPSSELERHLRSAGSRLTSGRGSNYGSLLTTEGQ FQVFAKTAYYKGNLVAVKRVNRKRIELTRKVLPELKHMRDVQNEHLTRF VGACTDPPNICILTEYCPRGSQDILENESITLDWMFRYSLTNDI- VKGMLFL	

TABLE 1-continued

NPR1 Protein Sequences	
SEQ ID NO: Animal	Sequence (amino acid)
	HNGAICSHGNLKSSNCVVDGRFVLKITDYGLSEFRDPEPEQGHTLFAKKL WTAPELLRRMASPPARGSQAGDVYSFGIILQEIARSGVFVYVEGLDL- SPKEII ERVTRGEQPPFRPSMDLQSHLEELGQLMQRCAEDPQERPPFQQIRLALR KFNKENSNNILDNLLSRMEQYANNLEELVEERTQAYLEEKRAEALLYQI LPHSVAEQLKRGETVQAEAFDSVTIYFSDIVGFTALSASTPMQVVTLLND LYTCFDAVIDNFDVYKVETIGDAYMVVSGLPVRNGQLHAREVARMALAL LDAVRSFRIRHRPQEQRLRLRIGIHTGPVCAGVVLKMPRYCLFGDTVNTA SRMESNGEALKIHLSSSETKAVLEEFDFGELELRGDVEMKGKGVRTYWL LGERGCSTRG

In various embodiments, the anti-NPR1 antibodies and antigen binding fragments disclosed herein are capable of binding to NPR1 and activating NPR1 in the absence of ANP. By virtue of this activity, the disclosed anti-NPR1 antibodies and antigen binding fragments may be useful in treating undesirable conditions, diseases and disorders including cardiovascular disorders (e.g., hypertension, peripheral vascular disease, heart failure (including but not limited to heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure), coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy (e.g., ventricular hypertrophy), diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, or myocardial infarction (MI)), hypertension (e.g., resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, or pulmonary arterial hypertension), preeclampsia, asthma, glaucoma, cytokine release syndrome, and/or a kidney disorder (e.g., diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD)).

The term "antibody" as used herein refers to a whole antibody or antigen binding fragment thereof. A whole antibody is a glycoprotein comprising at least two heavy (H) chains and two light (L) chains inter-connected by disulfide bonds. Each heavy chain is comprised of a heavy chain variable region (abbreviated herein as VH) and a heavy chain constant region. The heavy chain constant region is comprised of three domains, CH1, CH2 and CH3. Each light chain is comprised of a light chain variable region (abbreviated herein as VL) and a light chain constant region. The light chain constant region is comprised of one domain, CL. The VH and VL regions can be further subdivided into regions of hypervariability, termed complementarity determining regions (CDR), interspersed with regions that are more conserved, termed framework regions (FR). Each VH and VL is composed of three CDRs and four FRs arranged from amino-terminus to carboxy-terminus in the following

order: FR1, CDR1, FR2, CDR2, FR3, CDR3, FR4. The variable regions of the heavy and light chains contain a binding domain that interacts with an antigen. The constant regions of the antibodies may mediate the binding of the immunoglobulin to host tissues or factors, including various cells of the immune system (e.g., effector cells) and the first component (C1q) of the classical complement system. The term "antibody" includes, but is not limited to, monoclonal antibodies, human antibodies, humanized antibodies, camelised antibodies, and chimeric antibodies. The antibodies can be of any isotype/class (e.g., IgG, IgE, IgM, IgD, IgA and IgY) or subclass (e.g., IgG₁, IgG₂, IgG₃, IgG₄, IgA₁, and IgA₂).

The term "antigen binding fragment" refers to a fragment of an intact antibody that retains the ability to specifically bind to a given antigen (e.g., NPR1) and/or provide a function of the intact antibody. Such fragments include Fab fragments, Fab' fragments, monovalent fragments consisting of the VL, VH, CL and CHI domains; F(ab')₂ fragments, bivalent fragments comprising two Fab fragments linked by a disulfide bridge at the hinge region; an Fd fragment consisting of the VH and CHI domains; an Fv fragment consisting of the VL and VH domains, a single chain Fv fragment (scFv) consisting of the VL and VH domains connected by a linker sequence; and a single domain antibody (dAb) fragment (Ward et al., 1989 Nature 341:544-546), which consists of a VH domain or a VL domain.

The term "single chain antibody", "single chain Fv" or "scFv" is refers to a molecule comprising an antibody heavy chain variable domain (or region; VH) and an antibody light chain variable domain (or region; VL) connected by a linker. Such scFv molecules can have the general structures: NH₂-VL-linker-VH-COOH or NH₂-VH-linker-VL-COOH. Any suitable linker may be used. A non-limiting set of linkers that can be used in such single chain antibodies are described by Holliger et al. (1993), Proc. Natl. Acad. Sci. USA 90:6444-6448, Alfthan et al. (1995), Protein Eng. 8:725-731, Choi et al. (2001), Eur. J. Immunol. 31:94-106, Hu et al. (1996), Cancer Res. 56:3055-3061, Kipriyanov et al. (1999), J. Mol. Biol. 293:41-56 and Roovers et al. (2001), Cancer Immunol; the contents of each of which are herein incorporated by reference for this purpose. Such single chain antibodies are also intended to be encompassed within the term "antigen binding fragment" of an antibody. These antibody fragments are obtained using techniques known to those of skill in the art, and the fragments are screened for utility in the same manner as are intact antibodies. Without limitation, an antigen binding fragment can be produced by any suitable method known in the art. For instance, the various antigen binding fragments described herein can be produced by enzymatic or chemical modification of intact antibodies,

synthesized de novo using recombinant DNA methodologies (e.g., single chain Fv), or identified using phage display libraries (see, e.g., Pini and Bracci, *Curr Protein Pept Sci* 2000; 1(2):155-69, the contents of which are herein incorporated by reference for this purpose). Antigen binding fragments are screened for utility (e.g., binding affinity, activity) in the same manner as are intact antibodies.

Antigen binding fragments can also be incorporated into single domain antibodies, maxibodies, minibodies, intrabodies, diabodies, triabodies, tetrabodies, v-NAR and bis-scFv (see, e.g., Hollinger and Hudson, 2005, *Nature Biotechnology*, 23, 9, 1126-1136, the contents of which are herein incorporated by reference for this purpose). Antigen binding portions of antibodies can be grafted into scaffolds based on polypeptides such as Fibronectin type III (Fn3) (see e.g., U.S. Pat. No. 6,703,199, which describes fibronectin polypeptide monobodies, the contents of which are herein incorporated by reference for this purpose).

Antigen binding fragments can be incorporated into single chain molecules comprising a pair of tandem Fv segments (VH-CH1-VH-CH1) which, together with complementary light chain polypeptides, form a pair of antigen binding regions (see Zapata et al., 1995 *Protein Eng.* 8 (10): 1057-1062; and U.S. Pat. No. 5,641,870; the contents of each of which are herein incorporated by reference for this purpose).

The term "isolated" means throughout this specification, that the immunoglobulin, antibody or polynucleotide, as the case may be, exists in a physical milieu distinct from that in which it may occur in nature. For example, a naturally-occurring polynucleotide or polypeptide present in a living organism is not isolated, but the same polynucleotide or polypeptide separated from some or all of the coexisting materials in the living organism, is isolated.

The term "isolated antibody," as used herein, refers to an antibody that has been identified and separated from one or more (e.g., the majority) of the components (by weight) of its source environment, e.g., from the components of a hybridoma cell culture or a different cell culture that was used for its production (e.g., producer cells including but not limited to the exemplary host cells described herein that recombinantly express the antibody). The separation is performed such that it sufficiently removes components that may otherwise interfere with the suitability of the antibody for the desired applications (e.g., for therapeutic use of an anti-NPR1 antibody). Methods for preparing isolated antibodies are known in the art and include Protein A chromatography, anion exchange chromatography, cation exchange chromatography, virus retentive filtration, and ultrafiltration.

Throughout this specification, complementarity determining regions ("CDR") are defined according to the Kabat definition unless specified that the CDR are defined according to another definition. The precise amino acid sequence boundaries of a given CDR can be determined using any of a number of well-known schemes, including those described by Kabat et al. (1991), "Sequences of Proteins of Immunological Interest," 5th Ed. Public Health Service, National Institutes of Health, Bethesda, MD ("Kabat" numbering scheme), Al-Lazikani et al., (1997) *JMB* 273, 927-948 ("Chothia" numbering scheme) and ImMunoGenTics (IMGT) numbering (Lefranc, M.-P., *The Immunologist*, 7, 132-136 (1999); Lefranc, M.-P. et al., *Dev. Comp. Immunol.*, 27, 55-77 (2003) ("IMGT" numbering scheme); the contents of each of which are herein incorporated by reference for this purpose. For example, for classic formats, under Kabat, the CDR amino acid residues in the heavy chain variable domain (VH) are numbered 31-35 (HCDR1), 50-65 (HCDR2), and 95-102 (HCDR3); and the CDR amino

acid residues in the light chain variable domain (VL) are numbered 24-34 (LCDR1), 50-56 (LCDR2), and 89-97 (LCDR3). Under Chothia the CDR amino acids in the VH are numbered 26-32 (HCDR1), 52-56 (HCDR2), and 95-102 (HCDR3); and the amino acid residues in VL are numbered 26-32 (LCDR1), 50-52 (LCDR2), and 91-96 (LCDR3). By combining the CDR definitions of both Kabat and Chothia, the CDRs consist of amino acid residues 26-35 (HCDR1), 50-65 (HCDR2), and 95-102 (HCDR3) in human VH and amino acid residues 24-34 (LCDR1), 50-56 (LCDR2), and 89-97 (LCDR3) in human VL. Under IMGT the CDR amino acid residues in the VH are numbered approximately 26-35 (CDR1), 51-57 (CDR2) and 93-102 (CDR3), and the CDR amino acid residues in the VL are numbered approximately 27-32 (CDR1), 50-52 (CDR2), and 89-97 (CDR3) (numbering according to "Kabat"). Under IMGT, the CDR regions of an antibody can be determined using the program IMGT/DomainGap Align.

By convention, the CDR regions in the heavy chain are typically referred to as HCDR1, HCDR2 and HCDR3 and in the light chain as LCDR1, LCDR2 and LCDR3. They are numbered sequentially in the direction from the amino terminus to the carboxy terminus.

The term "antibody framework" as used herein refers to the part of the variable domain, either VL or VH, which serves as a scaffold for the antigen binding loops (CDRs) of this variable domain. In essence, it is the variable domain without the CDRs.

The terms "constant region" or "constant domain" refer to a carboxy terminal portion of the light and heavy chain which is not directly involved in binding of the antibody to antigen but exhibits various effector functions, such as interaction with the Fc receptor. The terms refer to the portion of an immunoglobulin molecule having a more conserved amino acid sequence relative to the other portion of the immunoglobulin, the variable domain, which contains the antigen binding site. The constant domain contains the CH1, CH2 and CH3 domains of the heavy chain and the CHL domain of the light chain.

The term "epitope" or "antigenic determinant" refers to a site on an antigen to which an immunoglobulin or antibody specifically binds (e.g., a specific site on the target molecule). An epitope typically includes at least 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 consecutive or non-consecutive amino acids in a unique spatial conformation. See, e.g., *Epitope Mapping Protocols in Methods in Molecular Biology*, Vol. 66, G. E. Morris, Ed. (1996), the contents of which are herein incorporated by reference for this purpose. In addition, as used herein, an epitope can comprise one or more monosaccharide units of a polysaccharide to which an antibody specifically binds. In specific aspects, an epitope can be a conformational epitope. See, e.g., Thompson et al., 2009, *J. of Biol. Chem.* 51:35621-35631, the contents of which are herein incorporated by reference for this purpose.

The terms "monoclonal antibody" or "monoclonal antibody composition" as used herein refers to an antibody obtained from a population of substantially homogeneous antibodies produced by a particular cell or cell line, wherein the individual antibodies comprising the population are essentially identical in sequence except for possible naturally-occurring mutations that may be present in minor amounts. A monoclonal antibody preparation displays a single binding specificity and affinity for a particular epitope. In contrast, conventional (polyclonal) antibody preparations typically include a multitude of antibodies directed against or specific for different epitopes. The modifier "monoclonal" indicates the character of the antibody as

being obtained from a substantially homogeneous population of antibodies and is not to be construed as requiring production of the antibody by any particular method. Monoclonal antibodies (mAbs) can be produced by a variety of techniques, including conventional methodology, e.g., the standard somatic cell hybridization technique of Kohler and Milstein (Nature 1975; 256(5517):495-7), the contents of which are herein incorporated by reference for this purpose. A monoclonal antibody may also be obtained from other suitable methods, including phage display techniques such as those described in Clackson et al. (Nature 1991; 352 (6336):624-8) or Marks et al. (J Mol Biol 1991; 222(3):581-97), the contents of each of which are herein incorporated by reference for this purpose. The term "monoclonal antibody" is also not limited to antibody sequences from particular species of origin or from one single species of origin. Thus, the meaning of the term "monoclonal antibody" encompasses chimeric monoclonal antibodies such as humanized monoclonal antibodies.

The term "chimeric antibody," as used herein, refers to antibodies in which (a) the constant region is altered, replaced, or exchanged such that the antigen binding site (variable region) is linked to a constant region of a different or altered class, effector function, and/or species; or (b) the variable region, or a portion thereof, is altered, replaced, or exchanged with a variable region, or a portion thereof, having a different or altered antigen specificity. To create a chimeric antibody, the variable region sequences from a non-human donor antibody (e.g., a mouse, rabbit, or rat donor antibody) can be linked to human constant regions using methods known in the art (see, e.g., U.S. Pat. No. 4,816,567 (Cabilly et al.), the contents of which are herein incorporated by reference for this purpose). For instance, a mouse anti-NPR1 antibody can be modified by replacing its constant region with the constant region from a human immunoglobulin. Due to the replacement with a human constant region, the chimeric antibody can retain its specificity in recognizing human NPR1 while having reduced immunogenicity in human as compared to the original mouse antibody.

As used herein, the term "humanized antibody" refers to forms of antibodies that contain at least some human sequence and at least some non-human sequence. Typically, the antibody contains human sequences and a minor portion of non-human sequences which confer binding specificity to the target antigen. Such antibodies are chimeric antibodies which contain minimal sequence derived from a non-human immunoglobulin and retain the reactivity of a non-human antibody while being less immunogenic in humans. Typically, humanized antibodies are generated by replacing hypervariable region sequences from a human acceptor antibody with hypervariable region sequences from a non-human donor antibody (e.g., a mouse, rabbit, or rat donor antibody) that binds to an antigen of interest (e.g., NPR1). In some cases, framework region sequences of the acceptor antibody may also be replaced with the corresponding sequences of the donor antibody (e.g., via affinity maturation). In addition to the sequences derived from the donor and acceptor antibodies, the humanized antibody can be further modified by the substitution of residues, either in the framework region and/or within the replaced non-human residues to refine and optimize antibody specificity, affinity, and/or activity, as discussed herein. Methods to generate humanized antibodies are known in the art. See, e.g., Riechmann et al. (Nature 1988; 332(6162):323-7); Jones et al. (Nature 1986; 321(6069):522-5); U.S. Pat. No. 5,225,539 (Winter); and U.S. Pat. Nos. 5,530,101; 5,585,089; 5,693,

762, and 6,180,370 (Queen et al.), the contents of each of which are herein incorporated by reference for this purpose.

The term "human antibody", as used herein, is intended to include antibodies having variable regions in which both the framework and CDR regions are derived from sequences of human origin. Furthermore, if the antibody contains a constant region, the constant region also is derived from such human sequences, e.g., human germline sequences, or mutated versions of human germline sequences or antibody containing consensus framework sequences derived from human framework sequences analysis, for example, as described in Knappik, et al., (2000) J Mol Biol; 296:57-86, the contents of which are herein incorporated by reference for this purpose). Human antibodies may include amino acid residues not encoded by human sequences (e.g., mutations introduced by random or site-specific mutagenesis in vitro or by somatic mutation in vivo).

The antibodies or antigen binding fragments of the disclosure may include amino acid residues not encoded by human sequences (e.g., mutations introduced by random or site-specific mutagenesis in vitro or by somatic mutation in vivo). However, the term "human antibody", as used herein, is not intended to include antibodies in which CDR sequences derived from the germline of another mammalian species, such as a mouse, have been grafted onto human framework sequences.

The terms "peptide," "polypeptide," and "protein" are used interchangeably herein to refer to a polymer of amino acid residues. The terms encompass amino acid polymers in which one or more amino acid residues is an artificial chemical mimetic of a corresponding naturally-occurring amino acid, as well as naturally-occurring amino acid polymers and non-naturally-occurring amino acid polymers. Unless otherwise indicated, a particular polypeptide sequence also implicitly encompasses conservatively modified variants thereof.

The term "conservatively modified variant" applies to both amino acid and nucleic acid sequences. For nucleic acid sequences, conservatively modified variants refer to those nucleic acids which encode identical or essentially identical amino acid sequences, or where the nucleic acid does not encode an amino acid sequence, to essentially identical sequences. Because of the degeneracy of the genetic code, a large number of functionally identical nucleic acids encode any given protein. For instance, the codons GCA, GCC, GCG, and GCU all encode the amino acid alanine. Thus, at every position where an alanine is specified by a codon, the codon can be altered to any of the corresponding codons described without altering the encoded polypeptide. Such nucleic acid variations are "silent variations," which are one species of conservatively modified variations. Every nucleic acid sequence herein which encodes a polypeptide also describes every possible silent variation of the nucleic acid. One of skill will recognize that each codon in a nucleic acid (except AUG, which is ordinarily the only codon for methionine, and TGG, which is ordinarily the only codon for tryptophan) can be modified to yield a functionally identical molecule. Accordingly, each silent variation of a nucleic acid that encodes a polypeptide is implicit in each described sequence. For polypeptide sequences, conservatively modified variants include individual substitutions, deletions, or additions to a polypeptide sequence which result in the substitution of an amino acid with a chemically similar amino acid. Conservative substitution tables providing functionally similar amino acids are well known in the art. The following eight groups contain amino acids that are conservative substitutions for one another:

- 1) Alanine (A), Glycine (G);
- 2) Aspartic acid (D), Glutamic acid (E);
- 3) Asparagine (N), Glutamine (Q);
- 4) Arginine (R), Lysine (K);
- 5) Isoleucine (I), Leucine (L), Methionine (M), Valine (V);
- 6) Phenylalanine (F), Tyrosine (Y), Tryptophan (W);
- 7) Serine(S), Threonine (T); and
- 8) Cysteine (C), Methionine (M).

The term "identity" or "homology" refers to a relationship between the sequences of two or more polypeptides, as determined by comparing the sequences. "Identity" also means the degree of sequence relatedness between polypeptides, as determined by the number of matches between strings of two or more amino acid residues. The percent "identity" between the two sequences is a function of the number of identical positions shared by the sequences (i.e., percent identity equals number of identical positions/total number of positions $\times$ 100), taking into account the number of gaps, and the length of each gap, which need to be introduced for optimal alignment of the two sequences. The comparison of sequences and determination of percent identity between two sequences can be accomplished using a mathematical algorithm. For sequence comparison, typically one sequence acts as a reference sequence, to which test sequences are compared. When using a sequence comparison algorithm, test and reference sequences are entered into a computer, subsequence coordinates are designated, if necessary, and sequence algorithm program parameters are designated. Default program parameters can be used, or alternative parameters can be designated. The sequence comparison algorithm then calculates the percent sequence identities for the test sequences relative to the reference sequence, based on the program parameters. Additionally, or alternatively, the protein sequences of the present disclosure can further be used as a "query sequence" to perform a search against public databases to, for example, identify related sequences. For example, such searches can be performed using the BLAST program of Altschul et al. (J Mol Biol 1990; 215(3):403-10), the contents of which are herein incorporated by reference for this purpose.

Two sequences are "substantially identical" if two sequences have a specified percentage of amino acid residues or nucleotides that are the same (e.g., 60% identity, 65% identity, 70% identity, 75% identity, 80% identity, 85% identity, 90% identity, 95% identity, or 99% identity over a specified region, or, when not specified, over the entire sequence), when compared and aligned for maximum correspondence over a comparison window, or designated region as measured using one of the following sequence comparison algorithms or by manual alignment and visual inspection. Optionally, the identity exists over a region that is at least about 50 nucleotides (or 10 amino acids) in length, or exists over a region that is 100 to 500 or 1000 or more nucleotides (or 20, 50, 200 or more amino acids) in length.

Binding "affinity" refers to the strength of interaction between antibody and antigen at single antigenic sites. Within each antigenic site, the variable region of the antibody "arm" interacts through weak non-covalent forces with antigen at numerous sites. In general, the more interactions, the stronger the affinity. Generally, such determinations can be made using a cell-based assay.

The term "K_{assoc}" or "K_a", as used herein, is intended to refer to the association rate of a particular binding molecule-antigen interaction, whereas the term "K_{dis}" or "K_d", as used herein, is intended to refer to the dissociation rate of a particular binding molecule-antigen interaction. The term

"K_d", as used herein, is intended to refer to the equilibrium dissociation constant, which is obtained from the ratio of K_d to K_a (i.e., K_d/K_a) and is expressed as a molar concentration (M). K_d values for antibodies can be determined using methods well established in the art. A method for determining the K_d of an antibody is by using surface plasmon resonance, such as a Biacore® system, or solution equilibrium titration (SET) (see Friguier et al., (1985) J. Immunol. Methods, 77 (2): 305-319, and Hanel et al., (2005) Anal. Biochem., 339 (1): 182-184), the contents of each of which are herein incorporated by reference for this purpose.

As used herein, the term "specific," "specifically binds," and "binds specifically" refers to a binding reaction between an antibody or antigen binding fragment (e.g., an anti-NPR1 antibody) and a target antigen (e.g., NPR1) in a heterogeneous population of proteins and other biologics. Antibodies can be tested for specificity of binding by comparing binding to an appropriate antigen to binding to an irrelevant antigen or antigen mixture under a given set of conditions. If the antibody binds to the appropriate antigen with at least 2, 5, 7, and preferably 10 or more times more affinity than to the irrelevant antigen or antigen mixture, then it is considered to be specific. A "specific antibody" or a "target-specific antibody" is one that only binds the target antigen (e.g., NPR1), but does not bind (or exhibits minimal binding) to other antigens. In certain embodiments, an antibody or antigen binding fragment that specifically binds the target antigen (e.g., NPR1) has a K_d of less than 1 $\times$ 10⁻⁶ M, less than 1 $\times$ 10⁻⁷ M, less than 1 $\times$ 10⁻⁸ M, less than 1 $\times$ 10⁻⁹ M, less than 1 $\times$ 10⁻¹⁰ M, less than 1 $\times$ 10⁻¹¹ M, less than 1 $\times$ 10⁻¹² M, or less than 1 $\times$ 10⁻¹³ M. In certain embodiments, the K_d is about 1 pM to about 600 pM. In certain embodiments, the K_d is between 600 pM to 1 μ M, 1 μ M to 100 nM, or 100 nM to 10 nM (inclusive).

In some embodiments, the antibodies or antigen binding fragments thereof act as non-competitive agonists. A "non-competitive agonist" refers to a molecule which binds to an enzyme or receptor at a site distant from the binding sites of its natural ligands. The non-competitive or allosteric agonism is generally independent of the association or concentration of the natural ligands for the enzyme or receptor. Such non-competitive agonists can, for example, provide for a level of activation that can be substantially independent of natural ligands. In a specific embodiment, the anti-NPR1 antibodies or antigen binding fragments described herein are ANP non-competitive, meaning that the antibody or antigen binding fragment acts as an agonist which binds at site away from ANP binding site of NPR1 and effect agonistic activity regardless of whether or not NPR1 is bound to ANP.

In some embodiments, the antibodies or antigen binding fragments thereof act as competitive agonists. A "competitive agonist" refers to an agonist which interferes or competes with a natural ligand for its binding site on an enzyme or receptor. In a specific embodiment, the anti-NPR1 antibodies or antigen binding fragments described herein are ANP competitive, meaning that the antibody or antigen binding fragment acts as an agonist which competes with ANP at the ANP binding site of NPR1.

In some embodiments, the activation of NPR1 by an antibody or antigen binding fragment may be determined by any suitable assay. An exemplary assay for determination of NPR1 activation is the production of cGMP by mammalian cells (e.g., CHO cells or a human cell line) expressing hNPR1.

In some embodiments, the stabilization of the ANP-NPR1 complex may be determined by any suitable assay. An

exemplary assay for determination of the stability of the ANP-NPR1 complex is the FRET assay described herein (see, e.g., FIG. 3).

The term “about” in relation to a numerical value x means, for example, $x \pm 10\%$.

Antibodies of the Disclosure

Below are disclosed certain specific anti-NPR1 antibody sequences of the disclosure. As used herein, the term “anti-NPR1 antibody” or “antibody that binds to NPR1” refers to any form of an antibody or antigen binding fragment that specifically binds to NPR1, e.g., those binding with a K_D of less than 1×10^{-8} M, as determined by, e.g., surface plasmon resonance (SPR) spectroscopy (using Biacore™) or solution equilibrium titration (SET). The term encompasses monoclonal antibodies (including intact monoclonal antibodies), polyclonal antibodies, and biologically functional antigen binding fragments so long as they specifically bind to NPR1.

Amino acid and nucleic acid sequences of exemplary anti-NPR1 antibodies of the present disclosure are set forth in Table 2. In some embodiments, the antibody has the heavy and light chain CDRs, VH and VL sequence, and/or the heavy and light chain sequence of any of the antibodies described in Table 2. In some embodiments, the anti-NPR1 antibody is a four-chain antibody (also referred to as an intact antibody), comprising two heavy chains and two light chains. In some embodiments, the anti-NPR1 antibody is an antigen binding fragment of an intact antibody, e.g., a functional fragment of an intact antibody selected from any of those set forth in Table 2 that retains the ability to bind NPR1 and/or provide a function of the intact antibody (e.g., activating NPR1 in the absence of ANP). In some embodiments, the anti-NPR1 antibody is an antibody having the CDRs of any heavy chain variable region and light chain variable region pair shown in Table 2. In some embodiments, the anti-NPR1 antibody is an antibody having the CDRs of any heavy and light chain pair shown in Table 2.

TABLE 2

Exemplary anti-NPR1 antibody sequences			
WW01_			
LALA_			
SEQ ID	HCDR1	GFTFNTHYIH	
NO: 4	(Combined)		
SEQ ID	HCDR2	SISGSGSNTYYADSVKG	
NO: 5	(Combined)		
SEQ ID	HCDR3	ERGVVYHMFDP	
NO: 6	(Combined)		
SEQ ID	HCDR1	THYIH	
NO: 7	(Kabat)		
SEQ ID	HCDR2	SISGSGSNTYYADSVKG	
NO: 5	(Kabat)		
SEQ ID	HCDR3	ERGVVYHMFDP	
NO: 6	(Kabat)		
SEQ ID	HCDR1	GFTFNTH	
NO: 8	(Chothia)		
SEQ ID	HCDR2	SGSGSN	
NO: 9	(Chothia)		
SEQ ID	HCDR3	ERGVVYHMFDP	
NO: 6	(Chothia)		
SEQ ID	HCDR1	GFTFNTHY	
NO: 10	(IMGT)		
SEQ ID	HCDR2	ISGSGSNT	
NO: 11	(IMGT)		
SEQ ID	HCDR3	ARERGVVYHMFDP	
NO: 12	(IMGT)		
SEQ ID	VH	QVQLLESGLVQPGGSLRLSCAASGFTFNTHYIHWVRQAPGKLEWVSSI	
NO: 13		SGSGSNTYYADSVKGRFTISRDNKNTLYLQMNSLAEDTAVYYCARERG	
		YVYHMFDPWGQGLTVTVSS	
SEQ ID	DNA VH	cagggtgcaattgctggaaagcggcggtggcctggtgcagccgggtggcagcctgcgtctgagctgcgcggcggtccg	
NO: 14		gattcacctttaacactcattacatccattgggtgcgccaggccccggcgaaggtctcgagtggggttccctctatctctgg	
		ttctggttctaacacctactatgcggatagcgtgaaaggccgctttaccatcagccgcgataattcgaaaaacacctgtg	
		tctgcaaatgaacagcctgcgtgcggaagatacggccgtgtattattgcgcgctgaacgtgggttacgtttactaccatat	
		gttcgatccgtggggccaaggcacccctggtgactgttagctca	
SEQ ID	Heavy	QVQLLESGLVQPGGSLRLSCAASGFTFNTHYIHWVRQAPGKLEWVSSI	
NO: 15	Chain	SGSGSNTYYADSVKGRFTISRDNKNTLYLQMNSLAEDTAVYYCARERG	
		YVYHMFDPWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVK	
		DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYI	

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		CNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDT LMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTT LPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSD GSFFFLYSLKLTVDKSRWQQGNVFSCSVMEALHNHYTQKSLSLSPGK
SEQ ID NO: 16	DNA Heavy Chain	cagggtgcaattgctggaaagcggcggtggcctgggtgcagccgggtggcagcctgcgtctgagctgcgcggtgcgtccg gattcacctttaacactcattacatccattgggtgcgccaggcccccggaagggtctcgagtggtttcctctatctctgg ttctgggtcttaacacctactatgcggatagcgtgaaaggcgcgtttaccatcagccgcgataatcgaaaaacacctgta tctgcaaatgaacagcctgcgtgcggaagatcggcgcgtgtattattgcgcgcgtgaacgtggttacgtttactaccat gtccgatccgtggggccaaggcaccctgggtgactgttagctcagcctccaccaagggtccatcggtcttccccctggca cctcctccaagagcacctctgggggcacagcggcctgggctgcctggtaaggactacttccccgaaccggtgac ggtgtcgtggaactcaggcgcctgaccagcggcgtgcacaccttcccggtgtcctacagtcctcaggactctactcc ttcagcagcgtggtgaccgtgccctccagcagcttgggcacccagacctacatctgcaacgtgaatcacaagccag caacaccaagggtggacaagagagttgagcccaaatcttgtgacaaaactcacacatgccaccgtgccagcacctg aagcagcggggggaccgtcagtccttctcttcccccaaaacccaaggacacctcatgatctcccgaccctcgagg tcacatgcgtgggtggacgtgagccacgaagacctgaggtcaagttcaactggtacgtggacggcgtggagggtg cataatgccaagacaaagccggggaggagcagtaaacagcacgtaccgggtggtcagcgtcctcacctgcctgc accaggactggctgaatggcaaggagtacaagtgcaaggtctccaacaaagccctccagccccatcgagaaaaac catctccaaagccaaagggcagccccgagaaccacaggtgtacacctgcccccatcccgaggagagatgaccaag aaccaggtcagcctgacctgcctgggtcaaaggcttctatcccagcgacatcgccgtggagtgggagagcaatgggca gccggagaacaactacaagaccagcctcccggtgctggactccgacggctccttcttctctacagcaagctcacctg ggacaagagcaggtggcagcaggggaacgtcttctcatgctccgtgcatgaggtcctgcacaaacctacacgc agaagagcctctcctgtctccgggtaaa
SEQ ID NO: 17	LCDR1 (Combined)	RASQSITRNYLA
SEQ ID NO: 18	LCDR2 (Combined)	GASSRAT
SEQ ID NO: 19	LCDR3 (Combined)	QQHSMYPRT
SEQ ID NO: 17	LCDR1 (Kabat)	RASQSITRNYLA
SEQ ID NO: 18	LCDR2 (Kabat)	GASSRAT
SEQ ID NO: 19	LCDR3 (Kabat)	QQHSMYPRT
SEQ ID NO: 20	LCDR1 (Chothia)	SQSITRNY
SEQ ID NO: 21	LCDR2 (Chothia)	GAS
SEQ ID NO: 22	LCDR3 (Chothia)	HSMYPR
SEQ ID NO: 23	LCDR1 (IMGT)	QSITRNY
SEQ ID NO: 21	LCDR2 (IMGT)	GAS
SEQ ID NO: 19	LCDR3 (IMGT)	QQHSMYPRT
SEQ ID NO: 24	VL	DIVLTQSPATLSLSPGERATLSCRASQSITRNYLAWYQQKPGQAPRLLIYGA SSRATGIPARFSGSGSDFTLTISLLEPEDFAVYYCQQHSMYPRTFGQGTK VEIK
SEQ ID NO: 25	DNA VL	gatatcgctgtgacccagagcccgcgacacctgagcctgagcccggtgaaagtgccacctgagctgcagagcga gccagtcctatcactcgtaactacctggcttggtaccagcagaaacccggccaggccccgcgtctattaatctacggtgct tcttctcgtgcgacggcattccggcgcggtttagcggcagcggatccggcaccgatttcacctgaccattagcagcct ggaaccggaagactttcggtgtattattgccagcagcattctatgtaccgcgctacctttggccaggggcacgaaagtgg aaattaaa
SEQ ID NO: 26	Light Chain	DIVLTQSPATLSLSPGERATLSCRASQSITRNYLAWYQQKPGQAPRLLIYGA SSRATGIPARFSGSGSDFTLTISLLEPEDFAVYYCQQHSMYPRTFGQGTK VEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNAL QSGNSQESVTEQDSKIDSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPV TKSFNRGEC

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 27	DNA Light Chain	gatatcgtgctgacccagagcccgccgacccctgagcctgagcccggtgaaactgcccacccctgagctgcagagcga gccagtcctatcactcgtaactacctggcttggtaccagcagaaacccggccaggccccgcgtctattaatctacgggtgct tcttctcgtgcgaccggcattccggcgcgttttagcggcagcggatccggcaccgatttcaccctgaccattagcagcct ggaaccggaagactttgcgggtgattattgccagcagcattctatgtaccgcgtacctttggccagggcacgaaagttag aaattaaacgtacggtagcggcgtcccagcgtgttcattctccccccagcgaacgagcagctgaagagcggcaccgcc agcgtggtgctgctgctgaacaacttctacccccgggaggccaaggtgcagtggaggtggacaacgccctgcaga gcggcaacagccaggaaagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccctgaccc tgagcaaggccgactacgagaagcacaaggtgtacgcctgcgaggtgacccaccagggcctgtccagccccgtgac caagagcttcaaccgggagcgtgt
WW03_ LALA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 29	HCDR2 (Combined)	AISSDGSYTTYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 29	HCDR2 (Kabat)	AISSDGSYTTYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 33	HCDR2 (Chothia)	SSDGSY
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSY
SEQ ID NO: 35	HCDR2 (IMGT)	ISSDGSY
SEQ ID NO: 36	HCDR3 (IMGT)	ARDYSMIYSYGAGAFDY
SEQ ID NO: 37	VH	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS AISSDGSYTTYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVIYCARD RYSMIYSYGAGAFDYWGQGLTVTVSS
SEQ ID NO: 38	DNA VH	caggtgcaattgctggaagcggcggtggcctggtgcagccgggtggcagcctgcgtctgagctgcgcggcgctccg gattcaccttttcttcttactggatgaactgggtgcgccaggccccgggcaagggtctcgagtggtttccgctatctcttc tgacgggttcttacacctactatgcggatagcgtgaaaggccgctttaccatcagccgcgataatcgaaaaacaccctgt atctgcaaatgaacagcctgcgtgcggaagatacggccggtgtattattgcgcgctgaccgttactctatgatctactctt acggtgctggtgctttcgattactggggccaaggcacccctggtgactgttagctca
SEQ ID NO: 39	Heavy Chain	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS AISSDGSYTTYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVIYCARD RYSMIYSYGAGAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSL GTQTYICNVNHKPSNTKVDKRVPEPKSCDKHTHTCPPCPAPEAAGGPSVFLFP PKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNATKPREE QYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI SKAKGQPR EPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTP PVLDSDSGFFLYSKLTVDKSRWQQGNVFSQSVMHREALHNYTQKSLSLSP GK
SEQ ID NO: 40	DNA Heavy Chain	caggtgcaattgctggaagcggcggtggcctggtgcagccgggtggcagcctgcgtctgagctgcgcggcgctccg gattcaccttttcttcttactggatgaactgggtgcgccaggccccgggcaagggtctcgagtggtttccgctatctcttc tgacgggttcttacacctactatgcggatagcgtgaaaggccgctttaccatcagccgcgataatcgaaaaacaccctgt atctgcaaatgaacagcctgcgtgcggaagatacggccggtgtattattgcgcgctgaccgttactctatgatctactctt acggtgctggtgctttcgattactggggccaaggcacccctggtgactgttagctcagcctccaccaagggtccatcgggt cttccccctggcacccctcctccaagagcacctctgggggcacagcggccctgggctgcctgggtcaaggactactccc cgaaccggtgacgggtgctggtggaactcaggccgctgaccagcggcgtgcacaccttccgggtgctctacagtcctc aggactctactccctcagcagcgtggtgaccgtgccctccagcagcttggggacccagacctacatctgcaacgtgaat

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 53	HCDR2 (Combined)	IIYPDVSYTRYSPSFQG
SEQ ID NO: 54	HCDR3 (Combined)	YWSEAYTFDY
SEQ ID NO: 55	HCDR1 (Kabat)	NYWIG
SEQ ID NO: 53	HCDR2 (Kabat)	IIYPDVSYTRYSPSFQG
SEQ ID NO: 54	HCDR3 (Kabat)	YWSEAYTFDY
SEQ ID NO: 56	HCDR1 (Chothia)	GYSFSNY
SEQ ID NO: 57	HCDR2 (Chothia)	YPDVSY
SEQ ID NO: 54	HCDR3 (Chothia)	YWSEAYTFDY
SEQ ID NO: 58	HCDR1 (IMGT)	GYSFSNYW
SEQ ID NO: 59	HCDR2 (IMGT)	IYPDVSYT
SEQ ID NO: 60	HCDR3 (IMGT)	ARYWSEAYTFDY
SEQ ID NO: 61	VH	QVQLVQSGAEVKKPGESLKISCKGSGYSFSNYWIGWVRQMPGKLEWMG IIYPDVSYTRYSPSFQGVTTISADKSI STAYLQWSSLKASDTAMYCARYWS EAYTFDYWGQGLTVTVSS
SEQ ID NO: 62	DNA VH	caggtgcaatttggtgcagagcgggtgcggaagtgaataaacccggggcgaagcctgaaaattagctgcaaaaggctccg gatatagcttctctaaactactggatcggttgggtgcgccagatgccgggcaaaaggctctcgagtggaatgggcatcatctac ccggacgcttagctacaccgcttatagcccgagctttcagggccaggtgaccattagcgcggataaaaagcatcagcacc gcgtatctgcaatggagcagcctgaaagcgagcgataccgcgatgtattattgcgcgcgttactggctgaagcttacac tttcgattactggggccaaggcacccctggtagctttagctca
SEQ ID NO: 63	Heavy Chain	QVQLVQSGAEVKKPGESLKISCKGSGYSFSNYWIGWVRQMPGKLEWMG EAYTFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF PEPVTWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNV NHKPSNTKVDKRVPEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMIS RTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVV SVLTVLHQDWLNGKEYCKVSNKALPAPIEKTI SKAKGQPREPQVYTLPPS REEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSF LYSKLTVDKSRWQQGNVFSVSMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 64	DNA Heavy Chain	caggtgcaatttggtgcagagcgggtgcggaagtgaataaacccggggcgaagcctgaaaattagctgcaaaaggctccg gatatagcttctctaaactactggatcggttgggtgcgccagatgccgggcaaaaggctctcgagtggaatgggcatcatctac ccggacgcttagctacaccgcttatagcccgagctttcagggccaggtgaccattagcgcggataaaaagcatcagcacc gcgtatctgcaatggagcagcctgaaagcgagcgataccgcgatgtattattgcgcgcgttactggctgaagcttacac tttcgattactggggccaaggcacccctggtagctttagctcagcctccaccaagggtccatcggtcttccccctggcac cctcctccaagagcacctctgggggcacagcggccctgggctgcctggtaggactacttccccgaaccggtagc gtgtcgtggaactcaggcgccctgaccagcggcgtgcacaccttccccgctgtcctacagctcctcaggactctactccc tcagcagcgtggtagccgtgcctccagcagcttggggcaccagacctacatctgcaacgtgaatcacaagccagc aacaccaagggtggacaagagagttgagcccaaatcttgtgacaaaactcacacatgccaccgtgcccagcacctga agcagcggggggaccgtcagcttctctcttcccccaaaacccaaggacacctcatgatctccgggacccctgaggt cacatgcgtggtagccgtgcctccagcagcttggggcaccagacctacatctgcaacgtgaatcacaagccagc ataatgccaaagacaagcccgaggaggagcagtagacaacagcacgtaccgggtggtagcgtcctcaccgtcctgca ccaggactggctgaatggcaaggagtacaagtgaaggtctcccaaaagccctcccagccccatcgagaaaacc atctccaaagccaaaggcagccccgagaaacacaggtgtacacctgccccatccgggaggagatgaccaaga accaggtcagcctgacctgcctggtagaagccttctatccagcgacatcgccgtggagtgaggagagcaatgggag ccggagaaactacaagaccagcctcccgctgctggactccgacggctccttctcctctacagcaagctcaccgtg gacaagagcaggtggcagcaggggaacgtcttctcatgctcgtgatgcattgaggctctgcacaaccactacacgca gaagagcctctccctgtctccgggtaaa
SEQ ID NO: 65	LCDR1 (Combined)	SGDNIRKKYVF
SEQ ID NO: 66	LCDR2 (Combined)	GDNDRPS

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 67	LCDR3 (Combined)	GTYTLLFTSKV
SEQ ID NO: 65	LCDR1 (Kabat)	SGDNIRKKYVF
SEQ ID NO: 66	LCDR2 (Kabat)	GDNDRPS
SEQ ID NO: 67	LCDR3 (Kabat)	GTYTLLFTSKV
SEQ ID NO: 68	LCDR1 (Chothia)	DNIRKKY
SEQ ID NO: 69	LCDR2 (Chothia)	GDN
SEQ ID NO: 70	LCDR3 (Chothia)	YTLLFTSK
SEQ ID NO: 71	LCDR1 (IMGT)	NIRKKY
SEQ ID NO: 69	LCDR2 (IMGT)	GDN
SEQ ID NO: 67	LCDR3 (IMGT)	GTYTLLFTSKV
SEQ ID NO: 72	VL	DIELTQPPSVSVSPGQTASITCSGDNIRKKYVFWYQQKPGQAPVLIYGDND RPSGIPERFSGSNSGNTATLTISGTQAEDEADYYCGTYTLLFTSKVFGGGTK LTVL
SEQ ID NO: 73	DNA VL	gat atcg aactg acccag ccg ccg agcgt g agcgt g agtcc ggg ccag accgc g agc attac cctg tagc ggc gata acatcc gtaaaaaa tacgt tttct ggtacc agcagaaa accggg ccaggc gcggtg ctggt gatct acggtg acaacg accgtcc gagcgg catcccg gaacgt tttagc ggtcca acagcgg caacacc gcgacc ctgacc attagcggc acc caggcgg aagacga agcgg attatt actg cggta ctta cactct gctgtt cactt ctaa agtgtt tggcgg ccgacgaa gttaacc gtccta
SEQ ID NO: 74	Light Chain	DIELTQPPSVSVSPGQTASITCSGDNIRKKYVFWYQQKPGQAPVLIYGDND RPSGIPERFSGSNSGNTATLTISGTQAEDEADYYCGTYTLLFTSKVFGGGTK LTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADSS PVKAGVETTPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHEGSTVEK TVAPTECS
SEQ ID NO: 75	DNA Light Chain	gat atcg aactg acccag ccg ccg agcgt g agcgt g agtcc ggg ccag accgc g agc attac cctg tagc ggc gata acatcc gtaaaaaa tacgt tttct ggtacc agcagaaa accggg ccaggc gcggtg ctggt gatct acggtg acaacg accgtcc gagcgg catcccg gaacgt tttagc ggtcca acagcgg caacacc gcgacc ctgacc attagcggc acc caggcgg aagacga agcgg attatt actg cggta ctta cactct gctgtt cactt ctaa agtgtt tggcgg ccgacgaa gttaacc gtcctc aggtc agccca aggtgc cccctc ggtc actct gttcc cgcctc cctc tggagg agctt caagcc aaca aggccac actggtgt gctc ataagt gacttc acccgg gaggcgt gacagt ggctt ggaaggc agat agcagccc gtcaagg cggg agtgg agacc accac cccctcca aacaa agcaaca agtac gcggc cagc agctat ctgagcc tgacgc ctgagc agtgg aagtccc acaga agctac agctg ccaggt cagc eatga aggg agcacc gttg agaagac agtggccc ctacaga atgttca
WW06_ LALA		
SEQ ID NO: 76	HCDR1 (Combined)	GYSFTSYWIA
SEQ ID NO: 77	HCDR2 (Combined)	RIDPDNSYTRYSPSFQG
SEQ ID NO: 78	HCDR3 (Combined)	WLSPGYALGEQFAGMDH
SEQ ID NO: 79	HCDR1 (Kabat)	SYWIA
SEQ ID NO: 77	HCDR2 (Kabat)	RIDPDNSYTRYSPSFQG
SEQ ID NO: 78	HCDR3 (Kabat)	WLSPGYALGEQFAGMDH

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 80	HCDR1 (Chothia)	GYSFTSY
SEQ ID NO: 81	HCDR2 (Chothia)	DPDNSY
SEQ ID NO: 78	HCDR3 (Chothia)	WLSPGYALGEQPGMDH
SEQ ID NO: 82	HCDR1 (IMGT)	GYSFTSYW
SEQ ID NO: 83	HCDR2 (IMGT)	IDPDNSYT
SEQ ID NO: 84	HCDR3 (IMGT)	ARWLSPGYALGEQPGMDH
SEQ ID NO: 85	VH	QVQLVQSGAEVKKPGESLKISCKSGSYFTSYWIAWVRQMPGKGLEWMG RIDPDNSYTRYSPSPQGGVTISADKSI STAYLQWSSSLKASDTAMYICARWL SPGYALGEQPGMDHWGGTLVTVSS
SEQ ID NO: 86	DNA VH	caggtgcaattgggtgcagagcgggtgcggaagtgaacccggcgaaagcctgaaaattagctgcaaaggctccg gatatagcttcacttcttactggatcgcttgggtgcgcagatgccgggcaaaggctctcgagtggatgggccgtatcgac ccggacaacagctacaccggttatagcccgagctttcagggccaggtgaccattagcgcggataaaaagcatcagcac cgcgtatctgcaatggagcagcctgaaagcgagcgataaccgcgatgtattattgcgcgcgttggctgtctccgggttac gctctgggtgaacagccggctggatggatcattggggccaaggcaccctgggtgactgttagctca
SEQ ID NO: 87	Heavy Chain	QVQLVQSGAEVKKPGESLKISCKSGSYFTSYWIAWVRQMPGKGLEWMG RIDPDNSYTRYSPSPQGGVTISADKSI STAYLQWSSSLKASDTAMYICARWL SPGYALGEQPGMDHWGGTLVTVSSASTKGPSVFP LAPSSKSTSGGTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSL GTQTYICNVNHKPSNTKVDKRVKPSCKDTHCTPCPAPAEAGGSPVFLFP PKPKDITLMISRTEVTVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREE QYNSTYRVVSVLTVTLHQDWLNGKEYKCKVSNKALPAPIEKTI SKAKGQPR EPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTP PVLDSGDSFFLYSKLTVDKSRWQQGNVPSCSVMHEALHNHYTQKSLSLSP GK
SEQ ID NO: 88	DNA Heavy Chain	caggtgcaattgggtgcagagcgggtgcggaagtgaacccggcgaaagcctgaaaattagctgcaaaggctccg gatatagcttcacttcttactggatcgcttgggtgcgcagatgccgggcaaaggctctcgagtggatgggccgtatcgac ccggacaacagctacaccggttatagcccgagctttcagggccaggtgaccattagcgcggataaaaagcatcagcac cgcgtatctgcaatggagcagcctgaaagcgagcgataaccgcgatgtattattgcgcgcgttggctgtctccgggttac gctctgggtgaacagccggctggatggatcattggggccaaggcaccctgggtgactgttagctcagcctccaccaag ggctccatcgggtcttccccctggcaccctcctccaagagcaccctctgggggacagcggccctgggtgctcctggaag gactacttccccgaaccggtgacgggtgctgctggaactcaggcgccctgaccagcggcggtgcacaccttccgggtgtc ctacagtcctcaggactctactcctcagcagcgtgggtgaccgtgccctccagcagcttgggcacccagacctacatct gcaacgtgaatcacaagcccagcaacaccaagggtggacaagagagttgagcccaaatcttgtgacaaaactcacaca tgcccaccgtgcccagcaccctgaagcagcggggggaccggtcagctctcctcttcccccaaaacccaaggacacct catgatctcccggaccctgaggtcacatgcgtgggtggaggagtgagccacgaagaccctgaggtcaagttcaactg gtacgtggacggcgtggaggtgcataatgccaagacaaagccgcgggaggagcagtagcaacagcacgtaccgggt ggtcagcgtcctcaccgtcctgcaccaggactggctgaatggcaaggagtagcaaggtctccaacaagccct ccagcccccatcgagaaaaccatctccaaagccaaagggcagccccgagaaaccacaggtgtacacctgccccca tccccggaggagatgaccaagaaccaggtcagcctgacctgcctgggtcaaaaggctctatcccagcgacatcgccgt ggagtgaggagcaatgggcagccggagaaacatacaagaccagcctcccgctgctggactccgagcggtcctctc tctcctacagcaagctcaccgtggacaagagcaggtggcagcaggggaacgtctctcatgctccgtgatgcatgag gctctgcacaaccactacacgcagaagagcctctcctctgctcctcggtaaa
SEQ ID NO: 89	LCDR1 (Combined)	TGSSSNIGAGYAVH
SEQ ID NO: 90	LCDR2 (Combined)	SNNKRPS
SEQ ID NO: 91	LCDR3 (Combined)	QSYDLQKSSRV
SEQ ID NO: 89	LCDR1 (Kabat)	TGSSSNIGAGYAVH
SEQ ID NO: 90	LCDR2 (Kabat)	SNNKRPS
SEQ ID NO: 91	LCDR3 (Kabat)	QSYDLQKSSRV
SEQ ID NO: 92	LCDR1 (Chothia)	SSSNIGAGYA

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 93	LCDR2 (Chothia)	SNN
SEQ ID NO: 94	LCDR3 (Chothia)	YDLQKSSR
SEQ ID NO: 95	LCDR1 (IMGT)	SSNIGAGYA
SEQ ID NO: 93	LCDR2 (IMGT)	SNN
SEQ ID NO: 91	LCDR3 (IMGT)	QSYDLQKSSRV
SEQ ID NO: 96	VL	DIVLTQPPSVSGAPGQRTISCTGSSSNIGAGYAVHWYQQLPGTAPKLLIYS NNKRPSGVPDFRSGSKSGTSASLAITGLQAEDEADYYCQSYDLQKSSRVFG GGTKLTVL
SEQ ID NO: 97	DNA VL	gat atcgtgctgaccagccgcccagcgtgagcgggtgcaccgggcccagcgcgtgaccattagctgtaccggcagca gcagcaacattgggtgctgggtacgctgtgcatgggtaccagcagctgccgggacggcgccgaaactgctgatctactc taacaacaaacgcccagcggcggtgccggatcgctttagcggatccaaaagcggcaccagcggcagcctggcgatt accggcctgcaagcagaagacgaagcggattattactgccagtcctacgacctgcagaaatcttctcgtgtgtttggcg cggcacgaagttaaccgtccta
SEQ ID NO: 98	Light Chain	DIVLTQPPSVSGAPGQRTISCTGSSSNIGAGYAVHWYQQLPGTAPKLLIYS NNKRPSGVPDFRSGSKSGTSASLAITGLQAEDEADYYCQSYDLQKSSRVFG GGTKLTVLQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWK ADSSPVKAGVETTPSKQSNKYAASSYLSLTPEQWKSHRSYSCQVTHEGS TVEKTVAPTECS
SEQ ID NO: 99	DNA Light Chain	gat atcgtgctgaccagccgcccagcgtgagcgggtgcaccgggcccagcgcgtgaccattagctgtaccggcagca gcagcaacattgggtgctgggtacgctgtgcatgggtaccagcagctgccgggacggcgccgaaactgctgatctactc taacaacaaacgcccagcggcggtgccggatcgctttagcggatccaaaagcggcaccagcggcagcctggcgatt accggcctgcaagcagaagacgaagcggattattactgccagtcctacgacctgcagaaatcttctcgtgtgtttggcg cggcacgaagttaaccgtcctaggtcagcccaaggtgcacctcgttcccgcctcctctgaggagctt caagccaaacagggccacactgggtgtgtctcataagtgaactctaccgggagccgtgacagtggcctggaaggcagat agcagcccggtcaaggcgggagtgagaccaccacacctccaaacaaagcaacaacagtacgcggccagcag ctatctgagcctgacgcctgagcagtggaagtcacacagaagctacagctgccaggtcacgcattgaagggagcaccg tggaagaagacagtggccctacagaatgttca
XX01_ LALA		
SEQ ID NO: 4	HCDR1 (Combined)	GFTFNTHYIH
SEQ ID NO: 100	HCDR2 (Combined)	SISSSQSTYYADSVKG
SEQ ID NO: 6	HCDR3 (Combined)	ERGVVYHMFDP
SEQ ID NO: 7	HCDR1 (Kabat)	THYIH
SEQ ID NO: 100	HCDR2 (Kabat)	SISSSQSTYYADSVKG
SEQ ID NO: 6	HCDR3 (Kabat)	ERGVVYHMFDP
SEQ ID NO: 8	HCDR1 (Chothia)	GFTFNTH
SEQ ID NO: 101	HCDR2 (Chothia)	SSSQS
SEQ ID NO: 6	HCDR3 (Chothia)	ERGVVYHMFDP
SEQ ID NO: 10	HCDR1 (IMGT)	GFTFNTHY
SEQ ID NO: 102	HCDR2 (IMGT)	ISSSQST

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 12	HCDR3 (IMGT)	ARERGVVYHMFDP
SEQ ID NO: 103	VH	EVQLLESGGGLVQPGGSLRLSCAASGFTFNTHYIHWVRQAPGKGLEWVSSI SSSGQSTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARERG VYVYHMFDPWGQGLTVTVSS
SEQ ID NO: 104	DNA VH	gagggtgcaattgctggaaagcggcggtggcctgggtgcagccgggtggcagcctgcgtctgagctgcgcggcgctccg gattcacctttaacactcattacatccattgggtgcccaggcccccggcaagggtctcgagtggtttcctctatctcttct tctggccagctctacttactatgcggatagcgtgaaaggccgctttaccatcagccgcgataaattcgaaaaacacctgtat ctgcaaatgaacagcctgcgtgcggaagatacggccgtgtattattgcgcgcgtgaacgtggttaccttactaccatat gttcgatccgtggggccaaggcaccctggtagctgttagctca
SEQ ID NO: 105	Heavy Chain	EVQLLESGGGLVQPGGSLRLSCAASGFTFNTHYIHWVRQAPGKGLEWVSSI SSSGQSTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARERG VYVYHMFDPWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVK DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYI CMVNHKPSNTKVDKRVPEPKCDKHTCTPPCPAPEAAGGPSVFLFPPKPKDT LMSRTPEVTVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVLHQDLNGKEYCKVSNKALPAPIEKTI SKAKGQPREPQVYIT LPDSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSD GSFFLYSKLTVDKSRWQQGNVPSCSVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 106	DNA Heavy Chain	gagggtgcaattgctggaaagcggcggtggcctgggtgcagccgggtggcagcctgcgtctgagctgcgcggcgctccg gattcacctttaacactcattacatccattgggtgcccaggcccccggcaagggtctcgagtggtttcctctatctcttct tctggccagctctacttactatgcggatagcgtgaaaggccgctttaccatcagccgcgataaattcgaaaaacacctgtat ctgcaaatgaacagcctgcgtgcggaagatacggccgtgtattattgcgcgcgtgaacgtggttaccttactaccatat gttcgatccgtggggccaaggcaccctggtagctgttagctcagcctccaccaagggtccatcggtcttccccctggca ccctcctccaagagcacctctgggggacacagcggccctgggctgcctgggtcaaggactacttccccgaaccggtgac gggtgctgctggaactcaggcgccctgaccagcggcggtgcacaccttccccggctgtcctacagtcctcaggactctactcc ctcagcagcgtggtgacctgcccctccagcagcttggggcaccagacctacatctgcaacgtgaatcacagcccgag caacccaagggtggacaagagagttgagcccaaatcttgtgacaaaactcacacatgccaccgtgcccagcacctg aagcagcggggggacgtcagtccttccctcttccccccaaaacccaaggacacctcatgatctccccgaccctcgagg tcacatgcgtggtggtggagctgagccacgaagacctgagggtcaagttcaactggtacgtggagcggcggtggaggtg cataatgccaaagacaaagccgcgggaggagcagtaacaacagcacgtaccgggtggtcagcgtcctcaccgtcctgc accaggactggctgaatggcaaggagtaacaagtgcagggtctccaaacaaagccctccagcccccatcgagaaaac catctccaaagccaaagggcagccccgagaaccacaggtgtacacctgcccccatccccgggaggagatgaccaag aacagggtcagcctgacctgctggtcaaggcttctatcccagcgacatcgccgtggagtgaggagcaatgggca gcccggagaaacactacaagaccacgcctcccgctgctggactccgacggctccttcttctctacagcaagctcaccgt ggacaagagcaggtggcagcaggggaacctcttctatgctccgtgatgatgaggtctctgcacaaccactacacgc agaagagcctctccctgtctccgggtaaa
SEQ ID NO: 17	LCDR1 (Combined)	RASQSITRNYLA
SEQ ID NO: 18	LCDR2 (Combined)	GASSRAT
SEQ ID NO: 19	LCDR3 (Combined)	QQHSMYPRT
SEQ ID NO: 17	LCDR1 (Kabat)	RASQSITRNYLA
SEQ ID NO: 18	LCDR2 (Kabat)	GASSRAT
SEQ ID NO: 19	LCDR3 (Kabat)	QQHSMYPRT
SEQ ID NO: 20	LCDR1 (Chothia)	SQSITRNY
SEQ ID NO: 21	LCDR2 (Chothia)	GAS
SEQ ID NO: 22	LCDR3 (Chothia)	HSMYPR
SEQ ID NO: 23	LCDR1 (IMGT)	QSITRNY
SEQ ID NO: 21	LCDR2 (IMGT)	GAS
SEQ ID NO: 19	LCDR3 (IMGT)	QQHSMYPRT

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 24	VL	DIVLTQSPATLSLSPGERATLSCRASQSI TRNYLAWYQQKPGQAPRLLIYGA SSRATGIPARFSGSGSGTDFTLTISSELEPEDFAVYQCQHSMPRTFGQGTK VEIK
SEQ ID NO: 25	DNA VL	gatatcggtgctgaccagagcccgggcgaccctgagcctgagcccggtgaacgtgccaccctgagctgcagagcga gccagtctatcactcgtaactacctggcttggtaccagcagaaacggggccaggccccgcgtctattaatctacggtgct tcttctcggtgcgaccggcattccggcgcggtttagcggcagcggatccggcaccgatttcaccctgaccattagcagcct ggaaccggaagactttcggtgtattattgccagcagcattctatgtaccgcgtacctttggccaggggcacgaaagttag aaattaaa
SEQ ID NO: 26	Light Chain	DIVLTQSPATLSLSPGERATLSCRASQSI TRNYLAWYQQKPGQAPRLLIYGA SSRATGIPARFSGSGSGTDFTLTISSELEPEDFAVYQCQHSMPRTFGQGTK VEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNAL QSGNSQESVTEQDSKDYSLSSLTLSKADYERHKVYACEVTHQGLSSPV TKSFNRGEC
SEQ ID NO: 27	DNA Light Chain	gatatcggtgctgaccagagcccgggcgaccctgagcctgagcccggtgaacgtgccaccctgagctgcagagcga gccagtctatcactcgtaactacctggcttggtaccagcagaaacggggccaggccccgcgtctattaatctacggtgct tcttctcggtgcgaccggcattccggcgcggtttagcggcagcggatccggcaccgatttcaccctgaccattagcagcct ggaaccggaagactttcggtgtattattgccagcagcattctatgtaccgcgtacctttggccaggggcacgaaagttag aaattaaacgtacggtggccgctcccagcgtgttcatcttccccccagcgacgagcagctgaagagcgggcaccgcc agcgtggtgtgctctgtaacaacttctacccccgggaggccaaaggtgcagtgggaaggtggacaacgccctgcaga gcggcaacagccaggaaagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccctgaccc tgagcaaggccgactacgagaagcacaaggtgtacgcctgcgaggtgacccaccagggcctgtccagccccgtgac caagagcttcaaccggggcgagtg
XX01_ DAPA		
SEQ ID NO: 4	HCDR1 (Combined)	GFTFNTHYIH
SEQ ID NO: 100	HCDR2 (Combined)	SISSSGQSTYYADSVKG
SEQ ID NO: 6	HCDR3 (Combined)	ERGVVYHMFDP
SEQ ID NO: 7	HCDR1 (Kabat)	THYIH
SEQ ID NO: 100	HCDR2 (Kabat)	SISSSGQSTYYADSVKG
SEQ ID NO: 6	HCDR3 (Kabat)	ERGVVYHMFDP
SEQ ID NO: 8	HCDR1 (Chothia)	GFTFNTH
SEQ ID NO: 101	HCDR2 (Chothia)	SSSGQS
SEQ ID NO: 6	HCDR3 (Chothia)	ERGVVYHMFDP
SEQ ID NO: 10	HCDR1 (IMGT)	GFTFNTHY
SEQ ID NO: 102	HCDR2 (IMGT)	ISSSGQST
SEQ ID NO: 12	HCDR3 (IMGT)	ARERGVVYHMFDP
SEQ ID NO: 103	VH	EVQLLESQGLVQPGGSLRLSCAASGFTFNTHYIHWVRQAPGKGLEWVSSI SSSGQSTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARERG YVYVHMFDPWGQGLTVTVSS
SEQ ID NO: 107	DNA VH	gaagtgcagctgcttgagtcggggggtggactgggtgcagcccgaggatccctgcgcctgagctgcgctgcatccgg cttcaccttcaacacgcactacatccattgggtcagacaggccccaggaaaaggcctggaatgggtgtcctccatctcct cgtcggggcagtcacactactacgcggactccgtcaaggcccggtttaccattagccgggacaaacagcaagaatacc ctgtacctccaaatgaactcgtgagggccgaagataccgcccgtgtattactgtgcccgcgagagaggctacgtgtact accacatgttcgaccgtggggacagggtactctcgtgactgtgtctct
SEQ ID NO: 108	Heavy Chain	EVQLLESQGLVQPGGSLRLSCAASGFTFNTHYIHWVRQAPGKGLEWVSSI SSSGQSTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARERG YVYVHMFDPWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAAAGCLVK

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYI CNVNHKPSNTKVDKRVPEKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDT LMI SRTPEVTCVVAVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVLHQDLNGKEYKCKVSNKALAAPIEKTI SKAKGQPREPQVYT LPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSD GSFFLYSKLTVDKSRWQQGNVPSCSVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 109	DNA Heavy Chain	gaagtgcagctgcttgagtcgggggggtggactgggtgcagcccgaggatccctgcgcctgagctgcgctgcatccgg cttcaccttcaacacgcactacatccattgggtcagacaggcccccaggaaaaggcctggaatgggtgtcctccatctcct cgtcggggcagtcacactactacgcggactccgtcaaggccgggtttaccattagccgggacaacagcaagaatacc ctgtacctccaaatgaactcgctgagggcgaagataccgcctgtattactgtgcccgcgagagaggctacgtgtact accacatgttcgaccctggggacagggtagctctcgtagctgtgtcttctgcgagcactaaggcccgctcagtggtccc gctggctccatcgtagaagtcacctccggaggaaacgcagcactcgggtgcctgggtcaaggactacttccctgagcc agtgaccgtgtcgtggaaacagcggagccctgacttccggcgtgcacactttcccgcggtgctgcagtcctccgggtctg tactcccttctcgctgggtcaccgtgcctgctagcctgggcacccagacctacatctgcaacgtgaaccacaagcc gtccaacaccaaagtggataagcgggtggagccgaagtccctgcgat aagacacacagtcgccgccatgtccagcg cctgaattgcttggcggaccttccgtgttccgtgtcccgccctaaagcccaaggacaccttgatgattagccggactccga agtcaactgtgtggtgggtggcagtggtcccacgaggaccccgaggtcaagtttaattggtacgtggacggcgtcgaagt gcacaacgccaaagactaagcccccgggagggaacagtagacaacagcactaccgggtcgtgtccgtgctgacctgctg caccaggactggctgaatgggaaagagtacaagtgcaaaagtgtccaacaaggccttgcccgctcctatcgaaaaaact atcagcaaggctaaaggacagccgagggaaaccccaagtcacacccctgccccccttcacgcgaagagatgaccaaga atcaagtgctcgctgacctgctcgtcaagggtatctaccctccgacattgcggtggagtgagggtccaacggccagc ccgagaacaactacaagactactccgccgtgctggactccgacggcagcttcttctgtattccaagctgacctggga caagtcccggtggcagcaaggaaacgtgttctcctgctcggtcatgcacgaagccctgcacaaccactatacgcagaa gtccctgtccttgagccccgggaaa
SEQ ID NO: 17	LCDR1 (Combined)	RASQSITRNYLA
SEQ ID NO: 18	LCDR2 (Combined)	GASSRAT
SEQ ID NO: 19	LCDR3 (Combined)	QQHSMYPRT
SEQ ID NO: 17	LCDR1 (Kabat)	RASQSITRNYLA
SEQ ID NO: 18	LCDR2 (Kabat)	GASSRAT
SEQ ID NO: 19	LCDR3 (Kabat)	QQHSMYPRT
SEQ ID NO: 20	LCDR1 (Chothia)	SQSITRNY
SEQ ID NO: 21	LCDR2 (Chothia)	GAS
SEQ ID NO: 22	LCDR3 (Chothia)	HSMYPR
SEQ ID NO: 23	LCDR1 (IMGT)	QSITRNY
SEQ ID NO: 21	LCDR2 (IMGT)	GAS
SEQ ID NO: 19	LCDR3 (IMGT)	QQHSMYPRT
SEQ ID NO: 24	VL	DIVLTQSPATLSLSPGERATLSCRASQSITRNYLAWYQQKPGQAPRLLIYGA SSRATGIPARFSGSGSGTDFTLTISLLEPEDFAVYYCQQHSMYPRTFGQGTK VEIK
SEQ ID NO: 110	DNA VL	gacatcgtgctgactcagtcctccctgcgactctgagcctgtcaccgggagaaacgggcccacctctcttgcgcgcctcc caatccattactcggaactacctggcctgggtatcagcagaagccaggacaggccctaggcttctgatctacggggcc agctcaagagcaactggcatcccgctcgcttctccggttcgggaagcggcaccgacttcacctgacaatttctgccc tcgaacccgaggatttcccgctgtactactgccaacagcactccatgtacccccggacctttgggcagggaacccaaagt cgagatcaag
SEQ ID NO: 26	Light Chain	DIVLTQSPATLSLSPGERATLSCRASQSITRNYLAWYQQKPGQAPRLLIYGA SSRATGIPARFSGSGSGTDFTLTISLLEPEDFAVYYCQQHSMYPRTFGQGTK VEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNAL QSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPV TKSFNRGEC

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 111	DNA Light Chain	gacatcgtgctgactcagtcacctgcgactctgagcctgtcaccgggagaacggggccacctctcttgcgcgcctcc caatccattactcggaactacctggcctgggtatcagcagaagccaggacaggccctaggcttctgatctacggggcc agctcaagagcaactggcatcccggtcgcttctccgggttcgggaagcggcaccgacttcacctgacaatttcgtccc tcgaacccgaggtatttcgccgtgtactactgccaacagcactccatgtaccccggaactttgggcaggggaaccaaagt cgagatcaagcgtagcgtggccgctcccagcgtgttcatcttccccccagcgacgagcagctgaagagcggcaccg ccagcgtggtgtgcctgtgaacaacttctaccccgggaggccaaggtgcagtggaagtggaacgcctgcag agcggcaacagccaggagagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccctgacc ctgagcaaggccgactacgagaagcataaagggtgtacgcctgcgaggtgacccaccaggcctgtccagcccgtga ccaagagcttcaacagggcgagtgc
XX01 N30S DAPA		
SEQ ID NO: 112	HCDR1 (Combined)	GFTFSTHYIH
SEQ ID NO: 100	HCDR2 (Combined)	SISSSGQSTYYADSVKG
SEQ ID NO: 6	HCDR3 (Combined)	ERGVVYHMFDP
SEQ ID NO: 7	HCDR1 (Kabat)	THYIH
SEQ ID NO: 100	HCDR2 (Kabat)	SISSSGQSTYYADSVKG
SEQ ID NO: 6	HCDR3 (Kabat)	ERGVVYHMFDP
SEQ ID NO: 113	HCDR1 (Chothia)	GFTFSTH
SEQ ID NO: 101	HCDR2 (Chothia)	SSSGQS
SEQ ID NO: 6	HCDR3 (Chothia)	ERGVVYHMFDP
SEQ ID NO: 114	HCDR1 (IMGT)	GFTFSTHY
SEQ ID NO: 102	HCDR2 (IMGT)	ISSSGQST
SEQ ID NO: 12	HCDR3 (IMGT)	ARERGVVYHMFDP
SEQ ID NO: 115	VH	EVQLLESGGGLVQPGGSLRLSCAASGFTFSTHYIHWRQAPGKGLEWVSSI SSSGQSTYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARERG YVYVHMFDPWGQGLVTVSS
SEQ ID NO: 116	DNA VH	gaagtgcagctgcttgagtcggggggtggactgggtgcagcccgaggatccctgcgcctgagctgcgctgcatccgg cttcaccttcagcacgcactacatccattgggtcagacagggcccaggaaaaggcctggaatgggtgtcctccatctcc tcgtcggggcagtcacactactacgcggactccgtcaaggccgggttaccattagccgggacaacagcaagaatacc ctgtacctccaaatgaactcgtgagggccgaagataccgcctgtattactgtgcccgcgagagaggtacgtgtact accacatgttcgaccgtggggacagggtagctcctcgtgactgtgtctctt
SEQ ID NO: 117	Heavy Chain	EVQLLESGGGLVQPGGSLRLSCAASGFTFSTHYIHWRQAPGKGLEWVSSI SSSGQSTYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARERG YVYVHMFDPWGQGLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVK DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYI CNVNHKPSNTKVDKRVPEPKSCDKHTHTCPPCPAPELLGGPSVFLFPPKPKDT LMI SRTPETVTVVAVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVLHQDWLNGKEYKCKVSNKALAAPIEKTISKAKGQPREPQVYT LPPSRREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSD GSFFLYSKLTVDKSRWQQGNVFSQSVMHENHNTQKSLSLSPGK
SEQ ID NO: 118	DNA Heavy Chain	gaagtgcagctgcttgagtcggggggtggactgggtgcagcccgaggatccctgcgcctgagctgcgctgcatccgg cttcaccttcagcacgcactacatccattgggtcagacagggcccaggaaaaggcctggaatgggtgtcctccatctcc tcgtcggggcagtcacactactacgcggactccgtcaaggccgggttaccattagccgggacaacagcaagaatacc ctgtacctccaaatgaactcgtgagggccgaagataccgcctgtattactgtgcccgcgagagaggtacgtgtact accacatgttcgaccgtggggacagggtagctcctcgtgactgtgtctcttctgcgagcactaaggggcccgtagtgtccc gctggctccatcgtcgaagtcacctccggaggaacccgcagcactcgggtgcctgggtcaaggactacttcctgagcc agtgaccgtgtcgtggaacagcgagccctgaactccggcgtgcacactttcccgcgggtgctgcagtcctccgggtctg tactcccttctcgtccgtggtcaccgtgcctcgtctagcctgggcacccagacctacatctgcaacgtgaaccacaagcc

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		gtccaacaccaaagtggataaagcgggtggagccgaagtcctgcgataagacacacacgtgcccgccatgtccagcg cctgaattgcttggcggaccttccgtgttccgtgttccgcctaaagcccaaggacaccttgatgattagccggactccga agtcaacctgtgtggtggtggcagtggtcccacgaggaccccgaggtcaagtttaattggtacgtggacggcgtcgaagt gcacaacgccaagactaagccccgggaggaacagtagacaacagcacctaccgggtcgtgtccgtgctgacctgctg caccaggactggctgaatgggaaagagtacaagtgc aaagtgtccaacaaggccttgcccgctcctatcgaaaaaact atcagcaaggctaaaggacagccgagggaaaccccaagtctacacctgcccccttcacgcgaagagatgaccaaga atcaagtgtcgtgacctgctcgtcaagggattctaccctccgacattgcggtggagtgaggagccaacggccagc ccgagaacaactacaagactactccgccgtgctggactccgacggcagcttcttccgtattccaagctgacctggga caagtcccggtggcagcaaggaaacgtgttctcctgctcggtcatgcacgaagcctgcacaaccactatcgcagaa gtccctgtccttgagccccgggaaa
SEQ ID NO: 17	LCDR1 (Combined)	RASQSITRNYLA
SEQ ID NO: 18	LCDR2 (Combined)	GASSRAT
SEQ ID NO: 19	LCDR3 (Combined)	QQHSMYPRT
SEQ ID NO: 17	LCDR1 (Kabat)	RASQSITRNYLA
SEQ ID NO: 18	LCDR2 (Kabat)	GASSRAT
SEQ ID NO: 19	LCDR3 (Kabat)	QQHSMYPRT
SEQ ID NO: 20	LCDR1 (Chothia)	SQSITRNY
SEQ ID NO: 21	LCDR2 (Chothia)	GAS
SEQ ID NO: 22	LCDR3 (Chothia)	HSMYPR
SEQ ID NO: 23	LCDR1 (IMGT)	QSITRNY
SEQ ID NO: 21	LCDR2 (IMGT)	GAS
SEQ ID NO: 19	LCDR3 (IMGT)	QQHSMYPRT
SEQ ID NO: 24	VL	DIVLTQSPATLSLSPGERATLSCRASQSITRNYLAWYQQKPGQAPRLLIYGA SSRATGIPARFSGSGSGTDFTLTISLEPEDFAVYYCQQHSMYPRTFGQGTK VEIK
SEQ ID NO: 110	DNA VL	gacatcgtgctgactcagtcacctgctgactctgagcctgtcaccgggagaaacggggccacctctcttgcgcgcctcc caatccattactcggaactacctggcctgggtatcagcagaagccaggacaggccctaggcttctgatctacggggcc agctcaagagcaactggcatccccgctcgcttctccggttcgggaagcggcaccgacttcacctgacaatttctgtccc tcgaacccgaggatttccgcgtgtactactgccaacagcaactccatgtacccccggacctttgggcagggaaaccaaagt cgagatcaag
SEQ ID NO: 26	Light Chain	DIVLTQSPATLSLSPGERATLSCRASQSITRNYLAWYQQKPGQAPRLLIYGA SSRATGIPARFSGSGSGTDFTLTISLEPEDFAVYYCQQHSMYPRTFGQGTK VEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNAL QSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPV TKSFNRGEC
SEQ ID NO: 111	DNA Light Chain	gacatcgtgctgactcagtcacctgctgactctgagcctgtcaccgggagaaacggggccacctctcttgcgcgcctcc caatccattactcggaactacctggcctgggtatcagcagaagccaggacaggccctaggcttctgatctacggggcc agctcaagagcaactggcatccccgctcgcttctccggttcgggaagcggcaccgacttcacctgacaatttctgtccc tcgaacccgaggatttccgcgtgtactactgccaacagcaactccatgtacccccggacctttgggcagggaaaccaaagt cgagatcaagcgtacgggtggcgcgtcccagcgtgttcatcttccccccagcgacgagcagctgaagagcggcacgcg ccagcgtggtgtgcctgctgaacaacttctacccccgggagggccaaggtgcagtggaaggtggacaacgcctgcag agcggcaacagccaggagagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccctgacc ctgagcaaggccgactacgagaagcataaggtgtacgcctgcgaggtgacccaccaggccctgtccagccccgtga ccaagagcttcaacagggcgagtgcc
XX03_ LALA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 119	HCDR2 (Combined)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 119	HCDR2 (Kabat)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 120	HCDR2 (Chothia)	ESKGNV
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSY
SEQ ID NO: 121	HCDR2 (IMGT)	IESKGNV
SEQ ID NO: 36	HCDR3 (IMGT)	ARDYSMIYSYGAGAFDY
SEQ ID NO: 122	VH	EVQLLESQGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS VIESKGNIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLTVTVSS
SEQ ID NO: 123	DNA VH	gaggtgcaattgctggaagcggcggtggcctgggtgcagcgggtggcagcctgcgtctgagctgcgcggtgcgtccg gattcaccttttcttcttactggatgaactgggtgcgcccaggcccggaagggtctcgagtggtttccggttatcgaat ctaaaggcaactacatcttctatgctggtatgagcgtgaaaggccgctttaccatcagccgcgat aattcgaaaaacacctg tatctgcaaatgaacagcctgcgtgcggaagat acggcgcgtgattattgcgcgcgtgaccttactctatgatctactctt acgggtgctggtgctttcgattactggggccaaggcacccctggtagctgtagctca
SEQ ID NO: 124	Heavy Chain	EVQLLESQGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS VIESKGNIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALG CLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGT QTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLPPPKP KDTLMISRTPETVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY NSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREP QVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPV LDSGSEFPLYSKLTVDKSRWQQGNVFPSCVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 125	DNA Heavy Chain	gaggtgcaattgctggaagcggcggtggcctgggtgcagcgggtggcagcctgcgtctgagctgcgcggtgcgtccg gattcaccttttcttcttactggatgaactgggtgcgcccaggcccggaagggtctcgagtggtttccggttatcgaat ctaaaggcaactacatcttctatgctggtatgagcgtgaaaggccgctttaccatcagccgcgat aattcgaaaaacacctg tatctgcaaatgaacagcctgcgtgcggaagat acggcgcgtgattattgcgcgcgtgaccttactctatgatctactctt acgggtgctggtgctttcgattactggggccaaggcacccctggtagctgtagctcagcctccaccaagggtccatcggt cttccccctggcaccctctccaagagcacctctgggggacagcggccctgggtgcgtggtcaaggactacttccc cgaacccgtgacgggtgctgctggaactcaggcgcctgaccagcggcgtgcacaccttcccgggtgctctacagtcctc aggactctactccctcagcagcgtggtgacctgcccctccagcagcttgggcaaccagacctacatctgcaacgtgaat cacaagccagcaacaccaaaggtggacaagagagttgagcccaaatcttgtgacaaaactcacacatgccaccctg cccagcacctgaagcagcggggggacgctcagctcttctcttcccccaaaacccaaggacacctcatgatctcccg gacctcaggtcacatgcgtggtggtggagctgagccacgaagacctgaggtcaagttcaactggtacgtggagc gctgagaggtgcat aatgccaagacaaagccgaggagagcagtaacaacagcagct accgggtggtcagcgtcct caccgtctctgaccaggactggctgaatggcaaggagtacaagtgcaaggtctcccaaaaagccctcccagcccca tcgagaaaacacatctccaaagccaaagggcagcccgagaaacacaggtgtacacctgccccatcccgggagga gatgaccaagaacacaggtcagcctgacctgctggtcaaaggcttctatcccagcagcatcgccgtggagtgaggaga gcaatgggcagcggagaaactacaagaccagcctcccgtgctggactccgacggctccttcttctctacagca agctcaccgtggacaagagcaggtggcagcaggggaacgtctctcagctcgtgagtgatgaggtctgcacaacc actacacgcagaagacctctcctctgctccgggtaaa
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 43	LCDR3 (Combined)	MQSYEKPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 43	LCDR3 (Kabat)	MQSYEKPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 46	LCDR3 (Chothia)	SYEKPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 43	LCDR3 (IMGT)	MQSYEKPRT
SEQ ID NO: 48	VL	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCMQSYEKPRTFGQGTKV EIK
SEQ ID NO: 49	DNA VL	gatatccagatgaccagagcccgagcagcctgagcgcagcgtggcgcatcgcgtagaccattacctgcagagcca gccagggtatttcttcttacctggccttggtaccagcagaaacggggcaaaagcgccgaaactattaatctacactgcttcta ctctgcaaagcggcggtgccgagccgctttagcggcagcggatccggcacccgatttcaccctgaccattagctctctgca accggaagactttgcgacctattattgcatgcagtccttacgaaaaaccgcgtacctttggccagggcacgaaagtgtgaaa ttaaa
SEQ ID NO: 50	Light Chain	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCMQSYEKPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID NO: 51	DNA Light Chain	gatatccagatgaccagagcccgagcagcctgagcgcagcgtggcgcatcgcgtagaccattacctgcagagcca gccagggtatttcttcttacctggccttggtaccagcagaaacggggcaaaagcgccgaaactattaatctacactgcttcta ctctgcaaagcggcggtgccgagccgctttagcggcagcggatccggcacccgatttcaccctgaccattagctctctgca accggaagactttgcgacctattattgcatgcagtccttacgaaaaaccgcgtacctttggccagggcacgaaagtgtgaaa ttaaacgtacgggtggcgctcccagcgtgttcatcttccccccagcagcagcagctgaagagcggccaccgcccagc gtgggtgtgcctgctgaacaaacttctacccccggggaggccaaaggtgcagtggaagggtggacaacgcccctgcagagcg gcaacagccaggaaagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccctgaccctga gcaaggccgactacgagaagcacaaaggtgtacgcctgcgaggtgacccaccagggcctgtccagccccgtgacca agagcttcaaccggggcgagtg
XX04_ LALA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 29	HCDR2 (Combined)	AISSDGSYTTYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 29	HCDR2 (Kabat)	AISSDGSYTTYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 33	HCDR2 (Chothia)	SSDGSY
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 35	HCDR2 (IMGT)	ISSDGSYT
SEQ ID NO: 36	HCDR3 (IMGT)	ARDRYSMIYSYGAGAFDY
SEQ ID NO: 37	VH	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS AISSDGSYTYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARD RYSMIYSYGAGAFDYWGQGLTVTVSS
SEQ ID NO: 38	DNA VH	caggtgcaattgctggaaagcggcggtggcctgggtgcagccgggtggcagcctgcgtctgagctgcgcggcgctccg gattcaccttttcttcttactggatgaactgggtgcgcagggcccgggcaaaaggtctcgagtgagggtttccgctatctcttc tgacgggtctctacacctactatgcggatagcgtgaaagggcgcgtttaccatcagccgcgataatcgaaaaacaccctgt atctgcaaatgaacagcctgcgtgcggaagatacggcgcgtgtattattgcgcgcgtgacccgttactctatgatctactctt acgggtgctggtgctttcgattactggggccaaggcaccctggtgactgttagctca
SEQ ID NO: 39	Heavy Chain	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS AISSDGSYTYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARD RYSMIYSYGAGAFDYWGQGLTVTVSSASTKGPSVFLPAPSSKSTSGGTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSL GTQTYICNVNHKPSNTKVDKRVKPSCKDTHCTPPCPAPEAAGGPSVFLFP PKPKDITLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNATKPREE QYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTIISKAKGQPR EPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPP PVLDSDSGFFLYSKLTVDKSRWQQGNVPSCSVMHEALHNHYTQKSLSLSP GK
SEQ ID NO: 40	DNA Heavy Chain	caggtgcaattgctggaaagcggcggtggcctgggtgcagccgggtggcagcctgcgtctgagctgcgcggcgctccg gattcaccttttcttcttactggatgaactgggtgcgcagggcccgggcaaaaggtctcgagtgagggtttccgctatctcttc tgacgggtctctacacctactatgcggatagcgtgaaagggcgcgtttaccatcagccgcgataatcgaaaaacaccctgt atctgcaaatgaacagcctgcgtgcggaagatacggcgcgtgtattattgcgcgcgtgacccgttactctatgatctactctt acgggtgctggtgctttcgattactggggccaaggcaccctggtgactgttagctcagcctccaccaagggtccatcggt cttccccctggcaccctctccaagagcacctctgggggcaacagcggccctgggtgcctgggtcaaggactacttccc cgaacccgtgacgggtgctgctggaactcaggcgcctgaccagcggcgtgcacaccttccccgggtgctctacagtcctc aggactctactccctcagcagcgtggtgacccgtgccctccagcagcctggggcaccagacctacatctgcaacgtgaat cacaagccagcaacaccaaggtggacaagagagttgagcccaaatcttgtgacaaaactcacacatgccaccctg cccagcacctgaagcagcggggggaccgtcagctcttctcttcccccaaaacccaaggacacctcatgatctcccg gaccctgaggtcacatgcgtggtggtggacgtgagccacgaagacctgaggtcaagttcaactggtacgtggacg gcgtggaggtgcataatgccaagacaaagccgaggagagcagtaacaacagcagctaccgggtggtcagcgtcct cacctgctctgaccaggactggctgaatggcaaggagtacaagtgcaaggtctcccaaaaagccctcccagcccca tcgagaaaaacatctccaaagccaaagggcagcccgagaaaccacaggtgtacacccctgccccatcccgggaggga gatgaccaagaaccaggtcagcctgacctgctggtcaaaggcttctatccagcgcacatcgccgtggagtgaggaga gcaatgggcagccggagaaactacaagaccagcctcccggtgctggactccgacggctccttcttctctacagca agctcaccgtggacaagagcaggtggcagcaggggaacgtctctcatgctcggtgatgcatgaggctctgcacaacc actacacgcagaagacgtctcctctgctccgggtaaa
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 126	LCDR3 (Combined)	QQEWVKPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 126	LCDR3 (Kabat)	QQEWVKPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 127	LCDR3 (Chothia)	EWVKPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 126	LCDR3 (IMGT)	QQEWVKPRT
SEQ ID NO: 128	VL	DIQMTQSPSSLSASVGRVTITCRASQGISSYLAWYQQKPKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQEWVKPRTFGQGTKV EIK
SEQ ID NO: 129	DNA VL	gatatccagatgaccagagcccgagcagcctgagcgcagcgtggcgatcgcgtagaccattacctgcagagcca gccagggtatttcttcttacctggccttggtaccagcagaaacggggcaaaagcgcgaaactattaatctacctgcttcta ctctgcaaagcggcgtgccagccgctttagcggcagcggatccggcaccgatttcaccctgaccattagctctctgca accggaagactttgcgacctattattgccagcaggaatgggttaaaccgcgtacctttggccagggcacgaaagtga attaaa
SEQ ID NO: 130	Light Chain	DIQMTQSPSSLSASVGRVTITCRASQGISSYLAWYQQKPKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQEWVKPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSPVT KSFNRGEC
SEQ ID NO: 131	DNA Light Chain	gatatccagatgaccagagcccgagcagcctgagcgcagcgtggcgatcgcgtagaccattacctgcagagcca gccagggtatttcttcttacctggccttggtaccagcagaaacggggcaaaagcgcgaaactattaatctacctgcttcta ctctgcaaagcggcgtgccagccgctttagcggcagcggatccggcaccgatttcaccctgaccattagctctctgca accggaagactttgcgacctattattgccagcaggaatgggttaaaccgcgtacctttggccagggcacgaaagtga attaaacgtacgggtggcgcgtcccagcgtgttcattcttccccccagcagcagcagcgtgaagagcggcaccgccag cgtgggtgacctgctgaacaaacttctacccccgggaggccaaaggtgcagtggaaggtggacaacgcccctgcagagc ggcaacagccaggaaagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccctgaccctg agcaaggccgactacgagaagcacaaggtgtacgctgcgaggtgaccaccaggggcctgtccagccccgtgacc aagagcttcaaccggggcgagtgt
XX06_ LALA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 29	HCDR2 (Combined)	AISSDGSYTYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 29	HCDR2 (Kabat)	AISSDGSYTYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 33	HCDR2 (Chothia)	SSDGSY
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSY
SEQ ID NO: 35	HCDR2 (IMGT)	ISSDGSYT

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 36	HCDR3 (IMGT)	ARDRYSMIYSYGAGAFDY
SEQ ID NO: 37	VH	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS AISSDGSYTYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARD RYSMIYSYGAGAFDYWGQGLVTVSS
SEQ ID NO: 132	DNA VH	caggtgcaattgctggaaagcggcggtggcctgggtgcagcgggtggcagcctgcgtctgagctgcgcggcgctccg gattcaccttttcttcttactggatgaactgggtgcgcagggccccaggcaagggtctcgagtggtttccgctatctcttc tgacgggtctttacacctactatgcggatagcgtgaaagggcgcgtttaccatcagccgcgataatcgaaaaacacctgt atctgcaaatgaacagcctgcgtgcggaagatacggcgcgtgtattattgcgcgcgtgacccgttactctatgatctactctt acgggtgctggtgcttctcgattactggggccaaggcaccctggtgactgttagctca
SEQ ID NO: 39	Heavy Chain	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS AISSDGSYTYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARD RYSMIYSYGAGAFDYWGQGLVTVSSASTKGPSVFLAPSSKSTSGGTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSL GTQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFP PKPKDTLMISRTPEVTVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREE QYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTIISKAKGQPR EPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPP PVLDSGDSFPLYSKLTVDKSRWQQGNVFSQVMHEALHNHYTQKSLSLSP GK
SEQ ID NO: 133	DNA Heavy Chain	caggtgcaattgctggaaagcggcggtggcctgggtgcagcgggtggcagcctgcgtctgagctgcgcggcgctccg gattcaccttttcttcttactggatgaactgggtgcgcagggccccaggcaagggtctcgagtggtttccgctatctcttc tgacgggtctttacacctactatgcggatagcgtgaaagggcgcgtttaccatcagccgcgataatcgaaaaacacctgt atctgcaaatgaacagcctgcgtgcggaagatacggcgcgtgtattattgcgcgcgtgacccgttactctatgatctactctt acgggtgctggtgcttctcgattactggggccaaggcaccctggtgactgttagctcagcctccaccaagggtccatcggt cttccccctggcaccctctccaagagcacctctgggggcacagcggccctgggtgctggtcaaggactacttccc cgaacccgtgacgggtgctgctggaactcaggcgcctgaccagcggcgtgcacaccttcccgggtgctctacagtcctc aggactctactccctcagcagcgtggtgaccgtgccctccagcagccttgggcacccagacctacatctgcaacgtgaat cacaagcccagcaacaccaagggtggacaagagagttgagcccaaatcttgtgacaaaactcacacatgccaccctg cccagcacctgaagcagcggggggaccgtcagctcttctcttccccccaaaacccaaggacacctcatgatctcccg gaccctgaggtcacatgcgtggtggtggacgtgagccacgaagacctgaggtcaagttcaactggtagctggacg gcgtggaggtgcataatgccaaagacaaagccgcgggagggagcagtaacacagcagctaccgggtggtcagcgtcct caccgtctctgaccaggactggctgaatggcaaggagtaacagtgcaaggtctccaaacaaagccctcccagccccc tcgagaaaaccatctccaaagccaaagggcagcccgagaaaccaaggtgtacacctgccccatcccgggagga gatgaccaagaaccaggtcagcctgacctgcctggtcaaaggctctatcccagcgacatcgccgtggagtgaggaga gcaatgggcagcggagaaactacaagaccacgcctcccgtgctggactccgacggctccttcttctctacagca agctcaccgtggacaagagcaggtggcagcaggggaacgtctctcatgctcgtgatgcatgaggctctgcacaacc actacacgcagaagacgtctctcctgctctccgggtaaa
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 134	LCDR3 (Combined)	QQTWRKPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 134	LCDR3 (Kabat)	QQTWRKPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 135	LCDR3 (Chothia)	TWRKPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 134	LCDR3 (IMGT)	QQTWRKPRT

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 136	VL	DIQMTQSPSSLSASVGDRVTITCRASQGISSYLAWYQQKPGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQTWRKPRTFGQGTKV EIK
SEQ ID NO: 137	DNA VL	gatatccagatgaccagagcccgagcagcctgagcgccagcgtggcgcatcgcgtagaccattacctgcagagcca gccagggtatttcttcttacctggcttggtaccagcagaaaccgggcaaaagcgcgaaactattaatctacactgcttcta ctctgcaaaagcggcgtgcccagccgctttagcggcagcggatccggcaccgatttcaccctgaccattagctctctgca accggaagactttgcgacctattatgcccagcagacttggcgtaaacccggtacctttggccagggcacgaaagtga attaaa
SEQ ID NO: 138	Light Chain	DIQMTQSPSSLSASVGDRVTITCRASQGISSYLAWYQQKPGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQTWRKPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID NO: 139	DNA Light Chain	gatatccagatgaccagagcccgagcagcctgagcgccagcgtggcgcatcgcgtagaccattacctgcagagcca gccagggtatttcttcttacctggcttggtaccagcagaaaccgggcaaaagcgcgaaactattaatctacactgcttcta ctctgcaaaagcggcgtgcccagccgctttagcggcagcggatccggcaccgatttcaccctgaccattagctctctgca accggaagactttgcgacctattatgcccagcagacttggcgtaaacccggtacctttggccagggcacgaaagtga attaaacgtacggtggcgctcccagcgtgttcattcttccccccagcagcagcagcgtgaagagcggcaccgccag cgtggtgtgcctgctgaacaacttctacccccgggagggccaaggtgagtggaaggtggacaacgccctgcagagc ggcaacagccaggaagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccctgaccctg agcaaggccgactacgagaagcacaaggtgtacgctgagaggtgaccaccaggggcctgtccagccccgtgacc aagagcttcaaccggggcgagtgt
XX06_ DAPA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 29	HCDR2 (Combined)	AISSDGSYTYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 29	HCDR2 (Kabat)	AISSDGSYTYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 33	HCDR2 (Chothia)	SSDGSY
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 35	HCDR2 (IMGT)	ISSDGSYT
SEQ ID NO: 36	HCDR3 (IMGT)	ARDRYSMIYSYGAGAFDY
SEQ ID NO: 37	VH	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS AISSDGSYTYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARD RYSMIYSYGAGAFDYWGQGLTVTVSS
SEQ ID NO: 140	DNA VH	caagtgcagctgcttgagagcgggtggcggaactggtgcagccagggggatccttgcgctgtcatgcgtgcgtcggg gttcaccttctcgtcctactggatgaactgggtcagacaggtccggggaagggaactcgaatgggtgtccgccatttctc ccgacggctcctacacttactacgccgatagcgtcaaggccgggttcaccatctccccgggacaattcgaagaacacc tgtacctccaaatgaactcactgcgcgccgaggacactgcgggtgtattactgtgccgggatagggtacagcatgatcta ctcctacgggtgccggagcctttgactactggggacaggaacccctgtgaccgtgtctagc
SEQ ID NO: 141	Heavy Chain	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS AISSDGSYTYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARD

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		RYSMIYSYGAGAFDYWGQGLTVTVSSASTKGPSVFPPLAPSSKSTSGGTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSL GTQTYICNVNHKPSNTKVDKRVEPKSCDKHTCTPCPCPAPELLGGPSVFLFPP KPKDTLMISRTPEVTCVVAVVSHEDPEVKFNWYVDGVEVHNAKTKPREEQ YNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALAAPIEKTISKAKGQPRE PQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTP VLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPG K
SEQ ID NO: 142	DNA Heavy Chain	caagtgcagctgcttgagagcgggtggcgggactgggtgcagccagggggatccttgcgctgtcatgcgctgcgtcggg gttcaccccttctcgtcctactggatgaactgggtcagacaggtcccggggaagggaactcgaatgggtgtccgccatttcct ccgacggctcctacacttactacgcgatagcgtcaagggccggttcaccatctcccgggacaattcgaagaacaccc tgtacctccaaatgaactcactgcgcgcgaggacactgcggtgtattactgtgcccgggataggtagacatgatcta ctcctacgggtgcgggagcctttgactactggggacagggaacccttgtgacggtgtctagcgcgtccactaagggcc gtcagtggtcccgctggctccatcgtcgaagtcacactccggaggaaaccgcagcactcggtgcctgggtcaaggactac ttccctgagccagtgacccgtgtcgtggaacagcggagccctgacttccggcgtgcacacttttcccgcggtgctgcagt cctccggtctgtactccctttcgtccgtgggtcaccgtgccgtcgtctagcctgggcaccagacctacatctgcaacgtg aaccacaagcgtccaacaccaaagtggataaaggcgggtggagccgaagtcctgcgataagacacacacgtgccgcg catgtccagcgcctgaattgcttggcggaccttcggtgttcctgttcccgccaaagcccaaggacacctgatgattagcc ggactcccgaaagtcacctgtgtggtggtggcagtggtcccacgaggaccccgaggtcaagtttaattggtacgtggacg gcgtcgaagtgcaaacgccaagactaagccccgggaggaaacagtacaacagcactaccgggtcgtgtccgtgct gacccgtgctgcaccaggactggctgaatgggaaagagtaacagtgcaaaagtgtccaaacaggccttggccgctcctat cgaaaaaactatcagcaaggctaaggggacagccgagggaaacccaagtctacaccctgcccccttcacgcgaagag atgaccaagaatcaagtgtcgtgacccctgctcgtcaagggtattctaccctccgacattgcggtggagtgaggagtgcca acggccagccccgagaacaactacaagactactccgccgtgctggactccgacggcagcttcttccctgtattccaagct gaccgtggacaagtcccggtggcagcaaggaaacgtgttctcctgctcggtcatgcacgaagccctgcacaaccacta tacgcagaagtcctgtccttgagcccggggaaa
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 134	LCDR3 (Combined)	QQTWRKPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 134	LCDR3 (Kabat)	QQTWRKPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 135	LCDR3 (Chothia)	TWRKPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 134	LCDR3 (IMGT)	QQTWRKPRT
SEQ ID NO: 136	VL	DIQMTQSPSSLSASVGDRVTITCRASQGISSYLAWYQQKPKGKAPKLLIYTAS TLQSGVPSRFGSGSGTDFTLTISLQPEDFATYYCQQTWRKPRTFGQGTKV EIK
SEQ ID NO: 143	DNA VL	gacatccagatgaccagctctccgtcctccctgtccgcatcagtgggggacagagtgaccatcacttgtcgggcctccc aaggcatctcgtcatacctggcctggatcagcagaaacccggaaaggctccaaagctgctcatcacaccgcctcga ctctgcaatccggagtgccttcccgccttctccggatccgggttcgggaaccgaacttaccctcaccattagcagccttcag ccggaagatttcgcgacctactactgccagcaaacctggcggaagcccaggacatttggccagggcactaaggtcga gattaag
SEQ ID NO: 138	Light Chain	DIQMTQSPSSLSASVGDRVTITCRASQGISSYLAWYQQKPKGKAPKLLIYTAS TLQSGVPSRFGSGSGTDFTLTISLQPEDFATYYCQQTWRKPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPV

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
KSFNRGEC		
SEQ ID NO: 144	DNA Light Chain	gacatccagatgaccagtcctccgtcctccctgtccgcatcagtgggggacagagtgaccatcacttgtcgggctccc aaggcatctcgtcatacctggcctggatcagcagaaacccggaaaggctccaaagctgctcatctacacgcctcga ctctgcaatccggagtgcttcccgcttctccggatccgggttcgggaaccgacttcacctcaccattagcagccttcag ccggaagatttcgcgacctactactgccagcaaacctggcgaagcccaggacatttggccaggggcactaaggctga gat taagcgtacgggtggcgcctcccagcgtgttcatcttccccccagcgacgagcagctgaagagcggcaccgcca gcgtgggtgctgctgaacaacttctacccccgggaggccaagggtgcagtggaaagtggaacaacgcctgcagag cggcaacagccaggagagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcacccctgaccct gagcaaggccgactacgagaagcataagggtgtacgcctgcgaggtgacccaccagggcctgtccagccccgtgacc aagagcttcaacaggggcgagtgc
XX07_ LALA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 29	HCDR2 (Combined)	AISSDGSYTYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 29	HCDR2 (Kabat)	AISSDGSYTYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 33	HCDR2 (Chothia)	SSDGSY
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 35	HCDR2 (IMGT)	ISSDGSYT
SEQ ID NO: 36	HCDR3 (IMGT)	ARDRYSMIYSYGAGAFDY
SEQ ID NO: 37	VH	QVQLLESGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKLEWVS AISSDGSYTYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARD RYSMIYSYGAGAFDYWGQTLTVTSS
SEQ ID NO: 38	DNA VH	cagggtgcaattgctggaaagcggcggtggcctgggtgcagccgggtggcagcctgcgtctgagctgcgcggcgctccg gattcaccttttcttcttactggatgaactgggtgcgccaggccccgggcaagggtctcgagtggggttcccgctatctcttc tgacgggttcttacacctactatgcggatagcgtgaaaggccgctttaccatcagccgcgataattcgaaaaacaccctgt atctgcaaatgaacagcctgcgtgcggaagatacggccgtgtattattgcgcgcgtgacccgttactctatgatctactctt acggtgctggtgctttcgattactggggccaaggcaccctggtagctgttagctca
SEQ ID NO: 39	Heavy Chain	QVQLLESGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKLEWVS AISSDGSYTYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARD RYSMIYSYGAGAFDYWGQTLTVTSSASTKGPSVFPLAPSSKSTSGGTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSL GTQTYICNVNHKPSNTKVDKRVPEPKSCDKTHTCPPCPAPEAAGGPSVFLFP PKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREE QYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI SKAKGQPR EPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTP PVLDSGDSFFLYSKLTVDKSRWQQGNVFC SVMHEALHNNHYTKSLSLSP GK
SEQ ID NO: 40	DNA Heavy Chain	cagggtgcaattgctggaaagcggcggtggcctgggtgcagccgggtggcagcctgcgtctgagctgcgcggcgctccg gattcaccttttcttcttactggatgaactgggtgcgccaggccccgggcaagggtctcgagtggggttcccgctatctcttc tgacgggttcttacacctactatgcggatagcgtgaaaggccgctttaccatcagccgcgataattcgaaaaacaccctgt atctgcaaatgaacagcctgcgtgcggaagatacggccgtgtattattgcgcgcgtgacccgttactctatgatctactctt acgggtgctggtgctttcgattactggggccaaggcaccctggtagctgttagctcagcctccaccaagggtccatcggt cttccccctggcaccctcctccaagagcacctctgggggacacagcggccctgggctgctgggtcaaggactacttccc

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		cgaaccggtgacggtgtcgtggaactcaggcgccctgaccagcggtgacacaccttcccggtgtcctacagtctc aggactctactccctcagcagcgtggtgacggtgacctccagcagcttgggcaccagacctacatctgcaacgtgaat cacaagcccagcaacaccaaggtggacaagagagttgagcccaaatcttgtgacaaaactcacacatgccaccgtg cccagcacctgaagcagcggggggaccgtcagtcctctcttcccccaaaacccaaggacacctcatgatctcccg gacctctgaggtcacatgctggtggtggacgtgagccacgaagacctgaggtcaagtccaactggtacgtggacg gctgtggaggtgcataatgccaagacaaagccggggaggagcagtagacaacagcagcgtaccgggtggtcagcgtcct cacgctcctgcaccaggactggctgaatggcaaggagtagaagtgcaaggtctccaaacaaagccctccagcccca tcgagaaaaccatctccaaagccaaagggcagcccgagaaaccacaggtgtacacctgcccccatccggggagga gatgaccaagaaccaggtcagcctgacctgctggtcaaaggcttctatcccagcgacatcgccgtggagtgaggaga gcaatgggcagccggagaacaactacaagaccagcctcccggtgctggactccgacggtccttctctctacagca agctcaccgtggacaagagcaggtggcagcaggggaacgtcttctcatgctccgtgatgcatgaggctctgcacaacc actacacgcagaagagcctctccctgtctccgggtaaa
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 145	LCDR3 (Combined)	QQIWTVPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 145	LCDR3 (Kabat)	QQIWTVPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 146	LCDR3 (Chothia)	IWTVPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 145	LCDR3 (IMGT)	QQIWTVPRT
SEQ ID NO: 147	VL	DIQMTQSPSSLSASVGRVTITCRASQGISSYLAWYQQKPKGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQIWTVPRTFGQGTKV EIK
SEQ ID NO: 148	DNA VL	gatatccagatgacccagagcccagcagcctgagcgcagcgtggcgatcgcgtagcattacctgcagagcca gccagggtatttcttcttacctggcttggtaccagcagaaacggggcaagcgccgaaactattaatctacactgcttcta ctctgcaaaagcggcgtgccgagccgcttagcggcagcggatccggcaccgatttcacctgaccattagctctctgca accggaagactttgcgacctattatggcagcagatctggactgttccgcgtacctttggccagggcacgaaagtggaaa ttaaa
SEQ ID NO: 149	Light Chain	DIQMTQSPSSLSASVGRVTITCRASQGISSYLAWYQQKPKGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQIWTVPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVCLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKDSSTLSLSTLTLKADYEKHKVYACEVTHQGLSSPVV KSFNRGEC
SEQ ID NO: 150	DNA Light Chain	gatatccagatgacccagagcccagcagcctgagcgcagcgtggcgatcgcgtagcattacctgcagagcca gccagggtatttcttcttacctggcttggtaccagcagaaacggggcaagcgccgaaactattaatctacactgcttcta ctctgcaaaagcggcgtgccgagccgcttagcggcagcggatccggcaccgatttcacctgaccattagctctctgca accggaagactttgcgacctattatggcagcagatctggactgttccgcgtacctttggccagggcacgaaagtggaaa ttaaacgtacggtggcgcctcccagcgtgttcattcttccccccagcagcagcgtgaagagcggcacccggcagc gtggtgtgctgtgaacaactctacccccgggaggccaaaggtgcagtggaaggtggacaacgcctgcagagcg gcaacagccaggaaagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcacccctgacctga gcaaggccgactacgagaagcacaaggtgtacgcctgcgaggtgacccaccagggtgtccagccccgtgacca agagcttcaaccggggcgagtg

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
XX08_ LALA		
SEQ ID NO: 4	HCDR1 (Combined)	GFTFNTHYIH
SEQ ID NO: 151	HCDR2 (Combined)	SIGGQGGMTLYADSVKG
SEQ ID NO: 6	HCDR3 (Combined)	ERGVVYHMFDP
SEQ ID NO: 7	HCDR1 (Kabat)	THYIH
SEQ ID NO: 151	HCDR2 (Kabat)	SIGGQGGMTLYADSVKG
SEQ ID NO: 6	HCDR3 (Kabat)	ERGVVYHMFDP
SEQ ID NO: 8	HCDR1 (Chothia)	GFTFNTH
SEQ ID NO: 152	HCDR2 (Chothia)	GGQGGM
SEQ ID NO: 6	HCDR3 (Chothia)	ERGVVYHMFDP
SEQ ID NO: 10	HCDR1 (IMGT)	GFTFNTHY
SEQ ID NO: 153	HCDR2 (IMGT)	IGGQGGMT
SEQ ID NO: 12	HCDR3 (IMGT)	ARERGVVYHMFDP
SEQ ID NO: 154	VH	EVQLLES GGLVQPGGSLRLSCAASGFTFNTHYIHWRQAPGKGLEWVSSI GGQGGMTLYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARER GYVYHMFDPWGQGLTVTVSS
SEQ ID NO: 155	DNA VH	gagggtgcaattgctggaagcgggcggtggcctggtgcagcggggtggcagcctgcgtctgagctgcgcgggcgctccg gattcacctttaacactcattacatccattgggtgcgccaggcccccggaagggtctcgagtggttttccctctatcggtg gtcagggcggtatgactctgtatgcggatagcgtgaaaggccgctttaccatcagccgcgataattcgaaaaacacct gtatctgcaaatgaacagcctgcgtgcggaagatacggccgtgtattatgcgcgcggtgaacgtggttacgtttactacc atatgttcgatccgtggggccaaggcacctcggtgactgttagctca
SEQ ID NO: 156	Heavy Chain	EVQLLES GGLVQPGGSLRLSCAASGFTFNTHYIHWRQAPGKGLEWVSSI GGQGGMTLYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARER GYVYHMFDPWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLV KDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTY ICNVN HKPSNTKVDKRV EPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDT LMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLT V L HQD W L N G E Y K C K V S N K A L P A I E K T I S K A G Q P R E P Q V Y T L P P S R E E M T K N Q V S L T C L V K G F Y P S D I A V E W E S N G Q P E N N Y K T T P P V L D S D G S F F L Y S K L T V D K S R W Q Q G N V F S C S V M H E A L H N H Y T Q K S L S L S P G K
SEQ ID NO: 157	DNA Heavy Chain	gagggtgcaattgctggaagcgggcggtggcctggtgcagcggggtggcagcctgcgtctgagctgcgcgggcgctccg gattcacctttaacactcattacatccattgggtgcgccaggcccccggaagggtctcgagtggttttccctctatcggtg gtcagggcggtatgactctgtatgcggatagcgtgaaaggccgctttaccatcagccgcgataattcgaaaaacacct gtatctgcaaatgaacagcctgcgtgcggaagatacggccgtgtattatgcgcgcggtgaacgtggttacgtttactacc atatgttcgatccgtggggccaaggcacctcggtgactgttagctcagcctccaccaagggtccatcggtcttccccctg gcacctctccaagagcacctctgggggcacagcgccctgggctgcctgggtcaaggactacttcccgaaacgggt gacgggtgctggaactcaggcgccctgaccagcgcggtgcacaccttccccggctgtcctacagtcctcaggactcta ctccctcagcagcgtggtgacctgcctccagcagcttgggcacccagacctacatctgcaacgtgaatcacaaaggc cagcaacaccaaagggtggacaagagagttgagcccaaatcttgtagacaaaactcacacatgccaccggtgccagcac ctgaagcagcgggggacgtcagtcctcctcttcccccaaaacccaaggacacctcatgatctcccggaacctgtg agggtcacatgcgtggtggtggagctgagccacgaagacctgagggtcaagttcaactggtacgtggacggcggtggag gtgcataatgccaaagacaaagccgggagagcagtagcaacagcagctacccgggtggtcagcgtcctcaccgtcct gcaccaggactggctgaatggcaaggagtagcaagtgcaaggtctccaacaaagccctccagcccccatcgagaaa accatctccaaagccaaagggcagccccgagaaccacaggtgtacacctgcccccatccccggaggagatgacca agaaccagggtcagcctgacctgcctggtaaaaggcttctatcccagcgacatcgccgtggagtgaggagacaaatggg cagccggagaacaactacaagaccacgcctccccgtgctggactccgacggctccttcttctctacagcaagctcaac gtggacaagagcaggtggcagcaggggaacgtcttctcatgctccgtgatgcatgaggtctgcacaaccactacacg cagaagagcctctccctgtctccgggtaaa

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 17	LCDR1 (Combined)	RASQSITRNYLA
SEQ ID NO: 18	LCDR2 (Combined)	GASSRAT
SEQ ID NO: 19	LCDR3 (Combined)	QQHSMYPRT
SEQ ID NO: 17	LCDR1 (Kabat)	RASQSITRNYLA
SEQ ID NO: 18	LCDR2 (Kabat)	GASSRAT
SEQ ID NO: 19	LCDR3 (Kabat)	QQHSMYPRT
SEQ ID NO: 20	LCDR1 (Chothia)	SQSITRNY
SEQ ID NO: 21	LCDR2 (Chothia)	GAS
SEQ ID NO: 22	LCDR3 (Chothia)	HSMYPR
SEQ ID NO: 23	LCDR1 (IMGT)	QSITRNY
SEQ ID NO: 21	LCDR2 (IMGT)	GAS
SEQ ID NO: 19	LCDR3 (IMGT)	QQHSMYPRT
SEQ ID NO: 24	VL	DIVLTQSPATLSLSPGERATLSCRASQSITRNYLAWYQQKPGQAPRLLIYGA SSRATGIPARFSGSGGTDFLTLSISLEPEDFAVYYCQQHSMYPRTFGQGTK VEIK
SEQ ID NO: 25	DNA VL	gat atcgtgctgacccagagcccgcgacccctgagcctgagcccggtgaacgtgccaccctgagctgcagagcga gccagtctatcactcgtaactacctggcttggtaccagcagaaacggggccaggcccgctctattaatctacggtgct tcttctcgtgcgacccggcattccggcgcggtttageggcagcggatccggcaccgatttcaccctgaccattagcagcct ggaaccggaagactttgcgggtgtattattgccagcagcattctatgtaccgcgtacctttggccagggcacgaaagttag aaattaaa
SEQ ID NO: 26	Light Chain	DIVLTQSPATLSLSPGERATLSCRASQSITRNYLAWYQQKPGQAPRLLIYGA SSRATGIPARFSGSGGTDFLTLSISLEPEDFAVYYCQQHSMYPRTFGQGTK VEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNAL QSGNSQESVTEQDSKDSSTLSLTLSKADYEKHKVYACEVTHQGLSSPV TKSFNRGEC
SEQ ID NO: 27	DNA Light Chain	gat atcgtgctgacccagagcccgcgacccctgagcctgagcccggtgaacgtgccaccctgagctgcagagcga gccagtctatcactcgtaactacctggcttggtaccagcagaaacggggccaggcccgctctattaatctacggtgct tcttctcgtgcgacccggcattccggcgcggtttageggcagcggatccggcaccgatttcaccctgaccattagcagcct ggaaccggaagactttgcgggtgtattattgccagcagcattctatgtaccgcgtacctttggccagggcacgaaagttag aaattaaacgtacggtggccgctcccagcgtgttcattccccccagcgacgagcagctgaagagcggcaccgcc agcgtggtgtgctcgtgaacaacttctacccccgggaggccaaggtgcagtggaaggtggacaacgccctgcaga gctggcaacagccaggaaagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccctgaccc tgagcaaggccgactacgagaagcacaaggtgtacgcctgcgaggtgacccaccagggcctgtccagcccgtagc caagagcttcaacggggcgaggtgt
XX08_ DAPA		
SEQ ID NO: 4	HCDR1 (Combined)	GFTFNTHYIH
SEQ ID NO: 151	HCDR2 (Combined)	SIGGQGGMTLYADSVKG
SEQ ID NO: 6	HCDR3 (Combined)	ERGVVYHMFDP
SEQ ID NO: 7	HCDR1 (Kabat)	THYIH

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 151	HCDR2 (Kabat)	SIGGGGGMPLYADSVKG
SEQ ID NO: 6	HCDR3 (Kabat)	ERGVVYHMFDP
SEQ ID NO: 8	HCDR1 (Chothia)	GFTFNTH
SEQ ID NO: 152	HCDR2 (Chothia)	GGQGGM
SEQ ID NO: 6	HCDR3 (Chothia)	ERGVVYHMFDP
SEQ ID NO: 10	HCDR1 (IMGT)	GFTFNTHY
SEQ ID NO: 153	HCDR2 (IMGT)	IGGQGGMT
SEQ ID NO: 12	HCDR3 (IMGT)	ARERGVVYHMFDP
SEQ ID NO: 154	VH	EVQLLESGGGLVQPGGSLRLSCAASGFTFNTHYIHWRQAPGKGLEWVSSI GGQGGMTLYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARER GYVYHMFDPWGQGLVTVSS
SEQ ID NO: 158	DNA VH	gaagtgcagctcctggagtcgggtggcggactggtgcagcctggcggatcactgcggctgtcatgtgccgcgagcg ggtttactttcaacacccactacatccactgggtccgccaaagctcccgaaagggaactcgaatgggtgtcctccattggt ggacagggcgccgatgacctttacgcggatagcgtgaaggggaggttcaccatctccgcgacaacagcaagaaca ccctgtacctccaaatgaactcgcttcgggcccagggacactgccgtgtactattgcgcaagagagcggggctacgtgt actaccacatgttcgacccatggggacagggaaacgtggtcaccgtgtcctcc
SEQ ID NO: 159	Heavy Chain	EVQLLESGGGLVQPGGSLRLSCAASGFTFNTHYIHWRQAPGKGLEWVSSI GGQGGMTLYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARER GYVYHMFDPWGQGLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLV KDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTY ICNVNHKPSNTKVDKRVKPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDT LMI SRTPEVTCVVVAVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVLHQDLNGKEYKCKVSNKALAAP IEKTI SKAKGQPREPQVYT LPSPSREMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPVLDSD GSFPLYSKLTVDKSRWQQGNVFPSCSVMH EALHNHYTQKSLSLSPGK
SEQ ID NO: 160	DNA Heavy Chain	gaagtgcagctcctggagtcgggtggcggactggtgcagcctggcggatcactgcggctgtcatgtgccgcgagcg ggtttactttcaacacccactacatccactgggtccgccaaagctcccgaaagggaactcgaatgggtgtcctccattggt ggacagggcgccgatgacctttacgcggatagcgtgaaggggaggttcaccatctccgcgacaacagcaagaaca ccctgtacctccaaatgaactcgcttcgggcccagggacactgccgtgtactattgcgcaagagagcggggctacgtgt actaccacatgttcgacccatggggacagggaaacgtggtcaccgtgtcctccgcctccactaagggcccgctcagtggt cccgctggctccatcgtcgaagtcacactccggaggaaacgcagcactcggttgctgggtcaagggaactactccctga gccagtgaccgtgtcggtgaacagcggagccctgacttcggcgctgcacactttccgcggtgtcgcagtcctccggt ctgtactcccttcgtccgtggtcaccgtgcccgtcgctagcctgggacaccagacctacatctgcaacgtgaaccacaa gcccgtccaacaccaaagtgataaagcgggtggagccgaagtcctgcgat aagacacacacgtgcccgccatgtcca gcgcctgaattgcttggcggaccttcggtgttcctgttcccgccctaagcccaaggacaccttgatgat tagccggactcc cgaagtacacctgtgtggtgggtggcagtggtcccacgaggaccccgaggtcaagtttaattggtacgtggacggcgtcga agtgacaaacgccaagactaagccccgggagggaacagtaaacacagcacctaccgggtcggtgcgtccgtgctgaccgtg ctgcaccaggactggctgaatgggaaagagtacaagtgc aaagtgtccaacaaggccttgccgctcctatcgaaaaa actatcagcaaggctaaaggacagccgagggaacccccaaagtctacacccctgcccccttcacgcgaagagatgacca agaatcaagtgctcgctgacctgcctcgtaagggtattctacccctccgacattgcgggtggagtgggaggtccaacggcc agcccggagaacaactacaagactactccgcccgtgctggactccgacggcagcttcttctgtattccaagctgaccgt ggacaagtcccggtggcagcaaggaaacgtgttctcctgctcggtcatgcacgaagccctgcacaaccactatacgca gaagtcctgtccttgagccggggaaa
SEQ ID NO: 17	LCDR1 (Combined)	RASQSITRNYLA
SEQ ID NO: 18	LCDR2 (Combined)	GASSRAT
SEQ ID NO: 19	LCDR3 (Combined)	QQHSMYPRT
SEQ ID NO: 17	LCDR1 (Kabat)	RASQSITRNYLA
SEQ ID NO: 18	LCDR2 (Kabat)	GASSRAT

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 19	LCDR3 (Kabat)	QQHSMYPRT
SEQ ID NO: 20	LCDR1 (Chothia)	SQSITRNY
SEQ ID NO: 21	LCDR2 (Chothia)	GAS
SEQ ID NO: 22	LCDR3 (Chothia)	HSMYPR
SEQ ID NO: 23	LCDR1 (IMGT)	QSITRNY
SEQ ID NO: 21	LCDR2 (IMGT)	GAS
SEQ ID NO: 19	LCDR3 (IMGT)	QQHSMYPRT
SEQ ID NO: 24	VL	DIVLTQSPATLSLSPGERATLSCRASQSITRNYLAWYQQKPGQAPRLLIYGA SSRATGIPARFSGSGSGTDFTLTISSELEPEDFAVYYCQQHSMYPRTFGQGTK VEIK
SEQ ID NO: 110	DNA VL	gacatcgtgctgactcagtcacctcgactctgagcctgtcaccgggagaaacggggccacctctcttgcgcgcctcc caatccattactcggaactacctggcctggtatcagcagaagccaggacaggccctaggcttctgatctacggggcc agctcaagagcaactggcatcccggtcgtctctcgggttcgggaagcggcaccgacttcacctgacaatttcgtccc tcgaaccggaggatttcgcccgtgtactactgccaacagcactccatgtaccccggaaccttggggcagggaaaccaaagt cgagatcaag
SEQ ID NO: 26	Light Chain	DIVLTQSPATLSLSPGERATLSCRASQSITRNYLAWYQQKPGQAPRLLIYGA SSRATGIPARFSGSGSGTDFTLTISSELEPEDFAVYYCQQHSMYPRTFGQGTK VEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNAL QSGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPV TKSFNRGEC
SEQ ID NO: 111	DNA Light Chain	gacatcgtgctgactcagtcacctcgactctgagcctgtcaccgggagaaacggggccacctctcttgcgcgcctcc caatccattactcggaactacctggcctggtatcagcagaagccaggacaggccctaggcttctgatctacggggcc agctcaagagcaactggcatcccggtcgtctctcgggttcgggaagcggcaccgacttcacctgacaatttcgtccc tcgaaccggaggatttcgcccgtgtactactgccaacagcactccatgtaccccggaaccttggggcagggaaaccaaagt cgagatcaagcgtacgggtggcgcgtcccagcgtgttcatcttccccccagcgacgagcagctgaagagcggcaccg ccagcgtggtgtgcctgctgaacaacttctaccccgggaggccaaggtgcagtggaaggtggacaacgccttcgag agcggcaacagccaggagagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcacccctgacc ctgagcaaggccgactacgagaagcataaaggtgtacgcctgcgaggtgacccaccaggccctgtccagccccgtga ccaagagcttcaacaggggagagtc
XX08_ N30S_ DAPA		
SEQ ID NO: 112	HCDR1 (Combined)	GFTFSTHYIH
SEQ ID NO: 151	HCDR2 (Combined)	SIGGGGGMPLYADSVKG
SEQ ID NO: 6	HCDR3 (Combined)	ERGVVYHMFDP
SEQ ID NO: 7	HCDR1 (Kabat)	THYIH
SEQ ID NO: 151	HCDR2 (Kabat)	SIGGGGGMPLYADSVKG
SEQ ID NO: 6	HCDR3 (Kabat)	ERGVVYHMFDP
SEQ ID NO: 113	HCDR1 (Chothia)	GFTFSTH
SEQ ID NO: 152	HCDR2 (Chothia)	GGQGGM
SEQ ID NO: 6	HCDR3 (Chothia)	ERGVVYHMFDP

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 114	HCDR1 (IMGT)	GFTFSTHY
SEQ ID NO: 153	HCDR2 (IMGT)	IGGQGGMT
SEQ ID NO: 12	HCDR3 (IMGT)	ARERGVVYHMFDP
SEQ ID NO: 161	VH	EVQLLESGGGLVQPGGSLRLSCAASGFTFSTHYIHWRQAPGKGLEWVSSI GGQGGMTLYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVVYCARER GYVYYHMFDPWGQGLVTVSS
SEQ ID NO: 162	DNA VH	gaagtgcagctcctggagtcgggtggcggactggtgcagcctggcggatcactgcggctgtcatgtgccgcgagcg ggtttactttctccaccactacatccactgggtccgccaagctcccggaaagggaactcgaatgggtgtcctccattggt ggacagggcgccatgacctttacgcggatagcgtgaaggggaggttcaccatctccgcgacaacagcaagaaca ccctgtacctccaaatgaactcgcttcgggcccaggacactgccgtgtactattgcgcaagagagcggggctacgtgt actaccacatgttcgacccatggggacagggaacgctggtcaccgtgtcctcc
SEQ ID NO: 163	Heavy Chain	EVQLLESGGGLVQPGGSLRLSCAASGFTFSTHYIHWRQAPGKGLEWVSSI GGQGGMTLYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVVYCARER GYVYYHMFDPWGQGLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLV KDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTY ICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDT LMI SRTPEVTVVAVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVHLQDNLNGKEYKCKVSNKALAAPIEKTI SKAKGQPREPQVYT LPDSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSD GSFFLYSKLTVDKSRWQQGNVFPSCSVMEALHNHYTQKSLSLSPGK
SEQ ID NO: 164	DNA Heavy Chain	gaagtgcagctcctggagtcgggtggcggactggtgcagcctggcggatcactgcggctgtcatgtgccgcgagcg ggtttactttctccaccactacatccactgggtccgccaagctcccggaaagggaactcgaatgggtgtcctccattggt ggacagggcgccatgacctttacgcggatagcgtgaaggggaggttcaccatctccgcgacaacagcaagaaca ccctgtacctccaaatgaactcgcttcgggcccaggacactgccgtgtactattgcgcaagagagcggggctacgtgt actaccacatgttcgacccatggggacagggaacgctggtcaccgtgtcctccgcctccactaagggcccgctcagtggt cccgtggctccatcgtcgaagtcacacctccggaggaaaccgcagcactcgggttgctgggtcaaggactacttccctga gccagtgaccgtgtcgtggaacagcggagccctgacttcggcgctgcacacttttccgcggtgctgcagtcctccggt ctgtactccctttcgtccgtggtcaccgtgcccgtcgtctagcctgggacaccagacctacatctgcaacgtgaaccacaa gccgtccaaacaccaaagtggataagcgggtggagccgaagtcctgcgat aagacacacacgtgcccgccatgtcca gcgcctgaattgcttggcggaccttcctgtgtcctgttcccgcctaagcccaaggacaccttgatgat tagccggactcc cgaagtccactgtgtggtggtggcagtgctccacagaggaccccgaggtcaagtttaattggtacgtggacggcgtcga agtgacaaacgccaagactaagccccgggagggaacagtaacaacagcaccacccgggtcgtgtccgtgctgaccgtg ctgcaccaggactggctgaatgggaaagagtacaagtgc aaagtgtccaacaaggcccttgcccgctcctatcgaaaaa actatcagcaaggctaaaggacagccgagggaaccccccaagtcacacccctgcccccttcacgcgaagagatgacca agaatcaagtgtcgctgacctgcctcgtcaagggtattctacccctccgacattgcgggtggagtgggagtcacaaggcc agcccagaaactacaagactactccgcccgtgctggactccgacggcagcttcttccgtgattccaagctgaccgt ggacaagtcccgggtggcagcaaggaaacgtgttctcctgctcgggtcatgcacgaagccctgcacaaccactatacgca gaagtcctgtccttgagccccgggaaa
SEQ ID NO: 17	LCDR1 (Combined)	RASQSITRNYLA
SEQ ID NO: 18	LCDR2 (Combined)	GASSRAT
SEQ ID NO: 19	LCDR3 (Combined)	QQHSMYPRT
SEQ ID NO: 17	LCDR1 (Kabat)	RASQSITRNYLA
SEQ ID NO: 18	LCDR2 (Kabat)	GASSRAT
SEQ ID NO: 19	LCDR3 (Kabat)	QQHSMYPRT
SEQ ID NO: 20	LCDR1 (Chothia)	SQSITRNY
SEQ ID NO: 21	LCDR2 (Chothia)	GAS
SEQ ID NO: 22	LCDR3 (Chothia)	HSMYPR
SEQ ID NO: 23	LCDR1 (IMGT)	QSITRNY

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 21	LCDR2 (IMGT)	GAS
SEQ ID NO: 19	LCDR3 (IMGT)	QQHSMYPRT
SEQ ID NO: 24	VL	DIVLTQSPATLSLSPGERATLSCRASQSI TRNYLAWYQQKPGQAPRLLIYGA SSRATGIPARFSGSGSGTDFTLTISSELPEDFAVYYCQQHSMYPRTFGQGTK VEIK
SEQ ID NO: 110	DNA VL	gacatcgtgctgactcagtcacctgcgactctgagcctgtcaccgggagaaacggggccacctctcttgcgcgcctcc caatccattactcggaactacctggcctggtatcagcagaagccaggacaggccctaggcttctgatctacggggcc agctcaagagcaactggcatcccggtcgtctctcgggttcgggaagcggcaccgacttcacctgacaatttcgtccc tcgaaccgcgaggtattcgccgtgtactactgccaacagcactccatgtaccccgagcctttgggcagggaaccaaaagt cgagatcaag
SEQ ID NO: 26	Light Chain	DIVLTQSPATLSLSPGERATLSCRASQSI TRNYLAWYQQKPGQAPRLLIYGA SSRATGIPARFSGSGSGTDFTLTISSELPEDFAVYYCQQHSMYPRTFGQGTK VEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNAL QSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPV TKSFNRGEC
SEQ ID NO: 111	DNA Light Chain	gacatcgtgctgactcagtcacctgcgactctgagcctgtcaccgggagaaacggggccacctctcttgcgcgcctcc caatccattactcggaactacctggcctggtatcagcagaagccaggacaggccctaggcttctgatctacggggcc agctcaagagcaactggcatcccggtcgtctctcgggttcgggaagcggcaccgacttcacctgacaatttcgtccc tcgaaccgcgaggtattcgccgtgtactactgccaacagcactccatgtaccccgagcctttgggcagggaaccaaaagt cgagatcaagcgtacgggtggcgcctcccagcgtgttcatcttccccccagcagcagcagctgaagagcggcaccg ccagcgtggtgtgcctgctgaacaacttctaccccgggaggccaaggtgcagtggaaggtggacaacgccttgag agcggcaacagccaggagagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccttgacc ctgagcaaggccgactacgagaagcataaaggtgtacgcctgcgaggtgaccaccaggggcctgtccagcccgctga ccaagagcttcaacaggggagtgctg
XX08_ N30Q_ DAPA		
SEQ ID NO: 165	HCDR1 (Combined)	GFTFQTHYIH
SEQ ID NO: 151	HCDR2 (Combined)	SIGGGGGMTLYADSVKG
SEQ ID NO: 6	HCDR3 (Combined)	ERGVVYHMFDP
SEQ ID NO: 7	HCDR1 (Kabat)	THYIH
SEQ ID NO: 151	HCDR2 (Kabat)	SIGGGGGMTLYADSVKG
SEQ ID NO: 6	HCDR3 (Kabat)	ERGVVYHMFDP
SEQ ID NO: 166	HCDR1 (Chothia)	GFTFQTH
SEQ ID NO: 152	HCDR2 (Chothia)	GGQGGM
SEQ ID NO: 6	HCDR3 (Chothia)	ERGVVYHMFDP
SEQ ID NO: 167	HCDR1 (IMGT)	GFTFQTHY
SEQ ID NO: 153	HCDR2 (IMGT)	IGGQGGMT
SEQ ID NO: 12	HCDR3 (IMGT)	ARERGVVYHMFDP
SEQ ID NO: 168	VH	EVQLLESGGGLVQPGGSLRLSCAASGFTFQTHYIHWRQAPGKGLEWVSSI GGQGGMTLYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARER GYVYHMFDPWGQGLVTVSS
SEQ ID NO: 169	DNA VH	gaagtgcagctcctggagtcgggtggcggactggtgcagcctggcggatcactgcggctgtcatgtgcgcgagcg ggtttactttccagaccactacatccactgggtccgccaagctcccggaaagggactcgaatgggtgtcctccattggt

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		ggacagggcgccatgacctttacgcggatagcgtgaaggggaggttcaccatctccgcgacaacagcaagaaca ccctgtacctccaaatgaactcgcttcgggcccaggacactgccgtgtactattgcgcaagagagcggggctacgtgt actaccacatgttcgacccatggggacaggaacgctgggtcaccgtgtcctcc
SEQ ID NO: 170	Heavy Chain	EVQLLESQGGGLVQPGGSLRLSCAASGFTFQTHYIHWVRQAPGKGLEWVSSI GGQGGMTLYADSVKGRFTISRDNKNTLYLQMNSLRADTAVVYCARER GYVYHMFDPWQGGLVTVSSASTKGPSVFPLAPSSKSTSGGTAAALGCLV KDYFFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTY ICNVNHKPSNTKVDKRVPEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDT LMSIRTPEVTCVVAVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVLHQDLNGKEYKCKVSNKALAAPIEKTI SKAKGQPREPQVYT LPSPREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSD GSFFLYSKLTVDKSRWQQGNVPSFCSVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 171	DNA Heavy Chain	gaagtgcagctcctggagtcgggtggcggaactgggtgcagcctggcggaatcactgcggctgtcatgtgccgcgagcg gggttactttccagaccactacatccactgggtccgccaagctcccgaaagggaactcgaatgggtgtcctccattgggt ggacagggcgccatgacctttacgcggatagcgtgaaggggaggttcaccatctccgcgacaacagcaagaaca cctgtacctccaaatgaactcgcttcgggcccaggacactgccgtgtactattgcgcaagagagcggggctacgtgt actaccacatgttcgacccatggggacaggaacgctgggtcaccgtgtcctccgcctccactaagggcccgctcagtggt cccgctggctccatcgtcgaagtccacctccggaggaaaccgcagcactcgggttgctgggtcaaggactacttccctga gccagtgcacgtgtcggtgaacagcggagccctgacttccggcgtgcacacttttccgcgggtgctgcagtcctccgggt ctgtactcccttccgtccgtgggtcaccgtgcgtcgtctagcctgggcaaccagacctacatctgcaacgtgaaccacaa gccgtccacacacaaagtggataagcgggtggagccgaagtcctgcgat aagacacacacgtgcccgccatgtcca ggcctcgaattgcttggcggaaccttccgtgttccctgttcccgccctaagcccagggaaccttgatgatagccggactcc cgaagtccactgtgtgggtgggtggcagtgctccacagaggaccccgaggtcaagtttaattggtagctggacggcgtcga agtgcaaacgcgaagactaagccccgggagggaacagtaaacacagcactaccgggtcggtccgtgctgacccgtg ctgcaccaggactggctgaatgggaaagagtaaacgtgcaaaagtgtccaacaaggccttggccgctcctatcgaaaaa actatcagcaaggctaaaggacagccgagggaaacccccaaagtctacacccctgcccccttcacgcgaagagatgacca agaatcaagtgtcgctgacctgcctcgtaagggtattctacccctccgacattgcgggtggagtgaggagttccaacggcc agcccagaaactacaagactactccgcctgctggactccgacggcagcttcttccgtattccaagctgacctgt ggacaagtcccggtggcagcaaggaaacgtgttctcctgctcggtcatgcacgaagccctgcacaaccactatacgca gaagtccctgtccttgagcccggggaaa
SEQ ID NO: 17	LCDR1 (Combined)	RASQSITRNYLA
SEQ ID NO: 18	LCDR2 (Combined)	GASSRAT
SEQ ID NO: 19	LCDR3 (Combined)	QQHSMYPRT
SEQ ID NO: 17	LCDR1 (Kabat)	RASQSITRNYLA
SEQ ID NO: 18	LCDR2 (Kabat)	GASSRAT
SEQ ID NO: 19	LCDR3 (Kabat)	QQHSMYPRT
SEQ ID NO: 20	LCDR1 (Chothia)	SQSITRNY
SEQ ID NO: 21	LCDR2 (Chothia)	GAS
SEQ ID NO: 22	LCDR3 (Chothia)	HSMYPR
SEQ ID NO: 23	LCDR1 (IMGT)	QSITRNY
SEQ ID NO: 21	LCDR2 (IMGT)	GAS
SEQ ID NO: 19	LCDR3 (IMGT)	QQHSMYPRT
SEQ ID NO: 24	VL	DIVLTQSPATLSLSPGERATLSCRASQSITRNYLAWYQQKPGQAPRLLIYGA SSRATGIPARFSGSGSDFTLTISSELEPEDFAVYYCQQHSMYPRTFGQGTK VEIK
SEQ ID NO: 110	DNA VL	gacatcggtgctgactcagtcacctcgactctgagcctgtcaccgggagaaacggggccacctctcttgcgcgcctcc caatccattactcggaactacctggcctggatcagcagaagccaggacaggccctaggcttctgatctacggggcc agctcaagagcaactggcatcccggtcgcttctccgggtcggaagcggcaccgacttccacctgacaatttcgtccc tcgaacccgaggatttcggcgtgtactactgccaacagcaactccatgtaccccggaaccttggggcagggaacccaagt cgagatcaag

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 26	Light Chain	DIVLTQSPATLSLSPGERATLSCRASQSI TRNYLAWYQQKPGQAPRLLIYGA SSRATGIPARFSGSGSGTDFTLTISSELPEDFAVYQCQHSMPRTFGQGTK VEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNAL QSGNSQESVTEQDSKDSYLSSTLTLSKADYEKHKVYACEVTHQGLSSPV TKSFNRGEC
SEQ ID NO: 111	DNA Light Chain	gacatcgtgctgactcagtcacctgctgactctgagcctgtcaccgggagaaacggggccacctctcttgcgcgcctcc caatccattactcggaactacctggcctggtatcagcagaagccaggacaggccctaggcttctgatctacggggcc agctcaagagcaactggcatcccggtcgttctccggttcgggaagcggcaccgacttcacctgacaatttctgctcc tcgaaccccgaggatttcgccgtgtactactgccaaacagcactccatgtaccccccggaccttgggcaggggaaccaaagt cgagatcaagcgtacggtggcgcctcccagcgtgttcatcttccccccagcagcagcagctgaagagcggcaccg ccagcgtggtgtgctgctgaacaacttctacccccgggaggccaagtgagtggaagtggaacacgcctgcag agcggcaacagccaggagagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcacctgacc ctgagcaaggccgactacgagaagcataaaggtgtacgcctgaggtgaccaccaggccctgtccagccccgtga ccaagagcttcaacagggcgagtg
XX09_ LALA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 29	HCDR2 (Combined)	AISSDGSYTTYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 29	HCDR2 (Kabat)	AISSDGSYTTYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 33	HCDR2 (Chothia)	SSDGSY
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 35	HCDR2 (IMGT)	ISSDGSYT
SEQ ID NO: 36	HCDR3 (IMGT)	ARDYSMIYSYGAGAFDY
SEQ ID NO: 37	VH	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS AISSDGSYTTYADSVKGRFTISRDNKNTLYLQMNLSRAEDTAVIYCARD RYSMIYSYGAGAFDYWGQGLTVTVSS
SEQ ID NO: 38	DNA VH	caggtgcaattgctggaagcggcggtggcctggtgcagccgggtggcagcctgcgtctgagctgcgcggcgctccg gattcaccttttcttcttactggatgaactgggtgcgccaggccccgggcaagggtctcgagtggtttccgctatctcttc tgacgggttcttacacctactatgcggatagcgtgaaaggccgctttaccatcagccgcgataattcgaaaaacacctgt atctgcaaatgaacagcctgcgtgcggaagatacggccgtgtattattgcgcgcgtgaccgttactctatgatctactctt acggtgctggtgcttctcgattactggggccaaggcaccctggtgactgttagtca
SEQ ID NO: 39	Heavy Chain	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS AISSDGSYTTYADSVKGRFTISRDNKNTLYLQMNLSRAEDTAVIYCARD RYSMIYSYGAGAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSL GTQTYICNVNHKPSNTKVDKRVPEKSCDKHTHTCPPCPAPEAAGGSPVFLFP PKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREE QYNSTYRVVSVLTVQLQDNLNGKEYCKVSNKALPAPIEKTI SKAKGQPR EPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTP PVLDSGDSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSP GK

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 40	DNA Heavy Chain	caggtgcaattgctggaagcggcggtggcctggtgcagccgggtggcagcctgcgtctgagctgcgcggcgctccg gattcaccttttcttcttactggatgaactgggtgcgcagcccgccgggcaagggtctcgagtggtttccgctatctctt tgacgggtctttacacctactatgcggatagcgtgaaaggcgcgtttaccatcagccgcgataattcgaaaaacacctgt atctgcaaatgaacagcctgcgtgcggaagatacggcgtgtattattgcgcgcgtgacgttactctatgatctactctt acgggtgctgggtgcttctcgattactggggccaaggcaccctggtagctgttagctcagcctccaccaagggtccatcggt cttccccctggcaccctctccaagagcacctctgggggcacagcggccctgggtgcttgggtcaaggactactccc cgaaccggtgacgggtgctgctggaactcaggcgcctgaccagcggcgtgcacaccttcccggtgctctacagtcctc aggactctactccctcagcagcgtggtgaccgtgccctccagcagcttgggcaccagacctacatctgcaacgtgaat cacaagcccgcaacaccaaggtggacaagagagttgagcccaaatcttgtgacaaaactcacacatgcccaccgtg cccagcacctgaagcagcggggggaccgtcagctcttctcttcccccaaaacccaaaggacacctcatgatctcccg gacccctgaggtcacatgcgtggtggtggacgtgagccacgaagacctgaggtcaagttcaactggtacgtggacg gcgtggaggtgcataatgccaaagacaagccgcgggaggagcagtaacaacagcacgtaccgggtggtcagcgtcct cacctgctctgaccaggactggctgaatggcaaggagtaacagtgcaaggtctccaaacaaagccctccagccccc tcgagaaaaccatctccaaagccaaagggcagcccgagaccaggtgtacacctgcccccatcccgggagga gatgaccaagaaccaggtcagcctgacctgctggtcaaaggcttctatccagcgacatcgccgtggagtgggaga gcaatgggcagcggagaaactacaagaccacgcctcccggtgctggactccgacggctccttctctctacagca agctcacctgggacaagagcaggtggcagcaggggaacgtcttctcatgctccgtgatgcatgaggtctcgcacaacc actacacgcagaagacgtctcctctgctccgggtaaa
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 172	LCDR3 (Combined)	QQEWAKPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 172	LCDR3 (Kabat)	QQEWAKPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 173	LCDR3 (Chothia)	EWAKPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 172	LCDR3 (IMGT)	QQEWAKPRT
SEQ ID NO: 174	VL	DIQMTQSPSSLSASVGRVITITCRASQGISSYLAWYQQKPKGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQEWAKPRTFGQGTKV EIK
SEQ ID NO: 175	DNA VL	gatatccagatgaccagagcccgagcagcctgagcgcagcgtggcgatcgcgtagaccattacctgcagagcca gccaggggtatttcttcttacctggcttggtagcagcagaaacccgggcaaaagcgcgaaactattaatctacactgcttcta ctctgcaaaagcggcgtgcccagccgctttagcggcagcggatccggcaccgatttcacctgaccattagctctctgca accggaagactttgcgacctattattgccagcaggaatgggttaaaccgcgtacctttggccagggcacgaaagttaa attaaa
SEQ ID NO: 176	Light Chain	DIQMTQSPSSLSASVGRVITITCRASQGISSYLAWYQQKPKGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQEWAKPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKDSSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVIT KSFNRGEC
SEQ ID NO: 177	DNA Light Chain	gatatccagatgaccagagcccgagcagcctgagcgcagcgtggcgatcgcgtagaccattacctgcagagcca gccaggggtatttcttcttacctggcttggtagcagcagaaacccgggcaaaagcgcgaaactattaatctacactgcttcta ctctgcaaaagcggcgtgcccagccgctttagcggcagcggatccggcaccgatttcacctgaccattagctctctgca accggaagactttgcgacctattattgccagcaggaatgggttaaaccgcgtacctttggccagggcacgaaagttaa attaaacgtacgggtggcgcgtcccagcgtgttcatcttccccccagcgcagcagcagctgaagagcggcaccgcca cgtggtgctgctgtaacaaacttctacccccgggagggccaaggtgcagtggaaaggtggacaacgcccctgcagagc ggcaacagccaggaagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccctgacctg

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		agcaaggccgactacgagaagcacaaggtgtacgcctgagaggtgacccaccagggcctgtccagccccgtgacc aagagcttcaaccggggcgagtg
XX11_ LALA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 29	HCDR2 (Combined)	AISSDGSYTYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 29	HCDR2 (Kabat)	AISSDGSYTYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 33	HCDR2 (Chothia)	SSDGSY
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSY
SEQ ID NO: 35	HCDR2 (IMGT)	ISSDGSYT
SEQ ID NO: 36	HCDR3 (IMGT)	ARDYSMIYSYGAGAFDY
SEQ ID NO: 37	VH	QVQLLESggglvqpqgslrslscAASGFTFSSYWMNWVRQAPGKGLEWVS AISSDGSYTYADSVKGRFTISRDNskNTLYLQMNSLRAEDTAVYYCARD RYSMIYSYGAGAFDYWGQGLTVTVSS
SEQ ID NO: 38	DNA VH	caggtgcaattgctggaaagcggcggtggcctgggtgcagccgggtggcagcctgcgtctgagctgcgcggcgctccg gattcaccttttcttcttactggatgaactgggtgcgcagggcccgggcaagggtctcgagtggtttccgctatctcttc tgacgggtctttacacctactatgcggatagcgtgaaaggccgctttaccatcagccgcgataaattcgaaaaacacctgt atctgcaaatgaacagcctgcgtgcggaagatacggccgtgtattattgcgcgcgtgaccggttactctatgatctactctt acgggtgctggtgctttcgattactggggccaaggcacccctgggtgactgttagctca
SEQ ID NO: 39	Heavy Chain	QVQLLESggglvqpqgslrslscAASGFTFSSYWMNWVRQAPGKGLEWVS AISSDGSYTYADSVKGRFTISRDNskNTLYLQMNSLRAEDTAVYYCARD RYSMIYSYGAGAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAAL GCLVLDYFPEPVTWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSL GTQTYICNVNHKPSNTKVDKRVKPSCKDTHCTPCPAPAEAGGSPVFLFP PKPKDTLMISRTPEVTVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREE QYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI SKAKGQPR EPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTP PVLDSGDSFPLYSKLTVDKSRWQQGNVFS CSVMHEALHNHYTQKSLSLSP GK
SEQ ID NO: 40	DNA Heavy Chain	caggtgcaattgctggaaagcggcggtggcctgggtgcagccgggtggcagcctgcgtctgagctgcgcggcgctccg gattcaccttttcttcttactggatgaactgggtgcgcagggcccgggcaagggtctcgagtggtttccgctatctcttc tgacgggtctttacacctactatgcggatagcgtgaaaggccgctttaccatcagccgcgataaattcgaaaaacacctgt atctgcaaatgaacagcctgcgtgcggaagatacggccgtgtattattgcgcgcgtgaccggttactctatgatctactctt cgggtgctggtgctttcgattactggggccaaggcacccctgggtgactgttagctcagcctccaccaagggtccatcggt cttccccctggcaccctctccaagagcacctctgggggcacagcggccctgggctgcctgggtcaaggactacttccc cgaacccgtgacgggtgctggtggaactcaggcgccctgaccagcggcggtgcacaccttccccgggtgctctacagtcctc aggactctactccctcagcagcgtggtgaccggtgccctccagcagcttgggcaccacagacctacatctgcaacgtgaat cacaagcccgagcaacaccaaggtggacaagagagttgagcccaaatcttgtgacaaaactcacacatgccaccgtg cccagcacctgaagcagcggggggacggctcagctcttctcttccccccaaaacccaaggacacctcatgatctcccg gacctcgaggtcacatgcgtggtggtggacgtgagccacgaagacctgaggtcaagttcaactggtacgtggacg gcgtggaggtgcataatgccaagacaaagccgcgaggagcagtaacaacagcagctaccgggtggtcagcgtcct cacggtctctgcaccaggactggctgaatggcaaggagtacaaggtctccaacaaagccctcccagccccc tcgagaaaaccatctccaaagccaaaggcagcccgagaaacacaggtgtacacctgcccccatccgggaggga gatgaccaagaaccaggtcagcctgacctgcctggtcaaaggcttctatccagcgacatcgccgtggagtgaggaga

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		gcaatgggcagccggagaaactacaagaccacgcctcccgctgctggactccgacggctccttcttctctacagca agctcaccgtggacaagagcaggtggcagcaggggaacgtcttctctatgctccgtgatgcatgaggtctctgcacaacc actacacgcagaagagcctctccctgtctccgggtaaa
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 178	LCDR3 (Combined)	QQSWTRPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 178	LCDR3 (Kabat)	QQSWTRPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 179	LCDR3 (Chothia)	SWTRPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 178	LCDR3 (IMGT)	QQSWTRPRT
SEQ ID NO: 180	VL	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQSWTRPRTFGQGTKV EIK
SEQ ID NO: 181	DNA VL	gatatccagatgaccagagcccgagcagcctgagcgcagcgtggcgatccgctgaccattacctgcagagcca gccaggggtatttcttcttacctggcttggtagcagcagaaacgggcaaaagcgcgaaactattaatctacctgcttcta ctctgcaaaagcggcgtgccagccgctttagcggcagcggatccggcaccgatttcacctgaccattagctctctgca accggaagactttgcgacctattattgccagcagctcttggaactcgtccgcgtacctttggccagggcacgaaagtgtaaa ttaaa
SEQ ID NO: 182	Light Chain	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQSWTRPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID NO: 183	DNA Light Chain	Gatatccagatgaccagagcccgagcagcctgagcgcagcgtggcgatccgctgaccattacctgcagagcca gccaggggtatttcttcttacctggcttggtagcagcagaaacgggcaaaagcgcgaaactattaatctacctgcttcta ctctgcaaaagcggcgtgccagccgctttagcggcagcggatccggcaccgatttcacctgaccattagctctctgca accggaagactttgcgacctattattgccagcagctcttggaactcgtccgcgtacctttggccagggcacgaaagtgtaaa ttaaacgtacgggtggcgctcccagcgtgttcatcttccccccagcagcagcagctgaagagcggcaccgcccagc gtgggtgtcctgctgaacaaacttctaccccgggagggcgaaggtgcagtggagggtggacaacgcccctgcagagcg gcaacagccaggaaagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccctgacctga gcaagggccgactacgagaagcacaaggtgtacgcctgcgaggtgacccaccagggcctgtccagccccgtgacca agagcttcaaccggggcgagtg
XX12_ LALA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 29	HCDR2 (Combined)	AISSDGSYTYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 29	HCDR2 (Kabat)	AISSDGSYTYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 33	HCDR2 (Chothia)	SSDGSY
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 35	HCDR2 (IMGT)	ISSDGSYT
SEQ ID NO: 36	HCDR3 (IMGT)	ARDRYSMIYSYGAGAFDY
SEQ ID NO: 37	VH	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS AISSDGSYTYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARD RYSMIYSYGAGAFDYWGQGLVTVSS
SEQ ID NO: 38	DNA VH	caggtgcaattgctggaaagcggcggtggcctgggtgcagccgggtggcagcctgcgtctgagctgcgcggcgctccg gattcaccttttcttcttactggatgaactgggtgcgcaggcccccgggcaagggtctcgagtggtttccgctatctcttc tgacgggtctctacacctactatgcggatagcgtgaaagggcgctttaccatcagccgcgataattcgaaaaacaccctgt atctgcaaatgaacagcctgcgtgcggaagatacggccgtgtattattgcgcgctgacccgttactctatgatctactctt acgggtgctggtgctttcgattactggggccaaggcacccctggtgactgttagctca
SEQ ID NO: 39	Heavy Chain	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS AISSDGSYTYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARD RYSMIYSYGAGAFDYWGQGLVTVSSASTKGPSVFP LAPSSKSTSGGTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVPSSSL GTQTYICNVNHKPSNTKVDKRVPEPKSCDKTHTCPPCPAPEAAGGPSVFLFP PKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREE QYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR EPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPP PVLDSGDSFFLYSLKLTVDKSRWQQGNVPSCSVMHEALHNHYTQKSLSLSP GK
SEQ ID NO: 40	DNA Heavy Chain	caggtgcaattgctggaaagcggcggtggcctgggtgcagccgggtggcagcctgcgtctgagctgcgcggcgctccg gattcaccttttcttcttactggatgaactgggtgcgcaggcccccgggcaagggtctcgagtggtttccgctatctcttc tgacgggtctctacacctactatgcggatagcgtgaaagggcgctttaccatcagccgcgataattcgaaaaacaccctgt atctgcaaatgaacagcctgcgtgcggaagatacggccgtgtattattgcgcgctgacccgttactctatgatctactctt acgggtgctggtgctttcgattactggggccaaggcacccctggtgactgttagctcagccctccaccaagggtccatcggt cttccccctggcaccctctccaagagcacctctgggggcaacagccctgggctgcctggtcaaggactacttccc cgaacccgtgacgggtgctgctggaactcaggcgccctgaccagcggtgacacacttccccggctgctctacagtcctc aggactctactccctcagcagcgtggtgacccgtgccctccagcagcttgggcaaccagacctacatctgcaacgtgaat cacaagcccagcaacaccaaggtggacaagagagttgagcccaaatcttgtgacaaaactcacacatgccaccctg cccagcacctgaagcagcggggggaccgtcagctcttctcttcccccaaaacccaaggacacctcatgatctcccg gaccctgaggtcacatgcgtggtggtggacgtgagccacgaagacctgaggtcaagttcaactggtacgtggacg gcgtggaggtgcataatgccaagacaaagccgaggagagcagtaaacagcacgtaccgggtggtcagcgtcct caccgtctctgcaccaggactggctgaatggcaaggagtacaagtgcaagggtctccaaacaaagccctcccagcccca tcgagaaaaccatctccaaagccaaagggcagcccccagagaccacaggtgtacacccctgccccatcccgggagga gatgaccaagaaccaggtcagcctgacctgcctggtcaaaggctctctatcccagcgacatcgccgtggagtgaggaga gcaatgggcagccggagaaactacaagaccacgcctcccggtgctggactccgacggctccttcttctctacagca agctcaccgtggacaagagcaggtggcagcaggggaacgtctctcatgctcgtgatgcatgaggtctgcacaacc actacacgcagaagacgtctctcctgctctccgggtaaa
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 184	LCDR3 (Combined)	QQIWMAPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 184	LCDR3 (Kabat)	QQIWMAPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 185	LCDR3 (Chothia)	IWMAPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 184	LCDR3 (IMGT)	QQIWMAPRT
SEQ ID NO: 186	VL	DIQMTQSPSSLSASVGDRVTITCRASQGISSYLAWYQQKPGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQIWMAPRTFGQGTKV EIK
SEQ ID NO: 187	DNA VL	gatatccagatgaccagagcccgagcagcctgagcgccagcgtggcgcatcgcgtagaccattacctgcagagcca gccaggggtatttcttcttacctggccttggtaccagcagaaacggggcaagcgccgaaactattaatctacactgcttcta ctctgcaaagcggcgtgccgagccgctttagcggcagcggatccggcaccgatttcaccctgaccattagctctctgca accggaagactttgcgacctattattgccagcagatctggatggctccgctacctttggccagggcacgaaagtggaa attaaa
SEQ ID NO: 188	Light Chain	DIQMTQSPSSLSASVGDRVTITCRASQGISSYLAWYQQKPGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQIWMAPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSPVT KSFNRGEC
SEQ ID NO: 189	DNA Light Chain	gatatccagatgaccagagcccgagcagcctgagcgccagcgtggcgcatcgcgtagaccattacctgcagagcca gccaggggtatttcttcttacctggccttggtaccagcagaaacggggcaagcgccgaaactattaatctacactgcttcta ctctgcaaagcggcgtgccgagccgctttagcggcagcggatccggcaccgatttcaccctgaccattagctctctgca accggaagactttgcgacctattattgccagcagatctggatggctccgctacctttggccagggcacgaaagtggaa attaaacgtacgggtggcgctcccagcgtgtcatcttccccccagcgacgagcagctgaagagcggcaccgccag cgtgggtgtgcctgctgaacaacttctaccccgggaggccaaaggtgcagtggaaggtggacaacgccctgcagagc ggcaacagccaggaagcgtcaccgagcaggacagcaaggaactccacctacagcctgagcagcaccctgacctg agcaaggccgactacgagaagcacaaggtgtacgctgcgaggtgaccaccaggggcctgtccagccccgtgacc aagagcttcaacggggcgagtgt
XX13_ LALA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 190	HCDR2 (Combined)	AISSKGSYTTYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 190	HCDR2 (Kabat)	AISSKGSYTTYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 191	HCDR2 (Chothia)	SSKGSY

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 192	HCDR2 (IMGT)	ISSKGSYT
SEQ ID NO: 36	HCDR3 (IMGT)	ARDRYSMIYSYGAGAFDY
SEQ ID NO: 193	VH	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS AISSKGSYTYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARD RYSMIYSYGAGAFDYWGQGLTVTVSS
SEQ ID NO: 194	DNA VH	caagttcagctccttgagctctggggggggcctgggtgcaacctgggggctctctgcggctttcatgtgcggcctcaggggt tcactttcagctcatactggatgaattgggtacgccaaagctccaggcaaggactcgaatgggttaagcgctatatccagc aaagggagctatacctattacgcggattccgttaagggcaggttcactatatcccgcgacaactccaaaaatactttgtat ctgcaaatgaattccctccgagccgaagataccgcagtatattactgtgcgagggacaggtactccatgatttacagcta cgggtccgggtgctttcgattattggggacaggggacacttgtgaccgtcagttct
SEQ ID NO: 195	Heavy Chain	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS AISSKGSYTYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARD RYSMIYSYGAGAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSL GTQTYICNVNHKPSNTKVDKRVPEKSCDKTHTCPPCPAPEAAGGPSVFLFP PKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREE QYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI SKAKGQPR EPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTP PVLDSGDSFPLYSKLTVDKSRWQQGNVFSQVMHEALHNHYTQKSLSLSP GK
SEQ ID NO: 196	DNA Heavy Chain	caagttcagctccttgagctctggggggggcctgggtgcaacctgggggctctctgcggctttcatgtgcggcctcaggggt tcactttcagctcatactggatgaattgggtacgccaaagctccaggcaaggactcgaatgggttaagcgctatatccagc aaagggagctatacctattacgcggattccgttaagggcaggttcactatatcccgcgacaactccaaaaatactttgtat ctgcaaatgaattccctccgagccgaagataccgcagtatattactgtgcgagggacaggtactccatgatttacagcta cgggtccgggtgctttcgattattggggacaggggacacttgtgaccgtcagttctgcaagtaccaaagggcctctgtttt cccatggctccctcatccaagagcacagagtgaggccaccgcgcgtgggatgcttgtgaaagactatttcccgga gcccgtgaccgttagctggaacagcggcgtcttaccagtgccgttcacacattccagctgtttgcagtcacccgggc tctactctctctcatccgtgggtcaccgtgcccgtctagtcttggggcaccagacctacatctgtaacgtaaatcacaaacct agtaatactaaggtggacaagcaggttgaaacgaagagctgtgataagacacatacttgtccaccatgtccggcacc gaggcagcggggggccaggtgttttctcttccacccaaagccaaagacacattgatgatctcacgaacccagag gttaacttgtgtcgtggttagatgaagccatgaggacccgaagtaagttcaattggtatgttgacggtgttagaggtgcac aatgccccaaactaaacccgggaggagcaatacaactcaacttacagagtcgtatccgtgctgaccgttttgaccagg attggttgaatggttaaggaatacaaatgaagtgaagcaataaagctctccagcgcctatcgagaagaccattagcaa agccaaaggtcaacccaggggaacccaggtatatacgctgccaccctcaagggaagagatgacaaagaatcaagtgt cactgacgtgtcttgtcaaggggtttctatcctagcgacattgcggtggaatgggagtcacaaatgggcaacccgagaacaa ctacaagactactctccgctcctggacagcgacggctccttcttctctgtatagtaaacgtgacgctcgataaaagtaggtg gcagcaggggaatgtctttagttgctctgtcatgcatgaggcgtccataaaccctacacccaaaaatctttgagcttgag ccttgga
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 134	LCDR3 (Combined)	QQTWRKPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 134	LCDR3 (Kabat)	QQTWRKPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 135	LCDR3 (Chothia)	TWRKPRT

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 134	LCDR3 (IMGT)	QQTWRKPRT
SEQ ID NO: 136	VL	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPKGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQTWRKPRTFGQGTKV EIK
SEQ ID NO: 197	DNA VL	gacattcaaatgacacaaagtcggtccagtccttagtgcttctgtggcgcatagggtcaccatcacttgtcgggcgtctcag gggatcagctcttacttggcatgggtatcaacaaaagccaggaaaagcacctaaattgcttatttatacagcggtccaccctc cagtcaggagtgccctagtaggttctcagggtctgggtccggtactgacttcacgctgactatatacaagcttgcaaccga agattttgcaacatactactgccaacagacatggaggaagccaagaactttcggtcagggaaacgaaagttgagataaa g
SEQ ID NO: 138	Light Chain	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPKGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQTWRKPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSPVT KSFNRGEC
SEQ ID NO: 198	DNA Light Chain	gacattcaaatgacacaaagtcggtccagtccttagtgcttctgtggcgcatagggtcaccatcacttgtcgggcgtctcag gggatcagctcttacttggcatgggtatcaacaaaagccaggaaaagcacctaaattgcttatttatacagcggtccaccctc cagtcaggagtgccctagtaggttctcagggtctgggtccggtactgacttcacgctgactatatacaagcttgcaaccga agattttgcaacatactactgccaacagacatggaggaagccaagaactttcggtcagggaaacgaaagttgagataaa ggcactgtgcgcagacacttccgtgttcattttcccgcttccgacgagcagcttaaatcagggaccgcgagtggtgtgttg cttgcttaataacttttaccacgggaagccaagttcagtggaaggtggacaatgcactccaaagcgggaatagtcag gagtcagttactgagcaagatagtaaagactctacttactctttgagttcaaccttgaccctctcaaaagcggactacgag aagcataaagtgtacgcctgcgaggtgacgcatcaaggtttgtcttccccgggtacgaagtcctttaataggggggaatg t
XX14_ LALA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 190	HCDR2 (Combined)	AISSKGSYTYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 190	HCDR2 (Kabat)	AISSKGSYTYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 191	HCDR2 (Chothia)	SSKGSY
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 192	HCDR2 (IMGT)	ISSKGSYT
SEQ ID NO: 36	HCDR3 (IMGT)	ARDRYSMIYSYGAGAFDY
SEQ ID NO: 193	VH	QVQLLESQGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS AISSKGSYTYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARD RYSMIYSYGAGAFDYWGQGLTVTVSS

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 194	DNA VH	caagttcagctccttgagctctgggggggctgggtgcaacctgggggctctctgcggctttcatgtgcggcctcaggggt tcactttcagctcatactggatgaattgggtacgccaaagctccaggcaaggactcgaatgggt aagcgctatatccagc aaagggagctat acctattacgcggattccgttaagggcaggtt cactatatcccgcgacaactccaaaaat actttgtat ctgcaaatgaattccctccgagccgaagataccgcagtatattactgtgcgagggacaggtactccatgatttacagcta cgggtgccggtgctttcgattattggggacaggggacacttgtgaccgtcagttct
SEQ ID NO: 195	Heavy Chain	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKLEWVS AISSSKGSYTYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARD RYSMIYSYGAGAFDYWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSL GTQTYICNVNHKPSNTKVDKRVPEKSCDKTHTCPPCPAPEAAGGPSVFLFP PKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREE QYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR EPQVYITLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTP PVLDSGDSFFLYSKLTVDKSRWQQGNV FSCSVMHEALHNHYTQKSLSLSP GK
SEQ ID NO: 196	DNA Heavy Chain	caagttcagctccttgagctctgggggggctgggtgcaacctgggggctctctgcggctttcatgtgcggcctcaggggt tcactttcagctcatactggatgaattgggtacgccaaagctccaggcaaggactcgaatgggt aagcgctatatccagc aaagggagctat acctattacgcggattccgttaagggcaggtt cactatatcccgcgacaactccaaaaat actttgtat ctgcaaatgaattccctccgagccgaagataccgcagtatattactgtgcgagggacaggtactccatgatttacagcta cgggtgccggtgctttcgattattggggacaggggacacttgtgaccgtcagttctgcaagtaccaaagggcctctgtttt cccatggctccctcatccaagagcacgagtgaggccaccgcgcgtgggatgcttgtgaaagactatttcccgga gccccgtgaccgttagctggaacagcggcgctcttaccagtggttccacacattccagctggtttgcagtcacccgggc tctactctctctcatccgtgggtcaccgtgcccgtctagtctcttggggcaccagacctacatctgtaacgtaaatcacaaccc agttaactaaggtggacaagcgagttgaaccgaagagctgtgataagacacat acttgtccaccatgtccggcacc gaggcagcggggggcccgagtttttctcttccaccccaagcccaagacacatgatgatctcacgaacccagag gttaacttgtgtcgtggtagatgtaagccatgaggaccccgaaagttaagttcaattgggtatgttgacggtgttagaggtgcac aatgccaaaactaaaccccgaggagcaatacaactcaacttacagagtcgtatccgtgctgaccgttttgcaccagg attggttgaatggt aaggaatacaaatgt aaagtgagcaataaagctctcccagcggccatcgagaagaccattagcaa agcccaagggtcaacccaggggaacccaggtatatacgctgccaccctcaagggaagagatgacaaagaatcaagtg cactgacgtgtcttgcagggtttctatcctagcgacattgcggtggaatgggagtc aaatgggcaacccgagaacaa ctacaagactactcctccgctcctggacagcgacggctccttcttccctgtatagtaaacgtgacgctcgataaaagtaggtg gcagcaggggaatgtctttagttgctctgtcatgcatgaggcgctccataaccactacaccccaaaaatctttgagcttgag ccttgggaaa
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 172	LCDR3 (Combined)	QQEWAKPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 172	LCDR3 (Kabat)	QQEWAKPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 173	LCDR3 (Chothia)	EWAKPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 172	LCDR3 (IMGT)	QQEWAKPRT
SEQ ID NO: 174	VL	DIQMTQSPSSLSASVGRVITITCRASQGISSYLA WYQQKPKGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQEWAKPRTFGQGTKV EIK

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 199	DNA VL	gatatacagatgacgcaaaagtcctctagctctttctgcaagtgctcggggacagagttaccattacctgcagagcggtcaca aggcacatctctagttatctcgcgtggtaccaacagaagccaggtaaagcacctaaactgttgatttacacggcatcaacatt gcagtcagggtgtccctcccgatttagtgccagtggtagcgggtacagatttactcttaccatttcattctcttcagccagaa gatattgtactgtactactgtcaacaagaatgggctaaaccacgaacctttggacagggtacgaaggtcgaaataaaa
SEQ ID NO: 176	Light Chain	DIQMTQSPSSLSASVGDRVTITCRASQGISSYLAWYQQKPKAPKLLIYTAS TIQSGVPSRFGSGSGTDFTLTISLQPEDFATYYCQQEWAKPRTFGQGTKV EIKRITVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNISQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID NO: 200	DNA Light Chain	gatatacagatgacgcaaaagtcctctagctctttctgcaagtgctcggggacagagttaccattacctgcagagcggtcaca aggcacatctctagttatctcgcgtggtaccaacagaagccaggtaaagcacctaaactgttgatttacacggcatcaacatt gcagtcagggtgtccctcccgatttagtgccagtggtagcgggtacagatttactcttaccatttcattctcttcagccagaa gatattgtactgtactactgtcaacaagaatgggctaaaccacgaacctttggacagggtacgaaggtcgaaataaaac ggaccgttgccgccccctcgtcttcatcttcccccgctctgacgagcagctcaaatccggcacagcttctgtagctctgct tgctgaataacttctaccacaagagaagccaaagttcagtggaaggtcgataatgcattgcaatctggtaaatagtcaggaa tctgtgactgagcaggatagcaaaagactcaacttacagcctctcttcaaccttgacgttgccaaagcggattatgagaaa cacaaggtgtacgcttgcgagggtgacgcatcaagggttagttccccggtaaccaaatctttcaaccgaggtgaatgc
XX15_ LALA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 119	HCDR2 (Combined)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 119	HCDR2 (Kabat)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 120	HCDR2 (Chothia)	ESKGNV
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 121	HCDR2 (IMGT)	IESKGNVI
SEQ ID NO: 36	HCDR3 (IMGT)	ARDRYSMIYSYGAGAFDY
SEQ ID NO: 201	VH	QVQLLESQGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS VIESKGNIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYICARDR YSMIYSYGAGAFDYWGQGLTVTVSS
SEQ ID NO: 202	DNA VH	caagttcaattgctggaaagcggaggtggacttgccaacctggagggctcactccgactgtcttgccgtgcacagcagatt cacttttagtagctattggatgaactgggtccggcaggctcctgggaaagggcttgagtgaggtaagtgctcattgaatcaa agggcaactacatctttatgctgattctgtaaagggtaggttcaccatctccagggacaattcaaaaaatactttgtatctg cagatgaactctctcagggcagaagacagccgctttattactgcgcccgcatcgatacagcatgatatactcctacg gcgcaggagcttttgactactgggtcaggcacacttgtagtgcagtagc
SEQ ID NO: 203	Heavy Chain	QVQLLESQGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS VIESKGNIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYICARDR YSMIYSYGAGAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALG CLVKDYFPEPTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGT QTYICNVNHKPSNTKVDKRVPEPKCDKTHCTCPPEAPEAAGGSPVFLPPPKP KDTLMISSRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY NSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREP QVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPPV LDDSGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 204	DNA Heavy Chain	caagttcaattgctggaaagcggaggtggacttgtccaacctggagggctcactccgactgtcttgcgctgcatcaggatt caccttttagtagctatttgatgaactgggtccggcaggtctctgggaaagggttgagtgggttaagtgtcattgaatcaa agggcaactacatctttttagctgattctgtaaagggtaggttcaccatctccagggaacattcaaaaaaactttgtatctg cagatgaactctctcagggcagaagacacggccgtttattactgcgcccgcgacgatacagcatgatatactcctacg gcgcaggagcttttgactactgggttcaaggcacacttgttactgtcagtagcgctcaacgaaaggacccgtccgtgttt cctcttgctcctagctccaaatccacctcaggtggaaacggccgctgggtgctggttaaggactatttccagagc cagttactgtgtcttggaattctggtgcattgacaagtggcgtacacactttcccgcggtcctccaatctagtggtctgtac tcaactgtcctccgttgtgactgtcccaagtagctcacttggcacacagacttacatctgtaagtgttaatcataagccgtcaaa cacgaaggtggataaagaggttagaacctaaagtcatgtgacaaaacgcatacttgcctccctgcctgcgcccgaag ccgctggcggacccctccgtattcttgttccctccaaagccaaaggacactctgatgattagccggacacccggaggtcac ttgtgtttagttgacgtcagccatgaggatcctgaggtgaaatttaattggtacgtggacggggttgaagtcacaatgc taaaactaaacctaggggaagagcaatataatagtacacaggggttgcagtggtgacccgttcccatcaggactggc tgaacggcgaaggaatacaagtgcagggtcagcaacaaggccttgccggcccccacgagaagacgatctccaaagc caagggggcaaccccgagaaccgcaggtatacacgtcccccttagtagagaagagatgacaaagaatcaagtttccct gacgtgcttgtgaaagcttctaccctagtacatcgagtcgaatgggagagcaacgggcagccggagaaataacta taaaacaacccccccgtgcttgactcagacgggtcatttttctgtatagcaaatgactgttgataaatcaggtgggcaa caaggaaacgtgttagttgcagcgtaatgcacgaagctctccacaatcactatactcaaaagtcactgtcactctccct ggcaag
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 126	LCDR3 (Combined)	QQEWVKPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 126	LCDR3 (Kabat)	QQEWVKPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 127	LCDR3 (Chothia)	EWVKPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 126	LCDR3 (IMGT)	QQEWVKPRT
SEQ ID NO: 128	VL	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPKAPKLLIYTAS TLQSGVPSRFSGSGSDFTLTISLQPEDFATYYCQQEWVKPRTFGQGTKV EIK
SEQ ID NO: 205	DNA VL	gacatacaaatgacgcaatctccgagtagcttgtcagcgtccgtaggcgaccgagtaacgattacgtgtagagcgagc cagggaaatttcatcttatttggcttggtatcagcaaaaagccgggaaaagcaccacaaactcctcatttatactgccagcacg ttgcaaaagcggcgttccgagtcggttctctggatcaggggtccgggacggacttcaccttgacgatttcatctttgcaacct gaagattttgcaacatactactgtcaacaggagtggtggaagccaaggaccttcggacaaggcacgaaggtcgaatc aag
SEQ ID NO: 130	Light Chain	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPKAPKLLIYTAS TLQSGVPSRFSGSGSDFTLTISLQPEDFATYYCQQEWVKPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPV TKSFNREGC
SEQ ID NO: 206	DNA Light Chain	gacatacaaatgacgcaatctccgagtagcttgtcagcgtccgtaggcgaccgagtaacgattacgtgtagagcgagc cagggaaatttcatcttatttggcttggtatcagcaaaaagccgggaaaagcaccacaaactcctcatttatactgccagcacg ttgcaaaagcggcgttccgagtcggttctctggatcaggggtccgggacggacttcaccttgacgatttcatctttgcaacct gaagattttgcaacatactactgtcaacaggagtggtggaagccaaggaccttcggacaaggcacgaaggtcgaatc aagcgaacccgtggcagctccgtccgtgtttattttccgccttccgacgaacaacttaaaagtggaaacagcctctgtcgtc tgtctccttaacaacttctacccaggggaagctaagtagcagtggaaggtagataacgctctgcaaaagtggttaattctcag

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		gagagcgtcacggaacaggactccaaagactccacctattctctgagctctacactgacgtcagcaaggcagactac gaaaagcacaaaagtatatgcgtgtgaggtgacgcatcaaggccttagcagtcacagttacaaaaagttttaacaggggag aatgc
XX15_ DAPA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 119	HCDR2 (Combined)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 119	HCDR2 (Kabat)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 120	HCDR2 (Chothia)	ESKGNV
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSY
SEQ ID NO: 121	HCDR2 (IMGT)	IESKGNVI
SEQ ID NO: 36	HCDR3 (IMGT)	ARDRYSMIYSYGAGAFDY
SEQ ID NO: 122	VH	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKLEWVS VIESKGNIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLTVTVSS
SEQ ID NO: 207	DNA VH	gaagtgcagctgctggaatccggcggaggtctgggtccagcctggaggtccctgcgcctgtcatgcgcagcctccgg attcaccttttcgtcgactggatgaactgggtcagacaggctcctggaaaggcctggaatgggtgtctgtgatgaaac caaggggaactacatcttctacgcggacagcgtgaagggccgggttcactatcagcagagacaacagcaagaacaccc tgtacctccaaatgaactcgctgagggccgaagatactgccgtgtactactgtgcccgcatcgctactcgatgatctac agctatggtgccggagcgttcgattactggggacagggaaacctcgtagccgtcagctcc
SEQ ID NO: 208	Heavy Chain	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKLEWVS VIESKGNIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALG CLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGT QTYICNVNHKPSNTKVDKRVEPKSCDKHTHTCPPCPAPELLGGPSVFLFPPPKP KDTLMISRTPEVTCVVAVVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY NSTRYRVVSVLTVLHQDWLNGKEYKCKVSNKALAAPIEKTISKAKGQPREP QVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPPV LDSDSGFFLYSKLTVDKSRWQQGNVFPSCVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 209	DNA Heavy Chain	gaagtgcagctgctggaatccggcggaggtctgggtccagcctggaggtccctgcgcctgtcatgcgcagcctccgg attcaccttttcgtcgactggatgaactgggtcagacaggctcctggaaaggcctggaatgggtgtctgtgatgaaac caaggggaactacatcttctacgcggacagcgtgaagggccgggttcactatcagcagagacaacagcaagaacaccc tgtacctccaaatgaactcgctgagggccgaagatactgccgtgtactactgtgcccgcatcgctactcgatgatctac agctatggtgccggagcgttcgattactggggacagggaaacctcgtagccgtcagctccgcctcaaccaaggcccc gtcagtggttcccgctgggtccatcgctgaagttccacctccggaggaaaccgcagcactcggttgcctgggtcaaggactac ttccctgagccagtgaacctgtcggtgaacagcggagccctgaactccggcggtgcacacttttcccgcggtgtgcagct cctccgggtctgtactccctttcgctccgtgggtcaccgtgccgtcgctcagcctgggcacccagacctacatctgcaacgtg aaccacaagcgtccaacaccaaagtggataagcgggtggagccgaagtccctgcgatgaagacacacacgtgcccgcc catgtccagcgcctgaattgcttggcggaccttccgtgttccgtgttcccgccctaaagcccaaggacaccttgatgattagcc ggactccgaagtcacctgtgtggtgggtggcagtgctcccagaggaccccgaggtcaagtttaattgggtacgtggacg gcgtcgaagtgacacacccaagactaagccccgggaggaacagtaacaacagcactaccgggtcggtgcctgctgct gacctgtgtgcaccaggtggtgaatgggaaagagtacaagtgcgaagtgtccaaacaggccttggcgctcctat cgaaaaaactatcagcaaggctaagggaacagccgagggaaacccaagtctacacccctgcccccttcacgcgaagag atgaccaagaatcaagtgtcgctgacctgctcgctcaagggtctacacctccgacattgcggtggagtgaggagtgcca

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		acggccagcccgagaaactacaagactactccgcegtgctggactccgacggcagctttcttctgtattccaagct gaccgtggacaagtcccggtggcagcaaggaaactgttctctctgctcggtcatgcacgaagccctgcacaaccacta tacgcagaagtcctgtccttgagcccggggaaa
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 126	LCDR3 (Combined)	QQEWVKPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 126	LCDR3 (Kabat)	QQEWVKPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 127	LCDR3 (Chothia)	EWVKPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 126	LCDR3 (IMGT)	QQEWVKPRT
SEQ ID NO: 128	VL	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPKGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQEWVKPRTFGQGTKV EIK
SEQ ID NO: 210	DNA VL	gacatccagatgaccagtcctccgtcctccctgtccgcacagtgaggggacagagtgaccatcacttgtcgggcctccc aaggcatctcgtcatacctggcctgggtatcagcagaaacccggaaaggctccaaagctgctcatctacacgcctoga ctctgcaatccggagtgcttccccgtctctccggatccgggttcgggaaccgacttcacctcaccattagcagccttcag ccggaagatttcgcgacctactactgccagcaagaatgggtgaagcccaggacatttggccagggcactaaggctga gattaag
SEQ ID NO: 130	Light Chain	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPKGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQEWVKPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID NO: 211	DNA Light Chain	gacatccagatgaccagtcctccgtcctccctgtccgcacagtgaggggacagagtgaccatcacttgtcgggcctccc aaggcatctcgtcatacctggcctgggtatcagcagaaacccggaaaggctccaaagctgctcatctacacgcctoga ctctgcaatccggagtgcttccccgtctctccggatccgggttcgggaaccgacttcacctcaccattagcagccttcag ccggaagatttcgcgacctactactgccagcaagaatgggtgaagcccaggacatttggccagggcactaaggctga gattaagcgtacgggtggcgcctcccagcgtgttcatctccccccagcagcagcagctgaagagcggcaccgccca gcgtgggtgtcctgctgaacaacttctacccccgggaggccaagggtgcagtggagggtggacaacgcctgcagag cggcaacagccaggagagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcacctgaccct gagcaagggccgactacgagaagcatagggtgtacgcctgcgaggtgacccaccagggcctgtccagccccgtgacc aagagcttcaacagggcgagtgc
XX16_ LALA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 119	HCDR2 (Combined)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 119	HCDR2 (Kabat)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 120	HCDR2 (Chothia)	ESKGNV
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 121	HCDR2 (IMGT)	IESKGNVI
SEQ ID NO: 36	HCDR3 (IMGT)	ARDRYSMIYSYGAGAFDY
SEQ ID NO: 201	VH	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS VIESKGNIFYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQTLTVSS
SEQ ID NO: 202	DNA VH	caagttcaattgctggaaagcggaggtggacttgccaacctggagggctcactccgactgtcttgcgctgcatcaggatt caccttttagtagctattggatgaactgggtccggcaggctcctgggaaagggcttgagtggttaagtgtcattgaatcaa agggcaactacatctttatgctgattctgtaaagggtaggttcaccatctccagggaacaattcaaaaaatactttgtatctg cagatgaactctctcagggcagaagacacggccgtttattactgcgcccgcgatacagcatgatatactcctacg gcgcaggagcttttgactactgggtcaaggcacacttgttactgtcagtagc
SEQ ID NO: 203	Heavy Chain	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS VIESKGNIFYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQTLTVSSASTKGPSVFPLAPSSKSTSGGTAALG CLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGT QTYICNVNHKPSNTKVDKRVPEKSCDKTHTCPPCPAPEAAGGPSVFLFPPKP KDTLMISRTEPVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY NSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI SKAKGQPREP QVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPV LDSGDSFPLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 204	DNA Heavy Chain	caagttcaattgctggaaagcggaggtggacttgccaacctggagggctcactccgactgtcttgcgctgcatcaggatt caccttttagtagctattggatgaactgggtccggcaggctcctgggaaagggcttgagtggttaagtgtcattgaatcaa agggcaactacatctttatgctgattctgtaaagggtaggttcaccatctccagggaacaattcaaaaaatactttgtatctg cagatgaactctctcagggcagaagacacggccgtttattactgcgcccgcgatacagcatgatatactcctacg gcgcaggagcttttgactactgggtcaaggcacacttgttactgtcagtagcgctcaacgaaggagccgtccgtgttt cctctgtcctcagctccaaatccacctcagggtggaacggccgcctggggtgcctggtaaaggactatttccagagc cagttactgtgtcttggaattctgggtgcattgacaagtggcgtaacacactttcccgcggtcctccaatctagtgtgtac tcactgtcctccgttgtgactgtcccaagtagctcacttggcacacagacttacatctgtaattgttaataaagccgtcaaa cacgaaggtggataaagagggtagaacctaagtcattgtgacaaaacgcatacttgccccctgcctgcgcgggaag ccgctggcggaccctccgtattcttgttccctccaaagccaaaggacactctgatgattagccggacacccggaggtcac ttgtgtttagttgacgtcagccatgaggatcctgaggtgaaatttaattggtagctggacgggggtgaaagtcacaaatgc taaaactaaacctaggggaagacaataataatagtagacatcaggggttgtagtggtgacctgtccatcaggactggc tgaacggcaagggaatacaagtgcaaggtcagcaacaaggccttgccggccccatcgagaagacgatctccaaagc caaggggcaaccccgagaacccgaggtatcacgctccccctagttagagaagagatgacaaagaatacaagtttccct gacgtgccttgtagaaggcttctaccctagtacatcgcagtcgaatgggagagcaacgggcagccggagaataacta taaaacaacccccccgtgcttgactcagacgggtcatttttctgtatagcaaatgactgttgataaatcacggtggcaa caagggaacgtgttagttgcagcgttaatgcacgaagctctccacaatcactatactcaaaagtcaactgtcactctccct ggcaag
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 134	LCDR3 (Combined)	QQTWRKPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 134	LCDR3 (Kabat)	QQTWRKPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 135	LCDR3 (Chothia)	TWRKPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 134	LCDR3 (IMGT)	QQTWRKPRT
SEQ ID NO: 136	VL	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPKGAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQTWRKPRTFGQGTKV EIK
SEQ ID NO: 197	DNA VL	gacattcaaatgacacaaagtcctgagtccttagtgcttctgtggcgatagggtcaccatcacttgctcgggcgctctcag gggattcagctcttacttgccatgggtatcaacaaaagccaggaaaagcacctaaattgcttattatcacagcgtccaccctc cagtcaggagtgccctagtaggttctcaggctctgggtccggtactgacttcacgctgactatatacaagcttgcaaccgga agattttgcaacatactactgccacagacatggaggaagccaagaacttccggtcaggggaacgaaagttagataaaa g
SEQ ID NO: 138	Light Chain	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPKGAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQTWRKPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKDSSTLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID NO: 198	DNA Light Chain	gacattcaaatgacacaaagtcctgagtccttagtgcttctgtggcgatagggtcaccatcacttgctcgggcgctctcag gggattcagctcttacttgccatgggtatcaacaaaagccaggaaaagcacctaaattgcttattatcacagcgtccaccctc cagtcaggagtgccctagtaggttctcaggctctgggtccggtactgacttcacgctgactatatacaagcttgcaaccgga agattttgcaacatactactgccacagacatggaggaagccaagaacttccggtcaggggaacgaaagttagataaaa gcgcactgtcgcagcaccttccgtgttcattttcccgcccttcgcacgagcagcttaaatcagggaccgcgagtggtgtttg cttgcttaataacttttaccacagggaagccaaagtccagtggaaggtggacaatgcactccaaagcgggaatagtcag gagtcagttactgagcaagatagtaagactctacttactctttgagttcaaccttgacctctcaaaagcgggactacgag aagcataaagtgtacgcctgaggtgacgcatcaaggtttgtcttccccggttacgaagtcctttaataggggggaatg t
XX16_ DAPA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 119	HCDR2 (Combined)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 119	HCDR2 (Kabat)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 120	HCDR2 (Chothia)	ESKGN

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 121	HCDR2 (IMGT)	IESKGNFY
SEQ ID NO: 36	HCDR3 (IMGT)	ARDRYSMIYSYGAGAFDY
SEQ ID NO: 122	VH	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS VIESKGNFYIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLTVTVSS
SEQ ID NO: 207	DNA VH	gaagtgcagctgctggaatccggcggaggtctggtccagcctggaggttccctgcgcctgtcatgcgcagcctccgg attcaccttttcgtcgactggatgaactgggtcagacaggtcctcgaaagggcctggaatgggtgtctgtgatgaaac caaggggaactacatcttctacgcggacagcgtgaagggccggttcactatcagcagagacaacagcaagaacaccc tgtacctccaaatgaactcgctgagggccgaagatactgccgtgtactactgtgcccgcgatcgctactcgatgatctac agctatggtgccggagcgttcgattactggggacagggaaacctcgtagccgtcagctcc
SEQ ID NO: 208	Heavy Chain	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS VIESKGNFYIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLTVTVSSASTKGPSVFPPLAPSSKSTSGGTAALG CLVKDYFPEPVTWSWNSGALTSGVHTFPAVLQSSSLYSLSSVTVPSSSLGT QTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKP KDTLMISRTEPVTCVVAVVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY NSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALAAPIEKTIISKAKGQPREP QVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPV LDSGDSFPLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 209	DNA Heavy Chain	gaagtgcagctgctggaatccggcggaggtctggtccagcctggaggttccctgcgcctgtcatgcgcagcctccgg attcaccttttcgtcgactggatgaactgggtcagacaggtcctcgaaagggcctggaatgggtgtctgtgatgaaac caaggggaactacatcttctacgcggacagcgtgaagggccggttcactatcagcagagacaacagcaagaacaccc tgtacctccaaatgaactcgctgagggccgaagatactgccgtgtactactgtgcccgcgatcgctactcgatgatctac agctatggtgccggagcgttcgattactggggacagggaaacctcgtagccgtcagctccgcctcaaccaagggccc gtcagtggttcccgctggctccatcgctcgaagtcacacctcgggaggaaccgcagcactcggttgctggtcagggaactac ttccctgagccagtgaccgtgtcggtggaacagcggagccctgacttccggcgtgcacactttcccgcggtgctgcagtg cctccggtctgtactcccttctcgctcggtggtcaccgtgcccgtcgctcagcctgggcaaccagacctacatctgcaacgtg aaccacaagccgtccaacaccaaagtggataagcgggtggagccgaagtccgtcgataagacacacacgtgcccg catgtccagcgcctgaattgcttggcggaccttccgtgttccgttcccgctcgaagcccaaggacaccttgatgattagcc ggactcccgaagtcacctgtgtggtggtggcagtggtcccacagaggaccccgaggtcaagtttaattggtacgtggacg gcgtcgaagtgcaacaagccaagactaagccccgggaggaacagtaacaacagacctaccgggtgtgtccgtgct gaccgtgtgtgaccaggactggctgaatgggaaagagtacaagtgcgaagtgtccaacaaggccttggccgctcctat cgaaaaaactatcagcaaggctaaagggacagccgagggaaacccaagtctacacctgcccccttcacgcgaagag atgaccaagaatcaagtgctcgctgacctgctcgtaagggattctaccctccgacattgcggtggagtgaggagtgca acggccagcccgagaaactacaagactactccgcccgtgctggactccgacggcagcttcttctgtattccaagct gaccgtggacaagtcccggtggcagcaaggaaactgttctcctgctcggtcatgcacgaagccctgcacaacctata tcgcagaagtcctctgctccttgagcccggggaaa
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 134	LCDR3 (Combined)	QQTWRKPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 134	LCDR3 (Kabat)	QQTWRKPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 135	LCDR3 (Chothia)	TWRKPRT

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 134	LCDR3 (IMGT)	QQTWRKPRT
SEQ ID NO: 136	VL	DIQMTQSPSSLSASVGDRVTITCRASQGISSYLAWYQQKPGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQTWRKPRTFGQGTKV EIK
SEQ ID NO: 143	DNA VL	gacatccagatgaccagctctccgtcctccctgtccgcatcagtgaggacagagtgaccatcacttgctcgggcctccc aaggcatctcgtcatacctggcctgggtatcagcagaaaacccggaaggctccaaagctgctcatctacaccgcctcga ctctgcaatccggagtgctctcccgcttctccggatccgggttcgggaaccgacttcaccctcaccattagcagccttcag ccggaagatttcgcgacctactactgccagcaaacctggcggaagcccaggacatttgccaggggcactaaggtcga gattaag
SEQ ID NO: 138	Light Chain	DIQMTQSPSSLSASVGDRVTITCRASQGISSYLAWYQQKPGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQTWRKPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID NO: 144	DNA Light Chain	gacatccagatgaccagctctccgtcctccctgtccgcatcagtgaggacagagtgaccatcacttgctcgggcctccc aaggcatctcgtcatacctggcctgggtatcagcagaaaacccggaaggctccaaagctgctcatctacaccgcctcga ctctgcaatccggagtgctctcccgcttctccggatccgggttcgggaaccgacttcaccctcaccattagcagccttcag ccggaagatttcgcgacctactactgccagcaaacctggcggaagcccaggacatttgccaggggcactaaggtcga gattaagcgtacgggtggcgcgtcccagcgtgttcatcttccccccagcgacgagcagctgaagagcggcaccgcca gcgtgggtgtcctgtgaacaacttctacccccgggaggccaagggtgcagtggaagggtggacaacgcctgcagag cggcaacagccaggagagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccctgacct gagcaaggccgactacgagaagcatagggtgtacgcctgcgaggtgacccaccagggcctgtccagccccgtgacc aagagcttcaacaggggcgagtg
XX17_ LALA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 119	HCDR2 (Combined)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 119	HCDR2 (Kabat)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 120	HCDR2 (Chothia)	ESKGNV
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 121	HCDR2 (IMGT)	IESKGNVI
SEQ ID NO: 36	HCDR3 (IMGT)	ARDYSMIYSYGAGAFDY
SEQ ID NO: 201	VH	QVQLLESQGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKLEWVS VIESKGNIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLTVTVSS

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 202	DNA VH	caagttcaattgctggaaagcggaggtggacttgtccaacctggagggtcactccgactgtcttgcgctgcatcaggatt caccttttagtagctattggatgaactgggtccggcagggtcctgggaaagggttgagtgggtaagtgtcattgaatcaa agggcaactacatctttatgctgattctgtaaagggtagggtccaccatctccagggaacaattcaaaaaatactttgtatctg cagatgaactctctcagggcagaagacacggccgtttattactgcgccgcgatacgatacagcatgatatactcctacg gcgcaggagcttttgactactggggtcaaggcacacttgttactgtcagtagc
SEQ ID NO: 203	Heavy Chain	QVQLLESGGGLVQPGGSLRLSCAASGFTSSYWMNWVRQAPGKGLEWVS VIESKGNIFYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSGAGAFDYWGQGLVTVSSASTKGPSVFPLAPSSKSTSGGTAAALG CLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGT QTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKP KDTLMISRTEPVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY NSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREP QVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPV LDSGGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 204	DNA Heavy Chain	caagttcaattgctggaaagcggaggtggacttgtccaacctggagggtcactccgactgtcttgcgctgcatcaggatt caccttttagtagctattggatgaactgggtccggcagggtcctgggaaagggttgagtgggtaagtgtcattgaatcaa agggcaactacatctttatgctgattctgtaaagggtagggtccaccatctccagggaacaattcaaaaaatactttgtatctg cagatgaactctctcagggcagaagacacggccgtttattactgcgccgcgatacgatacagcatgatatactcctacg gcgcaggagcttttgactactggggtcaaggcacacttgttactgtcagtagcgctcaacgaaaggaccgtccgtgttt cctcttgctcctagctccaaatccacctcagggtggaacggccgcctggggtgcctggtaaaggactatttccagagc cagttactgtgtcttggaattctgggtgcattgacaagtggcgtagacactttcccgcggtcctccaatctagtggtctgtac tactgtcctccgttgtgactgtcccaagtagctcacttggcacacagacttacatctgtaattgtaatacagaacgctcaaa cacgaaggtggataaagagggtagaacctaaagtcattgtgacaaaacgcatacttgccccctgccctgcgcgggaag ccgctggcggaaccctccgtattcttgttccctccaaagccaaaggacactctgatgattagccgggacacggagggtcac ttgtgttgtagtgacgtcagccatgaggatcctgaggtgaaatttaattggtagctggacggggttgaagtcacacatgc taaaactaaacctagggaagagcaatataatagtagacacaggggtgtcagtggtgctgacggttcccatcaggactggc tgaacggcaaggaatacaagtgcaaggtcagcaacaaggccttgccggccccatcgagaagacgatctccaaagc caaggggcaaccccgagaaccgcaggtatcacgctccccctagt agagaagagatgacaaagaatcaagt ttcctt gacgtgccttgtgaaaggcttctaccctagt gacatcgacgtcgaatgggagagcaacgggcagccggagaataacta taaaacaacccccccgtgcttgactcagacgggtcattttttctgtatagcaaatgactgttgataaatcacggtggcaa caaggaaacgtgttttagttgcagcgtaatgcacgaagctctccacaatcactatactcaaaagt cactgtcactctccct ggcaag
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 145	LCDR3 (Combined)	QQIWTVPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 145	LCDR3 (Kabat)	QQIWTVPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 146	LCDR3 (Chothia)	IWTVPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 145	LCDR3 (IMGT)	QQIWTVPRT
SEQ ID NO: 147	VL	DIQMTQSPSSLSASVGRVITICRASQGISSYLAWYQQKPKAPKLLIYTAS TLQSGVPSRFGSGSGTDFTLTISLQPEDFATYYCQQIWTVPRTFGQGTKV EIK

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 212	DNA VL	gacatacagatgactcagagtccttccctccctcagtgcttcagtggggtgatcgcgtagcatcacgtgcagagcctcaca agggatctccagttacctggcctgggtatcaacaaaaaccaggcaaggcgccctaagctgttgatatatacggcatctacat tgcagttctggggtaccagtcgattcagtggttctggctcaggcactgactttacccttacaatatcaagtcctcagccgg aggatttcgcaacttactattgccagcagatttggacggtgccgcgcactttcggtcagggaacaaaaggtggaataaaa a
SEQ ID NO: 149	Light Chain	DIQMTQSPSSLSASVGDRVTITCRASQGISSYLAWYQQKPKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQIWTVPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKDSITYLSSTLTLSKADYEKHKVYACEVTHQGLSPVPT KSFNRGEC
SEQ ID NO: 213	DNA Light Chain	gacatacagatgactcagagtccttccctccctcagtgcttcagtggggtgatcgcgtagcatcacgtgcagagcctcaca agggatctccagttacctggcctgggtatcaacaaaaaccaggcaaggcgccctaagctgttgatatatacggcatctacat tgcagttctggggtaccagtcgattcagtggttctggctcaggcactgactttacccttacaatatcaagtcctcagccgg aggatttcgcaacttactattgccagcagatttggacggtgccgcgcactttcggtcagggaacaaaaggtggaataaaa aagaacggtcgagcaccgagtggtttcatcttccctccctccgacgagcagcttaaaagcggtacagccagcgtagtgtg tgtttgtgaataaattttatccacgcgaagcaaaagttcagtggaaggtagacaacgcattgcaaagcggaattcccaa gaaagtgttacggagcaagacagtaaggactctacatatccctgtcatcaacactcacccttagtaagcagattacga gaaacacaaggtctatgcatgtgaggttaacgcacagggcctctccagtcctcgtcaccaagtccttcaacaggggtga gtgc
XX17_		
DAPA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 119	HCDR2 (Combined)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 119	HCDR2 (Kabat)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 120	HCDR2 (Chothia)	ESKGNV
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 121	HCDR2 (IMGT)	IESKGNVI
SEQ ID NO: 36	HCDR3 (IMGT)	ARDYSMIYSYGAGAFDY
SEQ ID NO: 122	VH	EVQLLESQGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKLEWVS VIESKGNIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQTLTVSS
SEQ ID NO: 207	DNA VH	gaagtgcagctgctggaatccggcggaggtctggtccagcctggaggttccctgcgcctgtcatgcgcagcctccgg attcaccttttcgtcgtagtggaactgggtcagacaggtcctggaaagggcctggaatgggtgtctgtgatgaaac caaggggaactacatcttctacgcggacagcgtgaaggccgggttcaactatcagcagagacaacagcaagaacaccc tgtacctccaaatgaactcgctgagggccgaagatactgccgtgtactactgtgccccgcgatcgtactcgatgatctac agctatggtgccggagcgttcgattactggggacagggaaaccctcgtgaccgtcagctcc
SEQ ID NO: 208	Heavy Chain	EVQLLESQGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKLEWVS VIESKGNIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQTLTVSSASTKGPSVFPLAPSSKSTSGGTAALG CLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGT QTYICNVNHKPSNTKVDKRVPEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKP KDTLMISRTPEVTCVAVVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY NSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALAAPIEKTISKAKGQPREP

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 209	DNA Heavy Chain	QVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPV LDSGGSFFLYSKLTVDKSRWQQGNVFCFVMSHHEALHNHYTQKSLSLSPGK
SEQ ID NO: 209	DNA Heavy Chain	gaagtgcagctgctggaatccggcggaggtctgggtccagcctggaggttcctgcgcctgtcatgcgcagcctccgg attcaccttttctgctgctactggatgaactgggtcagacaggtctcctggaaagggtcctggaatgggtgtctgtgattgaatc caaggggaactacatcttctacgcggacagcgtgaagggcgggttactatcagcagagacaacagcaagaacaccc tgtacctccaaatgaactcgctgagggccgaagatactgcccgtgtactactgtgcccgcgatcgctactcgatgatctac agctatgggtgccggagcgttcgattactggggacagggaaacctcgtgaccgtcagctccgcctcaaccaagggtcc gtcagtggtcccgctgggtccatcgtcgaagtccacctccggaggaaaccgcagcactcggttgctgggtcaaggactac ttccctgagccagtgaccgtgtcgtggaacagcggagccctgacttccggcgtgcacacttttcccggtgtgtgcagt cctccgggtctgtactccctttcgtccgtgggtcaccgtgcccgtcgtctagcctgggcacccagacctacatctgcaactg aaccacaagcgtccaacaccaaagtggataagcgggtggagccgaagtctgcgataagacacacagtgcccg catgtccagcgcctgaattgcttggcggaccttccgtgttccgttcccgctaaagcccaaggacacctgatgattagcc ggactcccgaagtcacctgtgtgggtgggtggcagtggtcccacgaggaccccgaggtcaagtttaattgggtacgtggacg gcgtcgaagtgcacaacccaagactaagccccgggaggaaacagtacaacagcacctaccgggtcgtgtccgtgct gaccgtgtgcaccaggactgggtgaatgggaaagagtaacagtgcaagtgctcaacaaaggccttggcgcctcctat cgaaaaaactatcagcaaggctaagggaacagccgagggaacccaagtctacacctgcccccttcacgcgaagag atgaccaagaatcaagtgtcgtgacctgctcgtcaagggtattctaccctccgacattgcggtggagtgaggagtgcca acggccagcccgagaaactacaagactactccgcctgctggactccgacggcagcttcttccgtattccaagct gaccgtggacaagtcccggtggcagcaaggaaactgttctcctgctcgggtcatgcacgaagcctgcacaacctata tacgcagaagtcctgtccttgagccccgggaaa
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 145	LCDR3 (Combined)	QQIWTVPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 145	LCDR3 (Kabat)	QQIWTVPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 146	LCDR3 (Chothia)	IWTVPRT
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 145	LCDR3 (IMGT)	QQIWTVPRT
SEQ ID NO: 147	VL	DIQMTQSPSSLSASVGRVITITCRASQGISSYLAHWYQQKPKAPKLLIYTAS TLQSGVPSRFGSGSGTDFTLTITSSLPEDFATYYCQQIWTVPRTFGQGTKV EIK
SEQ ID NO: 214	DNA VL	gacatccagatgaccagctctccgtcctccctgtccgcacagtggtggggacagagtgaccatcacttgtcgggcctccc aaggcatctcgtcatacctggcctgggtatcagcagaaacccggaagggtccaaagctgctcatcacaccgcctcga ctctgcaatccggagtgcttcccgtctctccggatccgggttcgggaaccgacttcaacctcaccattagcagccttcag ccggaagatttcgcgacctactactgccagcaaatctggaccgtgccccaggacatttggccagggcactaaggctcgag attaag
SEQ ID NO: 149	Light Chain	DIQMTQSPSSLSASVGRVITITCRASQGISSYLAHWYQQKPKAPKLLIYTAS TLQSGVPSRFGSGSGTDFTLTITSSLPEDFATYYCQQIWTVPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVIT KSFNRGEC
SEQ ID NO: 215	DNA Light Chain	gacatccagatgaccagctctccgtcctccctgtccgcacagtggtggggacagagtgaccatcacttgtcgggcctccc aaggcatctcgtcatacctggcctgggtatcagcagaaacccggaagggtccaaagctgctcatcacaccgcctcga ctctgcaatccggagtgcttcccgtctctccggatccgggttcgggaaccgacttcaacctcaccattagcagccttcag ccggaagatttcgcgacctactactgccagcaaatctggaccgtgccccaggacatttggccagggcactaaggctcgag

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		attaagcggtacgggtggcgcgtcccagcgtgttcattcttccccccagcgcagcagcagctgaagagcggcaccgccag cgtgggtgtgcctgctgaacaacttctacccccgggaggccaaggtgcagtggaaggtggacaacgccctgcagagc ggcaacagccaggagagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccctgacctg agcaaggccgactacgagaagcataaggtgtacgcctgcgaggtgacccaccagggcctgtccagccccgtgacca agagcttcaacaggggcgagtgc
XX18_ LALA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 119	HCDR2 (Combined)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 119	HCDR2 (Kabat)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 120	HCDR2 (Chothia)	ESKGNV
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSY
SEQ ID NO: 121	HCDR2 (IMGT)	IESKGNV
SEQ ID NO: 36	HCDR3 (IMGT)	ARDYSMIYSYGAGAFDY
SEQ ID NO: 201	VH	QVQLLESGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKLEWVS VIESKGNIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLTVTVSS
SEQ ID NO: 202	DNA VH	caagtccaattgctggaagcggaggtggactgtgtccaaacctggagggctcactccgactgtcttgcgctgcatcaggatt cacccttagtagctatttgatgaactgggtccggcaggtcctgggaaagggcttgagtggtgaagtgtcattgaatcaa agggcaactacatctttatgctgattctgtaaagggtaggttcaccatctccagggacaattcaaaaaatactttgtatctg cagatgaactctctcagggcagaagacacggccgtttattactgcgcccgcgatcgatacagcatgatatactcctacg gcgcaggagcttttgactactgggtcaaggcacacttgttactgtcagtagc
SEQ ID NO: 203	Heavy Chain	QVQLLESGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKLEWVS VIESKGNIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALG CLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGT QTYICNVNHKPSNTKVDKRVKPKCDKTHTCPPCPAPEAAGGPSVFLFPPKP KDTLMISRTEPVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY NSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTIISKAKGQPREP QYVYLPSSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPV LDDSGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 204	DNA Heavy Chain	caagtccaattgctggaagcggaggtggactgtgtccaaacctggagggctcactccgactgtcttgcgctgcatcaggatt cacccttagtagctatttgatgaactgggtccggcaggtcctgggaaagggcttgagtggtgaagtgtcattgaatcaa agggcaactacatctttatgctgattctgtaaagggtaggttcaccatctccagggacaattcaaaaaatactttgtatctg cagatgaactctctcagggcagaagacacggccgtttattactgcgcccgcgatcgatacagcatgatatactcctacg gcgcaggagcttttgactactgggtcaaggcacacttgttactgtcagtagcgcctcaacgaagaggacctccgtgttt cctctgtcctagctccaaatccacctcaggtggaacggccgcctgggggtgcctggtaaaggactatttccagagc cagttactgtgtcttggaattctggtgcattgacaagtggcgtagacacactttcccgcggtcctccaatctagtgtgtac tcactgtcctccgtgtgtgactgtccaaagttagctcacttggcacacagacttacatctgtaattgtaataataagccgtcaaa cacgaaggtggataagagggtagaacctaagtcagtgacaaaacgcatacttgccccctgcctgcccgggaag ccgctggcggaccctccgtattcttgttccctccaaagccaaaggacactctgatgattagccggacacggaggtcac ttgtgtgttagtgacgtcagccatgaggatcctgaggtgaaatttaattggtagctggacgggggtgaagtcacaatgc taaaactaaacctagggaagagcaataataatgacacacaggggtgtcagtggtgctgaccttccatcaggactggc tgaacggcaaggaatacaagtgcgaaggtcagcaacaaggccttgccggcccccatcgagaagacgatctccaaagc

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		caaggggcaaccccgagaaccgcaggtatacacgctccccctagtagagaagagatgacaaagaatcaagtttccctt gacgtgccttggtgaaaggcttctaccctagtgacatcgacgtcgaaatgggagagcaacgggcagccggagagaataacta taaaacaacccccccgtgcttgactcagacgggtcatTTTTTctgtatagcaattgactgttgataaatcacggtggcaa caaggaaacgtgttagttgcagcgtaatgcacgaagctctccacaatcactatactcaaaagtcactgtcactctccct ggcaag
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 172	LCDR3 (Combined)	QQEWAKPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 172	LCDR3 (Kabat)	QQEWAKPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 173	LCDR3 (Chothia)	EWAKPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 172	LCDR3 (IMGT)	QQEWAKPRT
SEQ ID NO: 174	VL	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQEWAKPRTFGQGTKV EIK
SEQ ID NO: 199	DNA VL	gatatacagatgacgcaaaagtcctctagctcttctgcaagtgctggggacagagttaccattacctgcagagcgtcaca aggcatctctagttatctcgctgggtaccaaagcagaagccaggtaaaagcacctaaactgttgatttacacggcatcaacatt gcagtcagggtgtccctcccgatttagtgccagtggttagcggtacagattttactcttaccatttcatctcttcagccagaa gattttgctacgtactactgtcaacaagaatgggctaaaccacgaaccttggacagggtacgaaggtcgaaataaaa
SEQ ID NO: 176	Light Chain	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQEWAKPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID NO: 200	DNA Light Chain	gatatacagatgacgcaaaagtcctctagctcttctgcaagtgctggggacagagttaccattacctgcagagcgtcaca aggcatctctagttatctcgctgggtaccaaagcagaagccaggtaaaagcacctaaactgttgatttacacggcatcaacatt gcagtcagggtgtccctcccgatttagtgccagtggttagcggtacagattttactcttaccatttcatctcttcagccagaa gattttgctacgtactactgtcaacaagaatgggctaaaccacgaaccttggacagggtacgaaggtcgaaataaaac ggaccgttgccgccccctccgtcttcatcttcccccgctctgacgagcagctcaaactccggcacagcttctgtagctctgct tgctgaataaacttctaccaagagaagccaaagttcagtggaaggtcgataatgcattgcaatctggttaatagtcaggaa tctgtgactgagcaggatagcaaaagactcaacttacagcctctcttcaaccttgacgttggtccaaagcggattatgagaaa cacaagggtgacgcttgcgagggtgacgcatcaagggttagttccccggtaaccaaatcttcaaccagggtgaatgc
XX18_ DAPA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 119	HCDR2 (Combined)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 119	HCDR2 (Kabat)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 120	HCDR2 (Chothia)	ESKGNV
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 121	HCDR2 (IMGT)	IESKGNVI
SEQ ID NO: 36	HCDR3 (IMGT)	ARDRYSMIYSYGAGAFDY
SEQ ID NO: 122	VH	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS VIESKGNIFYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLTVTVSS
SEQ ID NO: 207	DNA VH	gaagtgcagctgctggaatccggcggaggtctggtccagcctggaggttcctgcgcctgtcatgcgcagcctccgg attcaccttttctgctgactggatgaactgggtcagacaggtcctggaaagggcctggaatgggtgtctgtgatgaatc caaggggaactacatcttctacgcggacagcgtgaagggccggttcactatcagcagagacaacagcaagaacaccc tgtacctccaaatgaactcgctgagggccgaagatactgccgtgtactactgtgcccgcatcgctactcgatgatctac agctatggtgccggagcgttcgattactggggacagggaaacctcgtagccgtcagctcc
SEQ ID NO: 208	Heavy Chain	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS VIESKGNIFYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALG CLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGT QTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKP KDTLMISRTEPVTCVVVAVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY NSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALAAPIEKTI SKAKGQPREP QVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPV LDSGDSFPLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 209	DNA Heavy Chain	gaagtgcagctgctggaatccggcggaggtctggtccagcctggaggttcctgcgcctgtcatgcgcagcctccgg attcaccttttctgctgactggatgaactgggtcagacaggtcctggaaagggcctggaatgggtgtctgtgatgaatc caaggggaactacatcttctacgcggacagcgtgaagggccggttcactatcagcagagacaacagcaagaacaccc tgtacctccaaatgaactcgctgagggccgaagatactgccgtgtactactgtgcccgcatcgctactcgatgatctac agctatggtgccggagcgttcgattactggggacagggaaacctcgtagccgtcagctccgcctcaaccaagggccc gtcagtggtcccgctggctccatcgctcgaagtcacacctccggaggaaaccgcagcactcggttgctcgtggaaggaact ttccctgagccagtgacccgtgctggtgaacacggagccctgacttccggcgtgcacacttttcccgcggtgctgcagt cctccggtctgtactcccttctgctcgtggtcaccgtgcccgtgctcagcctgggcaaccagacctacatctgcaacgtg aaccacaagccgtccaacaccaaagtggataagcgggtggagccgaagt cctgcgataagacacacacgtgcccgcc catgtccagccctgaattgcttggcggaccttccgtgttctcgttcccgccctaaagcccaaggacaccttgatgattagcc ggactcccgaagtcacctgtgtggtgggtggcagtggtcccacgaggaccccagaggtcaagtttaattggtagctggacg gcgtcgaagtgcaaacgccaagactaagccccgggagggaacagtagcaaacagacctaccgggtcggtgcccgtgct gaccgtgctgcaccaggaactggctgaatgggaaagagtacaagtgcaaaagtgtccaacaaggccttggccgctcctat cgaaaaaactatcagcaaggctaaagggacagccgagggaaacccaagtctacacctgcccccttcacgcgaagag atgaccaagaatcaagtgctcgctgacctgctcgaaggattctaccctccgacattgcggtggagtgaggagtgcca acggccagcccagagaacaactacaagactactccgcccgtgctggactccgacggcagcttcttccgtgattccaagct gaccgtggacaagtcccgggtggcagcaaggaaacgtgttctcctgctcggtcatgcacgaagccctgcacaacctata tcgcagaagtcctctgctccttgagccccggggaaa
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 172	LCDR3 (Combined)	QQEWAKPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 172	LCDR3 (Kabat)	QQEWAKPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 173	LCDR3 (Chothia)	EWAKPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 172	LCDR3 (IMGT)	QQEWAKPRT
SEQ ID NO: 174	VL	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPGKAPKLLIYTAS TLQSGVPSRFRSGSGSDFTLTISLQPEDFATYYCQQEWAKPRTFGQGTKV EIK
SEQ ID NO: 216	DNA VL	gacatccagatgaccagtcctccgtcctccctgtccgcatcagtgggggacagagtgaccatcacttgctcgggcctccc aaggcatctcgtcatacctggcctgggtatcagcagaaaacccggaaaggctccaaagctgctcatctacaccgcctcga ctctgcaatccggagtgccctcccgtctctcgggatccgggttcgggaaccgacttcaccctcaccattagcagccttcag ccggaagatttcgcgacctactactgccagcaagaatgggccaagcccaggacatttgccaggggcactaaggtcga gattaag
SEQ ID NO: 176	Light Chain	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPGKAPKLLIYTAS TLQSGVPSRFRSGSGSDFTLTISLQPEDFATYYCQQEWAKPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKDSSTLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID NO: 217	DNA Light Chain	gacatccagatgaccagtcctccgtcctccctgtccgcatcagtgggggacagagtgaccatcacttgctcgggcctccc aaggcatctcgtcatacctggcctgggtatcagcagaaaacccggaaaggctccaaagctgctcatctacaccgcctcga ctctgcaatccggagtgccctcccgtctctcgggatccgggttcgggaaccgacttcaccctcaccattagcagccttcag ccggaagatttcgcgacctactactgccagcaagaatgggccaagcccaggacatttgccaggggcactaaggtcga gattaagcgtacgggtggcgcgtcccagcgtgttcatcttccccccagcgacgagcagctgaagagcggcaccgcca gcgtgggtgtgcctgtgaacaacttctacccccgggaggccaagggtgcagtggaaggtgggacaacgcctgcagag cggcaacagccaggagagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccctgacct gagcaaggccgactacgagaagcat aagggtgtacgcctgcgaggtgacccaccagggcctgtccagccccgtgacc aagagcttcaacaggggagtgctgc
XX19_ LALA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 119	HCDR2 (Combined)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 119	HCDR2 (Kabat)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 120	HCDR2 (Chothia)	ESKGN

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 121	HCDR2 (IMGT)	IESKGNYY
SEQ ID NO: 36	HCDR3 (IMGT)	ARDRYSMIYSYGAGAFDY
SEQ ID NO: 201	VH	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS VIESKGNYYIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLTVTVSS
SEQ ID NO: 202	DNA VH	caagttcaattgctggaaagcggaggtggacttgtccaacctggagggtcactccgactgtcttgcgctgcatcaggatt caccttttagtagctattggatgaactgggtccggcaggctcctgggaagggttgagtgggttaagtgtcattgaatcaa agggcaactacatcttttatgctgattctgtaaagggttaggttcaccatctccagggaacaattcaaaaaatactttgtatctg cagatgaactctctcagggcagaagacacggccgtttattactgcgcccgcgacgatacagcatgatatactcctacg gcgcaggagcttttgactactgggtcaaggcacacttgttactgtcagtagc
SEQ ID NO: 203	Heavy Chain	QVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS VIESKGNYYIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAAALG CLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGT QTYICNVNHKPSNTKVDKRVEPKCDKHTHTCPPCPAPEAAGGPSVFLFPPKP KDTLMISRTPETCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY NSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTIISKAKGQPREP QVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPV LDSGDSFPLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 204	DNA Heavy Chain	caagttcaattgctggaaagcggaggtggacttgtccaacctggagggtcactccgactgtcttgcgctgcatcaggatt caccttttagtagctattggatgaactgggtccggcaggctcctgggaagggttgagtgggttaagtgtcattgaatcaa agggcaactacatcttttatgctgattctgtaaagggttaggttcaccatctccagggaacaattcaaaaaatactttgtatctg cagatgaactctctcagggcagaagacacggccgtttattactgcgcccgcgacgatacagcatgatatactcctacg gcgcaggagcttttgactactgggtcaaggcacacttgttactgtcagtagcgctcaacgaaaggaccgtccgtgttt cctcttgctcctagctccaaatccacctcagggtggaacggccgcctggggtgcctggtaaaggactatttccagagc cagttactgtgtcttggaattctggtgcattgacaagtggcgtaacacactttcccgcggtcctccaatctagtgtgtac tcactgtcctccgttgtagtgcctcaagtagctcacttggcacacagacttacatctgtaattgtaatacaagccgtcaaa cacgaaggtggataaagagggtagaacctaaagtcagtgacaaaaacgcatacttgcctccctgcctgcgcgggaag ccgctggcgaccctccgtattcttgttccctccaaagcgaaggacactctgatgattagccggacaccggagggtcac ttgtgttgtagttgacgtcagccatgaggatcctgaggtgaaatttaattggtagcgtggacggggtgaaagtcacaatgc taaaactaaacctaggggaagagcaataataatagtagacacaggggttgtagtgctgaccgttctccatcaggactggc tgaaacggcaaggaatacaagtgcaaggtcagcaacaaggccttgccggcccccacgagaagacgatctccaaagc caaggggcaaccccgagaaccgcaggtatcacgctccccctagt agagaagagatgacaaagaatcaagtttccct gacgtgccttgtagaaaggcttctaccctagtgacatcgcagtcgaatggagagcaacgggcagccggagaataacta taaaacaacccccccgtgcttgactcagacgggtcatttttctgtatagcaaatgactgttgataaatcacggtggcaa caaggaaacgtgttagttgcagcgttaatagcacgaagctctcccaatcactatactcaaaagtcactgtcactctccct ggcaag
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 178	LCDR3 (Combined)	QQSWTRPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 178	LCDR3 (Kabat)	QQSWTRPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 179	LCDR3 (Chothia)	SWTRPR

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 178	LCDR3 (IMGT)	QQSWTRPRT
SEQ ID NO: 180	VL	DIQMTQSPSSLSASVGDRVTITCRASQGISSYLAWYQQKPGKAPKLLIYTAS TLQSGVPSRFRSGSGSDFTLTISLQPEDFATYYCQQSWTRPRTFGQGTKV EIK
SEQ ID NO: 218	DNA VL	gatattcagatgacgcaatctccgtcttcttctgtcagctagtgtaggagaccgcgtcaccaattacctgtagagccagcca ggggatttctctataccttgcatgggtaccagcaaaagccaggcaaaagcccccactgctgatctacaccgcgtctacc ttgcaatctggtgtgccgtcacgcttttccggctctggctcaggtactgatttcacattgacgatctcaagtctccagccgg aagacttcgcaacttactactgccaacaatcctggacgagccgaggactttcgggcagggaacaaagggtgaaatta aa
SEQ ID NO: 182	Light Chain	DIQMTQSPSSLSASVGDRVTITCRASQGISSYLAWYQQKPGKAPKLLIYTAS TLQSGVPSRFRSGSGSDFTLTISLQPEDFATYYCQQSWTRPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKDSSTLSLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID NO: 219	DNA Light Chain	gatattcagatgacgcaatctccgtcttcttctgtcagctagtgtaggagaccgcgtcaccaattacctgtagagccagcca ggggatttctctataccttgcatgggtaccagcaaaagccaggcaaaagcccccactgctgatctacaccgcgtctacc ttgcaatctggtgtgccgtcacgcttttccggctctggctcaggtactgatttcacattgacgatctcaagtctccagccgg aagacttcgcaacttactactgccaacaatcctggacgagccgaggactttcgggcagggaacaaagggtgaaatta aaagaacagtcgcagcaccagtggttttattttccaccctcagacgagcagctcaagctctggcaccgcgagcgtagta tggttgttgaataattttaccctagggaagctaaaggtacagtggaagtggaataatgctctccaaagtggaactccag gaatcagtgactgagcaagattcaaaggacgacgctattctcttctctacgcttactctcttaaggccgactacgaaa aacacaaagttacgcttgcgaggttaccaccaggggctgtcctcaccagtaacgaaaagtttaaccggggcgagtg t
XX19_ DAPA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 119	HCDR2 (Combined)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 119	HCDR2 (Kabat)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 120	HCDR2 (Chothia)	ESKGNV
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 121	HCDR2 (IMGT)	IESKGNVI
SEQ ID NO: 36	HCDR3 (IMGT)	ARDYSMIYSYGAGAFDY
SEQ ID NO: 122	VH	EVQLLESQGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKLEWVS VIESKGNIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLVTVSS

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 207	DNA VH	gaagtgcagctgctggaatccggcggaggtctggtccagcctggaggttcctgcgcctgtcatgcgcagcctccgg attcaccttttcgtcgactggatgaactgggtcagacaggtcctcgaaagggcctggaatgggtgtctgtgattgaatc caaggggaactacatcttctacgcggacagcgtgaagggcgggttcaactatcagcagagacaacagcaagaacaccc tgtacctccaaatgaactcgctgagggccgaagatactgccgtgtactactgtgcccgcatcgctactcgatgatctac agctatgggtgccggagcgttcgattactggggacagggaaacctcgtgacctgcagctcc
SEQ ID NO: 208	Heavy Chain	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWVS VIESKGNVIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLVTVSSASTKGPSVFPLAPSSKSTSGGTAAALG CLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGT QTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKP KDTLMISRTEPEVTCVVVAVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY NSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALAAPIEKTISKAKGQPREP QVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPV LDSGSGFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 209	DNA Heavy Chain	gaagtgcagctgctggaatccggcggaggtctggtccagcctggaggttcctgcgcctgtcatgcgcagcctccgg attcaccttttcgtcgactggatgaactgggtcagacaggtcctcgaaagggcctggaatgggtgtctgtgatgaatc caaggggaactacatcttctacgcggacagcgtgaagggcgggttcaactatcagcagagacaacagcaagaacaccc tgtacctccaaatgaactcgctgagggccgaagatactgccgtgtactactgtgcccgcatcgctactcgatgatctac agctatgggtgccggagcgttcgattactggggacagggaaacctcgtgacctgcagctccgcctcaaccaagggccc gtcagtggttcccgctgggtccatcgtcgaagtccacctccggaggaaaccgcagcactcggttgctgggtcaaggactac ttccctgagccagtgaacctgtcgtggaacagcggagccctgacttccggcgtgcacacttttcccgcggtgctgcagt cctccggtctgtactcccttttcgtccgtggtcaccgtgccgtcgtctagcctgggcacccagacctacatctgcaacgtg aaccacaagccgtccaaaccaaagtggataagcgggtggagccgaagtcctgcgataagacacacacgtgcccgcc catgtccagcgccctgaattgcttggcggaaccttccgtgttccgttcccgccctaaagcccaaggacaccttgatgattagcc ggactcccgaagtcacctgtgtggtggtggcagtggtccacagaggaccccgaggtcaagtttaattggtacgtggacg gcgtcgaagtgcacaacgccaagactaaagcccgagggaacagtaacaacagcacctaccgggtcgtgtccgtgct gacctgtgtgcaccaggactgggtgaatgggaaagagtacaagtgcgaagtgtccaacaaggccttggccgctcctat cgaaaaaactatcagcaaggctaagggacagccgagggaaacccaagtcacacctgccccccttcacgcgaagag atgaccaagaatcaagtgtcgctgacctgcctcgtcaagggattctacctcccgacattgcggtggagtgaggagcca acggccagccccgagaacaactacaagactactccgcccggtgctggactccgacggcagcttcttctcgtattccaagct gaccgtggacaagtcccggtggcagcaaggaaactgttctcctgctcggtcatgcacgaagccctgcacaacctata tacgcagaagtcctgtccttgagcccgggaaa
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 178	LCDR3 (Combined)	QQSWTRPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 178	LCDR3 (Kabat)	QQSWTRPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 179	LCDR3 (Chothia)	SWTRPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 178	LCDR3 (IMGT)	QQSWTRPRT
SEQ ID NO: 180	VL	DIQMTQSPSSLSASVGRVITITCRASQGISSYLAWYQQKPKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQSWTRPRTFGQGTKV EIK

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 220	DNA VL	gacatccagatgaccagctctccgtcctccctgtccgcacagtgggggacagagtgaccatcacttgtcgggcctccc aaggcatctcgtcatacctggcctgggtatcagcagaaaaccggaaaggctccaaagctgctcatctacaccgcctcga ctctgcaatccggagtgccctccccgcttctccggatccgggtcgggaaccgacttcaccctcaccattagcagccttcag ccggaagatttcgcgacctactactgccagcaaaagctggaccaggcccaggacatttggccagggcactaaggctga gattaag
SEQ ID NO: 182	Light Chain	DIQMTQSPSSLSASVGDRVTITCRASQGISSYLAWYQQKPKAPKLLIYTAS TLQSGVPSRFRSGSGSDFTLTISLQPEDFATYYCQSWTRPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSPVPT KSFNRGEC
SEQ ID NO: 221	DNA Light Chain	gacatccagatgaccagctctccgtcctccctgtccgcacagtgggggacagagtgaccatcacttgtcgggcctccc aaggcatctcgtcatacctggcctgggtatcagcagaaaaccggaaaggctccaaagctgctcatctacaccgcctcga ctctgcaatccggagtgccctccccgcttctccggatccgggtcgggaaccgacttcaccctcaccattagcagccttcag ccggaagatttcgcgacctactactgccagcaaaagctggaccaggcccaggacatttggccagggcactaaggctga gattaagcgtacgggtggcgcctcccagcgtgttcatttccccccagcagcagcagctgaagagcggcaccgcca gcgtgggtgtcctgtgaacaacttctacccccgggaggccaaggtgcagtggaaggtggacaacgcctgcagag cggcaacagccaggagagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcacctgacct gagcaaggccgactacgagaagcataagggtgtacgcctgcaggtgaccaccagggcctgtccagccccgtgacc aagagcttcaacaggggcgagtgcc
XX20_ LALA		
SEQ ID NO: 28	HCDR1 (Combined)	GFTFSSYWMN
SEQ ID NO: 119	HCDR2 (Combined)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Combined)	DRYSMIYSYGAGAFDY
SEQ ID NO: 31	HCDR1 (Kabat)	SYWMN
SEQ ID NO: 119	HCDR2 (Kabat)	VIESKGNIFYADSVKG
SEQ ID NO: 30	HCDR3 (Kabat)	DRYSMIYSYGAGAFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 120	HCDR2 (Chothia)	ESKGNV
SEQ ID NO: 30	HCDR3 (Chothia)	DRYSMIYSYGAGAFDY
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 121	HCDR2 (IMGT)	IESKGNVI
SEQ ID NO: 36	HCDR3 (IMGT)	ARDYSMIYSYGAGAFDY
SEQ ID NO: 201	VH	QVQLLESGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKLEWVS VIESKGNIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLTVTVSS
SEQ ID NO: 202	DNA VH	caagttcaattgctggaagcggaggtggacttgtccaacctggaggggtcactccgactgtcttgcgctgcatcaggatt caccttttagtagctattggatgaactgggtccggcaggctcctgggaagggttgagtgggttaagtgtcattgaatcaa agggaactacatcttttatgctgattctgtaaaagggtaggttcacctctccagggaacaattcaaaaaatactttgtatctg cagatgaactctctcagggcagaagacacggccgtttattactgcgcccgcgatcgatacagcatgatatactcctaag gcgcaggagcttttgactactgggttcaaggcacacttgttactgtcagtagc
SEQ ID NO: 203	Heavy Chain	QVQLLESGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKLEWVS VIESKGNIFYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARDR YSMIYSYGAGAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALG CLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGT QTYICNVNHKPSNTKVDKREPKSCDKHTCTPPCPAPEAAGGPSVFLFPPKP KDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY NSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREP

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 204	DNA Heavy Chain	QVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPV LDSGGSFPLYSKLTVDKSRWQQGNVFSVSMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 204	DNA Heavy Chain	caagttcaattgctggaagcggaggtggacttgtccaacctggaggggcactccgactgtcttgcgctgcatcaggatt caccttttagtagctattggatgaactgggtccggcagggtcctgggaaagggttgagtgggtaagtgtcattgaatcaa agggcaactacatcttttatgctgattctgtaaagggtagggttcaccatctccagggaacaattcaaaaaatactttgtatctg cagatgaactctctcagggcagaagacacggccgtttattactgcgcccgcgatcgatacagcatgatatactcctacg gcgcaggagcttttgactactgggggtcaaggcacacttggtactgtcagtagcgccctcaacgaaaggaccgtccgtgttt cctcttgcctcctagctccaatccacctcaggtggaacggccgcccctggggtgccttggtaaaggactattcccagagc cagttactgtgtcttggaattctgggtcattgacaagtggcgtagcacacttttcccgcggtcctccaatctagtggtctgtac tcactgtcctccgttggtgactgtcccaagtagctcacttggcacacagacttacatctgtaattgtaataacataagccgtcaaa cacgaaggtggataaaggggtagaacctaatgtcatgtgacaaaaacgcatacttgccccccctgcctgcgcgggaag ccgtcggcggaccctccgtattcttgttccctccaaagccaaaggacactctgatgattagccggacacccggaggtcac ttgtgtttgtagtgcagctcagccatgaggatcctgaggtgaaatttaattggtagctggacggggttgaagtccacaatgc taaaactaaacctagggaagagcaatataatagtacatacaggggttgtagtgcgtgacctgttcccatcaggactggc tgaacggcaaggaatacaagtgcagggtcagcaacaaggccttgccggcccccatcgagaagacgatctccaaagc caaggggcaaccccgagaaccgcaggtatacacgctccccctagtagagaagagatgacaaagaatcaagtttccct gacgtgccttgtagaaggcttctaccctagtacatcgacgtcgatgggagagcaacgggcagccggagagaataacta taaaacaacccccccgtgcttgactcagacgggtcatttttctgtatagcaaatgactgttgataaatcacggtggcaa caaggaaacgtgttttagttgacgctaagtcacgaagctctccacaatactactcaaaagtcactgtcactctccct ggcaag
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 184	LCDR3 (Combined)	QQIWMAPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 184	LCDR3 (Kabat)	QQIWMAPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 185	LCDR3 (Chothia)	IWMAPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 184	LCDR3 (IMGT)	QQIWMAPRT
SEQ ID NO: 186	VL	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPKGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQIWMAPRTFGQGTKV EIK
SEQ ID NO: 222	DNA VL	gacattcaaatgactcagtcctccctcatctttgtcagcatcagttggggacaggggtgacaatcacatgccgagcctcaca ggggatttctagctatcttgcattggtaccaacagaagcccggaagcccccaagcttttgatatatacggcatccactct tcagagcggagtagccagtaggtttagtggctccgggagtggtacggactttactctgacgatttccctcccttcaacctga agactttgcaacgtattactgtcagcaaatatggatggctcccagaacgtttggtcaaggtactaaagttgaaataaag
SEQ ID NO: 188	Light Chain	DIQMTQSPSSLSASVGDRTITCRASQGISSYLAWYQQKPKGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQIWMAPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNFFYPREAKVQWKVDNALQ SNSQESVTEQDSKDSYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID NO: 223	DNA Light Chain	gacattcaaatgactcagtcctccctcatctttgtcagcatcagttggggacaggggtgacaatcacatgccgagcctcaca ggggatttctagctatcttgcattggtaccaacagaagcccggaagcccccaagcttttgatatatacggcatccactct tcagagcggagtagccagtaggtttagtggctccgggagtggtacggactttactctgacgatttccctcccttcaacctga agactttgcaacgtattactgtcagcaaatatggatggctcccagaacgtttggtcaaggtactaaagttgaaataaagc aactgtagcagcacctagtgtatttatctccccctctctgatgaacagttgaagtcgggacgggtccgtcgtatgtctc

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
ctgaacaacttttaccgaaggaggcaaaaggtgcaatggaaggtggataatgcactccagagtggcaatagccaaga atcagtaaccgaacaggattccaaggattctacctacagcctttcctctacgcttacattgagcaaggcggactatgaaa gcataaggtgtatgcgtgcgaagtaacacaccagggtctcagcagtcagttacgaagtccttcaatcggggagaatgt		
XX20_		
DAPA		
SEQ ID	HCDR1	GFTFSSYWMN
NO: 28	(Combined)	
SEQ ID	HCDR2	VIESKGNIFYADSVKG
NO: 119	(Combined)	
SEQ ID	HCDR3	DRYSMIYSYGAGAFDY
NO: 30	(Combined)	
SEQ ID	HCDR1	SYWMN
NO: 31	(Kabat)	
SEQ ID	HCDR2	VIESKGNIFYADSVKG
NO: 119	(Kabat)	
SEQ ID	HCDR3	DRYSMIYSYGAGAFDY
NO: 30	(Kabat)	
SEQ ID	HCDR1	GFTFSSY
NO: 32	(Chothia)	
SEQ ID	HCDR2	ESKGNV
NO: 120	(Chothia)	
SEQ ID	HCDR3	DRYSMIYSYGAGAFDY
NO: 30	(Chothia)	
SEQ ID	HCDR1	GFTFSSY
NO: 34	(IMGT)	
SEQ ID	HCDR2	IESKGNVI
NO: 121	(IMGT)	
SEQ ID	HCDR3	ARDRYSMIYSYGAGAFDY
NO: 36	(IMGT)	
SEQ ID	VH	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKLEWVS
NO: 122		VIESKGNIFYADSVKGRFTISRDNKNTLYLQMNLSRAEDTAVYYCARDR
		YSMIYSYGAGAFDYWGQGLTVTVSS
SEQ ID	DNA VH	gaagtgcagctgctggaatccggcggaggtctgggtccagcctggaggtccctgcgcctgtcatgcgcagcctccgg
NO: 207		attcaccttttcgtcgactggatgaactgggtcagacaggctcctggaaaggcctggaatgggtgtctgtgatgaaac
		caaggggaactacatcttctacgcggacagcgtgaaggccgggttcactatcagcagagacaacagcaagaacaccc
		tgtacctccaaatgaactcgctgagggccgaagatactgccgtgtactactgtgcccgcatcgctactcgatgatctac
		agctatggtgccggagcgttcgattactggggacagggaaacctcgtagccgtcagctcc
SEQ ID	Heavy	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKLEWVS
NO: 208	Chain	VIESKGNIFYADSVKGRFTISRDNKNTLYLQMNLSRAEDTAVYYCARDR
		YSMIYSYGAGAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALG
		CLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGT
		QTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPPKP
		KDTLMISRTPEVTCVVAVVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY
		NSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALAAPIEKTISKAKGQPREP
		QVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPPV
		LDSDGSFFLYSKLTVDKSRWQQGNVFPSCVMHEALHNHYTQKSLSLSPGK
SEQ ID	DNA	gaagtgcagctgctggaatccggcggaggtctgggtccagcctggaggtccctgcgcctgtcatgcgcagcctccgg
NO: 209	Heavy	attcaccttttcgtcgactggatgaactgggtcagacaggctcctggaaaggcctggaatgggtgtctgtgatgaaac
	Chain	caaggggaactacatcttctacgcggacagcgtgaaggccgggttcactatcagcagagacaacagcaagaacaccc
		tgtacctccaaatgaactcgctgagggccgaagatactgccgtgtactactgtgcccgcatcgctactcgatgatctac
		agctatggtgccggagcgttcgattactggggacagggaaacctcgtagccgtcagctccgctcccaacaaaggcc
		gtcagtggtcccgctgggtccatcgctgaagtcacacctccggaggaaaccgcagcactcggttgccgtggtcaaggactac
		ttccctgagccagtgaccgtgtcggtggaacagcggagccctgactccggcggtgcacacttttcccgcggtgtgcagct
		cctccggtctgtactccctttcgctccgtgggtcaccgtgccgtcgctcagcctgggcacccagacctacatctgcaacgtg
		aaccacaagccgtccaacaccaaagtgataagcgggtggagccgaagtcctgcgatcaagacacacacgtgcccg
		catgtccagcgcctgaattgcttggcggaccttccgtgttccgtgtcccgccctaaagcccaaggacaccttgatgattagcc
		ggactcccggaagtcacctgtgtggtgggtggcagtgctcccagaggaccccgaggtcaagtttaattggtagctggacg
		gcgtcgaagtgacacacccaagactaagccccgggaggaacagtaacaacagcactaccgggtcggtgccgtgct
		gaccgtgtgcaccaggaactggctgaatgggaaagagtaacaagtgtccaaacaggccttggccgctcctat
		cgaaaaaactatcagcaaggctaagggaacagccgagggaaacccaagtcacacccctgcccccttcacgcgaagag
		atgaccaagaatcaagtgtcgctgacctgctcgtaagggtctacccctccgacattgcggtggagtgaggagtgca

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		acggccagcccgagaaactacaagactactccgcectgctggactccgacggcagcttcttctctgtattccaagct gaccgtggacaagtcccgggtggcagcaaggaaacgtgttctctctgctcggtcatgcacgaagccctgcacaaccacta tacgcagaagtcctgtccttgagccccgggaaa
SEQ ID NO: 41	LCDR1 (Combined)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Combined)	TASTLQS
SEQ ID NO: 184	LCDR3 (Combined)	QQIWMAPRT
SEQ ID NO: 41	LCDR1 (Kabat)	RASQGISSYLA
SEQ ID NO: 42	LCDR2 (Kabat)	TASTLQS
SEQ ID NO: 184	LCDR3 (Kabat)	QQIWMAPRT
SEQ ID NO: 44	LCDR1 (Chothia)	SQGISSY
SEQ ID NO: 45	LCDR2 (Chothia)	TAS
SEQ ID NO: 185	LCDR3 (Chothia)	IWMAPR
SEQ ID NO: 47	LCDR1 (IMGT)	QGISSY
SEQ ID NO: 45	LCDR2 (IMGT)	TAS
SEQ ID NO: 184	LCDR3 (IMGT)	QQIWMAPRT
SEQ ID NO: 186	VL	DIQMTQSPSSLSASVGDRVTITCRASQGISSYLAHWYQQKPKGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQIWMAPRTFGQGTKV EIK
SEQ ID NO: 224	DNA VL	gacatccagatgaccagtcctccgtcctccctgtccgcatcagtgggggacagagtgaccatcacttgtcgggcctccc aaggcatctcgtcatacctggcctgggtatcagcagaaacccggaaaggctccaaagctgctcatctacacgcctcga ctctgcaatccggagtgcttccccgtctctccggatccgggttcgggaaccgacttcacctcaccattagcagccttcag ccggaagatttcgcgacctactactgccagcaaatctggatggccccccaggacatttgccaggggcactaaggctcag attaag
SEQ ID NO: 188	Light Chain	DIQMTQSPSSLSASVGDRVTITCRASQGISSYLAHWYQQKPKGKAPKLLIYTAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQIWMAPRTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKDSSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID NO: 225	DNA Light Chain	gacatccagatgaccagtcctccgtcctccctgtccgcatcagtgggggacagagtgaccatcacttgtcgggcctccc aaggcatctcgtcatacctggcctgggtatcagcagaaacccggaaaggctccaaagctgctcatctacacgcctcga ctctgcaatccggagtgcttccccgtctctccggatccgggttcgggaaccgacttcacctcaccattagcagccttcag ccggaagatttcgcgacctactactgccagcaaatctggatggccccccaggacatttgccaggggcactaaggctcag attaagcgtacgggtggcgcctcccagcgtgttcatcttccccccagcgacgagcagctgaagagcggcaccgccag cgtgggtgctcgtgaacaacttctacccccgggaggccaagggtgcagtggaagggtggacaacgcccctgcagagc ggcaacagccaggagagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccctgaccctg agcaaggccgactacgagaagcataagggtgtacgcctgcgaggtgaccaccaggggcctgtccagccccgtgacca agagcttcaacaggggcgagtg
YY01_ LALA		
SEQ ID NO: 226	HCDR1 (Combined)	GFTFSSYWIS
SEQ ID NO: 227	HCDR2 (Combined)	NIKQSGSETYYVESVKG
SEQ ID NO: 228	HCDR3 (Combined)	SLRRRSTEHAGFDV

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 229	HCDR1 (Kabat)	SYWIS
SEQ ID NO: 227	HCDR2 (Kabat)	NIKQSGSETYYVESVKG
SEQ ID NO: 228	HCDR3 (Kabat)	SLRRRSTEHAGFDV
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 230	HCDR2 (Chothia)	KQSGSE
SEQ ID NO: 228	HCDR3 (Chothia)	SLRRRSTEHAGFDV
SEQ ID NO: 34	HCDR1 (IMGT)	GFTFSSYW
SEQ ID NO: 231	HCDR2 (IMGT)	IKQSGSET
SEQ ID NO: 232	HCDR3 (IMGT)	ARSLRRRSTEHAGFDV
SEQ ID NO: 233	VH	EVQLVESGGGLVQPGGSLRLSCAASGFTFSSYWISWVRQAPGKGLEWVAN IKQSGSETYYVESVKGRFTISRDNKNSLYLQMNSLRAEDTAVYYCARSLR RRSTEHAGFDVWGQTLTVSS
SEQ ID NO: 234	DNA VH	gaagtgcagctgggtggaagcggcggtggcctgggtgcagccaggtggtagcctgcgcctgagctgcgcgcgcagc ggcctttaccttttagcagctattggattagctgggttcgccaggccccaggcaaaaggcctggaatgggtggcgaacatca aacagagcggcagcgagacctactatgtggagagcgtgaaaggccgctttaccattagccgcgataacgccccaaaac agcctgtatctgcaaatgaacagcctgcgggcccgaagataccgcccgtgtattattgcgcgcgtagcctgcgtcgta gcactgagcacgcaggattcgacgtttggggccagggcaccctggttactgtctcgagc
SEQ ID NO: 235	Heavy Chain	EVQLVESGGGLVQPGGSLRLSCAASGFTFSSYWISWVRQAPGKGLEWVAN IKQSGSETYYVESVKGRFTISRDNKNSLYLQMNSLRAEDTAVYYCARSLR RRSTEHAGFDVWGQTLTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLV KDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTY ICNVNHKPSNTKVDKRVPEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDT LMI SRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVHLQDNLNGKEYKCKVSNKALPAPIEKTI SKAKGQPREPQVY LPPEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSD GSFFLYSKLTVDKSRWQQGNVPSCSVMHEALHNYHTQKSLSLSPGK
SEQ ID NO: 236	DNA Heavy Chain	gaagtgcagctgggtggaagcggcggtggcctgggtgcagccaggtggtagcctgcgcctgagctgcgcgcgcagc ggcctttaccttttagcagctattggattagctgggttcgccaggccccaggcaaaaggcctggaatgggtggcgaacatca aacagagcggcagcgagacctactatgtggagagcgtgaaaggccgctttaccattagccgcgataacgccccaaaac agcctgtatctgcaaatgaacagcctgcgggcccgaagataccgcccgtgtattattgcgcgcgtagcctgcgtcgta gcactgagcacgcaggattcgacgtttggggccagggcaccctggttactgtctcgagcgcgtcgacccaaaggcccc agcctggttccctctggccccagcagcaagagcacctctggcggaaacagcgcctgggctgctggtcaaggacta cttccccgagcccgtagcctgtctctggaactctggcgccctgaccagcggcgtagcacacctttccagccgtgctcca gagcagcggcctgtacagcctgagcagcgtcgtagcctgcccagcagcagcctgggcaaccagacctacatctgc aacgtgaaccacaagcccagcaacacaaggtggacaagcgggtggaacccaagagctgcgacaagaccacac ctgtccccctgcccctgcccctgaagcggcgaggccccctcctgtgttccctgttcccccaagcctaaggacaccct gatgatcagcgggacccccgaagtgcacctgcgtggtggtggaagctgtcccacgaggacacctgaagtgaagtttaattg gtacgtggacggcggtggaagtgcacaacgccaagccaagccagagaggaacagtaacaacagacctaccgggt ggtgtccgtgctgacctgctgcaccaggactggctgaacggcaagagtaacaagtgaaggtgtccaacaaaggccc tgccctgcccccatcgagaaaacctcagcaaggccaaaggccagccccgcgagccccaggtgtacacactgcccc tagccgggaagagatgaccaagaaccagggtgctccctgacctgctcgtgaagggttctaccccagcgacattgcccgt ggaatgggagagcaacggccagcccgagaaactacaagaccacccccctgtgctggacagcgacggctcattc ttcctgtacagcaagctgacctggacaagagccggtggcagcagggcaacctgttcagctgctccgtgatgcacga ggcctgtcacaaacctacaccagaagtccctgagcctgagccccggcaag
SEQ ID NO: 237	LCDR1 (Combined)	RASQGISNYLA
SEQ ID NO: 238	LCDR2 (Combined)	AASTLQS
SEQ ID NO: 239	LCDR3 (Combined)	QQADKFPYT
SEQ ID NO: 237	LCDR1 (Kabat)	RASQGISNYLA

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 238	LCDR2 (Kabat)	AAS
SEQ ID NO: 239	LCDR3 (Kabat)	QQADKFPYT
SEQ ID NO: 240	LCDR1 (Chothia)	SQGISNY
SEQ ID NO: 241	LCDR2 (Chothia)	AAS
SEQ ID NO: 242	LCDR3 (Chothia)	ADKFPY
SEQ ID NO: 243	LCDR1 (IMGT)	QGISNY
SEQ ID NO: 241	LCDR2 (IMGT)	AAS
SEQ ID NO: 239	LCDR3 (IMGT)	QQADKFPYT
SEQ ID NO: 244	VL	DIQMTQSPSSLSASVGDRTITCRASQGISNYLAWYQQKPKGVKLLIYAAS TLQSGVPSRFRSGSGTDFTLTISLQPEDVATYYCQQADKFPYTFGGGTKV EIK
SEQ ID NO: 245	DNA VL	gatattcagatgaccagagcccgagcagcctgagcgcaagcgtggcgatcgcgtagaccattacctgccgcgcca gccagggcattagcaactatctggcctggtatcagcagaaaacgggcaaaagtgccgaaactgctgatctatgccgcca gcacctgcaaaagcggcgtgccaagtcgcttagcggcagcggtagcggcaccgatttcacctgaccattagcagc ctgcaaccggaagacgtggcgacctattattgccagcaggctgacaaattcccgtagaccttcggccagggtacccaa gtggaatcaag
SEQ ID NO: 246	Light Chain	DIQMTQSPSSLSASVGDRTITCRASQGISNYLAWYQQKPKGVKLLIYAAS TLQSGVPSRFRSGSGTDFTLTISLQPEDVATYYCQQADKFPYTFGGGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID NO: 247	DNA Light Chain	gatattcagatgaccagagcccgagcagcctgagcgcaagcgtggcgatcgcgtagaccattacctgccgcgcca gccagggcattagcaactatctggcctggtatcagcagaaaacgggcaaaagtgccgaaactgctgatctatgccgcca gcacctgcaaaagcggcgtgccaagtcgcttagcggcagcggtagcggcaccgatttcacctgaccattagcagc ctgcaaccggaagacgtggcgacctattattgccagcaggctgacaaattcccgtagaccttcggccagggtacccaa gtggaatcaagcggaccgtggcgctccctccgtgttcatttcccccagcgagcagcgtgaagtccggcaca gccagcgtcgtgtgctgctgaacaacttctaccccgcgaggccaaagtgcagtggaaggtggacaacgccctcca gagcggcaacagccaggaaagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcacctgac cctgagcaaggccgactacgagaagcacaaggtgtacgcctgcgaagtgacccaccagggcctgtccagccccgtg accaagagcttcaaccggggcgagtgt
YY02_ LALA		
SEQ ID NO: 248	HCDR1 (Combined)	GFTFSSYSMN
SEQ ID NO: 249	HCDR2 (Combined)	SISSSSSIYYADSVKG
SEQ ID NO: 250	HCDR3 (Combined)	SGYRGVYGFY
SEQ ID NO: 251	HCDR1 (Kabat)	SYSMN
SEQ ID NO: 249	HCDR2 (Kabat)	SISSSSSIYYADSVKG
SEQ ID NO: 250	HCDR3 (Kabat)	SGYRGVYGFY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 252	HCDR2 (Chothia)	SSSSSY

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 250	HCDR3 (Chothia)	SGYRGVYGFYD
SEQ ID NO: 253	HCDR1 (IMGT)	GFTFSSYS
SEQ ID NO: 254	HCDR2 (IMGT)	ISSSSSYI
SEQ ID NO: 255	HCDR3 (IMGT)	ARSGYRGVYGFYD
SEQ ID NO: 256	VH	EVQLVESGGGLVQPKGSLRSLSCAASGFTFSSYSMNWVRQAPGKLEWVSS ISSSSYIYYADSVKGRFTISRDNKNSLYLQMNSLRAEDTAVYYCARSGY RGVYGFYDYGQTLVTVSS
SEQ ID NO: 257	DNA VH	gaagtgcagctgggtggaagcgcggtggcctgggtgaaaccaggcggtagcctgcgcctgagctgcgcgccagc ggctttaccttttagcagctatagcatgaactgggttcgccaggccccaggcaaggcctggaatgggttagcagcatca gcagcagtagcagctatatctattacgcgatagcgtgaaaggccgctttaccattagccgcgataacgcaaaaacag cctgtatctgcaaatgaacagcctgcgggcccgaagataccgcctgtattattgcgcgcgaagcggatattcgtggagtt tacggattgattattggggccagggcaccctgggtactgtctcgcgc
SEQ ID NO: 258	Heavy Chain	EVQLVESGGGLVQPKGSLRSLSCAASGFTFSSYSMNWVRQAPGKLEWVSS ISSSSYIYYADSVKGRFTISRDNKNSLYLQMNSLRAEDTAVYYCARSGY RGVYGFYDYGQTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKD YFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC NVNHHKPSNTKVDKRVKPKSCDKHTHTCPPCPAPEAAGGPSVFLFPPKPKDTL MISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR VVSIVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTKAKGQPREPQVYTLTP PSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGS FFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 259	DNA Heavy Chain	gaagtgcagctgggtggaagcgcggtggcctgggtgaaaccaggcggtagcctgcgcctgagctgcgcgccagc ggctttaccttttagcagctatagcatgaactgggttcgccaggccccaggcaaggcctggaatgggttagcagcatca gcagcagtagcagctatatctattacgcgatagcgtgaaaggccgctttaccattagccgcgataacgcaaaaacag cctgtatctgcaaatgaacagcctgcgggcccgaagataccgcctgtattattgcgcgcgaagcggatattcgtggagtt tacggattgattattggggccagggcaccctgggtactgtctcgcgcgctgcacaaaggccccagcgtgttccctct ggccccagcagcaagagcacctctggcggaacagccgccccgggtgcctggtaaggactacttccccgagccc gtgaccgtgtcctggaactctggcgccccgaccagcgcggtgcacacctttccagccgtgctccagagcagcgccctg tacagcctgagcagcgtcgtgaccgtgcccagcagcagcctgggcaaccagacctacatctgcaacgtgaaccacaa gcccagcaacacaaagggtggacaagcggtggaaaccaagagctgcgacaagaccacacctgtccccctgcct gccccctgaagcgcgaggccccctccgtgttccctgttcccccaaaagcctaaggacacctgatgatcagccggac ccccgaagtgaacctgcgtgggtggtagcgtgtccacagaggacctgaagtgaagtttaatttggtacgtggacggcgt ggaagtgcacaacgccaagaccagccagagaggaaacagtaacaacgacacctaccgggtgggtgcctgctgacc gtgctgcaccaggactggctgaacggcaaaagagtaacaagtgaaggtgtccacaagggccctgcctgccccatcga gaaaaccatcagcaaggccaaaggccagccccgcgagccccaggtgtacacactgccccctagccgggaagagat gaccaagaaccaggtgtccctgacctgcctcgtgaagggttctaccccagcgacattgccgtggaatgggagagca acggccagccccgagaacaactacaagaccacccccctgtgctggacagcagcgtcattcttccgttacagcaag ctgaccgtggacaagagccgtggcagcagggaacgtgtcagctgctccgtgatgcacagggccctgcacaaacc actacaccagaagtccctgagcctgagccccggcaag
SEQ ID NO: 260	LCDR1 (Combined)	RASQGISSWLA
SEQ ID NO: 261	LCDR2 (Combined)	AASSLQS
SEQ ID NO: 262	LCDR3 (Combined)	QYYHSPLT
SEQ ID NO: 260	LCDR1 (Kabat)	RASQGISSWLA
SEQ ID NO: 261	LCDR2 (Kabat)	AASSLQS
SEQ ID NO: 262	LCDR3 (Kabat)	QYYHSPLT
SEQ ID NO: 263	LCDR1 (Chothia)	SQGISSW
SEQ ID NO: 241	LCDR2 (Chothia)	AAS
SEQ ID NO: 264	LCDR3 (Chothia)	YYHSPL

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 265	LCDR1 (IMGT)	QGISSW
SEQ ID NO: 241	LCDR2 (IMGT)	AAS
SEQ ID NO: 262	LCDR3 (IMGT)	QQYYHSPLT
SEQ ID NO: 266	VL	DIQMTQSPSSVSASVGDRTITCRASQGISSWLAWYQQKPGKAPKLLIYAA SSLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQYYHSPLTFGQGTK VEIK
SEQ ID NO: 267	DNA VL	gatattcagatgaccagagcccgagcagcgttagcgccagcgtggcgatcgcgtagaccattacctgccgcgccag tcagggcattagcagctggctggcctggatcagcagaaacggggcaagccccgaaactgctgatctatgccgcc gcagcctgcaaagcggcgtgccaagtgcgttagcggcagcggtagcggcaccgatttcaccctgaccattagcagtc tgcaaccggaagactttgccacctattattgccagcagtagtaccattctcgctgaccttcggccagggtaccaaagt gaaatcaag
SEQ ID NO: 268	Light Chain	DIQMTQSPSSVSASVGDRTITCRASQGISSWLAWYQQKPGKAPKLLIYAA SSLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQYYHSPLTFGQGTK VEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNAL QSGNSQESVTEQDSKDSTYSLSSTLTLSKADYERHKVYACEVTHQGLSSPV TKSFNRGEC
SEQ ID NO: 269	DNA Light Chain	gatattcagatgaccagagcccgagcagcgttagcgccagcgtggcgatcgcgtagaccattacctgccgcgccag tcagggcattagcagctggctggcctggatcagcagaaacggggcaagccccgaaactgctgatctatgccgcc gcagcctgcaaagcggcgtgccaagtgcgttagcggcagcggtagcggcaccgatttcaccctgaccattagcagtc tgcaaccggaagactttgccacctattattgccagcagtagtaccattctcgctgaccttcggccagggtaccaaagt gaaatcaagcggcagcgtggcgcctccctcgctgttcattctccaccagcgcagcagcagctgaagtccggcacagc cagcgtcgtgtgcctgctgaacaattctaccccgcgaggccaaagtgcagtggaaggtggacaacgcccctccaga gcggcaacagccaggaaagcgtcacccagcaggacagcaaggactccacctacagcctgagcagcacccctgaccc tgagcaaggccgactacgagaagcacaaggtgtacgcctgcgaagtgacccaccagggcctgtccagccccgtgac caagagcttcaaccggggcgagtg
YY03_ LALA		
SEQ ID NO: 270	HCDR1 (Combined)	GFTFSSYAIS
SEQ ID NO: 271	HCDR2 (Combined)	AISGSGGSTYYAESVKG
SEQ ID NO: 272	HCDR3 (Combined)	ESGYVYYLKFDY
SEQ ID NO: 273	HCDR1 (Kabat)	SYAIS
SEQ ID NO: 271	HCDR2 (Kabat)	AISGSGGSTYYAESVKG
SEQ ID NO: 272	HCDR3 (Kabat)	ESGYVYYLKFDY
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 274	HCDR2 (Chothia)	SGSGGS
SEQ ID NO: 272	HCDR3 (Chothia)	ESGYVYYLKFDY
SEQ ID NO: 275	HCDR1 (IMGT)	GFTFSSYA
SEQ ID NO: 276	HCDR2 (IMGT)	ISGSGGST
SEQ ID NO: 277	HCDR3 (IMGT)	ARESGYVYYLKFDY
SEQ ID NO: 278	VH	EVQLLESQGLVQPGGSLRLSCAASGFTFSSYAIISWVRQAPGKGLEWVSAI SGSGGSTYYAESVKGRFTISRDN SKNTLYLQMNSLR AEDTAVYYCARESG YVYYLKFDYWGQGLTVTVSS

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 279	DNA VH	gaagtgcagctgctggaaagcgggtggcggtctggtgcagccaggtggtagcctgcgcctgagctgtgccgcaagcg gctttaccttttagcagctatgccattagctgggtgcgccaagcaccaggcaaggcctggaatgggtgagcgccattag cggcagcgggtggcagcacctattatgccgagagcgtgaaaggctcgtttaccattagtcgcgataacagcaaaaacac cctgtatctgcaaatgaacagcctgcgggcagaagataccgcagtttatattgcgcgcgtgagagcggatacgtttact atctgaaattcgattatttggggccagggcaccctggttactgtctcgagc
SEQ ID NO: 280	Heavy Chain	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAISWVRQAPGKGLEWWSAI SGSGGSTYYAESVKGRFTISRDNKNTLYLQMNSLRAEDTAVVYCARESG YVYYLKFYWGQGTLLTVSSASTKGPSVFPLAPSSKSTSGGTALGCLVK DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYI CNVNHKPSNTKVDKRVEPKSCDKHTHTCPPCPAPEAAGGPSVFLFPPKPKDT LMI SRTPETVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTT LPPSRREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSD GSFFLYSKLTVDKSRWQQGNVPSCSVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 281	DNA Heavy Chain	gaagtgcagctgctggaaagcgggtggcggtctggtgcagccaggtggtagcctgcgcctgagctgtgccgcaagcg gctttaccttttagcagctatgccattagctgggtgcgccaagcaccaggcaaggcctggaatgggtgagcgccattag cggcagcgggtggcagcacctattatgccgagagcgtgaaaggctcgtttaccattagtcgcgataacagcaaaaacac cctgtatctgcaaatgaacagcctgcgggcagaagataccgcagtttatattgcgcgcgtgagagcggatacgtttact atctgaaattcgattatttggggccagggcaccctggttactgtctcgagcgcgtcgaccaaaaggccccagcgtgttccct ctggccccagcagcaagagcacctctggcggaacagcgcgcctgggtgcctgggtcaaggactacttccccgagc cgtgacccgtgtcctggaactctggcgccctgaccagcggcgtgcacacctttccagccgtgctccagagcagcggc ctgtacagcctgagcagcgtcgtgaccctgcccagcagcagcctgggcaccagacctacatctgcaacgtgaacca caagcccagcaacacaaagggtggacaagcgggtggaacccaagagctgcgacaagaccacacctgtccccctg ccctgccccgaagcgggaggaggccctccgtgttccgtgttcccccaaggcctaaggacacctgatgatcagccg gacccccgaagtacctgcgtggtggtggacgtgtcccacgaggacctgaagtgaagttaaattggtacgtggacgg cgtggaagtgcacaacgccaagaccaagcccagagaggaacagtaacaacagcacctaccgggtggtgtccgtgctg accgtgctgcaccaggactggtgaacggcgaagagtacaagtgcagggtgtccaacaaggccctgcctgcccccat cgagaaaaccatcagcaaggccaaaggccagccccgcgagccccaggtgtacacactgccccctagccgggaaga gatgaccaagaaccaggtgtccctgacctgcctcgtgaagggcttctaccccagcgacattgcccgtggaatgggagag caacggccagcccagagaacaactacaagaccacccccctgtgctggacagcgacggctcatcttctctgtacagca agctgaccgtggacaagagccggtggcagcagggcaacgtgttcagctgctccgtgatgcacgaggccctgcacaa ccactacaccagaagtccctgagcctgagccccggcgaag
SEQ ID NO: 282	LCDR1 (Combined)	RASQSISSYLN
SEQ ID NO: 261	LCDR2 (Combined)	AASSLQS
SEQ ID NO: 283	LCDR3 (Combined)	QQHVRVPIT
SEQ ID NO: 282	LCDR1 (Kabat)	RASQSISSYLN
SEQ ID NO: 261	LCDR2 (Kabat)	AASSLQS
SEQ ID NO: 283	LCDR3 (Kabat)	QQHVRVPIT
SEQ ID NO: 284	LCDR1 (Chothia)	SQSISSY
SEQ ID NO: 241	LCDR2 (Chothia)	AAS
SEQ ID NO: 285	LCDR3 (Chothia)	HVRVPI
SEQ ID NO: 286	LCDR1 (IMGT)	QSISSY
SEQ ID NO: 241	LCDR2 (IMGT)	AAS
SEQ ID NO: 283	LCDR3 (IMGT)	QQHVRVPIT
SEQ ID NO: 287	VL	DIQMTQSPSSLSASVGRVITITCRASQSISSYLNWYQQKPKGKAPKLLIYAAS SLQSGVPSRFSGSGSDFTLTITSLQPEDFATYYCQQHVRVPITFGQGTKVE IK
SEQ ID NO: 288	DNA VL	gatattcagatgacccagagccccgagcagcctgagcgcagcgtgggtgatcgcgtagcattacctgtcgcgcaag ccagagcattagcagctatctgaactggtatcagcagaaaccaggcaaaagccccaaaactgctgatttatgccgcaag cagcctgcaaaagcgggtgtgccgagccgctttagcggcagcggtagcggcaccgattttaccctgaccattagtagcct

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		gcaaccggaagactttgccacctattattgccagcagcatgttcgtgttccgatcaccttcggccagggtaccaaagtgg aatcaag
SEQ ID NO: 289	Light Chain	DIQMTQSPSSLSASVGDRVTITCRASQSISSYLNWYQQKPKAPKLLIYAAS SLQSGVPSRFRSGSGSDFTLTISLQPEDFATYYCQQHVRVPIITFGQGTKVE IKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQS GNSQESVTEQDSKDSYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTK SFNRGEC
SEQ ID NO: 290	DNA Light Chain	gatattcagatgacccagagcccgagcagcctgagcgcagcgtgggtgatcgcgtagaccattacctgtcgcgcaag ccagagcattagcagctatctgaactggatcacgagaaaccaggcaaaagcccaaaactgctgatttatgccgcaag cagcctgcaaagcgggtgtgccgagccgctttagcggcagcggtagcggcaccgattttacctgaccattagtagcct gcaaccggaagactttgccacctattattgccagcagcatgttcgtgttccgatcaccttcggccagggtaccaaagtgg aatcaagcggagccgtggcgcctccctccgtgttcattcttccaccagcagcagcagctgaagtccggcacagcc agcgtcgtgtgcctgctgaacaacttctacccccgcgagggccaaagtgcagtggaggtggacaacgccctccagag cggcaacagccaggaaagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccctgaccct gagcaaggccgactacgagaagcacaagggtgtacgcctgcgaagtgaaccaccagggcctgtccagccccgtgac caagagcttcaaccggggcgagtgt
YY04_ LALA		
SEQ ID NO: 291	HCDR1 (Combined)	GFTFSNYWIS
SEQ ID NO: 292	HCDR2 (Combined)	RIKSKTYGGTTDYAEPVKG
SEQ ID NO: 293	HCDR3 (Combined)	EKYSIRARGHGDYGFVDV
SEQ ID NO: 294	HCDR1 (Kabat)	NYWIS
SEQ ID NO: 292	HCDR2 (Kabat)	RIKSKTYGGTTDYAEPVKG
SEQ ID NO: 293	HCDR3 (Kabat)	EKYSIRARGHGDYGFVDV
SEQ ID NO: 295	HCDR1 (Chothia)	GFTFSNY
SEQ ID NO: 296	HCDR2 (Chothia)	KSKTYGGT
SEQ ID NO: 293	HCDR3 (Chothia)	EKYSIRARGHGDYGFVDV
SEQ ID NO: 297	HCDR1 (IMGT)	GFTFSNYW
SEQ ID NO: 298	HCDR2 (IMGT)	IKSKTYGGTT
SEQ ID NO: 299	HCDR3 (IMGT)	AREKYSIRARGHGDYGFVDV
SEQ ID NO: 300	VH	EVQLVESGGGLVKPGGSLRLSCAASGFTFSNYWISWVRQAPGKGLEWVGR IKSKTYGGTTDYAEPVKGRFTISRDDSKNTLYLQMNSLKTEDTAVYYCARE KYSIRARGHGDYGFVDVWGQGLTVTVSS
SEQ ID NO: 301	DNA VH	gaagtgcagctgggtggaagcggcggtggcctgggtgaaaccaggcggtagcctgcgcctgagctgcgcgccagc ggccttacctttagcaactattggattagctgggttcgccaggccccaggcaaaaggcctggaatgggttgcccgcatca aaagcaaaacctatggcggcaccaccgatgatgccgagccagtgaaaggccgctttaccattagccgcgacgatagc aaaaacaccctgtacctgcaaatgaacagcctgaaaaccgaagataccgccgtgtattattgcgcgctgagaaatatt ccatccgtgcacgtgggtcacggagactacggattttagtgtgtggggccagggcaccctggttactgtctcgagc
SEQ ID NO: 302	Heavy Chain	EVQLVESGGGLVKPGGSLRLSCAASGFTFSNYWISWVRQAPGKGLEWVGR IKSKTYGGTTDYAEPVKGRFTISRDDSKNTLYLQMNSLKTEDTAVYYCARE KYSIRARGHGDYGFVDVWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAA LGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSL GTQTYICNVNHKPSNTKVDKRVPEKSCDKHTHTCPPCPAPEAAGGPSVFLFP PKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREE QYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTIISKAKGQPR EPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTP PVLDSDDGSFFLYSKLTVDKSRWQQGNVFSQSVMHREALHNNHTQKSLSLSP GK

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 303	DNA Heavy Chain	gaagtgcagctggtggaaagcggcggtggcctgggtgaaccaggcggtagcctgcgcctgagctgcgcgcgcagc ggctttacctttagcaactattggattagctgggttcgccaggccccaggcgaaggcctggaatgggttggccgcac aaagcaaacctatggcgccaccaccgatgatgcgagccagtgaaagggcgctttaccattagccgcgacgatagc aaaaacaccctgtacctgcaaatgaacagcctgaaaaccgaagataccgccgtgtattattgcgcgcgtgagaaatatt ccattccgtgcacgtggtcacggagactacggatttgatgtgtggggccagggcaccctggttactgtctcgagcgctc gaccaaagggccccagcgtgttccctctggccccagcagcaagagcacctctggcggaacagcgcgcctgggtgc ctggtcaaggactacttccccgagcccgtagccgtgtcctggaaactctggcgccctgaccagcggcggtgcacaccttc cagccgtgctccagagcagcggcctgtacagcctgagcagcgtcgtgacctgcccagcagcagcctgggcaccca gacctacatctgcaacgtgaaccacaagcccagcaacacaagggtggacaagcgggtggaacccaagagctgcga caagaccacacacctgtccccctgcccctgcccctgaagcggcgggaggccccctccgtgttccctgttcccccaagcc taaggacacctgatgatcagccggacccccgaagtgaactgcgtggtggtggacgtgtcccacgaggacctgaag tgaagtttaattggtagctggcggcggtggaagtgcacaacggcgaagaccaagccagagagggaacagtacaacagc acctaccgggtggtgtccgtgctgacctgctgcaccaggactggctgaacggcgaagagtaacaagtgaaggtgtc caacaagggcctgctgcccccatcgagaaaaccatcagcaaggccaaaggccagccccgcgagccccaggtgt cacactgccccctagccgggaagagatgaccaagaaccaggtgtccctgacctgctcgtgaagggtcttacccca ggcacattgccgtggaatgggagagcaacggccagcccgagaacaactacaagaccacccccctgtgctggacag cgacggctcattctcctgtacagcaagctgacctgggacaagagcgggtggcagcagggcaacgtgttcagctgctc cgtgatgcacgagggccctgcacaaccactacaccagaagtccctgagcctgagccccggcgaag
SEQ ID NO: 237	LCDR1 (Combined)	RASQGISNYLA
SEQ ID NO: 238	LCDR2 (Combined)	AASTLQS
SEQ ID NO: 304	LCDR3 (Combined)	QQGYHAPFT
SEQ ID NO: 237	LCDR1 (Kabat)	RASQGISNYLA
SEQ ID NO: 238	LCDR2 (Kabat)	AASTLQS
SEQ ID NO: 304	LCDR3 (Kabat)	QQGYHAPFT
SEQ ID NO: 240	LCDR1 (Chothia)	SQGISNY
SEQ ID NO: 241	LCDR2 (Chothia)	AAS
SEQ ID NO: 305	LCDR3 (Chothia)	GYHAPF
SEQ ID NO: 243	LCDR1 (IMGT)	QGISNY
SEQ ID NO: 241	LCDR2 (IMGT)	AAS
SEQ ID NO: 304	LCDR3 (IMGT)	QQGYHAPFT
SEQ ID NO: 306	VL	DIQMTQSPSSLSASVGDRTITCRASQGISNYLAWYQQKPKGVKLLIYAAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDVATYYCQQGYHAPFTFGQGTKV EIK
SEQ ID NO: 307	DNA VL	gatattcagatgacccagagccccgagcagcctgagcgcgaagcgtgggcgatcgcgtagaccattacctgccgcgcca gccagggcattagcaactatctggcctggtatcagcagaaaaccgggcaaaagtgcggaactgctgatctatgccgcca gcacctgcaaaagcggcggtgccaagtgcgttttagcggcagcggtagcggcaccgatttcacctgaccattagcagc ctgcaaccggaagacgtggcgacctattattgccagcagggttaccatgctccgttcaccttcggccaggggtaccaaag tggaaatcaag
SEQ ID NO: 308	Light Chain	DIQMTQSPSSLSASVGDRTITCRASQGISNYLAWYQQKPKGVKLLIYAAS TLQSGVPSRFSGSGSGTDFTLTISLQPEDVATYYCQQGYHAPFTFGQGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPV TKSFNREGC
SEQ ID NO: 309	DNA Light Chain	gatattcagatgacccagagccccgagcagcctgagcgcgaagcgtgggcgatcgcgtagaccattacctgccgcgcca gccagggcattagcaactatctggcctggtatcagcagaaaaccgggcaaaagtgcggaactgctgatctatgccgcca gcacctgcaaaagcggcggtgccaagtgcgttttagcggcagcggtagcggcaccgatttcacctgaccattagcagc ctgcaaccggaagacgtggcgacctattattgccagcagggttaccatgctccgttcaccttcggccaggggtaccaaag tggaaatcaagcggacgtggcgctccctccgtgtcatctcccaccagcgacgagcagctgaagtcgggcacag ccagcgtcgtgtgctgctgaacaactctacccccgcgaggccaaagtgcagtggaaggtggacaacgcctccag

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
agcggcaacagccaggaaagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcaccctgacc ctgagcaaggccgactacgagaagcacaaggtgtacgcctgcgaagtgaccaccagggcctgtccagccccgtga ccaagagcttcaaccggggcgagtgt		
YY05_ LALA		
SEQ ID NO: 310	HCDR1 (Combined)	GYSFTSYWIS
SEQ ID NO: 311	HCDR2 (Combined)	IIYPGTSYTRYSPSFQG
SEQ ID NO: 312	HCDR3 (Combined)	GAVAGQLGFDH
SEQ ID NO: 229	HCDR1 (Kabat)	SYWIS
SEQ ID NO: 311	HCDR2 (Kabat)	IIYPGTSYTRYSPSFQG
SEQ ID NO: 312	HCDR3 (Kabat)	GAVAGQLGFDH
SEQ ID NO: 80	HCDR1 (Chothia)	GYSFTSY
SEQ ID NO: 313	HCDR2 (Chothia)	YPGTSY
SEQ ID NO: 312	HCDR3 (Chothia)	GAVAGQLGFDH
SEQ ID NO: 82	HCDR1 (IMGT)	GYSFTSYW
SEQ ID NO: 314	HCDR2 (IMGT)	IYPGTSYT
SEQ ID NO: 315	HCDR3 (IMGT)	ARGAVAGQLGFDH
SEQ ID NO: 316	VH	EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWISWVRQMPGKLEWMGII YPGTSYTRYSPSFQGVITISADKSI STAYLQWSSLKASDTAMYYCARGAVA GQLGFDHWGQGLTVTVSS
SEQ ID NO: 317	DNA VH	gaagtgcagctggtgcagagcgggtgccgaagtgaataaacccggggcgaagcctgaaatcagctgcaaaggcagc ggctatagctttaccagctattggattagctgggttcgccagatgccggggcaaggcctggaatggatgggcattatcta tcggggcaccagctataccgcctatagcccgagctttcaggggcagggtacaattagcgccgacaaaagcatcagcac cgctatctgcaatggagcagcctgaaagccagcgataccgccatgtattattgcgcgcgtggtgcagttgcaggaca actgggatttgatcactggggccaggggcacctgggttactgtctcgagcgcgtcgaccaaaggccccagcgtgtccct ctggccccagcagcaagagcacctctggcggaacagccgcctgggctgcctgggtcaaggactactccccagagc ccgtgaccgtgtcctggaactctggcgccctgaccagcggcggtgcacacctttccagccgtgtccagagcagcggc ctgtacagcctgagcagcgtcgtgaccgtgcccagcagcagcctgggcacccagacctacatctgcaacgtgaacca caagcccagcaacacaaaaggtggacaaggggtggaacccaagagctcgcgacaagaccacacctgtccccctg ccctgcccctgaagcggggaggggccccctcggtgttctctgttccccccaaagcctaaggacacctgatgatcagccg gacccccgaagtgaactgcgtgggtggtggagctgtcccagaggacctgaagtgaagttaaattggtacgtggacgg cgtggaagtgcacaacccaagaccaagcccagagaggaacagtacaacagcactaccgggtggtgtccgtgctg accgtgctgcaccaggactggctgaacggcaagagtgacaagtgaaggtgtccaacaaggccctgctgtcccccat cgagaaaacatcagcaaggccaaaggccagcccccgagccccaggtgtacacactgccccctagccgggaaga gatgaccaagaaccaggtgtccctgacctgcctcgtgaagggttctaccgccagcagattgcccgtggaatgggagag
SEQ ID NO: 318	Heavy Chain	EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWISWVRQMPGKLEWMGII YPGTSYTRYSPSFQGVITISADKSI STAYLQWSSLKASDTAMYYCARGAVA GQLGFDHWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF PEPVTWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNV NHKPSNTKVDKRVPEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMIS RTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVV SVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI SKAKGQPREPQVYITLPPS REEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPPVLDSDGSFF LYSKLTVDKSRWQQGNVFSVSMHEALHNHYTQKLSLSLSPGK
SEQ ID NO: 319	DNA Heavy Chain	gaagtgcagctggtgcagagcgggtgccgaagtgaataaacccggggcgaagcctgaaatcagctgcaaaggcagc ggctatagctttaccagctattggattagctgggttcgccagatgccggggcaaggcctggaatggatgggcattatcta tcggggcaccagctataccgcctatagcccgagctttcaggggcagggtacaattagcgccgacaaaagcatcagcac cgctatctgcaatggagcagcctgaaagccagcgataccgccatgtattattgcgcgcgtggtgcagttgcaggaca actgggatttgatcactggggccaggggcacctgggttactgtctcgagcgcgtcgaccaaaggccccagcgtgtccct ctggccccagcagcaagagcacctctggcggaacagccgcctgggctgcctgggtcaaggactactccccagagc ccgtgaccgtgtcctggaactctggcgccctgaccagcggcggtgcacacctttccagccgtgtccagagcagcggc ctgtacagcctgagcagcgtcgtgaccgtgcccagcagcagcctgggcacccagacctacatctgcaacgtgaacca caagcccagcaacacaaaaggtggacaaggggtggaacccaagagctcgcgacaagaccacacctgtccccctg ccctgcccctgaagcggggaggggccccctcggtgttctctgttccccccaaagcctaaggacacctgatgatcagccg gacccccgaagtgaactgcgtgggtggtggagctgtcccagaggacctgaagtgaagttaaattggtacgtggacgg cgtggaagtgcacaacccaagaccaagcccagagaggaacagtacaacagcactaccgggtggtgtccgtgctg accgtgctgcaccaggactggctgaacggcaagagtgacaagtgaaggtgtccaacaaggccctgctgtcccccat cgagaaaacatcagcaaggccaaaggccagcccccgagccccaggtgtacacactgccccctagccgggaaga gatgaccaagaaccaggtgtccctgacctgcctcgtgaagggttctaccgccagcagattgcccgtggaatgggagag

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		caacggccagccccgagaacaactacaagaccacccccctgtgctggacagcgagcggtcattcttctgtacagca agctgaccgtggacaagagccggtggcagcagggcaacgtgttcagctgctccgtgatgcacgagccctgcacaa ccactacaccagaagtccctgagcctgagccccggcaag
SEQ ID NO: 320	LCDR1 (Combined)	TGSSSNIGAGYDVH
SEQ ID NO: 321	LCDR2 (Combined)	GNSNRPS
SEQ ID NO: 322	LCDR3 (Combined)	QSYTSSHGVP
SEQ ID NO: 320	LCDR1 (Kabat)	TGSSSNIGAGYDVH
SEQ ID NO: 321	LCDR2 (Kabat)	GNSNRPS
SEQ ID NO: 322	LCDR3 (Kabat)	QSYTSSHGVP
SEQ ID NO: 323	LCDR1 (Chothia)	SSSNIGAGYD
SEQ ID NO: 324	LCDR2 (Chothia)	GNS
SEQ ID NO: 325	LCDR3 (Chothia)	YTTSSHGP
SEQ ID NO: 326	LCDR1 (IMGT)	SSNIGAGYD
SEQ ID NO: 324	LCDR2 (IMGT)	GNS
SEQ ID NO: 322	LCDR3 (IMGT)	QSYTSSHGVP
SEQ ID NO: 327	VL	QSVLTQPPSVSGAPGQRVTI SCTGSSSNIGAGYDVHWYQQLPGTAPKLLIYG NSNRPSGVDPDRFSGSGKSGTSASLAITGLQAEDEADYYCQSYTSSHGVPVFG GGTKLTVL
SEQ ID NO: 328	DNA VL	cagagcgtgctgaccagccaccaagcgtgagcgggtgcaccaggtcagcgcgtgaccattagctgcaccggcagca gcagcaacattggcgcaggctatgatgtgcattggtatcagcagctgccaggccaccgcaccgaaactgctgatttatgg caacagcaatcgcccaagcgggtgtgccggatcgctttagcggcagcaaaagcggcaccagcgccagcctggcgatt accggctctgcaagccgaagacgaagccgattattactgccagtcttactacacttcttctcatgggtccgggtgttggcggc ggtaccaagctgaccgtgctg
SEQ ID NO: 329	Light Chain	QSVLTQPPSVSGAPGQRVTI SCTGSSSNIGAGYDVHWYQQLPGTAPKLLIYG NSNRPSGVDPDRFSGSKSGTSASLAITGLQAEDEADYYCQSYTSSHGVPVFG GGTKLTVLQPKAAPSVTLPFSSSEELQANKATLVCLISDFYPGAVTVAWK ADSSPVKAGVETTPSKQSNKYAASSYLSLTPEQWKSHRSYSCQVTHEGS TVEKTVAPTECS
SEQ ID NO: 330	DNA Light Chain	cagagcgtgctgaccagccaccaagcgtgagcgggtgcaccaggtcagcgcgtgaccattagctgcaccggcagca gcagcaacattggcgcaggctatgatgtgcattggtatcagcagctgccaggccaccgcaccgaaactgctgatttatgg caacagcaatcgcccaagcgggtgtgccggatcgctttagcggcagcaaaagcggcaccagcgccagcctggcgatt accggctctgcaagccgaagacgaagccgattattactgccagtcttactacacttcttctcatgggtccgggtgttggcggc ggtaccaagctgaccgtgctgggccagcccaagcggccctagcgtgacctgttcccccaagcagcagaggaac tccaggcccaacaaggccacctcggtgtgcctgatcagcgacttctacctggcgccgtgaccgtggcctggaaggcc gatagcagccctgtgaaggcggcgtggaaaccaccaccccccagcaagcagagcaacaacaaatacgcccagc agctacctgagcctgacccccgagcagtggaagtcccacagatcctacagctgccaggtcacacacaggggcagca ccgtggaaaagaccgtggccccccaccagatgcagc
YY06_ LALA		
SEQ ID NO: 270	HCDR1 (Combined)	GFTFSSYAIS
SEQ ID NO: 271	HCDR2 (Combined)	AISGSGGSTYYAESVKG
SEQ ID NO: 331	HCDR3 (Combined)	PYLGDRRSYGFDH

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 273	HCDR1 (Kabat)	SYAIS
SEQ ID NO: 271	HCDR2 (Kabat)	AISGSGGSTYYAESVKG
SEQ ID NO: 331	HCDR3 (Kabat)	PYLGDRRSYGFDH
SEQ ID NO: 32	HCDR1 (Chothia)	GFTFSSY
SEQ ID NO: 274	HCDR2 (Chothia)	SGSGGS
SEQ ID NO: 331	HCDR3 (Chothia)	PYLGDRRSYGFDH
SEQ ID NO: 275	HCDR1 (IMGT)	GFTFSSYA
SEQ ID NO: 276	HCDR2 (IMGT)	ISGSGGST
SEQ ID NO: 332	HCDR3 (IMGT)	ARPYLGDRRSYGFDH
SEQ ID NO: 333	VH	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAIISWVRQAPGKGLEWWSAI SGSGGSTYYAESVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARPYL GDRRSYGFHDHWGQGLTVTVSS
SEQ ID NO: 334	DNA VH	gaagtgcagctgctggaagcgggtggcgggtctgggtgcagccaggtggttagcctgcgcctgagctgtgcccgaagcg gctttaccttttagcagctatgccattagctgggtgcgccaagcaccaggcgaaggcctggaatgggtgagcgccattag cggcagcgggtggcagcacctattatgccgagagcgtgaaaggctcgctttaccattagtcgcgataaacagcaaaacac cctgtatctgcaaatgaacagcctgcgggcagaagataccgcagtttatattgcgcgcgaccttatctgggtgaccgtg gtagctatggtttcgaccactggggccagggcacctgggtactgtctcgagc
SEQ ID NO: 335	Heavy Chain	EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAIISWVRQAPGKGLEWWSAI SGSGGSTYYAESVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARPYL GDRRSYGFHDHWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAAALGCLVK DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYI CMVNHKPSNTKVDKRVPEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDT LMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY LPPEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSD GSFFLYSKLTVDKSRWQQGNVPSCSVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 336	DNA Heavy Chain	gaagtgcagctgctggaagcgggtggcgggtctgggtgcagccaggtggttagcctgcgcctgagctgtgcccgaagcg gctttaccttttagcagctatgccattagctgggtgcgccaagcaccaggcgaaggcctggaatgggtgagcgccattag cggcagcgggtggcagcacctattatgccgagagcgtgaaaggctcgctttaccattagtcgcgataaacagcaaaacac cctgtatctgcaaatgaacagcctgcgggcagaagataccgcagtttatattgcgcgcgaccttatctgggtgaccgtg gtagctatggtttcgaccactggggccagggcacctgggtactgtctcgagcgcgtcgaccaaggcccccagcgtgtt ccctctggccccagcagcaagagcacctctggcggaacagccgcgcctgggctgctgggtcaaggactacttccccg agccccgtgaccgtgtcctggaactctggcgccctgaccagcggcgtgcacaccttccagccgtgctccagagcagc ggcctgtacagcctgagcagcgtcgtgaccgtgcccagcagcagcctgggcccagcacctacatctgcaacgtgaa ccacaagcccagcaacacaaagggtggacaagcgggtggaacccaagagctgcgacaagaccacacctgtcccc ctgccctgccccgaagcggcgggaggccccctccgtgttccctgttcccccaaggcctaaggacacctgatgatcag ccggacccccgaagtgaactgcgtgggtggtagcgtgtcccacgaggacctgaagtgaagttaaattggtagcgtgga cggcgtggaagtgcacaacgccaagaccaagcccagagaggaacagtaacaacagcacctaccgggtgggtgctcgt gctgaccgtgctgcaccaggactggctgaacggcacaagagtacaagtgaagggtgtccaacaaggccctgcctgcc ccatcgagaaaacctcagcaaggccaaaggccagccccgcgagccccaggtgtacacactgccccctagccggg aagagatgaccaagaaccaggtgtccctgacctgacctgtaagggttctaccccagcgacattgacctgggaatggg agagcaacggccagcccagaaacaactacaagaccacccccctgtgctggacagcagcgctcattcttctgtac agcaagctgaccgtggacaagagccgggtggcagcagggaacgtgtcagctgctccgtgatgcacaggccctgc acaacactacaccagaagtccctgagcctgagccccggcaag
SEQ ID NO: 337	LCDR1 (Combined)	TGTSSDVGSYNLVS
SEQ ID NO: 338	LCDR2 (Combined)	EGSKRPS
SEQ ID NO: 339	LCDR3 (Combined)	SSYGFHIVVVV
SEQ ID NO: 337	LCDR1 (Kabat)	TGTSSDVGSYNLVS

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 348	HCDR3 (Chothia)	GSLPGLLGFDH
SEQ ID NO: 82	HCDR1 (IMGT)	GYSFTSYW
SEQ ID NO: 314	HCDR2 (IMGT)	IYPGTSYT
SEQ ID NO: 349	HCDR3 (IMGT)	ARGSLPGLLGFDH
SEQ ID NO: 350	VH	EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWISWVRQMPGKGLEWMGII YPGTSYTRYSPSFQGGVTISADKSI STAYLQWSSLKASDTAMYVCARGSLP GLLGFDHWGQGTLVTVSS
SEQ ID NO: 351	DNA VH	gaagtgcagctgggtgcagagcgggtgccgaagtgaataaccggcgaaagcctgaaatcagctgcaaaggcagc ggctatagctttaccagctattggattagctgggttcgccagatgccgggcaaaggcctggaatggatgggcattatcta tcggggcaccagctataccgcctatagcccgagctttcagggccaggttacaattagcgccgacaaaagcatcagcac cgctatctgcaatggagcagcctgaaagccagcgataccgccatgtattattgcgcgctggaagcctgctgggtctg ctgggttttgatcactggggccagggcaccctgggtactgtctcgagc
SEQ ID NO: 352	Heavy Chain	EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWISWVRQMPGKGLEWMGII YPGTSYTRYSPSFQGGVTISADKSI STAYLQWSSLKASDTAMYVCARGSLP GLLGFDHWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAAALGCLVKDYF PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNV NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMIS RTPEVTCVVDVSHEDPEVKFKNWYVDGVEVHNAKTKPREEQYNSTYRVV SVLTVLHQDWLNGKEYCKVSNKALPAPIEKTI SKAKGQPREPQVYTLPPS REEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFF LYSKLTVDKSRWQQGNVPSCSVMHEALHNYTQKSLSLSPGK
SEQ ID NO: 353	DNA Heavy Chain	gaagtgcagctgggtgcagagcgggtgccgaagtgaataaccggcgaaagcctgaaatcagctgcaaaggcagc ggctatagctttaccagctattggattagctgggttcgccagatgccgggcaaaggcctggaatggatgggcattatcta tcggggcaccagctataccgcctatagcccgagctttcagggccaggttacaattagcgccgacaaaagcatcagcac cgctatctgcaatggagcagcctgaaagccagcgataccgccatgtattattgcgcgctggaagcctgctgggtctg ctgggttttgatcactggggccagggcaccctgggtactgtctcgagcgcgtcgaccaaaggccccagcgtgttcctc tgccccccagcagcaagagcacctctggcggaacagccgccctgggtgcttggtcaaggactacttccccgagcc cgtgaccgtgtcctggaactctggcgccctgaccagcggcgtgcacacctttccagccgtgctccagagcagcggcct gtacagcctgagcagcgtcgtgaccgtgccagcagcagcctgggcaaccagacctacatctgcaacgtgaaccaca agccccagcaacacaaaaggtggacaaggggtggaacccaagagctgcgacaagaccacacctgtccccctgcc ctgccccctgaagcgggaggagggccccctccgtgttccctgttcccccaagcctaaaggacacctgatgatcagccgga cccccgaaagtgaacctgctggtggtggacgtgtcccacgaggacctgaagtgaagttaattggtacgtggacggcg tggaagtgcacaacgccaagccaagcccagagaggaacagtaacaacagacctaccgggtggtgtccgtgctgac cgtgctgcaccaggactgggtgaacggcaagagtaacagtgcaaggtgtccaacaaggccctgctgcccccatcg agaaaacctcagcaaggccaaaggccagccccgcgagccccaggtgtacacactgccccctagccgggaagaga tgaccaagaaccaggtgtccctgacctgctcgtgaagggtctctacccagcgacattgccgtggaatgggagagca acggccagcccgagaaactacaagaccacccccctgtgctggacagcagcggtcattcttccgttacagcaag ctgaccgtggacaagagccgtggcagcagggaacgtgtcagctgctccgtgatgcacagggccctgcacaaac actacaccagaagtccctgagcctgagccccggcaag
SEQ ID NO: 320	LCDR1 (Combined)	TGSSSNIGAGYDVH
SEQ ID NO: 354	LCDR2 (Combined)	GNSNRPN
SEQ ID NO: 355	LCDR3 (Combined)	QSYDSPTSSSV
SEQ ID NO: 320	LCDR1 (Kabat)	TGSSSNIGAGYDVH
SEQ ID NO: 354	LCDR2 (Kabat)	GNSNRPN
SEQ ID NO: 355	LCDR3 (Kabat)	QSYDSPTSSSV
SEQ ID NO: 323	LCDR1 (Chothia)	SSSNIGAGYD
SEQ ID NO: 324	LCDR2 (Chothia)	GNS
SEQ ID NO: 356	LCDR3 (Chothia)	YDSPTSSS

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 326	LCDR1 (IMGT)	SSNIGAGYD
SEQ ID NO: 324	LCDR2 (IMGT)	GNS
SEQ ID NO: 355	LCDR3 (IMGT)	QSYDSPTSSSV
SEQ ID NO: 357	VL	QSVLTQPPSVSGAPGQRTISCTGSSSNIGAGYDVHWYQQLPGTAPKLLIYG NSNRPNGVPDFRSGSGSGTSASLAI TGLQAEDEADYYCQSYDSPTSSSVFGG GTKLTVL
SEQ ID NO: 358	DNA VL	cagagcgtgctgaccagccaccaagcgtgagcgggtgcaccaggtcagcgcgtgaccattagctgcaccggcagca gcagcaacattggcgcaggctatgatgtgcattggtatcagcagctgccaggcaccgcaccgaaactgctgatttatgg caacagcaatcgcccaaacgggtgtgccggatcgctttagcggcagcaaaagcggcaccagcgcagcctggcgatt accggctctgcaagccgaagacgaagccgattattactgccagtcttacgactctccgacttcttcttctgtgtttggcggc ggtaccaagctgaccgtgctg
SEQ ID NO: 359	Light Chain	QSVLTQPPSVSGAPGQRTISCTGSSSNIGAGYDVHWYQQLPGTAPKLLIYG NSNRPNGVPDFRSGSGSGTSASLAI TGLQAEDEADYYCQSYDSPTSSSVFGG GTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKA DSSPVKAGVETTPSKQSNNKYAASSYLSTPEQWKSHRSYSCQVTHEGST VEKTVAPTECS
SEQ ID NO: 360	DNA Light Chain	cagagcgtgctgaccagccaccaagcgtgagcgggtgcaccaggtcagcgcgtgaccattagctgcaccggcagca gcagcaacattggcgcaggctatgatgtgcattggtatcagcagctgccaggcaccgcaccgaaactgctgatttatgg caacagcaatcgcccaaacgggtgtgccggatcgctttagcggcagcaaaagcggcaccagcgcagcctggcgatt accggctctgcaagccgaagacgaagccgattattactgccagtcttacgactctccgacttcttcttctgtgtttggcggc ggtaccaagctgaccgtgctgggcccagccaaagccgcccctagcgtgacctgttcccccaagcagcagggaaac tccagggccaacaaggccaccctcggtgtgcctgatcagcgacttctacctggcgccgtgacctggcctggaaggcc gatagcagccctgtgaaggccggcgtggaaaccaccaccccccagcaagcagagcaacaacaaatacgcgccagc agctacctgagcctgacccccgagcagtggaagtccacagatcctacagctgccaggtcacacacgagggcagca ccgtggaaaagaccgtggccccaccgagtgacg
ZZ05_ LALA		
SEQ ID NO: 310	HCDR1 (Combined)	GYSFYSYWIS
SEQ ID NO: 311	HCDR2 (Combined)	IIYPGTSYTRYSPSFQG
SEQ ID NO: 348	HCDR3 (Combined)	GSLPGLLGFDH
SEQ ID NO: 229	HCDR1 (Kabat)	SYWIS
SEQ ID NO: 311	HCDR2 (Kabat)	IIYPGTSYTRYSPSFQG
SEQ ID NO: 348	HCDR3 (Kabat)	GSLPGLLGFDH
SEQ ID NO: 80	HCDR1 (Chothia)	GYSFYSY
SEQ ID NO: 313	HCDR2 (Chothia)	YPGTSY
SEQ ID NO: 348	HCDR3 (Chothia)	GSLPGLLGFDH
SEQ ID NO: 82	HCDR1 (IMGT)	GYSFYSY
SEQ ID NO: 314	HCDR2 (IMGT)	IYPGTSYT
SEQ ID NO: 349	HCDR3 (IMGT)	ARGSLPGLLGFDH
SEQ ID NO: 350	VH	EVQLVQSGAEVKKPGESLKISCKGSGYSFYSYISWVRQMPGKGLEWMGII YPGTSYTRYSPSFQGVTVISADKSISTAYLQWSSSLKASD TAMYYCARGSLP GLLGFDHWGQGLTVTVSS

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 351	DNA VH	gaagtgcagctggtgcagagcgggtgccgaagtgaaaaaaccgggcgaaagcctgaaaatcagctgcaaaggcagc ggctatagctttaccagctattggattagctgggttcgccagatgccgggcaaaggcctggaatggatgggcattatcta tccgggcaccagctataccgctatagcccagctttcagggccaggttacaattagcgcgcacaaaagcatcagcac cgctatctgcaatggagcagcctgaaagccagcgataccgccatgtattattgcgcgcgtggaagcctgctggtctg ctgggttttgatcactggggccagggcaccctggttactgtctcgcagc
SEQ ID NO: 352	Heavy Chain	EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWISWVRQMPGKGLEWMGII YPGTSYTRYSPSFQGGVTISADKSI STAYLQWS SLKASDTAMY YCARGSLP GLLGPDHWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAAALGLVKDYF PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNV NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMIS RTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVV SVLTVLHQDWLNGKEYKCKVSNKALPAPI EKTISKAKGQPREPQVYTLPPS REEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFF LYSKLTVDKSRWQQGNVSCSVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 353	DNA Heavy Chain	gaagtgcagctggtgcagagcgggtgccgaagtgaaaaaaccgggcgaaagcctgaaaatcagctgcaaaggcagc ggctatagctttaccagctattggattagctgggttcgccagatgccgggcaaaggcctggaatggatgggcattatcta tccgggcaccagctataccgctatagcccagctttcagggccaggttacaattagcgcgcacaaaagcatcagcac cgctatctgcaatggagcagcctgaaagccagcgataccgccatgtattattgcgcgcgtggaagcctgctggtctg ctgggttttgatcactggggccagggcaccctggttactgtctcgcagcgcgtcgaccaaaaggccccagcgtgttccctc tggccccccagcagcaagagcacctctggcgggaacagcgcgcctgggctgectggtcaaggactacttccccgagcc cgtgaccgtgtcctggaaactctggcgccctgaccagcggcgtgcacacctttccagccgtgctccagagcagcggcct gtacagcctgagcagcgtcgtgaccgtgccagcagcagcctgggcacccagacctacatctgcaacgtgaaccaca agccccagcaacacaaggtggacaagcgggtggaacccaagagctgcgacaagacccacacctgtccccctgcc ctgccccctgaagcggcgggaggccccctcgctgttccctgttccccccaaagcctaaggacacctgatgatcagccgga cccccgaagtgcctgctggtggtggacgtgtcccacgaggacctgaagtgaagttaaattggtacgtggacggcg tggaaagtgcacaacgccaagaccagccagagaggaaacagtacaacagcacctaccgggtggtgtcgctgctgac cgtcgtgcaccaggactggctgaacggcacaagagtagaagtgcaaggtgtccaacaaggccctgctgcccccatcg agaaaaccatcagcaagggccaaaggccagccccgcgagccccaggtgtacacactgccccctagccgggaagaga tgaccaagaaccaggtgtccctgacctgcctcgtgaagggtcttaccacagcgacattgccgtggaatgggagagca acggccagccccgagaaacaactacaagaccacccccctgtgctggacagcgagcgtcattcttccgtacagcaag ctgacctggacaagagccggtggcagcagggcaacgtgtcagctgctccgtgatgcacgagccctgcacaacc actacaccagaagtccccctgagcctgagccccggcaag
SEQ ID NO: 320	LCDR1 (Combined)	TGSSSNIGAGYDVH
SEQ ID NO: 354	LCDR2 (Combined)	GNSNRPN
SEQ ID NO: 361	LCDR3 (Combined)	QSYGAFPRFVV
SEQ ID NO: 320	LCDR1 (Kabat)	TGSSSNIGAGYDVH
SEQ ID NO: 354	LCDR2 (Kabat)	GNSNRPN
SEQ ID NO: 361	LCDR3 (Kabat)	QSYGAFPRFVV
SEQ ID NO: 323	LCDR1 (Chothia)	SSSNIGAGYD
SEQ ID NO: 324	LCDR2 (Chothia)	GNS
SEQ ID NO: 362	LCDR3 (Chothia)	YGAFPRFV
SEQ ID NO: 326	LCDR1 (IMGT)	SSNIGAGYD
SEQ ID NO: 324	LCDR2 (IMGT)	GNS
SEQ ID NO: 361	LCDR3 (IMGT)	QSYGAFPRFVV
SEQ ID NO: 363	VL	QSVLTQPPSVSGAPGQRVTISCTGSSSNIGAGYDVHWYQQLPGTAPKLLIYG NSNRPNGVDFRPSGSKSGTSASLAI TGLQAEDEADYYCQSYGAFPRFVVFG GGTKLTVL
SEQ ID NO: 364	DNA VL	cagagcgtgctgacccagccaccaagcgtgagcgggtgcaccaggtcagcgcgtgaccattagctgcaccggcagca gcagcaacattggcgaggctatgatgtgcatgtggtatcagcagctgccaggcaccgcaccgaaactgctgatttatgg caacagcaatcgcccaaacggtgtgccggatcgctttagcggcagcaaaagcggcaccagcgcagcctggcgatt

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		accggtctgcaagccgaagacgaagccgattattactgccaatcctatggtgccttcctcggttcgtgtgttttggcgcg gtaccaagctgaccgtgctg
SEQ ID NO: 365	Light Chain	QSVLTQPPSVSGAPGQRTISCTGSSSNIGAGYDVHWYQQLPGTAPKLLIYG NSNRPNGVPDFRSGSKSGTSASLAI TGLQAEDEADYYCQSYGAFPRFVFG GGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWK ADSSPVKAGVETTPPSKQSNKNYAASSYLSLTPEQWKSHRSYSCQVTHEGS TVEKTVAPTECS
SEQ ID NO: 366	DNA Light Chain	cagagcgtgctgacccagccaccaagcgtgagcgggtgcaccaggtcagcgcgtgaccattagctgcaccggcagca gcagcaacattggcgagcgtatgatgtgcattggtatcagcagctgccaggcaccgcaccgaaactgctgatttatgg caacagcaatcgccaaacgggtgtgccggtatcgctttagcggcagcaaaagcggcaccagcgcagcctggcgatt accggtctgcaagccgaagacgaagccgattattactgccaatcctatggtgccttcctcggttcgtgtgttttggcgcg gtaccaagctgaccgtgctggggccagccaaagccgcccctagcgtgacctgttcccccaagcagcgaggaactc caggccaacaaggccaccctcgtgtgcctgatcagcgaacttctaccctggcgccgtgaccgtggcctggaaggccga tagcagccctgtgaaggccggcgtggaaccaccacccccagcaagcagagcaacaacaaat acgcccagcagcag ctacctgagcctgacccccgagcagtggaagtcacacagatcctacagctgccaggtcacacacgagggcagcacc gtggaagaccgtggccccccaccgagtgacg
ZZ12_ LALA		
SEQ ID NO: 367	HCDR1 (Combined)	GFSFSKYYLN
SEQ ID NO: 368	HCDR2 (Combined)	SIHQQAHEKKYVESVKG
SEQ ID NO: 228	HCDR3 (Combined)	SLRRRSTEHAGFDV
SEQ ID NO: 369	HCDR1 (Kabat)	KYYLN
SEQ ID NO: 368	HCDR2 (Kabat)	SIHQQAHEKKYVESVKG
SEQ ID NO: 228	HCDR3 (Kabat)	SLRRRSTEHAGFDV
SEQ ID NO: 370	HCDR1 (Chothia)	GFSFSKY
SEQ ID NO: 371	HCDR2 (Chothia)	HQQAHE
SEQ ID NO: 228	HCDR3 (Chothia)	SLRRRSTEHAGFDV
SEQ ID NO: 372	HCDR1 (IMGT)	GFSFSKYY
SEQ ID NO: 373	HCDR2 (IMGT)	IHQQAHEK
SEQ ID NO: 232	HCDR3 (IMGT)	ARSLRRRSTEHAGFDV
SEQ ID NO: 374	VH	EVQLVESGGGLVQPGGSLRLSCAASGFSFSKYYLNWVRQAPGKGLEWVAS IHQQAHEKKYVESVKGRFTISRDNKNSLYLQMNSLRAEDTAVYYCARSL RRRSTEHAGFDVWGQGLTVTVSS
SEQ ID NO: 375	DNA VH	gaagtgcagctggtggaagcggcggtggcctggtgcagccaggtggtagcctgcgcctgagctgcgcgccagc ggccttagcttcagcaaatattactgaaactgggttcgccaggccccaggcaaaaggcctggaatgggtggccagcattc accagcaagcacacgagaaaaaatacgtggagtcctgtgaaaggccgctttaccattagccgcgataacgcaaaaac agcctgtatctgcaaatgaacagcctcgccggccgaagataccgccgtgtattattgcgcgcgtagcctgcgtcgtcgta gcactgagcacgcaggattcgacgtttggggccagggcacccctggttactgtctcgagc
SEQ ID NO: 376	Heavy Chain	EVQLVESGGGLVQPGGSLRLSCAASGFSFSKYYLNWVRQAPGKGLEWVAS IHQQAHEKKYVESVKGRFTISRDNKNSLYLQMNSLRAEDTAVYYCARSL RRRSTEHAGFDVWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCL VRDYFPEPVTWSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQT YICNVNHKPSNTKVDKRVPEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPK DFLMTSRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYN TYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI SKAKGQPREPQV YTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLD SDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNNHYTQKSLSLSPGK

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 377	DNA Heavy Chain	gaagtgcagctgggtggaagcggcggtggcctgggtgcagccaggtggtagcctgcgcctgagctgcgcgcgcagc ggctttagcttcagcaaatattacttgaactgggttcgccaggccccaggcgaagcctggaatgggtggccagcattc accagcaagcacacgagaaaaatacgtggagtcctgtgaaaggccgctttaccattagccgcgataaacgcaaaaac agcctgtatctgcaaatgaacagcctgcgggcccgaagataccgcctgtattattgcgcgcgtagcctgcgtcgtcgta gcactgagcacgcaggattcgacgtttggggccaggccaccctgggttactgtctcgagcgcgtcgaccaaaggcccc agcgtgttccctctggccccagcagcaagagcacctctggcggaaacagccgcctgggctgctgggtcaaggacta ctccccagagcccgtagcctgtcctggaactctggcgcctgaccagcggcgtgcacaccttccagcctgctcca gagcagcggcctgtacagcctgagcagcgtcgtgacctgcccagcagcagcctgggcacccagacctacatctgc aacgtgaaccacaagccagcaacacaaaggtggacaagcgggtggaacccaagagctgcgacaagacccacac ctgtccccctgacctgccccgaagcggcgaggccccctcgtgttctgttcccccaaaagcctaaggacacct gatgatcagccggacccccgaagtgaacctgcgtgggtggtagcgtgtccacgaggacctgaagtgaagtttaattg gtactgtggacggcgtggaagtgcacaacgcgaagaccagccagagaggaaacagtacaacagcacctaccgggt ggtgtcctgctgacctgctgcaccaggactggctgaacggcgaagagtacaagtgcaaggtgtccaacaaggccc tgcctgcccccatcgagaaaaccatcagcaaggccaaaggccagccccgcgagccccaggtgtacacactgcccc tagccgggaagagatgaccaagaaccaggtgtcctgacctgcctcgtgaagggttctaccccagcgaccttgcct ggaatgggagagcaacggccagcccgagaacaactacaagaccacccccctgtgctggacagcgacggctcattc ttcctgtacagcaagctgacctgggacaagagccgggtggcagcagggcaacgtgttcagctgctcctgtagtcacga ggccctgcacaaccactacaccagaagtccctgagcctgagccccggcaag
SEQ ID NO: 237	LCDR1 (Combined)	RASQGISNYLA
SEQ ID NO: 238	LCDR2 (Combined)	AASTLQS
SEQ ID NO: 239	LCDR3 (Combined)	QQADKFPYT
SEQ ID NO: 237	LCDR1 (Kabat)	RASQGISNYLA
SEQ ID NO: 238	LCDR2 (Kabat)	AASTLQS
SEQ ID NO: 239	LCDR3 (Kabat)	QQADKFPYT
SEQ ID NO: 240	LCDR1 (Chothia)	SQGISNY
SEQ ID NO: 241	LCDR2 (Chothia)	AAS
SEQ ID NO: 242	LCDR3 (Chothia)	ADKFPY
SEQ ID NO: 243	LCDR1 (IMGT)	QGISNY
SEQ ID NO: 241	LCDR2 (IMGT)	AAS
SEQ ID NO: 239	LCDR3 (IMGT)	QQADKFPYT
SEQ ID NO: 244	VL	DIQMTQSPSSLSASVGRVITITCRASQGISNYLAWYQQKPKGVKPLLIYAAS TLQSGVPSRFSGSGSGTDFTLTITSLQPEDVATYYCQQADKFPYTFGGGTKV EIK
SEQ ID NO: 245	DNA VL	gatattcagatgaccagagccccagcagcctgagcgcgaagcgtggcgatcgcgtagaccattacctgccgcgcca gccagggcatttagcaactatctggcctggatcagcagaaaacggggcaaaagtgcggaactgctgatctatgccgcca gcacctgcgaagcggcgtgccaagtgcctttagcggcagcggtagcggcacccgatttcacctgaccttagcagc ctgcaaccggaagacgtggcgacctattattgccagcaggctgacaaaatccccgtacaccttcggccagggtaccaa gtggaatcaag
SEQ ID NO: 246	Light Chain	DIQMTQSPSSLSASVGRVITITCRASQGISNYLAWYQQKPKGVKPLLIYAAS TLQSGVPSRFSGSGSGTDFTLTITSLQPEDVATYYCQQADKFPYTFGGGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKDSSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVIT KSFNRGEC
SEQ ID NO: 247	DNA Light Chain	gatattcagatgaccagagccccagcagcctgagcgcgaagcgtggcgatcgcgtagaccattacctgccgcgcca gccagggcatttagcaactatctggcctggatcagcagaaaacggggcaaaagtgcggaactgctgatctatgccgcca gcacctgcgaagcggcgtgccaagtgcctttagcggcagcggtagcggcacccgatttcacctgaccttagcagc ctgcaaccggaagacgtggcgacctattattgccagcaggctgacaaaatccccgtacaccttcggccagggtaccaa gtggaatcaagcggacgtggcgctccctccgtgttcattcttccccaccagcagcagcagcagctgaagtccggcaca gccagcgtcgtgtgcctgtgaacaaattctaccccccgaggccaaagtgcagtggaagtggaacacgcccctcca gagcggcaacagccaggaaagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcacctgac

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		cctgagcaaggccgactacgagaagcacaaggtgtacgcctgcgaagtgacccaccagggcctgtccagccccgtg accaagagcttcaaccggggcgagtgt
ZZ13_ LALA		
SEQ ID NO: 378	HCDR1 (Combined)	GFTFSRYIIN
SEQ ID NO: 379	HCDR2 (Combined)	SIHQHGLETRYVESVKG
SEQ ID NO: 228	HCDR3 (Combined)	SLRRRSTEHAGFDV
SEQ ID NO: 380	HCDR1 (Kabat)	RYIIN
SEQ ID NO: 379	HCDR2 (Kabat)	SIHQHGLETRYVESVKG
SEQ ID NO: 228	HCDR3 (Kabat)	SLRRRSTEHAGFDV
SEQ ID NO: 381	HCDR1 (Chothia)	GFTFSRY
SEQ ID NO: 382	HCDR2 (Chothia)	HQHGLE
SEQ ID NO: 228	HCDR3 (Chothia)	SLRRRSTEHAGFDV
SEQ ID NO: 383	HCDR1 (IMGT)	GFTFSRY
SEQ ID NO: 384	HCDR2 (IMGT)	IHQHGLE
SEQ ID NO: 232	HCDR3 (IMGT)	ARSLRRRSTEHAGFDV
SEQ ID NO: 385	VH	EVQLVESGGGLVQPGGSLRLSCAASGFTFSRYINWVRQAPGKGLEWVASI HQHGLETRYVESVKGRFTISRDNAKNSLYLQMNSLRAEDTAVYYCARSLR RRSTEHAGFDVWGQGLTVVSS
SEQ ID NO: 386	DNA VH	gaagtgcagctgggtggaagcggcggtggcctggtgcagccaggtggtagcctgcgcctgagctgcgcgccagc gggtttactttttccagatattacattaattgggttcgccaggccccaggcaaggcctggaatgggtggcgagcatcca ccagcacggcctggagaccagat atgtggaatctgtcaaaagggcgctttaccattagccgcgataacgccccaaaacag cctgtatctgcaaatgaacagcctgcgggcccgaagataccgccgtgtattattgcgcgcgtagcctgcgtcgtcgtagc actgagcacgcaggattcgacgtttggggccaggccaccctggttactgtctcgagc
SEQ ID NO: 387	Heavy Chain	EVQLVESGGGLVQPGGSLRLSCAASGFTFSRYINWVRQAPGKGLEWVASI HQHGLETRYVESVKGRFTISRDNAKNSLYLQMNSLRAEDTAVYYCARSLR RRSTEHAGFDVWGQGLTVVSSASTKGPSVFPLAPSSKSTSGGTAAALGCLV KDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTY ICNVNHKPSNTKVDKRVKPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDT LMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVHLQDNLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY LPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSD GSFFLYSKLTVDKSRWQQGNVPSCSVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 388	DNA Heavy Chain	gaagtgcagctgggtggaagcggcggtggcctggtgcagccaggtggtagcctgcgcctgagctgcgcgccagc gggtttactttttccagatattacattaattgggttcgccaggccccaggcaaggcctggaatgggtggcgagcatcca ccagcacggcctggagaccagat atgtggaatctgtcaaaagggcgctttaccattagccgcgataacgccccaaaacag cctgtatctgcaaatgaacagcctgcgggcccgaagataccgccgtgtattattgcgcgcgtagcctgcgtcgtcgtagc actgagcacgcaggattcgacgtttggggccaggccaccctggttactgtctcgagcgcgtcgaccaaaggccccag cgtgttcctctggccccagcagcaagacacacctctggcggaacagccgccctgggctgcctgggtcaaggactactt ccccgagccccgtgaccgtgtcctggaactctggcgccctgaccagcggcgtgcacacctttccagccgtgctccaga gcagcggcctgtacagcctgagcagcgtcgtgaccgtgcccagcagcagcctgggcaaccagacctacatctgcaa cgtgaaccacaaagcccagcaacacaaaggtggacaagcgggtggaacccaagagctgcgacaagacccacacctg tccccctgccccctgccccgaagcggcgaggccccctccgtgttccctgttcccccaagcctaaaggacacctgat gatcagccgggacccccgaagtgaacctgcgtgggtggtagctgtcccacgaggaccctgaagtgaagttaaattggta cgtggacggcgtggaagtgcacaacgccaagcaagccagagaggaacagtaacaacagcacctaccgggtggt gtccgtgtgacccgtgtgcaccaggactggctgaacggcgaagagtaacaagtgcgaaggtgtccaacaaggccctgc ctgcccccatgagaaaaccatcagcaaggccaaaggccagccccgcgagccccaggtgtacacactgccccctag ccgggaagagatgaccaagaaccaggtgtccctgacctgacctgtaagggtctctaccccagcgacattgccgtgg aatgggagagcaacggccagccccgagaacaactacaagaccacccccctgtgctggacagcgacggctcatctctc

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		ctgtacagcaagctgaccgtggacaagagccggtggcagcagggcaacgtgttcagctgctccgtgatgcacgagg ccctgcacaaccactacaccagaagtccctgagcctgagccccggcaag
SEQ ID NO: 237	LCDR1 (Combined)	RASQGISNYLA
SEQ ID NO: 238	LCDR2 (Combined)	AASTLQS
SEQ ID NO: 239	LCDR3 (Combined)	QQADKFPYT
SEQ ID NO: 237	LCDR1 (Kabat)	RASQGISNYLA
SEQ ID NO: 238	LCDR2 (Kabat)	AASTLQS
SEQ ID NO: 239	LCDR3 (Kabat)	QQADKFPYT
SEQ ID NO: 240	LCDR1 (Chothia)	SQGISNY
SEQ ID NO: 241	LCDR2 (Chothia)	AAS
SEQ ID NO: 242	LCDR3 (Chothia)	ADKFPY
SEQ ID NO: 243	LCDR1 (IMGT)	QGISNY
SEQ ID NO: 241	LCDR2 (IMGT)	AAS
SEQ ID NO: 239	LCDR3 (IMGT)	QQADKFPYT
SEQ ID NO: 244	VL	DIQMTQSPSSLSASVGDRTITCRASQGISNYLAWYQQKPGKVPKLLIYAAS TLQSGVPSRFRSGSGTDFTLTISLQPEDVATYYCQQADKFPYTFGGGTKV EIK
SEQ ID NO: 245	DNA VL	gatattcagatgacccagagcccgagcagcctgagcgcaagcgtggcgcatcgcgtagaccattacctgccgcgcca gccagggcattagcaactatctggcctggatcagcagaaaacgggcaaaagtgcggaactgctgatctatgccgcca gcacctgcaaaagcggcgctgccaagtgccttagcggcagcggtagcggcacccgatttcacctgaccattagcagc ctgcaaccggaagacgtggcgacctattattgccagcaggctgacaaattcccgtagaccttcggccagggtacccaa gtggaatcaag
SEQ ID NO: 246	Light Chain	DIQMTQSPSSLSASVGDRTITCRASQGISNYLAWYQQKPGKVPKLLIYAAS TLQSGVPSRFRSGSGTDFTLTISLQPEDVATYYCQQADKFPYTFGGGTKV EIKRTVAAPSVFIFPPSDEQLKSGTASVCLLNNFYFPREKVKVDNALQ SGNSQESVTEQDSKDSSTLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
SEQ ID NO: 247	DNA Light Chain	gatattcagatgacccagagcccgagcagcctgagcgcaagcgtggcgcatcgcgtagaccattacctgccgcgcca gccagggcattagcaactatctggcctggatcagcagaaaacgggcaaaagtgcggaactgctgatctatgccgcca gcacctgcaaaagcggcgctgccaagtgccttagcggcagcggtagcggcacccgatttcacctgaccattagcagc ctgcaaccggaagacgtggcgacctattattgccagcaggctgacaaattcccgtagaccttcggccagggtacccaa gtggaatcaagcggacgtggcgctccctccgtgttcattctccacccagcagcagcagctgaagtccggcaca gccagcgtcgtgtgctgctgaacaactctacccccgcgaggccaaagtgcagtggaaggtggacaacgcccctoca gagcggcaacagccaggaagcgtcaccgagcaggacagcaaggactccacctacagcctgagcagcacccctgac cctgagcaaggccgactacgagaagcacaaggtgtacgcctgcgaagtgaccaccagggcctgtccagccccctg accaagagcttcaaccggggcgagtgt
ZZ14_ LALA		
SEQ ID NO: 270	HCDR1 (Combined)	GFTFSSYAIS
SEQ ID NO: 389	HCDR2 (Combined)	SISSHGYYTRYAESVKG
SEQ ID NO: 331	HCDR3 (Combined)	PYLGDRRSYGFDH

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 331	HCDR3 (Chothia)	PYLGDRRSYGFDH
SEQ ID NO: 401	HCDR1 (IMGT)	GFTFASYA
SEQ ID NO: 402	HCDR2 (IMGT)	ISGSGVYT
SEQ ID NO: 332	HCDR3 (IMGT)	ARPYLGDRRSYGFDH
SEQ ID NO: 403	VH	EVQLLESGGGLVQPGGSLRLSCAASGFTFASYAITWVRQAPGKGLEWVSTI SGSGVITYYAESVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARPYL GDRRSYGFDHWGQGLTVTVSS
SEQ ID NO: 404	DNA VH	gaagtgcagctgctggaaagcgggtggcgggtctggtgcagccaggtggtagcctgcgcctgagctgtgccgcaagcg ggttcacattcgcatcctatgcaattacttgggtgcgccaagcaccaggcgaaggcctggaatgggtgagcaccatttc cgggtccgggtgtgtacacctattacgccgagtcctgcaaaagccgctttaccattagtcgcgataaacagcaaaaacacc ctgtatctgcaaatgaacagcctgcgggcagaagataccgcagtttattattgcgcgcgaccttatctgggtgaccgtcg tagctatgggttcgaccactggggccaggccaccctgggtactgtctcgagc
SEQ ID NO: 405	Heavy Chain	EVQLLESGGGLVQPGGSLRLSCAASGFTFASYAITWVRQAPGKGLEWVSTI SGSGVITYYAESVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARPYL GDRRSYGFDHWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAAALGCLVK DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYI CNVNHKPSNTKVDKRVPEPKSCKDTHCTCPPCPAPEAAGGPSVFLFPPKPKDT LMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI SKAKGQPREPQVYT LPDSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSD GSFFLYSKLTVDKSRWQQGNVPSCSVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 406	DNA Heavy Chain	gaagtgcagctgctggaaagcgggtggcgggtctggtgcagccaggtggtagcctgcgcctgagctgtgccgcaagcg ggttcacattcgcatcctatgcaattacttgggtgcgccaagcaccaggcgaaggcctggaatgggtgagcaccatttc cgggtccgggtgtgtacacctattacgccgagtcctgcaaaagccgctttaccattagtcgcgataaacagcaaaaacacc ctgtatctgcaaatgaacagcctgcgggcagaagataccgcagtttattattgcgcgcgaccttatctgggtgaccgtcg tagctatgggttcgaccactggggccaggccaccctgggtactgtctcgagcgcgctcgaccaaaggccccagcgtgttc cctctggccccccagcagcaagagcacctctggcggaacagccgccccgggtgcctgggtcaaggactacttccccga gccccgtgaccgtgtcctggaactctggcgccccctgaccagcggcgtgcacaccttccagccgtgctccagagcagcg gacctgtacagcctgagcagcgtcgtgacctgcccagcagcagcctgggcaccccagacctacatctgcaacgtgaac cacaagccccagcaacacaaagggtggacaagcgggtggaacccaagagctgcgacaagacccaacacctgtcccccc tgccctgccccctgaagcggcgggaggccccctccgtgttccctgttccccccaaagcctaaggacacctgatgatcagc cggacccccgaagtgaacctgctggtggtggacctgtcccacgaggacctgaagtgaagtttaattgggtacgtggac ggcgtggaagtgcacacgccaagaccaagccccagagaggaacagtaacaacagcacctaccgggtgggtgtccgtg ctgacctgtgtgcaccaggactggctgaacggcgaagagtacaagtgcagggtgtccacaaggccccctgctgtcccc catcgagaaaaccatcagcaaggccaaaggccagccccgcgagccccaggtgtacacactgcccccttagccggga agagatgaccaagaaccaggtgtccctgacctgacctgtgaagggtcttaccacagcgacattgcccgtggaatggga gagcaacggccagcccgagaacaactacaagaccacccccctgtgctggacagcgacggctcattcttctgtaca gcaagctgacctgtggaagaagaccgggtggcagcagggcaacgtgttcagctgctccgtgatgcacgaggccctgca caaccactacacccagaagtccctgagcctgagccccggcaag
SEQ ID NO: 337	LCDR1 (Combined)	TGTSSDVGSYNLVS
SEQ ID NO: 338	LCDR2 (Combined)	EGSKRPS
SEQ ID NO: 339	LCDR3 (Combined)	SSYGFIHVVVV
SEQ ID NO: 337	LCDR1 (Kabat)	TGTSSDVGSYNLVS
SEQ ID NO: 338	LCDR2 (Kabat)	EGSKRPS
SEQ ID NO: 339	LCDR3 (Kabat)	SSYGFIHVVVV
SEQ ID NO: 340	LCDR1 (Chothia)	TSSDVGSYNL
SEQ ID NO: 341	LCDR2 (Chothia)	EGS
SEQ ID NO: 342	LCDR3 (Chothia)	YGFHIVVV

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 343	LCDR1 (IMGT)	SSDVGSYNL
SEQ ID NO: 341	LCDR2 (IMGT)	EGS
SEQ ID NO: 339	LCDR3 (IMGT)	SSYGFHIVVVV
SEQ ID NO: 344	VL	QSALTQPASVSGSPGQSITISCTGTSSDVGSYNLVSQHPGKAPKLMIE GSKRPSGVSNRFGSGKSGNTASLTISGLQAEDEADYYCSSYGFHIVVVVFGG GTKLTVL
SEQ ID NO: 345	DNA VL	cagagcgccctgaccagccagccagcgttagcggtagcccaggccagagcattaccattagctgcaccggcacca gcagcgacgtgggcagctataacctggtagctggtagcagcatccgggcaaagccccgaaactgatgatctatg aaggcagcaaacgcccagcggcgtagcaaccgcttagtgggcagcaaaagcggcaacaccgcccagcctgacat tagcggcctgcaagccgaagacgaagccgattattactgctcctcttacgggtttccatcggtgtgtgtgtgtgtggcggc ggtaccaagctgaccgtgctg
SEQ ID NO: 346	Light Chain	QSALTQPASVSGSPGQSITISCTGTSSDVGSYNLVSQHPGKAPKLMIE GSKRPSGVSNRFGSGKSGNTASLTISGLQAEDEADYYCSSYGFHIVVVVFGG GTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKA DSSPVKAGVETTPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHEGST VEKTVAPTECS
SEQ ID NO: 347	DNA Light Chain	cagagcgccctgaccagccagccagcgttagcggtagcccaggccagagcattaccattagctgcaccggcacca gcagcgacgtgggcagctataacctggtagctggtagcagcatccgggcaaagccccgaaactgatgatctatg aaggcagcaaacgcccagcggcgtagcaaccgcttagtgggcagcaaaagcggcaacaccgcccagcctgacat tagcggcctgcaagccgaagacgaagccgattattactgctcctcttacgggtttccatcggtgtgtgtgtgtgtggcggc ggtaccaagctgaccgtgctgggcccagccaaagccgcccctagcgtgacctgttcccccaagcagcagggaaac tcagggccaacaaggccaccctcgtagcctgatcagcgacttctacctggcgccgtgacctggcctggaaggcc gatagcagccctgtgaaggccggcgtagaaaccaccacccagcaagcagagcaacacaaatacgcgccagc agctacctgagcctgacccccgagcagtggaagtccacagatcctacagctgccaggtcacacacgagggcagca ccgtggaaaagaccgtggccccaccgagtgacg
ZZ16_ LALA		
SEQ ID NO: 407	HCDR1 (Combined)	GFTFGTYAMT
SEQ ID NO: 408	HCDR2 (Combined)	SISASGYANYAGSVKG
SEQ ID NO: 331	HCDR3 (Combined)	PYLGDRRSYGFHDH
SEQ ID NO: 409	HCDR1 (Kabat)	TYAMT
SEQ ID NO: 4080	HCDR2 (Kabat)	SISASGYANYAGSVKG
SEQ ID NO: 331	HCDR3 (Kabat)	PYLGDRRSYGFHDH
SEQ ID NO: 410	HCDR1 (Chothia)	GFTFGTY
SEQ ID NO: 411	HCDR2 (Chothia)	SASGY
SEQ ID NO: 331	HCDR3 (Chothia)	PYLGDRRSYGFHDH
SEQ ID NO: 412	HCDR1 (IMGT)	GFTFGTYA
SEQ ID NO: 413	HCDR2 (IMGT)	ISASG
SEQ ID NO: 332	HCDR3 (IMGT)	ARPYLGDRRSYGFHDH
SEQ ID NO: 414	VH	EVQLLESQGLVQPGGSLRLSCAASGFTFGTYAMTWVRQAPGKLEWVSS ISASGYANYAGSVKGRFTISRDNKNTLYLQMNSLRADTAVYYCARPY LGDRRSYGFHDHWGQGLTVTVSS

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 415	DNA VH	gaagtgcagctgctggaaagcgggtggcggtctggtgcagccaggtggttagcctgcgcctgagctgtgccgcaagcg gggtttacattcggcacctatgcaatgacttgggtgcgccaagcaccaggcaaggcctggaatgggtgagtagcattag cgcataccggatattacgctaactacgcaggcagcgtcaaaggccgctttaccattagtcgcgataaacagcaaaaacacc ctgtatctgcaaatgaacagcctgcgggcagaagataccgcagtttattatgcgcgcgaccttattctgggtgaccgtcg tagctatgggttcgaccactggggccagggcaccctggttactgtctcgagc
SEQ ID NO: 416	Heavy Chain	EVQLLESGGGLVQPGGSLRLSCAASGFTFGTYAMTWVRQAPGKLEWVSS ISASGYANYAGSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYYCARPY LDRRSYGFDPHNGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLV KDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTY ICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDT LMI SRTPEVTVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVLHQDLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYT LPSPREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSD GSFFLYSKLTVDKSRWQQGNVPSCSVMHEALHNHYTQKSLSLSPGK
SEQ ID NO: 417	DNA Heavy Chain	gaagtgcagctgctggaaagcgggtggcggtctggtgcagccaggtggttagcctgcgcctgagctgtgccgcaagcg gggtttacattcggcacctatgcaatgacttgggtgcgccaagcaccaggcaaggcctggaatgggtgagtagcattag cgcataccggatattacgctaactacgcaggcagcgtcaaaggccgctttaccattagtcgcgataaacagcaaaaacacc ctgtatctgcaaatgaacagcctgcgggcagaagataccgcagtttattatgcgcgcgaccttattctgggtgaccgtcg tagctatgggttcgaccactggggccagggcaccctggttactgtctcgagcgcgtcgaccaaaaggccccagcgtgttc cctctggccccccagcagcaagagcacctctggcggaacagccgcctgggtgacctggtcaaggactacttccccga gccccgtgacctgtcctggaaactctggcgccctgaccagcggcggtgcacaccttccagccgtgctccagagcagcg gccccgtacagcctgagcagcgtcgtgacctgccccagcagcagcctgggcaaccagacctacatctgcaacgtgaac cacaagccagcaacacaaaggtggacaagcgggtggaaacccaagagctgcgacaagaccacacacctgtcccccc tgccctgccccctgaagcggcgggaggccccctccgtgttccgtgtccccccaaagcctaaaggacacctgatgatcagc cggacccccgaagtgcctgctggtggtggacctgtcccacgaggacctgaagtgaagtttaattggtacctggac ggcgtggaagtgcacaacgccaagaccaagcccagagaggaaacagtacaacagcacctaccgggtggtgtccgtg ctgacccgtgctgcaccaggactggctgaacggcgaagagtacaagtgcaagggtgtccaaacaggccccctgctgcccc catcgagaaaaccatcagcaaggccaaaggccagcccccgagccccaggtgtacacactgccccctagccggga agagatgaccaagaaccaggtgtcctgacctgctcgtgaagggttcttaccacagcgacattgcccgtggaatggga gagcaacggccagcccagagaacactacaagaccacccccctgtgctggacagcgacggctcattctcctgtaca gcaagctgacctggacaagagccggtggcagcagggaacacgtgttcagctgctccgtgatgcacaggccctgca caaccactacaccagaagtcctgagcctgagccccggcaag
SEQ ID NO: 337	LCDR1 (Combined)	TGTSSDVGSYNLVS
SEQ ID NO: 338	LCDR2 (Combined)	EGSKRPS
SEQ ID NO: 339	LCDR3 (Combined)	SSYGFHIVVVV
SEQ ID NO: 337	LCDR1 (Kabat)	TGTSSDVGSYNLVS
SEQ ID NO: 338	LCDR2 (Kabat)	EGSKRPS
SEQ ID NO: 339	LCDR3 (Kabat)	SSYGFHIVVVV
SEQ ID NO: 340	LCDR1 (Chothia)	TSSDVGSYNL
SEQ ID NO: 341	LCDR2 (Chothia)	EGS
SEQ ID NO: 342	LCDR3 (Chothia)	YGFHIVVV
SEQ ID NO: 343	LCDR1 (IMGT)	SSDVGSYNL
SEQ ID NO: 341	LCDR2 (IMGT)	EGS
SEQ ID NO: 339	LCDR3 (IMGT)	SSYGFHIVVVV
SEQ ID NO: 344	VL	QSALTQPASVSGSPGQSITISCTGTSSDVGSYNLVSWYQQHPGKAPKLMIE GSKRPSGVSNRPSGSKSGNTASLTISGLQAEDEADYYC SSYGFHIVVVVFGG GTKLTVL
SEQ ID NO: 345	DNA VL	cagagcgccttgacccagccagccagcgttagcggtagccagccagagcattaccattagctgcaccggcacca gcagcagctgggcagctataacctgggttagctggtatcagcagcatccgggcaaaagccccgaaactgatgatctatg aaggcagcaaacgcccagcggcgttagcaaccgctttagtggcagcaaaagcggcaacaccgcccagcctgacct

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
		tagcggcctgcaagccgaagacgaagccgattattactgctcctcttacggtttccatatacggtgtgtgtgtgtttggcggc ggtaccaagctgaccgtgctg
SEQ ID NO: 346	Light Chain	QSALTQPASVSGSPGQSITISCTGTSSDVGSYNLVSQHPGKAPKLMIE GSKRPSGVSNRFGSKSGNTASLTISGLQAEDEADYICSSYGFHVVVFGG GTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKA DSSPVKAGVETTTPSKQSNKYAASSYLSTPEQWKSHRSYSCQVTHEGST VEKTVAPTECS
SEQ ID NO: 347	DNA Light Chain	cagagcgccctgacccagccagccagcgttagcggtagccagccagagcattaccattagctgcaccggcacc gcagcgacgtgggcagctataacccgttagctggtagcagcagcaccgggcaagccccgaaactgatgatctatg aaggcagcaaacgcccagcggcggttagcaaccgctttagtggcagcaaacggcggaacaccgccagcctgacat tagcggcctgcaagccgaagacgaagccgattattactgctcctcttacggtttccatatacggtgtgtgtgtttggcggc ggtaccaagctgaccgtgctgggcccagcccaagccgcccctagcgtgacctgttcccccaagcagcaggaac tccagggcaacaagccaccctcgtgtgctgatcagcgaacttctaccctggcgccgtgaccgtggcctggaaggcc gatagcagccctgtgaaggccggcgtggaaaccaccaccccccagcaagcagagcaacaacaaatcgcgcgcagc agctacctgagcctgacccccgagcagtggaagtcccacagatcctacagctgccaggtcacacacgagggcagca ccgtggaaaagaccgtggccccaccagagtgcagc
ZZ17_ LALA		
SEQ ID NO: 418	HCDR1 (Combined)	GFTFSDYAIS
SEQ ID NO: 419	HCDR2 (Combined)	SISGGGYHTQYAGSVKG
SEQ ID NO: 331	HCDR3 (Combined)	PYLGDRRSYGFDH
SEQ ID NO: 420	HCDR1 (Kabat)	DYAIS
SEQ ID NO: 419	HCDR2 (Kabat)	SISGGGYHTQYAGSVKG
SEQ ID NO: 331	HCDR3 (Kabat)	PYLGDRRSYGFDH
SEQ ID NO: 421	HCDR1 (Chothia)	GFTFSDY
SEQ ID NO: 422	HCDR2 (Chothia)	SGGGYH
SEQ ID NO: 331	HCDR3 (Chothia)	PYLGDRRSYGFDH
SEQ ID NO: 423	HCDR1 (IMGT)	GFTFSDYA
SEQ ID NO: 424	HCDR2 (IMGT)	ISGGGYHT
SEQ ID NO: 332	HCDR3 (IMGT)	ARPYLGDRRSYGFDH
SEQ ID NO: 425	VH	EVQLLESQGLVQPGGSLRLSCAASGFTFSDYAIISWVRQAPGKLEWVSSI SGGGYHTQYAGSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARPYL GDRRSYGFDHWGQGLTVTVSS
SEQ ID NO: 426	DNA VH	gaagtgcagctgctggaagcggtggcggtctggtgcagccaggtgtagcctgcgcctgagctgtgccgcaagcg gctttacctttccgactatgcaatcagctgggtgcgcccaagcaccaggcaaggcctgggaatgggtgagcagcatttc cgggggggggtatcatcacaaatagcaggatccgtgaaaggccgcttaccattagtcgcgataacagcaaaaacac cctgtatctgcaaatgaacagcctgcgggcagaagataccgcagtttatattgcgcgcgacctatctgggtgaccgtc gtagctatggtttcgaccactggggccagggcaccctggttactgtctcgagc
SEQ ID NO: 427	Heavy Chain	EVQLLESQGLVQPGGSLRLSCAASGFTFSDYAIISWVRQAPGKLEWVSSI SGGGYHTQYAGSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARPYL GDRRSYGFDHWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVK DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYI CNVNHKPSNTKVDKRVPEKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDT LMI SRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVLHQDLNGKEYCKVSNKALPAPIEKTI SKAKGQPREPQVY LPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSD GSFPLYSKLTVDKSRWQQGNVPSFSVMHEALHNHYTQKSLSLSPGK

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences		
SEQ ID NO: 428	DNA Heavy Chain	gaagtgcagctgctggaagcggtggcggtctggtgcagccaggtggtagcctgcgcctgagctgtgccgcaagcg gctttaccttttccgactatgcaatcagctgggtgcgccaagcaccaggcgaaggcctggaatgggtgagcagcatttc cgggggggggtatcatcacaaatcgaggatccgtgaaaggccgctttaccattagtcgcgataacagcaaaaacac cctgtatctgcaaatgaacagcctgcgggcagagataaccgcagtttatattgcgcgcgaccttatctgggtgaccgtc gttagctatggtttcgaccactggggccagggcaccctgggtactgtctcgagcgcgtcgaccaaggccccagcgtgtt cctcttgccccccagcagcaagagcacctctggcggaacagccgctgggctgctggtaaggactacttccccg agcccgtagccgtgtcctggaactctggcgccctgaccagcggcgtgcacacctttccagccgtgctccagagcagc ggccctgtacagcctgagcagcgtcgtgaccgtgcccagcagcagcctgggcacccagacctacatctgcaacgtgaa ccacaagcccagcaacacaaaggtggacaaggggtggaacccaagagctgcgacaagaccacacctgtcccc ctgcccctgccccgaagcggcgggaggccccctcgtgttcctgttcccccaaggcctaaggacacctgatgatcag ccggacccccgaagtgcctgcgtgggtgggacgtgtccacgaggacctgaagtgaagttaaattggtaacgtgga cggcgtggaagtgcacaacgccaaagaccagccagagaggacagtaaacagcacctaccgggtgggtgtccgt gctgaccgtgctgcaccaggactggctgaacggcgaagagtacaagtgaagggtgtccacaaggccctgctgccc ccatacgagaaaaccatcagcaaggccaaaggccagccccgcgagccccaggtgtacacactgccccctagccggg aagagatgaccaagaaccaggtgtccctgacctgctcgtgaagggtcttaccgccagcacattgcccgtggaatggg agagcaacggccagcccgagaacaactacaagacccccccctgtgctggacagcgacgctcatcttctctgtac agcaagctgaccgtggacaagagccggtggcagcaggggcaacgtgtcagctgctccgtgatgcacgaggccctgc acaaccctacaccagaagtccttgagcctgagccccggcaag
SEQ ID NO: 337	LCDR1 (Combined)	TGTSSDVGSYNLVS
SEQ ID NO: 338	LCDR2 (Combined)	EGSKRPS
SEQ ID NO: 339	LCDR3 (Combined)	SSYGFHIVVV
SEQ ID NO: 337	LCDR1 (Kabat)	TGTSSDVGSYNLVS
SEQ ID NO: 338	LCDR2 (Kabat)	EGSKRPS
SEQ ID NO: 339	LCDR3 (Kabat)	SSYGFHIVVV
SEQ ID NO: 340	LCDR1 (Chothia)	TSSDVGSYNL
SEQ ID NO: 341	LCDR2 (Chothia)	EGS
SEQ ID NO: 342	LCDR3 (Chothia)	YGFHIVVV
SEQ ID NO: 343	LCDR1 (IMGT)	SSDVGSYNL
SEQ ID NO: 341	LCDR2 (IMGT)	EGS
SEQ ID NO: 339	LCDR3 (IMGT)	SSYGFHIVVV
SEQ ID NO: 344	VL	QSALTQPASVSGSPGQSITISCTGTSSDVGSYNLWSWYQQHPGKAPKLMIE GSKRPSGVSNRPSGSKSGNTASLTISGLQAEDEADYYCSSYGFHIVVVVFGG GTKLTLV
SEQ ID NO: 345	DNA VL	cagagcgccctgaccagccagccagcgttagcggtagccagggccagagcattaccattagctgcaccggcacca gcagcgacgtgggcagctataacctgggttagctggtatcagcagcatccgggcaaaagccccgaaactgatgatctatg aaggcagcaaacgcccagcggcgttagcaaccgctttagtggcagcaaaagcggcaacaccgccagcctgacctat tagcggcctgcaagccgaagacgaagccgattattactgctcctcttacgggtttccatatacgttgtgtgtgttggcggc ggtaccaagctgaccgtgctg
SEQ ID NO: 346	Light Chain	QSALTQPASVSGSPGQSITISCTGTSSDVGSYNLWSWYQQHPGKAPKLMIE GSKRPSGVSNRPSGSKSGNTASLTISGLQAEDEADYYCSSYGFHIVVVVFGG GTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKA DSSPVKAGVETTPSKQSNKYYAASSYLSLTPEQWKSHRYSYSCQVTHEGST VEKTVAPTECS
SEQ ID NO: 347	DNA Light Chain	cagagcgccctgaccagccagccagcgttagcggtagccagggccagagcattaccattagctgcaccggcacca gcagcgacgtgggcagctataacctgggttagctggtatcagcagcatccgggcaaaagccccgaaactgatgatctatg aaggcagcaaacgcccagcggcgttagcaaccgctttagtggcagcaaaagcggcaacaccgccagcctgacctat tagcggcctgcaagccgaagacgaagccgattattactgctcctcttacgggtttccatatacgttgtgtgtgttggcggc ggtaccaagctgaccgtgctggggcagcccaagccgcccctagcgtgacccctgttcccccaagcagcaggaac tccaggccaacaaggccacctcgtgtgctgacagcgaactctaccctggcgccgtgaccgtggcctggaaggcc gatagcagccctgtgaaggccggcgtggaaaccaccacccccagcaagcagagcaacaacaataccgcccgaccg

TABLE 2-continued

Exemplary anti-NPR1 antibody sequences

agctacctgagcctgacccccgagcagtggaagtccacagatcctacagctgccaggtcacacacgagggcagca
ccgtggaaaagaccgtggccccccacgagtgacgc

For example, WW01_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 4 (HCDR1), SEQ ID NO: 5 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (II) SEQ ID NO: 7 (HCDR1), SEQ ID NO: 5 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (III) SEQ ID NO: 8 (HCDR1), SEQ ID NO: 9 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3); or (IV) SEQ ID NO: 10 (HCDR1), SEQ ID NO: 11 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, WW01_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 4 (HCDR1), SEQ ID NO: 5 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, WW01_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 7 (HCDR1), SEQ ID NO: 5 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, WW01_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 8 (HCDR1), SEQ ID NO: 9 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3). In some embodiments, WW01_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 10 (HCDR1), SEQ ID NO: 11 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, WW01_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 13, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 24. In some embodiments, WW01_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 15, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 26.

For example, WW03_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 46 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47

(LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 43 (LCDR3). In some embodiments, WW03_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3). In some embodiments, WW03_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3). In some embodiments, WW03_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 46 (LCDR3). In some embodiments, WW03_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 43 (LCDR3). In some embodiments, WW03_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 37, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 48. In some embodiments, WW03_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 39, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 50.

For example, WW05_LALA may be defined as having three heavy chain complementarity regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 52 (HCDR1), SEQ ID NO: 53 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 65 (LCDR1), SEQ ID NO: 66 (LCDR2), and SEQ ID NO: 67 (LCDR3); (II) SEQ ID NO: 55 (HCDR1), SEQ ID NO: 53 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 65 (LCDR1), SEQ ID NO: 66 (LCDR2), and SEQ ID NO: 67 (LCDR3); (III) SEQ ID NO: 56 (HCDR1), SEQ ID NO: 57 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 68 (LCDR1), SEQ ID NO: 69 (LCDR2), and SEQ ID NO: 70 (LCDR3); or (IV) SEQ ID NO: 58 (HCDR1), SEQ ID NO: 59 (HCDR2), SEQ ID NO: 60 (HCDR3), SEQ ID NO: 71 (LCDR1), SEQ ID NO: 69 (LCDR2), and SEQ ID NO: 67 (LCDR3). In some embodiments, WW05_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 52 (HCDR1), SEQ ID NO: 53 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 65 (LCDR1), SEQ ID NO: 66 (LCDR2), and SEQ ID NO: 67 (LCDR3). In some embodiments, WW05_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 55 (HCDR1), SEQ ID NO: 53 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 65 (LCDR1), SEQ ID NO: 66 (LCDR2), and SEQ ID NO: 67 (LCDR3). In some embodiments, WW05_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 56 (HCDR1), SEQ ID NO: 57 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 68 (LCDR1), SEQ ID NO: 69 (LCDR2), and SEQ ID NO: 70 (LCDR3). In some embodiments, WW05_LALA may be defined as comprising

For example, XX01_DAPA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 4 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (II) SEQ ID NO: 7 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (III) SEQ ID NO: 8 (HCDR1), SEQ ID NO: 101 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3); or (IV) SEQ ID NO: 10 (HCDR1), SEQ ID NO: 102 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX01_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 4 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX01_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 7 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX01_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 8 (HCDR1), SEQ ID NO: 101 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3). In some embodiments, XX01_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 10 (HCDR1), SEQ ID NO: 102 (HCDR2),

For example, XX08_DAPA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 4 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (II) SEQ ID NO: 7 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (III) SEQ ID NO: 8 (HCDR1), SEQ ID NO: 152 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3); or (IV) SEQ ID NO: 10 (HCDR1), SEQ ID NO: 153 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19

For example, XX08_N30S_DAPA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 112 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (II) SEQ ID NO: 7 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (III) SEQ ID NO: 113 (HCDR1), SEQ ID NO: 152 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3); or (IV) SEQ ID NO: 114 (HCDR1), SEQ ID NO: 153 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_N30S_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 112 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_N30S_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 7 (HCDR1), SEQ ID NO: 151 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_N30S_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 113 (HCDR1), SEQ ID NO: 152 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3). In some embodiments, XX08_N30S_DAPA may be defined as comprising or having amino acid sequences of SEQ ID NO: 114 (HCDR1), SEQ ID NO: 153 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3). In some embodiments, XX08_N30S_DAPA may be defined as comprising or having a heavy chain variable region comprising or having an

For example, XX11_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 179 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 178 (LCDR3). In some embodiments, XX11_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3). In some embodiments, XX11_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3). In some embodiments, XX11_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 179 (LCDR3). In some embodiments, XX11_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 178 (LCDR3). In some embodiments, XX11_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ

For example, XX14_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 191 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 173 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 192 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX14_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX14_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX14_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 191 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 173 (LCDR3). In some embodiments, XX14_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 192 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 172 (LCDR3). In some embodiments, XX14_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 193, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 174. In some

For example, XX16_LALA may be defined as having three heavy chain complementarity regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX16_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX16_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX16_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3). In some embodiments, XX16_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3). In some embodiments, XX16_LALA may be defined as comprising or having a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 136. In some embodiments, XX16_LALA may be defined as comprising or having a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 203, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 138.

For example, YY02_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 248 (HCDR1), SEQ ID NO: 249 (HCDR2), SEQ ID NO: 250 (HCDR3), SEQ ID NO: 260 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 262 (LCDR3); (II) SEQ ID NO: 251 (HCDR1), SEQ ID NO: 249 (HCDR2), SEQ ID NO: 250 (HCDR3), SEQ ID NO: 260 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 262 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 252 (HCDR2), SEQ ID NO: 250 (HCDR3), SEQ ID NO: 263 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 264 (LCDR3); or (IV) SEQ ID NO: 253 (HCDR1), SEQ ID NO: 254 (HCDR2), SEQ ID NO: 255 (HCDR3), SEQ ID NO: 265 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 262 (LCDR3). In some embodiments, YY02_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 248 (HCDR1), SEQ ID NO: 249 (HCDR2), SEQ ID NO: 250 (HCDR3), SEQ ID NO: 260 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 262 (LCDR3). In some embodiments, YY02_LALA may be defined as comprising or having amino acid sequences of SEQ ID NO: 251 (HCDR1), SEQ ID NO: 249 (HCDR2), SEQ ID NO: 250 (HCDR3), SEQ ID NO: 260 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 262 (LCDR3). In some embodiments, YY02_LALA may be defined as comprising or having amino acid sequences of

For example, YY04_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from: (I) SEQ ID NO: 291 (HCDR1), SEQ

For example, YY07_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and

For example, ZZ13_LALA may be defined as having three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain

In some embodiments, the antibody or antigen binding fragment comprises a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 37 or

In some embodiments, the antibody or antigen binding fragment comprises a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 193 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 193, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 136 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 136. In some embodiments, the antibody or antigen binding fragment

In some embodiments, the antibody or antigen binding fragment comprises a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 201 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 201, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 174 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 174. In some embodiments, the antibody or antigen binding fragment comprises a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 203 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 203, and a light chain comprising or having an amino

In some embodiments, the antibody or antigen binding fragment comprises a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 300

In some embodiments, the antibody or antigen binding fragment comprises a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 403 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 403, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 344 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 344. In some embodiments, the antibody or antigen binding fragment

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comprises a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 405 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 405, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 346 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 346.

In some embodiments, the antibody or antigen binding fragment comprises a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 414 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 414, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 344 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 344. In some embodiments, the antibody or antigen binding fragment comprises a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 416 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 416, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 346 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 346.

In some embodiments, the antibody or antigen binding fragment comprises a heavy chain variable region comprising or having an amino acid sequence of SEQ ID NO: 425 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 425, and a light chain variable region comprising or having an amino acid sequence of SEQ ID NO: 344 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 344. In some

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embodiments, the antibody or antigen binding fragment comprises a heavy chain comprising or having an amino acid sequence of SEQ ID NO: 427 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 427, and a light chain comprising or having an amino acid sequence of SEQ ID NO: 346 or a sequence having at least 95%, 96%, 97%, 98%, or 99% identity with SEQ ID NO: 346.

Groups of exemplary anti-NPR1 antibodies of the present disclosure are set forth in Table 3 or Table 4 by CDR (i.e., numerical values in these tables represent sequence identifiers such that “28” represents “SEQ ID NO: 28”) or amino acid consensus sequences. When multiple numerical values are presented in Table 3, they may be used in the alternative for that CDR (i.e., when SEQ ID Nos: 29, 119, and 190 are listed for HCDR2, they may be used in the alternative for that CDR). In Table 4, those amino acids presented in parentheses and separated by a slash represent alternative amino acids in that position (e.g., “(A/V)” represents a position at which the amino acid may be alanine or valine). In some embodiments, the antibody has the heavy and light chain CDRs of any of the antibodies described in Table 3 or Table 4. In some embodiments, the anti-NPR1 antibody is a four-chain antibody (also referred to as an intact antibody), comprising two heavy chains and two light chains. In some embodiments, the anti-NPR1 antibody is an antigen binding fragment of an intact antibody, e.g., a functional fragment of an intact antibody selected from any of those set forth in Table 3 or Table 4 that retains the ability to bind NPR1 and/or provide a function of the intact antibody (e.g., activating NPR1 in the absence of ANP).

TABLE 3

Exemplary anti-NPR1 antibody groups by CDR						
	HCDR1	HCDR2	HCDR3	LCDR1	LCDR2	LCDR3
ANP non-competitive group 1						
Combined	28	29 119 190	30	41	42	43 126 134 145 172 178 184
Kabat	31	29 119 190	30	41	42	43 126 134 145 172 178 184
Chothia	32	33 120 191	30	44	45	46 127 135 146 173 179 185
IMGT	34	35 121 192	36	47	45	43 126 134 145

TABLE 3-continued

Exemplary anti-NPR1 antibody groups by CDR						
	HCDR1	HCDR2	HCDR3	LCDR1	LCDR2	LCDR3
ANP non-competitive group 2						172
						178
						184
	Combined	28	119	30	41	42
						126
						134
						145
						172
						178
						184
Kabat	31	119	30	41	42	126
						134
						145
						172
						178
						184
	Chothia	32	120	30	44	45
						127
						135
						146
IMGT	34	121	36	47	45	173
						179
						185
						126
						134
						145
						172
						178
						184
ANP non-competitive group 3	Combined	4	5	6	17	18
		112	100			19
		165	151			
	Kabat	7	5	6	17	18
			100			19
			151			
	Chothia	8	9	6	20	21
		113	101			22
		166	152			
	IMGT	10	11	12	23	21
ANP non-competitive group 4		114	102			19
		167	153			
	Combined	4	5	6	17	18
		112	100			19
		165				
	Kabat	7	5	6	17	18
			100			19
			151			
	Chothia	8	9	6	20	21
		113	101			22
ANP non-competitive group 5		166				
	IMGT	10	11	12	23	21
		114	102			19
		167				
	Combined	226	227	228	237	238
		367	368			239
		378	379			
	Kabat	229	227	228	237	238
		369	368			239
		380	379			

TABLE 3-continued

Exemplary anti-NPR1 antibody groups by CDR						
	HCDR1	HCDR2	HCDR3	LCDR1	LCDR2	LCDR3
Chothia	32 370	230 371	228	240	241	242
IMGT	381 34 372 383	382 231 373 384	232	243	241	239
ANP non-competitive group 6						
Combined	226 378	227 368 379	228	237	238	239
Kabat	229 380	227 368 379	228	237	238	239
Chothia	32 381	230 371 382	228	240	241	242
IMGT	34 383	231 373 384	232	243	241	239
ANP non-competitive group 7						
Combined	226 367 378	368 379	228	237	238	239
Kabat	229 369 380	368 379	228	237	238	239
Chothia	32 370 381	371 382	228	240	241	242
IMGT	34 372 383	373 384	232	243	241	239
ANP non-competitive group 8						
Combined	226 378	368 379	228	237	238	239
Kabat	229 380	368 379	228	237	238	239
Chothia	32 381	371 382	228	240	241	242
IMGT	34 372 383	373 384	232	243	241	239
ANP competitive group 1						
Combined	310	311	312 348	320	321 354	322 355 361
Kabat	229	311	312 348	320	321 354	322 355 361
Chothia	80	313	312 348	323	324	325 356 362
IMGT	82	314	315 349	326	324	322 355 361
ANP competitive group 2						
Combined	270 407	271 389 408	331	337	338	339
Kabat	273 409	271 389 408	331	337	338	339

TABLE 3-continued

Exemplary anti-NPR1 antibody groups by CDR						
	HCDR1	HCDR2	HCDR3	LCDR1	LCDR2	LCDR3
Chothia	32	274	331	340	341	342
	410	390				
		411				
IMGT	275	276	332	343	341	339
	412	391				
		413				
ANP competitive group 3						
Combined	270	389	331	337	338	339
	407	408				
Kabat	273	389	331	337	338	339
	409	408				
Chothia	32	390	331	340	341	342
	410	411				
IMGT	275	276	332	343	341	339
	412	391				
		413				

TABLE 4

Exemplary anti-NPR1 antibody groups by consensus sequence						
	HCDR1	HCDR2	HCDR3	LCDR1	LCDR2	LCDR3
ANP non-competitive group A						
Combined	28	(A/V)I(S/E)S(D/K) G(S/N)Y(I/T)(Y/F) YADSVKG (SEQ ID NO: 429)	30	41	42	(M/Q)Q(S/E/T/I) (Y/W)(E/V/R/A/T/M) (K/V/R/A)PRT (SEQ ID NO: 430)
Kabat	31	(A/V)I(S/E)S(D/K) G(S/N)Y(I/T)(Y/F) YADSVKG (SEQ ID NO: 429)	30	41	42	(M/Q)Q(S/E/T/I) (Y/W)(E/V/R/A/T/M) (K/V/R/A)PRT (SEQ ID NO: 430)
Chothia	32	(E/S)S(D/K) G(S/N)Y (SEQ ID NO: 431)	30	44	45	(S/E/T/I)(Y/W) (E/V/R/A/T/M) (K/V/R/A)PR (SEQ ID NO: 432)
IMGT	34	I(S/E)S(D/K)G (S/N)Y(I/T) (SEQ ID NO: 433)	36	47	45	(M/Q)Q(S/E/T/I) (Y/W)(E/V/R/A/T/M) (K/V/R/A)PRT (SEQ ID NO: 430)
ANP non-competitive group B						
Combined	28	(A/V)I(S/E)S(D/K) G(S/N)Y(I/T)(Y/F) YADSVKG (SEQ ID NO: 429)	30	41	42	QQ(S/E/T/I)W (V/R/A/T/M) (K/V/R/A)PRT (SEQ ID NO: 434)
Kabat	31	(A/V)I(S/E)S(D/K) G(S/N)Y(I/T)(Y/F) YADSVKG (SEQ ID NO: 429)	30	41	42	QQ(S/E/T/I)W (V/R/A/T/M) (K/V/R/A)PRT (SEQ ID NO: 434)
Chothia	32	(E/S)S(D/K) G(S/N)Y (SEQ ID NO: 431)	30	44	45	(S/E/T/I)W (V/R/A/T/M) (K/V/R/A)PR (SEQ ID NO: 435)

TABLE 4-continued

Exemplary anti-NPR1 antibody groups by consensus sequence						
	HCDR1	HCDR2	HCDR3	LCDR1	LCDR2	LCDR3
IMGT	34	I(S/E)S(D/K) G(S/N)Y(I/T) (SEQ ID NO: 433)	36	47	45	QQ(S/E/T/I)W (V/R/A/T/M) (K/V/R/A)PRT (SEQ ID NO: 434)
ANP non-competitive group C						
Combined	28	119	30	41	42	QQ(S/E/T/I)W (V/R/A/T/M) (K/V/R/A)PRT (SEQ ID NO: 434)
Kabat	31	119	30	41	42	QQ(S/E/T/I) W(V/R/A/T/M) (K/V/R/A)PRT (SEQ ID NO: 434)
Chothia	32	120	30	44	45	(S/E/T/I)W (V/R/A/T/M) (K/V/R/A)PR (SEQ ID NO: 435)
IMGT	34	121	36	47	45	QQ(S/E/T/I) W(V/R/A/T/M) (K/V/R/A)PRT (SEQ ID NO: 434)
ANP non-competitive group D						
Combined	GFTF(N/S/Q) THYIH (SEQ ID NO: 436)	SI(S/G)(S/G) (S/Q)G(S/Q/G) (S/N/M)T(Y/L) YADSVKG (SEQ ID NO: 437)	6	17	18	19
Kabat	7	SI(S/G)(S/G)(S/Q) G(S/Q/G)(S/N/M)T (Y/L)YADSVKG (SEQ ID NO: 437)	6	17	18	19
Chothia	GFTF (N/S/Q)TH (SEQ ID NO: 438)	(S/G)(S/G)(S/Q) G(S/Q/G)(S/N/M) (SEQ ID NO: 439)	6	20	21	22
IMGT	GFTF (N/S/Q)THY (SEQ ID NO: 440)	I(S/G)(S/G) (S/Q)G(S/Q/G) (S/N/M)T (SEQ ID NO: 441)	12	23	21	19
ANP non-competitive group E						
Combined	GFTF(N/S/Q) THYIH (SEQ ID NO: 436)	SIS(S/G)SG (S/Q)(S/N) TYYADSVKG (SEQ ID NO: 442)	6	17	18	19
Kabat	7	SIS(S/G)SG (S/Q)(S/N) TYYADSVKG (SEQ ID NO: 442)	6	17	18	19
Chothia	GFTF (N/S/Q)TH (SEQ ID NO: 438)	S(S/G)SG (S/Q)(S/N) (SEQ ID NO: 443)	6	20	21	22
IMGT	GFTF (N/S/Q)THY (SEQ ID NO: 440)	IS(S/G)SG (S/Q)(S/N)T (SEQ ID NO: 444)	12	23	21	19
ANP non-competitive group F						
Combined	GF(S/T)FS (S/K/R)Y(W/Y) (I/L)(S/N) (SEQ ID NO: 445)	(N/S)I(K/H)Q (S/Q/H)(G/A) (S/H/L)E (T/K)(Y/K/R) YVESVKG (SEQ ID NO: 446)	228	237	238	239

TABLE 4-continued

Exemplary anti-NPR1 antibody groups by consensus sequence						
	HCDR1	HCDR2	HCDR3	LCDR1	LCDR2	LCDR3
Kabat	(S/K/R)Y(W/Y) (I/L)(S/N) (SEQ ID NO: 447)	(N/S)I(K/H)Q(S/Q/H) (G/A)(S/H/L)E (T/K)(Y/K/R) YVESVKG (SEQ ID NO: 446)	228	237	238	239
Chothia	GF(S/T)FS (S/K/R)Y (SEQ ID NO: 448)	(K/H)Q(S/Q/H) (G/A)(S/H/L)E (SEQ ID NO: 449)	228	240	241	242
IMGT	GF(S/T)FS (S/K/R)Y(W/Y) (SEQ ID NO: 450)	I(K/H)Q(S/Q/H) (G/A)(S/H/L)E(T/K) (SEQ ID NO: 451)	232	243	241	239
ANP non-competitive group G						
Combined	GFTFS(S/R)Y (W/Y)I(S/N) (SEQ ID NO: 452)	(N/S)I(K/H)Q (S/Q/H)(G/A)(S/H/L) E(T/K)(Y/K/R) YVESVKG (SEQ ID NO: 446)	228	237	238	239
Kabat	(S/R)Y (W/Y)I(S/N) (SEQ ID NO: 454)	(N/S)I(K/H)Q (S/Q/H)(G/A)(S/H/L) E(T/K)(Y/K/R) YVESVKG (SEQ ID NO: 446)	228	237	238	239
Chothia	GFTFS(S/R)Y (SEQ ID NO: 455)	(K/H)Q(S/Q/H) (G/A)(S/H/L)E (SEQ ID NO: 449)	228	240	241	242
IMGT	GFTFS(S/R) Y(W/Y) (SEQ ID NO: 457)	I(K/H)Q(S/Q/H) (G/A)(S/H/L)E(T/K) (SEQ ID NO: 451)	232	243	241	239
ANP non-competitive group H						
Combined	GF(S/T)FS (S/K/R)Y(W/Y) (I/L)(S/N) (SEQ ID NO: 445)	SIHQ(Q/H)(G/A) (H/L)E(T/K)(K/R) YVESVKG (SEQ ID NO: 453)	228	237	238	239
Kabat	(S/K/R)Y (W/Y)(I/L)(S/N) (SEQ ID NO: 447)	SIHQ(Q/H)(G/A) (H/L)E(T/K)(K/R) YVESVKG (SEQ ID NO: 453)	228	237	238	239
Chothia	GF(S/T)FS (S/K/R)Y (SEQ ID NO: 448)	HQ(Q/H) (G/A)(H/L)E (SEQ ID NO: 456)	228	240	241	242
IMGT	GF(S/T)FS (S/K/R)Y(W/Y) (SEQ ID NO: 450)	IHQ(Q/H)(G/A) (H/L)E(T/K) (SEQ ID NO: 458)	232	243	241	239
ANP non-competitive group I						
Combined	GFTFS(S/R) Y(W/Y)I(S/N) (SEQ ID NO: 452)	SIHQ(Q/H)(G/A) (H/L)E(T/K)(K/R) YVESVKG (SEQ ID NO: 453)	228	237	238	239
Kabat	(S/R)Y(W/Y) I(S/N) (SEQ ID NO: 454)	SIHQ(Q/H)(G/A) (H/L)E(T/K)(K/R) YVESVKG (SEQ ID NO: 453)	228	237	238	239
Chothia	GFTFS(S/R)Y (SEQ ID NO: 455)	HQ(Q/H) (G/A)(H/L)E (SEQ ID NO: 456)	228	240	241	242
IMGT	GFTFS(S/R) Y(W/Y) (SEQ ID NO: 457)	IHQ(Q/H)(G/A) (H/L)E(T/K) (SEQ ID NO: 458)	232	243	241	239
ANP competitive group A						
Combined	310	311	G(A/S)(V/L) (A/P)G(Q/L)L GFDH (SEQ ID NO: 459)	320	GNSNRP (S/N) (SEQ ID NO: 460)	QSY(Y/D/G) (T/S/A)(S/P/F) (S/T/P)(H/S/R) (G/S/F)(P/S/V)V (SEQ ID NO: 461)

TABLE 4-continued

Exemplary anti-NPR1 antibody groups by consensus sequence						
	HCDR1	HCDR2	HCDR3	LCDR1	LCDR2	LCDR3
Kabat	229	311	G(A/S)(V/L) (A/P)G(Q/L) LGFDH (SEQ ID NO: 459)	320	GNSNRP (S/N) (SEQ ID NO: 460)	QSY(Y/D/G)(T/S/A) (S/P/F)(S/T/P)(H/S/R) (G/S/F)(P/S/V)V (SEQ ID NO: 461)
Chothia	80	313	G(A/S)(V/L) (A/P)G(Q/L) LGFDH (SEQ ID NO: 459)	323	324	Y(Y/D/G)(T/S/A) (S/P/F)(S/T/P)(H/S/R) (G/S/F)(P/S/V) (SEQ ID NO: 462)
IMGT	82	314	ARG(A/S)(V/L) (A/P)G(Q/L) LGFDH (SEQ ID NO: 463)	326	324	QSY(Y/D/G)(T/S/A) (S/P/F)(S/T/P)(H/S/R) (G/S/F)(P/S/V)V (SEQ ID NO: 461)
ANP competitive group B						
Combined	GFTF(S/G) (S/T)YA(I/M)(S/T) (SEQ ID NO: 464)	(A/S)IS(A/S/G) (S/H)G(G/Y)(S/Y) (T/A)(Y/R/N) YA(E/G)SVKG (SEQ ID NO: 465)	331	337	338	339
Kabat	(S/T)YA (I/M)(S/T) (SEQ ID NO: 466)	(A/S)IS(A/S/G) (S/H)G(G/Y)(S/Y) (T/A)(Y/R/N) YA(E/G)SVKG (SEQ ID NO: 465)	331	337	338	339
Chothia	GFTF(S/G) (S/T)Y (SEQ ID NO: 467)	S(A/S/G)(S/H) G(G/Y)(S/Y) (SEQ ID NO: 468)	331	340	341	342
IMGT	GFTF(S/G) (S/T)YA (SEQ ID NO: 469)	IS(S/G)(S/H) G(G/Y)(S/Y)T (SEQ ID NO: 470)	332	343	341	339
ANP competitive group C						
Combined	GFTF(S/G)(S/T) YA(I/M)(S/T) (SEQ ID NO: 464)	SIS(A/S)(S/H)GY (T/A)(R/N)YA (E/G)SVKG (SEQ ID NO: 471)	331	337	338	339
Kabat	(S/T)YA(I/M) (S/T) (SEQ ID NO: 466)	SIS(A/S)(S/H)GY (T/A)(R/N) YA(E/G)SVKG (SEQ ID NO: 471)	331	337	338	339
Chothia	GFTF(S/G)(S/T)Y (SEQ ID NO: 467)	S(A/S)(S/H)GY (SEQ ID NO: 472)	331	340	341	342
IMGT	GFTF(S/G) (S/T)YA (SEQ ID NO: 469)	IS(A/S/G)(S/H)G (S/T)YA (SEQ ID NO: 473)	332	343	341	339

In some embodiments, an antibody or antigen-binding fragment thereof as provided herein binds to (a) human NPR1; and (b) mouse NPR1 and/or rat NPR1.

In some embodiments, an antibody or antigen-binding fragment thereof as provided herein binds to (a) human NPR1; and (b) cyno NPR1. In some embodiments, the antibody or antigen binding fragment thereof is therapeutic. A therapeutic antibody, as defined herein, is an antibody that is both efficacious and stable. Antibodies that Bind to the Same Epitope as Anti-NPR1 Antibodies of the Disclosure

In another embodiment, the disclosure provides antibodies or antigen-binding fragments thereof that bind to the same epitope as one or more of the anti-NPR1 antibodies described herein (e.g., WW06). Such antibodies:

- (i) bind NPR1;
 - (ii) are agonists of NPR1;
 - (iii) are ANP competitive; and
- bind the same epitope in NPR1 as antibody WW06.

In another embodiment, the disclosure provides antibodies or antigen-binding fragments thereof that bind to the

same epitope as one or more of the anti-NPR1 antibodies described herein (e.g., XX16). Such antibodies:

- (i) bind NPR1;
 - (ii) are agonists of NPR1;
 - (iii) are ANP non-competitive; and
- bind the same epitope in NPR1 as XX16.

In another embodiment, the disclosure provides antibodies or antigen-binding fragments thereof that bind to the same epitope as one or more of the anti-NPR1 antibodies described herein (e.g., WW03). Such antibodies:

- (i) bind NPR1;
 - (ii) are agonists of NPR1;
 - (iii) are ANP non-competitive; and
- bind the same epitope in NPR1 as WW03.

Following the crystallisation and structure determination, the binding regions of the preferred antibodies of the disclosure have been more clearly defined. Such binding is defined herein as being inclusive of both covalent and non-covalent bonds.

Thus, the disclosure provides an ANP competitive antibody that binds the same epitope as WW06. In some

embodiments, the disclosure provides an antibody that binds to an epitope of human NPR1 protein (Accession no. P16066; SEQ ID NO: 1) comprising amino acids 188, 192, 194, 197, 201, 208, and 219. In some embodiments, the disclosure provides an antibody that binds to an epitope of human NPR1 protein (Accession no. P16066; SEQ ID NO: 1) comprising amino acids 188, 192, 194, 197, 201, 208, 219, and 295. In some embodiments, the disclosure provides an antibody that binds to an epitope within amino acid numbers 188-198 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2, 3, or 4 amino acid residues within amino acid numbers 188-198 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to an epitope within amino acid numbers 201-208 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2 amino acids within amino acid numbers 201-208 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2, 3, or 4 amino acid residues within amino acid numbers 188-198 of SEQ ID NO: 1, and binds to at least 2 amino acids within amino acid numbers 201-208 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to an epitope comprising at least one amino acid residue within each of (i) amino acids 188-198 of SEQ ID NO: 1, (ii) amino acids 201-208 of SEQ ID NO: 1, (iii) amino acids 215-238 of SEQ ID NO: 1, and (iv) amino acids 294-297 of SEQ ID NO: 1.

The disclosure provides an ANP non-competitive antibody that binds the same epitope as WW03. In some embodiments, the disclosure provides an antibody that binds to an epitope of human NPR1 protein (Accession no. P16066; SEQ ID NO: 1) comprising amino acids 82, 102, 103, 105, 106, 109, 132, and 375. In some embodiments, the disclosure provides an antibody that binds to an epitope of human NPR1 protein (Accession no. P16066; SEQ ID NO: 1) comprising amino acids 79, 82, 99, 102, 103, 105, 106, 109, 131, 132, and 375. In some embodiments, the disclosure provides an antibody that binds to an epitope within amino acid numbers 99-111 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2, 3, 4, 5, or 6 amino acids within amino acid numbers 99-111 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2, 3, 4, 5, 6, 7, or 8 amino acid residues within amino acid numbers 99-133 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to an epitope within amino acid numbers 131-134 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2 amino acids within amino acid numbers 131-134 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2, 3, 4, 5, or 6 amino acids within amino acid numbers 99-111 of SEQ ID NO: 1, and binds to at least 2 amino acids within amino acid numbers 131-134 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to an epitope comprising at least one amino acid residue within each of (i) amino acids 99-111 of SEQ ID NO: 1, (ii) amino acids 131-134 of SEQ ID NO: 1, and (iii) amino acids 374-375 of SEQ ID NO: 1. Optionally, the antibody may additionally bind to amino acids 79 and/or 82 of SEQ ID NO: 1.

The disclosure additionally provides an ANP non-competitive antibody that binds the same epitope as XX16. In some embodiments, the disclosure provides an antibody that binds to an epitope of human NPR1 protein (Accession no. P16066; SEQ ID NO: 1) comprising amino acids 82, 102,

103, 105, 106, 109, 132, and 375. In some embodiments, the disclosure provides an antibody that binds to an epitope of human NPR1 protein (Accession no. P16066; SEQ ID NO: 1) comprising amino acids 34, 82, 102, 103, 105, 106, 107, 109, 132, 133, 375, and 378. In some embodiments, the disclosure provides an antibody that binds to an epitope within amino acid numbers 102-111 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2, 3, 4, 5, or 6 amino acids within amino acid numbers 102-111 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to an epitope within amino acid numbers 131-134 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2 amino acids within amino acid numbers 131-134 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to at least 2, 3, 4, 5, or 6 amino acids within amino acid numbers 102-111 of SEQ ID NO: 1, and binds to at least 2 amino acids within amino acid numbers 131-134 of SEQ ID NO: 1. In some embodiments, the disclosure provides an antibody that binds to an epitope comprising at least one amino acid residue within each of (i) amino acids 102-111 of SEQ ID NO: 1, (ii) amino acids 131-134 of SEQ ID NO: 1, and (iii) amino acids 374-378 of SEQ ID NO: 1. Optionally, the antibody may additionally bind to amino acids 34, 76, and/or 82 of SEQ ID NO: 1.

Additional antibodies can be identified based on their ability to cross-compete (e.g., to competitively inhibit the binding of, in a statistically significant manner) with other antibodies of the disclosure in standard NPR1 binding assays (e.g., XX16, WW06, or WW03). The ability of a test antibody to inhibit the binding of antibodies of the present disclosure to human NPR1 demonstrates that the test antibody can compete with that antibody for binding to human NPR1; such an antibody may, according to non-limiting theory, bind to the same or a related (e.g., a structurally similar or spatially proximal) epitope on human NPR1 as the antibody with which it competes. In a certain embodiment, the antibody that binds to the same epitope on human NPR1 as the antibodies of the present disclosure is a human antibody (e.g., a human monoclonal antibody or antigen binding fragment thereof). Such antibodies can be prepared and isolated as described herein.

Engineered and Modified Antibodies

An antibody of the disclosure can be prepared using an antibody having one or more of the VH and/or VL sequences shown herein as starting material to engineer a modified antibody, which modified antibody may have altered properties from the starting antibody. An antibody can be engineered by modifying one or more residues within one or both variable regions (i.e., VH and/or VL), for example within one or more CDR regions and/or within one or more framework regions. Additionally or alternatively, an antibody can be engineered by modifying residues within the constant region(s), for example to alter the effector function(s) of the antibody.

One type of variable region engineering that can be performed is antibody binding region/paratope or CDR grafting. Because paratope sequences are responsible for most antibody-antigen interactions, it is possible to express recombinant antibodies that mimic the properties of specific naturally occurring antibodies by constructing expression vectors that include CDR/paratope sequences from the specific naturally occurring antibody grafted onto framework sequences from a different antibody with different properties (see, e.g., Riechmann, L. et al., 1998 *Nature* 332:323-327; Jones, P. et al., 1986 *Nature* 321:522-525; Queen, C. et al.,

1989 Proc. Natl. Acad. Sci. U.S.A. 86:10029-10033; U.S. Pat. Nos. 5,225,539, and 5,530,101; 5,585,089; 5,693,762 and 6,180,370; the contents of each of which are herein incorporated by reference for this purpose).

Accordingly, another embodiment of the disclosure pertains to an isolated anti-NPR1 antibody, or a antigen-binding fragment thereof, comprising an antigen binding portion thereof, comprising a heavy chain variable region comprising the CDR sequences of an antibody or group of antibodies shown in Table 2, Table 3, or Table 4. Thus, such antibodies contain the VH and VL CDR sequences of monoclonal antibodies, yet may contain different framework sequences from these antibodies.

Such framework sequences can be obtained from public DNA databases or published references that include germline antibody gene sequences. For example, germline DNA sequences for human heavy and light chain variable region genes can be found in the "VBase" human germline sequence database (available on the Internet at [www \[dot\] mrc-cpe \[dot\] cam \[dot\] ac \[dot\] uk/vbase](http://www.mrc-cpe.cam.ac.uk/vbase)), as well as in Kabat, E. A., et al., 1991 Sequences of Proteins of Immunological Interest, Fifth Edition, U.S. Department of Health and Human Services, NIH Publication No. 91-3242; Tomlinson, I. M., et al., 1992 J. Mol. Biol. 227:776-798; and Cox, J. P. L. et al., 1994 Eur. J Immunol. 24:827-836; the contents of each of which are herein incorporated by reference for this purpose.

An example of framework sequences for use in the antibodies or antigen binding fragments of the disclosure are those that are structurally similar to the framework sequences used by selected antibodies of the disclosure, e.g., consensus sequences and/or framework sequences used by the antibodies or antigen binding fragments of the disclosure. The VH CDR1, 2 and 3 sequences, and the VL CDR1, 2 and 3 sequences, can be grafted onto framework regions that have an identical sequence to that found in the germline immunoglobulin gene from which the framework sequence derive, or the CDR sequences can be grafted onto framework regions that contain one or more mutations as compared to the germline sequences. For example, it has been found that in certain instances it is beneficial to mutate residues within the framework regions to maintain or enhance the antigen binding ability of the antibody (see e.g., U.S. Pat. Nos. 5,530,101; 5,585,089; 5,693,762 and 6,180,370; the contents of each of which are herein incorporated by reference for this purpose).

Another type of variable region modification is to mutate amino acid residues within the VH and/or VL CDR1, CDR2 and/or CDR3 regions to thereby improve one or more binding properties (e.g., affinity) of the antibody of interest, known as "affinity maturation." Site-directed mutagenesis or PCR-mediated mutagenesis can be performed to introduce the mutation(s) and the effect on antibody binding, or other functional property of interest, can be evaluated in vitro or in vivo assays as described herein. Conservative modifications (as discussed above) can also be introduced. The mutations may be amino acid substitutions, additions or deletions. Moreover, typically no more than one, two, three, four or five residues within a CDR region are altered. Grafting Antigen Binding Domains into Alternative Frameworks or Scaffolds

A wide variety of antibody/immunoglobulin frameworks or scaffolds can be employed so long as the resulting polypeptide includes at least one binding region which specifically binds to NPR1. Such frameworks or scaffolds include the 5 main idiotypes of human immunoglobulins, or fragments thereof (such as those disclosed elsewhere

herein), and include immunoglobulins of other animal species, preferably having humanized aspects. Single heavy-chain antibodies such as those identified in camelids are of particular interest in this regard. Novel frameworks, scaffolds and fragments continue to be discovered and developed by those skilled in the art.

In one aspect, the disclosure pertains to generating non-immunoglobulin based antibodies using non-immunoglobulin scaffolds onto which CDRs of the disclosure can be grafted. Known or future non-immunoglobulin frameworks and scaffolds may be employed, as long as they comprise a binding region specific for NPR1, e.g., such as those disclosed for an antibody described herein including, but not limited to, XX16, WW03, or WW06. Such compounds are known herein as "polypeptides comprising a target-specific binding region". Examples of non-immunoglobulin framework are further described in the sections below (camelid antibodies and non-antibody scaffold).

Camelid Antibodies

Antibody proteins obtained from members of the camel and dromedary (*Camelus bactrianus* and *Camelus dromedarius*) family including new world members such as llama species (*Lama paccos*, *Lama glama*, and *Lama vicugna*) have been characterized with respect to size, structural complexity and antigenicity for human subjects. Certain IgG antibodies from this family of mammals as found in nature lack light chains, and are thus structurally distinct from the typical four chain quaternary structure having two heavy and two light chains, for antibodies from other animals, see WO94/04678, the contents of which are herein incorporated by reference for this purpose.

A region of the camelid antibody which is the small single variable domain identified as VHH can be obtained by genetic engineering to yield a small protein having high affinity for a target, resulting in a low molecular weight antibody-derived protein known as a "camelid nanobody". See U.S. Pat. No. 5,759,808; see also Stijlemans, B. et al., 2004 J Biol Chem 279:1256-1261; Dumoulin, M. et al., 2003 Nature 424:783-788; Pleschberger, M. et al. 2003 Bioconjugate Chem 14:440-448; Cortez-Retamozo, V. et al. 2002 Int J Cancer 89:456-62; and Lauwereys, M. et al. 1998 EMBO J 17:3512-3520; the contents of each of which are herein incorporated by reference for this purpose. Engineered libraries of camelid antibodies and antibody fragments are commercially available, for example, from Abl-yx, Ghent, Belgium. As with other antibodies of non-human origin, an amino acid sequence of a camelid antibody can be altered recombinantly to obtain a sequence that more closely resembles a human sequence, i.e., the nanobody can be "humanized". Thus the natural low antigenicity of camelid antibodies to humans can be further reduced.

The camelid nanobody has a molecular weight approximately one-tenth that of a human IgG molecule, and the protein has a physical diameter of only a few nanometers. One consequence of the small size is the ability of camelid nanobodies to bind to antigenic sites that are functionally invisible to larger antibody proteins, i.e., camelid nanobodies are useful as reagents detect antigens that are otherwise cryptic using classical immunological techniques, and as possible therapeutic agents. Thus yet another consequence of small size is that a camelid nanobody can inhibit as a result of binding to a specific site in a groove or narrow cleft of a target protein, and hence can serve in a capacity that more closely resembles the function of a classical low molecular weight drug than that of a classical antibody.

The low molecular weight and compact size further result in camelid nanobodies being extremely thermostable, stable

to extreme pH and to proteolytic digestion, and poorly antigenic. Another consequence is that camelid nanobodies readily move from the circulatory system into tissues, and even cross the blood-brain barrier and can treat disorders that affect nervous tissue. Nanobodies can further facilitated drug transport across the blood brain barrier, see US2004/0161738, the contents of which are herein incorporated by reference for this purpose. These features combined with the low antigenicity to humans indicate great therapeutic potential. Further, these molecules can be fully expressed in prokaryotic cells such as *E. coli* and may be expressed as functional fusion proteins with bacteriophage.

Accordingly, a feature of the present disclosure is a camelid antibody or nanobody having high affinity for NPR1. In one embodiment, the camelid antibody or nanobody is obtained by grafting the CDRs sequences of the heavy or light chain of the human antibodies of the disclosure into nanobody or single domain antibody framework sequences as described, for example, in WO94/04678 (the contents of which are herein incorporated by reference for this purpose).

Framework or Fc Engineering

Engineered antibodies of the disclosure include those in which modifications have been made to framework residues within VH and/or VL, e.g., to improve one or more properties of the antibody. Typically such framework modifications are made to decrease the immunogenicity of the antibody. Antibodies of the disclosure may be modified in one or more ways, including each of the ways described herein.

For example, one approach is to "backmutate" one or more framework residues to the corresponding germline sequence. More specifically, an antibody that has undergone somatic mutation may contain framework residues that differ from the germline sequence from which the antibody is derived. Such residues can be identified by comparing the antibody framework sequences to the germline sequences from which the antibody is derived. To return the framework region sequences to their germline configuration, the somatic mutations can be "backmutated" to the germline sequence by, for example, site-directed mutagenesis or PCR-mediated mutagenesis. Such "backmutated" antibodies and additional modifications described herein are also intended to be encompassed by the disclosure.

Another type of framework modification involves mutating one or more residues within the framework region, or even within one or more CDR regions, to remove T-cell epitopes to thereby reduce the potential immunogenicity of the antibody. This approach is also referred to as "deimmunization" and is described in further detail in US2003/0153043, the contents of which are herein incorporated by reference for this purpose.

In addition or alternative to modifications made within the framework or CDR regions, antibodies of the disclosure may be engineered to include modifications within the Fc region, typically to alter one or more functional properties of the antibody, such as serum half-life, complement fixation, Fc receptor binding, and/or antigen-dependent cellular cytotoxicity. Furthermore, an antibody of the disclosure may be chemically modified (e.g., one or more chemical moieties can be attached to the antibody) or be modified to alter its glycosylation, again to alter one or more functional properties of the antibody. Each of these embodiments is described in further detail below. The numbering of residues in the Fc region is that of the EU index of Kabat.

In one embodiment, the hinge region of CHI is modified such that the number of cysteine residues in the hinge region

is altered, e.g., increased or decreased. This approach is described further in U.S. Pat. No. 5,677,425, the contents of which are herein incorporated by reference for this purpose. The number of cysteine residues in the hinge region of CHI is altered to, for example, facilitate assembly of the light and heavy chains or to increase or decrease the stability of the antibody.

In another embodiment, the Fc hinge region of an antibody is mutated to decrease the biological half-life of the antibody. More specifically, one or more amino acid mutations are introduced into the CH2-CH3 domain interface region of the Fc-hinge fragment such that the antibody has impaired Staphylococcal protein A (SpA) binding relative to native Fc-hinge domain SpA binding. This approach is described in further detail in U.S. Pat. No. 6,165,745, the contents of which are herein incorporated by reference for this purpose.

In another embodiment, the antibody is modified to increase its biological half-life. Various approaches are possible. For example, one or more of the following mutations can be introduced: T252L, T254S, T256F, as described in U.S. Pat. No. 6,277,375, the contents of which are herein incorporated by reference for this purpose. Alternatively, to increase the biological half life, the antibody can be altered within the CHI or CL region to contain a salvage receptor binding epitope taken from two loops of a CH2 domain of an Fc region of an IgG, as described in U.S. Pat. Nos. 5,869,046 and 6,121,022, the contents of each of which are herein incorporated by reference for this purpose.

In yet other embodiments, the Fc region is altered by replacing at least one amino acid residue with a different amino acid residue to alter the effector functions of the antibody. For example, one or more amino acids can be replaced with a different amino acid residue such that the antibody has an altered affinity for an effector ligand but retains the antigen-binding ability of the parent antibody. The effector ligand to which affinity is altered can be, for example, an Fc receptor or the C1 component of complement. This approach is described in further detail in U.S. Pat. Nos. 5,624,821 and 5,648,260, the contents of each of which are herein incorporated by reference for this purpose.

In order to minimize the ADCC activity of an antibody, specific mutations in the Fc region result in "Fc silent" antibodies that have minimal interaction with effector cells. In general, the "IgG Fc region" is used to define the C-terminal region of an immunoglobulin heavy chain, including native sequence Fc region and variant Fc regions. The human IgG heavy chain Fc region is generally defined as comprising the amino acid residue from position C226 or from P230 to the carboxyl-terminus of the IgG antibody. The numbering of residues in the Fc region is that of the EU index of Kabat. The C-terminal lysine (residue K447) of the Fc region may be removed, for example, during production or purification of the antibody.

Silenced effector functions can be obtained by mutation in the Fc region of the antibodies. See, for example, LALA and N297A (Strohl, W., 2009, Curr. Opin. Biotechnol. vol. 20(6):685-691); and D265A (Baudino et al., 2008, J. Immunol. 181:6664-69) see also Heusser et al., WO2012065950, the contents of each of which are herein incorporated by reference for this purpose. In particular, residues 234 and/or 235 may be mutated, optionally to alanine. Thus, in one embodiment, an antibody according to the disclosure has a mutation in the Fc region at one or both of amino acids 234 and 235. Such substitution of both amino acids 234 and 235 results in reduced ADCC activity. One example of such a mutation is the LALA mutant comprising L234A and L235A

mutation in the IgG1 Fc amino acid sequence. Another example of a silent IgG1 antibody is the DAPA (D265A, P329A) mutation (U.S. Pat. No. 6,737,056, the contents of which are herein incorporated by reference for this purpose). Another silent IgG1 antibody comprises the N297A mutation, which results in aglycosylated/non-glycosylated antibodies. Fc silent antibodies result in no or low ADCC activity, meaning that an Fc silent antibody exhibits an ADCC activity that is below 50% specific cell lysis. No ADCC activity means that the Fc silent antibody exhibits an ADCC activity (specific cell lysis) that is below 1%.

In another embodiment, one or more amino acids selected from amino acid residues can be replaced with a different amino acid residue such that the antibody has altered C1q binding and/or reduced or abolished complement dependent cytotoxicity (CDC). This approach is described in further detail in U.S. Pat. No. 6,194,551, the contents of which are herein incorporated by reference for this purpose.

In another embodiment, one or more amino acid residues are altered to thereby alter the ability of the antibody to fix complement. This approach is described further in WO94/29351, the contents of which are herein incorporated by reference for this purpose.

In yet another embodiment, the Fc region is modified to increase the ability of the antibody to mediate antibody dependent cellular cytotoxicity (ADCC) and/or to increase the affinity of the antibody for an Fcγ receptor by modifying one or more amino acids. This approach is described further in WO00/42072, the contents of which are herein incorporated by reference for this purpose. Moreover, the binding sites on human IgG₁ for FcγRI, FcγRII, FcγRIII and FcRn have been mapped and variants with improved binding have been described (see Shields, R. L. et al., 2001 J. Biol. Chem. 276:6591-6604, the contents of which are herein incorporated by reference for this purpose).

In still another embodiment, the glycosylation of an antibody is modified. For example, an aglycosylated antibody can be made (i.e., the antibody lacks glycosylation). Glycosylation can be altered to, for example, increase the affinity of the antibody for "antigen". Such carbohydrate modifications can be accomplished by, for example, altering one or more sites of glycosylation within the antibody sequence. For example, one or more amino acid substitutions can be made that result in elimination of one or more variable region framework glycosylation sites to thereby eliminate glycosylation at that site. Such aglycosylation may increase the affinity of the antibody for antigen. Such an approach is described in further detail in U.S. Pat. Nos. 5,714,350 and 6,350,861, the contents of each of which are herein incorporated by reference for this purpose.

Additionally or alternatively, an antibody can be made that has an altered type of glycosylation, such as a hypofucosylated antibody having reduced amounts of fucosyl residues or an antibody having increased bisecting GlcNAc structures. Such altered glycosylation patterns have been demonstrated to increase the ADCC ability of antibodies. Such carbohydrate modifications can be accomplished by, for example, expressing the antibody in a host cell with altered glycosylation machinery. Cells with altered glycosylation machinery have been described in the art and can be used as host cells in which to express recombinant antibodies of the disclosure to thereby produce an antibody with altered glycosylation. For example, EP 1,176,195 (the contents of which are herein incorporated by reference for this purpose) describes a cell line with a functionally disrupted FUT8 gene, which encodes a fucosyl transferase, such that antibodies expressed in such a cell line exhibit hypofuco-

sylation. WO03/035835 describes a variant CHO cell line, Lec13 cells, with reduced ability to attach fucose to Asn (297)-linked carbohydrates, also resulting in hypofucosylation of antibodies expressed in that host cell (see also Shields, R. L. et al., 2002 J. Biol. Chem. 277:26733-26740). WO99/54342 describes cell lines engineered to express glycoprotein-modifying glycosyl transferases (e.g., beta(1,4)-N acetylglucosaminyltransferase III (GnTIII)) such that antibodies expressed in the engineered cell lines exhibit increased bisecting GlcNAc structures which results in increased ADCC activity of the antibodies (see also Umana et al., 1999 Nat. Biotech. 17:176-180). The contents of each of the foregoing applications and references are herein incorporated by reference for this purpose.

Another modification of the antibodies herein that is contemplated by the disclosure is pegylation. An antibody can be pegylated, for example, to increase the biological (e.g., serum) half-life of the antibody. To pegylate an antibody, the antibody, or fragment thereof, typically is reacted with polyethylene glycol (PEG), such as a reactive ester or aldehyde derivative of PEG, under conditions in which one or more PEG groups become attached to the antibody or antibody fragment. The pegylation can be carried out by an acylation reaction or an alkylation reaction with a reactive PEG molecule (or an analogous reactive water-soluble polymer). As used herein, the term "polyethylene glycol" is intended to encompass any of the forms of PEG that have been used to derivatize other proteins, such as mono (C1-C10) alkoxy- or aryloxy-polyethylene glycol or polyethylene glycol-maleimide. In certain embodiments, the antibody to be pegylated is an aglycosylated antibody. Methods for pegylating proteins are known in the art and can be applied to the antibodies of the disclosure. See, for example, EP0154316 and EP0401384, the contents of each of which are herein incorporated by reference for this purpose.

Another modification of the antibodies that is contemplated by the disclosure is a conjugate or a protein fusion of at least the antigen-binding region of the antibody of the disclosure to serum protein, such as human serum albumin or a fragment thereof to increase half-life of the resulting molecule. Such an approach is described, for example, in EP0322094, the contents of which are herein incorporated by reference for this purpose.

Another possibility is a fusion of at least the antigen-binding region of the antibody of the disclosure to proteins capable of binding to serum proteins, such as human serum albumin to increase half-life of the resulting molecule. Such approach is described, for example, in EP0486525, the contents of which are herein incorporated by reference for this purpose.

Nucleic Acid Molecules Encoding Antibodies of the Disclosure

Another aspect of the disclosure pertains to nucleic acid molecules that encode the antibodies of the disclosure. The term "nucleic acid" is used herein interchangeably with the term "polynucleotide," and refers to deoxyribonucleotides or ribonucleotides and polymers thereof in either single- or double-stranded form. The term encompasses nucleic acids containing known nucleotide analogs or modified backbone residues or linkages, which are synthetic, naturally-occurring, and non-naturally-occurring, which have similar binding properties as the reference nucleic acid, and which are metabolized in a manner similar to the reference nucleotides. Examples of such analogs include, without limitation, phosphorothioates, phosphoramidates, methyl phosphonates, chiral-methyl phosphonates, 2-O-methyl ribonucleotides,

peptide-nucleic acids (PNAs). In some embodiments, the nucleic acid may be an mRNA.

Unless otherwise indicated, a particular nucleic acid sequence also implicitly encompasses conservatively modified variants thereof (e.g., degenerate codon substitutions) and complementary sequences, as well as the sequence explicitly indicated. Specifically, degenerate codon substitutions may be achieved by generating sequences in which the third position of one or more selected (or all) codons is substituted with mixed-base and/or deoxyinosine residues (See: Batzer et al., *Nucleic Acids Res* 1991; 25(19):5081; Ohtsuka et al., *J Biol Chem* 1985; 260(5):2605-8; Rossolini et al., *Mol Cell Probes* 1994; 8(2):91-8; the contents of each of which are herein incorporated by reference for this purpose).

Provided herein are exemplary full length heavy and light chain nucleotide sequences of anti-NPR1 antibodies. In some embodiments, the nucleic acid molecules are one or more of those identified in Table 2, e.g., those encoding an anti-NPR1 antibody or antigen binding fragment thereof. In some other embodiments, the nucleic acid molecules described herein comprise nucleotide sequences that are substantially identical (e.g., at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%) to the nucleotide sequences of those identified in Table 2. When expressed from appropriate expression vectors, polypeptides encoded by these polynucleotides are capable of binding to a NPR1 protein (e.g., human NPR1).

Also provided herein are polynucleotides which encode at least one CDR region, and usually all three CDR regions, from the heavy and/or light chain of an anti-NPR1 antibody or antigen binding fragment of the disclosure. Further provided herein are polynucleotides which encode all or substantially all of the variable region sequence of the heavy chain and/or the light chain of an exemplary anti-NPR1 antibody or antigen binding fragment of the disclosure. Because of the degeneracy of the genetic code, a variety of nucleic acid sequences will encode each of the immunoglobulin amino acid sequences.

In some embodiments, the nucleic acid molecules disclosed herein encode both a variable region and a constant region of an antibody. In some embodiments, the nucleic acid molecules disclosed herein comprise nucleotides encoding a full-length heavy chain sequence that is substantially identical (e.g., at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%) to the heavy chain sequence of one of the antibodies described herein including those in Table 2. In some embodiments, the nucleic acid molecules disclosed herein comprise nucleotides encoding a full-length light chain sequence that is substantially identical (e.g., at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%) to the light chain sequence of one of the antibodies described herein including those in Table 2.

The nucleic acids may be present in whole cells, in a cell lysate, or may be nucleic acids in a partially purified or substantially pure form. A nucleic acid is "isolated" or "rendered substantially pure" when purified away from other cellular components or other contaminants, e.g., other cellular nucleic acids or proteins, by standard techniques, including alkaline/SDS treatment, CsCl banding, column chromatography, agarose gel electrophoresis and others well known in the art. See, F. Ausubel, et al., ed. 1987 *Current Protocols in Molecular Biology*, Greene Publishing and Wiley Interscience, New York, the contents of which are herein incorporated by reference for this purpose. A nucleic acid of the disclosure can be, for example, DNA or RNA and

may or may not contain intronic sequences. In an embodiment, the nucleic acid is a cDNA molecule. The nucleic acid may be present in a vector such as a phage display vector, or in a recombinant plasmid vector.

Nucleic acids of the disclosure can be obtained using standard molecular biology techniques. For antibodies expressed by hybridomas (e.g., hybridomas prepared from transgenic mice carrying human immunoglobulin genes as described further herein), cDNAs encoding the light and heavy chains of the antibody made by the hybridoma can be obtained by standard PCR amplification or cDNA cloning techniques. For antibodies obtained from an immunoglobulin gene library (e.g., using phage display techniques), nucleic acid encoding the antibody can be recovered from various phage clones that are members of the library.

The polynucleotide sequences can be produced by de novo solid-phase DNA synthesis or by PCR mutagenesis of an existing sequence (e.g., sequences as described herein in, for example, Table 2). Direct chemical synthesis of nucleic acids can be accomplished by methods known in the art, such as the phosphotriester method of Narang et al., 1979, *Meth. Enzymol.* 68:90; the phosphodiester method of Brown et al., *Meth. Enzymol.* 68:109, 1979; the diethylphosphoramidite method of Beaucage et al., *Tetra. Lett.*, 22:1859, 1981; and the solid support method of U.S. Pat. No. 4,458,066 (the contents of each of which are herein incorporated by reference for this purpose). Introducing mutations to a polynucleotide sequence by PCR can be performed as described in, e.g., *PCR Technology: Principles and Applications for DNA Amplification*, H. A. Erlich (Ed.), Freeman Press, NY, N.Y., 1992; *PCR Protocols: A Guide to Methods and Applications*, Innis et al. (Ed.), Academic Press, San Diego, Calif., 1990; Mattila et al., *Nucleic Acids Res.* 19:967, 1991; and Eckert et al., *PCR Methods and Applications* 1:17, 1991.

Once DNA fragments encoding VH and VL segments are obtained, these DNA fragments can be further manipulated by standard recombinant DNA techniques, for example to convert the variable region genes to full-length antibody chain genes, to Fab fragment genes or to an scFv gene. In these manipulations, a VL- or VH-encoding DNA fragment is operatively linked to another DNA molecule, or to a fragment encoding another protein, such as an antibody constant region or a flexible linker. The term "operatively linked", as used in this context, is intended to mean that the two DNA fragments are joined in a functional manner, for example, such that the amino acid sequences encoded by the two DNA fragments remain in-frame, or such that the protein is expressed under control of a desired promoter.

The isolated DNA encoding the VH region can be converted to a full-length heavy chain gene by operatively linking the VH-encoding DNA to another DNA molecule encoding heavy chain constant regions (CH1, CH2 and CH3). The sequences of human heavy chain constant region genes are known in the art (see e.g., Kabat, E. A., et al., 1991 *Sequences of Proteins of Immunological Interest*, Fifth Edition, U.S. Department of Health and Human Services, NIH Publication No. 91-3242, the contents of which are herein incorporated by reference for this purpose) and DNA fragments encompassing these regions can be obtained by standard PCR amplification. The heavy chain constant region can be an IgG1, IgG2, IgG3, IgG4, IgA, IgE, IgM, or IgD constant region. In some embodiments, the heavy chain constant region is an IgG1 isotype. For a Fab fragment heavy chain gene, the VH-encoding DNA can be operatively linked to another DNA molecule encoding only the heavy chain CHI constant region.

The isolated DNA encoding the VL region can be converted to a full-length light chain gene (as well as to a Fab light chain gene) by operatively linking the VL-encoding DNA to another DNA molecule encoding the light chain constant region, CL. The sequences of human light chain constant region genes are known in the art (see, e.g., Kabat, E. A., et al., 1991 Sequences of Proteins of Immunological Interest, Fifth Edition, U.S. Department of Health and Human Services, NIH Publication No. 91-3242, the contents of which are herein incorporated by reference for this purpose) and DNA fragments encompassing these regions can be obtained by standard PCR amplification. The light chain constant region can be a kappa or a lambda constant region.

To create an scFv gene, the VH- and VL-encoding DNA fragments are operatively linked to another fragment encoding a flexible linker, such that the VH and VL sequences can be expressed as a contiguous single-chain protein, with the VL and VH regions joined by the flexible linker (see e.g., Bird et al., 1988 Science 242:423-426; Huston et al., 1988 Proc. Natl. Acad. Sci. USA 85:5879-5883; McCafferty et al., 1990 Nature 348:552-554; the contents of each of which are herein incorporated by reference for this purpose).

Vectors

Various expression vectors can be employed to express the polynucleotides encoding the antibody of the disclosure or antigen-binding fragment thereof. Both viral-based and nonviral expression vectors can be used to produce the antibodies in a mammalian host cell. Nonviral vectors and systems include plasmids, episomal vectors, typically with an expression cassette for expressing a protein or RNA, and human artificial chromosomes (see, e.g., Harrington et al., Nat Genet. 15:345, 1997, the contents of which are herein incorporated by reference for this purpose). For example, nonviral vectors useful for expression of the polynucleotides and polypeptides of the multispecific antibody of the disclosure or domains thereof in mammalian (e.g., human) cells include pThioHis A, B and C, pcDNA3.1/His, pEBVHis A, B and C, (Invitrogen, San Diego, Calif.), MPS V vectors, and numerous other vectors known in the art for expressing other proteins. Useful viral vectors include vectors based on retroviruses, adenoviruses, adenoassociated viruses, herpes viruses, vectors based on SV40, papilloma virus, HBP Epstein Barr virus, vaccinia virus vectors and Semliki Forest virus (SFV). See, Brent et al., supra; Smith, Annu. Rev. Microbiol. 49:807, 1995; and Rosenfeld et al., Cell 68:143, 1992, the contents of each of which are herein incorporated by reference for this purpose.

The choice of expression vector depends on the intended host cells in which the vector is to be expressed. Expression vectors for mammalian host cells can include expression control sequences, such as an origin of replication, a promoter, and an enhancer (see, e.g., Queen, et al., Immunol. Rev. 89:49-68, 1986, the contents of which are herein incorporated by reference for this purpose), and necessary processing information sites, such as ribosome binding sites, RNA splice sites, polyadenylation sites, and transcriptional terminator sequences. These expression vectors usually contain promoters derived from mammalian genes or from mammalian viruses. Suitable promoters may be constitutive, cell type-specific, stage-specific, and/or modulatable or regulatable. Useful promoters include, but are not limited to, the metallothionein promoter, the constitutive adenovirus major late promoter, the dexamethasone-inducible MMTV promoter, the SV40 promoter, the MRP polIII promoter, the constitutive MPS V promoter, the tetracycline-inducible CMV promoter (such as the human immediate-early CMV

promoter), the constitutive CMV promoter, and known promoter-enhancer combinations.

Cultures of transformed organisms can be expanded under non-inducing conditions without biasing the population for coding sequences whose expression products are better tolerated by the host cells. In addition to promoters, other regulatory elements may also be required or desired for efficient expression of the antibody of the disclosure or fragments thereof. These elements typically include an ATG initiation codon and adjacent ribosome binding site or other sequences. In addition, the efficiency of expression may be enhanced by the inclusion of enhancers appropriate to the cell system in use (see, e.g., Scharf et al., Results Probl. Cell Differ. 20:125, 1994; and Bittner et al., Meth. Enzymol., 153:516, 1987; the contents of each of which are herein incorporated by reference for this purpose). For example, the SV40 enhancer or CMV enhancer may be used to increase expression in mammalian host cells.

Accordingly, the disclosure provides a cloning or expression vector comprising one or more of the nucleic acid sequences of the antibodies shown in Table 2. Furthermore, the disclosure provides a cloning or expression vector comprising a nucleic acid encoding one or more of the nucleotide sequences shown in Table 2.

Host Cells

For expression of the light and heavy chains, the expression vector or expression vectors encoding the heavy and light chains may be transferred into a host cell by standard techniques.

Methods for introducing expression vectors containing the polynucleotide sequences of interest vary depending on the type of cellular host. For example, calcium chloride transfection is commonly utilized for prokaryotic cells, whereas calcium phosphate treatment or electroporation may be used for other cellular hosts. (See generally Sambrook, et al., supra, the contents of which are herein incorporated by reference for this purpose). Other methods include, e.g., electroporation, calcium phosphate treatment, liposome-mediated transformation, injection and microinjection, ballistic methods, virosomes, immunoliposomes, polycationnucleic acid conjugates, naked DNA, artificial virions, fusion to the herpes virus structural protein VP22 (Elliot and O'Hare, Cell 88:223, 1997, the contents of which are herein incorporated by reference for this purpose), agent-enhanced uptake of DNA, and ex vivo transduction.

It is theoretically possible to express the antibodies of the disclosure in either prokaryotic or eukaryotic host cells. Expression of antibodies in eukaryotic cells, in particular mammalian host cells, is discussed because such eukaryotic cells, and in particular mammalian cells, are more likely than prokaryotic cells to assemble and secrete a properly folded and immunologically active antibody. Prokaryotic expression of antibody genes has been reported to be ineffective for production of high yields of active antibody (Boss, M. A. and Wood, C. R., 1985 Immunology Today 6:12-13, the contents of which are herein incorporated by reference for this purpose).

For long-term, high-yield production of recombinant proteins, stable expression will often be desired. For example, cell lines which stably express the antibodies or antigen-binding fragments thereof of the disclosure can be prepared using expression vectors of the disclosure which contain viral origins of replication or endogenous expression elements and a selectable marker gene. Following the introduction of the vector, cells may be allowed to grow for 1-2 days in an enriched media before they are switched to selective media. The purpose of the selectable marker is to

confer resistance to selection, and its presence allows growth of cells which successfully express the introduced sequences in selective media. Resistant, stably transfected cells can be proliferated using tissue culture techniques appropriate to the cell type. The present disclosure thus provides a method of producing the antibodies or antigen-binding fragments of the disclosure, wherein said method comprises the step of culturing a host cell comprising a nucleic acid encoding the antibodies or antigen-binding fragments.

In some embodiments, mammalian host cells are used to express and produce the anti-NPR1 antibodies or antigen binding fragments of the present disclosure. For example, they can be either a hybridoma cell line expressing endogenous immunoglobulin genes or a mammalian cell line harboring an exogenous expression vector. These include any normal mortal or normal or abnormal immortal animal or human cell. For example, a number of suitable host cell lines capable of secreting intact immunoglobulins have been developed including the CHO cell lines, various COS cell lines, HeLa cells, myeloma cell lines, transformed B-cells, and hybridomas. Exemplary host cells include but are not limited to Chinese hamster ovary (CHO) cells, human embryonic kidney (HEK) cells (e.g., HEK293, HEK293T, HEK293F), monkey kidney (COS) cells (e.g., COS-1, COS-7), baby hamster kidney (BHK) cells (e.g., BHK-21), African green monkey kidney cells (e.g. BSC-1), HeLa cells, human hepatocellular carcinoma cells (e.g., Hep G2), myeloma cells (e.g., NS0, 653, SP2/0), lymphoma cells, oocyte cells, and cells from a transgenic animal (e.g., mammary epithelial cells), or any derivative, immortalized, or transformed cell thereof. In particular, for use with NS0 myeloma cells, another expression system is the GS gene expression system shown in WO87/04462, WO89/01036 and EP0338841, the contents of each of which are herein incorporated by reference for this purpose. When recombinant expression vectors encoding antibody nucleic acid are introduced into mammalian host cells, the antibodies are produced by culturing the host cells for a period of time sufficient to allow for expression of the antibody in the host cells or secretion of the antibody into the culture medium in which the host cells are grown. Antibodies can be recovered from the culture medium using standard protein purification methods. Such purified antibodies of the disclosure may be used for any purpose including, but not limited to, the methods and uses described herein, and/or as part of a pharmaceutical composition as described herein.

In a further alternative, the host cell may be a yeast or a filamentous fungi engineered for mammalian-like glycosylation pattern, and capable for producing antibodies lacking fucose as glycosylation pattern (see, for example, EP1297172, the contents of which are herein incorporated by reference for this purpose).

Accordingly, the disclosure provides a host cell comprising one or more of the vectors, or nucleic acid sequences of the disclosure described above.

Generation of Monoclonal Antibodies of the Disclosure

Monoclonal antibodies (mAbs) can be produced by a variety of techniques, including conventional monoclonal antibody methodology e.g., the somatic cell hybridization technique of Kohler and Milstein, 1975 *Nature* 256:495, the contents of which are herein incorporated by reference for this purpose. Many techniques for producing monoclonal antibody can be employed e.g., viral or oncogenic transformation of B lymphocytes.

Hybridomas may be prepared using, for example, the murine system. Immunization protocols and isolation of immunized splenocytes for fusion may be performed

according to any appropriate procedure. Chimeric or humanized antibodies can be prepared based on the sequence of a murine monoclonal antibody prepared as described herein. DNA encoding the heavy and light chain immunoglobulins can be obtained from the murine hybridoma of interest and engineered to contain non-murine (e.g., human) immunoglobulin sequences using standard molecular biology techniques. For example, to create a chimeric antibody, the murine variable regions can be linked to human constant regions using any known methods (see e.g., U.S. Pat. No. 4,816,567, the contents of which are herein incorporated by reference for this purpose). To create a humanized antibody, the murine CDR regions can be inserted into a human framework using any known methods. See e.g., U.S. Pat. Nos. 5,225,539, and 5,530,101; 5,585,089; 5,693,762 and 6,180,370, the contents of each of which are herein incorporated by reference for this purpose.

Human monoclonal antibodies can also be generated using transgenic or transchromosomal mice carrying parts of the human immune system rather than the mouse system. These transgenic and transchromosomal mice include mice referred to herein as HuMAb Mice® and KM mice, respectively, and are collectively referred to herein as "human Ig mice."

The HuMAb Mouse® (Medarex, Inc.) contains human immunoglobulin gene miniloci that encode un-rearranged human heavy (u and y) and K light chain immunoglobulin sequences, together with targeted mutations that inactivate the endogenous u and K chain loci (see e.g., Lonberg, et al., 1994 *Nature* 368(6474):856-859, the contents of which are herein incorporated by reference for this purpose). Accordingly, the mice exhibit reduced expression of mouse IgM or K, and in response to immunization, the introduced human heavy and light chain transgenes undergo class switching and somatic mutation to generate high affinity human IgGK monoclonal (Lonberg, N. et al., 1994 *supra*; reviewed in Lonberg, N., 1994 *Handbook of Experimental Pharmacology* 113:49-101; Lonberg, N. and Huszar, D., 1995 *Intern. Rev. Immunol.* 13:65-93, and Harding, F. and Lonberg, N., 1995 *Ann. N. Y. Acad. Sci.* 764:536-546; the contents of each of which are herein incorporated by reference for this purpose). The preparation and use of HuMAb Mice®, and the genomic modifications carried by such mice, is further described in Taylor, L. et al., 1992 *Nucleic Acids Research* 20:6287-6295; Chen, J. et al., 1993 *International Immunology* 5:647-656; Tuailon et al., 1993 *Proc. Natl. Acad. Sci. USA* 94:3720-3724; Choi et al., 1993 *Nature Genetics* 4:117-123; Chen, J. et al., 1993 *EMBO J.* 12:821-830; Tuailon et al., 1994 *J. Immunol.* 152:2912-2920; Taylor, L. et al., 1994 *International Immunology* 579-591; and Fishwild, D. et al., 1996 *Nature Biotechnology* 14:845-851, the contents of each of which are hereby incorporated by reference for this purpose. See further, U.S. Pat. Nos. 5,545,806; 5,569,825; 5,625,126; 5,633,425; 5,789,650; 5,877,397; 5,661,016; 5,814,318; 5,874,299; and 5,770,429; U.S. Pat. No. 5,545,807; WO92/103918, WO93/12227, WO94/25585, WO97/113852, WO98/24884 and WO99/45962; and WO01/14424; the contents of each of which are hereby incorporated by reference for this purpose.

In another embodiment, human antibodies can be raised using a mouse that carries human immunoglobulin sequences on transgenes and transchromosomes such as a mouse that carries a human heavy chain transgene and a human light chain transchromosome. Such mice, referred to herein as "KM mice", are described in detail in WO02/43478, the contents of which are hereby incorporated by reference for this purpose.

Still further, alternative transgenic animal systems expressing human immunoglobulin genes are available in the art and can be used to raise antibodies of the disclosure. For example, an alternative transgenic system referred to as the Xenomouse (Abgenix, Inc.) can be used. Such mice are described in, e.g., U.S. Pat. Nos. 5,939,598; 6,075,181; 6,114,598; 6,150,584 and 6,162,963, the contents of each of which are hereby incorporated by reference for this purpose.

Moreover, alternative transchromosomal animal systems expressing human immunoglobulin genes are available in the art and can be used to raise antibodies of the disclosure. For example, mice carrying both a human heavy chain transchromosome and a human light chain transchromosome, referred to as "TC mice" can be used; such mice are described in Tomizuka et al., 2000 Proc. Natl. Acad. Sci. USA 97:722-727, the contents of which are hereby incorporated by reference for this purpose. Furthermore, cows carrying human heavy and light chain transchromosomes have been described in the art (Kuroiwa et al., 2002 Nature Biotechnology 20:889-894, the contents of which are hereby incorporated by reference for this purpose) and can be used to raise antibodies of the disclosure.

Human monoclonal antibodies of the disclosure can also be prepared using phage display methods for screening libraries of human immunoglobulin genes. Such phage display methods for isolating human antibodies are established in the art or described in the examples below. See for example: U.S. Pat. Nos. 5,223,409; 5,403,484; and 5,571,698; 5,427,908 and 5,580,717; 5,969,108 and 6,172,197; and 5,885,793; 6,521,404; 6,544,731; 6,555,313; 6,582,915 and 6,593,081, the contents of each of which are hereby incorporated by reference for this purpose.

Human antibodies or antigen binding fragments of the disclosure can also be prepared using SCID mice into which human immune cells have been reconstituted such that a human antibody response can be generated upon immunization. Such mice are described in, for example, U.S. Pat. Nos. 5,476,996 and 5,698,767, the contents of each of which are hereby incorporated by reference for this purpose.

Antibodies of the disclosure may be prepared by any of the methods described herein.

Generation of Hybridomas Producing Antibodies or Antigen Binding Fragments of the Disclosure

To generate hybridomas producing the antibodies or antigen binding fragments of the disclosure, splenocytes and/or lymph node cells from immunized mice can be isolated and fused to an appropriate immortalized cell line, such as a mouse myeloma cell line. The resulting hybridomas can be screened for the production of antigen-specific antibodies. For example, single cell suspensions of splenic lymphocytes from immunized mice can be fused to one-sixth the number of P3X63-Ag8.653 nonsecreting mouse myeloma cells (ATCC, CRL 1580) with 50% PEG. Cells are plated at approximately 2x145 in flat bottom microtiter plates, followed by a two week incubation in selective medium containing 20% fetal Clone Serum, 18% "653" conditioned media, 5% origen (IGEN), 4 mM L-glutamine, 1 mM sodium pyruvate, 5 mM HEPES, 0.055 mM 2-mercaptoethanol, 50 units/ml penicillin, 50 mg/ml streptomycin, 50 mg/ml gentamycin and 1X HAT (Sigma; the HAT is added 24 hours after the fusion). After approximately two weeks, cells can be cultured in medium in which the HAT is replaced with HT. Individual wells can then be screened by ELISA for human monoclonal IgM and IgG antibodies. Once extensive hybridoma growth occurs, medium can be observed usually after 10-14 days. The antibody secreting hybridomas can be replated, screened again, and if still

positive for human IgG, the monoclonal antibodies can be subcloned at least twice by limiting dilution. The stable subclones can then be cultured in vitro to generate small amounts of antibody in tissue culture medium for characterization.

To purify antibodies or antigen binding fragments thereof, selected hybridomas can be grown in two-liter spinner-flasks for antibody purification. Supernatants can be filtered and concentrated before affinity chromatography with protein A-sepharose (Pharmacia, Piscataway, N.J.). Eluted IgG can be checked by gel electrophoresis and high performance liquid chromatography to ensure purity. The buffer solution can be exchanged into PBS, and the concentration can be determined by OD280 using 1.43 extinction coefficient. The antibodies or antigen binding fragments can be aliquoted and stored at -80° C.

Hybridomas producing the antibodies or antigen binding fragments of the disclosure may be produced, for example, using the methods described herein.

Generation of Transfectomas Producing Antibodies or Antigen Binding Fragments of the Disclosure

Antibodies or antigen binding fragments of the disclosure can also be produced in a host cell transfectoma using, for example, a combination of suitable recombinant DNA techniques and gene transfection methods (e.g., Morrison, S. (1985) Science 229:1202, the contents of which are incorporated herein by reference for this purpose).

For example, to express the antibodies, or antigen-binding fragments thereof, DNAs encoding partial or full-length light and heavy chains, can be obtained by standard molecular biology techniques (e.g., PCR amplification or cDNA cloning using a hybridoma that expresses the antibody of interest) and the DNAs can be inserted into expression vectors such that the genes are operatively linked to transcriptional and translational control sequences. In this context, the term "operatively linked" is intended to mean that an antibody gene is ligated into a vector such that transcriptional and translational control sequences within the vector serve their intended function of regulating the transcription and translation of the antibody gene. The expression vector and expression control sequences are chosen to be compatible with the expression host cell used. The antibody light chain gene and the antibody heavy chain gene can be inserted into separate vector or both genes may be inserted into the same expression vector. The antibody genes are inserted into the expression vector by standard methods (e.g., ligation of complementary restriction sites on the antibody gene fragment and vector, or blunt end ligation if no restriction sites are present). The light and heavy chain variable regions of the antibodies described herein can be used to create full-length antibody genes of any antibody isotype by inserting them into expression vectors already encoding heavy chain constant and light chain constant regions of the desired isotype such that the VH segment is operatively linked to the CH segment(s) within the vector and the VL segment is operatively linked to the CL segment within the vector. Additionally or alternatively, the recombinant expression vector can encode a signal peptide that facilitates secretion of the antibody chain from a host cell. The antibody chain gene can be cloned into the vector such that the signal peptide is linked in frame to the amino terminus of the antibody chain gene. The signal peptide can be an immunoglobulin signal peptide or a heterologous signal peptide (i.e., a signal peptide from a non-immunoglobulin protein).

In addition to the antibody chain genes, the recombinant expression vectors of the disclosure carry regulatory

sequences that control the expression of the antibody chain genes in a host cell. The term "regulatory sequence" is intended to include promoters, enhancers and other expression control elements (e.g., polyadenylation signals) that control the transcription or translation of the antibody chain genes. Such regulatory sequences are described, for example, in Goeddel (Gene Expression Technology; Methods in Enzymology 185, Academic Press, San Diego, CA 1990, the contents of which are incorporated herein by reference for this purpose). It will be appreciated by those skilled in the art that the design of the expression vector, including the selection of regulatory sequences, may depend on such factors as the choice of the host cell to be transformed, the level of expression of protein desired, etc. Regulatory sequences for mammalian host cell expression include viral elements that direct high levels of protein expression in mammalian cells, such as promoters and/or enhancers derived from cytomegalovirus (CMV), Simian Virus 40 (SV40), adenovirus (e.g., the adenovirus major late promoter (AdMLP)), and polyoma. Alternatively, nonviral regulatory sequences may be used, such as the ubiquitin promoter or P-globin promoter. Still further, regulatory elements composed of sequences from different sources, such as the SRa promoter system, which contains sequences from the SV40 early promoter and the long terminal repeat of human T cell leukemia virus type 1 (Takebe, Y. et al., 1988 Mol. Cell. Biol. 8:466-472, the contents of which are incorporated herein by reference for this purpose).

In addition to the antibody chain genes and regulatory sequences, the recombinant expression vectors of the disclosure may carry additional sequences, such as sequences that regulate replication of the vector in host cells (e.g., origins of replication) and selectable marker genes. The selectable marker gene facilitates selection of host cells into which the vector has been introduced (see, e.g., U.S. Pat. Nos. 4,399,216, 4,634,665 and 5,179,017, the contents of each of which are incorporated herein by reference for this purpose). For example, typically the selectable marker gene confers resistance to drugs, such as G418, hygromycin or methotrexate, on a host cell into which the vector has been introduced. Selectable marker genes include the dihydrofolate reductase (DHFR) gene (for use in dhfr-host cells with methotrexate selection/amplification) and the neo gene (for G418 selection).

For expression of the light and heavy chains, the expression vector(s) encoding the heavy and light chains is or are transfected into a host cell by standard techniques. The various forms of the term "transfection" are intended to encompass a wide variety of techniques commonly used for the introduction of exogenous DNA into a prokaryotic or eukaryotic host cell, e.g., electroporation, calcium-phosphate precipitation, DEAE-dextran transfection and the like. It is theoretically possible to express the antibodies of the disclosure in either prokaryotic or eukaryotic host cells. Expression of antibodies in eukaryotic cells, in particular mammalian host cells, is discussed because such eukaryotic cells, and in particular mammalian cells, are more likely than prokaryotic cells to assemble and secrete a properly folded and immunologically active antibody. Prokaryotic expression of antibody genes has been reported to be ineffective for production of high yields of active antibody (Boss, M. A. and Wood, C. R., 1985 Immunology Today 6:12-13, the contents of which are incorporated herein by reference for this purpose).

Mammalian host cells for expressing the antibodies of the disclosure are described elsewhere herein. When recombinant expression vectors encoding antibody genes are intro-

duced into mammalian host cells, the antibodies are produced by culturing the host cells for a period of time sufficient to allow for expression of the antibody in the host cells or secretion of the antibody into the culture medium in which the host cells are grown. Antibodies can be recovered from the culture medium using standard protein purification methods. Accordingly, the disclosure provides a process for the production of an anti-NPR1 antibody of the disclosure, or antigen-binding fragment thereof, comprising culturing a host cell of the disclosure and isolating the antibody or antigen-binding fragment thereof.

Uses and Methods of Treatment

Methods of Treatment

Provided herein are methods of treating a disease associated with NPR1 loss of function by using the anti-NPR1 antibodies or antigen binding fragments thereof disclosed herein (e.g., an antibody or group of antibodies as defined in Table 2, Table 3, or Table 4). In some embodiments, the antibody or antigen binding fragment thereof may be selected from WW01_LALA, WW03_LALA, WW05_LALA, WW06_LALA, XX01_LALA, XX01_DAPA, XX01_N30S_DAPA, XX03_LALA, XX04_LALA, XX06_LALA, XX06_DAPA, XX07_LALA, XX08_LALA, XX08_DAPA, XX08_N30S_DAPA, XX08_N30Q_DAPA, XX09_LALA, XX11_LALA, XX12_LALA, XX13_LALA, XX14_LALA, XX15_LALA, XX15_DAPA, XX16_LALA, XX16_DAPA, XX17_LALA, XX17_DAPA, XX18_LALA, XX18_DAPA, XX19_LALA, XX19_DAPA, XX20_LALA, XX20_DAPA, YY01_LALA, YY02_LALA, YY03_LALA, YY04_LALA, YY05_LALA, YY06_LALA, YY07_LALA, ZZ05_LALA, ZZ12_LALA, ZZ13_LALA, ZZ14_LALA, ZZ15_LALA, ZZ16_LALA, and ZZ17_LALA.

In some embodiments, the antibody or antigen binding fragment thereof may be selected from WW01_LALA, WW03_LALA, XX01_LALA, XX01_DAPA, XX01_N30S_DAPA, XX03_LALA, XX04_LALA, XX06_LALA, XX06_DAPA, XX07_LALA, XX08_LALA, XX08_DAPA, XX08_N30S_DAPA, XX08_N30Q_DAPA, XX09_LALA, XX11_LALA, XX12_LALA, XX13_LALA, XX14_LALA, XX15_LALA, XX15_DAPA, XX16_LALA, XX16_DAPA, XX17_LALA, XX17_DAPA, XX18_LALA, XX18_DAPA, XX19_LALA, XX19_DAPA, XX20_LALA, XX20_DAPA, YY01_LALA, YY03_LALA, YY04_LALA, ZZ12_LALA, and ZZ13_LALA. In some embodiments, the antibody or antigen binding fragment thereof may be selected from WW05_LALA, WW06_LALA, YY05_LALA, YY06_LALA, YY07_LALA, ZZ05_LALA, ZZ14_LALA, and ZZ16_LALA. In some embodiments, the antibody or antigen binding fragment thereof may be XX16_DAPA. In some embodiments, the antibody or antigen binding fragment thereof may be XX16_LALA.

In some embodiments, the disease associated with NPR1 loss of function is a cardiovascular disorder. In some embodiments, the cardiovascular disorder is selected from: hypertension, peripheral vascular disease, heart failure, coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy, diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, and myocardial infarction (MI). In some embodiments, the disease associated with NPR1 loss of function is heart failure, hypertrophic cardiomyopathy (HCM), hypertension, preeclampsia, asthma, glaucoma, or cytokine release syndrome.

In some embodiments, the heart failure is selected from a heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure. In some embodiments, the hypertrophic cardiomyopathy is ventricular hypertrophy. In some embodiments, the hypertension is selected from resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, and pulmonary arterial hypertension. In some embodiments, the hypertension is selected from resistant hypertension and hypertensive heart disease.

In some embodiments, the disease associated with NPR1 loss of function is a kidney disorder. In some embodiments, the kidney disorder is selected from: diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD).

NPR1-related disorders also include any other disorders which are directly or indirectly associated with aberrant NPR1 activity and/or expression. Provided herein are also methods of treating a NPR1 related disorder directly or indirectly associated with aberrant NPR1 activity and/or expression by using the anti-NPR1 antibodies or antigen binding fragments disclosed herein (e.g., from Table 2, Table 3, or Table 4, such as XX16_DAPA or XX16_LALA).

In some embodiments, the present disclosure provides methods of treating an undesirable condition, disease, or disorder associated with natriuretic peptide receptor activity in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of an antibody or antigen binding fragment disclosed herein. In some embodiments, the present disclosure provides a use of an antibody or antigen binding fragment disclosed herein for treatment of an undesirable condition, disease or disorder associated with natriuretic peptide receptor activity in a subject in need thereof. In some embodiments, the present disclosure provides an antibody or antigen binding fragment disclosed herein for use in a method for treating an undesirable condition, disease or disorder associated with natriuretic peptide receptor activity. In some embodiments, the present disclosure provides an antibody or antigen binding fragment disclosed herein for use in manufacturing a medicament for treating an undesirable condition, disease or disorder associated with natriuretic peptide receptor activity. Such conditions, diseases and disorders include, but are not limited to, cardiovascular disorders (e.g., hypertension, peripheral vascular disease, heart failure (including but not limited to heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure), coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy (e.g., ventricular hypertrophy), diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, or myocardial infarction (MI)), hypertension (e.g., resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated

systolic hypertension, resistant hypertension, or pulmonary arterial hypertension), preeclampsia, asthma, glaucoma, cytokine release syndrome, and/or a kidney disorder (e.g., diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD)).

In some embodiments, such methods include administering to a subject in need of treatment a therapeutically effective amount of an antibody or antigen-binding fragment thereof that specifically binds to the same epitope as one of the antibodies described herein. For example, such methods include administering to a subject in need of treatment a therapeutically effective amount of an antibody or antigen-binding fragment thereof that specifically binds to the same epitope as XX16. In another embodiment, such methods include administering to a subject in need of treatment a therapeutically effective amount of an antibody or antigen-binding fragment thereof that specifically binds to the same epitope as WW03. In another embodiment, such methods include administering to a subject in need of treatment a therapeutically effective amount of an antibody or antigen-binding fragment thereof that specifically binds to the same epitope as WW06.

All the aforementioned embodiments for the methods of protection and treatment according to the present invention are equally applicable to

- the use of any one of the antibodies or antigen binding fragments as described herein for the manufacture of a medicament for use according to the present invention,
- the use of any one of the antibodies or antigen binding fragments described herein according to the present invention,
- any one of the antibodies or antigen binding fragments described herein for use according to the present invention,
- the pharmaceutical compositions comprising any one of the antibodies or antigen binding fragments described herein for the use according to the present invention,
- the use of the pharmaceutical compositions comprising any one of the antibodies or antigen binding fragments described herein according to the present invention, and
- the use of the pharmaceutical compositions comprising any one of the antibodies or antigen binding fragments described herein for the manufacture of a medicament for use according to the present invention.

Combination Therapies

The various treatments described above can be combined with other treatment partners or therapeutic agents such as the current standard of care for a disease associated with NPR1 loss of function, e.g., the current standard of care for one or more of the diseases or disorders discussed herein. For example, the NPR1 antibodies or an antigen-binding fragment thereof described herein can be combined with one or more of an ACE (angiotensin-converting-enzyme) inhibitor, an angiotensin receptor blocker (ARB), a neprilysin inhibitor, a beta blocker, a diuretic, a calcium channel blocker, a cardiac glycoside, a sodium-glucose co-transporter 2 inhibitor (SGLT2i), or combinations thereof. As a non-limiting set of examples, the NPR1 antibody or antigen binding fragment may be combined with an additional thera-

peutic agent selected from enalapril, benazepril, captopril, fosinopril, lisinopril, moexipril, perindopril, quinapril, ramipril, trandolapril, valsartan, azilsartan, candesartan, eprosartan, irbesartan, losartan, olmesartan, telmisartan, sacubitril, bisoprolol, carvedilol, propanolol, metoprolol, metoprolol tartrate, metoprolol succinate, thiazide diuretics, loop diuretics, potassium-sparing diuretics, amlodipine, clevidipine, diltiazem, felodipine, isradipine, nifedipine, nisoldipine, verapamil, a digitalis glycoside, canagliflozin, dapagliflozin, empagliflozin, ertugliflozin, and combinations thereof. Exemplary diuretics and digitalis glycosides include, but are not limited to, chlorothiazide, chlorthalidone, hydrochlorothiazide, indapamide, metolazone, bumetanide, ethacrynic acid, furosemide, torsemide, amiloride, eplerenone, spironolactone, triamterene, digoxin, and combinations thereof. In some embodiments, the NPR1 antibodies or an antigen-binding fragment thereof described herein may be combined with an angiotensin receptor-neprilysin inhibitor (ARNi) such as a combination of sacubitril and valsartan (e.g., Entresto®). In some embodiments, the NPR1 antibodies or an antigen-binding fragment thereof described herein can be combined with one or more of a corticosteroid (e.g., an inhaled corticosteroid such as fluticasone, budesonide, mometasone, beclomethasone, ciclesonide, or fluticasone furoate; or an oral or intravenous corticosteroid such as prednisone or methylprednisolone), a leukotriene modifier (e.g., montelukast, zafirlukast, or zileuton), a bronchodilator (e.g., a long-acting beta agonist (e.g., salmeterol or formoterol), a short-acting beta agonist (e.g., albuterol or levalbuterol), theophylline or ipratropium), or combinations thereof (e.g., a combination of fluticasone and salmeterol, a combination of budesonide and formoterol, or a combination of formoterol and mometasone). In some embodiments, the NPR1 antibodies or an antigen-binding fragment thereof described herein can be combined with one or more of a beta-adrenoceptor antagonist (e.g., timolol, levobunolol, metipranolol, carteolol, or betaxolol), a carbonic anhydrase inhibitor (e.g., acetazolamide, dorzolamide, brinzolamide, or methazolamide), an alpha 2-adrenoceptor agonist (e.g., brimonidine or apraclonidine), a parasymphomimetic (e.g., cholinomimetics like pilocarpine), a prostaglandin analog (e.g., latanoprost, latanoprostene bunod, travoprost, bimatoprost, or tafluprost), a rho kinase inhibitor (e.g., netarsudil or ripasudil), or combinations thereof (e.g., a combination of rho kinase inhibitor and latanoprost).

Accordingly, the methods of treating a disease associated with NPR1 loss of function described herein can further include administering a second agent to the subject in need of treatment.

The term “combination” refers to either a fixed combination in one dosage unit form, or a combined administration where an anti-NPR1 antibody or antigen-binding fragment thereof described herein and a combination partner (e.g., another drug as explained below, also referred to as “therapeutic agent” or “co-agent”) may be administered independently at the same time or separately within time intervals, especially where these time intervals allow that the combination partners show a cooperative, e.g., synergistic effect. The single components may be packaged in a kit or separately. One or both of the components (e.g., powders or liquids) may be reconstituted or diluted to a desired dose prior to administration. The terms “co-administration” or “combined administration” or the like as utilized herein are meant to encompass administration of the selected combination partner to a single subject in need thereof (e.g., a patient), and are intended to include treatment regimens in

which the agents are not necessarily administered by the same route of administration and/or at the same time. The term “pharmaceutical combination” as used herein means a product that results from the mixing or combining of more than one therapeutic agent and includes both fixed and non-fixed combinations of the therapeutic agents. The term “fixed combination” means that the therapeutic agents, e.g., a compound of the present invention and a combination partner, are both administered to a patient simultaneously in the form of a single entity or dosage. The term “non-fixed combination” means that the therapeutic agents, e.g., a compound of the present invention and a combination partner, are both administered to a patient as separate entities either simultaneously, concurrently or sequentially with no specific time limits, wherein such administration provides therapeutically effective levels of the two compounds in the body of the patient. The latter also applies to cocktail therapy, e.g., the administration of three or more therapeutic agent.

The term “pharmaceutical combination” as used herein refers to either a fixed combination in one dosage unit form, or non-fixed combination or a kit of parts for the combined administration where two or more therapeutic agents may be administered independently at the same time or separately within time intervals, especially where these time intervals allow that the combination partners show a cooperative, e.g., synergistic effect.

The term “combination therapy” refers to the administration of two or more therapeutic agents to treat a therapeutic condition or disorder described in the present disclosure. Such administration encompasses co-administration of these therapeutic agents in a substantially simultaneous manner, such as in a single capsule having a fixed ratio of active ingredients. Alternatively, such administration encompasses co-administration in multiple, or in separate containers (e.g., tablets, capsules, powders, and liquids) for each active ingredient. Powders and/or liquids may be reconstituted or diluted to a desired dose prior to administration. In addition, such administration also encompasses use of each type of therapeutic agent in a sequential manner, either at approximately the same time or at different times. In either case, the treatment regimen will provide beneficial effects of the drug combination in treating the conditions or disorders described herein.

Pharmaceutical Compositions, Dosages, and Methods of Administration

Also provided herein are compositions, e.g., pharmaceutical compositions, for use in treatment of an NPR1-associated disease. Such compositions include one or more anti-NPR1 antibodies or an antigen-binding fragment thereof as described herein and may include a pharmaceutically acceptable carrier. Such compositions can further include another agent, e.g., a current standard of care for the disease to be treated.

Pharmaceutical compositions typically include a pharmaceutically acceptable carrier. As used herein the term “pharmaceutically acceptable carrier” refers to a carrier or a diluent that does not cause significant irritation to a subject and does not abrogate the biological activity and properties of the administered anti-NPR1 antibody or antigen binding fragment and/or any additional therapeutic agent in the composition. Pharmaceutically acceptable carriers may enhance or stabilize the composition or can be used to facilitate preparation of the composition. Pharmaceutically acceptable carriers may include saline, solvents, dispersion media, coatings, antibacterial and antifungal agents, isotonic and absorption delaying agents, and the like, compatible

with pharmaceutical administration. An adjuvant may also be included in any of these formulations. Pharmaceutical compositions are typically formulated to be compatible with its intended route of administration. Examples of routes of administration include parenteral (e.g., intravenous, intraarterial, intraperitoneal), oral, intracranial, intrathecal, or intranasal (e.g., inhalation), intradermal, subcutaneous, or transmucosal administration. In some embodiments, the pharmaceutical compositions are formulated to deliver anti-NPR1 antibodies or antigen-binding fragments thereof to cross the blood-brain barrier. The phrases "physiologically acceptable carrier" and "pharmaceutically acceptable carrier" may be used interchangeably.

As used herein, the term "excipient" refers to an inert substance added to a pharmaceutical composition to further facilitate administration of an active ingredient. Formulations for parenteral administration can, for example, contain excipients such as sterile water or saline, polyalkylene glycols such as polyethylene glycol, vegetable oils, or hydrogenated naphthalenes. Other exemplary excipients include, but are not limited to, calcium bicarbonate, calcium phosphate, various sugars and types of starch, cellulose derivatives, gelatin, ethylene-vinyl acetate co-polymer particles, and surfactants, including, for example, polysorbate 20.

A pharmaceutical composition of the present disclosure can be administered by a variety of methods known in the art. The route and/or mode of administration may vary depending upon the desired results. In some embodiments, the administration is intravitreal, intravenous, intramuscular, intraperitoneal, or subcutaneous. The pharmaceutically acceptable carrier should be suitable for intravitreal, intravenous, intramuscular, subcutaneous, parenteral, spinal, or epidermal administration (e.g., by injection or infusion). Depending on the route of administration, the active compound(s), i.e., the anti-NPR1 antibody or antigen binding fragment and optionally the additional therapeutic agent, may be coated in a material to protect the compound(s) from the action of acids and other natural conditions that may inactivate the compound(s).

Typically, a therapeutically effective dose or efficacious dose of the anti-NPR1 antibodies or antigen binding fragments is employed in the pharmaceutical compositions of the present disclosure. The anti-NPR1 antibodies or antigen binding fragments may be formulated into pharmaceutically acceptable dosage forms by conventional methods known to those of skill in the art.

Methods of formulating suitable pharmaceutical compositions are known in the art, see, e.g., Remington: The Science and Practice of Pharmacy, 21st ed., 2005; and the books in the series *Drugs and the Pharmaceutical Sciences: a Series of Textbooks and Monographs* (Dekker, NY), the contents of each of which are incorporated by reference herein for this purpose. For example, solutions or suspensions used for parenteral, intradermal, or subcutaneous application can include the following components: a sterile diluent such as water for injection, saline solution, fixed oils, polyethylene glycols, glycerin, propylene glycol or other synthetic solvents; antibacterial agents such as benzyl alcohol or methyl parabens; antioxidants such as ascorbic acid or sodium bisulfite; chelating agents such as ethylenediaminetetraacetic acid; buffers such as acetates, citrates or phosphates and agents for the adjustment of tonicity such as sodium chloride or dextrose. pH can be adjusted with acids or bases, such as hydrochloric acid or sodium hydroxide.

The parenteral preparation can be enclosed in ampoules, disposable syringes or multiple dose vials made of glass or plastic.

Dosage regimens for anti-NPR1 antibodies and antigen binding fragments with or without an additional therapeutic agent may be adjusted to provide the optimum desired response (e.g., a therapeutic response). For example, a single bolus of one or both agents may be administered at one time, several divided doses may be administered over a predetermined period of time, or the dose of one or both agents may be proportionally reduced or increased as indicated by the exigencies of the therapeutic situation. For any particular subject, specific dosage regimens may be adjusted over time according to the individual's need, and the professional judgment of the treating clinician. Parenteral compositions may be formulated in dosage unit form for ease of administration and uniformity of dosage. Dosage unit form as used herein refers to physically discrete units suited as unitary dosages for the subjects to be treated; each unit contains a predetermined quantity of active compound calculated to produce the desired therapeutic effect in association with the required pharmaceutical carrier.

An "effective amount" is an amount sufficient to effect beneficial or desired results. For example, a therapeutic amount is one that achieves the desired therapeutic effect. This amount can be the same or different from a prophylactically effective amount, which is an amount necessary to prevent onset of disease or disease symptoms. An effective amount can be administered in one or more administrations, applications or dosages. A therapeutically effective amount of a therapeutic compound (i.e., an effective dosage) depends on the therapeutic compounds selected. The skilled artisan (such as a medical doctor) will appreciate that certain factors may influence the dosage and timing required to effectively treat a subject, including but not limited to the severity of the disease or disorder, previous treatments, the general health and/or age of the subject, and other diseases present. Moreover, treatment of a subject with a therapeutically effective amount of the therapeutic compounds described herein can include a single treatment or a series of treatments.

Dosage, toxicity and therapeutic efficacy of the therapeutic compounds can be determined by standard pharmaceutical procedures in cell cultures or experimental animals, e.g., for determining the LD₅₀ (the dose lethal to 50% of the population) and the ED₅₀ (the dose therapeutically effective in 50% of the population). The dose ratio between toxic and therapeutic effects is the therapeutic index and it can be expressed as the ratio LD₅₀/ED₅₀. Compounds which exhibit high therapeutic indices are preferred. While compounds that exhibit toxic side effects may be used, care should be taken to design a delivery system that targets such compounds to the site of affected tissue in order to minimize potential damage to uninfected cells and, thereby, reduce side effects.

Dosage regimens for anti-NPR1 antibodies and antigen binding fragments alone or in combination with an additional therapeutic agent may be adjusted to provide the optimum desired response (e.g., a therapeutic response). For example, a single bolus of one or both agents may be administered at one time, several divided doses may be administered over a predetermined period of time, or the dose of one or both agents may be proportionally reduced or increased as indicated by the exigencies of the therapeutic situation. For any particular subject, specific dosage regimens may be adjusted over time according to the individual's need, and the professional judgment of the treating

clinician. Parenteral compositions may be formulated in dosage unit form for ease of administration and uniformity of dosage. Dosage unit form as used herein refers to physically discrete units suited as unitary dosages for the subjects to be treated; each unit contains a predetermined quantity of active compound calculated to produce the desired therapeutic effect in association with the required pharmaceutical carrier.

Dosage values for compositions comprising an anti-NPR1 antibody or antigen binding fragment, and/or any additional therapeutic agent(s), may be selected based on the unique characteristics of the active compound(s), and the particular therapeutic effect to be achieved. A physician or veterinarian can start doses of the antibodies of the disclosure employed in the pharmaceutical composition at levels lower than that required to achieve the desired therapeutic effect and gradually increase the dosage until the desired effect is achieved. In general, effective doses of the compositions of the present disclosure, for the treatment of obesity or another disorder described herein may vary depending upon many different factors, including means of administration, target site, physiological state of the patient, whether the patient is human or an animal, other medications administered, and whether treatment is prophylactic or therapeutic. The selected dosage level may also depend upon a variety of pharmacokinetic factors including the activity of the particular compositions of the present disclosure employed, or the ester, salt or amide thereof, the route of administration, the time of administration, the rate of excretion of the particular compound being employed, the duration of the treatment, other drugs, compounds and/or materials used in combination with the particular compositions employed, the age, sex, weight, condition, general health and prior medical history of the patient being treated, and like factors. Treatment dosages may be titrated to optimize safety and efficacy.

Kits

Also provided herein are kits including one or more of the compositions provided herein (e.g., an antibody or antigen binding fragment thereof described in Table 2, Table 3, or Table 4) and instructions for use. Instructions for use can include instructions for diagnosis or treatment of an NPR1-associated disease. Kits as provided herein may be used in accordance with any of the methods described herein. Those skilled in the art will be aware of other suitable uses for kits provided herein, and will be able to employ the kits for such uses. Kits as provided herein can also include a mailer (e.g., a postage paid envelope or mailing pack) that can be used to return the sample for analysis, e.g., to a laboratory. The kit can include one or more containers for the sample, or the sample can be in a standard blood collection vial. The kit can also include one or more of an informed consent form, a test requisition form, and instructions on how to use the kit in a method described herein. Methods for using such kits are also included herein. One or more of the forms (e.g., the test requisition form) and the container holding the sample can be coded, for example, with a bar code for identifying the subject who provided the sample.

The disclosure is further illustrated by the following examples and claims, which are illustrative and are not meant to be further limiting. One skilled in the art will recognize many methods and materials similar or equivalent to those described herein, which could be used in the practice of the described compositions and methods. Such equivalents are within the scope of the present disclosure and claims. The contents of all references, including issued patents and published patent applications, cited throughout this application are hereby incorporated by reference.

EMBODIMENTS

In more detail, the disclosure provides the following embodiments:

Embodiment 1. An isolated antibody or antigen binding fragment that (i) binds to natriuretic peptide receptor 1 (NPR1); and (ii) is capable of activating NPR1 in the absence of atrial natriuretic peptide (ANP). Embodiment 2. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen-binding fragment thereof of embodiment 1, which does not bind to and/or does not activate natriuretic peptide receptor 2 (NPR2) and/or natriuretic peptide receptor 3 (NPR3).

Embodiment 3. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of embodiment 1 or embodiment 2 which binds to (a) human NPR1; and (b) mouse NPR1 and/or rat NPR1.

Embodiment 4. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of embodiment 1 or embodiment 2 which binds to (a) human NPR1; and (b) cyno NPR1.

Embodiment 5. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen-binding fragment thereof of any one of embodiments 1-4, which is ANP non-competitive.

Embodiment 6. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen-binding fragment thereof of any one of embodiments 1, 2, or 4, which is ANP competitive.

Embodiment 7. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 which is capable of stabilizing the ANP-NPR1 complex.

Embodiment 8. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7, wherein the antibody or antigen binding fragment thereof binds to an epitope within amino acids 99-133 of SEQ ID NO: 1.

Embodiment 9. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5, 7, or 8, wherein the antibody or antigen binding fragment thereof binds to an epitope comprising at least two amino acid residues within amino acids 99-133 of SEQ ID NO: 1.

Embodiment 10. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-9, wherein the antibody or antigen binding fragment thereof binds to an epitope comprising at least 3, 4, 5, 6, 7, or 8 amino acid residues within amino acids 99-133 of SEQ ID NO: 1.

Embodiment 11. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-10, wherein the antibody or antigen binding fragment thereof binds to an epitope within amino acids 99-111 of SEQ ID NO: 1.

Embodiment 12. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-11, wherein the antibody or antigen binding fragment thereof binds to an epitope within amino acids 99-103 of SEQ ID NO: 1.

Embodiment 13. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen

binding fragment of any one of embodiments 1-5 or 7-12, wherein the antibody or antigen binding fragment thereof binds to an epitope within amino acids 105-111 of SEQ ID NO: 1.

Embodiment 14. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-13, wherein the antibody or antigen binding fragment thereof binds to an epitope comprising at least 2, 3, or 4 amino acid residues within amino acids 105-111 of SEQ ID NO: 1.

Embodiment 15. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-14, wherein the antibody or antigen binding fragment thereof binds to a conformational epitope of human NPR1, and wherein the conformational epitope comprises at least one amino acid residue within each of (i) amino acids 99-103 of SEQ ID NO: 1, (ii) 105-111 of SEQ ID NO: 1, (iii) 131-134 of SEQ ID NO: 1, and additionally binds to amino acid 375 and/or 378 of SEQ ID NO: 1.

Embodiment 16. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 8-14, wherein the epitope is a conformational epitope, and wherein the conformational epitope additionally comprises at least one amino acid residue selected from the group consisting of amino acids 33, 34, 76, 82, and 104 of SEQ ID NO: 1.

Embodiment 17. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of embodiment 15, wherein the conformational epitope additionally comprises at least one amino acid residue selected from the group consisting of amino acids 33, 34, 76, 82, 104, 374, and 375 of SEQ ID NO: 1.

Embodiment 18. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-17, wherein the antibody or antigen binding fragment thereof binds to at least amino acids 82, 102, 103, 105, 106, 109, 132, and 375 of SEQ ID NO: 1.

Embodiment 19. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-18, wherein the antibody or antigen binding fragment thereof binds to at least amino acids 34, 82, 102, 103, 105, 106, 107, 109, 132, 133, 375, and 378 of SEQ ID NO: 1.

Embodiment 20. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1-5 or 7-18, wherein the antibody or antigen binding fragment thereof binds to at least amino acids 79, 82, 99, 102, 103, 105, 106, 109, 131, 132, and 375 of SEQ ID NO: 1.

Embodiment 21. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1, 2, 4, or 6, wherein the antibody or antigen binding fragment thereof binds to an epitope within amino acids 188-219 of SEQ ID NO: 1.

Embodiment 22. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, or 21, wherein the antibody or antigen binding fragment thereof binds to an epitope comprising at least 2, 3, 4, 5, 6, or 7 amino acids within amino acids 188-219 of SEQ ID NO: 1.

Embodiment 23. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, 21,

or 22, wherein the antibody or antigen binding fragment thereof binds to a conformational epitope within NPR1, and wherein the conformational epitope comprises at least one amino acid residue within each of (i) amino acids 188-198 of SEQ ID NO: 1, (ii) 201-208 of SEQ ID NO: 1, (iii) 215-238 of SEQ ID NO: 1, and (iv) 294-297 of SEQ ID NO: 1.

Embodiment 24. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, or 21-23, wherein the antibody or antigen binding fragment thereof binds to at least amino acids 188, 192, 194, 197, 201, 208, and 219 of SEQ ID NO: 1.

Embodiment 25. An isolated anti-NPR1 antibody or antigen binding fragment; or the isolated antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, or 21-24, wherein the antibody or antigen binding fragment thereof binds to at least amino acids 188, 192, 194, 197, 201, 208, 219, and 295 of SEQ ID NO: 1.

Embodiment 26. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1-5 or 7-20, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), and wherein the antibody or antigen binding fragment comprises the CDRs of one of the ANP non-competitive groups described in Table 3 or Table 4.

Embodiment 27. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, or 21-25, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), and wherein the antibody or antigen binding fragment comprises the CDRs of one of the ANP competitive groups described in Table 3 or Table 4.

Embodiment 28. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1-5, 7-20, or 26, wherein the antibody or antigen binding fragment is WW01_LALA, WW03_LALA, XX01_LALA, XX01_DAPA, XX01_N30S_DAPA, XX03_LALA, XX04_LALA, XX06_LALA, XX06_DAPA, XX07_LALA, XX08_LALA, XX08_DAPA, XX08_N30S_DAPA, XX08_N30Q_DAPA, XX09_LALA, XX11_LALA, XX12_LALA, XX13_LALA, XX14_LALA, XX15_LALA, XX15_DAPA, XX16_LALA, XX16_DAPA, XX17_LALA, XX17_DAPA, XX18_LALA, XX18_DAPA, XX19_LALA, XX19_DAPA, XX20_LALA, XX20_DAPA, YY01_LALA, YY03_LALA, YY04_LALA, ZZ12_LALA, and ZZ13_LALA.

Embodiment 29. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, 21-25, or 27, wherein the antibody or antigen binding fragment is WW05_LALA, WW06_LALA, YY05_LALA, YY06_LALA, YY07_LALA, ZZ05_LALA, ZZ14_LALA, and ZZ16_LALA.

Embodiment 30. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1-5, 7-20, 26, or 28, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions

sists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239;

(g) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in $\text{GFX}_1\text{FSX}_2\text{YX}_3\text{X}_4\text{X}_5$ (SEQ ID NO: 445), wherein X_1 is S or T, X_2 is S, K, or R, X_3 is W or Y, X_4 is I or L, and X_5 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in $\text{SIHQY}_1\text{Y}_2\text{Y}_3\text{EY}_4\text{Y}_5\text{YVESVKG}$ (SEQ ID NO: 453), wherein Y_1 is Q or H, Y_2 is G or A, Y_3 is H or L, Y_4 is T or K, and Y_5 is K or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239;

(II) HCDR1 comprises or consists of an amino acid sequence as set forth in $\text{X}_1\text{YX}_2\text{X}_3\text{X}_4$ (SEQ ID NO: 447), wherein X_1 is S, K, or R, X_2 is W or Y, X_3 is I or L, and X_4 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in $\text{SIHQY}_1\text{Y}_2\text{Y}_3\text{EY}_4\text{Y}_5\text{YVESVKG}$ (SEQ ID NO: 453), wherein Y_1 is Q or H, Y_2 is G or A, Y_3 is H or L, Y_4 is T or K, and Y_5 is K or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239;

(III) HCDR1 comprises or consists of an amino acid sequence as set forth in $\text{GFX}_1\text{FSX}_2\text{Y}$ (SEQ ID NO: 448), wherein X_1 is S or T, and X_2 is S, K, or R, HCDR2 comprises or consists of an amino acid sequence as set forth in $\text{HQY}_1\text{Y}_2\text{Y}_3\text{E}$ (SEQ ID NO: 456), wherein Y_1 is Q or H, Y_2 is G or A, and Y_3 is H or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 240, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 242; or

(IV) HCDR1 comprises or consists of an amino acid sequence as set forth in $\text{GFX}_1\text{FSX}_2\text{YX}_3$ (SEQ ID NO: 450), wherein X_1 is S or T, X_2 is S, K, or R, and X_3 is W or Y, HCDR2 comprises or consists of an amino acid sequence as set forth in $\text{IHQY}_1\text{Y}_2\text{Y}_3\text{EY}_4$ (SEQ ID NO: 458), wherein Y_1 is Q or H, Y_2 is G or A, Y_3 is H or L, and Y_4 is T or K, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 232, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239;

(h) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in $\text{GFTFSX}_1\text{YX}_2\text{IX}_3$ (SEQ ID NO: 452), wherein X_1 is S or R, X_2 is W or Y, and X_3 is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in

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- Y₁IY₂QY₃Y₄Y₅EY₆Y₇YVESVKG (SEQ ID NO: 446), wherein Y₁ is S or N, Y₂ is K or H, Y₃ is S, Q, or H, Y₄ is G or A, Y₅ is S, H, or L, Y₆ is T or K, and Y₇ is Y, K, or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; 10
- (II) HCDR1 comprises or consists of an amino acid sequence as set forth in X₁YX₂IX₃ (SEQ ID NO: 454), wherein X₁ is S or R, X₂ is W or Y, and X₃ is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in Y₁IY₂QY₃Y₄Y₅EY₆Y₇YVESVKG (SEQ ID NO: 446), wherein Y₁ is S or N, Y₂ is K or H, Y₃ is S, Q, or H, Y₄ is G or A, Y₅ is S, H, or L, Y₆ is T or K, and Y₇ is Y, K, or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; 25
- (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX₁Y (SEQ ID NO: 455), wherein X₁ is S or R, HCDR2 comprises or consists of an amino acid sequence as set forth in Y₁QY₂Y₃Y₄E (SEQ ID NO: 449), wherein Y₁ is K or H, Y₂ is S, Q, or H, Y₃ is G or A, and Y₄ is S, H, or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 240, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 242; or 35
- (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX₁YX₂ (SEQ ID NO: 457), wherein X₁ is S or R, and X₂ is W or Y, HCDR2 comprises or consists of an amino acid sequence as set forth in IY₁QY₂Y₃Y₄EY₅ (SEQ ID NO: 451), wherein Y₁ is K or H, Y₂ is S, Q, or H, Y₃ is G or A, Y₄ is S, H, or L, and Y₅ is T or K, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 232, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; or 45
- (i) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX₁YX₂IX₃ (SEQ ID NO: 452), wherein X₁ is S or R, X₂ is W or Y, and X₃ is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in SIHQY₁Y₂Y₃EY₄Y₅YVESVKG (SEQ ID NO: 453), wherein Y₁ is Q or H, Y₂ is G or A, Y₃ is H or L, Y₄ is T or K, and Y₅ is K or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; 65

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- (II) HCDR1 comprises or consists of an amino acid sequence as set forth in X₁YX₂IX₃ (SEQ ID NO: 454), wherein X₁ is S or R, X₂ is W or Y, and X₃ is S or N, HCDR2 comprises or consists of an amino acid sequence as set forth in SIHQY₁Y₂Y₃EY₄Y₅YVESVKG (SEQ ID NO: 453), wherein Y₁ is Q or H, Y₂ is G or A, Y₃ is H or L, Y₄ is T or K, and Y₅ is K or R, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 237, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 238, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239; 5
- (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX₁Y (SEQ ID NO: 455), wherein X₁ is S or R, HCDR2 comprises or consists of an amino acid sequence as set forth in HQY₁Y₂Y₃E (SEQ ID NO: 456), wherein Y₁ is Q or H, Y₂ is G or A, and Y₃ is H or L, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 228, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 240, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 242; or 15
- (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFSX₁YX₂ (SEQ ID NO: 457), wherein X₁ is S or R, and X₂ is W or Y, HCDR2 comprises or consists of an amino acid sequence as set forth in IHQY₁Y₂Y₃EY₄ (SEQ ID NO: 458), wherein Y₁ is Q or H, Y₂ is G or A, Y₃ is H or L, and Y₄ is T or K, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 232, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 243, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 241, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 239. 20
- Embodiment 31. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, 21-25, 27, or 29, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), and wherein: 25
- (a) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 310, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 311, HCDR3 comprises or consists of an amino acid sequence as set forth in GX₁X₂X₃GX₄LGFDH (SEQ ID NO: 459), wherein X₁ is A or S, X₂ is V or L, X₃ is A or P, and X₄ is Q or L, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 320, LCDR2 comprises or consists of an amino acid sequence as set forth in GNSNRPY₁ (SEQ ID NO: 460), wherein Y₁ is S or N, and LCDR3 comprises or consists of an amino acid sequence as set forth in QSYZ₁Z₂Z₃Z₄Z₅Z₆Z₇V (SEQ ID NO: 461), wherein Z₁ is Y, D, or G, Z₂ is T, S, or A, Z₃ is S, P, or F, Z₄ is S, T, or P, Z₅ is H, S, or R, Z₆ is G, S, or F, and Z₇ is P, S, or V; 30
- (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 229, HCDR2 35

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- comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 311, HCDR3 comprises or consists of an amino acid sequence as set forth in $GX_1X_2X_3GX_4LGFDH$ (SEQ ID NO: 459), wherein X_1 is A or S, X_2 is V or L, X_3 is A or P, and X_4 is Q or L, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 320, LCDR2 comprises or consists of an amino acid sequence as set forth in $GNSNRPY_1$ (SEQ ID NO: 460), wherein Y_1 is S or N, and LCDR3 comprises or consists of an amino acid sequence as set forth in $QSYZ_1Z_2Z_3Z_4Z_5Z_6Z_7V$ (SEQ ID NO: 461), wherein Z_1 is Y, D, or G, Z_2 is T, S, or A, Z_3 is S, P, or F, Z_4 is S, T, or P, Z_5 is H, S, or R, Z_6 is G, S, or F, and Z_7 is P, S, or V;
- (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 80, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 313, HCDR3 comprises or consists of an amino acid sequence as set forth in $GX_1X_2X_3GX_4LGFDH$ (SEQ ID NO: 459), wherein X_1 is A or S, X_2 is V or L, X_3 is A or P, and X_4 is Q or L, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 323, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 324, and LCDR3 comprises or consists of an amino acid sequence as set forth in $YZ_1Z_2Z_3Z_4Z_5Z_6Z_7$ (SEQ ID NO: 462), wherein Z_1 is Y, D, or G, Z_2 is T, S, or A, Z_3 is S, P, or F, Z_4 is S, T, or P, Z_5 is H, S, or R, Z_6 is G, S, or F, and Z_7 is P, S, or V; or
- (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 82, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 314, HCDR3 comprises or consists of an amino acid sequence as set forth in $ARGX_1X_2X_3GX_4LGFDH$ (SEQ ID NO: 463), wherein X_1 is A or S, X_2 is V or L, X_3 is A or P, and X_4 is Q or L, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 326, LCDR2 comprises or consists of an amino acid sequence as set forth in $QSYZ_1Z_2Z_3Z_4Z_5Z_6Z_7V$ (SEQ ID NO: 461), wherein Z_1 is Y, D, or G, Z_2 is T, S, or A, Z_3 is S, P, or F, Z_4 is S, T, or P, Z_5 is H, S, or R, Z_6 is G, S, or F, and Z_7 is P, S, or V;
- (b) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in $GFTFX_1X_2YAX_3X_4$ (SEQ ID NO: 464), wherein X_1 is S or G, X_2 is S or T, X_3 is I or M, and X_4 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in $Y_1ISY_2Y_3GY_4Y_5Y_6Y_7YAY_8SVKVG$ (SEQ ID NO: 465), wherein Y_1 is A or S, Y_2 is A, S, or G, Y_3 is S or H, Y_4 is G or Y, Y_5 is S or Y, Y_6 is T or A, Y_7 is Y, R, or N, and Y_8 is E or G, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339;
- (II) HCDR1 comprises or consists of an amino acid sequence as set forth in $X_1YAX_2X_3$ (SEQ ID NO: 466), wherein X_1 is S or T, X_2 is I or M, and X_3 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in

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- $Y_1ISY_2Y_3GY_4Y_5Y_6Y_7YAY_8SVKVG$ (SEQ ID NO: 465), wherein Y_1 is A or S, Y_2 is A, S, or G, Y_3 is S or H, Y_4 is G or Y, Y_5 is S or Y, Y_6 is T or A, Y_7 is Y, R, or N, and Y_8 is E or G, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339;
- (III) HCDR1 comprises or consists of an amino acid sequence as set forth in $GFTFX_1X_2Y$ (SEQ ID NO: 467), wherein X_1 is S or G, and X_2 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in $SY_1Y_2GY_3Y_4$ (SEQ ID NO: 468), wherein Y_1 is A, S, or G, Y_2 is S or H, Y_3 is G or Y, and Y_4 is S or Y, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 340, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 342; or
- (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in $GFTFX_1X_2YA$ (SEQ ID NO: 469), wherein X_1 is S or G, and X_2 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in $ISY_1Y_2GY_3Y_4T$ (SEQ ID NO: 470), wherein Y_1 is S or G, Y_2 is S or H, Y_3 is G or Y, and Y_4 is S or Y, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 332, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 343, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; or
- (c) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in $GFTFX_1X_2YAX_3X_4$ (SEQ ID NO: 464), wherein X_1 is S or G, X_2 is S or T, X_3 is I or M, and X_4 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in $SISY_1Y_2GY_3Y_4YAY_5SVKVG$ (SEQ ID NO: 471), wherein Y_1 is A or S, Y_2 is S or H, Y_3 is T or A, Y_4 is R or N, and Y_5 is E or G, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339;
- (II) HCDR1 comprises or consists of an amino acid sequence as set forth in $X_1YAX_2X_3$ (SEQ ID NO: 466), wherein X_1 is S or T, X_2 is I or M, and X_3 is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in $SISY_1Y_2GY_3Y_4YAY_5SVKVG$ (SEQ ID NO: 471), wherein Y_1 is A or S, Y_2 is S or H, Y_3 is T or A, Y_4 is R or N, and Y_5 is E or G, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID

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NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX₁X₂Y (SEQ ID NO: 467), wherein X₁ is S or G, and X₂ is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in SY₁Y₂GYG (SEQ ID NO: 472), wherein Y₁ is A or S, and Y₂ is S or H, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 340, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 342; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in GFTFX₁X₂YA (SEQ ID NO: 469), wherein X₁ is S or G, and X₂ is S or T, HCDR2 comprises or consists of an amino acid sequence as set forth in ISY₁Y₂G (SEQ ID NO: 473), wherein Y₁ is A, S, or G, and Y₂ is S or H, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 332, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 343, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339.

Embodiment 32. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1-5, 7-20, 26, 28, or 30, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3), and wherein:

(a) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 28, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 29, 119, and 190, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 43, 126, 134, 145, 172, 178, and 184; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 31, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 29, 119, and 190, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 41, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 42, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 43, 126, 134, 145, 172, 178, and 184; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 32, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 33, 120, and 191, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 30, LCDR1 comprises or consists of an amino acid

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sequence as set forth in SEQ ID NO: 44, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 46, 127, 135, 146, 173, 179, and 185; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 34, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 35, 121, and 192, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 36, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 47, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 45, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 43, 126, 134, 145, 172, 178, and 184; (b) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 4, 112, and 165, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 5, 100, and 151, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 17, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 18, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 7, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 5, 100, and 151, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 17, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 18, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 8, 113, and 166, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 9, 101, and 152, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 6, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 20, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 22; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 10, 114, and 167, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 11, 102, and 153, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 12, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 23, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 21, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 19; or (c) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 226, 367, and 378, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 227, 368, and 379, HCDR3 comprises or consists

comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 321 and 354, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 322, 355, and 361; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 80, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 313, HCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 312 and 348, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 323, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 324, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 325, 356, and 362; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 82, HCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 314, HCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 315 and 349, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 326, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 324, and LCDR3 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 322, 355, and 361; or

(b) (I) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 270 and 407, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 271, 389, and 408, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (II) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 273 and 409, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 271, 389, and 408, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 337, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 338, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339; (III) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 32 and 410, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 274, 390, and 411, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 331, LCDR1 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 340, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 342; or (IV) HCDR1 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 275 and 412, HCDR2 comprises or consists of an amino acid sequence as set forth in any one of SEQ ID NOs: 276, 391, and 413, HCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 332. LCDR1 comprises or consists of an amino

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acid sequence as set forth in SEQ ID NO: 343, LCDR2 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 341, and LCDR3 comprises or consists of an amino acid sequence as set forth in SEQ ID NO: 339.

Embodiment 34. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1-5, 7-20, 26, 28, 30, or 32, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from:

- (a) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 134 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 135 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3);
- (b) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 126 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 127 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 126 (LCDR3);
- (c) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 145 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 146 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 145 (LCDR3);
- (d) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 173 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121

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- (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 172 (LCDR3);
- (e) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 178 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 179 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 178 (LCDR3);
- (f) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 184 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 184 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 185 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 121 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 184 (LCDR3);
- (g) (I) SEQ ID NO: 4 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (II) SEQ ID NO: 7 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (III) SEQ ID NO: 8 (HCDR1), SEQ ID NO: 101 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3); or (IV) SEQ ID NO: 10 (HCDR1), SEQ ID NO: 102 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3);
- (h) (I) SEQ ID NO: 112 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (II) SEQ ID NO: 7 (HCDR1), SEQ ID NO: 100 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (III) SEQ ID NO: 113 (HCDR1), SEQ ID NO: 101 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3); or (IV) SEQ ID NO: 114 (HCDR1), SEQ ID NO: 102 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3);
- (i) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 119 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 120 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 46 (LCDR3); or (IV) SEQ

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- (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 134 (LCDR3);
- (t) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 190 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 172 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 191 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 173 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 192 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 172 (LCDR3);
- (u) (I) SEQ ID NO: 4 (HCDR1), SEQ ID NO: 5 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (II) SEQ ID NO: 7 (HCDR1), SEQ ID NO: 5 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 17 (LCDR1), SEQ ID NO: 18 (LCDR2), and SEQ ID NO: 19 (LCDR3); (III) SEQ ID NO: 8 (HCDR1), SEQ ID NO: 9 (HCDR2), SEQ ID NO: 6 (HCDR3), SEQ ID NO: 20 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 22 (LCDR3); or (IV) SEQ ID NO: 10 (HCDR1), SEQ ID NO: 11 (HCDR2), SEQ ID NO: 12 (HCDR3), SEQ ID NO: 23 (LCDR1), SEQ ID NO: 21 (LCDR2), and SEQ ID NO: 19 (LCDR3);
- (v) (I) SEQ ID NO: 28 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3); (II) SEQ ID NO: 31 (HCDR1), SEQ ID NO: 29 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 41 (LCDR1), SEQ ID NO: 42 (LCDR2), and SEQ ID NO: 43 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 33 (HCDR2), SEQ ID NO: 30 (HCDR3), SEQ ID NO: 44 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 46 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 35 (HCDR2), SEQ ID NO: 36 (HCDR3), SEQ ID NO: 47 (LCDR1), SEQ ID NO: 45 (LCDR2), and SEQ ID NO: 43 (LCDR3);
- (w) (I) SEQ ID NO: 367 (HCDR1), SEQ ID NO: 368 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (II) SEQ ID NO: 369 (HCDR1), SEQ ID NO: 368 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (III) SEQ ID NO: 370 (HCDR1), SEQ ID NO: 371 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 242 (LCDR3); or (IV) SEQ ID NO: 372 (HCDR1), SEQ ID NO: 373 (HCDR2), SEQ ID NO: 232 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 239 (LCDR3);
- (x) (I) SEQ ID NO: 378 (HCDR1), SEQ ID NO: 379 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (II) SEQ ID NO: 380 (HCDR1), SEQ ID NO: 379 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (III) SEQ ID NO: 381 (HCDR1), SEQ ID NO: 382 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ

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- ID NO: 242 (LCDR3); or (IV) SEQ ID NO: 383 (HCDR1), SEQ ID NO: 384 (HCDR2), SEQ ID NO: 232 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 239 (LCDR3);
- (y) (I) SEQ ID NO: 226 (HCDR1), SEQ ID NO: 227 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 227 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 239 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 230 (HCDR2), SEQ ID NO: 228 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 242 (LCDR3); or (IV) SEQ ID NO: 34 (HCDR1), SEQ ID NO: 231 (HCDR2), SEQ ID NO: 232 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 239 (LCDR3);
- (z) (I) SEQ ID NO: 270 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 282 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 283 (LCDR3); (II) SEQ ID NO: 273 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 282 (LCDR1), SEQ ID NO: 261 (LCDR2), and SEQ ID NO: 283 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 274 (HCDR2), SEQ ID NO: 272 (HCDR3), SEQ ID NO: 284 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 285 (LCDR3); or (IV) SEQ ID NO: 275 (HCDR1), SEQ ID NO: 276 (HCDR2), SEQ ID NO: 277 (HCDR3), SEQ ID NO: 286 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 283 (LCDR3); or
- (aa) (I) SEQ ID NO: 291 (HCDR1), SEQ ID NO: 292 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 304 (LCDR3); (II) SEQ ID NO: 294 (HCDR1), SEQ ID NO: 292 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 237 (LCDR1), SEQ ID NO: 238 (LCDR2), and SEQ ID NO: 304 (LCDR3); (III) SEQ ID NO: 295 (HCDR1), SEQ ID NO: 296 (HCDR2), SEQ ID NO: 293 (HCDR3), SEQ ID NO: 240 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 305 (LCDR3); or (IV) SEQ ID NO: 297 (HCDR1), SEQ ID NO: 298 (HCDR2), SEQ ID NO: 299 (HCDR3), SEQ ID NO: 243 (LCDR1), SEQ ID NO: 241 (LCDR2), and SEQ ID NO: 304 (LCDR3).
- Embodiment 35. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, 21-25, 27, 29, 31, or 33, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from:
- (a) (I) SEQ ID NO: 52 (HCDR1), SEQ ID NO: 53 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 65 (LCDR1), SEQ ID NO: 66 (LCDR2), and SEQ ID NO: 67 (LCDR3); (II) SEQ ID NO: 55 (HCDR1), SEQ ID NO: 53 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 65 (LCDR1), SEQ ID NO: 66 (LCDR2), and SEQ ID NO: 67 (LCDR3); (III) SEQ ID NO: 56 (HCDR1), SEQ ID NO: 57 (HCDR2), SEQ ID NO: 54 (HCDR3), SEQ ID NO: 68 (LCDR1), SEQ ID NO: 69 (LCDR2), and SEQ ID NO: 70 (LCDR3); or (IV) SEQ ID NO: 58 (HCDR1), SEQ ID NO: 59 (HCDR2), SEQ ID NO: 60 (HCDR3), SEQ ID NO: 71 (LCDR1), SEQ ID NO: 69 (LCDR2), and SEQ ID NO: 67 (LCDR3);

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- (b) (I) SEQ ID NO: 76 (HCDR1), SEQ ID NO: 77 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 89 (LCDR1), SEQ ID NO: 90 (LCDR2), and SEQ ID NO: 91 (LCDR3); (II) SEQ ID NO: 79 (HCDR1), SEQ ID NO: 77 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 89 (LCDR1), SEQ ID NO: 90 (LCDR2), and SEQ ID NO: 91 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 81 (HCDR2), SEQ ID NO: 78 (HCDR3), SEQ ID NO: 92 (LCDR1), SEQ ID NO: 93 (LCDR2), and SEQ ID NO: 94 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 83 (HCDR2), SEQ ID NO: 84 (HCDR3), SEQ ID NO: 95 (LCDR1), SEQ ID NO: 93 (LCDR2), and SEQ ID NO: 91 (LCDR3);
- (c) (I) SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 361 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 361 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 362 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 349 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 361 (LCDR3);
- (d) (I) SEQ ID NO: 270 (HCDR1), SEQ ID NO: 389 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 273 (HCDR1), SEQ ID NO: 389 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 390 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 275 (HCDR1), SEQ ID NO: 391 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3);
- (e) (I) SEQ ID NO: 407 (HCDR1), SEQ ID NO: 408 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 409 (HCDR1), SEQ ID NO: 408 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 410 (HCDR1), SEQ ID NO: 411 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 412 (HCDR1), SEQ ID NO: 413 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3);
- (f) (I) SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 321 (LCDR2), and SEQ ID NO: 322 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 321 (LCDR2), and SEQ ID NO: 322 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 312 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 325 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO:

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- 315 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 322 (LCDR3);
- (g) (I) SEQ ID NO: 270 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (II) SEQ ID NO: 273 (HCDR1), SEQ ID NO: 271 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 337 (LCDR1), SEQ ID NO: 338 (LCDR2), and SEQ ID NO: 339 (LCDR3); (III) SEQ ID NO: 32 (HCDR1), SEQ ID NO: 274 (HCDR2), SEQ ID NO: 331 (HCDR3), SEQ ID NO: 340 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 342 (LCDR3); or (IV) SEQ ID NO: 275 (HCDR1), SEQ ID NO: 276 (HCDR2), SEQ ID NO: 332 (HCDR3), SEQ ID NO: 343 (LCDR1), SEQ ID NO: 341 (LCDR2), and SEQ ID NO: 339 (LCDR3); or
- (h) (I) SEQ ID NO: 310 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 355 (LCDR3); (II) SEQ ID NO: 229 (HCDR1), SEQ ID NO: 311 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 320 (LCDR1), SEQ ID NO: 354 (LCDR2), and SEQ ID NO: 355 (LCDR3); (III) SEQ ID NO: 80 (HCDR1), SEQ ID NO: 313 (HCDR2), SEQ ID NO: 348 (HCDR3), SEQ ID NO: 323 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 356 (LCDR3); or (IV) SEQ ID NO: 82 (HCDR1), SEQ ID NO: 314 (HCDR2), SEQ ID NO: 349 (HCDR3), SEQ ID NO: 326 (LCDR1), SEQ ID NO: 324 (LCDR2), and SEQ ID NO: 355 (LCDR3).
- Embodiment 36. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1-5, 7-20, 26, 28, 30, 32, or 34, wherein the antibody or antigen binding fragment comprises:
- (a) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 136;
- (b) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 136;
- (c) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 128;
- (d) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 128;
- (e) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 147;
- (f) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 147;
- (g) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 174;
- (h) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 174;

- [illegible]

- (y) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 193, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 136;
 - (z) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 193, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 174;
 - (aa) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 13, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 24;
 - (bb) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 37, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 48;
 - (cc) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 374, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 244;
 - (dd) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 385, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 244;
 - (ee) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 233, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 244;
 - (ff) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 278, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 287; or
 - (gg) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 300, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 306.
- Embodiment 37. An isolated anti-NPR1 antibody or antigen binding fragment; or the antibody or antigen binding fragment of any one of embodiments 1, 2, 4, 6, 21-25, 27, 29, 31, 33, or 35, wherein the antibody or antigen binding fragment comprises:
- (a) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 61, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 72;
 - (b) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 85, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 96;
 - (c) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 350, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 363;
 - (d) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 392, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 344;
 - (e) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 414, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 344;
 - (f) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 316, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 327;

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- (c) a heavy chain comprising an amino acid sequence of SEQ ID NO: 352, and a light chain comprising an amino acid sequence of SEQ ID NO: 365;
- (d) a heavy chain comprising an amino acid sequence of SEQ ID NO: 394, and a light chain comprising an amino acid sequence of SEQ ID NO: 346;
- (e) a heavy chain comprising an amino acid sequence of SEQ ID NO: 416, and a light chain comprising an amino acid sequence of SEQ ID NO: 346;
- (f) a heavy chain comprising an amino acid sequence of SEQ ID NO: 318, and a light chain comprising an amino acid sequence of SEQ ID NO: 329;
- (g) a heavy chain comprising an amino acid sequence of SEQ ID NO: 335, and a light chain comprising an amino acid sequence of SEQ ID NO: 346; or
- (h) a heavy chain comprising an amino acid sequence of SEQ ID NO: 352, and a light chain comprising an amino acid sequence of SEQ ID NO: 359.

Embodiment 40. The antibody or antigen binding fragment of any one of embodiments 1-39, which is an antigen binding fragment selected from the group consisting of a Fab, Fab', F(ab')₂, Fv, single domain antibody (dAb), and a single chain variable fragment (scFv), optionally wherein the antigen binding fragment is selected from the group consisting of a Fab, Fab', Fv, single domain antibody (dAb), and a single chain variable fragment (scFv).

Embodiment 41. The antibody or antigen binding fragment of any one of embodiments 1-40, which is monoclonal.

Embodiment 42. The antibody or antigen binding fragment of any one of embodiments 1-41, which is fully human.

Embodiment 43. The antibody or antigen binding fragment of any one of embodiments 1-42, which is an IgG antibody, optionally which is an IgG₁ antibody.

Embodiment 44. The antibody or antigen binding fragment of any one of embodiments 1-43, which is an IgG₁ antibody having a kappa light chain.

Embodiment 45. The antibody or antigen binding fragment of any one of embodiments 1-44, which is a fully human antibody of the IgG1 isotype and has a kappa light chain.

Embodiment 46. The antibody or antigen binding fragment of any one of embodiments 1-45, wherein the antibody or antigen binding fragment has further modifications as described herein, e.g., wherein the antibody or antigen binding fragment additionally has mutations in the Fc region according to the EU index of Kabat, wherein the mutations comprise at least D265A and P329A; or wherein the mutations comprise at least L234A and L235A.

Embodiment 47. The antibody or antigen binding fragment of any one of embodiments 1-40, wherein the antibody or antigen binding fragment is: a) monoclonal; and/or b) fully human; and/or c) an IgG antibody, optionally an IgG₁ antibody; and/or d) has a kappa light chain; and/or e) has mutations in the Fc region according to the EU index of Kabat, optionally wherein the mutations comprise at least D265A and P329A; and/or f) has mutations in the Fc region according to the EU index of Kabat, optionally wherein the mutations comprise at least L234A and L235A.

Embodiment 48. The antibody or antigen binding fragment of any one of embodiments 1-47, wherein the antibody or antigen binding fragment is therapeutic.

Embodiment 49. An isolated antibody or antigen binding fragment that binds to the same epitope on human NPR1 as the antibody or antigen binding fragment of any one of embodiments 1 to 48.

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Embodiment 50. An isolated antibody or antigen binding fragment that competes for binding to human NPR1 with the antibody or antigen binding fragment of any one of embodiments 1 to 49.

Embodiment 51. An isolated nucleic acid or nucleic acids encoding the amino acid sequence of the antibody or antigen binding fragment of any one of embodiments 1 to 50.

Embodiment 52. A vector comprising the isolated nucleic acid(s) of embodiment 51.

Embodiment 53. A host cell comprising the isolated nucleic acid(s) of embodiment 51 or the vector of embodiment 52.

Embodiment 54. A method of producing the antibody or antigen binding fragment of any one of embodiments 1 to 50, comprising culturing the host cell of embodiment 53 under conditions suitable to produce the antibody or antigen binding fragment.

Embodiment 55. The method of embodiment 54, wherein the method additionally comprises purification of the antibody or antigen binding fragment.

Embodiment 56. A pharmaceutical composition comprising a purified antibody or antigen binding fragment produced by the method of embodiment 55 and a pharmaceutically acceptable carrier.

Embodiment 57. A pharmaceutical composition comprising an antibody or antigen binding fragment of any one of embodiments 1 to 50 and a pharmaceutically acceptable carrier.

Embodiment 58. The pharmaceutical composition of embodiment 56 or 57 or a combination comprising an antibody or antigen binding fragment of any one of embodiments 1 to 50, wherein the composition further comprises an additional therapeutic agent.

Embodiment 59. The pharmaceutical composition or combination of embodiment 58, wherein the additional therapeutic agent is selected from an ACE (angiotensin-converting-enzyme) inhibitor, an angiotensin receptor blocker (ARB), a neprilysin inhibitor, a beta blocker, a diuretic, a calcium channel blocker, a cardiac glycoside, a sodium-glucose co-transporter 2 inhibitor (SGLT2i), and combinations thereof.

Embodiment 60. The pharmaceutical composition or combination of embodiment 58 or embodiment 59, wherein the additional therapeutic agent is selected from enalapril, benazepril, captopril, fosinopril, lisinopril, moexipril, perindopril, quinapril, ramipril, trandolapril, valsartan, azilsartan, candesartan, eprosartan, irbesartan, losartan, olmesartan, telmisartan, sacubitril, bisoprolol, carvedilol, propranolol, metoprolol, metoprolol tartrate, metoprolol succinate, thiazide diuretics, loop diuretics, potassium-sparing diuretics, amlodipine, clevidipine, diltiazem, felodipine, isradipine, nicardipine, nifedipine, nisoldipine, verapamil, a digitalis glycoside, canagliflozin, dapagliflozin, empagliflozin, ertugliflozin, and combinations thereof.

Embodiment 61. The pharmaceutical composition or combination of embodiment 58 or embodiment 59, wherein the additional therapeutic agent is selected from chlorothiazide, chlorthalidone, hydrochlorothiazide, indapamide, metolazone, bumetanide, ethacrynic acid, furosemide, torsemide, amiloride, eplerenone, spironolactone, triamterene, digoxin, and combinations thereof.

Embodiment 62. The pharmaceutical composition or combination of any one of embodiments 58-61, wherein the additional therapeutic agent is an angiotensin receptor-neprilysin inhibitor (ARNi).

Embodiment 63. The pharmaceutical composition or combination of embodiment 58, wherein the additional

therapeutic agent is selected from a corticosteroid, a leukotriene modifier, a bronchodilator, and combinations thereof.

Embodiment 64. The pharmaceutical composition or combination of embodiment 63, wherein the additional therapeutic agent is selected from fluticasone, budesonide, mometasone, beclomethasone, ciclesonide, fluticasone furoate, prednisone, methylprednisolone, montelukast, zafirlukast, zileuton, a long-acting beta agonist, a short-acting beta agonist, theophylline and ipratropium, and combinations thereof.

Embodiment 65. The pharmaceutical composition or combination of embodiment 63 or embodiment 64, wherein the additional therapeutic agent is selected from salmeterol, formoterol, albuterol, and levalbuterol, and combinations thereof.

Embodiment 66. The pharmaceutical composition or combination of embodiment 58, wherein the additional therapeutic agent is selected from a beta-adrenoceptor antagonist, a carbonic anhydrase inhibitor, an alpha 2-adrenoceptor agonist, a parasympathomimetic, a prostaglandin analog, a rho kinase inhibitor, and combinations thereof, and combinations thereof.

Embodiment 67. The pharmaceutical composition or combination of embodiment 66, wherein the additional therapeutic agent is selected from timolol, levobunolol, metipranolol, carteolol, betaxolol, acetazolamide, dorzolamide, brinzolamide, methazolamide, brimonidine, apraclonidine, a cholinomimetic, latanoprost, latanoprostene bunod, travoprost, bimatoprost, tafluprost, netarsudil and ripasudil, and combinations thereof.

Embodiment 68. The pharmaceutical composition of embodiment 58, wherein the additional therapeutic agent is selected from an ACE (angiotensin-converting-enzyme) inhibitor, an angiotensin receptor blocker (ARB), a neprilysin inhibitor, a beta blocker, a diuretic, a calcium channel blocker, a cardiac glycoside, a sodium-glucose co-transporter 2 inhibitor (SGLT2i), an angiotensin receptor-neprilysin inhibitor (ARNi), a corticosteroid, a leukotriene modifier, a bronchodilator, a beta-adrenoceptor antagonist, a carbonic anhydrase inhibitor, an alpha 2-adrenoceptor agonist, a parasympathomimetic, a prostaglandin analog, a rho kinase inhibitor, and combinations thereof.

Embodiment 69. The pharmaceutical composition of embodiment 68, wherein the additional therapeutic agent is selected from enalapril, benazepril, captopril, fosinopril, lisinopril, moexipril, perindopril, quinapril, ramipril,trandolapril, valsartan, azilsartan, candesartan, eprosartan, irbesartan, losartan, olmesartan, telmisartan, sacubitril, bisoprolol, carvedilol, propranolol, metoprolol, metoprolol tartrate, metoprolol succinate, thiazide diuretics, loop diuretics, potassium-sparing diuretics, amlodipine, clevidipine, diltiazem, felodipine, isradipine, nifedipine, nisoldipine, verapamil, a digitalis glycoside, canagliflozin, dapagliflozin, empagliflozin, ertugliflozin, chlorothiazide, chlorthalidone, hydrochlorothiazide, indapamide, metolazone, bumetanide, ethacrynic acid, furosemide, torsemide, amiloride, eplerenone, spironolactone, triamterene, digoxin, fluticasone, budesonide, mometasone, beclomethasone, ciclesonide, fluticasone furoate, prednisone, methylprednisolone, montelukast, zafirlukast, zileuton, a long-acting beta agonist, a short-acting beta agonist, theophylline, ipratropium, salmeterol, formoterol, albuterol, and levalbuterol, timolol, levobunolol, metipranolol, carteolol, betaxolol, acetazolamide, dorzolamide, brinzolamide, methazolamide, brimonidine, apraclonidine, a cholinomimetic,

latanoprost, latanoprostene bunod, travoprost, bimatoprost, tafluprost, netarsudil and ripasudil, and combinations thereof.

Embodiment 70. The antibody or antigen binding fragment thereof of any of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition of any of embodiments 56-69 for use (i) in therapy, (ii) as a medicament or (iii) in the manufacture of a medicament for the treatment of a disease.

Embodiment 71. Use of the antibody or antigen binding fragment of any of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition of any of embodiments 56-69 for the manufacture of a medicament for the treatment of a disorder or disease associated with natriuretic peptide receptor activity in a subject in need of such treatment.

Embodiment 72. Use of the antibody or antigen binding fragment of any one of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition of any of embodiments 56-69 for the manufacture of a medicament for the treatment of a cardiovascular disorder in a subject in need of such treatment.

Embodiment 73. The use of embodiment 72, wherein the cardiovascular disorder is selected from: hypertension, peripheral vascular disease, heart failure, coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy, diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, and myocardial infarction (MI).

Embodiment 74. Use of the antibody or antigen binding fragment of any one of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition of any of embodiments 56-69, for the manufacture of a medicament for the treatment of heart failure, hypertrophic cardiomyopathy (HCM), hypertension, preeclampsia, asthma, glaucoma, and/or cytokine release syndrome in a subject in need of such treatment.

Embodiment 75. The use of embodiment 73 or embodiment 74, wherein the subject has heart failure, and wherein the heart failure is selected from a heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure.

Embodiment 76. The use of embodiment 73 or embodiment 74, wherein the subject has hypertrophic cardiomyopathy, and wherein the hypertrophic cardiomyopathy is ventricular hypertrophy.

Embodiment 77. The use of embodiment 73 or embodiment 74, wherein the subject has hypertension, and wherein the hypertension is selected from resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, and pulmonary arterial hypertension.

Embodiment 78. The use of any one of embodiments 73, 74, or 77, wherein the subject has hypertension, and wherein the hypertension is selected from resistant hypertension or hypertensive heart disease.

Embodiment 79. Use of the antibody or antigen binding fragment of any one of embodiments 1-50, the isolated

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nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition of any of embodiments 56-69, for the manufacture of a medicament for the treatment of a kidney disorder in a subject in need of such treatment.

Embodiment 80. The use of embodiment 79, wherein the kidney disorder is selected from: diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD).

Embodiment 81. The antibody or antigen binding fragment of any of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition or combination of any of embodiments 56-69 for use in the treatment of a disorder or disease associated with natriuretic peptide receptor activity in a subject in need of such treatment.

Embodiment 82. The antibody or antigen binding fragment of any one of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition or combination of any of embodiments 56-69 for use in the treatment of a cardiovascular disorder in a subject in need of such treatment.

Embodiment 83. The antibody or antigen binding fragment, isolated nucleic acid or nucleic acids, vector, host cell, pharmaceutical composition, or combination of embodiment 82, wherein the cardiovascular disorder is selected from: hypertension, peripheral vascular disease, heart failure, coronary artery disease (CAD), ischemic heart disease (IHD), mitral stenosis and regurgitation, angina, hypertrophic cardiomyopathy, diabetic cardiomyopathy, supraventricular and ventricular arrhythmias, cardiac dysrhythmia, atrial fibrillation (AF), new onset of atrial fibrillation, recurrent atrial fibrillation, cardiac fibrosis, atrial flutter, detrimental vascular remodeling, plaque stabilization, and myocardial infarction (MI).

Embodiment 84. The antibody or antigen binding fragment of any one of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition or combination of any of embodiments 56-69, for use in the treatment of heart failure, hypertrophic cardiomyopathy (HCM), hypertension, preeclampsia, asthma, glaucoma, and/or cytokine release syndrome in a subject in need of such treatment.

Embodiment 85. The antibody or antigen binding fragment, isolated nucleic acid or nucleic acids, vector, host cell, pharmaceutical composition, or combination of embodiment 83 or embodiment 84, wherein the subject has heart failure, and wherein the heart failure is selected from a heart failure with reduced ejection fraction (HFrEF), heart failure with preserved ejection fraction (HFpEF), heart failure after acute myocardial infarct, or acute decompensated heart failure.

Embodiment 86. The antibody or antigen binding fragment, isolated nucleic acid or nucleic acids, vector, host cell, pharmaceutical composition, or combination of embodiment 83 or embodiment 84, wherein the subject has hypertrophic cardiomyopathy, and wherein the hypertrophic cardiomyopathy is ventricular hypertrophy.

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Embodiment 87. The antibody or antigen binding fragment, isolated nucleic acid or nucleic acids, vector, host cell, pharmaceutical composition, or combination of embodiment 83 or embodiment 84, wherein the subject has hypertension, and wherein the hypertension is selected from resistant hypertension, hypertensive heart disease, pulmonary hypertension, pulmonary arterial hypertension, isolated systolic hypertension, resistant hypertension, and pulmonary arterial hypertension.

Embodiment 88. The antibody or antigen binding fragment, isolated nucleic acid or nucleic acids, vector, host cell, pharmaceutical composition, or combination of embodiments 83, 84, or 87, wherein the subject has hypertension, and wherein the hypertension is selected from resistant hypertension or hypertensive heart disease.

Embodiment 89. The antibody or antigen binding fragment of any one of embodiments 1-50, the isolated nucleic acid or nucleic acids of embodiment 51, the vector of embodiment 52, the host cell of embodiment 53 or pharmaceutical composition or combination of any of embodiments 56-69, for use in the treatment of a kidney disorder in a subject in need of such treatment.

Embodiment 90. The antibody or antigen binding fragment, isolated nucleic acid or nucleic acids, vector, host cell, pharmaceutical composition, or combination of embodiment 89, wherein the kidney disorder is selected from: diabetic renal insufficiency, non-diabetic renal insufficiency, renal failure, diabetic nephropathy, non-diabetic nephropathy, acute renal injury, contrast induced nephropathy, nephrotic syndrome, glomerulonephritis, scleroderma, glomerular sclerosis, proteinuria of primary renal disease, renal vascular hypertension, diabetic retinopathy and end-stage renal disease (ESRD), endothelial dysfunction, diastolic dysfunction, renal fibrosis, and polycystic kidney disease (PKD).

EXAMPLES

The following examples provide illustrative embodiments of the disclosure. One of ordinary skill in the art will recognize the numerous modifications and variations that may be performed without altering the spirit or scope of the disclosure. Such modifications and variations are encompassed within the scope of the disclosure. The examples provided do not in any way limit the disclosure.

This disclosure provides anti-NPR1 antibodies that specifically bind and activate NPR1, e.g., antibodies and antigen binding fragments that (i) bind to NPR1; and (ii) activate NPR1 in the absence of ANP. Antibodies that specifically bind and activate NPR1 could have different possible modes of action: (1) the antibody induces a conformational change within the NPR1 monomers to activate the receptor; (2) the antibody directly mimics the structure and function of the natural ligand ANP and activates the receptor by binding in the ANP binding pocket of NPR1; or (3) the antibody stabilizes the preformed functionally active complex of hNPR1 and ANP (NPR1-ANP-complex).

Example 1: HuCAL® Phage Library Panning and Screening

For the selection of NPR1-specific antibodies covering the described different methods of action, 13 different panning strategies were applied (see Table 5). Ten strategies were performed exclusively on protein (strategies 1-6 and 10-13). In addition, three differential cell pannings were performed (strategies 7-9). In total, four panning strategies (strategies 3 and 11-13) aimed for the enrichment of ANP

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competing antibodies (elution with ANP, pre-adsorption of phage on NPR1-ANP-complexes, and anti-idiotypic panning on murine anti-ANP antibodies).

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For solution panning, NPR1 was biotinylated and the retained activity of biotinylated NPR1 for ANP binding was confirmed. During solution panning, the Fab displaying

TABLE 5

Overview of HuCAL ® panning strategies				
Strategy	1 st round	2 nd round	3 rd round	comments
1	hNPR1 solution	hNPR1 solution	hNPR1 solution	Solution panning with human antigen only
2	hNPR1 capture	hNPR1 capture	hNPR1 capture	Fc capture panning with human antigen only
3	hNPR1 solution	hNPR1 solution + ANP elution	hNPR1 solution + ANP elution	Solution panning aiming for enrichment of ANP competitors
4	hNPR1 solution	hNPR1 solution + pH 5.8 elution	hNPR1 solution + pH 5.8 elution	Solution panning aiming for enrichment of pH dependent binders
5	hNPR1 solution	rNPR1 capture	hNPR1 solution	Solution/Fc capture panning aiming for rat crossreactivity
6	rNPR1 capture	hNPR1 capture	rNPR1 capture	Fc capture panning aiming for rat crossreactivity
7	hNPR1 capture	CHO-K1 NPR1 cells	rNPR1 capture	Fc capture/cell panning aiming for rat crossreactivity
8	hNPR1 solution	CHO-K1 NPR1 cells	hNPR1 solution	Solution/cell panning with human antigen only
9	CHO-K1 NPR1 cells	hNPR1 solution	CHO-K1 NPR1 cells	Solution/cell panning with human antigen only
10	hNPR1-ANP-complex solution	rNPR1-ANP-complex capture	hNPR1-ANP-complex solution	Solution panning aiming for enrichment of NPR-ANP complex stabilizers and rat crossreactivity
11	pre-adsorption on NPR1-ANP complex, save the unbound phage still in solution, bind to hNPR1 in solution without ANP and capture NPR1/phage complexes	pre-adsorption on NPR1-ANP complex, save the unbound phage still in solution, bind to hNPR1 in solution without ANP and capture NPR1/phage complexes	pre-adsorption on NPR1-ANP complex, save the unbound phage still in solution, bind to hNPR1 in solution without ANP and capture NPR1/phage complexes	Solution panning aiming for enrichment of ANP competitors
12	mouse anti-ANP-mAb capture	hNPR1 solution	mouse anti-ANP-mAb capture	Fc capture/cell panning with anti-ANP mAb
13	mouse anti-ANP mix of two mAbs capture	mouse anti-ANP mix of two mAbs capture + ANP elution	mouse anti-ANP mix of two mAbs capture + ANP elution	Fc capture panning with anti-ANP mAb aiming for enrichment of ANP competitors

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For Fc capture panning, NPR1-hFc was immobilized on a 96-well plate via an appropriate capture antibody (a goat or mouse anti-human Fc antibody). The antigen was immobilized in an appropriate number of wells of a 96-well plate and wells were subsequently blocked prior to the addition of phage-antibodies. In parallel to well preparation, phage-antibodies were blocked. During blocking of phage, additional blocking reagents were added to the blocking buffer to avoid selection of antibodies against the hFc-tag or the capture antibody (goat or mouse y globulin). Following the blocking procedure, two pre-adsorption steps on human y globulin and on the counter-target hNPR3-hFc were performed to avoid selection of antibodies against the Fc-tag or the counter-target. The pre-blocked and pre-adsorbed phage mix was added to each well with immobilized NPR1-hFc and the phage-antibodies were allowed to bind to the antigen. Intensive washing ensured removal of non-specifically bound phage, followed by elution of specifically bound phage. The second and third round of solid phase panning was performed according to the protocol of the first panning round. Amounts of antigen were decreased and washing conditions with increased stringency were applied.

phage and the biotinylated NPR1-hFc were incubated in solution, which facilitated the accessibility of the antigen by the phage. An appropriate amount of Streptavidin beads was blocked and, in parallel, an appropriate amount of phage-antibodies was blocked. During blocking of phage, human y globulin, the counter-target hNPR3-hFc and the Flag-TEV linker peptide were added to the blocking buffer to avoid selection of antibodies against the hFc-tag, the counter-target, or the linker peptide. For removal of Streptavidin-, Biotin-, or bead-binding phage, pre-adsorption steps of blocked phage particles were performed using blocked Streptavidin beads with and without coupled biotinylated irrelevant antigen. Subsequently, biotinylated NPR1-hFc/ NPR1-hFc-ANP-complex was added to the pre-adsorbed and blocked phage particles and the phage-antibodies were allowed to bind to the antigen in solution. For enrichment of antibody phage binding to the ANP-binding site of NPR1 (ANP competitive antibodies) the pre-formed NPR1-ANP-complex was added to the phage blocking solution or the ANP peptide was used for elution of the bound phage. Thereby, the ANP peptide was used at least in 250-fold

molar excess to the NPR1 antigen or the NPR1 expressing cells. The phage-antigen complexes were captured using blocked Streptavidin beads and phage particles bound to the Streptavidin beads were collected with a magnetic separator. Phage bound nonspecifically were washed off by several washing steps. Specifically bound phage were eluted from Streptavidin beads. The eluate was transferred to an *E. coli* culture for phage infection. The second and third round of bead-based solution panning was performed according to the protocol of the first panning round. Amounts of antigen were decreased and washing conditions with increased stringency were applied.

For whole cell panning, an appropriate amount of phage-antibodies was blocked. During blocking of phage, counter-target hNPR3-hFc was added to the blocking buffer to avoid selection of antibodies against the counter-target. In parallel, an appropriate amount of target cells expressing NPR1 and an appropriate amount of adsorption cells without expression of antigen (parental cells) per phage pool were blocked. The blocked target cells were spun down, resuspended in the pre-blocked phage particles and the phage-antibodies were allowed to bind to the NPR1 presented on the cell. The phage-cell complexes were washed several times. For enrichment of antibody phage binding to the ANP-binding site of NPR1 (ANP competitive antibodies) the pre-formed NPR1-ANP-complex was added to the phage blocking solution or the ANP peptide was used for elution of the bound phage. Thereby, the ANP peptide was used at least in 250-fold molar excess to the NPR1 antigen or the NPR1 expressing cells. Specifically bound phage were eluted from target cells. After centrifugation, the supernatant (eluate) was applied to adsorption cells for removal of phage binding to cell surface molecules other than the target antigen (post-adsorption). The final supernatant was transferred to an *E. coli* culture for phage infection. The second and third round of the whole cell panning were performed according to the protocol of the first panning round. Washing conditions with increased stringency were applied.

The outputs of the panning rounds were subsequently subcloned into bacterial expression vectors and bacterial lysates (BEL) were used for primary and secondary screening. The outputs were analyzed for binding to human and rat NPR1 during the primary screening (ELISA-based). Clones binding to human NPR3 were deselected. Secondary screening was performed on hNPR1 expressing CHO-K1 cells. Further screenings regarding ANP competition and binding solely in presence of ANP were performed. Approximately 1700 clones fulfilled the screening selection criteria and 760 clones were selected for sequencing. The sequencing of 760 clones resulted in 210 HCDR3 unique hits, whose binding properties are summarized in Table 6. Of these clones, 72 demonstrated significant ANP competition, while 7 clones bound only in presence of ANP.

TABLE 6

Binding properties of 210 HCDR3 unique hits (HuCAL®)						
Number of HCDR3 unique candidates				Binding to		
				hNPR1	rNPR1	hNPR1 expr. cells
210 of which 7 clones only bind in presence of ANP	47 human/	28 human/rat cross reactive cell binders	Yes	Yes	Yes	Yes
	rat cross	19 human/rat cross reactive (not binding to cells)	Yes	Yes	Yes	No
	reactive					
	156 human	53 human specific cell binders	Yes	No	No	Yes
	specific	103 human specific (not binding to cells)	Yes	No	No	No

Example 2: Antibody Reformatting, Expression, and Purification

After confirmation of binding, the VH and VL domains of the 210 HCDR3 unique clones were subcloned into a vector with a human IgG constant region. 180 of the 210 clones were selected for expression and 166 of the 180 passed the production quality control. They were characterized in regard to binding to relevant cell lines and functional activity. 40 of the 166 candidates were then selected for exploratory scale production, and 31 of these candidates were characterized in detail as shown below with respect to binding to relevant antigens and cell lines, ANP competition, and functionality in a cell based cGMP production assay.

For production of the IgG candidates, eukaryotic HKB11 cells were transfected with mammalian expression vector DNA encoding both heavy and light chains of IgG. Cell culture supernatants were harvested at appropriate times and subjected to Protein A affinity chromatography. If needed, a second purification step was performed to remove aggregates. Buffer exchange was performed to 1x Dulbecco's PBS (pH 7.2) and samples were sterile filtered (0.2 µm pore size).

Example 3: Antigen Binding, ANP Competition and Cellular cGMP Production-HuCAL® Candidates

The 31 IgGs which passed the exploratory scale production quality control were tested via ELISA for binding to the following antigens: human NPR1, constitutively active human NPR1 mutant (W74R), rat NPR1, and human NPR3 (counter target). The clones were also tested by flow cytometry for binding to human NPR1 expressing CHO K1 cells in the absence and presence of ANP and on parental CHO K1 cells. The binding properties of the five functional candidates are shown in FIG. 1 and FIG. 2.

The same 31 IgGs were tested for ANP competition using a Fluorescence Resonance Energy Transfer (FRET)-based assay in which the NPR1-specific antibodies competed with ANP for binding to NPR1. In this FRET based assay (see FIG. 3), Eu-labeled Streptavidin (for measurement of IgGs) or Eu-labeled anti-hFc antibody (for measurement of FabCys) was used as an energy donor, while Cy5-labeled ANP was used as an acceptor (for all measurements). The resulting fluorescent signal was decreased by ANP-competitive antibodies. The assay was performed as follows: antibodies were mixed with NPR1 and incubated for 5 minutes at room temperature. After addition of the Eu-labeled donor and incubation for 30 minutes at room temperature, the Cy5-ANP solution was added. After a further 60-minute incubation, readout was performed using a TECAN Infinite M1000 Pro using an excitation wavelength of 317 nm and an

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emission wavelength of 665 nm. Percentage of ANP competition was calculated according to the following formulae:

$$\text{Ratio}^* = [(A_{665} \text{ nm} / A_{620} \text{ nm}) * 10^4]$$

$$\text{Ratio} = (\text{Ratio}^* - \text{Ratio}_{neg})$$

$$\text{Competition \%} = [100 - (\text{Ratio} / (\text{Ratio}_{pos} / 100))]$$

Ratio_{neg}: mean Ratio*data values of control without NPR1

Ratio_{pos}: mean Ratio data values of control without agonist (reaction buffer)

15 of the 31 IgGs were ANP competitive, but only two of these candidates showed functionality in the cGMP assay (WW04 and WW06). The other three functional candidates WW01, WW02 and WW03 demonstrated a “negative” ANP competition in this assay indicating the stabilization of the NPR1-ANP-complex. The FRET assay results for the five functional candidates are depicted in FIG. 4.

Additionally, as discussed above, the 31 IgGs were tested for their functional activity in a cellular cGMP production assay using human NPR1 expressing CHO-K1 cells. For the functional characterization of the selected antibodies the production of cyclic guanosine 3',5'-cyclic monophosphate (cGMP) upon binding to and stimulation of NPR1 expressed on the cell surface of CHO-K1 cells was monitored. Cellular cGMP is a major second messenger that mediates cell activities and is synthesized by activated NPR1 triggered by ANP or NPR1-specific antibodies. Therefore, a commercial assay kit was used (Cisbio Bioassays CisBio HTRF Assay Kit CisBio (Cat. #62GM2PEB)). The assay was performed according to manufacturer's instructions with minor deviations. In brief, cells were adjusted to 1×10^5 cells/mL, 20 μ L/well were seeded in 96 well microtiter plates and were incubated overnight. After addition of 10 μ L/well of the antibodies in different concentrations, the plate was incubated for 30 min at 37° C. to allow for cGMP production. In parallel, a standard curve using a calibrator (contained in the kit) was generated. The cells were lysed and a mix of cGMP-d2 and anti-cGMP-Cryptate was added and incubated for 1 h at room temperature. The readout was performed using a Tecan M1000 Pro using an excitation wave-

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length of 317 nm and an emission wavelength of 665 nm. cGMP concentration (Delta F [%]) was calculated according to the following formulae:

$$\text{Ratio} = [(A_{665} \text{ nm} / B_{620} \text{ nm}) * 10^4]$$

$$\text{Mean Ratio} = (\Sigma \text{ratios} / 2)$$

$$CV = [(\text{Std deviation} / \text{Mean ratio}) * 100]$$

$$\text{Delta F} = [((\text{Calibrator or sample Ratio} - \text{Ratio}_{neg}) / \text{Ratio}_{neg}) * 100]$$

Ratio_{neg}: negative control

Five candidates with significant functional activity were identified using the cellular cGMP assay: WW01, WW02, WW03, WW04, and WW06. These five candidates were functionally active and could be assigned to different methods of action. WW01, WW02, and WW03 were able to stabilize the NPR1-ANP-complex, while WW06 was determined to be ANP competitive. These candidates were all derived from initial panning codes 10 and 11 (aiming for method of action 2 or 3). The results of the assay for the cellular production of cGMP in the absence or presence of 0.075 nM ANP induced by the five functional candidates (IgG format) are shown in FIG. 5.

Example 4: Production and Characterization of HuCAL® Candidates in FabCys Format

The functional clones were also tested for functionality in FabCys format. Eukaryotic HKB11 cells were transfected with mammalian expression vector DNA encoding both heavy and light chains of disulfide-bridged FabCys. Cell culture supernatants were harvested at appropriate times and subjected to metal ion affinity chromatography using a liquid handling station. Buffer exchange was performed to 1x Dulbecco's PBS (pH 7.2) and samples were sterile filtered (0.2 μ m pore size).

The five functional candidates WW01, WW02, WW03, WW04 and WW06 were additionally analyzed with regard to their monovalent affinities for human and rat NPR1 and the counter-target human NPR3 in absence and presence of ANP in monovalent FabCys format. The results of the affinity determination, epitope binning, and cGMP assay are summarized in Table 7.

TABLE 7

Summary of Affinity, Epitope, and cGMP data for WW01, WW02, WW03, WW04, and WW06 in FabCys Format							
Characterization in monovalent FabCys format							
Antibody	Affinity FabCys K _D [nM]				Epitope Bin on	cGMP assay	
	hNPR1	hNPR1 + ANP	rNPR1	rNPR1 + ANP		cGMP conc [nM] at 2 μ M FabCys	
					hNPR1	+0.075	
WW01	—	1.5	—	2.0	B	30	121
WW02	weak	weak	weak	weak	n.a.	4	20
WW03	1000	0.1	2600	1.0	B	32	108
WW04	0.8	0.4	66	73	D	4	0
WW06	5.3	12	weak	weak	D	281	251

For candidates WW01 and WW03 no or very weak binding to human and rat NPR1 was observed in the absence of ANP, while the affinities in the presence of ANP were in the low nanomolar to subnanomolar range. Both shared the same epitope bin "B". The affinity of candidates WW02 was too weak for adequate determination of K_D values and the epitope bin. WW04 and WW06 had affinities in the double-digit nanomolar to subnanomolar range, which were independent from the presence or absence of ANP. Both shared the same epitope bin "A". WW06 was the only candidate which did not exhibit rat cross-reactivity.

While for WW02, WW03 and WW04 no binding to the counter-target hNPR3 in the absence or presence of ANP was observed, additional binding to the counter-target was detected for WW01 as well as for WW06 at higher concentrations.

Example 5: Reformatting HuCAL® Candidates into IgG Format

Subcloning from the FabCys vector into an IgG₁_LALA vector for expression in mammalian cells was performed via amplification of the Fab-encoding insert using one biotinylated primer and one non-biotinylated primer. The amplified product was bound on streptavidin beads, digested using restriction enzymes, and washed, resulting in the release of the purified insert into the supernatant. The insert was cloned into the acceptor vector, the DNA was transformed and single clones were quality controlled via colony PCR and sequencing.

The five functional candidates WW01, WW02, WW03, WW04, and WW06 in IgG format were characterized as described above. Binding data (ELISA, flow cytometry, ANP competition) and functional data (cGMP assay) as well as affinities, and epitope bins are shown in Table 8. Interestingly, WW06 had a significantly increased functional activity in FabCys format compared to IgG format as shown in FIG. 6.

Example 6: Generation of HuCAL® Maturation Libraries

To increase affinity and biological activity of the selected antibody fragments (WW01, WW02, WW03, WW04, and WW06), LCDR3 and HCDR2 regions were exchanged in parallel by diversified cassettes/modules (Prassler et al. (2009): In vitro affinity maturation of HuCAL® antibodies: complementarity determining region exchange and Rap-MAT technology; Immunotherapy 1 (4), pp. 571-583, the contents of which are hereby incorporated by reference for this purpose), while the framework regions were kept constant. Parental Fab fragments were transferred from the corresponding expression vector into a library cloning vector for affinity maturation.

The generation of HuCAL® maturation libraries was performed for each maturation candidate individually. For LCDR3 optimization, an approximately 400 bp DNA fragment encoding for the LCDR3, framework 4 as well as the constant region of the light chain was removed from the sequence encoding the parental antibody by restriction digest. In order to reduce the background of the parental undiversified sequence the excised fragment was replaced by an approximately 520 bp dummy sequence via ligation, before a repertoire of DNA fragments encoding for diversified LCDR3 regions together with framework 4 and the constant domain (diversified LCDR3 cassette) was inserted via restriction digest and ligation.

In a second library set the HCDR2-encoding sequence was diversified, while the connecting framework regions were kept constant. In order to reduce the background of the parental undiversified sequence an approximately 150 bp DNA fragment containing the parental HCDR2 and the framework 3 sequences was replaced by an approximately 590 bp dummy sequence via restriction digest and ligation, before the diversified HCDR2 cassette (including framework 3) was inserted also via restriction digest and ligation.

TABLE 8

Summary of Affinity, Epitope, and cGMP data for WW01, WW02, WW03, WW04, and WW06 in IgG Format													
Characterization in bivalent IgG format													
Antibody	Elisa +/- 100 nM ANP IgG binding/EC ₅₀ [nM]								Flow Cytometry [S/BG] at 2 µM IgG		ANP competition assay		cGMP assay cGMP conc [nM] at 2 µM IgG
	hNPR1 +	W74R +	rNPR1 +	hNPR3 +	hNPR1 CHO	hNPR1 KI + 100 nM ANP	hNPR1 CHO KI	hNPR1 CHO KI	hNPR1 KI + 100 nM ANP	hNPR1 CHO KI	hNPR1 CHO KI	hNPR1 CHO KI	hNPR1 CHO KI
	100 nM ANP	100 nM ANP	100 nM ANP	100 nM ANP	100 nM ANP	100 nM ANP	100 nM ANP	100 nM ANP	100 nM ANP	100 nM ANP	100 nM ANP	100 nM ANP	100 nM ANP
WW01	no specific binding	1.0	no specific binding	no specific binding	no specific binding	no specific binding	no specific binding	no specific binding	19.9	222.2	No	36	197
WW02	no specific binding	531.6	no specific binding	no specific binding	no specific binding	no specific binding	no specific binding	no specific binding	2.0	17.4	No	6	85
WW03	42.7	1.4	no specific binding	no specific binding	no specific binding	no specific binding	no specific binding	no specific binding	2.2	214.3	No	31	192
WW04	2.5	4.2	3.5	1.8	no specific binding	no specific binding	no specific binding	no specific binding	1.9	46.6	Yes	8	107
WW06	2.5	1.1	0.9	0.8	no specific binding	no specific binding	no specific binding	no specific binding	4.6	1.1	Yes	13	38

The ten maturation libraries were successfully cloned and had library sizes between 9.2×10^8 and 2.2×10^9 cfu. Ligation mixtures were electroporated into *E. coli* cells yielding $>10^8$ independent colonies. Amplification of the library was performed as described previously (Rauchenberger et al. (2003): Human combinatorial Fab library yielding specific and functional antibodies against the human fibroblast growth factor receptor 3; *J Biol Chem* 278 (40), pp. 38194-38205, the contents of which are hereby incorporated by reference for this purpose). For quality control, approx. 10-20 single clones per library were picked randomly and sequenced.

For the selection of affinity improved candidates, phage derived from maturation libraries were subjected to three rounds of maturation panning as described further below. Panning stringency was increased by prolonged washing steps. In addition, off-rate selection was performed (Hawkins et al. (1992): Selection of phage antibodies by binding affinity. Mimicking affinity maturation. In *J. Mol. Biol.* 226 (3), pp. 889-896, the contents of which are hereby incorporated by reference for this purpose).

Example 7: Pannings and Screenings-HuCAL®

The maturation libraries were used for four different maturation panning strategies. Strategies #3 and #4 aimed for the enrichment of progenies with improved affinities compared to the parental clones. In addition, strategies #1 and #2 aimed for the enrichment of clones with improved affinities for NPR1 instead of NPR1-ANP-complex. The rationale behind that was the idea to generate candidates which are able to directly active NPR1 by a conformational change. During the panning process, all maturation libraries were kept separately. The panning strategies are summarized in Table 9 in detail. The outputs of the third panning rounds were subsequently sub-cloned into a bacterial expression vector and bacterial lysates (BEL) were used for SET screening.

TABLE 9

Overview of HuCAL® Maturation Panning Strategies					
Strategy	Parental antibodies	1 st round	2 nd round	3 rd round	comments
1	WW01 WW02 WW03 WW04 WW06	CHO-K1 NPR1 cells	hNPR1 solution	CHO-K1 NPR1 cells	Cell/solution panning aiming for improved candidates
2	WW01 WW02 WW03 WW04	CHO-K1 NPR1 cells	hNPR1-ANP-complex solution	CHO-K1 NPR1 cells	Cell/solution panning aiming for improved candidates
3	WW01 WW02 WW03 WW04	hNPR1-ANP-complex solution	CHO-K1 NPR1-ANP-complex Cells	hNPR1-ANP-complex solution	Solution/cell aiming for improved NPR1-ANP-complex
4	WW04 WW06	Preadsorption on NPR1-ANP complex CHO-K1 NPR1 cells	Preadsorption on NPR1-ANP complex hNPR1 solution	Preadsorption on NPR1-ANP complex CHO-K1 NPR1 cells	Cell/solution panning aiming for improved ANP competitors

The outputs of the 3rd panning rounds were used for Solution Equilibrium Titration (SET) screening. 88 clones per subcode (2640 clones in total) were analyzed in SET screening for improved affinity for hNPR1 and/or hNPR1-ANP-complex compared to the parental clones.

During SET screening, 82 HCDR2 or LCDR3 unique improved derivatives were identified. Compared to their parental clones, the affinities of WW01 and WW03 derivatives were improved up to 20-fold both for hNPR1 and hNPR1-ANP-complex. The affinities of the WW04 derivatives were not improved significantly, while the WW06 derivatives had up to 3-fold improved affinities compared to the parental clone. See FIG. 7 (which shows affinities for hNPR1) and FIG. 8 (which shows affinities for hNPR1-ANP complex). Affinities (K_D [pM]) are indicated at the x-axis below the parental clone name for both figures.

Example 8: Characterization of Matured Candidates-HuCAL®

74 of the 82 improved candidates were successfully subcloned into FabCys format and 61 of the 74 clones passed the production quality control and were characterized in regard to binding to relevant antigens, binding to relevant cell lines, ANP competition, and functional activity in the cGMP production assay in comparison to their parental clones. All 16 derivatives of WW01 and all 27 derivatives of WW03 had up to 20-fold improved binding and functional activity. The majority of the derivatives also showed improved binding and functionality in the presence of ANP. Some derivatives showed binding to W74R (constitutively active hNPR1 mutant), which was not true for the parental FabCys. One of the four derivatives of WW04 had two-fold improved binding and functional activity, while the rest behaved like the parental FabCys. All 14 derivatives of WW06 had improved binding for NPR1 and remained competitive with ANP. The functional activities of 10 of the 14 progenies were improved up to three-fold compared to the parental FabCys. Some derivatives displayed rat cross-reactivity, which was not true for the parental FabCys.

After FabCys characterization, a further 40 of the 61 derivatives were selected for IgG conversion and further characterization. Ten potential candidates shown in Table 10

were then assayed with respect to binding to relevant antigens, binding to relevant cell lines, ANP competition, and functional activity in the cGMP production assay in comparison to their parental clones. They were further analyzed via 3P assay and their affinities for human and rat NPR1 in absence and presence of ANP were determined via

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SET K_D measurement. WW01 and WW03 derivative antibodies were analyzed in IgG format and the WW06 derivatives in FabCys format.

For protein panel profiling (Frese et al. (2013): An automated immunoassay for early specificity profiling of antibodies; mAbs 5 (2), pp. 279-287, the contents of which are herein incorporated by reference for this purpose), 32 different proteins and controls were coated on two 384-well MSD standard plates at a concentration of 1.0 $\mu\text{g/mL}$ at 4° C. overnight. The coating solution was discarded and plates were blocked with 50 μL 3% (w/v) BSA in PBS for one hour at RT on a microtiter plate shaker (~500 rpm) followed by three washing steps with 50 μL washing buffer (PBS with 0.05% (v/v) Tween 20). Antibody samples were diluted to 100 nM and 10 nM in assay buffer (PBS with 0.5% (w/v) BSA, 0.05% (v/v) Tween 20). As controls, a reference antibody (Fab or IgG, depending on the sample format) and assay buffer were used. Samples and controls were added at 30 μL /well and incubated for three hours at RT on a microtiter plate shaker. The plates were washed three times and 30 μL detection antibody (ECL-labeled anti-human Fab) were added per well and incubated for one hour on a microtiter plate shaker (~500 rpm). After washing the MSD plate and adding 35 μL /well MSD Read Buffer T with surfactant, electrochemiluminescence signals were detected using a Sector Imager 6000 (Meso Scale Discovery; Gaithersburg, MD, USA). For evaluation, signals of the antibody sample on a certain protein were divided by the respective signals of the reference mAb resulting in a binding ratio (BR). The cumulative binding ratio (CBR) of all proteins except the controls (25 in total) was then calculated: CBR up to 150 represented an antibody or fragment thereof without detectable non-specific binding. Values above represented an antibody or fragment thereof with increased non-specific binding compared to a reference mAb.

TABLE 10

Overview of Matured HuCAL ® Candidates		
Matured antibody	Parental antibody	Matured CDR
XX01	WW01	HCDR2
XX02	WW06	HCDR2
XX03	WW03	HCDR2
XX04	WW03	LCDR3
XX05	WW06	HCDR2
XX06	WW03	LCDR3
XX07	WW03	LCDR3
XX08	WW01	HCDR2
XX10	WW06	HCDR2
XX12	WW03	LCDR3

Furthermore, the clones were tested via ELISA for binding to the following antigens: human NPR1, constitutively active human NPR1 mutant (W74R), rat NPR1, human NPR3 (counter target), each in the absence and presence of ANP, and BSA. The clones were also analyzed by flow cytometry for binding to human NPR1 expressing CHO K1 cells in absence and presence of ANP (100 nM) and to parental CHO K1 cells. The binding properties of the 10 candidates are shown in FIGS. 9 and 10. The ANP competition results for nine of the candidates are shown in FIG. 11.

Negative values for XX01, XX03, XX04, XX06, XX07, and XX12 suggest enhancement of ANP binding by these antibodies. The functional activity of the 10 candidates was analyzed using the cellular cGMP production assay and results are shown in FIG. 12. As expected from the func-

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tional data of the parental clone WW06, its derivatives XX02, XX05, and XX10 showed weak to no functional activity in the IgG format, whereas they had very high functionality in monovalent FabCys format.

The affinities for XX01-XX08, XX10, and XX12 in monovalent FabCys format were determined via SET K_D measurement. The results are summarized in Table 11 in comparison to the affinities of the parental clones (determined in another experiment via Biacore®). The affinities for human and rat NPR1 of WW01 and WW03 derivatives were improved up to 2,300-fold, while the affinities for the NPR1-ANP-complexes were only slightly improved (maximal 5-fold). They had affinities between 10 and 46 nM for hNPR1 and between 100 and 300 pM for hNPR1-ANP-complex. All WW01 and WW03 progenies displayed rat cross-reactivity with rat/human K_D ratios <5. The affinities of the WW06 derivatives for human NPR1 and hNPR1-ANP-complex were improved maximally 8-fold and had K_D values between 1 and 5 nM, while no binding to rat NPR1 or rat NPR1-ANP-complex could be observed.

TABLE 11

Affinities of Parental and Matured HuCAL ® Candidates				
Antibody	Affinity (SET measurement) FabCys K_D [nM]			
	hNPR1	hNPR1 + ANP	rNPR1	rNPR1 + ANP
WW01	weak	1.5	weak	2.0
XX01	43	0.3	1.4	0.5
XX08	46	0.3	1.8	0.2
WW03	1000	0.1	2600	1.0
XX03	22	0.2	1.1	0.1
XX04	32	0.1	3.3	0.3
XX06	10	0.1	2.3	0.3
XX07	21	0.2	2.9	0.5
XX12	16	0.2	3.0	0.5
WW06	5.3	12	—	—
XX02	1.0	1.4	—	—
XX05	3.2	5.1	—	—
XX10	2.2	4.0	—	—

Example 9: Cross-Cloning and PTM Removal

Parental clone WW03 had a 'DG' site in HCDR2. The majority of the WW03 derivatives (26 out of 27) were diversified in LCDR3. Only one candidate (XX03) was diversified in HCDR2 including the mutation of 'DG' into 'DK' at amino acid position 54 in the heavy chain variable region (see, e.g., position 54 of SEQ ID NO: 122). The light chains of the functional LCDR3 diversified clones were cross-cloned with the heavy chain of XX03 to engineer these clones without loss of functionality. Furthermore, the 'DG' to 'DK' mutation was inserted in the original heavy chains of several LCDR3 diversified derivatives. An overview of exemplary cross-cloned and D54K engineered candidates is shown in Table 12.

TABLE 12

Overview of VL-VH Cross-cloned and D54K Engineered Clones (WW03 derivatives)		
Antibody	Light chain origin	Heavy chain origin
XX13	XX06	WW03 D54K
XX14	XX09	
XX15	XX04	XX03

TABLE 12-continued

Overview of VL-VH Cross-cloned and D54K Engineered Clones (WW03 derivatives)		
Antibody	Light chain origin	Heavy chain origin
XX16	XX06	
XX17	XX07	
XX18	XX09	
XX19	XX11	
XX20	XX12	

Exemplary functional data of a cross-clone (XX16) compared to the original clone (XX06) and ANP are shown in FIG. 13. All cross-clones had similar or even better functional activity compared to their original clones. XX16 even had a maximal cGMP concentration comparable to the natural ligand ANP in an in vitro study analyzing their activity in hNPR1 transformed CHO cells (see FIG. 14).

Cross-cloned and PTM removed clones were tested for their specificity via 3P assay. Both D54K engineered clones XX13 and XX14 showed non-specific binding to several antigens and were deselected, while no cross-clone showed non-specific binding. Results are shown in Table 13.

TABLE 13

In vitro functional data for XX15 and XX16									
cGMP Generation for human, rat, cyno NPR1 and human NPR2 expressing CHO cells									
Antibody	hNPR1	hNPR1	rNPR1	rNPR1	cNPR1	cNPR1	hNPR2	hNPR2	
	EC50 (nM)	Y max (% of ANP)	EC50 (nM)	Y max (% of ANP)	EC50 (nM)	Y max (% of ANP)	EC50 (nM)	Y max (% of ANP)	
	+/-STDEV	+/-STDEV	+/-STDEV	+/-STDEV	+/-STDEV	+/-STDEV	+/-STDEV	+/-STDEV	
XX16	3.2 +/- 0.4	97 +/- 1	8 +/- 2	94 +/- 8	12 +/- 2	86 +/- 2	>500	<1	
XX15	9.4 +/- 0.4	98 +/- 7	23 +/- 6	100 +/- 10	30 +/- 2	95 +/- 3	>500	<1	
XX18	40 +/- 10	110 +/- 10	26 +/- 9	99 +/- 8	50 +/- 10	90 +/- 5	>500	<1	

Example 10: Crystal Structure of Anti-NPR1 Antibodies

Crystal structures for several molecules in complex with hNPR1 were created as described below.

For Fab03-WW03, the Fab construct of WW03 was complexed to the extracellular domain of hNPR1 (C264T) with a molar ratio of 2 Fab molecules for every 1 NPR1 molecule. The complex was incubated for 1 hour in the cold room rocking and then loaded onto a Superdex 200 16/60 column in the buffer 20 mM HEPES pH7.4, 100 mM NaCl. The complexed protein was separated from a small aggregate peak and the excess Fab and then concentrated to 19.8 mg/mL. The complex crystallized in space group P212121 and diffracted to a resolution of 2.89 Å. The model was built using molecular replacement with the hNPR1 structure and a Fab molecule, iteratively built in Coot and refined with Buster to an Rfree of 21.7%.

For Fab06-WW06, the Fab construct of WW06 was complexed to the extracellular domain of hNPR1 (C264T) with a molar ratio of 2 Fab molecules for every 1 NPR1 molecule. The complex was incubated for 1 hour in the cold room rocking and then loaded onto a Superdex 200 16/60 column in the buffer 20 mM HEPES pH7.4, 100 mM NaCl. The complexed protein was separated from a small aggregate peak and the excess Fab and then concentrated to approximately 20.0 mg/mL. The complex crystallized in

space group P212121 and diffracted to a resolution of 2.17 Å. The model was built using molecular replacement with the hNPR1 structure and a Fab molecule, iteratively built in Coot and refined with Buster to an Rfree of 20.9%.

For Fab16-XX16, the Fab construct of XX16 was complexed to the extracellular domain of hNPR1 (C264T) with a molar ratio of 2 Fab molecules for every 1 NPR1 molecule. The complex was incubated for 1 hour in the cold room rocking and then loaded onto a Superdex 200 16/60 column in the buffer 20 mM HEPES pH7.4, 100 mM NaCl. The complexed protein was separated from a small aggregate peak and the excess Fab and then concentrated to approximately 20.0 mg/mL. The complex crystallized in space group P212121 and diffracted to a resolution of 3.02 Å. The model was built using molecular replacement with the hNPR1 structure and a Fab molecule, iteratively built in Coot and refined with Buster to an Rfree of 24.4%.

The crystal structure of Fab06 in complex with hNPR1 is shown in FIG. 15. FIG. 16 shows the conformation of the hNPR1 ECD as it would be in complex with Fab06; the Fab06 were removed from the image to more clearly reveal the conformation of hNPR1 induced by Fab binding. The structures shown in FIGS. 15 and 16 explain the discrepancy between Fab and IgG functional data shown in FIG. 6. The

Fab C-termini are 180 degrees apart precluding a single IgG spanning a receptor dimer. The structure of hNPR1-ECD in complex with ANP (which was also determined as a part of this work) is shown for comparison in FIG. 17 (left), and the crystal structure of Fab16 (the XX16 antibody in Fab format) in complex with hNPR1 extracellular domain (right). The WW06 Fab was significantly more potent (cGMP production) when tested on CHO cells expressing hNPR1 W74R/C232T (constitutively active mutant) compared to WT hNPR1 (FIG. 18, panels A and B) suggesting that the mutation destabilizes the head-to-head conformation of NPR1 and facilitates antibody binding/receptor activation.

Example 11: Ylanthia® Phage Library Panning and Screening

Six panning strategies were performed, which reflected the most successful strategies from the initial pannings (HuCAL®) or modifications of these strategies aiming for specific methods of action. The panning strategies are summarized in Table 14 in detail. Strategies #2 and #5 were identical to strategies performed in the initial HuCAL® campaign and were selected because all five initial functional candidates were derived from these panning strategies. Strategies #3 and #4 were variations from initial strategies with focus on hNPR1 alternating with hNPR1

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expressing cells. In addition, a constitutively active mutant of NPR1 (W74R) was used as an antigen in strategies #1 and #6. Bacterial lysates (BEL) of the outputs of the 3rd panning rounds in phage display vector pYPDis were directly used for primary and secondary screenings.

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compared to IgG format, the functional screening was performed in FabCys format rather than IgG format.

The sub-cloning of 138 HCDR3 unique clones into the FabCys format was performed via YClone®. 111 out of 138 clones were successfully converted into FabCys format and

TABLE 14

Overview of panning strategies-Ylanthia ®				
Strategy	1 st round	2 nd round	3 rd round	Comments
1	W74R hNPR1 capture	W74R hNPR1 capture	W74R hNPR1 capture	Fc capture panning with active W74R mutant only
2	hNPR1-ANP-complex solution	rNPR1-ANP-complex capture	hNPR1-ANP-complex solution	Solution/Fc capture panning aiming for enrichment of NPR1-ANP-complex
3	CHO-K1 NPR-ANP-complex cell	hNPR1-ANP-complex solution	CHO-K1 NPR-ANP-complex cell	Cell/solution panning aiming for enrichment of NPR1-ANP-complex stabilizers
4	hNPR1 solution	CHO-K1 NPR1 cell + ANP elution	hNPR1 solution + ANP elution	Solution/cell aiming for enrichment of ANP competitors
5	pre-adsorption on NPR1-ANP complex, save the unbound phage still in solution, bind to hNPR1 in solution without ANP and capture NPR1/phage complexes	pre-adsorption on NPR1-ANP complex, save the unbound phage still in solution, bind to hNPR1 in solution without ANP and capture NPR1/phage complexes	pre-adsorption on NPR1-ANP complex, save the unbound phage still in solution, bind to hNPR1 in solution without ANP and capture NPR1/phage complexes	Solution panning aiming for enrichment of ANP competitors
6	CHO-K1 NPR1 cell	W74R hNPR1 capture	CHO-K1 NPR1 cell	Cell/Fc capture panning with human antigen and active W74R mutant

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The outputs of the 3rd panning rounds were analyzed for binding to relevant antigens and cell lines. 368 clones per subcode (in total 4416 clones) were screened in ELISA-based primary screening on human NPR1 in absence and presence of ANP, constitutively active hNPR1 mutant (W74R) and counter-target human hNPR3. The primary screening yielded 810 hits, which were analyzed with respect to binding of relevant cell lines (human NPR1 expressing CHO-K1 cells in absence and presence of ANP, parental CHO-K1 cells) and rat NPR1 in secondary screening. In total, 380 clones from primary and secondary screening were selected for sequencing with priority for exclusive binding to NPR1-ANP-complex, good cell binding, and rat cross-reactivity. The VL and VH sequencing resulted in 138 HCDR3 unique clones with different binding properties (Table 15). Of these clones six bound only in presence of ANP.

TABLE 15

Binding properties of HCDR3 unique hits-Ylanthia ®						
Number of unique candidates			Binding to:			
			hNPR1	rNPR1	hNPR1 expressing cells	
140 (of which 6 clones bound only in presence of ANP)	53 human/rat cross-reactive	2 human/rat cross-reactive cell binders	Yes	Yes	Yes	
		51 human/rat cross-reactive (not binding to cells)	Yes	Yes	No	
	79 human specific	14 human specific cell binders	Yes	No	Yes	
		65 human specific (not binding to cells)	Yes	No	No	

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Since candidate WW06 derived from the initial HuCAL® pannings was significantly more active in FabCys format

95 clones were selected for further analysis. 92 of the 95 FabCys passed the production quality control and were analyzed in detail. Afterwards, 30 of the 92 clones with the most promising properties were selected for IgG conversion via AmplyFly®, exploratory scale expression and S-DAS. 24 of the 30 IgGs passed the production quality control and were analyzed in detail.

All 92 FabCys and 24 IgGs were tested for binding to relevant antigens via ELISA and relevant cell lines by flow cytometry. Furthermore, the clones were tested for ANP competition and functional activity in the cellular cGMP production assay. Eight functional candidates were identified and analyzed for specificity in the Protein Panel Profiling assay (3P assay) in IgG format. For comparison, one of the functional candidates from the initial campaign (WW03) was analyzed. YY02 and YY03 showed low non-specific binding; and YY01, YY04, YY05, YY06, YY07 and

WW03 did not show non-specific binding in this assay. All 92 FabCys and 24 IgGs were tested via ELISA for binding

on the following antigens: human NPR1, constitutively active human NPR1 mutant (W74R), rat NPR1, human NPR3 (counter target) in the absence or presence of ANP (100 nM) and irrelevant antigens. The clones were also tested by flow cytometry for binding on human NPR1 expressing CHO K1 cells in the absence and presence of ANP and on parental CHO K1 cells. The binding properties of the seven functional candidates in IgG format are shown in FIGS. 19 and 20. All 92 FabCys and 24 IgGs were tested for ANP competition. 22 out of 92 FabCys and 12 out of 24 IgGs showed a significant ANP competition >70% at a concentration of 1 μ M FabCys/IgG. The results of the seven functional candidates in IgG format are shown in FIG. 21. YY02, YY05, YY06, and YY07 showed clear ANP competition. YY03 and YY04 showed a “negative” ANP competition in this assay indicating the stabilization of the NPR1-ANP-complex.

All 92 FabCys and 24 IgGs were tested for their functional activity in the cellular cGMP production assay using human NPR1 expressing CHO-K1 cells in presence and absence of ANP. 8 of the 92 FabCys and the same 8 out of 24 IgGs were functionally active and could be assigned to the different methods of action. Five out of eight clones showed much higher functional activity in presence of ANP, including YY01, YY02, YY03, and YY04. Three other clones behaved ANP independent in the functional assay, namely YY05, YY06, and YY07. All eight clones were derived from the panning strategies #2, #3 and #5, whereby #2 and #5 were exact repetitions of the initial HuCAL® panning strategies #10 and #11, which led to the identification of the five functional clones from the initial HuCAL® campaign. The results of the cGMP assay for seven functional candidates in IgG format are shown in FIG. 22. As seen before for candidate WW06 derived from the initial HuCAL® campaign, the functional activities of candidates YY05 and YY07 significantly increased in monovalent FabCys format compared to the bivalent IgG format (FIG. 23).

The seven functional candidates YY01-YY07 were analyzed with regard to their monovalent affinities for human and rat NPR1 in absence and presence of ANP in monovalent FabCys format. The results of the affinity determination and the epitope binning are summarized in Table 16.

TABLE 16

Characterization in monovalent FabCys format-Ylanthia®							
Characterization in monovalent FabCys format							
Antibody	Affinity FabCys K_D [nM]				Epitope	Binning FabCys Without ANP	cGMP assay cGMP conc [nM] at 1 μ M FabCys +0.075 nM ANP
	hNPR1	hNPR1 + ANP	rNPR1	rNPR1 + ANP	Bin on hNPR1 (-ANP- complex)		
YY01	—	16	—	14	B	15	46
YY02	45	1.1	280	4.4	B	—	—
YY03	—	0.1	0.1	0.1	B	—	—
YY04	4.9	1.1	—	1	B	18	67
YY05	0.9	350	700	16000	A	207	227
YY06	0.5	1	1600	1400	C	20	75
YY07	1.8	590	—	—	A	77	88

The affinities were in the low nM to low μ M range and strongly depended on the presence or absence of ANP. Four of the seven functional candidates showed significantly improved binding in presence of ANP (YY01, YY02, YY03, and YY04). YY05 and YY07 competed with ANP for binding to NPR1 and showed much higher affinities in absence of ANP. The affinities of YY06 were independent from ANP. Some candidates exhibited non-specific binding to the reference flow cell, while others had such high affinities that their K_D values approach the assay limit. YY01, YY02, YY03, and YY04 share one epitope bin, which is the same as for WW03 from the initial HuCAL® campaign. YY05 and YY07 share another epitope bin, which is the same as for WW06 from the initial HuCAL® campaign. YY06 binds to a single epitope bin.

Example 12: Reformatting into IgG-Ylanthia®

Subcloning of the Ylanthia® candidates from the FabCys vector into the IgG1_LALA vector for expression in mammalian cells was performed via amplification of the Fab-encoding insert using two biotinylated primers. The amplified product was bound on streptavidin beads, digested using restriction enzymes, and washed, resulting in the release of the purified insert into the supernatant. The insert was cloned into the acceptor vector, the DNA was transformed and single clones were quality controlled via colony PCR and sequencing.

Five Ylanthia® candidates were selected for affinity maturation. YY01 and YY04 stabilize the NPR1-ANP-complex, YY06 behaves in an ANP-independent manner, and YY05 and YY07 are ANP-competitive. Binding data (ELISA, flow cytometry, ANP competition), functional data (cGMP assay), affinities, and epitope bins are shown in Table 17.

TABLE 17

Summary of Affinity, Epitope, and cGMP data for YY01, YY04, YY08, YY06, and YY07 in IgG Format													
Characterization in bivalent IgG format													
Antibody	Elisa +/- 100 nM ANP IgG binding at 1 μ M IgG								Flow Cytometry [S/BG] at 1 μ M IgG				
	hNPR1 +		W74R +		rNPR1 +		hNPR3 +		hNPR1 CHO		ANP	cGMP assay cGMP	
	100 nM	W74R	100 nM	W74R	100 nM	W74R	100 nM	W74R	KI	competition assay	ANP	conc [nM] at 1 μ M IgG	+0.075 nM
	hNPR1	hNPR1	hNPR1	hNPR1	hNPR1	hNPR1	hNPR1	hNPR1	hNPR1	hNPR1	hNPR1	Without ANP	Without ANP
YY01	54	76	33	53	26	50	18	6	35	21	yes	5	14
YY04	31	60	11	13	8	54	69	18	2	33	no	10	15
YY05	81	32	69	23	11	16	7	4	15	5	yes	13	15
YY06	91	102	83	86	3	4	6	6	26	18	yes	12	15
YY07	68	24	53	20	3	3	6	5	00	44	yes	4	6

Example 13: Generation of Ylanthia® Maturation Libraries

To increase affinity and biological activity and to reduce non-specificity of selected antibody candidates, LCDR3 and HCDR1/HCDR2 regions were optimized in parallel using diversified Ylanthia® maturation modules (YMM) previously generated with Slonomics® technology (van den Brulle et al. (2008): A novel solid phase technology for high-throughput gene synthesis; *Biotechniques* 45 (3), pp. 340-343, the contents of which are herein incorporated by reference for this purpose).

Cloning of the maturation libraries was performed in vectors encoding the parental Fab fragments. The generation of the maturation libraries was performed for five parental antibodies (YY01, YY04, YY05, YY06, and YY07). For the library generation, all maturation candidates were treated individually. The maturation libraries were successfully cloned and had library sizes between 6.2×10^8 and 4.5×10^9 cfu.

In order to monitor the cloning efficiency, the parental HCDR1/2 and LCDR3 were replaced by MBP-stuffers prior to insertion of the diversified YMM. Digested vector fragments were ligated with a 2-fold molar excess of the insert fragments carrying the diversified HCDR1/2 or LCDR3s.

Ligation mixtures were electroporated in *E. coli* cells yielding in $>10^8$ independent colonies. Amplification of the library was performed as described in the literature (Tiller et al. (2013): A fully synthetic human Fab antibody library based on fixed VH/VL framework pairings with favorable biophysical properties; mAbs 5 (3), pp. 445-470, the contents of which are herein incorporated by reference for this purpose). For quality control, approx. 10-20 single clones per library were randomly picked and sequenced.

Example 14: Pannings and Screenings—Ylanthia®

The nine maturation libraries were used for four different maturation panning strategies, which aimed for the enrichment of progenies with improved affinities compared to the parental clones. Furthermore, rat material was included where appropriate and the pannings were performed with high stringencies concerning antigen concentration and washing conditions. During the panning process, the libraries of YY01 and YY04 (only LCDR3) as well as YY05 and YY07 were pooled, while the libraries of YY06 were kept separately. The panning strategies are summarized in Table 18 in detail. The bacterial lysates (BEL) of the outputs after the third panning rounds were directly used for an ELISA-based pre-screening and for SET screening.

TABLE 18

Overview of Ylanthia® Maturation Panning Strategies					
Strategy	Parental antibodies	1 st round	2 nd round	3 rd round	comments
1	YY01/YY04	hNPR1-ANP-complex solution	rNPR1-ANP-complex solution	hNPR1-ANP-complex solution	Solution panning aiming for enrichment of NPR1-ANP-complex stabilizers
2	YY06	hNPR1-ANP-complex solution	hNPR1-ANP-complex solution	hNPR1-ANP-complex solution	Solution panning aiming for enrichment of NPR1-ANP-complex stabilizers
3	YY06	Preadsorption on NPR1-ANP-complex hNPR1 solution	Preadsorption on NPR1-ANP-complex hNPR1 solution	Preadsorption on NPR1-ANP-complex hNPR1 solution	Solution panning aiming for enrichment of ANP competitors
4	YY05/YY07	Preadsorption on NPR1-ANP-complex hNPR1 solution	Preadsorption on NPR1-ANP-complex hNPR1 solution	Preadsorption on NPR1-ANP-complex rNPR1 solution	Solution panning aiming for enrichment of ANP competitors

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The outputs of the third panning rounds were used for an ELISA-based pre-screening to ensure that only clones binding to NPR1 and/or NPR1-ANP-complex were selected for further Solution Equilibrium Titration (SET) screening. 880 clones in total were analyzed in SET screening for improved affinity for hNPR1 and/or hNPR1-ANP-complex compared to the parental clones.

During SET screening, 263 HCDR1//2 or LCDR3 unique improved derivatives were identified, which resulted in 112 unique clones after sequencing and conversion to IgG₁_LALA format. Compared to their parental clones, the affinities of the YY05 and YY07 derivatives were not improved for NPR1 but were improved up to 200-fold for NPR1-ANP-complex. The derivatives of YY01 had similar affinities to the parental clone YY04, whose derivatives had only slightly improved affinities. The affinities of the YY06 derivatives were improved 4- to 40-fold for NPR1 and 7- to 70-fold for NPR1-ANP-complex. See FIG. 24 (which shows affinities for hNPR1) and FIG. 25 (which shows affinities for hNPR1-ANP complex). Affinities (K_D [pM]) are indicated at the x-axis below the parental clone name for both figures.

Example 15: Characterization of Matured
Candidates—Ylanthia®

95 of 112 improved candidates were selected for advanced production. 77 of the 95 clones passed the production quality control and were characterized in regard to binding to relevant antigens, binding to relevant cell lines, and functional activity in the cGMP production assay in comparison to their parental clones. After detailed IgG characterization, 17 candidates (detailed in Table 19) were

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selected, produced in exploratory scale IgG production, and further analyzed via 3P assay. Furthermore, they were converted to FabCys format for their affinity determination on human and rat NPR1 in absence and presence of ANP via SET K_D measurement.

TABLE 19

Overview of Matured Ylanthia® Candidates		
Matured antibody	Parental antibody	Matured CDR
ZZ01	YY07	LCDR3
ZZ02	YY05	LCDR3
ZZ03	YY07	LCDR3
ZZ04	YY07	LCDR3
ZZ05	YY07	LCDR3
ZZ06	YY07	LCDR3
ZZ07	YY07	LCDR3
ZZ08	YY07	LCDR3
ZZ09	YY05	HCDR1 + 2
ZZ10	YY05	HCDR1 + 2
ZZ11	YY04	LCDR3
ZZ12	YY01	HCDR1 + 2
ZZ13	YY01	HCDR1 + 2
ZZ14	YY06	HCDR1 + 2
ZZ15	YY06	HCDR1 + 2
ZZ16	YY06	HCDR1 + 2
ZZ17	YY06	HCDR1 + 2

Affinities for 16 of the matured Ylanthia® candidates (in monovalent FabCys format) were determined via SET measurement and/or via Octet. The results are summarized in Tables 20 and 21 below in comparison to the affinities of the parental clones.

TABLE 20

Summary of Affinity data (SET) for YY01, YY06, ZZ12, ZZ13, ZZ15, ZZ16, and ZZ17 in FabCys Format					
Affinity (SET measurement) FabCys K_D [nM]					
Antibody	hNPR1	hNPR1 + ANP	rNPR1	rNPR1 + ANP	comment
YY01	—	16.00	—	14.00	K_D determination via Octet
ZZ12	6.70	0.32	2.10	1.40	
ZZ13	9.90	1.30	2.30	1.40	
YY04	4.90	1.10	—	1.00	K_D determination via Octet
ZZ11	46.00	2.40	21.00	18.00	
YY05	0.90	350.00	700.00	16000.00	K_D determination via Octet; assay limit reached
ZZ02	0.37	0.58	—	—	
ZZ09	0.49	0.57	—	—	
ZZ10	0.52	1.40	—	—	
YY06	0.50	1.00	1600.00	1400.00	K_D determination via Octet
ZZ15	0.053	0.074	—	—	
ZZ16	0.040	0.034	62.00	51.00	
ZZ17	0.046	0.050	52.00	36.00	
YY07	1.80	590.0	—	—	
ZZ01	0.65	0.85	—	—	
ZZ03	0.75	1.30	—	—	
ZZ04	0.30	0.38	—	—	
ZZ05	0.74	1.20	—	—	
ZZ06	0.35	0.50	—	—	
ZZ07	0.62	1.50	—	—	
ZZ08	0.57	1.30	—	—	

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TABLE 21

Summary of Affinity data (Octet) for YY01, YY06, ZZ12, ZZ13, ZZ15, ZZ16, and ZZ17 in FabCys Format					
Antibody	Affinity (Octet measurement) FabCys K_D [nM]				comment
	hNPR1	hNPR1 + ANP	rNPR1	rNPR1 + ANP	
YY01	—	26.00	—	8.80	Heterogenous binding
ZZ12	—	0.04	—	0.10	
ZZ13	—	0.14	66.00	0.10	
YY06	0.62	0.80	100.00	38.00	
ZZ15	0.26	0.19	—	—	
ZZ16	0.14	0.11	—	—	
ZZ17	0.16	0.09	—	—	

Example 16: Epitope Determination for ANP
Competitive and ANP Non-Competitive Agonist
Antibodies

The crystal structure of the hNPR1 and XX16 Fab complex was used to identify the XX16 Fab epitope on hNPR1. The interaction surface on hNPR1 by XX16 Fab was formed by several continuous and discontinuous (i.e., noncontiguous) sequences as detailed in Table 22. These residues form the three-dimensional conformational epitope recognized by XX16 Fab.

Results of the epitope mapping of XX16 Fab (ANP non-competitive) are shown in Table 22. hNPR1 residues are numbered based upon SEQ ID NO: 1 (P16066). Fab residues are numbered based upon their linear amino acid sequence. hNPR1 residues shown have at least one atom within 5 Å of an atom in the XX16 Fab, to account for potential water mediated interactions.

TABLE 22

Epitope Mapping of XX16 Fab					
Antibody XX16 Fab			Antigen (NPR1)		
Amino acid	Number	Chain	Amino acid	Number	Chain
SER	31	H	TRP	106	A
TYR	32	H	TRP	106	A
			TRP	106	B
TRP	33	H	ASN	109	A
GLU	52	H	HIS	108	A
			ASN	109	A
SER	53	H	GLU	107	A
			ASN	109	A
LYS	54	H	GLU	107	A
ASN	56	H	GLY	33	A
			ASN	34	A
			THR	36	A
			TRP	76	A
TYR	57	H	ASN	34	A
			TRP	76	A
			ASN	109	A
			PRO	110	A
			ALA	111	A
			VAL	374	A
			THR	375	A
PHE	59	H	THR	375	A
ARG	100	H	TRP	106	A
			ASP	103	B
			GLU	107	B
TYR	101	H	VAL	102	A
			TRP	106	A
			VAL	102	B
			ASP	103	B

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TABLE 22-continued

Epitope Mapping of XX16 Fab					
Antibody XX16 Fab			Antigen (NPR1)		
Amino acid	Number	Chain	Amino acid	Number	Chain
			TRP	106	B
			GLU	107	B
SER	102	H	LYS	105	A
			TRP	106	A
MET	103	H	VAL	102	A
			LYS	105	A
			TRP	106	A
			TRP	132	A
			LEU	99	B
			VAL	102	B
			ASP	103	B
ILE	104	H	LYS	105	A
			HIS	131	A
			TRP	132	A
			LEU	99	B
			ALA	100	B
			ASP	103	B
TYR	105	H	LYS	105	A
			TRP	132	A
SER	106	H	LYS	105	A
			HIS	131	A
			TRP	132	A
			ARG	133	A
TYR	107	H	LYS	105	A
			ASN	109	A
			PRO	110	A
			ALA	111	A
			VAL	134	A
			THR	375	A
ASP	113	H	GLU	107	B
ARG	18	L	GLN	336	B
SER	30	L	GLU	86	B
			ASN	87	B
			ALA	88	B
SER	31	L	SER	84	B
			GLU	86	B
TYR	32	L	HIS	131	A
TYR	49	L	ASP	103	B
			LEU	104	B
			GLU	107	B
			HIS	108	B
SER	52	L	LEU	43	B
			LEU	82	B
THR	53	L	VAL	81	B
			LEU	82	B
			GLY	83	B
			LEU	104	B
			HIS	108	B
LEU	54	L	ARG	79	B
			VAL	81	B
			HIS	108	B
GLN	55	L	GLU	107	B
			HIS	108	B
SER	56	L	GLU	107	B
			HIS	108	B
VAL	58	L	ARG	79	B
PRO	59	L	ARG	79	B
SER	60	L	ARG	79	B
SER	65	L	THR	44	B

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TABLE 22-continued

Epitope Mapping of XX16 Fab					
Antibody XX16 Fab			Antigen (NPR1)		
Amino acid	Number	Chain	Amino acid	Number	Chain
GLY	66	L	THR	44	B
SER	67	L	THR	44	B
			GLU	86	B
			ASN	87	B
SER	76	L	GLN	336	B
TRP	92	L	HIS	131	A
			ARG	133	A
			ALA	88	B
ARG	93	L	ARG	133	A
			GLU	378	A
LYS	94	L	ASP	376	A

Critical epitope residues for the binding of XX16 Fab and NPR1, which were determined using structural analysis and affinity maturation data, include (first tier) 99-103, 105-111, 131-133, 378; and (second tier): 33-34, 76, 82, 104, 374-375. Amino acids 99-103 of NPR1 (SEQ ID NO: 1) comprise both a E106 backbone that enabled the affinity maturation by D54K in HCDR2, and W106 from both NPR1 protomers that clamp Y101 of HCDR3. Certain critical epitope residues are shown in Table 23 below. Regions of NPR1 encompassing these critical residues include N34, W76, L82, V102-A111, H131-V134, and V374-E378.

TABLE 23

Critical Epitope Residues of XX16 Fab (ANP non-competitive)		
Critical epitope residue (amino acid)	Critical epitope residue (number)	Explanation
N	34	H-bond with N56 (backbone) of VH
L	82	H-bond (chain B) with T53 (side chain) of VL
V	102	Packing against M103 (side chain) of VH
D	103	Salt bridge (chain B) with R100 (side chain) of VH
K	105	H-bond (backbone) with M103 (backbone) and Y105 (backbone) of VH
W	106	Packing between Y32 (side chain) and Y101 (side chain) of VH
E	107	Salt bridge (chain B) with R100 (side chain) of VH
E	107	H-bond (backbone) with K54 (side chain) of VH
N	109	H-bond with E52 (side chain), S53 (side chain) and W33 (side chain) of VH
W	132	Packing against M103 (side chain) of VH
R	133	H-bond with S106 (side chain) of VH
T	375	Packing against Y57 (side chain) of VH
E	378	Salt bridge with R93 (side chain) of VL

The crystal structure of the hNPR1 and WW03 Fab complex was used to identify the WW03 Fab epitope on hNPR1. The interaction surface on hNPR1 by WW03 Fab was formed by several continuous and discontinuous (i.e., noncontiguous) sequences as detailed in Table 24. These residues form the three-dimensional conformational epitope recognized by WW03 Fab.

Results of the epitope mapping of WW03 Fab (ANP non-competitive) are shown in Table 24. hNPR1 residues are numbered based upon SEQ ID NO: 1 (P16066). Fab residues are numbered based upon their linear amino acid sequence. hNPR1 residues shown have at least one atom within 5 Å of an atom in the WW03 Fab, to account for potential water mediated interactions.

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TABLE 24

Epitope Mapping of WW03 Fab					
Antibody WW03 Fab			Antigen (NPR1)		
Amino acid	Number	Chain	Amino acid	Number	Chain
SER	31	H	TRP	106	A
TYR	32	H	TRP	106	A
			TRP	106	B
TRP	33	H	ASN	109	A
SER	52	H	HIS	108	A
			ASN	109	A
SER	53	H	TRP	106	A
			GLU	107	A
			ASN	109	A
ASP	54	H	THR	36	A
			ARG	79	A
SER	56	H	ASN	34	A
TYR	57	H	ASN	109	A
			PRO	110	A
			ALA	111	A
			VAL	374	A
			THR	375	A
TYR	59	H	THR	375	A
ARG	100	H	ASP	103	B
			TRP	106	B
			GLU	107	B
TYR	101	H	VAL	102	A
			TRP	106	A
			VAL	102	B
			ASP	103	B

TABLE 24-continued

Epitope Mapping of WW03 Fab					
Antibody WW03 Fab			Antigen (NPR1)		
Amino acid	Number	Chain	Amino acid	Number	Chain
			TRP	106	B
			GLU	107	B
SER	102	H	LYS	105	A
			TRP	106	A
			ASN	109	A
MET	103	H	VAL	102	A
			LYS	105	A
			TRP	106	A
			TRP	132	A
			LEU	99	B
			VAL	102	B

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TABLE 24-continued

Epitope Mapping of WW03 Fab					
Antibody WW03 Fab			Antigen (NPR1)		
Amino acid	Number	Chain	Amino acid	Number	Chain
ILE	104	H	ASP	103	B
			HIS	131	A
			TRP	132	A
TYR	105	H	LEU	99	B
			ASP	103	B
			LYS	105	A
SER	106	H	TRP	132	A
			LYS	105	A
			TRP	132	A
TYR	107	H	ARG	133	A
			LYS	105	A
			ASN	109	A
ASP	113	H	PRO	110	A
			ALA	111	A
			VAL	134	A
ARG	18	L	THR	375	A
			GLU	107	B
			GLN	336	B
SER	30	L	ASN	87	B
			ALA	88	B
			SER	84	B

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TABLE 24-continued

Epitope Mapping of WW03 Fab					
Antibody WW03 Fab			Antigen (NPR1)		
Amino acid	Number	Chain	Amino acid	Number	Chain
TYR	32	L	HIS	131	A
TYR	49	L	ASP	103	B
SER	52	L	GLU	107	B
			HIS	108	B
			LEU	43	B
THR	53	L	LEU	82	B
			VAL	81	B
			LEU	82	B
LEU	54	L	GLY	83	B
			LEU	104	B
			HIS	108	B
GLN	55	L	ARG	79	B
			VAL	81	B
			HIS	108	B
SER	56	L	GLU	107	B
			HIS	108	B
			ARG	79	B
VAL	58	L	ARG	79	B
			ARG	79	B
			ARG	79	B
PRO	59	L	THR	44	B
			THR	44	B
			GLU	86	B
SER	60	L	ASN	87	B
			ARG	133	A
			ARG	133	A
GLY	66	L	ASP	376	A
TYR	92	L			
GLU	93	L			
LYS	94	L			

Critical epitope residues for the binding of WW03 Fab and NPR1, which were determined using structural analysis and affinity maturation data, include the residues shown in Table 25. Regions of NPR1 encompassing these critical residues include R79, L82, L99-A111, H131-V134, and V374-T375.

TABLE 25

Critical Epitope Residues of WW03 Fab (ANP non-competitive)		
Critical epitope residue (amino acid)	Critical epitope residue (number)	Explanation
R	79	Salt bridge with D54 (side chain) of VH; H-bond (chain B) with S60 (side chain) of VL
L	82	H-bond (backbone) with T53 (side chain) of VL
L	99	Packing against M103 (side chain) and I104 (side chain) of VH
V	102	Packing against M103 (side chain) of VH
D	103	Salt bridge with R100 (side chain) of VH
K	105	H-bond with M103 (backbone) and Y105 (backbone) of VH
W	106	Packing between Y32 (side chain) and Y101 (side chain) of VH; stacking (chain B) against Y101 (side chain) of VH
N	109	H-bond with S53 (side chain) of VH
H	131	H-bond with I104 (backbone) of VH
W	132	H-bond with M103 (side chain) of VH
T	375	H-bond with Y59 (side chain) of VH; pack against Y57 (side chain) of VH

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The crystal structure of the hNPR1 and WW06 Fab complex was used to identify the WW06 Fab epitope on hNPR1. The interaction surface on hNPR1 by WW06 Fab was formed by several continuous and discontinuous (i.e., noncontiguous) sequences as detailed in Table 26. These residues form the three-dimensional conformational epitope recognized by WW06 Fab.

Results of the epitope mapping of the WW06 Fab (ANP competitive) are shown in Table 26. hNPR1 residues are numbered based upon SEQ ID NO: 1 (P16066). WW06 Fab residues are numbered based upon their linear amino acid sequence. hNPR1 residues shown have at least one atom within 5 Å of an atom in the WW06 Fab, to account for potential water mediated interactions.

TABLE 26

Epitope Mapping of WW06 Fab (ANP competitive)					
Antibody WW06 Fab			Antigen (NPR1)		
Amino acid	Number	Chain	Amino acid	Number	Chain
Y	27	H	R	294	B
		H	G	295	B
S	28	H	R	294	B
		H	G	295	B
		H	D	296	B
S	31	H	R	230	A
		H	D	296	B
Y	32	H	G	295	B
		H	D	296	B
		H	G	297	B
W	33	H	Y	188	A
		H	E	219	A
R	50	H	F	197	A
		H	E	219	A
D	52	H	Y	188	A
		H	A	221	A
		H	H	227	A
D	54	H	D	224	A
N	55	H	Y	188	A
		H	P	190	A
		H	A	221	A
		H	D	223	A
Y	57	H	Y	188	A
		H	R	189	A
		H	P	190	A
		H	G	191	A
		H	D	192	A
		H	E	193	A
		H	E	194	A
		H	F	197	A
T	58	H	E	194	A
R	59	H	E	194	A
		H	F	197	A
		H	E	201	A
Q	65	H	D	94	A
		H	E	194	A
R	98	H	G	295	B
W	99	H	E	219	A
L	100	H	R	230	A
S	101	H	H	217	A
P	102	H	K	238	A
G	103	H	D	216	A
		H	H	217	A
		H	T	234	A
		H	K	238	A
Y	104	H	L	218	A
		H	E	219	A
		H	F	220	A
		H	H	227	A
		H	R	230	A
		H	L	231	A
		H	T	234	A
		H	K	238	A

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TABLE 26-continued

Epitope Mapping of WW06 Fab (ANP competitive)					
Antibody WW06 Fab			Antigen (NPR1)		
Amino acid	Number	Chain	Amino acid	Number	Chain
A	105	H	K	238	A
L	106	H	R	233	A
		H	T	234	A
		H	R	237	A
		H	K	238	A
G	107	H	R	237	A
Q	109	H	G	297	B
Q		H	V	300	B
I	29	L	R	208	A
G	30	L	R	208	A
A	31	L	F	204	A
		L	R	208	A
G	32	L	R	208	A
Y	33	L	Y	186	A
		L	E	201	A
		L	F	204	A
		L	H	217	A
Y	93	L	F	197	A
		L	E	201	A
L	95	L	L	144	A
		L	E	201	A
		L	F	204	A
		L	M	205	A
Q	96	L	L	144	A
		L	G	145	A
		L	V	148	A
		L	M	205	A
S	98	L	F	197	A
		L	F	198	A
		L	E	201	A
R	100	L	E	219	A

Critical epitope residues for the binding of WW06 and NPR1, which were determined using structural analysis and affinity maturation data, include the residues shown in Table 27. Regions of NPR1 encompassing these critical residues include Y188-F198, E201-R208, V215-K238, and R294-G297.

TABLE 27

Critical Epitope Residues of WW06 (ANP competitive)		
Critical epitope residue (amino acid)	Critical epitope residue (number)	Explanation
Y	188	Packing against W33 (side chain) of VH
D	192	H-bond with Y57 (side chain) of VH
E	194	Salt bridge with R59 (side chain) of VH
F	197	Stacking against R59 (side chain) of VH
E	201	H-bond with S98 (side chain) of VL
R	208	H-bond with G30 (backbone) of VL
E	219	Salt bridge with R50 (side chain) of VH
G	295	H-bond (backbone, chain B) with S28 (backbone) of VH

Example 17: Mouse In Vivo Characterization of Effect of WW06 on Plasma cGMP

WW06 FabCys was used in an in vivo study in hNPR1 transgenic mice to determine the effect of this antibody on plasma cGMP levels in vivo.

For analysis of plasma cGMP samples, the LC-MS/MS detection method using $^{15}\text{N}_2$, ^{13}C cGMP as an internal standard was adopted from Oeckl and Ferger, Journal of Neuroscience Methods 203 (2012) 338-343; and Zhang et

al., J. Chromatogr B: Analyt Technol Biomed Life Sci 2009; 877:513-20; the contents of each of which are hereby incorporated by reference for this purpose).

Plasma cGMP concentration in hNPR1 Tg mice which were intravenously administered the WW06 Fab increased at the 5 minute time point and the signal returned to baseline by 3 h. As expected and consistent with the data shown in FIGS. 26 and 27, the T1/2 of the FabCys antibody was <30 min. Each value shown is the average of three points collected from three individual animals. Dose response data are shown in FIG. 26, and PK data are shown in FIG. 27.

Example 18: Effect of XX16 on Heart Weight and NT-proBNP in ANP KO and WT Mice

XX16 was used in an in vivo study to determine its effect on heart weight and NT-proBNP levels in ANP knockout (KO) and wild-type (WT) mice.

ANP knockout mice are hypertensive and have cardiac hypertrophy (increased HW/BW ratio). NT-proBNP is a biomarker of cardiac dysfunction. XX16 was administered at 0.3 or 3 mg/kg subcutaneously once every two weeks for four weeks in ANP knockout and wild type mice.

Results are shown in FIG. 28. At two weeks after the second treatment, XX16 dose-dependently reduced heart weight/body weight ratio and NT-proBNP in both wild type and ANP KO mice in comparison to vehicle treated animals.

Example 19: Acute Effect of XX16 on Blood Pressure and Urinary Flow Rate in Hypertensive Rats

XX16 was used to determine its effect on blood pressure and urinary flow rate in hypertensive rats (spontaneous hypertensive rat stroke prone, SHRsp). Animals were administered 0.3 mg/kg XX16, 1 mg/kg XX16, or a vehicle control intravenously (one time). Blood pressure was measured using a femoral artery catheter. Measurements were taken three hours after the intravenous administration and results are shown in FIG. 29.

Intravenous XX16 treatment normalized mean arterial pressure and increased urinary flow rate acutely in hypertensive rats (SHRsp) in comparison to vehicle treated animals.

Example 20: Chronic Effects of XX16 on Blood Pressure and Plasma cGMP in Wistar-Han Rats

Chronic hemodynamic effects of XX16 in telemetry implanted normal rats were evaluated. XX16 (at doses of 0.1, 0.3, 1, 3, 10, and 30 mg/kg) was administered subcutaneously one time. Results are shown in FIG. 30. Subcutaneous administration of XX16 decreased mean arterial pressure (MAP) in all groups except for the 0.1 mg/kg group. There was a floor (around 95 mmHg) for the MAP reduction. A dose-dependent effect for the time to return baseline MAP was observed; this effect was greater than 70 days for the 30 mg/kg treatment group. Plasma cGMP concentration also had a ceiling around 90 nmol/L, after which the plasma cGMP gradually reduced toward baseline. This effect is very similar to the blood pressure response.

Example 21: Comparison of Extant Anti-NPR1 Antibodies with Antibodies of the Application

The WW03 antibody was compared with antibody 5591-IgG of PCT Application No. WO2010/065293A1 in terms of the ability of both antibodies to produce cGMP in human cells expressing hNPR1 or rat cells expressing rNPR1 (both in the presence or absence of 0.075 nM human or rat ANP, respectively). The WW03 antibody displayed superior potentiation in the absence of ANP on both cell lines. Additionally, the WW03 antibody demonstrated superior potentiation on rat cells expressing rNPR1 (both in the presence and absence of ANP).

In summary, previous antibodies (e.g., those of WO2010/065293, including 5591-IgG) demonstrate no activity in vivo (e.g., through analysis of activation using cGMP assays). In contrast, the antibodies of the instant application demonstrated in vivo activity in both mouse and rat. Further, the epitope binding of the antibodies described herein has been demonstrated to be dissimilar to the antibodies of WO2010/065293 using crystal structure data. The activity, cross-reactivity, and crystallographic data described herein demonstrate the differing and superior effects of the antibodies described in this application as compared to previous antibodies.

SEQUENCE LISTING

Sequence total quantity: 473
 SEQ ID NO: 1 moltype = AA length = 1061
 FEATURE Location/Qualifiers
 source 1..1061
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 1

MPGPRRPAGS	RLRLLLLLLL	PPLLLLLRGS	HAGNLTAVV	LPLANTSYPW	SWARVGPAAVE	60
LALAQVKARP	DLPLGWTVRT	VLGSSSENALG	VCSDTAAPLA	AVDLKWEHNP	AVFLGPQCVY	120
AAAPVGRFTA	HWRVPLLTAG	APALGFGVKD	EYALTTRAGP	SYAKLGDFVA	ALHRLGWER	180
QALMLYAYRP	GDEEHCFPLV	EGLFMRVRDR	LNITVDHLEF	AEDDLSHYTR	LLRTMPRKGR	240
VIYCSPDA	FRTLMLLAL	AGLCGEDYVF	FHLDFGQSL	QGGQGPAPRR	PWERDGGQDV	300
SARQAFQAAK	IITYKDPDNP	EYLEPLKQLK	HLAYEQFNFT	MEDGLVNTIP	ASFHDGLLLY	360
IQAVTETLAH	GGTVTDGENI	TQRMWNRSFQ	GVTGYLKIDS	SGDRETDLSL	WDMDPENGAF	420
RVVLNNGTS	QELVAVSGRK	LNWPLGYPPP	DIPKCGFDNE	DPACNQDHL	TLEVLALVGS	480
LSLLGILIVS	FFIYRKMQL	KELASELWRV	RWEDVEPSSL	ERHLRSAGSR	LTLSGRGSNY	540
GSLLTTEGQF	QVFAKTAYYK	GNLVAVKRVN	RKRIELTRKV	LFELKHMRDV	QNEHLTRFVG	600
ACTDPPNICI	LTEYCPRGSL	QDILENESIT	LDWMFVRYSLT	NDIVKGMLFL	HNGAICSHGN	660
LKSSNCVVDG	RFVLKITDYG	LESFRDLDP	QGHTVYAKKL	WTAPELLRMA	SPPVRGSQAG	720
DVYSFGIILQ	EIALRSGVFH	VEGLDLSPKE	IIERVTRGEQ	PPFRPSLALQ	SHLEELGLLM	780
QRCWAEDPQE	RPPFQQIRLT	LRKFNRENSS	NILDNLLSRM	EQYANNLEEL	VEERTQAYLE	840
EKRKAALLY	QILPHSVABQ	LKRGETVQAE	AFDSVTIYFS	DIVGFTALSA	ESTPMQVTL	900
LNDLYTCFDA	VIDNFDVYKV	ETIGDAYMVV	SGLPVRNGRL	HACEVARMAL	ALLDAVRSFR	960

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IRHRPQEQLR	LRIGIHTGPV	CAGVVGLKMP	RYCLFGDTVN	TASRMESNGE	ALKIHLSSSET	1020
KAVLEEFEGGF	ELELRGDVEM	KGKGKVRTYW	LLGERGSSTR	G		1061

SEQ ID NO: 2	moltype = AA	length = 1057
FEATURE	Location/Qualifiers	
source	1..1057	
	mol_type = protein	
	organism = Mus musculus	

SEQUENCE: 2						
MPGSRVRPR	LRALLLPPL	LLLRSGHASD	LTVAVVPLT	NTSYPWSWAR	VGPAVELALG	60
RVKARPDLLP	GWTVRMVLGS	SENAAGVCSD	TAAPLAAVDL	KWEHSPAVFL	GPGCVYSAP	120
VGRFTAHRV	PLLTAGAPAL	GIGVKDEYAL	TTRTGPSHVK	LGDFVTALHR	RLGWEHQALV	180
LYADRLGDDR	PCFFIVEGLY	MRVRERLNT	VNHQEFVEGD	PDHYTKLLRT	VQRKGRVIYI	240
CSSPDAPRNL	MLLALDAGLT	GEDYVFFHLD	VFGQSLQGAQ	GPVPRKPWER	DDGQDRRARQ	300
AFQAAKIITY	KEPDNPEYLE	FLKQLKLLAD	KKFNFTMEDG	LKNII PASFH	DGLLLYVQAV	360
TETLAQGGTV	TDGENITQRM	WNRSFQGVGT	YLKIDRNGDR	DTDFSLWDM	PETGAFRVVL	420
NFNGTSQELM	AVSEHRLYWP	LGYPDPDIPK	CGFDNEDPAC	NQDHFSTLEV	LALVGSLSLV	480
SFLIVSFFIY	RKMQLKELV	SELWRVRWED	LQPSLERHL	RSAGSRLTSL	GRGSNYGSLL	540
TTEGQFQVFA	KTAYYKGNLV	AVKRVNRKRI	ELTRKVL FEL	KHMRDVQNEH	LTRFVGACTD	600
PPNICILTEY	CPRGSLQDIL	ENESITLDWM	FRYSLTNDIV	KGMLFLHNGA	IGSHGNLKSS	660
NCVVDGRFVL	KITDYGLESF	RDPEPEQGHT	LFPAKLLWAP	ELLRMASPPA	RGSQAGDVYS	720
FGIILQEIAL	RSQVFYVEGL	DLSPKEIIER	VTRGEQPPFR	PSMDLQSHLE	ELGQLMQRCW	780
AEDPQERPPF	QQIRLALRKF	NKENSNIIL	NLLSRMEQYA	NNLEELVEER	TQAYLEEKRK	840
AEALLYQILP	HSVAEQLEKRG	ETVQAEAFDS	VTIYFSDIVG	FTALSAESTP	MQVVTLLNDL	900
YTCFDAVIDN	FDVYKVTIG	DAYMVVSGLP	VRNGQLHARE	VARMALALLD	AVRSFRIRHR	960
PQEQLRLRIG	IHTGPVCAGV	VGLKMPRYCL	FGDTVNTASR	MESNGEALRI	HLSSETKAVL	1020
EEFDGFLELE	RGDVEMKGGK	KVRTYWLLGE	RGCSSTRG			1057

SEQ ID NO: 3	moltype = AA	length = 1057
FEATURE	Location/Qualifiers	
source	1..1057	
	mol_type = protein	
	organism = Rattus norvegicus	

SEQUENCE: 3						
MPGSRVRPR	LRALLLPPL	LLLRGGHASD	LTVAVVPLT	NTSYPWSWAR	VGPAVELALA	60
RVKARPDLLP	GWTVRMVLGS	SENAAGVCSD	TAAPLAAVDL	KWEHSPAVFL	GPGCVYSAP	120
VGRFTAHRV	PLLTAGAPAL	GIGVKDEYAL	TTRTGPSHVK	LGDFVTALHR	RLGWEHQALV	180
LYADRLGDDR	PCFFIVEGLY	MRVRERLNT	VNHQEFVEGD	PDHYPKLLRA	VRRKGRVIYI	240
CSSPDAPRNL	MLLALNAGLT	GEDYVFFHLD	VFGQSLKSAQ	GLVPQKPWER	GDGQDRSARQ	300
AFQAAKIITY	KEPDNPEYLE	FLKQLKLLAD	KKFNFTVEDG	LKNII PASFH	DGLLLYVQAV	360
TETLAQGGTV	TDGENITQRM	WNRSFQGVGT	YLKIDRNGDR	DTDFSLWDM	PETGAFRVVL	420
NYNGTSQELM	AVSEHKLYWP	LGYPDPDVPK	CGFDNEDPAC	NQDHFSTLEV	LALVGSLSLI	480
SFLIVSFFIY	RKMQLKELV	SELWRVRWED	LQPSLERHL	RSAGSRLTSL	GRGSNYGSLL	540
TTEGQFQVFA	KTAYYKGNLV	AVKRVNRKRI	ELTRKVL FEL	KHMRDVQNEH	LTRFVGACTD	600
PPNICILTEY	CPRGSLQDIL	ENESITLDWM	FRYSLTNDIV	KGMLFLHNGA	ICSHGNLKSS	660
NCVVDGRFVL	KITDYGLESF	RDPEPEQGHT	LFPAKLLWAP	ELLRMASPPA	RGSQAGDVYS	720
FGIILQEIAL	RSQVFYVEGL	DLSPKEIIER	VTRGEQPPFR	PSMDLQSHLE	ELGQLMQRCW	780
AEDPQERPPF	QQIRLALRKF	NKENSNIIL	NLLSRMEQYA	NNLEELVEER	TQAYLEEKRK	840
AEALLYQILP	HSVAEQLEKRG	ETVQAEAFDS	VTIYFSDIVG	FTALSAESTP	MQVVTLLNDL	900
YTCFDAVIDN	FDVYKVTIG	DAYMVVSGLP	VRNGQLHARE	VARMALALLD	AVRSFRIRHR	960
PQEQLRLRIG	IHTGPVCAGV	VGLKMPRYCL	FGDTVNTASR	MESNGEALKI	HLSSETKAVL	1020
EEFDGFLELE	RGDVEMKGGK	KVRTYWLLGE	RGCSSTRG			1057

SEQ ID NO: 4	moltype = AA	length = 10
FEATURE	Location/Qualifiers	
REGION	1..10	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	

SEQUENCE: 4		
GFTFNTHYIH		10

SEQ ID NO: 5	moltype = AA	length = 17
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	

SEQUENCE: 5		
SISGSGSNTY YADSVKG		17

SEQ ID NO: 6	moltype = AA	length = 12
FEATURE	Location/Qualifiers	
REGION	1..12	
	note = source = /note="Description of Artificial Sequence:	

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source	Synthetic peptide"	
	1..12	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 6		
ERGVVYHMF DP		12
SEQ ID NO: 7	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 7		
THYIH		5
SEQ ID NO: 8	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 8		
GTFNTH		7
SEQ ID NO: 9	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 9		
SGSGSN		6
SEQ ID NO: 10	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 10		
GTFNTHY		8
SEQ ID NO: 11	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 11		
ISGSGSNT		8
SEQ ID NO: 12	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 12		
ARERGVVYH MFDP		14
SEQ ID NO: 13	moltype = AA length = 121	
FEATURE	Location/Qualifiers	
REGION	1..121	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic polypeptide"	
source	1..121	

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mol_type = protein
organism = synthetic construct

SEQUENCE: 13
QVQLLESQGGG LVQPGGSLRL SCAASGFTFN THYIHWVRQA PGKGLEWVSS ISGSGSNTYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARER GYVYYHMFDP WQGGTLVTVS 120
S 121

SEQ ID NO: 14      moltype = DNA length = 363
FEATURE           Location/Qualifiers
misc_feature       1..363
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polynucleotide"
source            1..363
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 14
cagggtgcaat tgcgtgaaag cggcggtggc ctggtgcagc cgggtggcag cctgcgtctg 60
agctgcgcgg cgctccgatt cacccttaac actcattaca tccattgggt gcgccaggcc 120
cccggcaaaag gtctcgagtg ggtttcctct atctctggtt ctggttctaa caccctactat 180
gcggatagcg tgaagggcgc ctttaccatc agccgcgata attcgaaaaa caccctgtat 240
ctgcaaatga acagcctcgc tgcggaagat acggccgtgt attattgcgc gcgtgaacgt 300
ggttacgttt actaccatat gttcgatccg tggggccaag gcaccctggt gactgttagc 360
tca 363

SEQ ID NO: 15      moltype = AA length = 451
FEATURE           Location/Qualifiers
REGION            1..451
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polypeptide"
source            1..451
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 15
QVQLLESQGGG LVQPGGSLRL SCAASGFTFN THYIHWVRQA PGKGLEWVSS ISGSGSNTYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARER GYVYYHMFDP WQGGTLVTVS 120
SASTKGPSVF PLAPSSKSTS GGTAAALGCLV KDYPPEPVTV SWNSGALTSG VHTFPAVLQS 180
SGLYSLSSVV TVPSSSLGTQ TYICNVNHKP SNTKVDKRVK PKSCDKTHTC PPCPAPEAAG 240
GPSVFLFPPK PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY 300
NSTYRVSVL TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE 360
EMTKNQVSLT CLVKGFYPSD IAVEWESNGQ PENNYKTTTP VLDSDGSFFL YSKLTVDKSR 420
WQQGNVFSCS VMHEALHNHY TQKLSLSLSPG K 451

SEQ ID NO: 16      moltype = DNA length = 1353
FEATURE           Location/Qualifiers
misc_feature       1..1353
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polynucleotide"
source            1..1353
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 16
cagggtgcaat tgcgtgaaag cggcggtggc ctggtgcagc cgggtggcag cctgcgtctg 60
agctgcgcgg cgctccgatt cacccttaac actcattaca tccattgggt gcgccaggcc 120
cccggcaaaag gtctcgagtg ggtttcctct atctctggtt ctggttctaa caccctactat 180
gcggatagcg tgaagggcgc ctttaccatc agccgcgata attcgaaaaa caccctgtat 240
ctgcaaatga acagcctcgc tgcggaagat acggccgtgt attattgcgc gcgtgaacgt 300
ggttacgttt actaccatat gttcgatccg tggggccaag gcaccctggt gactgttagc 360
tcagcctcca ccaagggtcc atcgggtcttc cccctggcac cctcctccaa gagcacctct 420
gggggacacg cggccctggg ctgctggttc aaggactact tcccgaacc ggtgacggtg 480
tcgtggaact caggcgccct gaccagcggc gtgcacacct tcccggtgt cctacagtcc 540
tcaggactct actccctcag cagcgtggtg accgtgccct ccagcagctt gggcacccag 600
acctacatct gcaacgtgaa tcacaagccc agcaacacca aggtggacaa gagagttgag 660
cccaaatctt gtgacaaaac tcacacatgc ccaccgtgcc cagcacctga agcagcgggg 720
ggaccgtcag tctcctctct ccccccacaa cccaaggaca cctcatgat ctcccggacc 780
ctgaggtgca catgctggtt ggtggacgtg agccacgaag accctgaggt caagtccaac 840
tggtacgtgg acggcgtgga ggtgcataat gccaaagaca agccgcggga ggagcagtac 900
aacagcacgt accgggtggt cagcgtcttc accgtctcgc accaggactg gctgaatggc 960
aaggagtaca agtgcaaggt ctccaacaaa gccctcccag ccccctcga gaaaaccatc 1020
tccaaagcca aagggcagcc ccgagaacca caggtgtaca ccctgcccc atcccgggag 1080
gagatgacca agaaccaggt cagcctgacc tgcctggtca aaggcttcta tcccagcgac 1140
atcgccgtgg agtgggagag caatgggcag ccggagaaca actacaagac cagcctccc 1200
gtgctggact ccgacggctc cttctctctc tacagcaagc tcaccgtgga caagagcagg 1260
tggcagcagg ggaacgtctt ctcatgctcc gtgatgcatg aggtctctga caaccactac 1320
acgcagaaga gcctctccct gtctccgggt aaa 1353

SEQ ID NO: 17      moltype = AA length = 12
FEATURE           Location/Qualifiers
REGION            1..12
                   note = source = /note="Description of Artificial Sequence:

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source	Synthetic peptide"	
	1..12	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 17		
RASQSITRNY LA		12
SEQ ID NO: 18	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 18		
GASSRAT		7
SEQ ID NO: 19	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 19		
QQHSMYPRT		9
SEQ ID NO: 20	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 20		
SQSITRNY		8
SEQ ID NO: 21	moltype = length =	
SEQUENCE: 21		
000		
SEQ ID NO: 22	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 22		
HSMYPR		6
SEQ ID NO: 23	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 23		
QSITRNY		7
SEQ ID NO: 24	moltype = AA length = 108	
FEATURE	Location/Qualifiers	
REGION	1..108	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic polypeptide"	
source	1..108	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 24		
DIVLTQSPAT LSLSPGERAT LSCRASQSIT RNYLAWYQQK PGQAPRLLIY GASSRATGIP		60
ARFSGSGSGT DFTLTISLE PEDFAVYICQ QHSMYPRTFG QGTKVEIK		108
SEQ ID NO: 25	moltype = DNA length = 324	

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FEATURE                Location/Qualifiers
misc_feature            1..324
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polynucleotide"
source                  1..324
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 25
gatatcgtgc tgacccagag cccggcgacc ctgagcctga gcccggtga acgtgccacc 60
ctgagctgca gagcgagcca gtctatcact cgtaactacc tggcttgga ccagcagaaa 120
ccgggccagg ccccgcgctc attaatctac ggtgcttctt ctggtgcgac cggcattccg 180
gcgcgtttta gcgcgagcgg atccggcacc gatttcaccc tgaccattag cagcctggaa 240
ccggaagact ttgcggtgta ttattgccag cagcattcta tgtaccgcg tacctttggc 300
cagggcacga aagttgaaat taaa                                     324

SEQ ID NO: 26           moltype = AA length = 215
FEATURE                Location/Qualifiers
REGION                  1..215
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polypeptide"
source                  1..215
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 26
DIVLTQSPAT LSLSPGERAT LSCRASQSIT RNYLAWYQQK PGQAPRLLIY GASSRATGIP 60
ARFSGSGSGT DFTLTISLE PEDFAVYCYQ QHSMYPRTFG QGKVEIKRT VAAPSVFIPP 120
PSDEQLKSGT ASVVCLLNNF YPREAKVQWK VDNLQSGNS QESVTEQDSK DSTYLSSTL 180
TLKADYKXH KVIACEVTHQ GLSSPVTKSF NRGEC                                     215

SEQ ID NO: 27           moltype = DNA length = 645
FEATURE                Location/Qualifiers
misc_feature            1..645
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polynucleotide"
source                  1..645
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 27
gatatcgtgc tgacccagag cccggcgacc ctgagcctga gcccggtga acgtgccacc 60
ctgagctgca gagcgagcca gtctatcact cgtaactacc tggcttgga ccagcagaaa 120
ccgggccagg ccccgcgctc attaatctac ggtgcttctt ctggtgcgac cggcattccg 180
gcgcgtttta gcgcgagcgg atccggcacc gatttcaccc tgaccattag cagcctggaa 240
ccggaagact ttgcggtgta ttattgccag cagcattcta tgtaccgcg tacctttggc 300
cagggcacga aagttgaaat taaacgtacg gtggccgctc ccagcgtgtt catcttcccc 360
ccagcgacgc agcagctgaa gagcggcacc gccagcgtgg tgtgcctgct gaacaacttc 420
taccgccggg aggccaaagg gcaagtgaag gtggacaacg ccctgcagag cggcaacagc 480
caggaaagcg tcaccgagca ggacagcaag gactccacct acagcctgag cagcaccctg 540
accctgagca aggccgacta cgagaagcac aaggtgtacg cctgcgaggt gaccaccag 600
ggcctgtcca gccccgtgac caagagcttc aaccggggcg agtgt                                     645

SEQ ID NO: 28           moltype = AA length = 10
FEATURE                Location/Qualifiers
REGION                  1..10
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic peptide"
source                  1..10
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 28
GFTFSSYWMN                                                    10

SEQ ID NO: 29           moltype = AA length = 17
FEATURE                Location/Qualifiers
REGION                  1..17
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic peptide"
source                  1..17
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 29
AISSDGSYTY YADSVKG                                           17

SEQ ID NO: 30           moltype = AA length = 16
FEATURE                Location/Qualifiers
REGION                  1..16
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic peptide"
source                  1..16
                        mol_type = protein

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SEQUENCE: 30	organism = synthetic construct	
DRYSMIYSYG AGAFDY		16
SEQ ID NO: 31	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 31		5
SYWMN		
SEQ ID NO: 32	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 32		7
GFTFSSY		
SEQ ID NO: 33	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 33		6
SSDGSY		
SEQ ID NO: 34	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 34		8
GFTFSSYW		
SEQ ID NO: 35	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 35		8
ISSDGSYT		
SEQ ID NO: 36	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
REGION	1..18	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..18	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 36		18
ARDYSMIYS YGAGAFDY		
SEQ ID NO: 37	moltype = AA length = 125	
FEATURE	Location/Qualifiers	
REGION	1..125	
	note = source = /note="Description of Artificial Sequence: Synthetic polypeptide"	
source	1..125	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 37		

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QVQLLESQGG LVQPGGSLRL SCAASGFTFS SYWMNWVRQA PGKGLEWVSA ISSDGSYTTY 60
 ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARDR YSMIYSYGAG AFDYWQGGTL 120
 VTVSS 125

SEQ ID NO: 38 moltype = DNA length = 375
 FEATURE Location/Qualifiers
 misc_feature 1..375
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
 source 1..375
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 38
 cagggtgcaat tgcgtgaaag cggcggtggc ctggtgcagc cgggtggcag cctgcgtctg 60
 agctgcgcgg cgtccgatt cacttttct tcttactgga tgaactgggt gcgccaggcc 120
 ccgggcaaaag gtctcgagtg ggtttccgct atctcttctg acggttctta cactactat 180
 gcggatagcg tgaaggccg ctttaccatc agccgcgata attcgaaaaa caccctgat 240
 ctgcaaatga acagcctgcg tgcggaagat acggccgtgt attattgcgc gcgtgaccgt 300
 tactctatga tctactctta cgggtgctgt gctttcgatt actggggcca aggcaccctg 360
 gtgactgtta gctca 375

SEQ ID NO: 39 moltype = AA length = 455
 FEATURE Location/Qualifiers
 REGION 1..455
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..455
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 39
 QVQLLESQGG LVQPGGSLRL SCAASGFTFS SYWMNWVRQA PGKGLEWVSA ISSDGSYTTY 60
 ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARDR YSMIYSYGAG AFDYWQGGTL 120
 VTVSSASTKG PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA 180
 VLQSSGLYSL SSVVTPSSS LGTQTYICNV NHKPSNTKVD KRVEPKSCDK THTCPPCPAP 240
 EAAGGPSVFL PPPKPKDTLM ISRTPEVTCV VVDVSHEDPE VKFNWYVDGV EVHNAKTKPR 300
 EEQYNSTYRV VSVLTVLHQD WLNKGKEYCK VSNKALPAPI EKTISKAKGQ PREPQVYTL 360
 PSREEMTKNQ VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFFLYSKLTV 420
 DKSRWQQGNV FSCSVMHEAL HNHYTQKSL SLPKG 455

SEQ ID NO: 40 moltype = DNA length = 1365
 FEATURE Location/Qualifiers
 misc_feature 1..1365
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
 source 1..1365
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 40
 cagggtgcaat tgcgtgaaag cggcggtggc ctggtgcagc cgggtggcag cctgcgtctg 60
 agctgcgcgg cgtccgatt cacttttct tcttactgga tgaactgggt gcgccaggcc 120
 ccgggcaaaag gtctcgagtg ggtttccgct atctcttctg acggttctta cactactat 180
 gcggatagcg tgaaggccg ctttaccatc agccgcgata attcgaaaaa caccctgat 240
 ctgcaaatga acagcctgcg tgcggaagat acggccgtgt attattgcgc gcgtgaccgt 300
 tactctatga tctactctta cgggtgctgt gctttcgatt actggggcca aggcaccctg 360
 gtgactgtta gctcagcctc caccgaagggt ccatcggtct tccccctggc accctcctcc 420
 aagagcacct ctggggggcac agcggccctg ggctgcctgg tcaaggacta cttccccgaa 480
 ccggtgacgg tgcgtggaa ctcaggcgcc ctgaccagcg gcgtgcacac cttcccggct 540
 gtctacagt cctcaggact ctactccctc agcagcgtgg tgaccgtgcc ctccagcagc 600
 ttgggcaccc agacctacat ctgcaacgtg aatcacaagc ccagcaacac caagggtggac 660
 aagagagttg agcccaaatc ttgtgacaaa actcacacat gccacccgtg cccagcacct 720
 gaagcagcgg ggggaccgtc agtcttctc tttcccccaa aacccaagga caccctcatg 780
 atctcccggg cccctgaggt cacatgcgtg gtggtggacg tgagccacga agaccctgag 840
 gtcaagttca actggtacgt ggacggcgtg gaggtgcata atgccaagac aaagccggcg 900
 gagagcagtg acaacagcac gtaccgggtg gtcagcgtcc tcaccgtcct gcaccaggac 960
 tgggtgaatg gcaaggagta caagtgcag gtctccaaca aagccctccc agcccccatc 1020
 gagaaaacca tctccaaagc caaagggcag ccccgagAAC cacaggtgta caccctgccc 1080
 ccattccggg aggagatgac caagaaccag gtcagcctga cctgcctggt caaaggcttc 1140
 tatccagcgg acatcgccgt ggagtgaggc agccggagaa caactacaag 1200
 accacgcctc ccgtgctgga ctccgacggc tccttcttcc tctacagcaa gctcaccgtg 1260
 gacaagagca ggtggcagca ggggaacgtc ttctcatgct ccgtgatgca tgaggctctg 1320
 cacaaccact acacgcagaa gagcctctcc ctgtctccgg gtaaa 1365

SEQ ID NO: 41 moltype = AA length = 11
 FEATURE Location/Qualifiers
 REGION 1..11
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
 source 1..11
 mol_type = protein

-continued

SEQUENCE: 41	organism = synthetic construct	
RASQGISSYL A		11
SEQ ID NO: 42	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 42		
TASTLQS		7
SEQ ID NO: 43	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 43		
MQSYEKPRT		9
SEQ ID NO: 44	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 44		
SQGISSY		7
SEQ ID NO: 45	moltype = length =	
SEQUENCE: 45		
000		
SEQ ID NO: 46	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 46		
SYEKPR		6
SEQ ID NO: 47	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 47		
QGISSY		6
SEQ ID NO: 48	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
REGION	1..107	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic polypeptide"	
source	1..107	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 48		
DIQMTQSPSS LSASVGRVT ITCRASQGIS SYLAWYQQKP GKAPKLLIYT ASTLQSGVPS		60
RFGSGSGGTD FTLTISSLQP EDFATYYCMQ SYEKPRTFGQ GTKVEIK		107
SEQ ID NO: 49	moltype = DNA length = 321	
FEATURE	Location/Qualifiers	
misc_feature	1..321	
	note = source = /note="Description of Artificial Sequence:	

-continued

source Synthetic polynucleotide"
1..321
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 49
gatatccaga tgacccagag cccgagcagc ctgagcgcca gcgtgggcga tcgcgtgacc 60
attacctgca gagccagcca gggattttct tcttacctgg cttggtagca gcagaaaccg 120
ggcaaagcgc cgaaactatt aatctacact gcttctactc tgcaaagcgg cgtgccgagc 180
cgcttttagcg gcagcggatc cggcaccgat ttcaccctga ccattagctc tctgcaaccg 240
gaagactttg cgacctatta ttgcagtgcag tcttacgaaa aaccgcgtac ctttgccag 300
ggcacgaaaag ttgaaattaa a 321

SEQ ID NO: 50 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"

source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 50
DIQMTQSPSS LSASVGDRTV ITCRASQGIS SYLAWYQQKP GKAPKLLIYT ASTLQSGVPS 60
RFGSGSGSTD FTLTISSLQP EDFATYYCMQ SYEKPRTFGQ GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 51 moltype = DNA length = 642
FEATURE Location/Qualifiers
misc_feature 1..642
note = source = /note="Description of Artificial Sequence:
Synthetic polynucleotide"

source 1..642
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 51
gatatccaga tgacccagag cccgagcagc ctgagcgcca gcgtgggcga tcgcgtgacc 60
attacctgca gagccagcca gggattttct tcttacctgg cttggtagca gcagaaaccg 120
ggcaaagcgc cgaaactatt aatctacact gcttctactc tgcaaagcgg cgtgccgagc 180
cgcttttagcg gcagcggatc cggcaccgat ttcaccctga ccattagctc tctgcaaccg 240
gaagactttg cgacctatta ttgcagtgcag tcttacgaaa aaccgcgtac ctttgccag 300
ggcacgaaaag ttgaaattaa acgtacgggtg gccgtcccca gcgtgttcat cttccccccc 360
agcgacgagc agctgaagag cggcaccgcc agcgtgggtg gcctgctgaa caacttctac 420
ccccgggagg ccaagggtgca gtggaagggtg gacaacgccc tgacagagcgg caacagccag 480
gaaagcgtca ccgagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgtccagcc ccgtgaccaa gagcttcaac cggggcgagt gt 642

SEQ ID NO: 52 moltype = AA length = 10
FEATURE Location/Qualifiers
REGION 1..10
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"

source 1..10
mol_type = protein
organism = synthetic construct

SEQUENCE: 52
GYSFSNYWIG 10

SEQ ID NO: 53 moltype = AA length = 17
FEATURE Location/Qualifiers
REGION 1..17
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"

source 1..17
mol_type = protein
organism = synthetic construct

SEQUENCE: 53
IIYPDVSYTR YSPSFQG 17

SEQ ID NO: 54 moltype = AA length = 10
FEATURE Location/Qualifiers
REGION 1..10
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"

source 1..10
mol_type = protein
organism = synthetic construct

SEQUENCE: 54
YWSEAYTFDY 10

-continued

SEQ ID NO: 55	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 55		
NYWIG		5
SEQ ID NO: 56	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 56		
GYSFSNY		7
SEQ ID NO: 57	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 57		
YPDVSY		6
SEQ ID NO: 58	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 58		
GYSFSNYW		8
SEQ ID NO: 59	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 59		
IYPDVSYT		8
SEQ ID NO: 60	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
REGION	1..12	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..12	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 60		
ARYWSEAYTF DY		12
SEQ ID NO: 61	moltype = AA length = 119	
FEATURE	Location/Qualifiers	
REGION	1..119	
	note = source = /note="Description of Artificial Sequence: Synthetic polypeptide"	
source	1..119	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 61		
QVQLVQSGAE VKKPGESLKI SCKGSGYSFS NYWIGWVRQM PGKGLEWMMGI IYPDVSYTRY		60
SPSPQQQVTI SADKSISTAY LQWSSLKASD TAMYYCARYW SEAYTFDYWG QGTLVTVSS		119

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SEQ ID NO: 62      moltype = DNA length = 357
FEATURE           Location/Qualifiers
misc_feature       1..357
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polynucleotide"
source            1..357
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 62
caggtgcaat tgggtcagag cgggtcggaa gtgaaaaaac cgggcgaaag cctgaaaatt 60
agctgcгааг gctcgggata tagcttctct aactactgga tcggttgggg gcgccagatg 120
ccgggcгааг gtctcgagtg gatgggcatc atctaccggg acgttagcta cacccgttat 180
agcccgagct ttcaggggcca ggtgaccatt agcgcggata aaagcatcag caccgcgtat 240
ctgcaatgga gcagcctgaa agcgcgcatg accgcgatgt attattgcgc gcgttactgg 300
tctgaagctt acactttcga ttactggggc caaggcaccg tggtgactgt tagctca 357

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SEQ ID NO: 63      moltype = AA length = 449
FEATURE           Location/Qualifiers
REGION            1..449
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polypeptide"
source            1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 63
QVQLVQSGAE VKKPGESLKI SCKGSGYSFS NYWIGWVRQM PGKGLEWMMGI IYPDVSYTRY 60
SPSFQGGQVTI SADKSISTAY LQWSSLKASD TAMYYCARYW SEAYTFDYWG QGTLVTVSSA 120
STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTWSW NSGALTSGVH TTPAVLQSSG 180
LYSLSSVTVV PSSSLGTQTY ICNVNHKPSN TKVDKRVPEPK SCDKTHTCPP CPAPEAAGGP 240
SVFLFPPPKPK DTLMISRTPE VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVSVSLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

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SEQ ID NO: 64      moltype = DNA length = 1347
FEATURE           Location/Qualifiers
misc_feature       1..1347
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polynucleotide"
source            1..1347
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 64
caggtgcaat tgggtcagag cgggtcggaa gtgaaaaaac cgggcgaaag cctgaaaatt 60
agctgcгааг gctcgggata tagcttctct aactactgga tcggttgggg gcgccagatg 120
ccgggcгааг gtctcgagtg gatgggcatc atctaccggg acgttagcta cacccgttat 180
agcccgagct ttcaggggcca ggtgaccatt agcgcggata aaagcatcag caccgcgtat 240
ctgcaatgga gcagcctgaa agcgcgcatg accgcgatgt attattgcgc gcgttactgg 300
tctgaagctt acactttcga ttactggggc caaggcaccg tggtgactgt tagctcagcc 360
tccaccaagg gtccatcggt cttcccccctg gcacccctct ccaagagcac ctctgggggc 420
acagcgggccc tgggctgcct ggtcaaggac tacttccccg aaccggtgac ggtgtcgtgg 480
aactcaggcg ccccgaccag cggcgtgcac accttccccg ctgtcctaca gtcctcagga 540
ctctactccc tcagcagcgt ggtgaccgtg cctccagca gcttgggcac ccagacctac 600
atctgcaacg tgaatcacia gccccagcaac accaaggtgg acaagagagt tgagcccaaa 660
tcttgtgaca aaactcacac atgccaccg tgcccagcac ctgaagcagc ggggggaccg 720
tcagttcttc tcttcccccc aaaacccaag gacaccctca tgatctcccc gacccctgag 780
gtcacatcgg tgggtggtgga cgtgagccac gaagaccctg aggtcaagtt caactggtac 840
gtggacggcg tggaggtgca taatgccaag acaaagccgc gggaggagca gtacaacagc 900
acgtaccggg tggtcagcgt cctcaccgtc ctgcaccagg actggctgaa tggcaaggag 960
tacaagtgca aggtctccaa caaagccctc ccagccccc tgcagaaaaa catctccaaa 1020
gccaaaggcg agccccgaga accacaggtg tacaccctgc cccatccccg ggaggagatg 1080
accaagaacc aggtcagcct gacctgcctg gtcaaaggct tctatcccag cgacatcgcc 1140
gtggagtggg agagcaatgg gcagccggag aacaactaca agaccacgcc tcccgtgctg 1200
gactccgacg gctccttctt cctctacagc aagctcaccg tggacaagag caggtggcag 1260
caggggaacg tcttctcatg ctccgtgatg catgaggctc tgcacaacca ctacacgcag 1320
aagagcctct cctgtctctc gggtaaaa 1347

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SEQ ID NO: 65      moltype = AA length = 11
FEATURE           Location/Qualifiers
REGION            1..11
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic peptide"
source            1..11
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 65
SGDNIRKKYV F

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SEQ ID NO: 66      moltype = AA length = 7

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FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 66		
GDNDRPS		7
SEQ ID NO: 67	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 67		
GTYTLLFTSK V		11
SEQ ID NO: 68	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 68		
DNIRKKY		7
SEQ ID NO: 69	moltype = length =	
SEQUENCE: 69		
000		
SEQ ID NO: 70	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 70		
YTLLFTSK		8
SEQ ID NO: 71	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 71		
NIRKKY		6
SEQ ID NO: 72	moltype = AA length = 108	
FEATURE	Location/Qualifiers	
REGION	1..108	
	note = source = /note="Description of Artificial Sequence: Synthetic polypeptide"	
source	1..108	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 72		
DIELTQPPSV SVSPGQTASI TCSGDNIRKK YVFWYQQKPG QAPVLVIYGD NDRPSGIPER		60
FSGSNSGNTA TLTISGTQAE DEADYYCGTY TLLFTSKVFG GGTKLTVL		108
SEQ ID NO: 73	moltype = DNA length = 324	
FEATURE	Location/Qualifiers	
misc_feature	1..324	
	note = source = /note="Description of Artificial Sequence: Synthetic polynucleotide"	
source	1..324	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 73		

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gatatcgaac tgacccagcc gccgagcgtg agcgtgagtc cgggccagac cgcgagcatt 60
acctgtagcg gcgataacat ccgtaaaaaa tacgttttct ggtaccagca gaaaccgggc 120
caggcgccgg tgctgggtgat ctacggtgac aacgaccgtc cgagcgccat cccggaacgt 180
tttagcggat ccaacagcgg caacaccgcy accctgacca ttagcggcac ccaggcgga 240
gacgaagcgg attattactg cggctacttac actctgctgt tcacttctaa agtgtttggc 300
ggcggcacga agttaaccgt ccta 324

SEQ ID NO: 74      moltype = AA  length = 214
FEATURE           Location/Qualifiers
REGION            1..214
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 74
DIELTQPPSV SVSPGQTASI TCSGDNIRKK YVFWYQQKPG QAPVLVIYGD NDRPSGIPER 60
FSGSNSGNTA TLTISGTQAE DEADYYCGTY TLLFTSKVFG GGTKLTVLGG PKAAPSVTLF 120
PPSSEELQAN KATVCLISD FYPGAVTVAV KADSSPVKAG VETTPSKQS NNKYAASSYL 180
SLTPEQWKSH RSYSCQVTHE GSTVEKTVAP TECS 214

SEQ ID NO: 75      moltype = DNA  length = 642
FEATURE           Location/Qualifiers
misc_feature       1..642
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polynucleotide"
source            1..642
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 75
gatatcgaac tgacccagcc gccgagcgtg agcgtgagtc cgggccagac cgcgagcatt 60
acctgtagcg gcgataacat ccgtaaaaaa tacgttttct ggtaccagca gaaaccgggc 120
caggcgccgg tgctgggtgat ctacggtgac aacgaccgtc cgagcgccat cccggaacgt 180
tttagcggat ccaacagcgg caacaccgcy accctgacca ttagcggcac ccaggcgga 240
gacgaagcgg attattactg cggctacttac actctgctgt tcacttctaa agtgtttggc 300
ggcggcacga agttaaccgt cctaggtcag cccaaggctg cccctcggt cactctgttc 360
ccgcctctct ctgaggagct tcaagccaac aaggccacac tgggtgtgtc cataagtgac 420
ttctaccggg gagccgtgac agtggcctgg aaggcagata gcagccccgt caaggcgga 480
gtggagacca ccacaccctc caaacaagc aacaacaagt acgcgccag cagctatctg 540
agcctgacgc ctgagcagtg gaagtccac agaagctaca gctgccaggt cacgcatgaa 600
gggagcacccg tggagaagac agtggccccct acagaatgtt ca 642

SEQ ID NO: 76      moltype = AA  length = 10
FEATURE           Location/Qualifiers
REGION            1..10
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..10
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 76
GYSFTSYWIA 10

SEQ ID NO: 77      moltype = AA  length = 17
FEATURE           Location/Qualifiers
REGION            1..17
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..17
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 77
RIDPDNSYTR YSPSFQG 17

SEQ ID NO: 78      moltype = AA  length = 17
FEATURE           Location/Qualifiers
REGION            1..17
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..17
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 78
WLSPGYALGE QPAGMDH 17

SEQ ID NO: 79      moltype = AA  length = 5
FEATURE           Location/Qualifiers
REGION            1..5
                  note = source = /note="Description of Artificial Sequence:

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source	Synthetic peptide"	
	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 79		
SYWIA		5
SEQ ID NO: 80	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 80		
GYSFTSY		7
SEQ ID NO: 81	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 81		
DPDNSY		6
SEQ ID NO: 82	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 82		
GYSFTSYW		8
SEQ ID NO: 83	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 83		
IDPDNSYT		8
SEQ ID NO: 84	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic peptide"	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 84		
ARWLSPGYAL GEQPAGMDH		19
SEQ ID NO: 85	moltype = AA length = 126	
FEATURE	Location/Qualifiers	
REGION	1..126	
	note = source = /note="Description of Artificial Sequence:	
	Synthetic polypeptide"	
source	1..126	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 85		
QQLVQSGAE VKKPGESLKI SCKGSGYSFT SYWIAWVRQM PGKGLEWMGR IDPDNSYTRY		60
SPSFQQQVTI SADKSISTAY LQWSSLKASD TAMYYCARWL SPGYALGEQP AGMDHWGQGT		120
LVTVSS		126
SEQ ID NO: 86	moltype = DNA length = 378	
FEATURE	Location/Qualifiers	
misc_feature	1..378	
	note = source = /note="Description of Artificial Sequence:	

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source Synthetic polynucleotide"
1..378
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 86
caggtgcaat tgggtcgagag cgggtcgga gtgaaaaaac cgggcgaaag cctgaaaatt 60
agctgcaaaag gctccggata tagcttcact tcttactgga tcgcttgggt gcgccagatg 120
ccgggcgaaag gtctcgagtg gatgggccgt atcgacccgg acaacagcta caccggttat 180
agcccgagct ttcagggcca ggtgaccatt agcgcggata aaagcatcag caccgcgtat 240
ctgcaatgga gcagcctgaa agcagcgcat accgcgatgt attattgcgc gcgttggctg 300
tctccgggtt acgctctggg tgaacagccg gctggtatgg atcattgggg ccaaggcacc 360
ctggtgactg ttagctca 378

SEQ ID NO: 87 moltype = AA length = 456
FEATURE Location/Qualifiers
REGION 1..456
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"

source 1..456
mol_type = protein
organism = synthetic construct

SEQUENCE: 87
QVQLVQSGAE VKKPGESLKI SCKGSGYSFT SYWIAWVRQM PGKGLEWMGR IDPDNSYTRY 60
SPSFQGVITI SADKSIISTAY LQWSSLKASD TAMYYCARWL SPGYALGEQP AGMDHWGQGT 120
LVTVSSASTK GPSVFPPLAPS SKSTSGGTAA LGCLVKDYFP EPVTVSWNSG ALTSGVHTFP 180
AVLQSSGLYS LSSVVTVPSS SLGTQTYICN VNHKPSNTKV DKRVEPKSCD KTHTCPPCPA 240
PEAAGGPSVF LPPPKPKDTL MISRTPEVTC VVVDVSHEDP EVKFNWYVDG VEVHNAKTKP 300
REEQYNSTYR VVSVLTVLHQ DWLNGKEYKC KVSNKALPAP IEKTISKAKG QPREPQVYTL 360
PPSREEMTKN QVSLTCLVKG FYPSDIAVEW ESNQQPENNY KTTPPVLDSD GSPFLYSKLT 420
VDKSRWQQGN VFSCSVMEHA LHNHYTQKSL SLSPGK 456

SEQ ID NO: 88 moltype = DNA length = 1368
FEATURE Location/Qualifiers
misc_feature 1..1368
note = source = /note="Description of Artificial Sequence:
Synthetic polynucleotide"

source 1..1368
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 88
caggtgcaat tgggtcgagag cgggtcgga gtgaaaaaac cgggcgaaag cctgaaaatt 60
agctgcaaaag gctccggata tagcttcact tcttactgga tcgcttgggt gcgccagatg 120
ccgggcgaaag gtctcgagtg gatgggccgt atcgacccgg acaacagcta caccggttat 180
agcccgagct ttcagggcca ggtgaccatt agcgcggata aaagcatcag caccgcgtat 240
ctgcaatgga gcagcctgaa agcagcgcat accgcgatgt attattgcgc gcgttggctg 300
tctccgggtt acgctctggg tgaacagccg gctggtatgg atcattgggg ccaaggcacc 360
ctggtgactg ttagctcagc ctccaccaag ggtccatcgg tcttccccct ggcacctccc 420
tccaagagca cctctggggg cacagcggcc ctgggctgcc tggteaagga ctacttcccc 480
gaaccggtga cgggtgctg gaaactcagg gccctgacca gcggcgtgca caccttcccc 540
gctgtcctac agtctcagg aactctactcc ctcagcagcg tggtgaccgt gccctccagc 600
agcttgggca cccagacctc catctgcaac gtgaatcaca agcccagcaa caccaaggtg 660
gacaagagag ttgagcccaa atcttgtgac aaaactcaca catgccacc gtgccagca 720
cctgaagcag cgggggggacc gtcagtcttc ctcttcccc caaaacccaa ggacacctc 780
atgatctccc ggaccctga ggtcacatgc gtgggtggtg acgtgagcca cgaagacct 840
gaggtcaagt tcaactggta ctgggacggc gtggaggtgc ataagccaa gacaaagccg 900
cgggaggagc agtacaacag cacgtaccgg tgggtcagcg tcctcaccgt cctgcaccag 960
gactggctga atggcaagga gtacaagtgc aaggtctcca acaaagccct cccagcccc 1020
atcgagaaaa ccatctccaa agccaaaggg cagccccgag aaccacaggt gtacacctg 1080
cccccatccc gggaggagat gaccaagaac caggtcagcc tgacctgctt ggtcaaaaggc 1140
ttctatccca gcgacatcgc cgtggagtgg gagagcaatg ggcagccgga gaacaactac 1200
aagaccacgc ctcccgctgt ggactccgac ggctccttct tcctctacag caagctcacc 1260
gtggacaaga gcaggtggca gcaggggaac gtcttctcat gctccgtgat gcatgaggct 1320
ctgcacaacc actacacgca gaagagcctc tccctgtctc cgggtaaa 1368

SEQ ID NO: 89 moltype = AA length = 14
FEATURE Location/Qualifiers
REGION 1..14
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"

source 1..14
mol_type = protein
organism = synthetic construct

SEQUENCE: 89
TGSSSNIGAG YAVH 14

SEQ ID NO: 90 moltype = AA length = 7
FEATURE Location/Qualifiers
REGION 1..7
note = source = /note="Description of Artificial Sequence:

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source          "Synthetic peptide"
                1..7
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 90
SNNKRPS                      7

SEQ ID NO: 91      moltype = AA  length = 11
FEATURE           Location/Qualifiers
REGION           1..11
                note = source = /note="Description of Artificial Sequence:
                Synthetic peptide"
source           1..11
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 91
QSYDLQKSSR V                      11

SEQ ID NO: 92      moltype = AA  length = 10
FEATURE           Location/Qualifiers
REGION           1..10
                note = source = /note="Description of Artificial Sequence:
                Synthetic peptide"
source           1..10
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 92
SSSNIGAGYA                      10

SEQ ID NO: 93      moltype =      length =
SEQUENCE: 93
000

SEQ ID NO: 94      moltype = AA  length = 8
FEATURE           Location/Qualifiers
REGION           1..8
                note = source = /note="Description of Artificial Sequence:
                Synthetic peptide"
source           1..8
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 94
YDLQKSSR                      8

SEQ ID NO: 95      moltype = AA  length = 9
FEATURE           Location/Qualifiers
REGION           1..9
                note = source = /note="Description of Artificial Sequence:
                Synthetic peptide"
source           1..9
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 95
SSNIGAGYA                      9

SEQ ID NO: 96      moltype = AA  length = 111
FEATURE           Location/Qualifiers
REGION           1..111
                note = source = /note="Description of Artificial Sequence:
                Synthetic polypeptide"
source           1..111
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 96
DIVLTQPPSV SGAPGQRVTI SCTGSSSNIG AGYAVHWYQQ LPGTAPKLLI YSNNKRPSGV 60
PRRFGSGKSG TSASLAITGL QAEDEADYYC QSYDLQKSSR VFGGGTKLTV L          111

SEQ ID NO: 97      moltype = DNA  length = 333
FEATURE           Location/Qualifiers
misc_feature      1..333
                note = source = /note="Description of Artificial Sequence:
                Synthetic polynucleotide"
source           1..333
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 97
gatatcgtgc tgaccagcc gccgagcgtg agcgggtgcac cgggccagcg cgtgaccatt 60
agctgtaccg gcagcagcag caacattggt gctggttacg ctgtgcattg gtaccagcag 120
ctgccgggca cggcgccgaa actgctgatc tactctaaca acaaacgcc gagcggcgtg 180

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ccggatcgct ttagcgggatc caaaagcggc accagcgcca gcctggcgat taccggcctg 240
caagcagaag acgaagcgga ttattactgc cagtcttacg acctgcagaa atcttctcgt 300
gtgtttggcg gcggcacgaa gttaaccgtc cta 333

SEQ ID NO: 98      moltype = AA  length = 217
FEATURE           Location/Qualifiers
REGION           1..217
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source           1..217
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 98
DIVLTQPPSV SGAPGQRVTI SCTGSSSNIG AGYAVHWYQQ LPGTAPKLLI YSNNKRPSGV 60
PRRFGSGSKG TSASLAITGL QAEDEADYYC QSYDLQKSSR VFGGGTKLTV LGQPKAAPSV 120
TLFPPSSEEL QANKATLVCL ISDFYPGAVT VAWKADSSPV KAGVETTTPS KQSNNKYAAS 180
SYLSLTPEQW KSHRSYSCQV THEGSTVEKT VAPTECS 217

SEQ ID NO: 99      moltype = DNA  length = 651
FEATURE           Location/Qualifiers
misc_feature      1..651
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polynucleotide"
source           1..651
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 99
gatatcgctg tgaccagcc gccgagcgtg agcgggtgcac cgggccagcg cgtgaccatt 60
agctgtaccg gcagcagcag caacattggt gctggttacg ctgtgcattg gtaccagcag 120
ctgccgggca cggcgccgaa actgctgatc tactctaaca acaaacgccc gagcggcgtg 180
ccggatcgct ttagcgggatc caaaagcggc accagcgcca gcctggcgat taccggcctg 240
caagcagaag acgaagcgga ttattactgc cagtcttacg acctgcagaa atcttctcgt 300
gtgtttggcg gcggcacgaa gttaaccgtc ctaggtcagc ccaaggctgc cccctcggtc 360
actctgttcc cgccctcctc tgaggagctt caagccaaca aggccacact ggtgtgtctc 420
ataagtgact tctaccggg agccgtgaca gtggcctgga aggcagatag cagccccgtc 480
aaggcgggag tggagaccac cacaccctcc aaacaaagca acaacaagta cgcggccagc 540
agctatctga gcctgacgcc tgagcagtggt aagtcccaca gaagctacag ctgccaggtc 600
acgcataaag ggagcacctg ggagaagaca gtggccccta cagaatgttc a 651

SEQ ID NO: 100     moltype = AA  length = 17
FEATURE           Location/Qualifiers
REGION           1..17
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source           1..17
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 100
SISSSGQSTY YADSVKG 17

SEQ ID NO: 101     moltype = AA  length = 6
FEATURE           Location/Qualifiers
REGION           1..6
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source           1..6
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 101
SSSGQS 6

SEQ ID NO: 102     moltype = AA  length = 8
FEATURE           Location/Qualifiers
REGION           1..8
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source           1..8
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 102
ISSSGQST 8

SEQ ID NO: 103     moltype = AA  length = 121
FEATURE           Location/Qualifiers
REGION           1..121
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source           1..121
                  mol_type = protein

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SEQ ID NO: 111      moltype = DNA length = 645
FEATURE            Location/Qualifiers
misc_feature        1..645
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..645
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 111
gacatcgtgc tgactcagtc cctgcgact ctgagcctgt caccgggaga acggggccacc 60
ctctcttgcc gcgcctccca atccattact cggaaactacc tggcctggta tcagcagaag 120
ccaggacagg cccctaggct tctgatctac ggggcccagct caagagcaac tggcatcccg 180
gctcgcttct cgggttcggg aagcggcacc gacttcaccc tgacaatttc gtccctcgaa 240
cccaggagatt tcgcogtcta ctactgccaa cagcactcca tgtacccccc gacctttggg 300
cagggaaacca aagtcgagat caagcgtacg gtggccgctc ccagcgtgtt catcttcccc 360
cccagcgacg agcagctgaa gagcggcacc gccagcgtgg tgtgcctgct gaacaatttc 420
tacccccggg aggcctaaggt gcagtgggag gtggacaacg ccctgcagag cggcaacagc 480
caggagagcg tcaccgagca ggacagcaag gactccacct acagcctgag cagcaccctg 540
acccctgagca aggcogacta cgagaagcat aaggtgtacg cctgcgaggt gaccaccag 600
ggcctgtcca gcccggtgac caagagcttc aacagggcg agtgc 645

SEQ ID NO: 112      moltype = AA length = 10
FEATURE            Location/Qualifiers
REGION              1..10
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..10
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 112
GFTFSTHYIH 10

SEQ ID NO: 113      moltype = AA length = 7
FEATURE            Location/Qualifiers
REGION              1..7
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..7
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 113
GFTFSTH 7

SEQ ID NO: 114      moltype = AA length = 8
FEATURE            Location/Qualifiers
REGION              1..8
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..8
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 114
GFTFSTHY 8

SEQ ID NO: 115      moltype = AA length = 121
FEATURE            Location/Qualifiers
REGION              1..121
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source              1..121
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 115
EVQLLESGGG LVQPGGSLRL SCAASGFTFS THYIHWVRQA PGKGLEWVSS ISSSGQSTYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARER GYVYYHMFDP WQGGLVTVS 120
S 121

SEQ ID NO: 116      moltype = DNA length = 363
FEATURE            Location/Qualifiers
misc_feature        1..363
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..363
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 116
gaagtgcagc tgcttgagtc cgggggtgga ctggtgcagc cgggaggatc cctgcgcctg 60
agctgcgctg catccggctt caccttcagc acgcactaca tccattgggt cagacaggcc 120
ccaggaaaag gcctggaatg ggtgtcctcc atctcctcgt cggggcagtc aacctactac 180

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gcggactcgc tcaagggccg gttaccatt agccgggaca acagcaagaa taccctgtac 240
ctccaaatga actcgcgtgag ggccgaagat accgcgctgt attactgtgc ccgcgagaga 300
ggctacgtgt actaccacat gttcgacccg tggggacagg gtactctcgt gactgtgtct 360
tct 363

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SEQ ID NO: 117      moltype = AA length = 451
FEATURE            Location/Qualifiers
REGION             1..451
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polypeptide"
source             1..451
                   mol_type = protein
                   organism = synthetic construct

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SEQUENCE: 117
EVQLLESGGG LVQPGGSLRL SCAASGFTFS THYIHWVRQA PGKGLEWVSS ISSSGQSTYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARER GYVYHMFDP WGQGTLVTVS 120
SASTKGPSVF PLAPSSKSTS GGTAAALGLV KDYFPEPVTV SWNSGALTSG VHTFPAVLQS 180
SGLYSLSSVV TVPSSSLGTQ TYICNVNHKP SNTKVDKRVK PKSCDKTHTC PPCPAPELLG 240
GPSVFLPPPK PKDTLMISRT PEVTCVVAVV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY 300
NSTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALAAPIEKTI SKAKGQPREP QVYTLPPSRE 360
EMTKNQVSLT CLVKGFYPSD IAEWESNGQ PENNYKTPP VLDSGDGFFL YSKLTVDKSR 420
WQQGNVFSCS VMHEALHNYH TQKSLSLSPG K 451

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SEQ ID NO: 118      moltype = DNA length = 1353
FEATURE            Location/Qualifiers
misc_feature        1..1353
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polynucleotide"
source             1..1353
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 118
gaagtgcagc tgcttgagtc cgggggtgga ctggtgcagc ccggaggatc cctgcgcctg 60
agctgcgctg catccggett caccctcagc acgcactaca tccattgggt cagacaggcc 120
ccaggaaaag gcctggaatg ggtgtcctcc atctcctcgt cggggcagtc aaactactac 180
gcggactcgc tcaagggccg gttaccatt agccgggaca acagcaagaa taccctgtac 240
ctccaaatga actcgcgtgag ggccgaagat accgcgctgt attactgtgc ccgcgagaga 300
ggctacgtgt actaccacat gttcgacccg tggggacagg gtactctcgt gactgtgtct 360
tctgcgagca ctaagggccc gtcagtgttc ccgctggctc catcgtcgaa gtccacctcc 420
ggaggaaacc gagcactcgg ttgctgggtc aaggactact tccctgagcc agtgaccgtg 480
tcgtggaaca cgggagccct gaattccggc gtgcacactt ttcccgcggt gctgcagtcc 540
tcggtctgt actcccttcc gtcctgggtc accgtgccgt cgtctagcct gggcaccacg 600
acctacatct gcaacgtgaa ccacaagccg tccaacacca aagtggataa gcgggtggag 660
ccgaagtccg gcgataagac acacacgtgc ccgccatgtc cagcgctga attgcttgcc 720
ggaccttcg tgctcctggt ccgcgctaag cccaaggaca ccttgatgat tagccggact 780
cccgaagtca cctgtgtggt ggtggcagtg tccacgagg acccgagggt caagtttaat 840
tggtacgtgg acggcgctga agtgacaca gcccaagacta agccccggga ggaacagtac 900
aacagcacct accgggtcgt gtcctgtgtg accgtgtgac accaggactg gctgaatggg 960
aaagagtaca agtgcaaaagt gtccaacaag gccttggccg ctctatcgaa aaaaactatc 1020
agcaaggcta agggacagcc gagggaaacc caagtctaca ccctgcccc ttacgcgaa 1080
gagatgacca agaataagat gtcgtgacac tgcctcgtca agggattcta cccctccgac 1140
attgcggtgg agtgggagtc caacggccag cccgagaaca actacaagac tactccgccc 1200
gtgctggact ccgacggcag cttctcctg tattccaagc tgaccgtgga caagtcccgg 1260
tggcagcaag gaaacgtgtt ctctgtctcg gtcatgcacg aagccctgca caaccactat 1320
acgcagaagt ccctgtcctt gagcccgggg aaa 1353

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SEQ ID NO: 119      moltype = AA length = 17
FEATURE            Location/Qualifiers
REGION             1..17
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic peptide"
source             1..17
                   mol_type = protein
                   organism = synthetic construct

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SEQUENCE: 119
VIESKGNYIF YADSVKG 17

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SEQ ID NO: 120      moltype = AA length = 6
FEATURE            Location/Qualifiers
REGION             1..6
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic peptide"
source             1..6
                   mol_type = protein
                   organism = synthetic construct

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SEQUENCE: 120
ESKGNV 6

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SEQ ID NO: 121      moltype = AA length = 8

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FEATURE Location/Qualifiers
 REGION 1..8
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
 source 1..8
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 121
 IESKGNFYI 8

SEQ ID NO: 122 moltype = AA length = 125
 FEATURE Location/Qualifiers
 REGION 1..125
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..125
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 122
 EVQLLESGGG LVQPGGSLRL SCAASGFTFS SYWMNWVRQA PGKGLEWVSV IESKGNFYIF 60
 ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARDR YSMIYSYGAG AFDYWGQGT 120
 VTSS 125

SEQ ID NO: 123 moltype = DNA length = 375
 FEATURE Location/Qualifiers
 misc_feature 1..375
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
 source 1..375
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 123
 gaggtgcaat tgctggaaag cggcggtggc ctggtgcagc cgggtggcag cctgcgtctg 60
 agctgcgcgg cgctcggatt caccctttct tcttactgga tgaactgggt gcgccaggcc 120
 ccgggcaaaag gtctcagatg ggtttccggt atcgaatcta aaggcaacta catcttctat 180
 gcggatagcg tgaaggccg ctttaccatc agccgcgata attcgaaaaa caccctgtat 240
 ctgcaaatga acagcctgcg tgcggaagat acggccgtgt attattgcgc gcgtgaccgt 300
 tactctatga tctactctta cgggtgctgt gctttcgatt actggggcca aggcaccctg 360
 gtgactgtta gctca 375

SEQ ID NO: 124 moltype = AA length = 455
 FEATURE Location/Qualifiers
 REGION 1..455
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..455
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 124
 EVQLLESGGG LVQPGGSLRL SCAASGFTFS SYWMNWVRQA PGKGLEWVSV IESKGNFYIF 60
 ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARDR YSMIYSYGAG AFDYWGQGT 120
 VTVSSASTKG PSVFPLAPSS KTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA 180
 VLQSSGLYSL SSVVTPSSS LGTQTYICNV NHKPSNTKVD KRVEPKSCDK THTCPPCPAP 240
 EAAGGSPSVFL FPPKPKDTLM ISRTPEVTCV VVDVSHEDPE VKFNWYVDGV EVHNAKTKPR 300
 EEQYNSTYRV VSVLTVLHQD WLNGKEYCK VSNKALPAPI EKTISKAKGQ PREPQVYTL 360
 PSREEMTKNQ VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFFLYSKLTV 420
 DKSRWQGGNV FSCSVMHEAL HNHYTQKSL LSPGK 455

SEQ ID NO: 125 moltype = DNA length = 1365
 FEATURE Location/Qualifiers
 misc_feature 1..1365
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
 source 1..1365
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 125
 gaggtgcaat tgctggaaag cggcggtggc ctggtgcagc cgggtggcag cctgcgtctg 60
 agctgcgcgg cgctcggatt caccctttct tcttactgga tgaactgggt gcgccaggcc 120
 ccgggcaaaag gtctcagatg ggtttccggt atcgaatcta aaggcaacta catcttctat 180
 gcggatagcg tgaaggccg ctttaccatc agccgcgata attcgaaaaa caccctgtat 240
 ctgcaaatga acagcctgcg tgcggaagat acggccgtgt attattgcgc gcgtgaccgt 300
 tactctatga tctactctta cgggtgctgt gctttcgatt actggggcca aggcaccctg 360
 gtgactgtta gctcagcctc caccagggtt ccatcggtct tccccctggc accctcctcc 420
 aagagcacct ctggggggcag agcggccctg ggctgcctgg tcaaggacta cttccccgaa 480
 ccggtgacgg tgctcgtggaa ctacggcgcc ctgaccagcg gcgtgcacac cttcccggt 540
 gtcctacagt cctcaggact ctactccctc agcagcgtgg tgaccgtgcc ctccagcagc 600
 ttgggcaccc agacctacat ctgcaacgtg aatcacaagc ccagcaacac caagggtggac 660
 aagagagttg agcccaaatc ttgtgacaaa actcacacat gccaccctg cccagcacct 720

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gaagcagcgg ggggaccgtc agtcttcttc tccccccaa aacccaagga caccctcatg 780
atctcccga cccctgaggt cacatgctg gtggtggacg tgagccacga agaccctgag 840
gtcaagtcca actggtacgt ggacggcgtg gaggtgcata atgccaagac aaagccgcgg 900
gaggagcagt acaacagcac gtaccgggtg gtcagcgtcc tcaccgtcct gcaccaggac 960
tggtggaatg gcaaggagta caagtgcagg gtctccaaca aagccctccc agccccatc 1020
gagaaaacca tctccaaagc caaagggcag ccccgagAAC cacaggtgta caccctgccc 1080
ccatcccggg aggagatgac caagaaccag gtcagcctga cctgcctggt caaaggcttc 1140
tatcccagcg acatcgccgt ggagtgggag agcaatgggc agcggagaa caactacaag 1200
accacgcctc ccgtgctgga ctccgacggc tccttcttcc tctacagcaa gtcaccctg 1260
gacaagagca ggtggcagca ggggaacgtc ttctcatgct ccgtgatgca tgaggctctg 1320
cacaaccact acacgcagaa gagcctctcc ctgtctccgg gtaaa 1365

SEQ ID NO: 126      moltype = AA length = 9
FEATURE            Location/Qualifiers
REGION            1..9
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..9
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 126
QQEWVKPRT                                     9

SEQ ID NO: 127      moltype = AA length = 6
FEATURE            Location/Qualifiers
REGION            1..6
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..6
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 127
EWVKPR                                             6

SEQ ID NO: 128      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION            1..107
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..107
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 128
DIQMTQSPSS LSASVGRVT ITCRASQGIS SYLAWYQQKP GKAPKLLIYT ASTLQSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ EWVKPRTFGQ GTKVEIK 107

SEQ ID NO: 129      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature      1..321
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polynucleotide"
source            1..321
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 129
gatatccaga tgaccagag cccgagcagc ctgagcgcca gcgtgggcga tcgcgtgacc 60
attacctgca gagccagcca gggatattct tcttacctgg cttggtacca gcagaaaccg 120
ggcaaaagcgc cgaaactatt aatctacact gcttctactc tgcaaaagcgg cgtgccgagc 180
cgctttagcg gcagcggatc cggcaccgat ttcaccctga ccattagctc tctgcaaccg 240
gaagactttg cgacctatta ttgccagcag gaatgggtta aaccgcgtac ctttgccagc 300
ggcacgaaag ttgaaattaa a 321

SEQ ID NO: 130      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 130
DIQMTQSPSS LSASVGRVT ITCRASQGIS SYLAWYQQKP GKAPKLLIYT ASTLQSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ EWVKPRTFGQ GTKVEIKRTV AAPSVFIFFP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 131      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature      1..642

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note = source = /note="Description of Artificial Sequence:
    Synthetic polynucleotide"
source      1..642
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 131
gatatccaga tgacccagag cccgagcagc ctgagcgcca gcgtggcgga tcgcgtgacc 60
attacctgca gagccagcca gggatattct tcttacctgg cttggtagca gcagaaccg 120
ggcaaacgcg cgaaactatt aatctacact gcttctactc tgcaaacgcg cgtgccgagc 180
cgcttttagc gcagcggatc cggcaccgat ttcacctga ccattagctc tctgcaaccg 240
gaagactttg cgacotatta ttgccagcag gaatgggtta aaccgcgtac ctttggccag 300
ggcacgaaag ttgaaattaa acgtacgggtg gccgctccca gcgtgttcat cttccccccc 360
agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caacttctac 420
ccccgggagc ccaagggtga gtggaagggt gacaacgccc tgcaagcgcg caacagccag 480
gaaagcgctca ccgagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgtccagcc ccgtgaccaa gagcttcaac cggggcgagt gt 642

SEQ ID NO: 132      moltype = DNA length = 375
FEATURE            Location/Qualifiers
misc_feature        1..375
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..375
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 132
caggtgcaat tgctggaaag cggcggtggc ctggtgcagc cgggtggcag cctgcgtctg 60
agctgcgcgg cgtccgatt  caccttttct tcttactgga tgaactgggt gcgccaggcc 120
ccaggcaaaag gtctcgagtg ggtttccgct atctcttctg acggttctta cactactat 180
gcggatagcg tgaaggcgcc ctttaccatc agccgcgata attcgaaaaa caccctgcat 240
ctgcaaatga acagcctgcg tgcggaagat acggccgtgt attattgcgc gcgtgaccgt 300
tactctatga tctactctta cgtgtctggt gctttcgatt actggggcca aggcaccctg 360
gtgactgtta gctca 375

SEQ ID NO: 133      moltype = DNA length = 1365
FEATURE            Location/Qualifiers
misc_feature        1..1365
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..1365
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 133
caggtgcaat tgctggaaag cggcggtggc ctggtgcagc cgggtggcag cctgcgtctg 60
agctgcgcgg cgtccgatt  caccttttct tcttactgga tgaactgggt gcgccaggcc 120
ccaggcaaaag gtctcgagtg ggtttccgct atctcttctg acggttctta cactactat 180
gcggatagcg tgaaggcgcc ctttaccatc agccgcgata attcgaaaaa caccctgcat 240
ctgcaaatga acagcctgcg tgcggaagat acggccgtgt attattgcgc gcgtgaccgt 300
tactctatga tctactctta cgtgtctggt gctttcgatt actggggcca aggcaccctg 360
gtgactgtta gctcagcctc acccaagggt ccactcggtct tccccctggc accctcctcc 420
aagagcaccct ctgggggcac agcggccctg ggctgcctgg tcaaggacta cttccccgaa 480
cgggtgacgg tgcgtggaa ctcaggcgcc ctgaccagcg gcgtgcacac cttcccggtc 540
gtcctacagt cctcaggact ctactccctc agcagcgtgg tgaccgtgcc ctccagcagc 600
ttgggcaccc agacotacat ctgcaacgtg aatcacaagc ccagcaacac caaggtggac 660
aagagagttg agcccaaatc ttgtgacaaa actcacacat gcccaccgtg cccagcacct 720
gaagcagcgg ggggaccgtc agtcttctc tccccccaa aacccaagga caccctcatg 780
atctccccga cccctgaggt cacatgcgtg gtggtggacg tgagccaaga agaccctgag 840
gtcaagttca actggtacgt ggacggcgtg gaggtgcata atgccaagac aaagccgcgg 900
gaggagcagt acaacagcac gtaccgggtg gtcagcgtcc tcaccgtcct gcaccaggac 960
tggctgaatg gcaaggagta caagtgcgaag gtctccaaca aagccctccc agcccccatc 1020
gagaaaacca tctccaaagc caaagggcag ccccgagaac cacaggtgta caccctgccc 1080
ccatccccgg aggagatgac caagaaccag gtcagcctga cctgcctggt caaaggcttc 1140
tatccagcgg acatcgccgt ggaagtggcg agcaatggcg agccggagaa caactacaag 1200
accacgcctc ccgtgctgga ctcgcagcgc tccttcttcc tctacagcaa gctcaccgtg 1260
gacaagagca ggtggcagca ggggaacgtc ttctcatgct ccgtgatgca tgaggctctg 1320
cacaaccact acacgcagaa gagcctctcc ctgtctccgg gtaaa 1365

SEQ ID NO: 134      moltype = AA length = 9
FEATURE            Location/Qualifiers
REGION            1..9
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source            1..9
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 134
QQTWRKPRT

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source          1..375
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 140
caagtgcagc tgcttgagag cgggtggcgga ctggtgcagc cagggggatc cttgcgcctg 60
tcatgcgctg cgtcgggggt caccctctcg tctactgga tgaactgggt cagacaggct 120
ccggggaagg gactcgaatg ggtgtccgcc atttctccg acggctccta cacttactac 180
gccgatagcg tcaagggccg gttcaccatc tcccgggaca attcgaagaa caccctgtac 240
ctccaaatga actcactcgc cgcgcaggac actgcgggtg attactgtgc ccgggatagg 300
tacagcatga tctactccta cggtgccgga gcctttgact actggggaca gggaaccctt 360
gtgaccgtgt ctagc                                     375

SEQ ID NO: 141      moltype = AA length = 455
FEATURE            Location/Qualifiers
REGION            1..455
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"

source            1..455
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 141
QVQLLESGGG LVQPGGSLRL SCAASGFTFS SYWMNWVRQA PGKGLEWVSA ISSDGSYTTY 60
ADSVKGRFTI SRDNSKNITLY LQMNSLRAED TAVYYCARDR YSMIYSYGAG AFDYWQGQTL 120
VTVSSASTKG PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA 180
VLQSSGLYSL SSVVTPPSSS LGTQTYICNV NHKPSNTKVD KRVEPKSCDK THTCPPCPAP 240
ELLGGPSVFL PPPPKKDTLM ISRTPEVTCV VVAVSHEDPE VKFNWYVDGV EVHNAKTKPR 300
EEQYNSTYRV VSVLTVLHQD WLNKEYKCK VSNKALAAPI EKTISKAKGQ PREPQVYTL 360
PSREEMTKNQ VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFLYSLKLT 420
DKSRWQQGNV FSCSVMHEAL HNHYTQKSLS LSPGK                                     455

SEQ ID NO: 142      moltype = DNA length = 1365
FEATURE            Location/Qualifiers
misc_feature       1..1365
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polynucleotide"

source            1..1365
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 142
caagtgcagc tgcttgagag cgggtggcgga ctggtgcagc cagggggatc cttgcgcctg 60
tcatgcgctg cgtcgggggt caccctctcg tctactgga tgaactgggt cagacaggct 120
ccggggaagg gactcgaatg ggtgtccgcc atttctccg acggctccta cacttactac 180
gccgatagcg tcaagggccg gttcaccatc tcccgggaca attcgaagaa caccctgtac 240
ctccaaatga actcactcgc cgcgcaggac actgcgggtg attactgtgc ccgggatagg 300
tacagcatga tctactccta cggtgccgga gcctttgact actggggaca gggaaccctt 360
gtgaccgtgt ctagcgcgtc cactaagggc ccgtcagtg tcccgctggc tccatcgtcg 420
aagtcacact ccggaggaa cgcagcactc ggttgccctg tcaaggacta cttccctgag 480
ccagtgcacg tgcgtggaa cagcggagcc ctgacttccg gcgtgcacac tttcccgcg 540
gtgctgcagt cctcgcgtct gtactccctt tcgtccgtgg tcacgtgccc gtcgtctagc 600
ctgggcaccc agacctacat gtgcaacgtg aaccacaagc cgtccaacac caaagtggat 660
aagcgggttg agccgaagtc ctgcgataag acacacacgt gcccgccatg tccagcgcc 720
gaattgcttg gccgaccttc cgtgttctgt ttcccgcccta agcccaagga caccctgatg 780
attagccgga tccccaagtc cactgtgtgt gtggtggcag tgtcccacga ggaccccgag 840
gtcaagttta attggtacgt ggaacggcgt gaagtgcaca acgccaagac taagcccgcg 900
gaggaacagt acaacagcac ctaccgggtc gtgtccgtgc tgacgtgctg gcaccaggac 960
tggctgaatg ggaagagata caagtgcata gtgtccaaca aggccttggc cgtcctatc 1020
gaaaaaacta tcagcaaggc taaggagacg ccgagggaa cccaagtcta caccctgccc 1080
ccttcacgcg aagagatgac caagaatcaa gtgtcgtgta cctgcctcgt caagggattc 1140
taccctccg acattgcggt ggagtgggag tccaacggcc agcccgagaa caactacaag 1200
actactccgc ccgtgctgga ctccgacggc agcttcttcc tgtattccaa gctgaccgtg 1260
gacaagtcgc ggtggcagca aggaacgtg ttctcctgct cggtcatgca cgaagccctg 1320
cacaaccact atacgcagaa gtccctgtcc ttgagcccg ggaaa                                     1365

SEQ ID NO: 143      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature       1..321
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polynucleotide"

source            1..321
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 143
gacatccaga tgaccacgac tccgtcctcc ctgtccgcat cagtggggga cagagtgacc 60
atcacttgtc gggcctccca aggcctctcg tcatacctgg cctggatca gcagaacccc 120
ggaaaggctc caaagctgct catctacacc gccctgactc tgcaatccgg agtgccttcc 180
cgcttctccg gatccggttc gggaaccgac ttcacctca ccattagcag ccttcagccg 240
gaagatttcc cgacctacta ctgccagcaa acctggcgga agccaggac atttgccag 300
ggcactaagg tcgagattaa g                                     321

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SEQ ID NO: 144      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature       1..642
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polynucleotide"
source            1..642
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 144
gacatccaga tgacccagtc tccgtcctcc ctgtccgcat cagtggggga cagagtgacc 60
atcaacttgtc gggcctccca aggcattctcg tcatacctgg cctgggtatca gcagaaaccc 120
ggaaaggctc caaagctgct catctacacc gcctcgactc tgcaatccgg agtgccttcc 180
cgcttctccg gatccggttc gggaaccgac ttcacctca ccattagcag ccttcagccg 240
gaagatttctg cgacotacta ctgccagcaa acctggcgga agcccaggac atttggccag 300
ggcactaagg tcgagattaa gcgtacgggtg gccgctccca gcgtgttcat cttccccccc 360
agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caacttctac 420
ccccgggagg ccaaggtgca gtggaagggtg gacaacgccc tgcagagcgg caacagccag 480
gagagcgtca ccgagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcataag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgtccagcc ccgtgaccaa gagcttcaac aggggcgagt gc 642

SEQ ID NO: 145      moltype = AA length = 9
FEATURE            Location/Qualifiers
REGION            1..9
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic peptide"
source            1..9
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 145
QQIWTVPRT 9

SEQ ID NO: 146      moltype = AA length = 6
FEATURE            Location/Qualifiers
REGION            1..6
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic peptide"
source            1..6
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 146
IWTVPR 6

SEQ ID NO: 147      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION            1..107
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polypeptide"
source            1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 147
DIQMTQSPSS LSASVGRVT ITCRASQGIS SYLAWYQQKP GKAPKLLIYT ASTLQSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFATYCCQQ IWTVPRTFGQ GTKVEIK 107

SEQ ID NO: 148      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature       1..321
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polynucleotide"
source            1..321
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 148
gatatccaga tgacccagag cccgagcagc ctgagcgcca gcgtggggcga tcgcgtgacc 60
attacctgca gagccagcca gggtatcttct tcttacctgg cttggtagca gcagaaaccc 120
ggcaaagcgc cgaactattt aatctacact gcttctactc tgcaaagcgg cgtgcccagc 180
cgcttttagcg gcagcggatc cggcaccgat ttcacctga ccattagctc tctgcaaccg 240
gaagactttg cgacctatta ttgccagcag atctggactg ttccgcgtac ctttggccag 300
ggcacgaaaag ttgaaattaa a 321

SEQ ID NO: 149      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polypeptide"
source            1..214
                   mol_type = protein
                   organism = synthetic construct

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SEQUENCE: 149
DIQMTQSPSS LSASVGRVT ITCRASQGIS SYLAWYQQKP GKAPKLLIYT ASTLQSGVPS 60
RFSGSGSGTD FTLTISLQP EDFATYYCQQ IWTVPRTFGQ GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSLSTLT 180
LSKADYERKHK VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 150 moltype = DNA length = 642
FEATURE Location/Qualifiers
misc_feature 1..642
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
source 1..642
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 150
gatatccaga tgacccagag cccgagcagc ctgagcgcca gcgtgggcca tcgctgacc 60
attacctgca gagccagcca gggattttct tcttacctgg cttggtagca gcagaaaccg 120
ggcaaacgcg cgaaactatt aatctacact gcttctactc tgcaaacgcg cgtgccgagc 180
cgctttagcg gcagcggatc cggcaccgat ttcacctga ccattagctc tctgcaaccg 240
gaagactttg cgacotatta ttgccagcag atctggactg ttccgcgtac ctttgccag 300
ggcacgaaag ttgaaattaa acgtacgggtg gccgctccca gcgtgttcat cttccccccc 360
agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caactttac 420
ccccgggagg ccaaggtgca gtggaagggtg gacaacgccc tgcaagcggg caacagccag 480
gaaagcgtca ccgagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgtccagcc ccgtgaccaa gagcttcaac cggggcgagt gt 642

SEQ ID NO: 151 moltype = AA length = 17
FEATURE Location/Qualifiers
REGION 1..17
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
source 1..17
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 151
SIGGQGGMTL YADSVKG 17

SEQ ID NO: 152 moltype = AA length = 6
FEATURE Location/Qualifiers
REGION 1..6
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
source 1..6
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 152
GGQGGM 6

SEQ ID NO: 153 moltype = AA length = 8
FEATURE Location/Qualifiers
REGION 1..8
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
source 1..8
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 153
IGGQGGMT 8

SEQ ID NO: 154 moltype = AA length = 121
FEATURE Location/Qualifiers
REGION 1..121
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
source 1..121
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 154
EVQLLESQGG LVQPGGSLRL SCAASGFTFN THYIHWVRQA PGKGLEWVSS IGGQGGMTLY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARER GYVYHMFDP WGQGTLLVTVS 120
S 121

SEQ ID NO: 155 moltype = DNA length = 363
FEATURE Location/Qualifiers
misc_feature 1..363
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
source 1..363

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 155
gaggtgcaat tgctggaaag cggcggtggc ctggtgcagc cgggtggcag cctgcgtctg 60
agctgcgcgg cgctcggatt cacccttaac actcattaca tccattgggt gcgccaggcc 120
cccggcaaaag gtctcagatg ggtttcctct atcgggtggc agggcggtat gactctgtat 180
gcggatagcg tgaaggcccg cttaccatc agccgcgata attcgaaaaa caccctgtat 240
ctgcaaatga acagcctgcg tgcggaagat acggcctgtg attattgcgc gcgtgaacgt 300
ggttacgttt actaccatat gttcgatccg tggggccaag gcaccctggt gactgttagc 360
tca 363

SEQ ID NO: 156      moltype = AA length = 451
FEATURE            Location/Qualifiers
REGION              1..451
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source              1..451
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 156
EVQLLESGGG LVQPGGSLRL SCAASGFTFN THYIHWVRQA PGKGLEWVSS IGGQGGMTLY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARER GYVYHMFDP WGQGTLVTVS 120
SASTKGPSVF PLAPSSKSTS GGTAAALGCLV KDYPPEPVTV SWNSGALTSG VHTFPAVLQS 180
SGLYSLSSVV TVPSSSLGTQ TYICNVNHKP SNTKVDKRVK PKSCDKTHTC PPCPAPEAAG 240
GPSVFLFPPK PKDTLMISRT PEVTCVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY 300
NSTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE 360
EMTKNQVSLT CLVKGFYPSD IAVEWESNGQ PENNYKTPP VLDSGDGSFFL YSKLTVDKSR 420
WQQGNVFPSC VMHEALHNHY TQKSLSLSPG K 451

SEQ ID NO: 157      moltype = DNA length = 1353
FEATURE            Location/Qualifiers
misc_feature        1..1353
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..1353
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 157
gaggtgcaat tgctggaaag cggcggtggc ctggtgcagc cgggtggcag cctgcgtctg 60
agctgcgcgg cgctcggatt cacccttaac actcattaca tccattgggt gcgccaggcc 120
cccggcaaaag gtctcagatg ggtttcctct atcgggtggc agggcggtat gactctgtat 180
gcggatagcg tgaaggcccg cttaccatc agccgcgata attcgaaaaa caccctgtat 240
ctgcaaatga acagcctgcg tgcggaagat acggcctgtg attattgcgc gcgtgaacgt 300
ggttacgttt actaccatat gttcgatccg tggggccaag gcaccctggt gactgttagc 360
tcagcctcca ccaagggtcc atcgggtctt cccctggcac cctcctccaa gagcacctct 420
gggggcacag cggcctctgg ctgcctggtc aaggactact tcccgaacc ggtgacggtg 480
tcgtggaaact caggcgccct gaccagcggc gtgcacacct tcccggtgt cctacagtcc 540
tcaggactct actcctcag cagcgtggtg accgtgccct ccagcagctt gggcaccagg 600
acctacatct gcaacgtgaa tcacaagccc agcaacacca aggtggacaa gagagttgag 660
cccaaatctt gtgacaaaac tcacacatgc ccaccgtgcc cagcacctga agcagcgggg 720
ggaccgtcag tcttctcttt cccccaaaaa cccaaggaca cctcctatgat ctcccgagcc 780
cctgaggtga catgcgtgtg ggtggacgtg agccacgaag accctgaggt caagtccaac 840
tggtacgtgg acggcgtgga ggtgcataat gccaaagaca agccgcggga ggagcagtac 900
aacagcacgt accgggtggt cagcgtcctc accgtcctgc accaggactg gctgaatggc 960
aaggagtaca agtgcaaggt ctccaacaaa gccctcccag ccccatcga gaaaaccatc 1020
tccaaagcca aaggcgagcc ccgagaacca caggtgtaca ccctgcccc atcccgagg 1080
gagatgacca agaaccaggt cagcctgacc tgcctggtca aaggcttcta tcccagcgac 1140
atcgccgtgg agtgggagag caatgggcag ccggagaaca actacaagac cagcctccc 1200
gtgctggact ccgacggctc cttctctctc tacagcaagc tcaccgtgga caagagcagg 1260
tggcagcagg ggaacgtctt ctcagtctcc gtgatgcatg aggtctctga caaccactac 1320
acgcagaaga gcctctccct gtctccgggt aaa 1353

SEQ ID NO: 158      moltype = DNA length = 363
FEATURE            Location/Qualifiers
misc_feature        1..363
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..363
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 158
gaagtgcagc tcctggagtc ggggtggcga ctggtgcagc ctggcggatc actgcggctg 60
tcatgtgccg cgagcggggt tactttcaac acccactaca tccactgggt ccgccaaagt 120
cccgaaagg gactcgaatg ggtgtcctcc attggtggac agggcggcac gaccctttac 180
gcggatagcg tgaaggggag gttcaccatc tcccgcgaca acagcaagaa caccctgtac 240
ctccaaatga actcgcttcg ggcgaggac actgctgtgt actattgcgc aagagagcgg 300
ggctacgtgt actaccacat gttcgacca tggggacagg gaacgtggt caccgtgtcc 360
tcc 363

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SEQ ID NO: 159      moltype = AA  length = 451
FEATURE            Location/Qualifiers
REGION            1..451
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..451
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 159
EVQLLESGGG LVQPGGSLRL SCAASGFTFN THYIHWVRQA PGKGLEWVSS IGGQGGMPLY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARER GYVYYHMFDP WGQGTTLVTVS 120
SASTKGPSVF PLAPSSKSTS GGTAAALGCLV KDYPPEPVTV SWNSGALTSG VHTFPAVLQS 180
SGLYSLSSVV TVPSSSLGTQ TYICNVNHKP SNTKVDKRVK PKSCDKTHTC PPCPAPELLG 240
GPSVFLFPKP PKDTLMISRT PEVTCVVVAV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY 300
NSTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALAAPIEKTI SKAKGQPREP QVYTLPPSRE 360
EMTKNQVSLT CLVKGFPYPSD IAVEWESNGQ PENNYKTPPP VLDSDGSFFL YSKLTVDKSR 420
WQQGNVFSCT VMHEALHNNH TQKSLSLSPG K 451

```

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SEQ ID NO: 160      moltype = DNA  length = 1353
FEATURE            Location/Qualifiers
misc_feature       1..1353
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polynucleotide"
source            1..1353
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 160
gaagtgcagc tctgggagtc ggggtggcgga ctggtgcagc ctggcggatc actgcggctg 60
tcatgtgccg cgagcggggt tactttcaac acccactaca tccactgggt ccgccaaagt 120
cccgaaagg gactcgaatg ggtgtcctcc attggtggac agggcggcat gaccctttac 180
gcggatagcg tgaaggggag gttcaccatc tcccgcgaca acagcaagaa caccctgtac 240
ctccaaatga actcgcttcg ggccgaggac actgcccgtg actattgcgc aagagagcgg 300
ggctacgtgt actaccacat gttcgaccca tggggacagg gaacgctggt caccgtgtcc 360
tccgcctcca ctaaggggcc gtcagtgttc ccgctggctc catcgtcgaa gtccacctcc 420
ggaggaaccg cagcactcgg ttgctgggtc aaggactact tccctgagcc agtgaccgtg 480
tcgtggaaca gcggagccct gacttccggc gtgcacactt ttcccgggt gctgcagtcc 540
tccggtctgt actcccttcc gtcgtgggtc accgtgccgt cgtctagcct gggcaccocg 600
acctacatct gcaacgtgaa ccacaagccg tcccaacacca aagtggataa gcgggtggag 660
ccgaagtcct gcgataagac acacacgtgc ccgccatgtc cagcgcctga attgcttggc 720
ggaccttcgg tgttctctgt cccgcctaag cccaaggaca ccttgatgat tagccggact 780
cccgaagtca cctgtgtggt ggtggcagtg tcccacgagg accccgaggt caagtttaat 840
tggtacgtgg acggcgtcga agtgacacac gcccaagacta agccccggga ggaacagtac 900
aacagcactc accgggtcgt gtcctgtctg accgtgtctg accaggactg gctgaatggg 960
aaagagtaca agtgcaaaagt gtcccaacaag gccttggcgg ctctatcgaa aaaaactatc 1020
agcaaggcta agggacagcc gagggaaacc caagtctaca ccctgcccc ttacgcgcaa 1080
gagatgacca agaataaagt gtccgtgacc tgccctcgta agggattcta cccctccgac 1140
attgcggtgg agtgggagtc caacggccag cccgagaaca actacaagac tactccgccc 1200
gtgctggact ccgacggcag cttcttctcg tattccaagc tgaccgtgga caagtcccgg 1260
tggcagcaag gaaacgtggt ctctgtctcg gtcattgcacg aagccctgca caaccactat 1320
acgcagaagt cccctgtcctt gagccccggg aaa 1353

```

```

SEQ ID NO: 161      moltype = AA  length = 121
FEATURE            Location/Qualifiers
REGION            1..121
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..121
                  mol_type = protein
                  organism = synthetic construct

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```

SEQUENCE: 161
EVQLLESGGG LVQPGGSLRL SCAASGFTFS THYIHWVRQA PGKGLEWVSS IGGQGGMPLY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARER GYVYYHMFDP WGQGTTLVTVS 120
S 121

```

```

SEQ ID NO: 162      moltype = DNA  length = 363
FEATURE            Location/Qualifiers
misc_feature       1..363
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polynucleotide"
source            1..363
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 162
gaagtgcagc tctgggagtc ggggtggcgga ctggtgcagc ctggcggatc actgcggctg 60
tcatgtgccg cgagcggggt tactttctcc acccactaca tccactgggt ccgccaaagt 120
cccgaaagg gactcgaatg ggtgtcctcc attggtggac agggcggcat gaccctttac 180
gcggatagcg tgaaggggag gttcaccatc tcccgcgaca acagcaagaa caccctgtac 240
ctccaaatga actcgcttcg ggccgaggac actgcccgtg actattgcgc aagagagcgg 300
ggctacgtgt actaccacat gttcgaccca tggggacagg gaacgctggt caccgtgtcc 360

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tcc                                     363

SEQ ID NO: 163           moltype = AA  length = 451
FEATURE                 Location/Qualifiers
REGION                 1..451
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polypeptide"
source                 1..451
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 163
EVQLLESGGG LVQPGGSLRL SCAASGFTFS THYIHVVRQA PGKGLEWVSS IGGQGGMTLY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARER GYVYHMFDP WGQGTLVTVS 120
SASTKGPSVF PLAPSSKSTS GGTAAALGCLV KDYFPEPVTV SWNSGALTSG VHTFPAVLQS 180
SGLYSLSSVV TVPSSSLGTQ TYICNVNHKP SNTKVDKRVK PKSCDKTHTC PPCPAPELLG 240
GPSVFLFPPK PKDTLMISRT PEVTCVVAVV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY 300
NSTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALAAPIEKTI SKAKGQPREP QVYTLPPSRE 360
EMTKNQVSLT CLVKGFYPSD IAEWESNGQ PENNYKTPP VLDSGDGSFFL YSKLTVDKSR 420
WQQGNVFPSC VMHEALHNYH TQKSLSLSPG K 451

SEQ ID NO: 164           moltype = DNA  length = 1353
FEATURE                 Location/Qualifiers
misc_feature            1..1353
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polynucleotide"
source                 1..1353
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 164
gaagtgcagc tcctggagtc ggggtggcga ctggtgcagc ctggcggatc actgcggctg 60
tcatgtgccg cgagcggggt tactttctcc acccactaca tccactgggt ccgccaaagt 120
cccgaaaagg gactcgaatg ggtgtcctcc attggtggac agggcggcat gaccctttac 180
gcggatagcg tgaaggggag gtccaccatc tcccgcgaca acagcaagaa caccctgtac 240
ctccaaatga actcgcttcg ggcgaggagc actgcccgtg actattgcgc aagagagcgg 300
ggctacgtgt actaccacat gttcgaccca tggggacagg gaacgctggt caccgtgtcc 360
tccgcctcca ctaagggccc gtccagtgtc ccgctggctc catcgtcgaa gtccacctcc 420
ggaggaaccc gcgactcggg ttgctgggtc aaggactact tccctgagcc agtgaccgtg 480
tcgtggaaca gcggagccct gaattccggc gtgcacactt ttcccgcggg gctgcagtcc 540
tccggtctgt actcccttcc gtccgtgggt accgtgccgt cgtctagcct gggcacccag 600
acctacatct gcaacgtgaa ccacaagccg tccaacacca aagtggataa gcgggtggag 660
ccgaagtccg gcgataagac acacacgtgc ccgccatgtc cagcgctga attgcttggc 720
ggaccttcgg tgttccgtgt ccgcgctaag cccaaggaca ccttgatgat tagccggact 780
ccgaagtcca cctgtgtggt ggtggcagtg tcccacgagg acccgagggt caagtttaat 840
tggtacgtgg acggcgctga agtgacaca gcccccggga ggaacagtac 900
aacagcacct accgggtcgt gtccgtgctg accgtgtgac accaggactg gctgaatggg 960
aaagagtaca agtgcaaaagt gtccaacaag gccttggccg ctccatcgca aaaaactatc 1020
agcaaggcta agggacagcc gagggaaacc caagtctaca ccctgcccc ttcacgcgaa 1080
gagatgacca agaatacaagt gtccgtgacc tgcctcgta agggattcta cccctccgac 1140
attgcggtgg agtgggagtc caacggccag ccgagaaca actacaagac tactccgccc 1200
gtgctggact ccgacggcag cttcttctcg tattccaagc tgaccgtgga caagtcccgg 1260
tggcagcaag gaaacgtgtt ctctgctcgt gtcatgcacg aagccctgca caaccactat 1320
acgcagaagt ccctgtcctt gagcccgggg aaa 1353

SEQ ID NO: 165           moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                 1..10
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic peptide"
source                 1..10
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 165
GFTFQTHYIH                                     10

SEQ ID NO: 166           moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                 1..7
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic peptide"
source                 1..7
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 166
GFTFQTH                                     7

SEQ ID NO: 167           moltype = AA  length = 8
FEATURE                 Location/Qualifiers
REGION                 1..8
                        note = source = /note="Description of Artificial Sequence:

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source          "Synthetic peptide"
                1..8
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 167
GFTFQTHY                               8

SEQ ID NO: 168          moltype = AA  length = 121
FEATURE                Location/Qualifiers
REGION                1..121
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source                1..121
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 168
EVQLLESGGG LVQPGGSLRL SCAASGFTFQ THYIHWVRQA PGKGLEWVSS IGGQGGMTLY 60
ADSVKGRFTI SRDNSKNTLY LQMSLRAED TAVYYCARER GYVYYHMFDP WGQGTLVTVS 120
S                                                    121

SEQ ID NO: 169          moltype = DNA  length = 363
FEATURE                Location/Qualifiers
misc_feature          1..363
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source                1..363
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 169
gaagtgcagc tcctggagtc ggggtggcgga ctggtgcagc ctggcggatc actgcggtcg 60
tcatgtgccg cgagcggggt tactttccag acccactaca tccactgggt cgcgcaagct 120
cccgaaagg gactcgaatg ggtgtcctcc attggtggac agggcggcat gaccctttac 180
gcggatagcg tgaaggggag gttcaccatc tcccgcgaca acagcaagaa caccctgtac 240
ctccaaatga actcgcttcg ggcgaggac actgccgtgt actattgcgc aagagagcgg 300
ggctacgtgt actaccacat gttcgaccca tggggacagg gaacgctggt caccgtgtcc 360
tcc                                                    363

SEQ ID NO: 170          moltype = AA  length = 451
FEATURE                Location/Qualifiers
REGION                1..451
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source                1..451
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 170
EVQLLESGGG LVQPGGSLRL SCAASGFTFQ THYIHWVRQA PGKGLEWVSS IGGQGGMTLY 60
ADSVKGRFTI SRDNSKNTLY LQMSLRAED TAVYYCARER GYVYYHMFDP WGQGTLVTVS 120
SASTKGPSVF PLAPSSKSTS GGTAAAGCLV KDYFPEPVTV SWNSGALTSG VHTFPAVLQS 180
SGLYSLSSVV TVPSSSLGTQ TYICNVNHPK SNTKVDKRVK PKSCDKHTHC PPCPAPELLG 240
GPSVFLFPPK PKDTLMISRT PEVTCVVAVV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY 300
NSTYRVVSVL TVLHQDWLNG KEYCKVSNK ALAAPIEKTI SKAKGQPREP QVYTLPPSRE 360
EMTKNQVSLT CLVKGFYPSD IAVEWESNGQ PENNYKTPPP VLDSGDGSFFL YSKLTVDKSR 420
WQQGNVFSKS VMHEALHNHY TQKSLSLSPG K                                                    451

SEQ ID NO: 171          moltype = DNA  length = 1353
FEATURE                Location/Qualifiers
misc_feature          1..1353
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source                1..1353
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 171
gaagtgcagc tcctggagtc ggggtggcgga ctggtgcagc ctggcggatc actgcggtcg 60
tcatgtgccg cgagcggggt tactttccag acccactaca tccactgggt cgcgcaagct 120
cccgaaagg gactcgaatg ggtgtcctcc attggtggac agggcggcat gaccctttac 180
gcggatagcg tgaaggggag gttcaccatc tcccgcgaca acagcaagaa caccctgtac 240
ctccaaatga actcgcttcg ggcgaggac actgccgtgt actattgcgc aagagagcgg 300
ggctacgtgt actaccacat gttcgaccca tggggacagg gaacgctggt caccgtgtcc 360
tcgcctcca ctaagggccccc gtcagtgttc ccgctggctc catcgctgaa gtccacctcc 420
ggaggaaacc cagcactcgg ttgcttggtc aaggactact tccctgagcc agtgaccgtg 480
tcgtggaaca gcggagccct gacttccggc gtgcacactt ttcccgggt gctgcagtcc 540
tccggtctgt actcccttcc gtccttggtc accgtgccgt cgtctagcct gggcaccacc 600
acctacatct gcaacgtgaa ccacaagccg tccaacacca aagtggataa gcgggtggag 660
cgaagtcctt gcgataagac acacacgtgc ccgccatgtc cagcgctga attgcttggc 720
ggaccttccg tgttccgtgt cccgcctaag cccaaggaca ccttgatgat tagccggact 780
ccgaagtca cctgtgtggt ggtggcagtg tcccacgagg accccgaggt caagttaaat 840
tggtacgtgg acggcgtcga agtgcacaa cccaagacta agccccggga ggaacagtac 900

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aacagcacct accgggtcgt gtcctgtctg accgtgtctg accaggactg gctgaatggg 960
aaagagtaca agtgcaaatg gtccaacaag gccttggccg ctctatcga aaaaactatc 1020
agcaaggcta agggacagcc gagggaaacc caagtctaca ccttgccccc ttcacgcgaa 1080
gagatgacca agaataaagt gtcgtgtgacc tgctctgtca agggattcta cccctccgac 1140
attgcgggtg agtgggagtc caacggccag cccgagaaca actacaagac tactccgccc 1200
gtgtctggact ccgacggcag cttcttctctg tattccaagc tgaccgtgga caagtcccgg 1260
tggcagcaag gaaacgtgtt ctctgtctcg gtcattgcacg aagccctgca caaccactat 1320
acgcagaagt ccctgtcctt gagccccggg aaa 1353

SEQ ID NO: 172      moltype = AA length = 9
FEATURE            Location/Qualifiers
REGION            1..9
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source            1..9
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 172
QQEWAKPRT                      9

SEQ ID NO: 173      moltype = AA length = 6
FEATURE            Location/Qualifiers
REGION            1..6
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source            1..6
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 173
EWAKPR                      6

SEQ ID NO: 174      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION            1..107
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source            1..107
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 174
DIQMTQSPSS LSASVGRVT ITCRASQGIS SYLAWYQQKP GKAPKLLIYT ASTLQSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ EWAKPRTFGQ GTKVEIK 107

SEQ ID NO: 175      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature      1..321
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source            1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 175
gatatccaga tgaccagag ccgagcagc ctgagcgcca gcgtgggcga tcgctgtgacc 60
attacctgca gagccagcca gggatattct tcttacctgg cttggtagca gcagaaccg 120
ggcaaaagcgc cgaaactatt aatctacact gcttctact tgcaaaagcgg cgtgccgagc 180
cgctttagcg gcagcggatc cggcaccgat ttcacctga ccattagctc tctgcaaccg 240
gaagactttg cgacctatta ttgccagcag gaatgggcta aaccgcgtac ctttgccag 300
ggcacgaaag ttgaaattaa a 321

SEQ ID NO: 176      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source            1..214
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 176
DIQMTQSPSS LSASVGRVT ITCRASQGIS SYLAWYQQKP GKAPKLLIYT ASTLQSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ EWAKPRTFGQ GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 177      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature      1..642
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source            1..642

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 177
gatatccaga tgacccagag cccgagcagc ctgagcgcca gcgtgggcga tcgcgtgacc 60
attacctgca gagccagcca gggatattct tcttacctgg cttggtagca gcagaaaccg 120
ggcaaacgcg cgaaactatt aatctacact gcttctactc tgcaaacgcg cgtgccgagc 180
cgctttagcg gcagcggatc cggcaccgat ttcaccctga ccattagctc tctgcaaccg 240
gaagactttg cgacctatta ttgccagcag gaatgggcta aaccgcgtac ctttggccag 300
ggcacgaaaag ttgaaattaa acgtacgggtg gccgctccca gcgtgttcat cttccccccc 360
agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caacttctac 420
ccccgggagg ccaagggtgca gtggaagggtg gacaacgccc tgcaagcgcg caacagccag 480
gaaagcgtca ccgagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgtccagcc ccgtgaccaa gagcttcaac cggggcgagt gt 642

SEQ ID NO: 178      moltype = AA length = 9
FEATURE            Location/Qualifiers
REGION            1..9
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..9
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 178
QQSWTRPRT 9

SEQ ID NO: 179      moltype = AA length = 6
FEATURE            Location/Qualifiers
REGION            1..6
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..6
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 179
SWTRPR 6

SEQ ID NO: 180      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION            1..107
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..107
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 180
DIQMTQSPSS LSASVGDVRT ITCRASQGIS SYLAWYQQKP GKAPKLLIYT ASTLQSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ SWTRPRTFGQ GTKVEIK 107

SEQ ID NO: 181      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature      1..321
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polynucleotide"
source            1..321
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 181
gatatccaga tgacccagag cccgagcagc ctgagcgcca gcgtgggcga tcgcgtgacc 60
attacctgca gagccagcca gggatattct tcttacctgg cttggtagca gcagaaaccg 120
ggcaaacgcg cgaaactatt aatctacact gcttctactc tgcaaacgcg cgtgccgagc 180
cgctttagcg gcagcggatc cggcaccgat ttcaccctga ccattagctc tctgcaaccg 240
gaagactttg cgacctatta ttgccagcag tcttggaactc gtccgcgtac ctttggccag 300
ggcacgaaaag ttgaaattaa a 321

SEQ ID NO: 182      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 182
DIQMTQSPSS LSASVGDVRT ITCRASQGIS SYLAWYQQKP GKAPKLLIYT ASTLQSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ SWTRPRTFGQ GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQV ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

```

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```

SEQ ID NO: 183      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature        1..642
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..642
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 183
gatatccaga tgaccagag cccgagcagc ctgagcgcca gcgtgggcga tcgcgtgacc 60
attacctgca gagccagcca gggatattct tcttacctgg cttggtacca gcagaaaccg 120
ggcaaacgcg cgaaactatt aatctacact gcttctactc tgcaaacgcg cgtgccgagc 180
cgcttttagcg gcagcggatc cggcaccgat ttcacctga ccattagctc tctgcaaccg 240
gaagactttg cgacctatta ttgccagcag tcttggactc gtccgcgtac ctttggccag 300
ggcacgaaaag ttgaaattaa acgtacgggtg gccgctccca gcgtgttcat cttccccccc 360
agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caacttctac 420
ccccgggagg ccaaggtgca gtggaagggt gacaacgccc tgcaagcgcg caacagccag 480
gaaagcggtca ccgagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgtccagcc ccgtgaccaa gagcttcaac cggggcgagt gt 642

```

```

SEQ ID NO: 184      moltype = AA length = 9
FEATURE            Location/Qualifiers
REGION              1..9
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..9
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 184
QQIWMAPRT 9

```

```

SEQ ID NO: 185      moltype = AA length = 6
FEATURE            Location/Qualifiers
REGION              1..6
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..6
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 185
IWMAPR 6

```

```

SEQ ID NO: 186      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION              1..107
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source              1..107
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 186
DIQMTQSPSS LSASVGRVT ITCRASQGIS SYLAWYQQKP GKAPKLLIYT ASTLQSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ IWMAPRTFGQ GTKVEIK 107

```

```

SEQ ID NO: 187      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 187
gatatccaga tgaccagag cccgagcagc ctgagcgcca gcgtgggcga tcgcgtgacc 60
attacctgca gagccagcca gggatattct tcttacctgg cttggtacca gcagaaaccg 120
ggcaaacgcg cgaaactatt aatctacact gcttctactc tgcaaacgcg cgtgccgagc 180
cgcttttagcg gcagcggatc cggcaccgat ttcacctga ccattagctc tctgcaaccg 240
gaagactttg cgacctatta ttgccagcag atctggatgg ctccgcgtac ctttggccag 300
ggcacgaaaag ttgaaattaa a 321

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SEQ ID NO: 188      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION              1..214
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source              1..214
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 188
DIQMTQSPSS LSASVGRVT ITCRASQGIS SYLAWYQQKP GKAPKLLIYT ASTLQSGVPS 60
RFSGSGSGTD FTLTISLQP EDFATYYCQQ IWMAPRTFGQ GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSLSTLT 180
LSKADYERKHK VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 189      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature        1..642
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..642
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 189
gatatccaga tgacccagag cccgagcagc ctgagcgcca gcgtgggcga tcgcgtgacc 60
attacctgca gagccagcca gggattttct tcttacctgg cttggtagca gcagaaaccg 120
ggcaaacgag cgaaactatt aatctacact gcttctactc tgcaaacgag cgtgccgagc 180
cgctttagcg gcagcggatc cggcaccgat ttcacctga ccattagctc tctgcaaccg 240
gaagactttg cgacotatta ttgccagcag atctggatgg ctccgcgtac ctttgccag 300
ggcacgaaag ttgaaattaa acgtacgggtg gccgctccca gcgtgttcat cttccccccc 360
agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caactttac 420
ccccgggagg ccaaggtgca gtggaagggt gacaacgccc tgcaagcggg caacagccag 480
gaaagcgta cagagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgtccagcc ccgtgaccaa gagcttcaac cggggcgagt gt 642

SEQ ID NO: 190      moltype = AA length = 17
FEATURE            Location/Qualifiers
REGION              1..17
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..17
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 190
AISKGSYTY YADSVKG 17

SEQ ID NO: 191      moltype = AA length = 6
FEATURE            Location/Qualifiers
REGION              1..6
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..6
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 191
SSKGSY 6

SEQ ID NO: 192      moltype = AA length = 8
FEATURE            Location/Qualifiers
REGION              1..8
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..8
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 192
ISSKGSYT 8

SEQ ID NO: 193      moltype = AA length = 125
FEATURE            Location/Qualifiers
REGION              1..125
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source              1..125
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 193
QVQLLESGLG LVQPGGSLRL SCAASGFTFS SYWMNWVRQA PGKGLEWVSA ISSKGSYTTY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARDR YSMIYSYGAG AFDYWGQGT 120
VTVSS 125

SEQ ID NO: 194      moltype = DNA length = 375
FEATURE            Location/Qualifiers
misc_feature        1..375
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..375

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 194
caagttcagc tccttgagtc tggggggggc ctggtgcaac ctgggggctc tctgcggctt 60
tcatgtgcgg cctcagggtt cactttcagc tcatactgga tgaattgggt acgccaaagt 120
ccaggcaaaag gactcgaatg ggtaagcgct atatccagca aaggagagcta tacctattac 180
gcggattccg ttaagggcag gtctactata tcccgcgaca actccaaaaa tactttgtat 240
ctgcaaatga attcctccg agccgaagat accgcagtat attactgtgc gagggacagg 300
tactccatga tttacagcta cggtgccggt gctttcgatt attggggaca ggggacactt 360
gtgaccgtca gttct 375

SEQ ID NO: 195      moltype = AA length = 455
FEATURE            Location/Qualifiers
REGION              1..455
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source              1..455
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 195
QVQLLESGGG LVQPGGSLRL SCAASGFTFS SYWMNWVRQA PGKGLEWVSA ISSKGSYTTY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARDR YSMIYSYGAG AFDYWGQGT 120
VTSSASTKG PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA 180
VLQSSGLYSL SSVVTPSSS LGTQTYICNV NHKPSNTKVD KRVEPKSCDK THTCPPCPAP 240
EAAGGPSVFL FPPKPKDTLM ISRTPEVTCV VVDVSHEDPE VKFNWYVDGV EVHNAKTKPR 300
EEQYNSTYRV VSVLTVLHQD WLNGKEYKCK VSNKALPAPI EKTISKAKGQ PREPQVYTL 360
PSREEMTKNQ VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFFLYSKLTV 420
DKSRWQQGNV PSCSVMEHAL HNHYTQKSLS LSPGK 455

SEQ ID NO: 196      moltype = DNA length = 1365
FEATURE            Location/Qualifiers
misc_feature        1..1365
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..1365
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 196
caagttcagc tccttgagtc tggggggggc ctggtgcaac ctgggggctc tctgcggctt 60
tcatgtgcgg cctcagggtt cactttcagc tcatactgga tgaattgggt acgccaaagt 120
ccaggcaaaag gactcgaatg ggtaagcgct atatccagca aaggagagcta tacctattac 180
gcggattccg ttaagggcag gtctactata tcccgcgaca actccaaaaa tactttgtat 240
ctgcaaatga attcctccg agccgaagat accgcagtat attactgtgc gagggacagg 300
tactccatga tttacagcta cggtgccggt gctttcgatt attggggaca ggggacactt 360
gtgaccgtca gttctgcaag taccaaaagg cgtctgttt tccattggc tccctcatcc 420
aagagcacga gtggaggcac cgccgcgctg gtagtccttg tgaagacta tttcccgag 480
cccgtagccg ttagctggaa cagcggcgct cttaccagtg gcgttcacac attcccagct 540
gttttgcaag catccgggct ctactctctc tcatccgtgg tcaccgtgcc gtcagtgtct 600
ttgggcaccc agactacat ctgtaacgta aatcacaaac ctagtaatac taaggtggac 660
aagcgagttg aaacgaagag ctgtgataag acacatactt gtccaccatg tccggcaccc 720
gaggcagcgg gggggcccag tgtttttctc tccccacca agcccaaaaga cacattgatg 780
atctcacgaa cccagaggtt aacttgtgtc gtggtagatg taagccatga ggacccgaa 840
gttaagtcca attggtatgt tgacggtgta gaggtgcaca atgccccaac taaaccccg 900
gaggagcaat acaactcaac ttacagagtc gtatccgtgc tgaccgtttt gcaccaggat 960
tggttgaatg gtaagggaata caaatgtaaa gtgagcaata aagctctccc agcgcctatc 1020
gagaagacca ttagcaaaagc caaggggtcaa cccagggaac cccagggtata tacgtgcgca 1080
ccctcaaggg aagagatgac aaagaatcaa gtgtcactga cgtgtcttgt caagggtttc 1140
tatctagcgc acattgcggt ggaatgggag tcaaatgggc aaccgcagaa caactacaag 1200
actactcttc ccgtcttgga cagcgacggc tctttcttcc tgtatagtaa actgaccgtc 1260
gataaaagta ggtggcagca ggggaatgtc tttagtgtct ctgtcatgca tgaggcgctc 1320
cataaccact acacccaaaa atctttgagc ttgagccctg ggaaa 1365

SEQ ID NO: 197      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 197
gacattcaaa tgacacaaag tccgtccagt cttagtgtct ctgtgggcga tagggtcacc 60
atcacttgtc gggcgctctc ggggatcagc tcttacttgg catggtatca aaaaagcca 120
ggaaaagcac ctaaattgct tatttatata cgcgtccacc tccagtccag agtgccctagt 180
aggttctcag gctctgggtc cggtactgac ttcacgtgta ctatatcaag cttgcaaccc 240
gaagattttg caacatacta ctgccaacag acatggagga agccaagaac ttccggtcag 300
ggaacgaaag ttgagataaa g 321

SEQ ID NO: 198      moltype = DNA length = 642

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FEATURE                Location/Qualifiers
misc_feature            1..642
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polynucleotide"
source                  1..642
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 198
gacattcaaa tgacacaaag tccgtccagt cttagtgcct ctgtgggcga tagggtcacc 60
atcacttgct gggcgctctca ggggatcagc tcttacttgg catggatatca acaaaagcca 120
ggaaaagcac ctaaattgct tatttataca gcgtccaccc tccagtcagg agtgccctagt 180
aggttctcag gctctgggtc cggtactgac ttcacgctga ctatatcaag cttgcaaccc 240
gaagattttg caacatacta ctgccaacag acatggagga agccaagaac tttcggtcag 300
ggaacgaaag ttgagataaa gcgcactgtc gcagcacctt ccgtgttcat ttccccgcct 360
tccgacgagc agcttaaatc agggaccgcg agtggtgttt gcttgcttaa taacttttac 420
ccacgggaag ccaaagtcca gtggaagggt gacaatgcac tccaaagcgg gaatagtcag 480
gagtcagtta ctgagcaaga tagtaagac tctacttact ctttgagttc aaccttgacc 540
ctctcaaaag cggactacga gaagcataaa gtgtacgcct gcgaggtgac gcatcaagg 600
ttgtcttccc cggttacgaa gtcctttaat aggggggaat gt 642

SEQ ID NO: 199          moltype = DNA length = 321
FEATURE                Location/Qualifiers
misc_feature            1..321
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polynucleotide"
source                  1..321
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 199
gatatacaga tgacgcaaag tccctctagt ctttctgcaa gtgtcgggga cagagttacc 60
attacctgca gagcgtcaca aggcattctc agttatctcg cgtgggtacca acagaagcca 120
ggtaaaagcac ctaaactgtt gatttacacg gcatacaatc tgcagtcagg tgtccccctcc 180
cgatttagtg gcagtggtag cggtacagat tttactctta ccatttcatac tcttcagcca 240
gaagattttg ctacgtacta ctgtcaacaa gaatgggcta aaccacgaac ctttgagacg 300
ggtacgaagg tcgaaataaa a
ggtacgaagg tcgaaataaa a 321

SEQ ID NO: 200          moltype = DNA length = 642
FEATURE                Location/Qualifiers
misc_feature            1..642
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polynucleotide"
source                  1..642
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 200
gatatacaga tgacgcaaag tccctctagt ctttctgcaa gtgtcgggga cagagttacc 60
attacctgca gagcgtcaca aggcattctc agttatctcg cgtgggtacca acagaagcca 120
ggtaaaagcac ctaaactgtt gatttacacg gcatacaatc tgcagtcagg tgtccccctcc 180
cgatttagtg gcagtggtag cggtacagat tttactctta ccatttcatac tcttcagcca 240
gaagattttg ctacgtacta ctgtcaacaa gaatgggcta aaccacgaac ctttgagacg 300
ggtacgaagg tcgaaataaa acggaccggt gccgccccc cgttcttcat cttccccccg 360
tctgacgagc agctcaaatc cggcacagct tctgtagtct gcttgctgaa taactttcac 420
ccaagagaa ccaaagtcca gtggaagggt gataatgcat tgcaatctgg taatagtcag 480
gaatctgtga ctgagcagga tagcaaaagac tcaacttaca gcctctcttc aaccttgacg 540
ttgtccaaag cggattatga gaaacacaag gtgtacgctt gcgaggtgac gcatcaagg 600
cttagttccc cggtaaccaa atctttcaac cgaggtgaat gc 642

SEQ ID NO: 201          moltype = AA length = 125
FEATURE                Location/Qualifiers
REGION                  1..125
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polypeptide"
source                  1..125
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 201
QVQLLES GGG LVQPGGSLRL SCAASGFTFS SYWMNWVRQA PGKGLEWVSV IESKGNIFY 60
ADSVKGRPTI SRDNSKNTLY LQMNSLRAED TAVYYCARDR YSMIYSYGAG AFDYWGQGT 120
VTSS 125

SEQ ID NO: 202          moltype = DNA length = 375
FEATURE                Location/Qualifiers
misc_feature            1..375
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polynucleotide"
source                  1..375
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 202

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caagttcaat tgctggaag cggaggtgga cttgtccaac ctggagggtc actccgactg 60
tcttgcgctg catcaggatt cacccttagt agctattgga tgaactgggt ccggcaggct 120
cctgggaaag ggcttgagtg ggtaagtgtc attgaatcaa agggcaacta catcttttat 180
gctgattctg taaagggtag gtccaccatc tccagggaca attcaaaaaa tactttgtat 240
ctgcagatga actctctcag ggcagaagac acggccggtt attactgcgc ccgcgatcga 300
tacagcatga tatactccta cggcgcagga gcttttgact actgggtgca aggcacactt 360
gttactgtca gtagc 375

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SEQ ID NO: 203      moltype = AA length = 455
FEATURE            Location/Qualifiers
REGION              1..455
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source              1..455
                    mol_type = protein
                    organism = synthetic construct

```

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SEQUENCE: 203
QVQLLESGGG LVQPGGSLRL SCAASGFTFS SYWMNWVRQA PGKGLEWVSV IESKGNIFYF 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVVYCARDR YSMIYSYGAG AFDYWGQGTL 120
VTSSASTKG PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA 180
VLQSSGLYSL SSVVTPVSSS LGTQTYICNV NHKPSNTKVD KRVEPKSCDK THTCPPCPAP 240
EAAGGFSVFL FPPKPKDTLM ISRTPEVTCV VVDVSHEDPE VKFNWYVDGV EVHNAKTKPR 300
EEQYNSTYRV VSVLTVLHQD WLNGKEYKCK VSNKALPAPI EKTISKAKGQ PREPQVYTL 360
PREEEMTKNQ VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFFLYSKLTV 420
DKSRWQQGNV FSCSVMEHAL HNHYTQKSL SLPKG 455

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SEQ ID NO: 204      moltype = DNA length = 1365
FEATURE            Location/Qualifiers
misc_feature        1..1365
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..1365
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 204
caagttcaat tgctggaag cggaggtgga cttgtccaac ctggagggtc actccgactg 60
tcttgcgctg catcaggatt cacccttagt agctattgga tgaactgggt ccggcaggct 120
cctgggaaag ggcttgagtg ggtaagtgtc attgaatcaa agggcaacta catcttttat 180
gctgattctg taaagggtag gtccaccatc tccagggaca attcaaaaaa tactttgtat 240
ctgcagatga actctctcag ggcagaagac acggccggtt attactgcgc ccgcgatcga 300
tacagcatga tatactccta cggcgcagga gcttttgact actgggtgca aggcacactt 360
gttactgtca gtagcgctc aacgaaagga ccgtccgtgt tctctctgc tcttagctcc 420
aaatccacct cagggtggaac ggcgcacctg ggggtgctgg taaaggacta tttccagag 480
ccagttactg tgtcttggaa ttctgggtga ttgacaagtg gcgtacacac ttttcccgcg 540
gtcctccaat ctagtgtgtc gtactcactg tctcccggtg tgactgtccc aagtagctca 600
cttggcacac agacttacct ctgtaatggt aatcataagc cgtcaaacac gaaggtggat 660
aagagggtag aacctaagtc atgtgacaaa acgcatactt gccccacctg cctgcgcgcg 720
gaagccgctg gccgaccctc cgtattcttg tccctcccaa agccaaagga cactctgatg 780
attagccgga caccggaggt cacttgtgtt gtagttgacg tcagccatga ggatcctgag 840
gtgaaattta attggtacgt ggcaggggtt gaagtccaca atgctaaaac taaacctagg 900
gaagagcaat ataatagtac atacagggtt gtcagtgtgc tgaccgttct ccatcaggac 960
tggctgaacg gcaaggaata caagtgcgca agcccttgcc ggcccccatc 1020
gagaagacga tctccaaagc caaggggcaa ccccagaaac cgcaggtata cagctcccc 1080
cctagtagag aagagatgac aaagaatcaa gtttcttga cgtgccttgt gaaaggcttc 1140
taccctagtg acatcgagtg cgaatgggag agcaacgggc agccggagaa taactataaa 1200
acaaccccc cgtgtcttga ctcagacggg tcatttttct tgatatagcaa attgactgtt 1260
gataaatcac ggtggcaaca aggaaacgtg ttagttgca gcgtaatgca cgaagctctc 1320
cacaatcact atactcaaaa gtcactgtca ctctcccctg gcaag 1365

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SEQ ID NO: 205      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 205
gacatacaaa tgacgcaatc tccagtagc ttgtcagct ccgtaggcga ccgagtaacg 60
attacgtgta gagcgagcca ggggaatttca tcttatttgg cttgggtatca gcaaaagccg 120
ggaaaagcac ccaaacctct catttatact gccagcacgt tgcaaaagcg cgttccgagt 180
cggttctctg gatcagggtc cgggacggac ttcaccttga cgatttcac tttgcaacct 240
gaagattttg caacatacta ctgtcaacag gagtgggtga agccaaggac cttcggacaa 300
ggcacgaagg tcgaatcaa g 321

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SEQ ID NO: 206      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature        1..642
                    note = source = /note="Description of Artificial Sequence:

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source          Synthetic polynucleotide"
                1..642
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 206
gacatacaaaa tgacgcaatc tccgagtagc ttgtcagcgt ccgtaggcga ccgagtaacg 60
attacgtgta gagcgagcca ggggaatttca tcttatttgg cttgggtatca gcaaaagccg 120
ggaaaagcac ccaaaactcct catttatact gccagcacgt tgcaaaagcgg cgttccgagt 180
cggttctctg gatcagggtc cgggacggac ttcaccttga cgatttcatc ttgcaacct 240
gaagattttg caacatacta ctgtcaacag gagtgggtga agccaaggac cttcgacaa 300
ggcacgaagg tcgaaatcaa gcgaaccgtg gcagctccgt ccgtgtttat ttttccgcct 360
tccgacgaac aacttaaaag tggaaacagc tctgtcgtct gtctccttaa caactttctac 420
cccagggaag ctaaagtaca gtggaaggta gataacgctc tgcaaaagtg taattctcag 480
gagagcgtca cggaacagga ctccaaagac tccacctatt ctctgagctc tacactgacg 540
ctcagcaagg cagactacga aaagcacaaa gtatatgcgt gtgaggtgac gcatcaaggc 600
cttagcagtc cagttacaaa aagttttaac aggggagaat gc 642

SEQ ID NO: 207      moltype = DNA length = 375
FEATURE            Location/Qualifiers
misc_feature        1..375
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"

source              1..375
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 207
gaagtgcagc tgctggaatc cggcggaggt ctggtccagc ctggagggtc cctgcgcctg 60
tcatgcgcag cctccggatt cacttttctg tctgactgga tgaactgggt cagacaggct 120
cctggaaaag gccctggaatg ggtgtctgtg attgaatcca aggggaacta catcttctac 180
gcggaacagc tgaagggccg gttcactatc agcagagaca acagcaagaa caccctgtac 240
ctccaaatga actcgtgag ggcgaagat actgccgtgt actactgtgc ccgcgatcgc 300
tactcgatga tctacagcta tggtgccgga gcgttcgatt actggggaca gggaaacctc 360
gtgaccgtca gctcc 375

SEQ ID NO: 208      moltype = AA length = 455
FEATURE            Location/Qualifiers
REGION              1..455
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"

source              1..455
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 208
EVQLLESGGG LVQPGGSLRL SCAASGFTFS SYWMNWRQA PGKGLEWVSV IESKGNIFY 60
ADSVKGRFTI SRDNSKNTLY LQMSLRAED TAVYYCARDR YSMIYSYGAG APDYWGQGT 120
VTVSSASTKG PSVFPLAPSS KSTSGGTAAL GCLVKDYFPE PVTVSWNSGA LTSGVHTFPA 180
VLQSSGLYSL SSVVTPSSS LGTQTYICNV NHKPSNTKVD KRVEPKSCDK THTCPPCPAP 240
ELLGGPSVFL FPPKPKDTLM ISRTPEVTCV VVAVSHEDPE VKFNWYVDGV EVHNAKTKPR 300
EEQYNSTYRV VSVLTVLHQD WLNKGKEYCK VSNKALAAPI EKTISKAKGQ PREPQVYTL 360
PSREEMTKNQ VSLTCLVKGF YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFFLYSKLTV 420
DKSRWQQGNV FSCSVMHEAL HNHYTQKSL SPSGK 455

SEQ ID NO: 209      moltype = DNA length = 1365
FEATURE            Location/Qualifiers
misc_feature        1..1365
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"

source              1..1365
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 209
gaagtgcagc tgctggaatc cggcggaggt ctggtccagc ctggagggtc cctgcgcctg 60
tcatgcgcag cctccggatt cacttttctg tctgactgga tgaactgggt cagacaggct 120
cctggaaaag gccctggaatg ggtgtctgtg attgaatcca aggggaacta catcttctac 180
gcggaacagc tgaagggccg gttcactatc agcagagaca acagcaagaa caccctgtac 240
ctccaaatga actcgtgag ggcgaagat actgccgtgt actactgtgc ccgcgatcgc 300
tactcgatga tctacagcta tggtgccgga gcgttcgatt actggggaca gggaaacctc 360
gtgaccgtca gctccgctc aaccaagggc ccgtcagtg tcccgctggc tccatcgtcg 420
aagtcacact ccggaggaaac gcgagcactc ggttgccctg tcaaggacta cttccctgag 480
ccagtgcacg tgcgtggaa cagcggagcc ctgacttccg gcgtgcacac ttttcccgcg 540
gtgctgcagt cctccggtct gtaactccct tctgctcgtg tcacccgtgc gtcgtctagc 600
ctgggcaccc agacctacat ctgcaacgtg aaccacaagc cgtccaacac caaagtggat 660
aagcgggttg agccgaagtc ctgcgataag acacacacgt gcccgccatg tccagcgcct 720
gaattgcttg gccgaccttc cgtgttctgt ttcccgcccta agcccaaggga cacttgatg 780
attagccgga ctccogaagt cactgtgtgt gtggtggcag tgtcccacga ggaccccgag 840
gtcaagttta attggtacgt ggacggcgct gaagtgcaca acgccaagac taagcccccg 900
gaggaacagt acaacagcac ctaccgggtc gtgtccgtgc tgacctgtgt gcaccaggac 960
tggctgaatg ggaaagagta caagtgcaaa gtgtccaaca aggccttggc cgctcctatc 1020
gaaaaaacta tcagcaaggc taaggagacg ccgagggaac cccaagtcta caccctgccc 1080

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ccttcacgcg aagagatgac caagaatcaa gtgtcgcgtga cctgcctcgt caagggatc 1140
taccctccgc acattgcggt ggagtgaggag tccaacggcc agcccgagaa caactacaag 1200
actactccgc ccgtgctgga ctccgacggc agcttcttcc tgtattccaa gctgaccgtg 1260
gacaagtcgc ggtggcagca aggaaacgtg ttctcctgct cggtcacgca cgaagccctg 1320
cacaaccact atacgcagaa gtccctgtcc ttgagcccg ggaaa 1365

SEQ ID NO: 210      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 210
gacatccaga tgaccagtc tccgtcctcc ctgtccgcat cagtggggga cagagtgacc 60
atcacttgtc gggcctccca aggcattctcg tcatacctgg cctgggtatca gcagaaaccc 120
ggaaaggctc caaagctgct catctacacc gcctcgactc tgcaatccgg agtgccttcc 180
cgcttctccg gatccggttc gggaaaccgac ttcacccctca ccattagcag ccttcagccg 240
gaagatttgc cgacctacta ctgccagcaa gaatgggtga agcccaggac atttggccag 300
ggcactaagg tcgagattaa g
gcactaagg tcgagattaa g 321

SEQ ID NO: 211      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature        1..642
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..642
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 211
gacatccaga tgaccagtc tccgtcctcc ctgtccgcat cagtggggga cagagtgacc 60
atcacttgtc gggcctccca aggcattctcg tcatacctgg cctgggtatca gcagaaaccc 120
ggaaaggctc caaagctgct catctacacc gcctcgactc tgcaatccgg agtgccttcc 180
cgcttctccg gatccggttc gggaaaccgac ttcacccctca ccattagcag ccttcagccg 240
gaagatttgc cgacctacta ctgccagcaa gaatgggtga agcccaggac atttggccag 300
ggcactaagg tcgagattaa gcgtacgggtg gccgctccca gcgtgttcat ctccccccc 360
agcgacgagc agctgaagag cggcacggcc agcgtggtgt gcctgctgaa caactcttac 420
ccccgggagc ccaagggtga gtggaagggt gacaacggcc tgcaagcggc caacagccag 480
gagagcgctc ccgagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcataag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgtccagcc ccgtgaccaa gagcttcaac aggggagcagt gc 642

SEQ ID NO: 212      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 212
gacatacaga tgactcagag tccttctctcc ctcaagtgtt cagtgggtga tcgcgtgacg 60
atcacgtgca gagcctcaca agggatctcc agttacctgg cctgggtatca acaaaaacca 120
ggcaaggcgc ctaagctggt gatataacg gcattctacat tgcagtctgg ggtaccaagt 180
cgattcagtg gttctggctc aggcactgac tttaccctta caatatcaag tcttcagccg 240
gaggatttgc caacttacta ttgccagcag atttggacgg tgccgcgcac tttcggtcag 300
ggaacaaagg tggaataaaa a
ggaacaaagg tggaataaaa a 321

SEQ ID NO: 213      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature        1..642
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..642
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 213
gacatacaga tgactcagag tccttctctcc ctcaagtgtt cagtgggtga tcgcgtgacg 60
atcacgtgca gagcctcaca agggatctcc agttacctgg cctgggtatca acaaaaacca 120
ggcaaggcgc ctaagctggt gatataacg gcattctacat tgcagtctgg ggtaccaagt 180
cgattcagtg gttctggctc aggcactgac tttaccctta caatatcaag tcttcagccg 240
gaggatttgc caacttacta ttgccagcag atttggacgg tgccgcgcac tttcggtcag 300
ggaacaaagg tggaataaaa aagaacggtc gcagcaccca gtgttttcat ctccccccc 360
tcgcagcagc agcttaaaag cggtacagcc agcgtagtgt gtttgttgaa taatttttat 420
ccacgcgaag caaaggttca gtggaaggta gacaacgcct tgcaaacggc aaattcccaa 480
gaaagtgtta cggagcaaga cagtaaggac tctacatatt ccttgtcatc aacactcacc 540
cttagttaaag cagattacga gaaacacaag gtctatgcat gtgaggtaac gcacagggc 600
ctctccagtc ccgtcaccaa gtccttcaac aggggtgagt gc 642

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SEQ ID NO: 214      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 214
gacatccaga tgaccagtc tccgtcctcc ctgtccgcat cagtggggga cagagtgacc 60
atcacttgtc gggcctccca aggcattctcg tcatacctgg cctgggatca gcagaaaccc 120
ggaaaggctc caaagctgct catctacacc gcctcgactc tgcaatccgg agtgccttcc 180
cgcttctccg gatccggttc gggaaaccgac ttcaccctca ccattagcag ccttcagccg 240
gaagatttcg cgacctacta ctgccagcaa atctggaccg tgcccaggac atttggccag 300
ggcactaagg tcgagattaa g                                     321

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SEQ ID NO: 215      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature        1..642
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..642
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 215
gacatccaga tgaccagtc tccgtcctcc ctgtccgcat cagtggggga cagagtgacc 60
atcacttgtc gggcctccca aggcattctcg tcatacctgg cctgggatca gcagaaaccc 120
ggaaaggctc caaagctgct catctacacc gcctcgactc tgcaatccgg agtgccttcc 180
cgcttctccg gatccggttc gggaaaccgac ttcaccctca ccattagcag ccttcagccg 240
gaagatttcg cgacctacta ctgccagcaa atctggaccg tgcccaggac atttggccag 300
ggcactaagg tcgagattaa gcgtacgggtg gccgtctcca gcgtgttcat cttccccccc 360
agcgacgagc agctgaagag cggcaccgcc agcgtgtgtg gcctgctgaa caactcttac 420
ccccgggagg ccaaggtgca gtggaagggtg gacaacgccc tgcagagcgg caacagccag 480
gagagcgtca ccgagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcataag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgtccagcc ccgtgaccaa gagcttcaac aggggcgagt gc                                     642

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SEQ ID NO: 216      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..321
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 216
gacatccaga tgaccagtc tccgtcctcc ctgtccgcat cagtggggga cagagtgacc 60
atcacttgtc gggcctccca aggcattctcg tcatacctgg cctgggatca gcagaaaccc 120
ggaaaggctc caaagctgct catctacacc gcctcgactc tgcaatccgg agtgccttcc 180
cgcttctccg gatccggttc gggaaaccgac ttcaccctca ccattagcag ccttcagccg 240
gaagatttcg cgacctacta ctgccagcaa gaatgggcca agcccaggac atttggccag 300
ggcactaagg tcgagattaa g                                     321

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SEQ ID NO: 217      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature        1..642
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..642
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 217
gacatccaga tgaccagtc tccgtcctcc ctgtccgcat cagtggggga cagagtgacc 60
atcacttgtc gggcctccca aggcattctcg tcatacctgg cctgggatca gcagaaaccc 120
ggaaaggctc caaagctgct catctacacc gcctcgactc tgcaatccgg agtgccttcc 180
cgcttctccg gatccggttc gggaaaccgac ttcaccctca ccattagcag ccttcagccg 240
gaagatttcg cgacctacta ctgccagcaa gaatgggcca agcccaggac atttggccag 300
ggcactaagg tcgagattaa gcgtacgggtg gccgtctcca gcgtgttcat cttccccccc 360
agcgacgagc agctgaagag cggcaccgcc agcgtgtgtg gcctgctgaa caactcttac 420
ccccgggagg ccaaggtgca gtggaagggtg gacaacgccc tgcagagcgg caacagccag 480
gagagcgtca ccgagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcataag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgtccagcc ccgtgaccaa gagcttcaac aggggcgagt gc                                     642

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SEQ ID NO: 218      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature        1..321
                    note = source = /note="Description of Artificial Sequence:

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source          Synthetic polynucleotide"
                1..321
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 218
gatattcaga tgacgcaatc tccgtcttcc ttgtcagcta gtgtaggaga ccgcgtcaca 60
attacctgta gagccagcca ggggatttcc tcataccttg catggtacca gcaaaagcca 120
ggcaaaagccc ccaaactgct gatctacacc gcgtctacct tgcaatctgg tgtgccgtca 180
cgcttttccg gctctggctc aggtactgat ttcacattga cgatctcaag tctccagccg 240
gaagacttcg caacttacta ctgccaaaca tcctggacga ggccgaggac ttccgggcag 300
ggaacaaaagg ttgaaattaa a
                                321

SEQ ID NO: 219      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature       1..642
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polynucleotide"

source            1..642
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 219
gatattcaga tgacgcaatc tccgtcttcc ttgtcagcta gtgtaggaga ccgcgtcaca 60
attacctgta gagccagcca ggggatttcc tcataccttg catggtacca gcaaaagcca 120
ggcaaaagccc ccaaactgct gatctacacc gcgtctacct tgcaatctgg tgtgccgtca 180
cgcttttccg gctctggctc aggtactgat ttcacattga cgatctcaag tctccagccg 240
gaagacttcg caacttacta ctgccaaaca tcctggacga ggccgaggac ttccgggcag 300
ggaacaaaagg ttgaaattaa aagaacagtc gcagcaccaa gtgtttttat ttttccaccc 360
tcagacgagc agctcaagtc tggcaccgcg agcgtagtat gttgttgaa taatttttac 420
cctagggaag ctaagggtaca gtggaaagtg gataatgctc tccaaagtgg caactcccag 480
gaatcagtga ctgagcaaga ttcaaaggac agcacgtatt ctctttcttc tacgcttact 540
ctctctaagg ccgactacga aaacacaaa gtttacgctt gcgaggttac ccaccagggg 600
ctgtcctcac cagtaacgaa aagttttaac cggggcgagt gt
                                642

SEQ ID NO: 220      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature       1..321
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polynucleotide"

source            1..321
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 220
gacatccaga tgacccagtc tccgtcctcc ctgtccgcat cagtggggga cagagtgacc 60
atcacttgtc gggcctccca aggcactctc tcataccttg cctgggtatca gcagaaaccc 120
ggaaaggctc caaagctgct catctacacc gcctcgactc tgcaatccgg agtgccttcc 180
cgcttctccg gatccggttc gggaaaccgac ttcaccctca ccattagcag ccttcagccg 240
gaagatttcg gcacctacta ctgccagcaa agctggacca ggcccaggac atttggccag 300
ggcactaagg tcgagattaa g
                                321

SEQ ID NO: 221      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature       1..642
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polynucleotide"

source            1..642
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 221
gacatccaga tgacccagtc tccgtcctcc ctgtccgcat cagtggggga cagagtgacc 60
atcacttgtc gggcctccca aggcactctc tcataccttg cctgggtatca gcagaaaccc 120
ggaaaggctc caaagctgct catctacacc gcctcgactc tgcaatccgg agtgccttcc 180
cgcttctccg gatccggttc gggaaaccgac ttcaccctca ccattagcag ccttcagccg 240
gaagatttcg gcacctacta ctgccagcaa agctggacca ggcccaggac atttggccag 300
ggcactaagg tcgagattaa cggtacgggtg gccgtcctcca gcgtgttcat cttccccccc 360
agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caactttctac 420
ccccgggagg ccaaggtgca gtggaagggt gacaacgccc tgacagagcgg caacagccag 480
gagagcgctc ccgagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcataag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgtccagcc ccgtgaccaa gagcttcaac aggggcgagt gc
                                642

SEQ ID NO: 222      moltype = DNA length = 321
FEATURE            Location/Qualifiers
misc_feature       1..321
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polynucleotide"

source            1..321
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 222

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gacattcaaa tgactcagtc tccctcatct ttgtcagcat cagttgggga cagggtgaca 60
atcacatgcc gagcctcaca ggggatttct agctatcttg catggtacca acagaagccc 120
ggcaaaagccc ccaagctttt gatataatcg gcattccactc ttcagagcgg agtaccagct 180
aggttttagtg gctccgggag ttgtacggac tttactctga cgatttcctc ccttcaacct 240
gaagactttg caacgtatta ctgtcagcaa atatggatgg ctcccagaac gtttggtcaa 300
ggtactaaag ttgaaataaa g                                     321

SEQ ID NO: 223          moltype = DNA length = 642
FEATURE                Location/Qualifiers
misc_feature            1..642
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polynucleotide"
source                  1..642
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 223
gacattcaaa tgactcagtc tccctcatct ttgtcagcat cagttgggga cagggtgaca 60
atcacatgcc gagcctcaca ggggatttct agctatcttg catggtacca acagaagccc 120
ggcaaaagccc ccaagctttt gatataatcg gcattccactc ttcagagcgg agtaccagct 180
aggttttagtg gctccgggag ttgtacggac tttactctga cgatttcctc ccttcaacct 240
gaagactttg caacgtatta ctgtcagcaa atatggatgg ctcccagaac gtttggtcaa 300
ggtactaaag ttgaaataaa gcgaactgta gcagcaccta gtgtatttat cttcccccct 360
tctgatgaac agttgaagtc cgggacggct tccgtcgtat gtctcctgaa caacttttac 420
ccaaggggagg caaagggtgca atggaagggtg gataatgcac tccagagtgg caatagccaa 480
gaatcagtaa ccgaacagga ttccaaggat tctacctaca gccttctctc tacgcttaca 540
ttgagcaagg cggactatga aaagcataag gtgtatgcgt gcgaagtaac acaccagggt 600
ctcagcagtc cagttacgaa gtctttcaat cggggagaat gt                                     642

SEQ ID NO: 224          moltype = DNA length = 321
FEATURE                Location/Qualifiers
misc_feature            1..321
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polynucleotide"
source                  1..321
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 224
gacatccaga tgaccagtc tccgtcctcc ctgtccgcat cagtggggga cagagtgacc 60
atcactttgtc gggcctccca aggcattctcg tcatacctgg cctgggtatca gcagaaaccc 120
ggaaaggctc caaagctgct catctacacc gcctcgactc tgcaatccgg agtgccttcc 180
cgcttctccg gatccggttc ggggaaccgac ttcaccctca ccattagcag ccttcagccg 240
gaagatttcg cgacctacta ctgccagcaa atctggatgg cccccaggac atttgccag 300
ggcactaagg tcgagattaa g                                     321

SEQ ID NO: 225          moltype = DNA length = 642
FEATURE                Location/Qualifiers
misc_feature            1..642
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polynucleotide"
source                  1..642
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 225
gacatccaga tgaccagtc tccgtcctcc ctgtccgcat cagtggggga cagagtgacc 60
atcactttgtc gggcctccca aggcattctcg tcatacctgg cctgggtatca gcagaaaccc 120
ggaaaggctc caaagctgct catctacacc gcctcgactc tgcaatccgg agtgccttcc 180
cgcttctccg gatccggttc ggggaaccgac ttcaccctca ccattagcag ccttcagccg 240
gaagatttcg cgacctacta ctgccagcaa atctggatgg cccccaggac atttgccag 300
ggcactaagg tcgagattaa gcgtacgggtg gccgctccca gcgtgttcatt cttccccccc 360
agcgacgagc agctgaagag cggcaccgcc agcgtggtgt gcctgctgaa caacttctac 420
ccccgggagg ccaagggtgca gtggaagggtg gacaacgccc tgcaagcgg caacagccag 480
gagagcgtca ccgagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcataag gtgtacgcct gcgaggtgac ccaccagggc 600
ctgtccagcc ccgtgaccaa gagcttcaac aggggcgagt gc                                     642

SEQ ID NO: 226          moltype = AA length = 10
FEATURE                Location/Qualifiers
REGION                  1..10
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic peptide"
source                  1..10
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 226
GFTFSSYWIS                                     10

SEQ ID NO: 227          moltype = AA length = 17
FEATURE                Location/Qualifiers
REGION                  1..17

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	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 227		
NIKQSGSETY YVESVKG		17
SEQ ID NO: 228	moltype = AA length = 14	
FEATURE	Location/Qualifiers	
REGION	1..14	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 228		
SLRRRSTHA GFDV		14
SEQ ID NO: 229	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 229		
SYWIS		5
SEQ ID NO: 230	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 230		
KQSGSE		6
SEQ ID NO: 231	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 231		
IKQSGSET		8
SEQ ID NO: 232	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 232		
ARSLRRRSTE HAGFDV		16
SEQ ID NO: 233	moltype = AA length = 123	
FEATURE	Location/Qualifiers	
REGION	1..123	
	note = source = /note="Description of Artificial Sequence: Synthetic polypeptide"	
source	1..123	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 233		
EVQLVESGGG LVQPGGSLRL SCAASGFTFS SYWISWVRQA PGKGLEWVAN IKQSGSETYY		60
VESVKGRTI SRDIAKNSLY LQMSLRAD TAVYYCARSL RRRSTEHAGF DVWGQGLVT		120
VSS		123
SEQ ID NO: 234	moltype = DNA length = 369	
FEATURE	Location/Qualifiers	
misc_feature	1..369	

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note = source = /note="Description of Artificial Sequence:
    Synthetic polynucleotide"
source      1..369
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 234
gaagtcgacg tgggtgaaag cggcgggtggc ctggtgcagc caggtggtag cctgcgcctg 60
agctgcgcgc ccagcgggett taccttttagc agctattgga ttagctgggt tcgccaggcc 120
ccaggcaaaag gcctgggaatg ggtggcggaac atcaaacaga gcggcagcga gacctactat 180
gtggagagcg tgaaaggccg ctttaccatt agccgcgata acgccccaaa cagcctgtat 240
ctgcaaatga acagcctgcg ggccgaagat accgcggtgt attattgcgc gcgtagcctg 300
cgctcgtcgt gcactgagca cgcaggattc gacgtttggg gccaggggcac cctggttact 360
gtctcgagc                                     369

SEQ ID NO: 235      moltype = AA length = 453
FEATURE            Location/Qualifiers
REGION            1..453
note = source = /note="Description of Artificial Sequence:
    Synthetic polypeptide"
source      1..453
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 235
EVQLVESGGG LVQPGGSLRL SCAASGFTFS SYWISWVRQA PGKGLEWVAN IKQSGSETYY 60
VESVKGRFTI SRDNAKNSLY LQMNSLRAED TAVVYCARSL RRRSTEHAGF DVWQGQTLVT 120
VSSASTKGPS VFPLAPSSKS TSGGTAALGC LVKDYFPEPV TVSWNSGALT SGVHTFPAVL 180
QSSGLYSLSS VVTVPSSSLG TQTYICNVNH KPSNTKVDKR VEPKSCDKTH TCPPCPAPEA 240
AGGPSVFLFP PKPKDTLMIS RTPEVTCVVV DVSHEDPEVK FNNWVDGVEV HNAKTKPREE 300
QYNSTYRVVS VLTVLHQDWL NGKEYKCKVS NKALPAPIEK TISKAKGQPR EPQVYTLPPS 360
REEMTKNQVS LTCLVKGFPY SDIAVEWESN GQPENNYKTT PPVLDSGGSF FLYSKLTVDK 420
SRWQQGNVFS CSVMHEALHN HYTKQSLSLS PGK                                     453

SEQ ID NO: 236      moltype = DNA length = 1359
FEATURE            Location/Qualifiers
misc_feature      1..1359
note = source = /note="Description of Artificial Sequence:
    Synthetic polynucleotide"
source      1..1359
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 236
gaagtcgacg tgggtgaaag cggcgggtggc ctggtgcagc caggtggtag cctgcgcctg 60
agctgcgcgc ccagcgggett taccttttagc agctattgga ttagctgggt tcgccaggcc 120
ccaggcaaaag gcctgggaatg ggtggcggaac atcaaacaga gcggcagcga gacctactat 180
gtggagagcg tgaaaggccg ctttaccatt agccgcgata acgccccaaa cagcctgtat 240
ctgcaaatga acagcctgcg ggccgaagat accgcggtgt attattgcgc gcgtagcctg 300
cgctcgtcgt gcactgagca cgcaggattc gacgtttggg gccaggggcac cctggttact 360
gtctcgagcg cgtcgaccaa agccccagc gtgttccctc tggccccagc cagcaagagc 420
acctctggcg gaacagccgc cctgggctgc ctgggtcaagg actacttccc cgagcccggtg 480
accgtgtcct ggaactctgg cgccctgacc agcggcggtg acacctttcc agcctgtgctc 540
cagagcagcg gcctgtacag cctgagcagc gtcgtgaccg tgcccagcag cagcctgggc 600
accagacctc acatctgcaa cgtgaaccac aagcccagca acacaaaggt ggacaagcgg 660
gtggaaccca agagctgcga caagaccac acctgtcccc cctgccctgc ccctgaagcg 720
gcgggagggc cctcctgtgt cctgttcccc ccaaagccta aggacacct gatgatcagc 780
cggacccccg aagtgcacctg cgtggtggtg gacgtgtccc acgaggaccc tgaagtgaag 840
ttaattggtg acgtggacgg cgtggaagtg cacaacgcca agaccaagcc cagagaggaa 900
cagtacaaca gcacctaccg ggtggtgtcc gtgctgaccg tgctgcacca ggactggctg 960
aacggcaaaag agtacaagtg caaggtgtcc aacaaggccc tgccctgccc catcgagaaa 1020
accatcagca agggccaaagg ccagccccgc gagccccagg tgtacacact gccccctagc 1080
cgggaagaga tgaccaagaa ccagggtgtc ctgacctgcc tcgtgaaggg cttctacccc 1140
agcgacattg ccgtggaatg ggagagcaac ggccagcccg agaacaacta caagaccacc 1200
ccccctgtgc tggacagcga cggctcattc ttctgtaca gcaagctgac cgtggacaag 1260
agccggtggc agcagggcaa cgtgttcagc tgctccgtga tgcacgaggc cctgcacaa 1320
cactacaccc agaagtcctt gagcctgagc cccggcaag                                     1359

SEQ ID NO: 237      moltype = AA length = 11
FEATURE            Location/Qualifiers
REGION            1..11
note = source = /note="Description of Artificial Sequence:
    Synthetic peptide"
source      1..11
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 237
RASQGISNYL A                                     11

SEQ ID NO: 238      moltype = AA length = 7
FEATURE            Location/Qualifiers
REGION            1..7

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note = source = /note="Description of Artificial Sequence:
  Synthetic peptide"
source      1..7
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 238
AASTLQS      7

SEQ ID NO: 239
FEATURE
REGION      moltype = AA length = 9
            Location/Qualifiers
            1..9
            note = source = /note="Description of Artificial Sequence:
              Synthetic peptide"
source      1..9
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 239
QQADKFPYT    9

SEQ ID NO: 240
FEATURE
REGION      moltype = AA length = 7
            Location/Qualifiers
            1..7
            note = source = /note="Description of Artificial Sequence:
              Synthetic peptide"
source      1..7
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 240
SQGISNY      7

SEQ ID NO: 241
SEQUENCE: 241
000          moltype = length =

SEQ ID NO: 242
FEATURE
REGION      moltype = AA length = 6
            Location/Qualifiers
            1..6
            note = source = /note="Description of Artificial Sequence:
              Synthetic peptide"
source      1..6
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 242
ADKFPY      6

SEQ ID NO: 243
FEATURE
REGION      moltype = AA length = 6
            Location/Qualifiers
            1..6
            note = source = /note="Description of Artificial Sequence:
              Synthetic peptide"
source      1..6
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 243
QGISNY      6

SEQ ID NO: 244
FEATURE
REGION      moltype = AA length = 107
            Location/Qualifiers
            1..107
            note = source = /note="Description of Artificial Sequence:
              Synthetic polypeptide"
source      1..107
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 244
DIQMTQSPSS LSASVGRVT ITCRASQGIS NYLAWYQQKP GKVPKLLIYA ASTLQSGVPS 60
RFSGSGSGTD PTLTISSLQP EDVATYYCQQ ADKFPYTFGQ GTKVEIK 107

SEQ ID NO: 245
FEATURE
misc_feature 1..321
            note = source = /note="Description of Artificial Sequence:
              Synthetic polynucleotide"
source      1..321
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 245
gatattcaga tgaccacagag cccgagcagc ctgagcgcaa gcgtggggcga tcgcgtgacc 60
attacctgcc gcgccagcca gggcattagc aactatctgg cctgggtatca gcagaaaccg 120

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ggcaaaagtgc cgaaactgct gatctatgcc gccagcaccg tgcaaaagcgg cgtgccaagt 180
cgcttttagcg gcagcggttag cggcacccgat ttcacccctga ccattagcag cctgcaaccg 240
gaagacgtgg cgacctatta ttgccagcag gctgacaaat tcccgtagac cttcggccag 300
ggtaccaaag tggaaatcaa g 321

SEQ ID NO: 246      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polypeptide"
source              1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 246
DIQMTQSPSS LSASVGRVIT ITCRASQGIS NYLAWYQQKP GKVPKLLIYA ASTLQSGVPS 60
RFSGSGSGTD PTLTISSLQP EDVATYYCQQ ADKFPYTFGQ GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSPVTKSFN RGEC 214

SEQ ID NO: 247      moltype = DNA length = 642
FEATURE            Location/Qualifiers
misc_feature        1..642
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic polynucleotide"
source              1..642
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 247
gatattcaga tgaccagag cccgagcagc ctgagcgcaa gcgtggggcga tcgctgtgacc 60
attacctgcc gcgccagcca gggcattagc aactatctgg cctgggtatca gcagaaaccg 120
ggcaaaagtgc cgaaactgct gatctatgcc gccagcaccg tgcaaaagcgg cgtgccaagt 180
cgcttttagcg gcagcggttag cggcacccgat ttcacccctga ccattagcag cctgcaaccg 240
gaagacgtgg cgacctatta ttgccagcag gctgacaaat tcccgtagac cttcggccag 300
ggtaccaaag tggaaatcaa gcggaccgtg gccgctccct cgtgttccat ctcccaccc 360
agcgacgagc agctgaagtc cggcacagcc agcgtcgtgt gcctgtgtaa caacttctac 420
ccccgcgagg ccaaagtgca gtggaaggtg gacaacgccc tccagagcgg caacagccag 480
gaaagcggtc cagagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaagtgc ccaccagggc 600
ctgtccagcc cctgaccaa gagcttcaac cggggcgagt gt 642

SEQ ID NO: 248      moltype = AA length = 10
FEATURE            Location/Qualifiers
REGION             1..10
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic peptide"
source              1..10
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 248
GFTFSSYSMN 10

SEQ ID NO: 249      moltype = AA length = 17
FEATURE            Location/Qualifiers
REGION             1..17
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic peptide"
source              1..17
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 249
SISSSSSIYI YADSVKG 17

SEQ ID NO: 250      moltype = AA length = 11
FEATURE            Location/Qualifiers
REGION             1..11
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic peptide"
source              1..11
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 250
SGYRGVYGFDF Y 11

SEQ ID NO: 251      moltype = AA length = 5
FEATURE            Location/Qualifiers
REGION             1..5
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic peptide"
source              1..5

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-continued

SEQUENCE: 251	mol_type = protein	
SYSMN	organism = synthetic construct	5
SEQ ID NO: 252	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 252		6
SSSSSY		
SEQ ID NO: 253	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 253		8
GFTFSSYS		
SEQ ID NO: 254	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 254		8
ISSSSSYI		
SEQ ID NO: 255	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
REGION	1..13	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 255		13
ARSGYRGVYG FDY		
SEQ ID NO: 256	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120	
	note = source = /note="Description of Artificial Sequence: Synthetic polypeptide"	
source	1..120	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 256		60
EVQLVESGGG LVKPGGSLRL SCAASGFTFS SYSMNWVRQA PGKGLEWVSS ISSSSSYIYY		
ADSVKGRFTI SRDIAKNSLY LQMNSLRAED TAVYYCARSG YRGVYGFDPYV GQGTLLTVSS		120
SEQ ID NO: 257	moltype = DNA length = 360	
FEATURE	Location/Qualifiers	
misc_feature	1..360	
	note = source = /note="Description of Artificial Sequence: Synthetic polynucleotide"	
source	1..360	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 257		360
gaagtgcagc tgggtggaag cggtcggtgc ctggtgaaac caggcggtag cctgcgcctg		
agctgcgcgc ccagcggcct tacctttagc agctatagca tgaactgggt tcgccaggcc		120
ccaggcaaaag gcctggaatg ggttagcagc atcagcagca gtagcagcta tatctattac		180
gccgatagcg tgaaggccgc ctttaccatt agccgcgata acgcaaaaaa cagcctgtat		240
ctgcaaatga acagcctgcg ggccgaagat accgcctgtg attattgcgc gcgaagcgga		300
tatcgtggag tttaacggatt tgattattgg ggccaggcca ccctggttac tgtctcgagc		360
SEQ ID NO: 258	moltype = AA length = 450	
FEATURE	Location/Qualifiers	

-continued

REGION 1..450
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"

source 1..450
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 258

EVQLVESGGG	LVKPGGSLRL	SCAASGFTFS	SYSMNWRQA	PGKGLEWVSS	ISSSSYIYY	60
ADSVKGRFTI	SRDNAKNSLY	LQMNSLRAED	TAVYYCARSG	YRGVYGFDPY	GQGTLLTVSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAAAGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	NTKVDKRVPE	KSCDKTHTCP	PCPAPEAAGG	240
PSVFLFPPKP	KDTLMISRT	EVTCTVVDVS	HEDPEVKFNW	YVDGVEVHNA	KTKPREEQYN	300
STYRVSVLT	VLHQDWLNGK	EYKCKVSNKA	LPAPIEKTIS	KAKGQPREPQ	VYTLPPSREE	360
MTKNQVSLTC	LVKGFYPSDI	AVEWESNGQP	ENNYKTTTPV	LDSDGSFFLY	SKLTVDKSRW	420
QQGNVFSCSV	MHEALHNHYT	QKSLSLSPGK				450

SEQ ID NO: 259 moltype = DNA length = 1350
 FEATURE Location/Qualifiers
 misc_feature 1..1350
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"

source 1..1350
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 259

gaagtgcagc	tgggtgaaag	cggcggtggc	ctgggtgaaac	caggcggtag	cctgcgctg	60
agctgcgcgc	ccagcggtt	tacctttagc	agctatagca	tgaactgggt	tcgccaggcc	120
ccaggcaaa	gcctggaatg	ggttagcagc	atcagcagca	gtagcagcta	tatctattac	180
gccgatagcg	tgaaaggcgc	ctttaccatt	agccgcgata	acgccccaaa	cagcctgtat	240
ctgcaaatga	acagcctgcg	ggccgaagat	accgcccgtg	attattgcgc	gcgaagcggg	300
tatcgtggag	tttacggatt	tgattattgg	ggccagggca	ccctggttac	tgtctcgagc	360
gcgtcgacca	aaggccccag	cgtgttccct	ctggccccc	gcagcaagag	cacctctggc	420
ggaacagccg	ccctgggctg	cctggtcaag	gactacttcc	ccgagcccgt	gaccgtgtcc	480
tggaactctg	gcgcctgcag	cagcggcggtg	cacaccttcc	cagcgtgtg	ccagagcagc	540
ggcctgtaca	gcctgagcag	cgctgtgacc	gtgcccagca	gcagcctggg	caccagacc	600
tacatctgca	acgtgaacca	caagcccagc	aacacaaagg	tggacaagcg	gggtggaacc	660
aagagctgcg	acaagaccca	caactgtccc	ccctgcccgt	cccctgaagc	ggcgggaggg	720
ccctccgtgt	tcctgttccc	cccaaaagc	aaggacaccc	tgatgatcag	ccggaccccc	780
gaagtgcacct	gcgtgggtgt	ggacgtgtcc	cacgaggacc	ctgaagtga	gtttaattgg	840
tacgtggacg	gcgtggaagt	gcacaacgcc	aagaccaagc	ccagagagga	acagtacaac	900
agcacctacc	gggtgggtgt	cgctgtgacc	gtgctgcacc	aggactgggt	gaacggcaaa	960
gagtacaagt	gcaagggtgc	caacaaggcc	ctgcctgccc	ccatcgagaa	aaccatcagc	1020
aaggccaaag	gccagccccg	cagccccag	gtgtacacac	tgccccctag	ccgggaagag	1080
atgaccaaga	accaggtgtc	cctgacctgc	ctcgtgaagg	gcttctaccc	cagcgacatt	1140
gccgtggaat	gggagagcaa	cggccagccc	gagaacaact	acaagaccac	ccccctgtg	1200
ctggacagcg	acggctcatt	cttctgtgac	agcaagctga	ccgtggacaa	gagccggtgg	1260
cagcagggca	acgtgttcag	ctgctccgtg	atgcacgagg	ccctgcacaa	ccactacacc	1320
cagaagtccc	tgagcctgag	ccccgcgaag				1350

SEQ ID NO: 260 moltype = AA length = 11
 FEATURE Location/Qualifiers
 REGION 1..11
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"

source 1..11
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 260

RASQGISSWL	A	11
------------	---	----

SEQ ID NO: 261 moltype = AA length = 7
 FEATURE Location/Qualifiers
 REGION 1..7
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"

source 1..7
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 261

AASSLQS	7
---------	---

SEQ ID NO: 262 moltype = AA length = 9
 FEATURE Location/Qualifiers
 REGION 1..9
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"

source 1..9
 mol_type = protein
 organism = synthetic construct

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SEQUENCE: 262
QYYHSPLT 9

SEQ ID NO: 263 moltype = AA length = 7
FEATURE Location/Qualifiers
REGION 1..7
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"
source 1..7
mol_type = protein
organism = synthetic construct

SEQUENCE: 263
SQGISSW 7

SEQ ID NO: 264 moltype = AA length = 6
FEATURE Location/Qualifiers
REGION 1..6
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"
source 1..6
mol_type = protein
organism = synthetic construct

SEQUENCE: 264
YYHSPL 6

SEQ ID NO: 265 moltype = AA length = 6
FEATURE Location/Qualifiers
REGION 1..6
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"
source 1..6
mol_type = protein
organism = synthetic construct

SEQUENCE: 265
QGISSW 6

SEQ ID NO: 266 moltype = AA length = 107
FEATURE Location/Qualifiers
REGION 1..107
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
source 1..107
mol_type = protein
organism = synthetic construct

SEQUENCE: 266
DIQMTQSPSS VSASVGDRVT ITCRASQGIS SWLAWYQQKP GKAPKLLIYA ASSLQSGVPS 60
RFGSGSGGTD FTLTISSLQP EDFATYYCQQ YYHSPLTFGQ GTKVEIK 107

SEQ ID NO: 267 moltype = DNA length = 321
FEATURE Location/Qualifiers
misc_feature 1..321
note = source = /note="Description of Artificial Sequence:
Synthetic polynucleotide"
source 1..321
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 267
gatattcaga tgacccagag cccgagcagc gttagcgcca gcgtggcgga tcgcgtgacc 60
attacctgcc gcgccagtca gggcattagc agctggctgg cctgggtatca gcagaaaccg 120
ggcaaagccc cgaaactgct gatctatgcc gccagcagcc tgcaaagcgg cgtgccaagt 180
cgcttttagcg gcagcggtag cggcaccgat ttcaccctga ccattagcag tctgcaaccg 240
gaagactttg ccacctatta ttgccagcag tactaccatt ctccgctgac cttcggccag 300
ggtaccaaag tggaaatcaa g 321

SEQ ID NO: 268 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 268
DIQMTQSPSS VSASVGDRVT ITCRASQGIS SWLAWYQQKP GKAPKLLIYA ASSLQSGVPS 60
RFGSGSGGTD FTLTISSLQP EDFATYYCQQ YYHSPLTFGQ GTKVEIKRTV AAPSVFIIPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 269 moltype = DNA length = 642

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FEATURE	Location/Qualifiers	
misc_feature	1..642	
	note = source = /note="Description of Artificial Sequence: Synthetic polynucleotide"	
source	1..642	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 269		
gatattcaga	tgacccagag	cccagcagc gttagcgcca gcgtgggcga tcgcgtgacc 60
attacctgcc	gcgccagtc	ggccattagc agctggctgg cctgggtatca gcagaaaccg 120
ggcaaaagccc	cgaaactgct	gatctatgcc gccagcagcc tgcaaaagcgg cgtgcccaagt 180
cgcttttagcg	gcagcggtag	cggcacccgat ttcaccctga ccattagcag tctgcaaccg 240
gaagactttg	ccacctatta	ttgccagcag tactaccatt ctccgctgac cttcggccag 300
ggtaccctaa	tggaatcaa	gcggaccgtg gccgtccct cctgtgtcat cttcccaccc 360
agcgacgagc	agctgaagtc	cggcacagcc agcgtcgtgt gcctgtgaa caacttctac 420
ccccgcgagg	ccaaagtgca	gtggaagggtg gacaacgccc tccagagcgg caacagccag 480
gaaagcgtca	ccgagcagga	cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg	ccgactacga	gaagcacaag gtgtacgcct gcgaagtgac ccaccagggc 600
ctgtccagcc	ccgtgaccaa	gagcttcaac cggggcgagt gt 642
SEQ ID NO: 270	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
REGION	1..10	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 270		
GFTFSSYAIS		10
SEQ ID NO: 271	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 271		
AISGSGGSTY YAESVKG		17
SEQ ID NO: 272	moltype = AA length = 12	
FEATURE	Location/Qualifiers	
REGION	1..12	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..12	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 272		
ESGYVYLLKF DY		12
SEQ ID NO: 273	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 273		
SYAIS		5
SEQ ID NO: 274	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 274		
SGSGGS		6
SEQ ID NO: 275	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	

-continued

```

source          Synthetic peptide"
1..8
mol_type = protein
organism = synthetic construct

SEQUENCE: 275
GFTFSSYA                               8

SEQ ID NO: 276      moltype = AA  length = 8
FEATURE            Location/Qualifiers
REGION            1..8
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..8
mol_type = protein
organism = synthetic construct

SEQUENCE: 276
ISGSGGST                               8

SEQ ID NO: 277      moltype = AA  length = 14
FEATURE            Location/Qualifiers
REGION            1..14
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..14
mol_type = protein
organism = synthetic construct

SEQUENCE: 277
ARESGYVYYL KFDY                               14

SEQ ID NO: 278      moltype = AA  length = 121
FEATURE            Location/Qualifiers
REGION            1..121
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..121
mol_type = protein
organism = synthetic construct

SEQUENCE: 278
EVQLLESGGG LVQPGGSLRL SCAASGFTFS SYAISWVRQA PGKGLEWVSA ISGSGGSTYY 60
AESVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARES GYVYYLKFDY WQGTLVTVS 120
S                                                    121

SEQ ID NO: 279      moltype = DNA  length = 363
FEATURE            Location/Qualifiers
misc_feature       1..363
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polynucleotide"
source            1..363
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 279
gaagtgcagc tgctggaaag cgggtggcggc ctggtgcagc caggtggtag cctgcgcctg 60
agctgtgcgc caagcggcct taccttagc agctatgcca ttagctgggt gcgccaaaca 120
ccaggcaaaag gcctggaatg ggtgagcgcc attagcggca gcggtggcag cacctattat 180
gccgagagcg tgaaaggtcg ctttaccatt agtcgcgata acagcaaaaa caccctgcat 240
ctgcaaatga acagcctcgc ggcagaagat accgcagttt attattgcgc gcgtgagagc 300
ggatacgttt actatctgaa attcgattat tggggccagg gcaccctggt tactgtctcg 360
agc                                                    363

SEQ ID NO: 280      moltype = AA  length = 451
FEATURE            Location/Qualifiers
REGION            1..451
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..451
mol_type = protein
organism = synthetic construct

SEQUENCE: 280
EVQLLESGGG LVQPGGSLRL SCAASGFTFS SYAISWVRQA PGKGLEWVSA ISGSGGSTYY 60
AESVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARES GYVYYLKFDY WQGTLVTVS 120
SASTKGPSVF PLAPSSKSTS GGTAALGCLV KDYFPEPVTW SWNSGALTSG VHTFPAVLQS 180
SGLYSLSSVV TVPSSSLGTQ TYICNVNHPK SNTKVDKRVK PKSCDKHTHC PPCPAPEAAG 240
GPSVFLFPPK PKDTLMISRT PEVTCVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY 300
NSTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE 360
EMTKNQVSLT CLVKGFYPSD IAVEWESNGQ PENNYKTTTPP VLDSGDSFPL YSKLTVDKSR 420
WQQGNVFCSS VMHEALHNNH YQKSLSLSPG K                                                    451

SEQ ID NO: 281      moltype = DNA  length = 1353
FEATURE            Location/Qualifiers

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misc_feature      1..1353
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polynucleotide"
source            1..1353
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 281
gaagtgcagc tgctggaaag cgggtggcggc ctggtgcagc caggtggtag cctgcgacctg 60
agctgtgccg caagcggctt taccttttagc agctatgcca ttagctgggt gcgccaagca 120
ccaggcaaaag gcttggaaatg ggtgagcgcc attagcggca gcggtggcag cacctattat 180
gccgagagcg tgaaaggctg ctttaccatt agtcgcgata acagcaaaaa caccctgtat 240
ctgcaaatga acagcctgcg gccagaagat accgcagttt attattgcgc gcgtgagagc 300
ggatacgttt actatctgaa attcagattat tggggccagg gcaccctggt tactgtctcg 360
agcgcgtcga ccaaaggccc cagcgtgttc cctctggccc ccagcagcaa gagcacctct 420
ggcggaaacag ccgcccctggg ctgcctggtc aaggactact tccccgagcc cgtgaccctg 480
tcctggaact ctggcgccct gaccagcggc gtgcacacct ttccagccgt gctccagagc 540
agcggcctgt acagcctgag cagcgtcgtg accgtgccca gcagcagcct gggcaccagc 600
acctacatct gcaacgtgaa ccacaagccc agcaacacaa aggtggacaa gcgggtggaa 660
cccaagagct gcgacaagac ccacacctgt cccccctgcc ctgcccctga agcggcgagg 720
ggccccctcg tgttccctgt ccccccaaa cctaaggaca cctgatgat cagccggacc 780
cccgaaatga cctgcgtggt ggtggacgtg tcccacgagg accctgaagt gaagtttaat 840
tggtagctgg acggcgtgga agtgacaca gccaaagacca agcccagaga ggaacagtac 900
aacagcacct accgggtggt gtccctgctg accgtgtgc accaggactg gctgaacggc 960
aaagagtaca agtgcaaggt gtccaacaag gccctgcctg ccccatcga gaaaaccatc 1020
agcaaggcca aaggccagcc ccgcgagccc caggtgtaca cactgcccc tagccgggaa 1080
gagatgacca agaaccaggt tgccctgacc tgcctcgtga agggcttcta cccagcgac 1140
attgccgtgg aatgggagag caacggccag cccgagaaca actacaagac cccccccct 1200
gtgctggaca gcgacggctc attcttcctg tacagcaagc tgaccgtgga caagagccgg 1260
tggcagcagg gcaacgtggt cagctgctcc gtgatgcacg aggccttgca caaccactac 1320
accagaagt ccctgagcct gagccccggc aag 1353

SEQ ID NO: 282      moltype = AA length = 11
FEATURE            Location/Qualifiers
REGION             1..11
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source             1..11
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 282
RASQSISSYL N                               11

SEQ ID NO: 283      moltype = AA length = 9
FEATURE            Location/Qualifiers
REGION             1..9
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source             1..9
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 283
QQHVRVPIT                               9

SEQ ID NO: 284      moltype = AA length = 7
FEATURE            Location/Qualifiers
REGION             1..7
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source             1..7
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 284
SQSISSY                               7

SEQ ID NO: 285      moltype = AA length = 6
FEATURE            Location/Qualifiers
REGION             1..6
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source             1..6
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 285
HVRVPI                               6

SEQ ID NO: 286      moltype = AA length = 6
FEATURE            Location/Qualifiers
REGION             1..6
                  note = source = /note="Description of Artificial Sequence:

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source          Synthetic peptide"
                1..6
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 286
QSISSY                                                6

SEQ ID NO: 287          moltype = AA  length = 107
FEATURE                Location/Qualifiers
REGION                1..107
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source                1..107
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 287
DIQMTQSPSS LSASVGDRVIT ITCRASQSSIS SYLNWYQQKP GKAPKLLIYA ASSLQSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ HVRVPITFGQ GTKVEIK 107

SEQ ID NO: 288          moltype = DNA  length = 321
FEATURE                Location/Qualifiers
misc_feature          1..321
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source                1..321
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 288
gatattcaga tgaccacagag cccgagcagc ctgagcgcca gcgtgggtga tcgcgtgacc 60
attacctgtc gcgcaagcca gagcattagc agctatctga actgggtatca gcagaaacca 120
ggcaaaagccc caaaactgct gatttatgcc gcaagcagcc tgcaaaagcgg tgtgccgagc 180
cgcttttagcg gcagcggtag cggcaccgat tttaccctga ccattagtag cctgcaaccg 240
gaagactttg ccacctatta ttgccagcag catgttcgtg ttccgatcac cttcggccag 300
ggtaccaaag tggaaatcaa g 321

SEQ ID NO: 289          moltype = AA  length = 214
FEATURE                Location/Qualifiers
REGION                1..214
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source                1..214
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 289
DIQMTQSPSS LSASVGDRVIT ITCRASQSSIS SYLNWYQQKP GKAPKLLIYA ASSLQSGVPS 60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ HVRVPITFGQ GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 290          moltype = DNA  length = 642
FEATURE                Location/Qualifiers
misc_feature          1..642
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source                1..642
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 290
gatattcaga tgaccacagag cccgagcagc ctgagcgcca gcgtgggtga tcgcgtgacc 60
attacctgtc gcgcaagcca gagcattagc agctatctga actgggtatca gcagaaacca 120
ggcaaaagccc caaaactgct gatttatgcc gcaagcagcc tgcaaaagcgg tgtgccgagc 180
cgcttttagcg gcagcggtag cggcaccgat tttaccctga ccattagtag cctgcaaccg 240
gaagactttg ccacctatta ttgccagcag catgttcgtg ttccgatcac cttcggccag 300
ggtaccaaag tggaaatcaa cgggaccgtg gccgctccct ccgtgttcat cttcccaccc 360
agcgacgagc agctgaagtc cggcacagcc agcgtcgtgt gcctgctgaa caactttac 420
ccccgcgagg ccaaagtga gtggaagggtg gacaacgccc tccagagcgg caacagccag 480
gaaagcgtca ccgagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaagtga ccaccagggc 600
ctgtccagcc ccgtgaccaa gagcttaac cggggcgagt gt 642

SEQ ID NO: 291          moltype = AA  length = 10
FEATURE                Location/Qualifiers
REGION                1..10
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source                1..10
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 291

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GFTFSNYWIS	10
SEQ ID NO: 292	moltype = AA length = 19
FEATURE	Location/Qualifiers
REGION	1..19
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..19
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 292	
RIKSKTYGGT TDYAEPVKG	19
SEQ ID NO: 293	moltype = AA length = 17
FEATURE	Location/Qualifiers
REGION	1..17
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..17
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 293	
EKYSIRARGH GDYGFVDV	17
SEQ ID NO: 294	moltype = AA length = 5
FEATURE	Location/Qualifiers
REGION	1..5
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..5
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 294	
NYWIS	5
SEQ ID NO: 295	moltype = AA length = 7
FEATURE	Location/Qualifiers
REGION	1..7
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..7
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 295	
GFTFSNY	7
SEQ ID NO: 296	moltype = AA length = 8
FEATURE	Location/Qualifiers
REGION	1..8
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..8
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 296	
KSKTYGGT	8
SEQ ID NO: 297	moltype = AA length = 8
FEATURE	Location/Qualifiers
REGION	1..8
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..8
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 297	
GFTFSNYW	8
SEQ ID NO: 298	moltype = AA length = 10
FEATURE	Location/Qualifiers
REGION	1..10
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..10
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 298	
IKSKTYGGTT	10

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SEQ ID NO: 299      moltype = AA  length = 19
FEATURE            Location/Qualifiers
REGION            1..19
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..19
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 299
AREKYSIRAR GHGDYGFVDV                                     19

SEQ ID NO: 300      moltype = AA  length = 128
FEATURE            Location/Qualifiers
REGION            1..128
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..128
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 300
EVQLVESGGG LVKPGGSLRL SCAASGFTFS NYWISWVRQA PGKGLEWVGR IKSKTYGGTT 60
DYAEPVKGRF TISRDDSKNT LYLQMNLSLKT EDTAVYYCAR EKYSIRARGH GDYGFDDVWGQ 120
GTLVTVSS                                     128

SEQ ID NO: 301      moltype = DNA  length = 384
FEATURE            Location/Qualifiers
misc_feature       1..384
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polynucleotide"
source            1..384
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 301
gaagtgcagc tgggtgaaag cggcgggtggc ctggtgaaac caggcggtag cctgcgcctg 60
agctgcgcgc ccagcgggctt tacctttagc aactattgga ttagctgggt tcgccaggcc 120
ccaggcaaaag gcctggaatg ggttgccgc atcaaaagca aaacctatgg cggcaccacc 180
gattatgccg agccagtgaag aggcgcgttt accattagcc gcgacgatag caaaaacacc 240
ctgtacctgc aaatgaacag cctgaaaacc gaagataccg ccgtgtatta ttgcgcgcgt 300
gagaaatatt ccatccgtgc acgtggtcac ggagactacg gattgatgt gtggggccag 360
ggcacccctg ttactgtctc gage                                     384

SEQ ID NO: 302      moltype = AA  length = 458
FEATURE            Location/Qualifiers
REGION            1..458
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..458
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 302
EVQLVESGGG LVKPGGSLRL SCAASGFTFS NYWISWVRQA PGKGLEWVGR IKSKTYGGTT 60
DYAEPVKGRF TISRDDSKNT LYLQMNLSLKT EDTAVYYCAR EKYSIRARGH GDYGFDDVWGQ 120
GTLVTVSSAS TKGPSVFPLA PSSKSTSGGT AALGCLVKDY FPEPVTVSWN SGALTSGVHT 180
FPAVLQSSGL YSLSSVTVTP SSSLGTQTYI CNVNHKPSNT KVDKRVEPKS CDKHTCPCPC 240
PAPEAAGGPS VFLFPPKPKD TLMISRTPEV TCVVVDVSHE DPEVKFNWYV DGEVHNAKT 300
KPREEQYNST YRVVSVLTVL HQDWLNGKEY KCKVSNKALP APIEKTISKA KGQPREPQVY 360
TLPPSREEMT KNQVSLTCLV KGFYPSDIAV EWESNGQPEN NYKTPPVLD SDGSFFLYSK 420
LTVDKSRWQQ GNVFSCSMVH EALHNNHYTQK SLSLSPGK                                     458

SEQ ID NO: 303      moltype = DNA  length = 1374
FEATURE            Location/Qualifiers
misc_feature       1..1374
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polynucleotide"
source            1..1374
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 303
gaagtgcagc tgggtgaaag cggcgggtggc ctggtgaaac caggcggtag cctgcgcctg 60
agctgcgcgc ccagcgggctt tacctttagc aactattgga ttagctgggt tcgccaggcc 120
ccaggcaaaag gcctggaatg ggttgccgc atcaaaagca aaacctatgg cggcaccacc 180
gattatgccg agccagtgaag aggcgcgttt accattagcc gcgacgatag caaaaacacc 240
ctgtacctgc aaatgaacag cctgaaaacc gaagataccg ccgtgtatta ttgcgcgcgt 300
gagaaatatt ccatccgtgc acgtggtcac ggagactacg gattgatgt gtggggccag 360
ggcacccctg ttactgtctc gagcgcgtcg accaaaggcc ccagcgtgtt ccctctggcc 420
cccagcagca agagcacctc tggcggaaca gccgccctgg gctgcctggt caaggactac 480
ttccccgagc ccgtgaccgt gtcctggaac tctggcgccc tgaccagcgg cgtgcacacc 540
tttcagccg tgctccagag cagcggcctg tacagcctga gcagcgtcgt gaccgtgccc 600
agcagcagcc tgggcaccca gacctacatc tgcaacgtga accacaagcc cagcaacaca 660

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aaggtggaca agcgggtgga acccaagagc tgcgacaaga cccacacctg tccccctgc 720
cctgcccctg aagcggcgagg agggccctcc gtgttctctg tcccccaaa gcctaaggac 780
accctgatga tcagccggac ccccgaaagt acctgcgtgg tgggtggacgt gtcccacgag 840
gacctgaag tgaagttaa ttggtacgtg gacggcggtg aagtgcacaa cgccaagacc 900
aagcccagag aggaacagta caacagcacc taccgggtgg tgtccgtgct gaccgtgctg 960
caccaggact ggctgaacgg caaagagtac aagtgcgaag tgtccaacaa ggccctgcct 1020
gcccccatcg agaaaacccat cagcaaggcc aaaggccagc cccgcgagcc ccagggtgac 1080
acactgcccc ctagccggga agagatgacc aagaaccagg tgtccctgac ctgcctcgtg 1140
aagggtcttct accccagcga cattgccgtg gaatgggaga gcaacggcca gcccgagaac 1200
aactacaaga ccaccccccc tgtgctggac agcgcgggct cattcttct gtacagcaag 1260
ctgaccgtgg acaagagcgg gtggcagcag ggcaacgtgt tcagctgctc cgtgatgcac 1320
gaggccctgc acaaccacta caccagaag tccctgagcc tgagcccgg caag 1374

SEQ ID NO: 304      moltype = AA  length = 9
FEATURE            Location/Qualifiers
REGION            1..9
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..9
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 304
QQGYHAPFT                      9

SEQ ID NO: 305      moltype = AA  length = 6
FEATURE            Location/Qualifiers
REGION            1..6
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..6
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 305
GYHAPF                          6

SEQ ID NO: 306      moltype = AA  length = 107
FEATURE            Location/Qualifiers
REGION            1..107
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..107
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 306
DIQMTQSPSS LSASVGRVT ITCRASQGIS NYLAWYQQKP GKVPKLLIYA ASTLQSGVPS 60
RFGSGSGGTD FTLTISSLQP EDVATYYCQQ GYHAPFTFGQ GTKVEIK 107

SEQ ID NO: 307      moltype = DNA  length = 321
FEATURE            Location/Qualifiers
misc_feature      1..321
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polynucleotide"
source            1..321
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 307
gatattcaga tgaccagag cccgagcagc ctgagcgcaa gcgtgggcca tcgctgacc 60
attacctgcc ggcgcagcca gggcattagc aactatctgg cctggatatc gcagaaaccg 120
ggcaaaagtgc cgaaactgct gatctatgcc gccagcacc cgcgaagcgg cgtgccaagt 180
cgctttagcg gcagcggtag cggcaccgat ttcaccctga ccattagcag cctgcaaccg 240
gaagacgtgg cgacctatta ttgccagcag ggttaccatg ctccgttcac ctgcggccag 300
ggtaccaaag tggaaatcaa g 321

SEQ ID NO: 308      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 308
DIQMTQSPSS LSASVGRVT ITCRASQGIS NYLAWYQQKP GKVPKLLIYA ASTLQSGVPS 60
RFGSGSGGTD FTLTISSLQP EDVATYYCQQ GYHAPFTFGQ GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 309      moltype = DNA  length = 642
FEATURE            Location/Qualifiers

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misc_feature      1..642
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polynucleotide"
source            1..642
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 309
gatattcaga tgaccagag cccgagcagc ctgagcgcaa gcgtgggcga tcgcgtgacc 60
attacctgcc gcgccagcca gggcattagc aactatctgg cctggatca gcagaaaccg 120
ggcaaagtgc cgaactgtct gatctatgcc gccagcacc tgcaaagcgg cgtgccaagt 180
cgctttagcg gcagcggtag cggcaccgat ttcacccgtga ccattagcag cctgcaaccg 240
gaagacgtgg cgacctatta ttgccagcag ggttaccatg ctccgttcac cttcggccag 300
ggtaccaaag tggaaatcaa gcggaccgtg gccgctccct ccgtgttcac cttcccaccc 360
agcgacgagc agctgaagtc cggcacagcc agcgtcgtgt gcctgctgaa caactttac 420
ccccgcgagg ccaaagtgca gtggaagggtg gacaacgccc tccagagcgg caacagccag 480
gaaagcgtca ccgagcagga cagcaaggac tccacctaca gcctgagcag caccctgacc 540
ctgagcaagg ccgactacga gaagcacaag gtgtacgcct gcgaagtgac ccaccagggc 600
ctgtccagcc ccgtgaccaa gagcttcaac cggggcgagt gt 642

SEQ ID NO: 310      moltype = AA length = 10
FEATURE            Location/Qualifiers
REGION            1..10
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..10
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 310
GYSFTSYWIS                                10

SEQ ID NO: 311      moltype = AA length = 17
FEATURE            Location/Qualifiers
REGION            1..17
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..17
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 311
IIYPGTSYTR YSPSFQG                                17

SEQ ID NO: 312      moltype = AA length = 11
FEATURE            Location/Qualifiers
REGION            1..11
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..11
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 312
GAVAGQLGFD H                                11

SEQ ID NO: 313      moltype = AA length = 6
FEATURE            Location/Qualifiers
REGION            1..6
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..6
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 313
YPGTSY                                6

SEQ ID NO: 314      moltype = AA length = 8
FEATURE            Location/Qualifiers
REGION            1..8
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..8
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 314
IYPGTSYT                                8

SEQ ID NO: 315      moltype = AA length = 13
FEATURE            Location/Qualifiers
REGION            1..13
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"

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source          1..13
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 315
ARGAVAGQLG FDH                                     13

SEQ ID NO: 316          moltype = AA length = 120
FEATURE                Location/Qualifiers
REGION                1..120
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source                1..120
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 316
EVQLVQSGAE VKKPGESLKI SCKGSGYSFT SYWISWVRQM PGKGLEWMGI IYPGTSYTRY 60
SPSPQGQVTI SADKSISTAY LQWSSLKASD TAMYYCARGA VAGQLGFDHW GQGLTIVTVSS 120

SEQ ID NO: 317          moltype = DNA length = 360
FEATURE                Location/Qualifiers
misc_feature          1..360
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source                1..360
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 317
gaagtgcagc tgggtgcagag cgggtgccgaa gtgaaaaaac cgggcgaaag cctgaaaatc 60
agctgcaaaag gcagcgggcta tagctttacc agctattgga ttagctgggt tcgccagatg 120
ccgggcaaaag gcctgggaatg gatgggcatt atctatccgg gcaccagcta taccgcctat 180
agcccgagct ttcaggggcca gggtacaatt agcgccgaca aaagcatcag caccgcctat 240
ctgcaatgga gcagcctgaa agccagcgat accgccatgt attattgcgc gcgtggtgca 300
gttgaggac aactgggatt tgatcactgg ggccagggca ccctggttac tgtctcgagc 360

SEQ ID NO: 318          moltype = AA length = 450
FEATURE                Location/Qualifiers
REGION                1..450
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source                1..450
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 318
EVQLVQSGAE VKKPGESLKI SCKGSGYSFT SYWISWVRQM PGKGLEWMGI IYPGTSYTRY 60
SPSPQGQVTI SADKSISTAY LQWSSLKASD TAMYYCARGA VAGQLGFDHW GQGLTIVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKRVPE KSCDKTHTCP PCPAPEAAGG 240
PSVFLFPPPK KDTLMISRTP EVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN 300
STYRVVSVLT VHLQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 360
MTKNQVSLTC LVKGFYPSDI AVEWESNGQP ENNYKTPPVV LDSGGSFFLY SKLTVDKSRW 420
QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 450

SEQ ID NO: 319          moltype = DNA length = 1350
FEATURE                Location/Qualifiers
misc_feature          1..1350
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source                1..1350
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 319
gaagtgcagc tgggtgcagag cgggtgccgaa gtgaaaaaac cgggcgaaag cctgaaaatc 60
agctgcaaaag gcagcgggcta tagctttacc agctattgga ttagctgggt tcgccagatg 120
ccgggcaaaag gcctgggaatg gatgggcatt atctatccgg gcaccagcta taccgcctat 180
agcccgagct ttcaggggcca gggtacaatt agcgccgaca aaagcatcag caccgcctat 240
ctgcaatgga gcagcctgaa agccagcgat accgccatgt attattgcgc gcgtggtgca 300
gttgaggac aactgggatt tgatcactgg ggccagggca ccctggttac tgtctcgagc 360
gcgtcgacca aaggccccag cgtgttccct ctggccccca gcagcaagag cactctggc 420
ggaaacagccg ccctgggctg cctggtcaag gactacttcc ccgagcccg gaccgtgtcc 480
tggaactctg gcgcccctag cagcggcgctg cacaccttcc cagccgtgct ccagagcagc 540
ggcctgtaca gcctgagcag cgtcgtgacc gtgccagca gcagcctggg caccagacc 600
tacatctgca acgtgaacca caagcccagc aacacaaaagg tggacaagcg ggtggaaccc 660
aagagctcgc acaagaccca cactgtcccc cctgccccctg cccctgaagc ggcgggaggc 720
ccctccgtgt tcctgttccc cccaaagcct aaggacaccc tgatgatcag ccggaccccc 780
gaagtgcact gcgtggtggt ggacgtgtcc cagcaggacc ctgaagtga gtttaattgg 840
tacgtggacg gcgtggaagt gcacaacgcc aagaccaagc ccagagagga acagtacaac 900
agcacctacc ggggtgtgtc cgtgctgacc gtgctgcacc aggactggct gaacggcaaa 960
gagtacaagt gcaaggtgtc caacaaggcc ctgcctgccc ccatcgagaa aaccatcagc 1020
aaggccaaaag gccagccccg cgagccccag gtgtacacac tgccccctag ccgggaagag 1080

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atgaccaaga accaggtgtc cctgacctgc ctctgtgaagg gcttctaccc cagcgacatt 1140
gccgtggaat gggagagcaa cggccagccc gagaacaact acaagaccac cccccctgtg 1200
ctggacagcg acggctcatt ctctctgtac agcaagctga ccgtggacaa gagccggtgg 1260
cagcagggca acgtgttcag ctgctccgtg atgcacgagg ccctgcacaa ccactacacc 1320
cagaagtccc tgagcctgag ccccggaag                                     1350

SEQ ID NO: 320      moltype = AA length = 14
FEATURE            Location/Qualifiers
REGION            1..14
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..14
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 320
TGSSSNIGAG YDVH                                     14

SEQ ID NO: 321      moltype = AA length = 7
FEATURE            Location/Qualifiers
REGION            1..7
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..7
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 321
GNSNRPS                                               7

SEQ ID NO: 322      moltype = AA length = 11
FEATURE            Location/Qualifiers
REGION            1..11
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..11
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 322
QSYYTSSHGP V                                         11

SEQ ID NO: 323      moltype = AA length = 10
FEATURE            Location/Qualifiers
REGION            1..10
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..10
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 323
SSSNIGAGYD                                           10

SEQ ID NO: 324      moltype = length =
SEQUENCE: 324
000

SEQ ID NO: 325      moltype = AA length = 8
FEATURE            Location/Qualifiers
REGION            1..8
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..8
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 325
YTTSSHGP                                             8

SEQ ID NO: 326      moltype = AA length = 9
FEATURE            Location/Qualifiers
REGION            1..9
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..9
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 326
SSNIGAGYD                                           9

SEQ ID NO: 327      moltype = AA length = 111
FEATURE            Location/Qualifiers
REGION            1..111

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note = source = /note="Description of Artificial Sequence:
  Synthetic polypeptide"
source      1..111
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 327
QSVLTQPPSV SGAPGQQRVTI SCTGSSSNIG AGYDVHWYQQ LPGTAPKLLI YGNSNRPSGV 60
PDRFSGSKSG TSASLAITGL QAEDEADYYC QSYTSSHGP VFGGGTKLTV L 111

SEQ ID NO: 328      moltype = DNA length = 333
FEATURE            Location/Qualifiers
misc_feature       1..333
note = source = /note="Description of Artificial Sequence:
  Synthetic polynucleotide"
source            1..333
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 328
cagagcgtgc tgacccagcc accaagcgtg agcgggtgcac caggtcagcg cgtgaccatt 60
agctgcaccg gcagcagcag caacattggc gcaggctatg atgtgcattg gtatcagcag 120
ctgccaggca cgcaccgcaa actgctgatt tatggcaaca gcaatcgccc aagcgggtgtg 180
ccggatcgct ttagcggcag caaaagcggc accagcgcca gcctggcgat taccggtctg 240
caagccgaag acgaagccga ttattactgc cagtcttact acattcttc tcatggtccg 300
gtgtttggcg gcggtaccaa gctgaccgtg ctg 333

SEQ ID NO: 329      moltype = AA length = 217
FEATURE            Location/Qualifiers
REGION            1..217
note = source = /note="Description of Artificial Sequence:
  Synthetic polypeptide"
source            1..217
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 329
QSVLTQPPSV SGAPGQQRVTI SCTGSSSNIG AGYDVHWYQQ LPGTAPKLLI YGNSNRPSGV 60
PDRFSGSKSG TSASLAITGL QAEDEADYYC QSYTSSHGP VFGGGTKLTV LGQPKAAPSV 120
TLFPPSSEEL QANKATLVCL ISDFYPGAVT VAWKADSSPV KAGVETTPS KQSNKYAAS 180
SYLSLTPEQW KSHRSYSCQV THEGSTVEKT VAPTECS 217

SEQ ID NO: 330      moltype = DNA length = 651
FEATURE            Location/Qualifiers
misc_feature       1..651
note = source = /note="Description of Artificial Sequence:
  Synthetic polynucleotide"
source            1..651
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 330
cagagcgtgc tgacccagcc accaagcgtg agcgggtgcac caggtcagcg cgtgaccatt 60
agctgcaccg gcagcagcag caacattggc gcaggctatg atgtgcattg gtatcagcag 120
ctgccaggca cgcaccgcaa actgctgatt tatggcaaca gcaatcgccc aagcgggtgtg 180
ccggatcgct ttagcggcag caaaagcggc accagcgcca gcctggcgat taccggtctg 240
caagccgaag acgaagccga ttattactgc cagtcttact acattcttc tcatggtccg 300
gtgtttggcg gcggtaccaa gctgaccgtg ctgggccagc ccaaagccgc ccctagcgtg 360
accctgttcc cccaagcag cgaggaaactc caggccaaca aggccacctc cgtgtgacct 420
atcagcgact tctaccctgg cgccgtgacc gtggcctgga agggccgatag cagccctgtg 480
aaggccggcg tggaaaccac cacccccagc aagcagagca acaacaata cgccgccagc 540
agctacctga gctgagcccc cgagcagttg aagtcaccac gatcctacag ctgccaggtc 600
acacacgagg gcagcaccgt ggaaaagacc gtggccccc cagagtgcag c 651

SEQ ID NO: 331      moltype = AA length = 13
FEATURE            Location/Qualifiers
REGION            1..13
note = source = /note="Description of Artificial Sequence:
  Synthetic peptide"
source            1..13
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 331
PYLGDRRSYG FDH 13

SEQ ID NO: 332      moltype = AA length = 15
FEATURE            Location/Qualifiers
REGION            1..15
note = source = /note="Description of Artificial Sequence:
  Synthetic peptide"
source            1..15
                 mol_type = protein
                 organism = synthetic construct

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SEQUENCE: 332
 ARPYLGDRRS YGFDH 15

SEQ ID NO: 333 moltype = AA length = 122
 FEATURE Location/Qualifiers
 REGION 1..122
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..122
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 333
 EVQLLESGGG LVQPGGSLRL SCAASGFTFS SYAISWVRQA PGKGLEWVSA ISGSGGSTYY 60
 AESVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARPY LGDRRSYGYD HWGQGTLLTV 120
 SS 122

SEQ ID NO: 334 moltype = DNA length = 366
 FEATURE Location/Qualifiers
 misc_feature 1..366
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
 source 1..366
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 334
 gaagtgcagc tgctggaaag cgggtggcggc ctggtgcagc caggtggtag cctgcgcctg 60
 agctgtgccg caagcggcgt taccttttagc agctatgccca ttagctgggt gcgccaagca 120
 ccaggcaaaag gcctggaatg ggtgagcggc attagcggca gcggtggcag cacctattat 180
 gccgagagcg tgaaaggtcg ctttaccatt agtcgcgata acagcaaaaa caccctgtat 240
 ctgcaaatga acagcctgcg gccagaagat accgcagttt attattgcgc gcgaccttat 300
 ctgggtgacc gtcgtagcta tggtttcgac cactggggcc agggcaccct ggttactgtc 360
 tcgagc 366

SEQ ID NO: 335 moltype = AA length = 452
 FEATURE Location/Qualifiers
 REGION 1..452
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..452
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 335
 EVQLLESGGG LVQPGGSLRL SCAASGFTFS SYAISWVRQA PGKGLEWVSA ISGSGGSTYY 60
 AESVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARPY LGDRRSYGYD HWGQGTLLTV 120
 SSASTKGPSV FPLAPSSKST SGGTAALGCL VKDYFPEPVT VSWNSGALTS GVHTFPAVLQ 180
 SSGLYSLSSV VTPVSSSLGT QTYICNVNHK PSNTKVDKRV EPKSCDKTHT CPPCPAPEAA 240
 GGPSVFLFPP KPKDTLMISR TPEVTCVVVD VSHEDPEVKF NWYVDGVEVH NAKTKPREEQ 300
 YNSTYRVVSV LTVLHQDWLN GKEYKCKVSN KALPAPIEKT ISKAKGQPRE PQVYTLPPSR 360
 EEMTKNQVSL TCLVKGFYPS DIAVEWESNG QPENNYKTP PVLDSGSEFF LYSKLTVDKS 420
 RWQQGNVFSC SVMHEALHNNH YTKSLSLSLP GK 452

SEQ ID NO: 336 moltype = DNA length = 1356
 FEATURE Location/Qualifiers
 misc_feature 1..1356
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
 source 1..1356
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 336
 gaagtgcagc tgctggaaag cgggtggcggc ctggtgcagc caggtggtag cctgcgcctg 60
 agctgtgccg caagcggcgt taccttttagc agctatgccca ttagctgggt gcgccaagca 120
 ccaggcaaaag gcctggaatg ggtgagcggc attagcggca gcggtggcag cacctattat 180
 gccgagagcg tgaaaggtcg ctttaccatt agtcgcgata acagcaaaaa caccctgtat 240
 ctgcaaatga acagcctgcg gccagaagat accgcagttt attattgcgc gcgaccttat 300
 ctgggtgacc gtcgtagcta tggtttcgac cactggggcc agggcaccct ggttactgtc 360
 tcgagcgcgt cgaccaaaag cccagcgtg ttccctctgg cccccagcag caagagcacc 420
 tctggcggaa cagcgcgcct gggctgcctg gtcaaggact acttcccga gcccgtagc 480
 gtgtcctgga actctggcgc cctgaccagc ggctgcacac cttttccagc cgtgctccag 540
 agcagcggcc tgtacagcct gaggcagcgtc gtgaccgtgc ccagcagcag cctggggcacc 600
 cagacctaca tctgcaacgt gaaccacaag cccagcaaca caaagggtga caagcgggtg 660
 gaacccaaga gctgcgacaa gacccacacc tgtccccctt gccctgcccc tgaagcggcg 720
 ggaggccctt ccgtgttctt gttcccccca aagcctaagg acacctgat gatcagccgg 780
 acccccgaag tgacctgcgt ggtggtggac gtgtcccacg aggacctga agtgaagttt 840
 aattggtagc tggacggcgt ggaagtgcac aacgccaaga ccaagcccag agaggaacag 900
 tacaacagca cctaccgggt ggtgtccgtg ctgaccgtgc tgcaccagga ctggctgaac 960
 ggcaaaagat acaagtgcaa ggtgtccaac aaggccctgc ctgcccccat cgagaaaacc 1020
 atcagcaagg ccaaaaggcca gcccccgcag ccccagggtg acacactgcc ccctagccgg 1080
 gaagagatga ccaagaacca ggtgtccctg acctgcctcg tgaagggtct ctaccccgag 1140

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gacattgccg tggaaatggga gagcaacggc cagcccgaga acaactacaa gaccaccccc 1200
cctgtgctgg acagcgacgg ctctattctt ctgtacagca agctgacggt ggacaagagc 1260
cgggtggcagc agggcaacgt gtccagctgc tccgtgatgc acgaggccct gcacaaccac 1320
tacaccaga agtccctgag cctgagcccc ggcaag 1356

SEQ ID NO: 337      moltype = AA length = 14
FEATURE            Location/Qualifiers
REGION            1..14
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..14
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 337
TGTSSDVGSY NLVS 14

SEQ ID NO: 338      moltype = AA length = 7
FEATURE            Location/Qualifiers
REGION            1..7
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..7
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 338
EGSKRPS 7

SEQ ID NO: 339      moltype = AA length = 11
FEATURE            Location/Qualifiers
REGION            1..11
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..11
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 339
SSYGFHIVVV V 11

SEQ ID NO: 340      moltype = AA length = 10
FEATURE            Location/Qualifiers
REGION            1..10
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..10
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 340
TSSDVGSYNL 10

SEQ ID NO: 341      moltype = length =
SEQUENCE: 341
000

SEQ ID NO: 342      moltype = AA length = 8
FEATURE            Location/Qualifiers
REGION            1..8
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..8
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 342
YGFHIVVV 8

SEQ ID NO: 343      moltype = AA length = 9
FEATURE            Location/Qualifiers
REGION            1..9
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..9
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 343
SSDVGSYNL 9

SEQ ID NO: 344      moltype = AA length = 111
FEATURE            Location/Qualifiers
REGION            1..111
                  note = source = /note="Description of Artificial Sequence:

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source          Synthetic polypeptide"
                1..111
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 344
QSALTQPASV SGSPGQSITI SCTGTSSDVG SYNLVSWYQQ HPGKAPKLMI YEGSKRPSGV 60
SNRFGSGSKG NTASLTISGL QAEDEADYYC SSYGFHIVVV VFGGGTKLTV L 111

SEQ ID NO: 345      moltype = DNA length = 333
FEATURE            Location/Qualifiers
misc_feature        1..333
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"

source              1..333
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 345
cagagcgccc tgacccagcc agccagcggt agcggtagcc caggccagag cattaccatt 60
agctgcaccg gcaccagcag cgacgtgggc agctataacc tggtagctg gtatcagcag 120
catccgggca aagccccgaa actgatgata tatgaaggca gcaaacgccc gagcggcgtt 180
agcaaccgct ttagtgccag caaaagcggc aacaccgcca gcctgacctat tagcggcctg 240
caagccgaag acgaagccga ttattactgc tcctcttacg gtttccatat cgttggtgtt 300
gtggttgccg gcggtaccaa gctgaccgtg ctg 333

SEQ ID NO: 346      moltype = AA length = 217
FEATURE            Location/Qualifiers
REGION              1..217
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"

source              1..217
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 346
QSALTQPASV SGSPGQSITI SCTGTSSDVG SYNLVSWYQQ HPGKAPKLMI YEGSKRPSGV 60
SNRFGSGSKG NTASLTISGL QAEDEADYYC SSYGFHIVVV VFGGGTKLTV LGQPKAAPSV 120
TLFPSPSEEL QANKATLVCL ISDFYPGAVT VAWKADSSPV KAGVETTPS KQSNNKYAAS 180
SYLSLTPEQW KSHRSYSCQV THEGSTVEKT VAPTECS 217

SEQ ID NO: 347      moltype = DNA length = 651
FEATURE            Location/Qualifiers
misc_feature        1..651
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"

source              1..651
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 347
cagagcgccc tgacccagcc agccagcggt agcggtagcc caggccagag cattaccatt 60
agctgcaccg gcaccagcag cgacgtgggc agctataacc tggtagctg gtatcagcag 120
catccgggca aagccccgaa actgatgata tatgaaggca gcaaacgccc gagcggcgtt 180
agcaaccgct ttagtgccag caaaagcggc aacaccgcca gcctgacctat tagcggcctg 240
caagccgaag acgaagccga ttattactgc tcctcttacg gtttccatat cgttggtgtt 300
gtggttgccg gcggtaccaa gctgaccgtg ctgggcccag ccaaagccgc ccctagcgtg 360
accctgttcc ccccaagcag cgagggaactc caggccaaca aggccaccct cgtgtgcctg 420
atcagcgact tctaccctgg cgccgtgacc gtggcctgga aggcgcatag cagccctgtg 480
aaggccggcg tggaaaccac cacccccagc aagcagagca acaacaaata cgcgccagc 540
agctacctga gcctgacccc cgagcagtggt aagtcccaca gatcctacag ctgccaggtc 600
acacacgagg gcagcaccgt ggaagagacc gtggccccca ccgagtgcag c 651

SEQ ID NO: 348      moltype = AA length = 11
FEATURE            Location/Qualifiers
REGION              1..11
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"

source              1..11
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 348
GSLPGLLGFD H 11

SEQ ID NO: 349      moltype = AA length = 13
FEATURE            Location/Qualifiers
REGION              1..13
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"

source              1..13
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 349

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ARGSLPGLLG FDH 13

SEQ ID NO: 350 moltype = AA length = 120
 FEATURE Location/Qualifiers
 REGION 1..120
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..120
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 350
 EVQLVQSGAE VKKPGESLKI SCKGSGYSFT SYWISWVRQM PGKGLEWMGI IYPGTSYTRY 60
 SPSFQGQVTI SADKSISTAY LQWSSLKASD TAMYYCARGS LPGLLGFDHW GQGTLLTVVSS 120

SEQ ID NO: 351 moltype = DNA length = 360
 FEATURE Location/Qualifiers
 misc_feature 1..360
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
 source 1..360
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 351
 gaagtgcagc tgggtgcagag cgggtgccgaa gtgaaaaaac cgggcgaaag cctgaaaatc 60
 agctgcaaaag gcagcggtcta tagctttacc agctattgga ttagctgggt tcgccagatg 120
 ccgggcaaaag gcctgggaatg gatgggcatt atctatccgg gcaccagcta taccgcctat 180
 agcccgagct ttcagggcca ggttacaatt agcgcgcaca aaagcatcag caccgcctat 240
 ctgcaatgga gcagcctgaa agccagcgat accgccatgt attattgcgc gcgtggaagc 300
 ctgctcgttc tgctgggttt tgatcactgg ggccagggca ccctgggttac tgtctcgagc 360

SEQ ID NO: 352 moltype = AA length = 450
 FEATURE Location/Qualifiers
 REGION 1..450
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..450
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 352
 EVQLVQSGAE VKKPGESLKI SCKGSGYSFT SYWISWVRQM PGKGLEWMGI IYPGTSYTRY 60
 SPSFQGQVTI SADKSISTAY LQWSSLKASD TAMYYCARGS LPGLLGFDHW GQGTLLTVVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSVVT VPSSSLGTQT YICNVNHKPS NTKVDRVEP KSCDKHTTCP PCPAPEAAGG 240
 PSVFLFPPPK KDTLMISRTPEVTCVVVDVS HEDPEVKFNW YVDGVEVHNA KTKPREEQYN 300
 STYRVVSVLT VLNQDNLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 360
 MTKNQVSLT LKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SKLTVDKSRW 420
 QQGNVFSCSV MHEALHNHYT QKSLSLSPGK 450

SEQ ID NO: 353 moltype = DNA length = 1350
 FEATURE Location/Qualifiers
 misc_feature 1..1350
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
 source 1..1350
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 353
 gaagtgcagc tgggtgcagag cgggtgccgaa gtgaaaaaac cgggcgaaag cctgaaaatc 60
 agctgcaaaag gcagcggtcta tagctttacc agctattgga ttagctgggt tcgccagatg 120
 ccgggcaaaag gcctgggaatg gatgggcatt atctatccgg gcaccagcta taccgcctat 180
 agcccgagct ttcagggcca ggttacaatt agcgcgcaca aaagcatcag caccgcctat 240
 ctgcaatgga gcagcctgaa agccagcgat accgccatgt attattgcgc gcgtggaagc 300
 ctgctcgttc tgctgggttt tgatcactgg ggccagggca ccctgggttac tgtctcgagc 360
 gcgtcgacca aagcggcgag cgtgttccct ctggcccccga gcagcaagag cacctctggc 420
 ggaacagccg ccctgggctg cctgggtcaag gactacttcc ccgagcccg gaccgtgtcc 480
 tggaaactctg gcgcctgac cagcggcgctg cacaccttcc cagcgtgct ccagagcagc 540
 ggctgtgaca gcctgagcag cgtcgtgacc gtgcccagca gcagcctggg caccagacc 600
 tacatctgca acgtgaacca caagcccagc aacacaaagg tggacaagcg ggtggaacc 660
 aagagctgcg acaagaccca cacctgtccc ccctgcccgt cccctgaagc ggccggaggc 720
 cctcctgtgt tcctgttccc cccaaagcct aaggaccccc tgatgatcag ccggaccccc 780
 gaagtgaact gcgtgggtgt ggacgtgtcc cagcaggacc ctgaagtga gtttaattgg 840
 tacgtggacg gcgtggaagt gcacaacgcc aagaccaagc ccagagagga acagtacaac 900
 agcacctacc ggggtgggtc cgtgctgacc gtgctgcacc aggaactggc gaacggcaaa 960
 gagtacaagt gcaaggtgtc caacaaggcc ctgcctgccc ccatcgagaa aacctcagc 1020
 aaggccaaaag gccagccccc cgagcccccag gtgtacacac tgccccctag ccgggaagag 1080
 atgaccaaga accaggtgtc cctgacctgc ctgctgaagg gcttctaccc cagcgacatt 1140
 gccgtggaat gggagagcaa cggccagccc gagaacaact acaagaccac cccccctgtg 1200
 ctggacagcg accgctcatt ctctcgtac agcaagctga ccgtggacaa gagccggtgg 1260
 cagcagggca acgtgttcag ctgctccgtg atgcacgagg ccctgcacaa ccactacacc 1320

-continued

cagaagtcctc tgagcctgag ccccggaag 1350

SEQ ID NO: 354 moltype = AA length = 7
 FEATURE Location/Qualifiers
 REGION 1..7
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
 source 1..7
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 354
 GNSNRPN 7

SEQ ID NO: 355 moltype = AA length = 11
 FEATURE Location/Qualifiers
 REGION 1..11
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
 source 1..11
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 355
 QSYDSPTSSS V 11

SEQ ID NO: 356 moltype = AA length = 8
 FEATURE Location/Qualifiers
 REGION 1..8
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
 source 1..8
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 356
 YDSPTSSS 8

SEQ ID NO: 357 moltype = AA length = 111
 FEATURE Location/Qualifiers
 REGION 1..111
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..111
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 357
 QSVLTQPPSV SGAPGQRTI SCTGSSSNIG AGYDVHWYQQ LPGTAPKLLI YGNSNRPNV 60
 PDRFSGSKSG TSASLAITGL QAEDADYYC QSYDSPTSSS VFGGGTKLV L 111

SEQ ID NO: 358 moltype = DNA length = 333
 FEATURE Location/Qualifiers
 misc_feature 1..333
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
 source 1..333
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 358
 cagagcgtgc tgaccagcc accaagcgtg agcggcgac caggtcagcg cgtgaccatt 60
 agctgcaccg gcagcagcag caacattggc gcagctatg atgtgcattg gtatcagcag 120
 ctgccaggca ccgcaccgaa actgctgatt tatggcaaca gcaatcgccc aaacgggtgtg 180
 ccggatcgct tttagcggcag caaaagcggc accagcgcca gcctggcgat taccggtctg 240
 caagccgaag acgaagccga ttattactgc cagtcttacg actctccgac ttcttctct 300
 gtgtttggcg gcggtaccaa gctgaccgtg ctg 333

SEQ ID NO: 359 moltype = AA length = 217
 FEATURE Location/Qualifiers
 REGION 1..217
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
 source 1..217
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 359
 QSVLTQPPSV SGAPGQRTI SCTGSSSNIG AGYDVHWYQQ LPGTAPKLLI YGNSNRPNV 60
 PDRFSGSKSG TSASLAITGL QAEDADYYC QSYDSPTSSS VFGGGTKLV LGQPKAAPSV 120
 TLFPPSSEEL QANKATLVCL ISDFYPGAVT VAWKADSSPV KAGVETTPS KQSNKYAAS 180
 SYLSLTPEQW KSHRSYSCQV THEGSTVEKT VAPTECS 217

SEQ ID NO: 360 moltype = DNA length = 651
 FEATURE Location/Qualifiers

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```

misc_feature      1..651
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polynucleotide"
source            1..651
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 360
cagagcgtgc tgaccagcc accaagcgtg agcgggtgcac caggtcagcg cgtgaccatt 60
agctgcaccg gcagcagcag caacattggc gcaggctatg atgtgcattg gtatcagcag 120
ctgccaggca ccgcaccgaa actgctgatt tatggcaaca gcaatcgccc aaacgggtgtg 180
ccggatcgct ttagcggcag caaaagcggc accagcgcca gcctggcgat taccggtctg 240
caagccgaag acgaagccga ttattactgc cagtcttacg actctccgac ttcttcttct 300
gtgtttggcg gcggtaccaa gctgaccgtg ctgggccagc ccaaagccgc ccctagcgtg 360
accctgttcc ccccaagcag cgaggaaact caggccaaca aggccacct cgtgtgacctg 420
atcagcgact tctacctggt cgccgtgacc gtggcctgga aggccgatag cagccctgtg 480
aaggccggcg tggaaaccac cacccccagc aagcagagca acaacaata cgccgccagc 540
agctacctga gcctgacccc cgagcagtggt aagtcccaca gatctacag ctgccaggtc 600
acacacgagg gcagcaccgt ggaaaagacc gtggccccc cagagtgcag c 651

SEQ ID NO: 361      moltype = AA length = 11
FEATURE            Location/Qualifiers
REGION            1..11
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..11
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 361
QSYGAPPRFV V 11

SEQ ID NO: 362      moltype = AA length = 8
FEATURE            Location/Qualifiers
REGION            1..8
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..8
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 362
YGAPPRFV 8

SEQ ID NO: 363      moltype = AA length = 111
FEATURE            Location/Qualifiers
REGION            1..111
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..111
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 363
QSVLTQPPSV SGAPGQRTI SCTGSSSNIG AGYDVHWYQQ LPGTAPKLLI YGNSNRPNGV 60
PDRFSGSKSG TSASLAITGL QAEDEADYYC QSYGAPPRFV VFGGGTKLTV L 111

SEQ ID NO: 364      moltype = DNA length = 333
FEATURE            Location/Qualifiers
misc_feature      1..333
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polynucleotide"
source            1..333
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 364
cagagcgtgc tgaccagcc accaagcgtg agcgggtgcac caggtcagcg cgtgaccatt 60
agctgcaccg gcagcagcag caacattggc gcaggctatg atgtgcattg gtatcagcag 120
ctgccaggca ccgcaccgaa actgctgatt tatggcaaca gcaatcgccc aaacgggtgtg 180
ccggatcgct ttagcggcag caaaagcggc accagcgcca gcctggcgat taccggtctg 240
caagccgaag acgaagccga ttattactgc caatcctatg gtgccttccc tcgtttcgtt 300
gtttttggcg gcggtaccaa gctgaccgtg ctg 333

SEQ ID NO: 365      moltype = AA length = 217
FEATURE            Location/Qualifiers
REGION            1..217
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic polypeptide"
source            1..217
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 365
QSVLTQPPSV SGAPGQRTI SCTGSSSNIG AGYDVHWYQQ LPGTAPKLLI YGNSNRPNGV 60

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```

PDRFSGSKSG TSASLAITGL QAEDEADYYC QSYGAFPRFV VFGGGTKLTV LGQPKAAPSV 120
TLFPSPSEEL QANKATLVCL ISDFYPGAVT VAWKADSSPV KAGVETTPS KQSNNKYAAS 180
SYLSLTPEQW KSHRSYSCQV THEGSTVEKT VAPTECS 217

SEQ ID NO: 366      moltype = DNA length = 651
FEATURE            Location/Qualifiers
misc_feature        1..651
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polynucleotide"
source              1..651
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 366
cagagcgtgc tgaccagcc accaagcgtg agcgggtgcac caggtcagcg cgtgaccatt 60
agctgcaccg gcagcagcag caacattggc gcaggctatg atgtgcattg gtatcagcag 120
ctgccaggca ccgaccgaa actgctgatt tatggcaaca gcaatcgccc aaacgggtgtg 180
ccggatcgct ttagcggcag caaaagcggc accagcgcca gcctggcgat taccggtctg 240
caagccgaag acgaagccga ttattactgc caatcctatg gtgccttccc tcgtttcggt 300
gtttttggcg gcggtaccaa gctgaccgtg ctgggccagc ccaaagccgc ccctagcgtg 360
accctgttcc ccccaagcag cgaggaaact caggccaaca aggccacct cgtgtgcctg 420
atcagcgact tctaccctgg cgccgtgacc gtggcctgga aggcgatag cagccctgtg 480
aaggccggcg tggaaaccac caccgccagc aagcagagca acaacaaata cgccgccagc 540
agctacctga gcctgacccc cgagcagtggt aagtcccaca gatcctacag ctgccagggtc 600
acacacgagg gcagcaccgt ggaaaagacc gtggccccc cagagtgcag c 651

SEQ ID NO: 367      moltype = AA length = 10
FEATURE            Location/Qualifiers
REGION              1..10
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..10
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 367
GFSFSKYYLN 10

SEQ ID NO: 368      moltype = AA length = 17
FEATURE            Location/Qualifiers
REGION              1..17
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..17
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 368
SIHQQAHEKK YVESVKG 17

SEQ ID NO: 369      moltype = AA length = 5
FEATURE            Location/Qualifiers
REGION              1..5
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..5
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 369
KYYLN 5

SEQ ID NO: 370      moltype = AA length = 7
FEATURE            Location/Qualifiers
REGION              1..7
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..7
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 370
GFSFSKY 7

SEQ ID NO: 371      moltype = AA length = 6
FEATURE            Location/Qualifiers
REGION              1..6
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..6
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 371
HQQAHE 6

```


SEQUENCE: 377						
gaagtgcagc	tgttggaaga	cggcggtggc	ctggtgcagc	caggttggta	cctgcgcctgc	60
agctgcgccg	ccagcgcgctt	tagcttcacg	aaataattact	tgaactgggt	tgccagaccg	120
ccaggcaaa	gacctggaatg	ggtggccagc	attcaccagc	aagcacacga	gaaaaaatac	180
gtggagctccg	tgaaaaggccg	ctttaccatt	agccgcgata	accgccaaaa	cagcctctgat	240
ctgcgaatatg	acacgcctgcg	ggcccagaat	accgcgctgt	attattgcgc	gcgtagcctg	300
cgctcgtcgta	gcactgagca	cgcaggattc	gacgtttggg	gccaggggcac	cctggttact	360
gtctcgagcg	cgctgaccaac	agggccccag	gtgttccctc	tgtgcccacg	cagcaagaagc	420
accctctggcg	gaacgcagcgc	cctgggctgc	ctgtgcgaag	actacttccc	cggcccgctg	480
accgtgtcct	ggaactctgg	cgcctgacc	agcggcgctgc	acacetttec	agccgtgtctc	540
cagacgacgc	gcctctacag	cctgtagcag	gtcgtgaecg	tggccacgag	cagcctggggc	600
accagacaet	acatctgcga	cgtgaaccac	agaccacga	acacaaagg	ggacaaggcg	660
gtggaacca	agagctgcga	caagaccac	acctgtccc	cctgcctcgc	cctgaagcg	720
cggggaggcc	ctctcgctgt	cttgttcccc	ccaaagctca	aggacacct	gatgatcag	780
cggacccccg	aagtgcactg	ctgtggtgtc	gacgtgtccc	accgaggacc	tgaagtgaag	840
tttaattggt	acgtggacgg	cgtggaagtg	cacaacgcca	agaccaagcc	cagagaggaa	900
cagtacaaac	gcacctaccg	ggctggtgtc	gtctgacgac	tgtctgacca	ggaactggctg	960
aacggcgaag	agtcacaagt	caaggtgtcc	atacgaagcc	tgctgcgcc	catcgaaaa	1020
accatcagca	aggccaaaag	ccagccccgc	gagccccagg	tgtaacact	gccccctagc	1080
cgggaagaga	tgcctagaac	ccagggtgtc	ctgcagtggc	ctctgaaggg	ctcttacccc	1140
agcacattg	cgctggaatg	ggagagcaac	ggccagcccg	agaaacatac	caagaccccc	1200
ccccctgtgc	tggacagcga	cggctcattc	ttctctgtac	gcaagctgac	cgtggacaag	1260
agccggtggc	agcagagccat	cgtgttcacg	tgtccctgta	tgcacagggc	cctgcacaa	1320
cactacaccc	agaagctcoa	qaqcttgaqc	cccqqcaaq			1359

```

SEQ ID NO: 378      moltype = AA  length = 10
FEATURE            Location/Qualifiers
REGION             1..10
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source             1..10
                  mol_type = protein
                  organism = synthetic construct

```

SEQUENCE: 378
GFTFSRYVIN 10

```

SEQ ID NO: 379      moltype = AA  length = 17
FEATURE            Location/Qualifiers
REGION             1..17
                   note = source = /note="Description of Artificial Sequence:
                   Synthetic peptide"
source             1..17
                   mol_type = protein
                   organism = synthetic construct

```

SEQUENCE: 379
SIHOHGLETR YVESVKG 17

```

SEQ ID NO: 380      moltype = AA  length = 5
FEATURE             Location/Qualifiers
REGION              1..5
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..5
                    mol_type = protein
                    organism = synthetic construct

```

SEQUENCE: 380
RYYIN

```
SEQ ID NO: 381      moltype = AA length = 7
FEATURE            Location/Qualifiers
REGION             1..7
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source             1..7
                  mol_type = protein
                  organism = synthetic construct
```

SEQUENCE: 381
GFTFSRY

```
SEQ ID NO: 382      moltype = AA  length = 6
FEATURE            Location/Qualifiers
REGION            1..6
                  note = source = /note="Description of Artificial Sequence:
                  Synthetic peptide"
source            1..6
                  mol_type = protein
                  organism = synthetic construct
```

SEQUENCE: 382
HOHGLE 6

```

SEQ ID NO: 383      moltype = AA  length = 8
FEATURE             Location/Qualifiers
REGION              1..8
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..8
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 383
GFTFSRYY

```

```

SEQ ID NO: 384      moltype = AA  length = 8
FEATURE             Location/Qualifiers
REGION              1..8
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic peptide"
source              1..8
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 384
IHOHGLET

```

```

SEQ ID NO: 385      moltype = AA  length = 123
FEATURE             Location/Qualifiers
REGION              1..123
                    note = source = /note="Description of Artificial Sequence:
                    Synthetic polypeptide"
source              1..123
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 385
EVQLVESGGG LVQPGGSLRL SCAASGFTFS RYYINWVRQA PGKGLEWVAS IHQHGLETRY   60
VESVKGKRFIT SRDPAKNSLY LQMNSLRAED TAVYYCARSL RRRSTEHAGF DVWGQGTLVLT 120
VSS                                                     123

```

```

SEQ ID NO: 386          multype = DNA   length = 369
FEATURE                Location/Qualifiers
misc_feature            1..369
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polynucleotide"
source                 1..369
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 386
gaagtgcagc tgggtgaaag cggcggtggc ctggtgcagc caggtggtag cctgcgctctg 60
agctgcgcgc ccagcggggt tacttttttc agatattaca ttaattgggt tcgccaggcc 120
ccaggcaaa gcttggaaatg ggtggcgagc atccaccagc acgcctctgc gaccagatat 180
gtggaatctg tcaaagggcg cttaccatt agcccgcata accgcaaaaa cagcctgtat 240
ctgcaaatga acagcctgcg gcgcgaagat acgcgcgtgt attattgcgc gcgtgacctg 300
cgtcgtcgta gcaactgagca cgcaggattc gacgtttggg gccaggggcac cctggttact 360
gtctcgagc

```

```

SEQ ID NO: 387          moltype = AA length = 453
FEATURE                Location/Qualifiers
REGION                 1..453
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polypeptide"
source                 1..453
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 387
EVQLVESGGG LVQPGGSLRL SCAASGFTFS RYYINWVRQA PGKGLEWVAS IHQHGLEYRY 60
VESYKGRFTI SRDNAKNSLY LQMSLSRAED TAYYYCARSL RRRSTEHAGF DVWGQGTLTVT 120
VSSASTKGPS VFPLAPSSKS TSGGTAALGC LVKDYFPEPV TVSWNSGALT SGVHTFPAVL 180
QSSGLYSLSS VVTVPSSSLG RTQYICNVNH KPSNTKVDKR VEPKSCDKTH TCPPCPAPEA 240
AGGPSVFLFP PKPDTLMIS TPTEVTCVVV DVSHEDEPVE FNMWYDGVVEV HNAKTKPREE 300
QYNSTYRVVS VLTVLHQDWL NGKEYKCKVS NKALPAPIEK TISKAKGQPR EPQVYTLPPS 360
REEMTKNQVS LTCLVKGFYP SDIAVESWESN GQPENNYKTT PPVLSDSGSF FLYSKLTVDK 420
SRWOGNVFS CSVMHEALHN HYTKSLSLGS PGK 453

```

```

SEQ ID NO: 388      moltype = DNA  length = 1359
FEATURE             Location/Qualifiers
misc_feature         1..1359
                     note = source = /note="Description of Artificial Sequence:
                     Synthetic polynucleotide"
source               1..1359
                     mol_type = other DNA
                     organism = synthetic construct

```



```

SEQUENCE: 393
gaagtgcagc tgcctgaaaag cgggtggcgggt ctggtgcagc caggtggttag cctgcgcctg 60
agctgtgcgc caagcggggtt tacattttcc agctatgcta tcagctgggt gcgccaagca 120
ccaggcaaaag gccttgaatg ggtgagcagc attagctcac atggatatta caccgcgtat 180
gccgagtcgc tgaagggtcg cttaccatt agtcgcgata acagcaaaaa caccctgtat 240
ctgcaaatga acagcctgcg gccagaagat accgcagttt attattgcgc gcgaccttat 300
ctgggtgacc gtctgtagcta tggtttcgac cactggggcc agggcacccct ggttactgtc 360
tcgagc 366

SEQ ID NO: 394 moltype = AA length = 452
FEATURE Location/Qualifiers
REGION 1..452
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
source 1..452
mol_type = protein
organism = synthetic construct

SEQUENCE: 394
EVQLLESGGG LVQPGGSLRL SCAASGFTFS SYAISVWRQA PGKGLEWVSS ISSHGYYTRY 60
AESVKGRTTI SRDNSKNTLY LQMNSLRAED TAVYYCARYP LGDRRSYGFDF HWQGGLTVTV 120
SSASTKGPSV FPLAPSSKST SGGTAAALGCL VKDYFPEPVT VSWNSGALTS GVHTFPAVLQ 180
SSGLYSLSV VTPSSSLGT QTYICNVNHK PSNTKVDKRV EPKSCDKTHT CPPCPAPEAA 240
GGPSVFLFPP KPKDTLMISR TPEVTCVVVD VSHEDPEVKF NWYVDGVEVH NAKTKPREEQ 300
YNSTYRVVSV LTVLHQDWLNL KKEYCKYVSN KALPAPIEKI ISKAKGQPRE PQVITLPPSR 360
EEMTKNQVSL TCLVKGFYPS DIAVEWESNG QPENNYKTTT PVLDSGDSFF LYSKLTVDKS 420
RWQGNVFSC SVMHEALHNNH YTQKSLSLSP GK 452

SEQ ID NO: 395 moltype = DNA length = 1356
FEATURE Location/Qualifiers
misc_feature 1..1356
note = source = /note="Description of Artificial Sequence:
Synthetic polynucleotide"
source 1..1356
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 395
gaagtgcagc tgcctgaaaag cgggtggcgggt ctggtgcagc caggtggttag cctgcgcctg 60
agctgtgcgc caagcggggtt tacattttcc agctatgcta tcagctgggt gcgccaagca 120
ccaggcaaaag gccttgaatg ggtgagcagc attagctcac atggatatta caccgcgtat 180
gccgagtcgc tgaagggtcg cttaccatt agtcgcgata acagcaaaaa caccctgtat 240
ctgcaaatga acagcctgcg gccagaagat accgcagttt attattgcgc gcgaccttat 300
ctgggtgacc gtctgtagcta tggtttcgac cactggggcc agggcacccct ggttactgtc 360
tcgagcgcgt cgaccaaagg cccagcgtg tcccccttgg cccccagcag caagagcacc 420
tctggcggaa cagccgcctt gggctgcctg gtcaaggact acttccccga gcccgtagc 480
gtgtccttga actctggcgc cctgaccagc ggctgtcacca cctttccagc cgtgtctcag 540
agcagcggcc tgtacagcct gaggcagctc gtgaccgtgc ccagcagcag cctggggcacc 600
cagacctaca tctgcaacgt gaaccacaag cccagcaaca caaaggtgga caagcgggtg 660
gaacccaaga gctgcgacaa gaccacacac tgtccccctt gccctgcccc tgaagcggcg 720
ggagggccct cctgtgttct gtccccca aagccttaagg acacctgat gatcagccgc 780
acccccgaag tgacctgcgt ggtgtgtgac gtgtccccag aggaccttga agtgaagttt 840
aatttggtacg tggacggcgt ggaagtgcac aacgccaaga ccaagcccag agaggaacag 900
tacaacagca cctaccgggt ggtgtccgtg ctgaccgtgc tgcaccagga ctggctgaac 960
ggcaaaagagt acaagtgcaa ggtgtccaac aaggccctgc ctgcccccat cgagaaaaacc 1020
atcagcaagg ccaaaggcca gccccgcgag ccccgaggtt acacactgcc ccttagccgg 1080
gaagagatga ccaagaacca ggtgtccctg acctgcctcg tgaagggcct ctaccccagc 1140
gacattgcgc tggaaatggga gagcaacggc cagcccgaga acaactacaa gaccaccccc 1200
cctgtgtcgc agcagcagcg ctctattctt ctgtacagca agctgaccgt ggacaagagc 1260
cgggtggcagc agggcaacgt gttcagctgc tccgtgatgc acgagggcct gcacaaccac 1320
tacaccaga agtccctgag cctgagcccc ggcaag 1356

SEQ ID NO: 396 moltype = AA length = 10
FEATURE Location/Qualifiers
REGION 1..10
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"
source 1..10
mol_type = protein
organism = synthetic construct

SEQUENCE: 396
GFTFASYAIT 10

SEQ ID NO: 397 moltype = AA length = 17
FEATURE Location/Qualifiers
REGION 1..17
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"
source 1..17
mol_type = protein
organism = synthetic construct

```

-continued

SEQUENCE: 397
TISGSGVYTY YAESVKG 17

SEQ ID NO: 398 moltype = AA length = 5
FEATURE Location/Qualifiers
REGION 1..5
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
source 1..5
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 398
SYAIT 5

SEQ ID NO: 399 moltype = AA length = 7
FEATURE Location/Qualifiers
REGION 1..7
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
source 1..7
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 399
GFTFASY 7

SEQ ID NO: 400 moltype = AA length = 6
FEATURE Location/Qualifiers
REGION 1..6
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
source 1..6
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 400
SGSGVY 6

SEQ ID NO: 401 moltype = AA length = 8
FEATURE Location/Qualifiers
REGION 1..8
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
source 1..8
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 401
GFTFASYA 8

SEQ ID NO: 402 moltype = AA length = 8
FEATURE Location/Qualifiers
REGION 1..8
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
source 1..8
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 402
ISGSGVYT 8

SEQ ID NO: 403 moltype = AA length = 122
FEATURE Location/Qualifiers
REGION 1..122
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
source 1..122
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 403
EVQLLES GGG LVQPGGSLRL SCAASGFTFA SYAITWVRQA PGKGLEWVST ISGSGVYTTY 60
AESVKGRFTI SRDNSKNTLY LQMNLSRAED TAVYYCARPY LGDRRSYGFH HWGQGLVTV 120
SS 122

SEQ ID NO: 404 moltype = DNA length = 366
FEATURE Location/Qualifiers
misc_feature 1..366
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
source 1..366
 mol_type = other DNA
 organism = synthetic construct

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SEQUENCE: 408
SISASGYAN YAGSVKG 17

SEQ ID NO: 409 moltype = AA length = 5
FEATURE Location/Qualifiers
REGION 1..5
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
source 1..5
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 409
TYAMT 5

SEQ ID NO: 410 moltype = AA length = 7
FEATURE Location/Qualifiers
REGION 1..7
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
source 1..7
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 410
GFTFGTY 7

SEQ ID NO: 411 moltype = AA length = 6
FEATURE Location/Qualifiers
REGION 1..6
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
source 1..6
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 411
SASGY 6

SEQ ID NO: 412 moltype = AA length = 8
FEATURE Location/Qualifiers
REGION 1..8
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
source 1..8
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 412
GFTFGTYA 8

SEQ ID NO: 413 moltype = AA length = 5
FEATURE Location/Qualifiers
REGION 1..5
 note = source = /note="Description of Artificial Sequence:
 Synthetic peptide"
source 1..5
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 413
ISASG 5

SEQ ID NO: 414 moltype = AA length = 122
FEATURE Location/Qualifiers
REGION 1..122
 note = source = /note="Description of Artificial Sequence:
 Synthetic polypeptide"
source 1..122
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 414
EVQLLESQGG LVQPGGSLRL SCAASGFTFG TYAMTWVRQA PGKGLEWVSS ISASGYANY 60
AGSVKGRFTI SRDNSKNTLY LQMNLSRAED TAVYYCARPY LGDRRSYGFQ HWGQGLVTV 120
SS 122

SEQ ID NO: 415 moltype = DNA length = 366
FEATURE Location/Qualifiers
misc_feature 1..366
 note = source = /note="Description of Artificial Sequence:
 Synthetic polynucleotide"
source 1..366
 mol_type = other DNA
 organism = synthetic construct

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SEQUENCE: 419
SISGGGYHTQ YAGSVKG 17

SEQ ID NO: 420 moltype = AA length = 5
FEATURE Location/Qualifiers
REGION 1..5
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"
source 1..5
mol_type = protein
organism = synthetic construct

SEQUENCE: 420
DYAIS 5

SEQ ID NO: 421 moltype = AA length = 7
FEATURE Location/Qualifiers
REGION 1..7
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"
source 1..7
mol_type = protein
organism = synthetic construct

SEQUENCE: 421
GFTFSDY 7

SEQ ID NO: 422 moltype = AA length = 6
FEATURE Location/Qualifiers
REGION 1..6
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"
source 1..6
mol_type = protein
organism = synthetic construct

SEQUENCE: 422
SGGGYH 6

SEQ ID NO: 423 moltype = AA length = 8
FEATURE Location/Qualifiers
REGION 1..8
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"
source 1..8
mol_type = protein
organism = synthetic construct

SEQUENCE: 423
GFTFSDYA 8

SEQ ID NO: 424 moltype = AA length = 8
FEATURE Location/Qualifiers
REGION 1..8
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"
source 1..8
mol_type = protein
organism = synthetic construct

SEQUENCE: 424
ISGGGYHT 8

SEQ ID NO: 425 moltype = AA length = 122
FEATURE Location/Qualifiers
REGION 1..122
note = source = /note="Description of Artificial Sequence:
Synthetic polypeptide"
source 1..122
mol_type = protein
organism = synthetic construct

SEQUENCE: 425
EVQLLES GGG LVQPGGSLRL SCAASGFTFS DYAIWVRQA PGKGLEWVSS ISGGGYHTQY 60
AGSVKGRFTI SRDNSKNTLY LQMNLSRAED TAVYYCARPY LGDRRSYGFH HWGQGTLLTV 120
SS 122

SEQ ID NO: 426 moltype = DNA length = 366
FEATURE Location/Qualifiers
misc_feature 1..366
note = source = /note="Description of Artificial Sequence:
Synthetic polynucleotide"
source 1..366
mol_type = other DNA
organism = synthetic construct

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SEQUENCE: 426
gaagtgcagc tgctggaaag cgggtggcggc ctggtgcagc caggtggtag cctgcgcctg 60
agctgtgccg caagcggcgt taccttttcc gactatgcaa tcagctgggt gcgccaagca 120
ccaggcaaaag gcctggaatg ggtgagcagc atttccgggg gggggtatca tacacaatat 180
gcaggatccg tgaaaggccg ctttaccatt agtcgcgata acagcaaaaa caccctgtat 240
ctgcaaatga acagcctgcg ggcagaagat accgcagttt attattgcgc gcgaccttat 300
ctgggtgacc gtcgtagcta tggtttcgac cactggggcc agggcaccct ggttactgtc 360
tcgagc                                     366

SEQ ID NO: 427          moltype = AA length = 452
FEATURE                Location/Qualifiers
REGION                1..452
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polypeptide"
source                1..452
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 427
EVQLLESGGG LVQPGGSLRL SCAASGFTFS DYAIWVRQA PGKGLEWVSS ISGGGYHTQY 60
AGSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARPY LGDRRSYGFH HWGQGTTLVTV 120
SSASTKGPSV FPLAPSSKST SGGTAALGCL VKDYFPEPVT VSWNSGALTS GVHTFPAVLQ 180
SSGLYSLSLV VTPVSSSLGT QTYICNVNHK PSNTKVDKRV EPKSCDKTHT CPPCPAPEAA 240
GGPSVFLFPP KPKDTLMISR TPEVTCVVVD VSHEDPEVKF NWYVDGVEVH NAKTKPREEQ 300
YNSTYRVVSV LTVLHQDWLN GKEYKCKVSN KALPAPIEKT ISKAKGQPRE PQVYTLPPSR 360
EMMTKNQVSL TCLVKGFYPS DIAVEWESNG QPENNYKTTP PVLDSGGSFF LYSKLTVDKS 420
RWQQGNVFSK SVMHEALHNNH YTKSLSLSP GK                                     452

SEQ ID NO: 428          moltype = DNA length = 1356
FEATURE                Location/Qualifiers
misc_feature          1..1356
                        note = source = /note="Description of Artificial Sequence:
                        Synthetic polynucleotide"
source                1..1356
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 428
gaagtgcagc tgctggaaag cgggtggcggc ctggtgcagc caggtggtag cctgcgcctg 60
agctgtgccg caagcggcgt taccttttcc gactatgcaa tcagctgggt gcgccaagca 120
ccaggcaaaag gcctggaatg ggtgagcagc atttccgggg gggggtatca tacacaatat 180
gcaggatccg tgaaaggccg ctttaccatt agtcgcgata acagcaaaaa caccctgtat 240
ctgcaaatga acagcctgcg ggcagaagat accgcagttt attattgcgc gcgaccttat 300
ctgggtgacc gtcgtagcta tggtttcgac cactggggcc agggcaccct ggttactgtc 360
tcgagcgcgt cgaccaaagg ccccgagcgtg ttccctctgg ccccgagcag caagagcacc 420
tctggcggaa cagcgcgcct ggggtgcctg gtcaaggact acttccccga gcccgtagc 480
gtgtcctgga actctggcgc cctgaccagc ggcgtgcaca cctttccagc cgtgtccag 540
agcagcggcc tgtacagcct gagcagcgtc gtgaccgtgc ccagcagcag cctgggcacc 600
cagacctaca tctgcaacgt gaaccacaag cccagcaaca caaagggtgga caagcgggtg 660
gaacccaaga gctgcgacaa gacccacacc tgtccccctt gccctgcccc tgaagcggcg 720
ggaggcccc cgtgttccct gttccccca aagcctaagg acaccctgat gatcagccgg 780
acccccgaag tgacctgcgt ggtggtggac gtgtcccacg aggacctga agtgaagttt 840
aattggtacg tggacggcgt ggaagtgcac aacgccaaga ccaagcccag agaggaaacag 900
tacaacagca cctaccgggt ggtgtccgtg ctgaccgtgc tgcaccagga ctggctgaac 960
ggcaaaagat acaagtgcac ggtgtccaac aaggccctgc ctgcccccat cgagaaaacc 1020
atcagcaagg ccaaaggcca gccccgcgag cccaggtgt acacactgcc ccctagccgg 1080
gaagagatga ccaagaacca ggtgtccctg acctgcctcg tgaagggtct ctaccccagc 1140
gacattgccg tgggaatggga gagcaacggc cagcccagga acaactacaa gaccaccccc 1200
cctgtgctgg acagcgacgg ctcatcttc ctgtacagca agctgaccgt ggacaagagc 1260
cgggtggcagc agggcaacgt gttcagctgc tccgtgatgc acgaggccct gcacaaccac 1320
tacaccagca agtccctgag cctgagcccc ggcaag                                     1356

SEQ ID NO: 429          moltype = AA length = 17
FEATURE                Location/Qualifiers
VARIANT              1
                        note = /replace="V"
VARIANT              3
                        note = /replace="E"
VARIANT              5
                        note = /replace="K"
VARIANT              7
                        note = /replace="N"
VARIANT              9
                        note = /replace="I"
VARIANT             10
                        note = /replace="P"
SITE                1..17
                        note = /note="Variant residues given in the sequence have
                        no preference with respect to those in the annotations for
                        variant positions"
REGION              1..17

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	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 429		
AISSDGSYIY YADSVKG		17
SEQ ID NO: 430	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
VARIANT	1	
	note = /replace="Q"	
VARIANT	3	
	note = /replace="E" or "T" or "I"	
VARIANT	4	
	note = /replace="W"	
VARIANT	5	
	note = /replace="V" or "R" or "A" or "T" or "M"	
VARIANT	6	
	note = /replace="V" or "R" or "A"	
SITE	1..9	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..9	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 430		
MQSYEKPR		9
SEQ ID NO: 431	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
VARIANT	1	
	note = /replace="E"	
VARIANT	3	
	note = /replace="K"	
VARIANT	5	
	note = /replace="N"	
SITE	1..6	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 431		
SSDGSY		6
SEQ ID NO: 432	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
VARIANT	1	
	note = /replace="E" or "T" or "I"	
VARIANT	2	
	note = /replace="W"	
VARIANT	3	
	note = /replace="V" or "R" or "A" or "T" or "M"	
VARIANT	4	
	note = /replace="V" or "R" or "A"	
SITE	1..6	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 432		
SYEKPR		6
SEQ ID NO: 433	moltype = AA length = 8	
FEATURE	Location/Qualifiers	

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VARIANT	2	
	note = /replace="E"	
VARIANT	4	
	note = /replace="K"	
VARIANT	6	
	note = /replace="N"	
VARIANT	8	
	note = /replace="T"	
SITE	1..8	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 433		
ISSDGSYI		8
SEQ ID NO: 434	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
VARIANT	3	
	note = /replace="E" or "T" or "I"	
VARIANT	5	
	note = /replace="R" or "A" or "T" or "M"	
VARIANT	6	
	note = /replace="V" or "R" or "A"	
SITE	1..9	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..9	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 434		
QQSWVKPRT		9
SEQ ID NO: 435	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
VARIANT	1	
	note = /replace="E" or "T" or "I"	
VARIANT	3	
	note = /replace="R" or "A" or "T" or "M"	
VARIANT	4	
	note = /replace="V" or "R" or "A"	
SITE	1..6	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 435		
SWVKPR		6
SEQ ID NO: 436	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
VARIANT	5	
	note = /replace="S" or "Q"	
SITE	1..10	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..10	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 436		
GPTFNTHYIH		10

SEQ ID NO: 437	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
VARIANT	3	
	note = /replace="G"	
VARIANT	4	
	note = /replace="G"	
VARIANT	5	
	note = /replace="Q"	
VARIANT	7	
	note = /replace="Q" or "G"	
VARIANT	8	
	note = /replace="N" or "M"	
VARIANT	10	
	note = /replace="L"	
SITE	1..17	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..17	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 437		
SISSSGSSTY YADSVKG		17
SEQ ID NO: 438	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
VARIANT	5	
	note = /replace="S" or "Q"	
SITE	1..7	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 438		
GFTFNTH		7
SEQ ID NO: 439	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
VARIANT	1	
	note = /replace="G"	
VARIANT	2	
	note = /replace="G"	
VARIANT	3	
	note = /replace="Q"	
VARIANT	5	
	note = /replace="Q" or "G"	
VARIANT	6	
	note = /replace="N" or "M"	
SITE	1..6	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 439		
SSSGSS		6
SEQ ID NO: 440	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
VARIANT	5	
	note = /replace="S" or "Q"	
SITE	1..8	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..8	

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	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8 mol_type = protein organism = synthetic construct	
SEQUENCE: 440		
GFTFNTHY		8
SEQ ID NO: 441	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
VARIANT	2 note = /replace="G"	
VARIANT	3 note = /replace="G"	
VARIANT	4 note = /replace="Q"	
VARIANT	6 note = /replace="Q" or "G"	
VARIANT	7 note = /replace="N" or "M"	
SITE	1..8 note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8 mol_type = protein organism = synthetic construct	
SEQUENCE: 441		
ISSSGSST		8
SEQ ID NO: 442	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
VARIANT	4 note = /replace="G"	
VARIANT	7 note = /replace="Q"	
VARIANT	8 note = /replace="N"	
SITE	1..17 note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..17	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 442		
SISSSGSSTY YADSVKG		17
SEQ ID NO: 443	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
VARIANT	2 note = /replace="G"	
VARIANT	5 note = /replace="Q"	
VARIANT	6 note = /replace="N"	
SITE	1..6 note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6 mol_type = protein organism = synthetic construct	
SEQUENCE: 443		
SSSGSS		6
SEQ ID NO: 444	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
VARIANT	3 note = /replace="G"	

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VARIANT	6	
	note = /replace="Q"	
VARIANT	7	
	note = /replace="N"	
SITE	1..8	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8	
	mol_type = protein organism = synthetic construct	
SEQUENCE: 444		
ISSSGSST		8
SEQ ID NO: 445	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
VARIANT	3	
	note = /replace="T"	
VARIANT	6	
	note = /replace="K" or "R"	
VARIANT	8	
	note = /replace="Y"	
VARIANT	9	
	note = /replace="L"	
VARIANT	10	
	note = /replace="N"	
SITE	1..10	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..10	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..10	
	mol_type = protein organism = synthetic construct	
SEQUENCE: 445		
GFSFSSYWIS		10
SEQ ID NO: 446	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
VARIANT	1	
	note = /replace="N"	
VARIANT	3	
	note = /replace="H"	
VARIANT	5	
	note = /replace="Q" or "H"	
VARIANT	6	
	note = /replace="A"	
VARIANT	7	
	note = /replace="H" or "L"	
VARIANT	9	
	note = /replace="K"	
VARIANT	10	
	note = /replace="K" or "R"	
SITE	1..17	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..17	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..17	
	mol_type = protein organism = synthetic construct	
SEQUENCE: 446		
SIKQSGSETY YVESVKG		17
SEQ ID NO: 447	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
VARIANT	1	
	note = /replace="K" or "R"	
VARIANT	3	
	note = /replace="Y"	
VARIANT	4	
	note = /replace="L"	

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VARIANT	5
	note = /replace="N"
SITE	1..5
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"
REGION	1..5
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..5
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 447	
SYWIS	5
SEQ ID NO: 448	moltype = AA length = 7
FEATURE	Location/Qualifiers
VARIANT	3
	note = /replace="T"
VARIANT	6
	note = /replace="K" or "R"
SITE	1..7
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"
REGION	1..7
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..7
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 448	
GFSFSSY	7
SEQ ID NO: 449	moltype = AA length = 6
FEATURE	Location/Qualifiers
VARIANT	1
	note = /replace="H"
VARIANT	3
	note = /replace="Q" or "H"
VARIANT	4
	note = /replace="A"
VARIANT	5
	note = /replace="H" or "L"
SITE	1..6
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"
REGION	1..6
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..6
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 449	
KQSGSE	6
SEQ ID NO: 450	moltype = AA length = 8
FEATURE	Location/Qualifiers
VARIANT	3
	note = /replace="T"
VARIANT	6
	note = /replace="K" or "R"
VARIANT	8
	note = /replace="Y"
SITE	1..8
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"
REGION	1..8
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..8
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 450	
GFSFSSYW	8
SEQ ID NO: 451	moltype = AA length = 8

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FEATURE	Location/Qualifiers	
VARIANT	2	
	note = /replace="H"	
VARIANT	4	
	note = /replace="Q" or "H"	
VARIANT	5	
	note = /replace="A"	
VARIANT	6	
	note = /replace="H" or "L"	
VARIANT	8	
	note = /replace="K"	
SITE	1..8	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 451		
IKQSGSET		8
SEQ ID NO: 452	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
VARIANT	6	
	note = /replace="R"	
VARIANT	8	
	note = /replace="Y"	
VARIANT	10	
	note = /replace="N"	
SITE	1..10	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..10	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 452		
GFTFSSYWIS		10
SEQ ID NO: 453	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
VARIANT	5	
	note = /replace="H"	
VARIANT	6	
	note = /replace="A"	
VARIANT	7	
	note = /replace="L"	
VARIANT	9	
	note = /replace="K"	
VARIANT	10	
	note = /replace="R"	
SITE	1..17	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..17	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 453		
SIHQQGHETK YVESVKG		17
SEQ ID NO: 454	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
VARIANT	1	
	note = /replace="R"	
VARIANT	3	
	note = /replace="Y"	
VARIANT	5	
	note = /replace="N"	
SITE	1..5	

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	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..5	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 454		5
SYWIS		
SEQ ID NO: 455	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
VARIANT	6	
	note = /replace="R"	
SITE	1..7	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 455		7
GTFSSY		
SEQ ID NO: 456	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
VARIANT	3	
	note = /replace="H"	
VARIANT	4	
	note = /replace="A"	
VARIANT	5	
	note = /replace="L"	
SITE	1..6	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..6	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 456		6
HQQGHE		
SEQ ID NO: 457	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
VARIANT	6	
	note = /replace="R"	
VARIANT	8	
	note = /replace="Y"	
SITE	1..8	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..8	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 457		8
GTFSSYW		
SEQ ID NO: 458	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
VARIANT	4	
	note = /replace="H"	
VARIANT	5	
	note = /replace="A"	
VARIANT	6	
	note = /replace="L"	
VARIANT	8	
	note = /replace="K"	

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SITE	1..8 note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..8 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8 mol_type = protein organism = synthetic construct	
SEQUENCE: 458 IHQQGHET		8
SEQ ID NO: 459 FEATURE VARIANT	moltype = AA length = 11 Location/Qualifiers 2 note = /replace="S"	
VARIANT	3 note = /replace="L"	
VARIANT	4 note = /replace="P"	
VARIANT	6 note = /replace="L"	
SITE	1..11 note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..11 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 459 GAVAGQLGFD H		11
SEQ ID NO: 460 FEATURE VARIANT	moltype = AA length = 7 Location/Qualifiers 7 note = /replace="N"	
SITE	1..7 note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..7 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 460 GNSNRPS		7
SEQ ID NO: 461 FEATURE VARIANT	moltype = AA length = 11 Location/Qualifiers 4 note = /replace="D" or "G"	
VARIANT	5 note = /replace="S" or "A"	
VARIANT	6 note = /replace="P" or "F"	
VARIANT	7 note = /replace="T" or "P"	
VARIANT	8 note = /replace="S" or "R"	
VARIANT	9 note = /replace="S" or "F"	
VARIANT	10 note = /replace="S" or "V"	
SITE	1..11 note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..11 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..11 mol_type = protein organism = synthetic construct	

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SEQUENCE: 461
QSYTSSHGP V 11

SEQ ID NO: 462 moltype = AA length = 8
FEATURE Location/Qualifiers
VARIANT 2
note = /replace="D" or "G"
VARIANT 3
note = /replace="S" or "A"
VARIANT 4
note = /replace="P" or "F"
VARIANT 5
note = /replace="T" or "P"
VARIANT 6
note = /replace="S" or "R"
VARIANT 7
note = /replace="S" or "F"
VARIANT 8
note = /replace="S" or "V"
SITE 1..8
note = /note="Variant residues given in the sequence have
no preference with respect to those in the annotations for
variant positions"
REGION 1..8
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"
source 1..8
mol_type = protein
organism = synthetic construct

SEQUENCE: 462
YYTSSHGP 8

SEQ ID NO: 463 moltype = AA length = 13
FEATURE Location/Qualifiers
VARIANT 4
note = /replace="S"
VARIANT 5
note = /replace="L"
VARIANT 6
note = /replace="P"
VARIANT 8
note = /replace="L"
SITE 1..13
note = /note="Variant residues given in the sequence have
no preference with respect to those in the annotations for
variant positions"
REGION 1..13
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"
source 1..13
mol_type = protein
organism = synthetic construct

SEQUENCE: 463
ARGAVAGQLG FDH 13

SEQ ID NO: 464 moltype = AA length = 10
FEATURE Location/Qualifiers
VARIANT 5
note = /replace="G"
VARIANT 6
note = /replace="T"
VARIANT 9
note = /replace="M"
VARIANT 10
note = /replace="T"
SITE 1..10
note = /note="Variant residues given in the sequence have
no preference with respect to those in the annotations for
variant positions"
REGION 1..10
note = source = /note="Description of Artificial Sequence:
Synthetic peptide"
source 1..10
mol_type = protein
organism = synthetic construct

SEQUENCE: 464
GFTFSSYAIS 10

SEQ ID NO: 465 moltype = AA length = 17

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FEATURE	Location/Qualifiers	
VARIANT	1	
	note = /replace="S"	
VARIANT	4	
	note = /replace="S" or "G"	
VARIANT	5	
	note = /replace="H"	
VARIANT	7	
	note = /replace="Y"	
VARIANT	8	
	note = /replace="Y"	
VARIANT	9	
	note = /replace="A"	
VARIANT	10	
	note = /replace="R" or "N"	
VARIANT	13	
	note = /replace="G"	
SITE	1..17	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..17	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 465		
AISASGGSTY YAESVKG		17
SEQ ID NO: 466	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
VARIANT	1	
	note = /replace="T"	
VARIANT	4	
	note = /replace="M"	
VARIANT	5	
	note = /replace="T"	
SITE	1..5	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..5	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 466		
SYAIS		5
SEQ ID NO: 467	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
VARIANT	5	
	note = /replace="G"	
VARIANT	6	
	note = /replace="T"	
SITE	1..7	
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..7	
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 467		
GFTFSSY		7
SEQ ID NO: 468	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
VARIANT	2	
	note = /replace="S" or "G"	
VARIANT	3	
	note = /replace="H"	
VARIANT	5	
	note = /replace="Y"	
VARIANT	6	

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SITE	note = /replace="Y" 1..6 note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..6 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..6 mol_type = protein organism = synthetic construct	
SEQUENCE: 468		
SASGGS		6
SEQ ID NO: 469	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
VARIANT	5 note = /replace="G"	
VARIANT	6 note = /replace="T"	
SITE	1..8 note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..8 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8 mol_type = protein organism = synthetic construct	
SEQUENCE: 469		
GFTFSSYA		8
SEQ ID NO: 470	moltype = AA length = 8	
FEATURE	Location/Qualifiers	
VARIANT	3 note = /replace="G"	
VARIANT	4 note = /replace="H"	
VARIANT	6 note = /replace="Y"	
VARIANT	7 note = /replace="Y"	
SITE	1..8 note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..8 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..8 mol_type = protein organism = synthetic construct	
SEQUENCE: 470		
ISSGGST		8
SEQ ID NO: 471	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
VARIANT	4 note = /replace="S"	
VARIANT	5 note = /replace="H"	
VARIANT	9 note = /replace="A"	
VARIANT	10 note = /replace="N"	
VARIANT	13 note = /replace="G"	
SITE	1..17 note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"	
REGION	1..17 note = source = /note="Description of Artificial Sequence: Synthetic peptide"	
source	1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 471		

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SISASGYTTR YAESVKG	17
SEQ ID NO: 472	moltype = AA length = 6
FEATURE	Location/Qualifiers
VARIANT	2
	note = /replace="S"
VARIANT	3
	note = /replace="H"
SITE	1..6
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"
REGION	1..6
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..6
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 472	
SASGYY	6
SEQ ID NO: 473	moltype = AA length = 5
FEATURE	Location/Qualifiers
VARIANT	3
	note = /replace="S" or "G"
VARIANT	4
	note = /replace="H"
SITE	1..5
	note = /note="Variant residues given in the sequence have no preference with respect to those in the annotations for variant positions"
REGION	1..5
	note = source = /note="Description of Artificial Sequence: Synthetic peptide"
source	1..5
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 473	
ISASG	5

The invention claimed is:

1. An isolated nucleic acid or nucleic acids encoding the amino acid sequence of an anti-NPR1 antibody or antigen binding fragment, wherein the antibody or antigen binding fragment comprises three heavy chain complementarity determining regions (HCDR1, HCDR2, and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2, and LCDR3) selected from:

(I)
SEQ ID NO: 28 (HCDR1),
SEQ ID NO: 119 (HCDR2),
SEQ ID NO: 30 (HCDR3),
SEQ ID NO: 41 (LCDR1),
SEQ ID NO: 42 (LCDR2), and
SEQ ID NO: 134 (LCDR3);

(II)
SEQ ID NO: 31 (HCDR1),
SEQ ID NO: 119 (HCDR2),
SEQ ID NO: 30 (HCDR3),
SEQ ID NO: 41 (LCDR1),
SEQ ID NO: 42 (LCDR2), and
SEQ ID NO: 134 (LCDR3);

(III)
SEQ ID NO: 32 (HCDR1),
SEQ ID NO: 120 (HCDR2),
SEQ ID NO: 30 (HCDR3),
SEQ ID NO: 44 (LCDR1),
SEQ ID NO: 45 (LCDR2), and
SEQ ID NO: 135 (LCDR3); or

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(IV)
SEQ ID NO: 34 (HCDR1),
SEQ ID NO: 121 (HCDR2),
SEQ ID NO: 36 (HCDR3),
SEQ ID NO: 47 (LCDR1),
SEQ ID NO: 45 (LCDR2), and
SEQ ID NO: 134 (LCDR3).

2. A vector comprising the isolated nucleic acid(s) of claim 1.

3. A host cell comprising the isolated nucleic acid(s) of claim 1.

4. A method of producing an isolated anti-NPR1 antibody or antigen binding fragment, comprising culturing the host cell of claim 3 under conditions suitable for producing the antibody or antigen binding fragment.

5. The isolated nucleic acid(s) of claim 1, wherein the antibody or antigen binding fragment comprises:

(a) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 201, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 136; or
(b) a heavy chain variable region comprising an amino acid sequence of SEQ ID NO: 122, and a light chain variable region comprising an amino acid sequence of SEQ ID NO: 136.

6. A vector comprising the isolated nucleic acid(s) of claim 5.

7. A host cell comprising the isolated nucleic acid(s) of claim 5.

8. A method of producing an isolated anti-NPR1 antibody or antigen binding fragment, comprising culturing the host

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cell of claim 7 under conditions suitable for producing the antibody or antigen binding fragment.

9. The isolated nucleic acid(s) of claim 1, wherein the antibody or antigen binding fragment comprises:

- (a) a heavy chain comprising an amino acid sequence of SEQ ID NO: 203, and a light chain comprising an amino acid sequence of SEQ ID NO: 138; or
- (b) a heavy chain comprising an amino acid sequence of SEQ ID NO: 208, and a light chain comprising an amino acid sequence of SEQ ID NO: 138.

10. A vector comprising the isolated nucleic acid(s) of claim 9.

11. A host cell comprising the isolated nucleic acid(s) of claim 9.

12. A method of producing an isolated anti-NPR1 antibody or antigen binding fragment, comprising culturing the host cell of claim 11 under conditions suitable for producing the antibody or antigen binding fragment.

* * * * *

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